ILLUSTRATED RESEARCH AND ENGINEERING SPECIFICATION

HAMMOND B-3 AND LESLIE

From original mechanisms to a digital instrument

Proposed interface art direction. Engineering specifications govern the final controls.

Circuit sources and service drawings
Tonewheel and valve modeling
Leslie mechanics and acoustics
Measurement, validation and build plans

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500 PAGES / 80 RESEARCH ASSIGNMENTS / 10 WAVES

Evidence edition • 12 November 2026

What this atlas establishes

This is a research and implementation specification for a faithful digital recreation of a vintage Hammond B-3 and a Leslie system. It connects primary service information, physical mechanisms, original research, numerical models, implementation contracts, measurement plans, and a proposed plugin interface. Its 80 chapters are organized into ten research waves, with six focused pages per chapter.

The chosen baseline is a vintage electromechanical B-3 with an identified AO-28 revision and a vintage Leslie 122. Other variants have explicit identities. A production B-3 is not a modern digital B-3, and a Leslie 147 interface is not interchangeable with a 122 interface. Precise revisions and an actual reference specimen must be fixed before calibrating a definitive model. [S001-01] [S001-02]

No physical instrument was measured during this research. There is no completed audio engine, plugin binary, captured reference dataset, or demonstrated one-to-one fidelity claim in this package. The deliverable is a large, actionable knowledge base and a plan that makes such claims testable. Missing measurements are recorded as unknowns rather than silently replaced with plausible numbers.

Use the PDF to understand mechanisms and decisions. Use the chapter JSON files, source catalog, source image attribution, mission prompt, and acceptance matrix as structured input to an implementation agent. Historical patent drawings and original conceptual figures are labeled separately from production evidence.

Evidence class	Meaning for implementation
Documented	A specific inspected primary source supports the claim; retain its variant and locator.
Measured	A recorded specimen measurement with calibration and uncertainty. None has been newly acquired here.
Derived	A calculation from stated assumptions; check units and applicability.
Proposed	An original model, interface, budget, or test target requiring validation.
Unknown	Do not invent a value. Preserve a parameter and an acquisition task.

Lock the reference configuration

A useful one-to-one target is a defined instrument in a defined condition, recorded through a defined signal chain. The word B-3 alone does not specify generator voicing, contact condition, preamplifier revision, cabinet driver replacement, microphone geometry, or room response. Configure those facts before optimizing a model.

The configuration manifest is a dependency of the audio engine, not a marketing preset. A change of specimen or circuit revision must create a new versioned profile. Keep a nominal engineering profile available while measurements are missing, with its limitations visible to developers. Chapters 1,14,38,57 and 58 define the evidence and data contracts.

Field	Required record
Organ identity	Model, serial, estimated manufacture date with source, restoration and modifications.
Electrical variant	Mains region, generator identity, AO-28 code/revision, wiring evidence.
Generator data	Frequency and phase relationships, per-wheel waveform/level/filter parameters, leakage topology.
Keyboard data	Upper/lower contact geometry, actuation and release captures, preset matrix.
Organ circuits	Scanner line and switch topology, percussion behavior, expression mechanism, valve stages.
Rotary cabinet	122 baseline;147 as separate profile; amplifier, crossover, driver and rotor identities.
Acoustic setup	Microphone coordinates and orientation, sample clock, gain calibration, room description.
Condition	Warmup, temperature, supply conditions, faults, mechanical wear, altered components.
Rights and provenance	Recording permission, image license, drawing source, file hashes and revision history.
Acceptance status	Measured, documented, modeled, unverified, or deliberately out of scope.

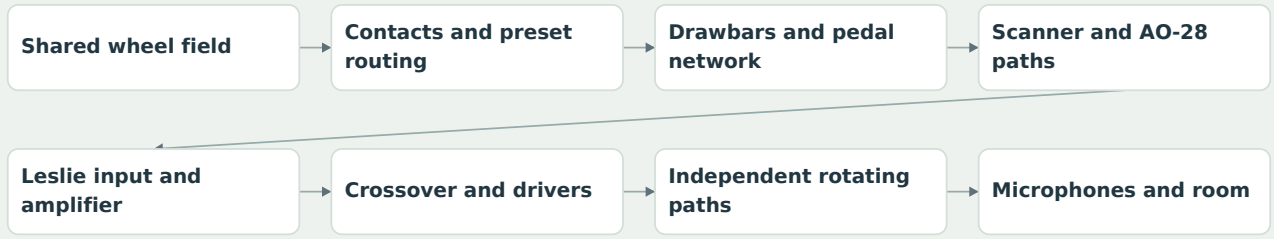
The system the agent must reproduce

Begin with shared, continuously running tone sources. Key contacts select and combine those sources through a wiring and drawbar network. Preserve the distinction between source generation, switching, linear networks, and nonlinear electronics. A voice-per-key synthesizer with restarted sine phases hides important mechanisms.

Trace the actual dry, vibrato, percussion and pedal paths on the selected schematic before ordering the DSP modules. The block view below is a responsibility map, not a replacement for the factory circuit. It deliberately groups internal routing whose ordering depends on the verified hardware revision.

The Leslie model starts with its own electrical input and amplifier behavior. It then separates the horn and bass paths, maintains independent rotor states, computes radiation and propagation to microphones, and includes room response only when the requested output represents a miked cabinet. Direct organ output and cabinet microphone output need distinct validation suites.

Original responsibility map



Conceptual software responsibilities. Internal branches and coupling must come from the selected circuit and measurements.

Build in dependency order

The research waves organize the book; implementation milestones organize the product. Begin with documentary and data contracts, build a deterministic dry engine, verify its behavior, and only then add the nonlinear and acoustic mechanisms. Establish measurable acceptance gates before producing a polished release interface.

A gate is passed by retained evidence: source provenance, tests, captures, plots, profile versions, and a reproducible build. A convincing demonstration or an attractive screenshot does not substitute for missing electrical or acoustical validation. Chapters 73,74 and 79 supply the work contract and detailed completion criteria.

Milestone	Deliverable and exit evidence
0 Reference lock	Specimen manifest, revision map, sources, unresolved facts and measurement permissions.
1 Testable skeleton	Framework-independent engine, sample-offset events, offline renderer, stable data schema.
2 Generator core	91 shared sources, explicit frequency/mapping tables, phase continuity and foldback tests.
3 Console behavior	Contact timing, drawbars, presets, pedals, percussion and expression tests.
4 Scanner model	Verified LC and switching topology, stable solver, six modes and direct-response comparisons.
5 Organ electronics	Revision-correct AO-28 networks, valve behavior, load interaction and alias protection.
6 Leslie electronics	Separate 122/147 profiles, crossover, driver and power-stage evidence.
7 Rotary acoustics	Independent rotor dynamics, directionality, propagation, microphones and mono tests.
8 Calibration	Retained training and held-out captures; fitting report and listening protocol.
9 Product and release	VST3/standalone host validation, reliable UI, automation/state compatibility and installers.

Use the visuals as evidence

Photographs identify shapes, finishes, markings and component relationships. Their evidential strength depends on provenance: a labeled vintage specimen is useful, an unknown replacement part is conditional, and a digitally reconstructed photograph cannot establish dimensions. Captions retain those limitations.

Historical patent plates explain inventions and construction options. A patented embodiment does not prove that a later factory B-3 used the same component values, dimensions or topology. Original conceptual diagrams and calculated plots explain proposed software responsibilities and mathematical behavior; they are not measured circuit responses.

The package contains reusable photographs with creator and license metadata, selected patent plates with source records, original figures, and an illustrative UI art direction. Full third-party service manuals remain linked references where redistribution permission has not been established. Exact printed-page and PDF-page locators make the original drawings easy to retrieve.

The companion image atlas and source catalog preserve longer descriptions, licenses, asset IDs and links. P005 and P020 have known retouching and must not support dimensional measurements. Sources showing a CV, M-series, modern B-3 or modern Leslie are contextual references unless exact applicability is established.

Visual type	Engineering use
Specimen photograph	Identify visible parts and surface detail; measure only with reliable scale and perspective.
Factory drawing	Transcribe the identified revision with independent checking; retain labels and page locator.
Patent plate	Understand mechanism and alternatives; verify production applicability separately.
Original technical plot	Follow stated equations and assumptions; illustrative parameters are not measurements.
Generated UI concept	Communicate art direction; implement controls from verified tables and interaction specifications.
Interactive prototype	Executable control study. Browser visual verification was blocked; no audio engine is included.

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Give the agent an evidence contract

Start by reading chapters 1,2,73,74,75 and 79. Then implement the next dependency-complete milestone and its tests. The companion mission prompt makes this instruction reusable; chapter JSON files let the agent retrieve a focused section instead of repeatedly ingesting the entire book.

For every physical constant, retain its unit, evidence class, source locator, applicable revision, uncertainty, and fitting history. For every subsystem, expose a deterministic offline test path. When a value is unknown, use an explicitly provisional parameter and create the acquisition task that will replace it. Never convert a plausible default into a factory fact.

Preserve the shared source field, wiring and contact behavior, upper-manual percussion trigger rules, scanner routing, and the distinction between organ and cabinet electronics. Use a modular circuit or behavioral representation with enough observability to compare intermediate signals. Do not tune downstream stages to conceal an upstream routing error.

Keep file access, allocation, locks, logging, image loading and rendering off the audio callback. Test the same musical event stream across sample rates, block sizes and supported host layouts. A successful build is necessary but insufficient: retain functional, numerical, musical and performance evidence for each release.

Build the visual product after the signal contracts stabilize. The UI must present verified control behavior, stable automation IDs, accessible alternatives, resizable layouts, and honest calibration status. Finish with a reproducible build, test report, source and asset notices, open issues, and an explicit statement of what has and has not been validated.

HARDWARE EVIDENCE AND PROPOSED SCOPE

Define the instrument before defining fidelity

Reference instrument and configuration contract

The reference is an original electromechanical B-3, paired with an AO-28 preamplifier and vintage Leslie 122. This is a proposed project configuration, not an assertion that every B-3 was sold with this particular speaker. The original service description identifies two 61-note manuals, a 25-note pedalboard, reverse-colored preset keys, and separate manual drawbar groups. [S001-01] The manufacturer later describes the B-3-type generator as having 91 active musical wheels, with five additional blank positions maintaining mechanical balance. [S001-03]

Do not search for a vintage B-3 chipset and then substitute a modern organ motherboard. The manufacturer's B-3 mk2 manual explicitly describes digital tonewheels and DSP effects. Its documentation can help identify modern interface conventions, but its processor architecture is outside this reference target. [S001-02]

ORIGINAL DESIGN PROPOSAL: define three independently testable results. The first is functional correspondence: controls, note routing and interaction rules. The second is specimen correspondence: audio behavior measured against a named physical instrument. The third is visual correspondence: cabinet proportions, materials and control placement. Passing one does not imply passing the other two. A beautiful case model cannot substantiate an audio-fidelity claim; a convincing sustained tone cannot establish correct percussion behavior.

Contract field	Initial project decision	Evidence status
Organ family	Vintage B-3, not New B-3 or B-3 mk2	Scope decision
Preamp	AO-28 revision to be photographed and recorded	Exact revision unknown
Speaker	Vintage Leslie 122 baseline	Scope decision
Mains configuration	60 Hz baseline; 50 Hz separate profile	Proposed baseline
Specimen identity	Serials and maintenance history required	Not yet acquired

P002 | Hammond-museum-B-3.png | Фіксер | CC BY-SA 4.0 Source and license

CHECKS AND DECISIONS

• No preset or UI label may claim a measured specimen match before a corresponding capture dataset exists.

SOURCES S001-01 Hammond Organ Company service manual scan with supplementary B3/C3 redraws / S001-02 Hammond B-3 mk2 C-3 mk2 and B-3P mk2 Owner's Manual / S001-03 Hammond SK PRO Custom Sets reference chapter

CONFIGURATION AND SOURCE PROVENANCE

Keep revisions and neighboring models separate

Reference instrument and configuration contract

A downloaded file title is insufficient evidence of circuit identity. The service scan inspected here contains historical Hammond pages followed by supplementary circuit redraws bearing 1995 dates. Preserve those layers separately in the source catalog. A redraw can improve legibility without becoming an original factory drawing; any discrepancy must remain visible until checked against the relevant hardware or an independently verified schematic. [S001-01]

The original Leslie service information identifies 122 and 147 as closely related cabinets with different amplifier input and switching arrangements. [S001-04] For this project, their common acoustic implementation may share software, while their input and amplifier profiles remain separately identified. A user-facing cabinet selector must not silently swap only a label or cosmetic cabinet skin.

ORIGINAL DESIGN PROPOSAL: create a configuration fingerprint from organ serial, generator identity, preamp code, scanner/line revision, modification list, Leslie serial, amplifier label, drivers and mains configuration. Unknown fields remain explicit null values. Do not infer an exact circuit revision from year alone. If a replaced generator or amplifier is discovered, revise the fingerprint and invalidate the affected calibration rather than overwriting the historical record.

Variant	What may be reused	What requires separate evidence
Different B-3 specimen	Module interfaces and control logic	Wheel levels, leakage, contacts, circuit state
50 Hz configuration	Software architecture	Drive gearing, motor/scanner observations
Leslie 147	Acoustic abstractions where verified	Electrical profile and calibration
Modern B-3 mk2	Optional workflow references	Digital engine is outside vintage baseline
Modified instrument	Declared extension interfaces	Every changed component or signal path

P013 | Hammond b3 con leslie 122.jpg | it:user:Salli | CC BY-SA 3.0 Source and license

CHECKS AND DECISIONS

- A profile with an unknown preamp revision must remain marked provisional.
- Every circuit-derived parameter must identify its source drawing and revision.

SOURCES S001-01 Hammond Organ Company service manual scan with supplementary B3/C3 redraws / S001-04 Leslie Models 122 and 147 Service Information

ORIGINAL DATA MODEL

Represent evidence as part of the model

Reference instrument and configuration contract

ORIGINAL DESIGN PROPOSAL: treat a parameter as a value plus provenance, rather than a number detached from its origin. A documented component marking, a measured transfer curve, a value derived from a circuit equation, and an engineering default can all be useful; they do not carry equal evidential weight. Preserve the distinction in exported profiles and in the advanced inspection interface.

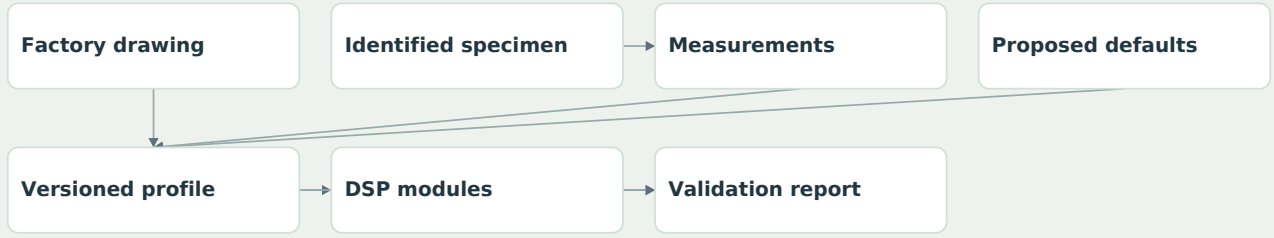
Use SI units internally and retain the original printed unit string in the evidence record. For a measurement, store method, instrument, stimulus, uncertainty estimate, operating conditions and asset hash. For a derived value, store the formula or fitting procedure and parent records. For a proposed default, record its rationale and the measurement that would replace it. Missing values must never be converted into zero automatically.

Keep mutable performance controls separate from immutable specimen identity. A musician can change drawbars or rotor speed without changing the reference instrument. A calibration edit creates a new profile version and an explicit difference report. This prevents a convenient sound-design adjustment from silently changing the meaning of a later validation result.

MODEL EQUATIONS

parameter = {value_SI, unit, evidence_kind, source_id, conditions, uncertainty, revision}

Evidence enters a versioned specimen profile



Original architecture proposal. Arrows describe evidence and data dependencies, not the organ's electrical wiring.

CHECKS AND DECISIONS

- Reject a numeric value whose unit is missing.
- Reject a measured tag without an associated acquisition record.
- Preserve unknown values across serialization without inventing defaults.

ORIGINAL IMPLEMENTATION CONTRACT

Make the first implementation executable and reviewable

Reference instrument and configuration contract

Start with a framework-independent engine and a machine-readable reference profile. The first implementation milestone is not a photorealistic plugin. It is an inspectable program that loads a profile, processes timestamped controls, renders offline audio, and writes a report stating which subsystems are implemented, approximated, disabled or unverified. Later chapters specify the detailed algorithms; this chapter establishes their accountability.

The engine must expose observation points at module boundaries so that an organ-only discrepancy can be distinguished from a speaker or room discrepancy. These are logical diagnostics, not instructions to attach test equipment to energized vintage circuitry. Preserve a trace of profile identity, software build, sample rate, event sequence and quality settings with every exported render.

ORIGINAL DESIGN PROPOSAL: ship a neutral development profile whose name includes 'unmeasured'. Selectable extensions such as velocity-sensitive level, altered foldback, extra reverb or modern rotor braking must identify themselves as extensions. They may be musically useful, but should not quietly redefine the historical reference. A reference preset must reset such options predictably, while retaining user audio-device preferences.

Initial deliverable	Review condition
Reference profile schema	Unknowns and units survive round trip
Offline renderer	Same deterministic input reproduces output
Module observation points	Each stage can be compared independently
Behavioral control script	Every baseline control has a test event
Status report	No unimplemented block appears as validated

```
profile = load_and_validate(profile_file)
assert profile.evidence_status_is_explicit()
engine.prepare(sample_rate, maximum_block)
engine.load_profile(profile)
for event_block in scripted_performance:
    engine.apply_sample_offset_events(event_block)
    audio, diagnostics = engine.render_block()
    capture.write(audio, diagnostics)
report.write(profile.hash, build.hash, capture.hash)
```

ORIGINAL MEASUREMENT PLAN

Acquire a specimen rather than an imagined average

Reference instrument and configuration contract

The manufacturer's own modern reference material names distinct vintage B-3 serial numbers as separate tonewheel profiles. [S001-03] This supports recording specimen identity, but it does not supply measurements for the instrument this project will emulate. No physical organ, serial number or capture session has been supplied for this atlas; the eventual reference dataset remains an acquisition task.

ORIGINAL DESIGN PROPOSAL: begin with an identification session. Photograph the console, serial plate, accessible chassis labels, speaker cabinet, amplifier label and driver markings. Record ownership permission, service history, known modifications, room, microphones, interface settings and the physical control positions. Obtain interior access and electrical measurements through a qualified vintage-organ technician; the software agent should request documented results rather than infer them from a cabinet photograph.

Capture quiet baseline recordings, isolated controls, sustained notes, overlapping note sequences, percussion retriggers, expression sweeps and rotor transitions. Repeat a subset at the beginning and end of the session to estimate within-session variation. Record direct organ and microphone signals as distinct datasets whenever the technician provides appropriate interfaces. Retain raw recordings; normalization, alignment and denoising belong to documented derived datasets.

Acquisition	Question it resolves
Identification photographs	Which physical assemblies are actually present?
Control-position photographs	Can another engineer reproduce the registration?
Repeated audio captures	How much does the reference vary by itself?
Independent module observations	Where does a mismatch enter the signal path?
Room and microphone coordinates	Which behavior belongs to recording geometry?
Maintenance and replacement record	Which characteristics are original or modified?

CHECKS AND DECISIONS

- Do not treat commercial recordings as isolated calibration signals.
- Do not replace missing direct recordings with filtered microphone audio and label it measured direct output.

SOURCES S001-03 Hammond SK PRO Custom Sets reference chapter

ORIGINAL ACCEPTANCE GATE

Define a defensible fidelity claim

Reference instrument and configuration contract

'One to one' must become a bounded engineering statement: which specimen, which controls, which signal levels, which recording geometry, which operating conditions and which tolerances? An unlimited identity claim cannot be established by a finite set of recordings. The useful project goal is traceable agreement across a declared operating envelope, with remaining uncertainty visible to the next engineer.

ORIGINAL DESIGN PROPOSAL: assign separate acceptance reports to behavior, audio, interface and deployment. Require held-out performances that were not used for parameter fitting. Compare repeated hardware captures before choosing audio tolerances, so that normal specimen variation is not mistaken for a model defect. Conversely, a generous perceptual preference score must not excuse a demonstrably wrong switching rule or an unstable real-time processor.

The first gate passes when scope, source provenance, profile schema and missing-data inventory are complete enough to implement. The final emulation gate passes only after subsystem and integrated measurements, listening evaluation, host tests and UI checks have been performed and recorded. Until then, describe the product as an evidence-informed model under development. Do not invent a percentage of authenticity, an ABX result or a production-readiness verdict.

Claim level	Evidence required
Document-informed architecture	Traceable primary references and explicit assumptions
Behaviorally verified model	Passing state and routing tests
Calibrated specimen profile	Identified hardware and reproducible measurements
Validated audio correspondence	Held-out comparisons and declared tolerances
Perceptually assessed recreation	Documented listening study and its limitations
Release-ready instrument	Host, performance, persistence and interface gates

CHECKS AND DECISIONS

- Reference identity is immutable within an acceptance report.
- Every advertised historical behavior has a matching acceptance test.
- Unmeasured defaults remain labeled after presets are saved and reloaded.
- The release statement identifies unresolved deviations instead of implying universal identity.

DOCUMENTARY EVIDENCE

Choose the drawing before choosing the circuit

Primary service manual and schematic source atlas

The most useful broad factory reference located is a 220-page service-manual scan. A second, B-3/C-3-focused PDF has 73 pages and ends with three redraws dated 1995. Keep these documents separate in the source catalog. The second file is neither merely three schematic pages nor a uniform original factory scan. [S002-01] [S002-02]

ORIGINAL DESIGN PROPOSAL: designate one drawing revision as the reference for each module before any coding begins. A source record describes the document; an applicability record states which specimen and module it supports. The latter requires evidence from the instrument being recreated. The project should allow a selected assembly to use a later revision without silently changing every neighboring assembly.

Treat the 220-page collection as a navigational foundation, then consult larger schematic images for transcription. Reading convenience is not authority. A clean redraw can help explain a connection, but the original drawing and a photograph of the selected chassis should resolve disagreement. Preserve all three readings when they differ and leave the affected parameter unresolved until reviewed.

Evidence layer	Required record	Engineering use
Factory sheet	Drawing identity and edition	Nominal topology and assembly scope
Later redraw	Author/date and source relationship	Readable cross-check, never silent replacement
Physical specimen	Serial, assembly code, photographs	Applicability and modifications
Measured capture	Setup, calibration, raw recording	Actual transfer and dynamic behavior

CHECKS AND DECISIONS

• A module cannot claim component-level fidelity while its drawing revision or specimen applicability is unknown.

SOURCES S002-01 Hammond Organ Service Manual Early Models, 220-page scan / S002-02 Hammond Organ B3 C3 Service Manual, 73-page compilation

VERIFIED SOURCE LOCATORS

A compact map into the original factory atlas

Primary service manual and schematic source atlas

The following locators use the 220-page scan. PDF page numbers are one-based viewer positions; printed numbers belong to the historical document. The two systems must be stored together. Scans of folded drawings, inserted supplements and parts plates make a single arithmetic page offset unreliable. Selected schematic and parts pages were rendered locally and visually inspected for their page identities. [S002-01]

ORIGINAL DESIGN PROPOSAL: navigate by subsystem rather than asking an agent to read hundreds of pages indiscriminately. Start with the source map, open the requested plate, then compare the associated description, wiring view and parts list. The circuit view establishes connectivity; the physical view identifies assemblies and orientation. Neither one alone proves that an undocumented replacement component matches the original.

Needed evidence	Printed locator	PDF page
Schematic index	3-1	80
B-3/C-3 system schematic	3-12	91
B-3/C-3/RT-3 preamplifier	3-50, Figure 24	129
Generator construction	5-1, Figure 5-1	170
Tonewheel tooth-count plate	5-2, Figure 5-2	171
Generator description	5-3	172
B-3 front-view parts plate	6-5, Figure 1	188
Manual breakdown	6-9, Figure 5	192

CHECKS AND DECISIONS

- Link opening must land on the intended one-based PDF page and show the expected printed number.
- Never substitute an adjacent model schematic just because its page is easier to read.

SOURCES S002-01 Hammond Organ Service Manual Early Models, 220-page scan

ORIGINAL DATA DESIGN

Source records that survive an engineering handoff

Primary service manual and schematic source atlas

ORIGINAL DESIGN PROPOSAL: store documentary evidence as immutable records and attach it to individual technical claims. A claim is more specific than a source: for example, a connection from one named terminal to another, a component marking, or the presence of a revision note. A citation to the whole manual is useful for discovery but insufficient for reproducible component extraction.

Use a document hash to distinguish downloaded versions with the same filename. Record the access date, original URL, local page dimensions, rotation, printed page label and a normalized crop rectangle. The crop rectangle provides a durable visual locator without embedding an entire unlicensed source image in every report. Keep source bytes in a restricted research collection where appropriate; distribute links and the original research conclusions separately.

The catalog must express uncertainty directly. Null means unknown, not zero. A candidate value stays outside a validated parameter profile until reviewed. If a higher-quality scan changes the reading, preserve the previous interpretation and explain the correction. This prevents a later developer from treating an early OCR guess as a factory specification.

Field	Meaning	Example form
source_id	Stable documentary identifier	S002-01
document_hash	Exact acquired file identity	SHA-256 hexadecimal
printed_page / pdf_page	Historical and viewer positions	3-12 / 91
crop_normalized	Unambiguous visual location	x0,y0,x1,y1 in [0,1]
reading_status	Evidence confidence	unreadable, candidate, verified
applies_to	Scope predicate	assembly code and revision
reviewer / date	Accountability	Recorded person or agent review

CHECKS AND DECISIONS

• Changing a parameter value must retain an explicit evidence record or an explicit modeling-assumption record.

IMPLEMENTATION SEQUENCE

Transcribe topology before interpreting the sound

Primary service manual and schematic source atlas

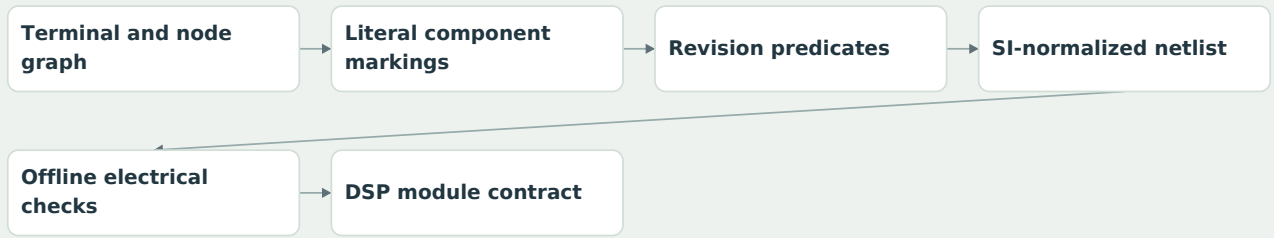
ORIGINAL DESIGN PROPOSAL: the first extraction pass captures connectivity only. Identify external terminals, ground symbols, transformer windings, valve sections, switch poles and junctions. Assign each a unique internal name. Do not yet merge nodes that merely appear close in a poor scan. A wire crossing and a connected junction have different meanings, and a switch drawing may depict only one mechanical position.

The second pass records component types and literal markings. Preserve the printed notation beside the normalized SI value. The third pass resolves revision notes and alternate components. Never average conflicting values into a fictional circuit. Create two explicit variants when the sources actually describe two configurations; leave an uncertainty ticket when the variant boundary is unknown.

Only then build a connectivity graph and an offline electrical model. A usable DSP approximation may eventually replace part of that network, but it should preserve the boundaries, loading assumptions and observable behavior identified here. The source graph therefore remains useful even if the final real-time engine uses state-space filters, wave digital elements or measured responses.

```
for sheet in selected_revision:
    capture_nodes_and_terminals(sheet)
    transcribe_literal_markings(sheet)
    attach_revision_predicates(sheet)
    flag_ambiguous_junctions(sheet)
assert unresolved_items_are_explicit()
netlist = normalize_verified_records()
validate_topology_before_fitting(netlist)
```

From drawing evidence to a model boundary



Original engineering workflow; a readable redraw is an input, not automatic validation.

MEASUREMENT AND DOCUMENT CONTROL

Image acquisition and transcription quality control

Primary service manual and schematic source atlas

The AO-28 schematic archive provides a separate image sheet and a linked underside layout. The schematic image opened successfully. The layout was located, but this chapter did not verify its pixels because the web parser rejected its GIF type; it remains an acquisition task. Do not present that layout as inspected evidence. [S002-04] [S002-05]

ORIGINAL DESIGN PROPOSAL: use lossless image intermediates for reading circuit labels, retain the original scan, and record any rotation, contrast adjustment or perspective correction. These operations can aid readability but cannot recover a digit that was never captured. Obtain a fresh photograph or another scan when a decimal point, connection dot, component suffix or pin number remains ambiguous.

Prepare a photographic comparison session for the selected instrument: one assembly overview, an orientation reference, terminal labels, valve socket identifiers and focused images of uncertain components. Include a scale where geometry matters. Keep private serial information out of public copies unless sharing was agreed. Assign invasive inspection and electrical measurements to a qualified technician; this research plan does not require an inexperienced user to expose energized chassis circuitry.

Possible defect	Required response
OCR merges label and value	Inspect original pixels and preserve literal text
Missing decimal point	Mark unresolved; acquire another view
Unclear winding or junction	Compare schematic and physical wiring
Wrong-side layout orientation	Record top/underside orientation explicitly
Anonymous redraw disagreement	Retain both readings pending primary verification

CHECKS AND DECISIONS

- No unverified OCR value enters a production calibration profile.
- Every edited research image retains a path to its untouched original.

SOURCES S002-04 AO-28 preamplifier factory schematic scan / S002-05 AO-28 underside component layout scan

ACCEPTANCE AND HANDOFF

The documentary gate for faithful emulation

Primary service manual and schematic source atlas

ORIGINAL DESIGN PROPOSAL: the documentary gate passes when another engineer can reproduce the chosen circuit interpretation from the recorded sources without guessing which version was intended. Passing this gate establishes traceability, not audible equivalence. Physical component tolerances, aging and undocumented repairs still require specimen inspection and measurements.

Deliver a source catalog, the selected revision matrix, a page-and-figure index, a connectivity graph, a component table and an uncertainty register. Each unresolved item needs an owner and a concrete next acquisition, such as a photograph of a named terminal or a higher-resolution scan of a specific printed page. Prioritize uncertainties that change topology or harmonic mapping ahead of cosmetic details.

The three 1995 redraw sheets are useful navigation assets but must remain labeled as later redraws. The alternate 73-page host is a redundant access route, not an independent historical witness. [S002-02] [S002-03] Public access to these files does not establish a reuse license. Supply their links and original analysis; do not automatically package every downloaded manual or image with the distributable instrument.

Gate	Concrete pass condition
Identity	Drawing edition and assembly applicability recorded
Connectivity	All external ports and ambiguous crossings reviewed
Components	Each value has units, source locator and status
Variants	Alternate notes compile into separate configurations
Reproducibility	Source file identity and printed/PDF locations retained
Rights	Every bundled asset has a recorded reuse basis
Honesty	Unmeasured fidelity claims remain explicitly unvalidated

CHECKS AND DECISIONS

• Reject a module specification that says only “according to the B-3 service manual” without identifying its drawing and revision.

• No percentage fidelity score follows from document completeness alone.

SOURCES S002-02 Hammond Organ B3 C3 Service Manual, 73-page compilation / S002-03 Hammond B3 Service Manual alternate-host compilation

HARDWARE EVIDENCE

Use invention records as an architecture map

Patents and design history relevant to emulation

The foundational Hammond filing is dated January 19, 1934, with publication on April 24, 1934. Its drawing set covers the console, generator assembly, controls, wiring and gearing. It is an early design record, not a B-3 service manual. Keep filing and publication dates as separate fields so a historical caption does not accidentally convert a grant date into an invention date. [S003-01]

ORIGINAL RESEARCH METHOD: Treat each patent as a searchable explanation of a mechanism and a source of questions for the production drawings. A patent can disclose several alternative implementations, including arrangements never installed in the selected organ. The implementation agent must name the exact figure and embodiment being studied, then link the resulting hypothesis to a service-manual circuit and a photographed specimen. A plausible historical relationship alone is insufficient to populate a production bill of materials.

Organize the patent index by subsystem rather than chronology alone. This makes the relevant invention available while implementing a module and exposes incorrect transfers between families. Preserve an explicit applicability status: conceptual ancestor, production match established, alternative embodiment, or unresolved. No patent in this chapter has been promoted to a measured specimen match.

Patent	Study purpose	Applicability limit
US1956350	Generator, gearing, contacts and mixing	Early architecture; predates B-3
US2382413	Scanned artificial-line vibrato	Several circuit alternatives
US2560568	Capacitive scanner construction	Patent line differs from selected production line
US2565512	Drawbar contact mechanism	Match physical revision first
US2953055	Percussion decay and triggering	Separate each embodiment
US2489653 / RE23323	Rotary sound channels and radiation	Not a 122 parts catalogue

SOURCES S003-01 US1956350A Electrical musical instrument — Laurens Hammond / S003-02 US2382413A Electrical musical apparatus — John M. Hanert / S003-03 US2560568A Vibrato apparatus — Laurens Hammond and John M. Hanert / S003-04 US2565512A Tone control apparatus for electrical musical instruments — Herbert E. Mei / S003-06 US2953055A Percussion tone electrical musical instrument — John M. Hanert / S003-07 US2489653A Rotatable tremulant sound producer — Donald J. Leslie / S003-08 USRE23323E Rotatable tremulant sound producer — reissue

ORIGINAL MODEL

Translate generator drawings into explicit hypotheses

Patents and design history relevant to emulation

The 1934 record explicitly describes shared sources for coincident musical frequencies and a drive system using mechanical gearing. Its final drawing sheets include gearing and wiring charts. The complex bass-generator patent supplies a separate historical lead for pedal voicing and routing. Neither source establishes the selected B-3 generator terminal map without a production cross-check. [S003-01] [S003-05]

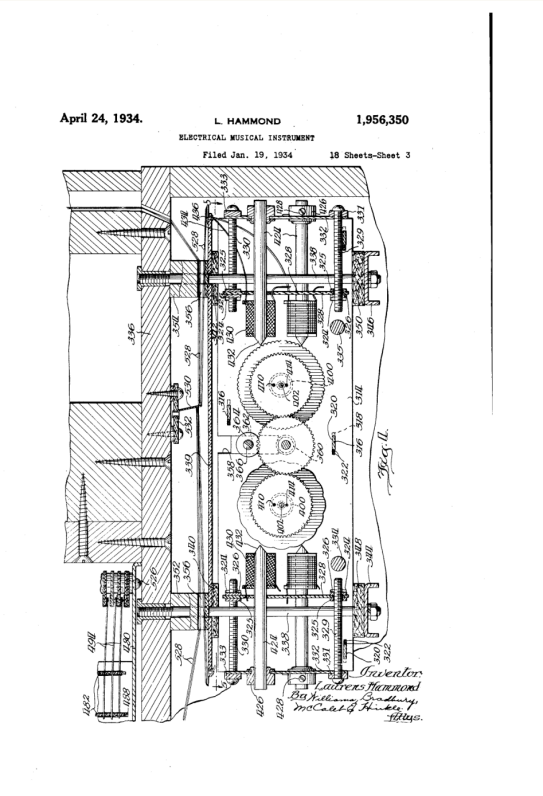
ORIGINAL DESIGN PROPOSAL: Build a graph whose vertices are motor, shaft, gear, wheel, pickup, filter and electrical terminal. Store mechanical ratios separately from musical note labels. This prevents a programmer from silently replacing gear-derived tuning with ideal equal temperament, or creating independent oscillators whenever two keys request the same source. The graph should identify which phases are shared, which couplings may introduce correlated disturbance, and which measured waveforms belong to a single physical pickup.

The derived equation below only expresses the kinematics: tooth passages per second equal shaft revolutions per second multiplied by teeth. It supplies a consistency check, not missing tooth counts. For each unknown gear, retain a symbolic ratio until a legible production chart or measurement resolves it. Begin with clean, continuously running sources; introduce harmonic residuals and leakage only when evidence supports their structure.

MODEL EQUATIONS

f_wheel = (rpm_motor / 60) * gear_ratio * tooth_count

gear_ratio = wheel_shaft_speed / motor_shaft_speed



Historical patent plate US1956350A_p03; invention embodiment, not a verified production drawing for the selected organ or cabinet. Source and license

CHECKS AND DECISIONS

- Reject a wheel record lacking a provenance field for frequency or gear ratio.
- Two routes to the same source must read the same phase state.

SOURCES S003-01 US1956350A Electrical musical instrument — Laurens Hammond / S003-05 US2508514A Electrical musical instrument — John M. Hanert and Laurens Hammond

HARDWARE EVIDENCE AND MODEL

Read scanner patents without importing the wrong circuit

Patents and design history relevant to emulation

Hanert’s US2382413 describes artificial-line phase modulation, including a capacitive scanning arrangement and alternatives. US2560568 supplies rotor, stator, output-contact and housing detail; its text identifies figures for the scanner and line box. It also describes a possible twenty-four-section line. That number must not be imported into a model whose selected production drawing specifies a different topology. [S003-02] [S003-03]

ORIGINAL DESIGN PROPOSAL: Separate three evidence objects: the line network, the angular pickup weighting, and the rotating assembly. Extract the line from the selected service drawing. Use the patent to understand how overlapping capacitance can blend neighboring taps and why output pickup construction matters. Fit the weights from geometry or measured transfer functions rather than replacing the rotor with abrupt switching. Record any simplifying high-impedance assumption explicitly.

A first digital experiment can hold the rotor angle constant and evaluate the linear network at every angle. Compare its frequency response, phase and magnitude with the measured scanner output before animating it. Once static behavior agrees, advance angle continuously and compare modulation sidebands. The formula is a proposed abstraction; actual loading may require solving the coupling capacitances together with the line. A normalized weighted sum is not automatically a circuit-exact result.

Evidence item	Model field	Do not infer
Production line schematic	Network topology and component values	Patent constants apply unchanged
Rotor drawings	Candidate overlap geometry	Drawing scale is metrology
Stationary-angle measurement	Tap weighting and loading	All specimens share the curve
Running measurement	Rate and sideband pattern	Nominal rate alone proves fidelity

MODEL EQUATIONS

$$y[n] = \sum_k (w_k(\theta[n]) * v_k[n])$$

$$\theta[n+1] = \mathrm{wrap}(\theta[n] + 2\pi f_{scan}/F_s)$$

SOURCES S003-02 US2382413A Electrical musical apparatus — John M. Hanert / S003-03 US2560568A Vibrato apparatus — Laurens Hammond and John M. Hanert

IMPLEMENTATION

Separate control mechanics from percussion behavior

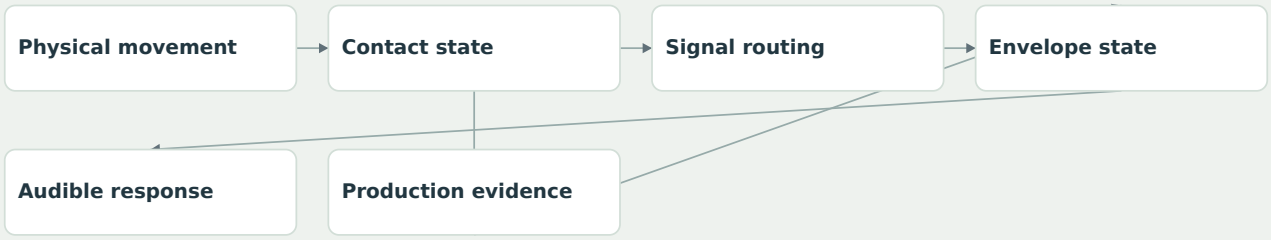
Patents and design history relevant to emulation

Meinema’s drawbar patent is a useful primary record for contactors and collector bars. Hanert’s percussion patent describes converting sustained tone signals into a sharp attack with controlled decay and presents multiple circuit arrangements. These records explain design intent, but an AO-28 implementation needs its own revision-specific schematic and switching evidence. [S003-04] [S003-06]

ORIGINAL DESIGN PROPOSAL: Represent physical motion, electrical connectivity, and resulting sound as separate layers. A drawbar position selects an electrical state; it should not merely move a graphical slider. A percussion control alters a state machine and signal path; it should not automatically add an independent per-note envelope. The production switching circuit determines whether a held key blocks retrigger, which harmonic path is selected, and which regular drawbar path is affected.

Use the patent’s alternate embodiments as deliberate negative test cases. For example, a circuit that detects signal onset and a circuit keyed by a dedicated contact may react differently to silence, attenuation or overlapping notes. Do not mix features from both to produce an attractive but undocumented hybrid. Write a short mapping note showing every software state, its production electrical cause, and its measurement evidence. If that link is missing, label the state provisional and keep it out of the faithful factory profile.

Evidence for a control behavior



Original implementation decomposition. Actual percussion state transitions must come from the chosen AO-28 revision.

CHECKS AND DECISIONS

- Run overlapping-note tests before accepting percussion retrigger behavior.
- Every graphical drawbar position resolves to a defined electrical/model state.

SOURCES S003-04 US2565512A Tone control apparatus for electrical musical instruments — Herbert E. Mei / S003-06 US2953055A Percussion tone electrical musical instrument — John M. Hanert

ORIGINAL ACOUSTIC MODEL

Use Leslie drawings to distinguish motion from sound source

Patents and design history relevant to emulation

Leslie’s US2489653 describes rotating sound-transmitting channels, different cabinet arrangements and radiation patterns. Its drawings include cabinet views, sections, modified embodiments and directional diagrams. The 1951 reissue is a distinct document and marks changes to the original specification. These records explain rotary sound mechanisms but are not interchangeable with a Model 122 assembly drawing. [S003-07] [S003-08]

ORIGINAL DESIGN PROPOSAL: Label every part as electrically driven transducer, rotating acoustic channel, stationary boundary, motor transmission, or listening position. This classification prevents the common modeling shortcut of assuming that every visibly rotating object is a loudspeaker diaphragm. For each acoustic branch, draw its propagation path and identify which part changes direction, distance, occlusion or radiation pattern over one revolution.

Use a path-delay model for changing distance and a separate frequency-dependent directivity model for radiation. With path length r in metres and sound speed c in metres per second, delay is r/c. The observed-frequency approximation below assumes source-time variation is slow enough for the chosen propagation model; it is an educational consistency check, not a complete rotating-horn solver. Stationary-angle measurements and moving recordings must validate the chosen geometry. Do not duplicate a measured Doppler effect with an additional arbitrary pitch LFO.

Drawing clue	DSP question	Required production check
Moving channel mouth	How does path length change?	122 rotor and microphone geometry
Cabinet opening	How does radiation vary with angle?	Measured angular transfer functions
Multiple embodiments	Which mechanism was actually built?	Model-specific assembly photos
Motor transmission	How does speed evolve?	Measured horn and drum ramps

MODEL EQUATIONS

$$\tau(t) = r(t) / c$$

$$f_{observed}(t) \approx f_{source} * (1 - d(\tau)/dt)$$

SOURCES S003-07 US2489653A Rotatable tremulant sound producer — Donald J. Leslie / S003-08 USRE23323E Rotatable tremulant sound producer — reissue

ACCEPTANCE AND HANDOFF

Deliver a traceable patent and drawing package

Patents and design history relevant to emulation

The USPTO explains that patent text and drawings are typically not subject to copyright restrictions, while acknowledging exceptions such as explicit notices and embedded trademarks. Record the notice check for each selected sheet; the presence of an accessible PDF alone is not a rights record. The rights of a contemporary scan, photograph or annotated redrawing must be evaluated separately. [S003-09]

ORIGINAL WORKFLOW: Preserve the original document identifier, exact download URL, retrieval date, file hash, PDF page index, printed sheet number and figure identifier. Keep annotated copies separate from untouched source files. Caption each image with inventor, patent number, figure, document version and a sentence stating its relevance to the chosen instrument. A page that illustrates a historical alternative must say so visibly. Never relabel an early Hammond patent plate as a photograph or exact schematic of a production B-3.

Build the engineering handoff from evidence records rather than an unstructured folder of images. Every adopted mechanism should point to a production confirmation or an unresolved acquisition task. US2828659 remains a source candidate because direct retrieval failed in this research pass; it must not support technical claims until examined. Completion means the source package is reproducible and its limits are explicit, not that every patented arrangement has been implemented.

```
for asset in patent_assets:
    verify_identifier_and_version(asset)
    record_original_url_and_hash(asset)
    record_page_sheet_and_figure(asset)
    review_notices_and_caption(asset)
    classify_production_applicability(asset)
    link_to_module_or_unknown(asset)
```

CHECKS AND DECISIONS

- No production parameter is justified solely by an unmatched historical patent.
- All reused plates retain a source caption and notice-review record.
- Original and reissue records remain distinguishable.
- Every unresolved source has a precise retrieval or measurement task.

SOURCES S003-09 Terms of Use for USPTO websites — Patent information / S003-10 US2828659A Electrical musical instruments — unresolved source candidate

HARDWARE EVIDENCE

Define the photographed object before modeling it

Console cabinet and physical appearance atlas

The cabinet is part of the reference contract. A convincing rendering can still depict the wrong instrument if it combines an early case, later controls, replacement bench, and another model's pedals. Record a specimen identifier for each photograph and distinguish an identified assembly from an attractive but unidentified example. The museum front and rear photographs provide useful visual observations, not an authenticated production history. [S004-02] [S004-03]

The factory service page gives an assembled footprint of 48-3/4 inches wide, 49-1/2 inches deep and 46 inches high with pedal keyboard and bench, and lists the B-3 finish as walnut-cherry. These are documented configuration bounds; they are not a complete bare-cabinet drawing or a wood-species analysis. [S004-01]

ORIGINAL DESIGN PROPOSAL: create a geometry ledger before sculpting. Each dimension needs its measurement endpoints, open/closed configuration, unit, specimen, uncertainty, and source photograph. Keep unverified dimensions empty. Use nominal overall proportions only for the first blockout, and flag that mesh as provisional. The closed cover, open playing position, removed rear cover, and separated pedalboard should be separate named capture states.

Record	Required meaning
Identity	Console serial plus independent bench and pedalboard identifiers
Envelope	Exactly which parts and cover positions are included
Material	Observed appearance separately from verified substrate
Geometry status	Documented, measured, inferred, or unverified
Modification	Replacement finish, hardware, wiring or furniture documented

P002 | Hammond-museum-B-3.png | Фікcер | CC BY-SA 4.0 Source and license

CHECKS AND DECISIONS

- The scene reports its specimen and capture state.
- An aggregate footprint is never used as the console-only bounding box.

SOURCES S004-01 Hammond Organ Service Manual for early models, factory cabinet specifications and B-3 / S004-02 Hammond-museum-B-3.png by Фікcер / S004-03 Hammond-museum-B-3-back.png by Фікcер

HARDWARE INVENTORY

Build a traceable cabinet parts hierarchy

Console cabinet and physical appearance atlas

Factory Figure 1 on printed page 6-5 identifies the walnut B-3 front assembly and its furniture hardware. Selected visually checked identifiers appear below. They identify listed service parts, not dimensions, materials certificates, or a guarantee of applicability to every serial number. [S004-01]

ORIGINAL DESIGN PROPOSAL: give every visible assembly a persistent scene identifier. Keep fastener instances attached to their parent part so an exploded view moves them coherently. The rear cover, music support, moving top, stationary structure, endblocks, legs, stretcher, bench and pedal furniture need individual objects where they affect silhouette or access. Tiny repeated fasteners can share a mesh but retain individual transforms and provenance.

Use the parts illustration to construct a completeness checklist, then verify the selected specimen. Dealer comparison photographs are useful leads for hidden supports, rear fixings and case differences, but their generalizations are not a substitute for a factory revision record. Record repairs without declaring the instrument fraudulent or inauthentic from a single cosmetic clue. [S004-07]

Factory listed part	Identifier
Cabinet rear cover	050-001745
Rear thumbscrew, brass	999-006006
Rear thumbscrew, brown	999-006007
Bench assembly	152-000012
Heel rest	070-000021
Stretcher bar	050-001619
Top hinge, long / short	032-022386 / 032-022387

P001 | Hammond-museum-B-3-back.png | Фіксер | CC BY-SA 4.0 Source and license

CHECKS AND DECISIONS

- Preserve printed part names and separately store normalized scene names.
- A low-confidence transcription cannot become a released BOM entry.

SOURCES S004-01 Hammond Organ Service Manual for early models, factory cabinet specifications and B-3 / S004-07 Vintage Organs first-hand comparison photographs of original and reconstructed B-3 ca

ORIGINAL MODEL

Separate material evidence from attractive shading

Console cabinet and physical appearance atlas

ORIGINAL DESIGN PROPOSAL: capture appearance as layered evidence. Store the visible substrate description, grain direction, finish color, gloss, wear, and observed repairs separately. A brown reflective photograph cannot establish solid walnut, veneer thickness, coating chemistry, or original finish. Those remain unknown until documentary or physical inspection supports them. The drawbar photograph has its own camera and lighting history; it is not an absolute color standard. [S004-04]

For rendering, build independent base-color, surface-normal, roughness and wear assets. Avoid baking room reflections into the base-color texture: those reflections should respond to the virtual lighting. Use a neutral-light capture for color, raking light for relief, and a color target with a scale. Preserve original images and write derived texture operations to a reproducible recipe.

ORIGINAL DESIGN PROPOSAL: define a right-handed local frame with x across the console, y toward its rear, and z upward. Choose an origin tied to a measurable structural point. Convert legacy dimensions to metres once at ingestion. Keep wood grain scale consistent between panels, and verify the visible grain direction in each photographic view rather than mirroring one generic texture everywhere.

MODEL EQUATIONS

length_m = length_in * 0.0254

p_world = T_console * T_part * p_local

P005 | Hammond B3 25-note pedalboard, Museum of Making Music.jpg | Hammond_B3,_Museum_of_Making_Music.jpg: doryfour derivative work: Clusternote | CC BY-SA 2.0 | Retouched/transformed; excluded from geometry evidence. Source and license

CHECKS AND DECISIONS

- No claimed material identity is inferred solely from image color.
- Metre conversion and transform composition have independent unit checks.

SOURCES S004-04 Hammond B3 and Drawbars by ArtBrom

IMPLEMENTATION

Translate the cabinet into an efficient inspectable scene

Console cabinet and physical appearance atlas

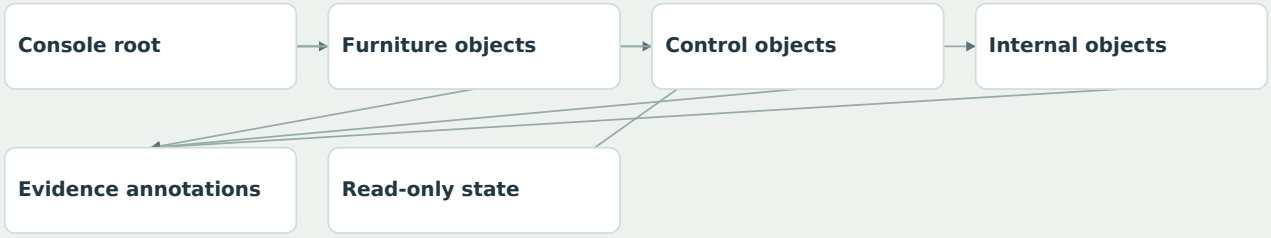
ORIGINAL DESIGN PROPOSAL: begin with a proportionally constrained blockout and a fixed orthographic review camera. Add silhouette-defining contours before screws or surface scratches. Maintain separate furniture, playing controls, internal assemblies, and presentation layers so the inspection view can reveal internals without changing the instrument state. A hidden back panel changes visibility, not audio processing.

Assign hinge pivots from measurements. Cover animation and key travel should use bounded transforms rather than arbitrary decorative motion. For details whose geometry remains unknown, attach a visible evidence-status tag in the service view and use a reversible placeholder mesh. Do not present a fabricated cross-section as a factory drawing.

The renderer consumes a read-only state snapshot. Audio events must not wait for mesh updates, texture loading, GPU synchronization or animation completion. A stalled render frame must leave sound generation unaffected. Preserve the same control identifiers in the perspective instrument view and a flat accessible control view, so visual elaboration never creates a second parameter system.

```
load geometry_manifest
validate units and parent links
create structural blockout
attach measured parts and pivots
attach evidence metadata per part
load appearance assets off audio thread
read instrument_state_snapshot
update visible control transforms
render inspection or player camera
```

Cabinet scene ownership



Original proposed scene graph. Acoustic behavior belongs to DSP modules, not meshes.

CHECKS AND DECISIONS

- Removing or hiding a mesh cannot change serialized audio state.
- All animated pivots have limits and an explicit evidence status.

MEASUREMENT

Acquire the missing views at useful scales

Console cabinet and physical appearance atlas

ORIGINAL DESIGN PROPOSAL: obtain orthogonal views of the same specimen, then oblique photographs for shape confirmation. Include a scale in the part's plane, capture the lens and camera distance, and photograph the rear both covered and uncovered. Interior access belongs to a qualified technician; a visual asset assignment does not authorize untrained work around energized valve electronics.

The popular Commons pedalboard derivative explicitly reconstructs missing front areas and reports a width alteration in its history. Therefore exclude it from dimensional and unseen-structure evidence. Its underlying original photograph is available and has a separate source record. The derivative may appear as an explicitly retouched contextual illustration. [S004-05] [S004-06]

Keep each shot attached to a measurement question. One beautiful overall photograph cannot resolve a key-edge radius, hinge offset, rear-panel attachment, or material construction. Prefer a small sharp macro showing a scale and part identifier to a large number of untraceable promotional images.

Capture group	Question resolved
Front, side, rear, top	Envelope and silhouette in one defined state
Cover open and closed	Hinge placement and clearance
Leg profiles and underside	Turned contours, joints and fixings
Bench all faces	Seat contour and independent geometry
Pedal front, rear, underside	Radiation, heel rest and covers
Endblocks and labels	Contours, engraving and switch placement
Fastener and hinge macros	Head shape, finish and attachment
Rear identity plates	Specimen and assembly traceability

P011 | Hammond B3 @ ONKIO HAUS (2014-06-01 by JacoTen @ Flickr 14356834232).jpg | JacoTen from Tokyo, Japan | CC BY-SA 2.0 Source and license

CHECKS AND DECISIONS

- Do not measure extrapolated or retouched regions.
- Capture coordinates and scale references accompany every accepted dimension.

SOURCES S004-05 Retouched Hammond B3 pedalboard image by doryfour and Clusternote / S004-06 Original Hammond B3 Museum of Making Music photograph by doryfour

ACCEPTANCE

Accept the visual reconstruction only against evidence

Console cabinet and physical appearance atlas

ORIGINAL DESIGN PROPOSAL: validate geometry before beauty rendering. Compare orthographic silhouettes, then project the model into calibrated photographs and inspect landmark residuals. Set tolerances from measurement uncertainty and intended display size; do not claim a universal factory tolerance for an undocumented cabinet. Separate geometric mismatch from lens distortion, camera pose error and a physically different specimen.

The handoff must include the native scene, exported meshes, material recipes, part manifest, source attribution, captured dimensions, pivot definitions, and an unresolved-geometry list. Every detailed visual should be classed as observed, documented, inferred, or artistic. The player view can omit invisible internal detail for performance while the service view retains a traceable assembly structure.

A close-looking console does not establish sound fidelity. Approve this chapter when the chosen physical reference is represented coherently and its uncertainty is visible. Reserve any full-recreation claim for the combined dimensional, electrical, behavioral and acoustic acceptance process described elsewhere in this atlas.

The companion physical inventory contains 319 factory-callout records,276 source-qualified manufacturer-number fields and 43 remaining nulls. A second-source review recovered 54 fields and separately flagged five discrepancies among earlier accepted numbers. These are catalog records, not quantities or a manufacturing-complete BOM. Read the additional-review data before selecting replacement numbers. [S004-99]

Gate	Pass evidence
Identity	One stated console configuration; mixed accessories disclosed
Dimensions	Endpoints, units and uncertainty retained
Parts	Factory callouts mapped or explicitly deferred
Images	Creator, license and alterations recorded
Geometry	Landmark residuals assessed at calibrated views
Performance	Render stalls cannot interrupt audio
Unknowns	No invented interior or material claim presented as measured

MODEL EQUATIONS

$$E_{reprojection} = \mathrm{sqrt}(\mathrm{mean}(\mathrm{norm}(\mathrm{project}(P_i) - q_i)^2))$$

CHECKS AND DECISIONS

- E_reprojection uses model landmarks P_i and image landmarks q_i in pixels; report camera calibration.
- Review both neutral technical lighting and the final player-view lighting.

SOURCES S004-99 Original Hammond repair-parts catalog in B3/C3 archival compilation

HARDWARE EVIDENCE

Treat each playing key as a mechanical switching assembly

Manual keyboard mechanism and contact assembly

Documented hardware: each manual has 61 playing keys. A playing key operates nine contact springs, supplied through resistance wires from generator terminals, against nine longitudinal busbars. The factory drawing identifies palladium-alloy contact wire; its text identifies bronze springs and precious-metal contact surfaces. The B-3 applicability paragraph follows the shared manual description. A selected preset or adjust key completes the onward circuit. [S005-01]

ORIGINAL DESIGN PROPOSAL: give the keyboard assembly its own module and evidence record. The audio engine should receive nine conductance states for each key, alongside the key identity and time. A graphical key animation is merely another consumer of the mechanical state. It must not become the source of audio timing. Derive the playing-contact count as 61 times 9, or 549 per manual; keep preset contacts separate from this count.

Freeze the identification vocabulary before taking photographs. Record upper or lower manual, note number, key color, keyboard assembly markings, restoration history and the photographed orientation. Distinguish a component that was installed at manufacture from a replacement. Treat undocumented material grades, wire resistance, dimensions and travel as unknown, even when a supplier sells a plausible substitute.

Object	Required model record	Evidence status
Playing key	Manual ID, note index, geometry ID	61 per manual documented
Contact spring	Contact ID, source mapping, closure state	Nine per playing key documented
Busbar	Harmonic bus ID and electrical connection	Nine per manual documented
Assembly materials	Photograph, marking, material claim source	Exact alloy composition unresolved
Preset network	Separate routing-state reference	Deferred to registration chapter

P006 | Hammond B3 manuals.png | Hammond_B3_at_RCA_Victor_Studio_B.jpg: Cliff from I now live in Arlington, VA (Outside Washington DC), USA derivative work: Clusternote (talk) | CC BY 2.0 | Source derivative is cropped. Source and license

CHECKS AND DECISIONS

- Fail ingestion if a keyboard has 548 or 550 playing-contact records while declaring 61 nine-contact keys.
- Do not interpret a rendered key depression as measured switch closure.

SOURCES S005-01 Hammond Organ Service Manual, factory manual compilation

MECHANISM AND VISUAL ATLAS

Map the visible action to a measured geometry hierarchy

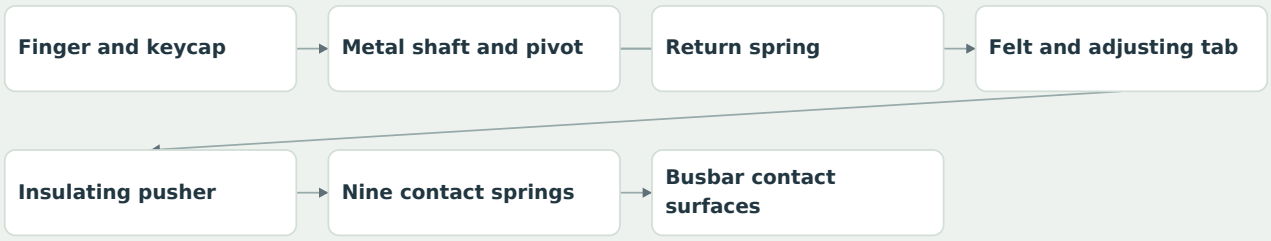
Manual keyboard mechanism and contact assembly

The 2017 investigation describes square-front keycaps on metal shafts, return springs, felt coupling, an adjusting tab and a bakelite pusher carrying the contact springs. Its measurements concern one serviced 1967 C-3: approximately 9 mm travel at a white-key front, 4 mm at the felt position and 2 mm resting clearance above the pusher. These are specimen observations, not universal B-3 manufacturing tolerances. [S005-02]

ORIGINAL DESIGN PROPOSAL: build the scene hierarchy around measured pivots and supports. Keep the keycap, shaft, felt and pusher as separate mesh objects. The player view needs only the external moving surface; the inspection view can expose the linkage and contact bank. Express geometry in millimetres and convert to rendering units in one documented import step. Leave uncertain dimensions as named parameters.

The available Commons keyboard image is explicitly a cropped derivative and is useful for appearance and layout, not scale recovery. Credit its photographer and crop author. [S005-04] A production asset should instead be fitted to front, side and underside photographs taken with a scale in the same plane. Lens correction must precede measurement. Never infer the hidden contact gap from the apparent size of a frontal photograph.

Original conceptual action hierarchy



Original mechanism map, not a dimensional drawing. Exact offsets, materials and clearances belong to a selected specimen record.

CHECKS AND DECISIONS

- Compare the mesh in side elevation against a scaled photograph before approving geometry.
- Retain the C-3 specimen label whenever displaying the published example dimensions.

SOURCES S005-02 Moro, McPherson and Sandler, Dynamic temporal behaviour of the keyboard action on the / S005-04 Hammond B3 manuals photograph, Cliff, cropped by Clusternote

ORIGINAL DIGITAL MODEL

Represent motion and nine contact states independently

Manual keyboard mechanism and contact assembly

ORIGINAL DESIGN PROPOSAL: define a continuous front-key displacement x(t), a pusher displacement p(t), and nine electrical states c_j(t). The simple relation below is an initial kinematic approximation. It separates the lever ratio from the lost motion before the felt engages. A release threshold can differ from a closure threshold; do not force them to agree until the measured trajectory supports that simplification.

The historical explanation in the B-3 mk2 manual describes partial key depression and an actuator moving through the contact stack. [S005-03] The empirical study reports that the closing pattern changes between keys. [S005-02] Therefore the model must store each contact threshold separately. Spatial order in a drawing does not justify assigning one universal time order to every physical key.

Build two input adapters. A continuous-position controller can supply x(t) directly after calibration. An ordinary MIDI note event can only initiate a declared synthetic trajectory. Such a trajectory is a compatibility model, not a reconstruction of the performer’s unobserved finger motion. Keep optional velocity-dependent transient shaping separate from sustained drawbar gain. Avoid silently turning the organ into a velocity-scaled synthesizer.

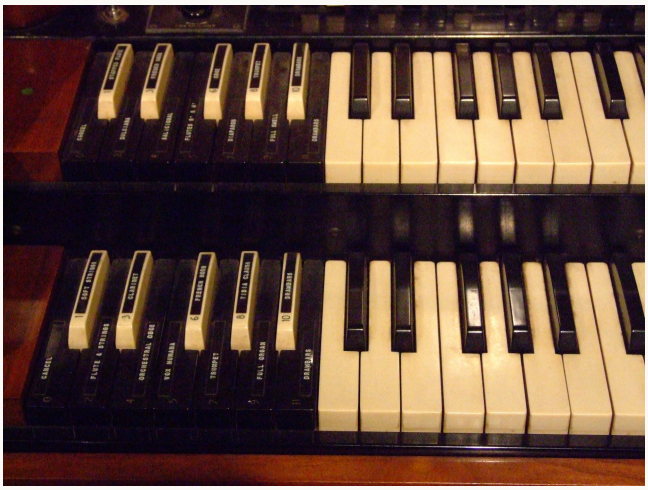
Symbol or field	Definition	Units / status
x, p	Key-front and pusher displacement	mm; measured or inferred
r	Effective lever ratio	Dimensionless; fitted
g	Pusher engagement clearance	mm; measured
theta_on, theta_off	Per-contact thresholds	mm; calibrated
c_j	Electrical contact state	Open, bouncing, closed
trajectory_source	Position capture or MIDI surrogate	Required provenance label

MODEL EQUATIONS

p(t) = max(0, r * x(t) - g)

close contact j when p(t) >= theta_on[j]

release contact j when p(t) <= theta_off[j]



P040 | Hammond preset keys.jpg | Les Chatfield from Brighton, England | CC BY 2.0 | Context specimen; exact model/revision requires confirmation. Source and license

SOURCES S005-02 Moro, McPherson and Sandler, Dynamic temporal behaviour of the keyboard action on the / S005-03 Hammond B-3 mk2 Owner Manual, Tone-Wheels and Multi-Contact Keys

IMPLEMENTATION SEQUENCE

Implement the switching layer before decorative animation

Manual keyboard mechanism and contact assembly

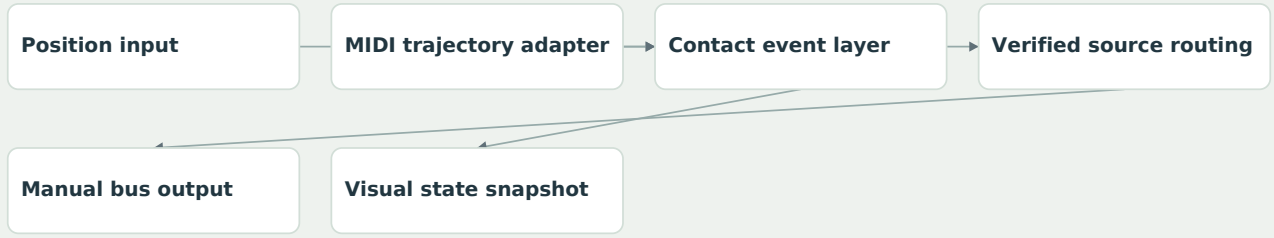
ORIGINAL DESIGN PROPOSAL: first implement a deterministic ideal-contact version. Create fixed-size arrays for two manuals, their playing keys and nine contact states. Load the verified source-to-contact routing table outside the audio callback. Keep source lookup, contact state and preset routing as separate data. This makes a wrong connection distinguishable from a wrong onset time during debugging.

Next add per-contact threshold crossing, then recorded or fitted bounce sequences. Generate events with audio-sample timestamps rather than GUI-frame timestamps. If a crossing lies between samples, retain its fractional position for the selected band-limited switching method. Do not add an arbitrary noise burst to hide an incorrect switching implementation; the later key-click chapter specifies the transient model.

Finally connect display state to the audio-owned event stream through a bounded message channel. The visual update can run more slowly than audio without changing the result. Keep release processing explicit so a rapid second press can begin from an incompletely returned key. Preset changes must update routing while the generator phase continues. All allocations, file reads and geometry updates belong outside the real-time path.

```
for each timed input event:
  update trajectory for manual and key
for each audio sample:
  evaluate key and pusher position
  for each of nine contacts:
    apply threshold or measured event
    update bounded bounce state
    route shared source through contact
  publish visual snapshot when due
```

Keep gesture inference outside audio identity



Original software boundary. Source oscillators remain shared and continuously advancing; this chapter owns the mechanical/contact interface.

CHECKS AND DECISIONS

- Identical timestamped input produces identical contact events at every tested host block size.
- No visual frame-rate change alters the audio render.

MEASUREMENT PLAN

Acquire the missing measurements without inventing tolerances

Manual keyboard mechanism and contact assembly

The authors' laboratory work demonstrates the usefulness of acquiring continuous key position together with contact states. [S005-05] ORIGINAL DESIGN PROPOSAL: arrange a technician-assisted capture session on the selected reference instrument. A qualified technician must determine any safe electrical access. The emulation specification should request synchronized data and instrumentation metadata, not give an unqualified operator live-chassis probing instructions.

Capture upper and lower manuals separately. Include slow partial presses, normal full presses, struck gestures, repeated notes before complete return, release-only trajectories and glissando passages. Use multiple keys across the range and include both key colors. Record enough repetitions to separate repeatable geometry from event-to-event variation. The number of repetitions is a project planning choice, not a manufacturer requirement.

Before fitting, align the position sensor and contact acquisition clock. Store calibration, sample periods, filtering and timestamp uncertainty. Preserve raw streams alongside processed events. Photograph each sensor location and reference axis. A final dataset must distinguish the mechanical strike sound recorded by a microphone from the electrical transient in the audio path; their causes and output routes are different.

Capture	Question resolved	Required output
Scaled side view	Pivot and travel geometry	Images plus dimensional uncertainty
Slow partial presses	Individual engagement thresholds	Position/contact pairs
Repeated full strokes	Variability and return behavior	Per-key event distributions
Held partial stroke	Stable incomplete closure	State occupancy record
Dry audio with timestamps	Electrical transient correlation	Calibrated audio/event bundle
Mechanical microphone	Acoustic action noise	Separately labeled optional layer

CHECKS AND DECISIONS

- Reject a timing dataset without a stated clock-alignment method.
- Keep a holdout set of keys and gestures out of the parameter fit.

SOURCES S005-05 Augmented Instruments Laboratory, Touch on Keyboard Instruments

ACCEPTANCE AND HANDOFF

Define what the keyboard subsystem must prove

Manual keyboard mechanism and contact assembly

ORIGINAL DESIGN PROPOSAL: accept the structural model when every playing key has the correct contact count, identity and routing references, the source phase remains continuous, and preset cancellation changes the onward route rather than destroying the oscillator state. Accept the mechanical model only against recorded key/pusher/contact traces. A convincing animation cannot substitute for that evidence.

Report timing error relative to the capture resolution and calibrated uncertainty; do not publish an arbitrary submillisecond tolerance as a historical specification. Examine contact order, first-to-last closure interval, release order, bounce occupancy and repeated-note recovery independently. Statistical agreement on one key is insufficient if the declared scope covers both manuals. Keep specimen calibration results distinct from general algorithm tests.

Handoff assets should comprise the keyboard manifest, source-map references, per-key geometry and threshold profiles, timestamped fixtures, fitting notes, an uncertainty register and an exploded scene hierarchy. Unknown return-force curves, contact resistance under load and serial-specific tolerances remain explicit open items. Release notes should state whether performance input came from continuous sensing or a MIDI surrogate, since that choice limits the available expressive information.

Test ID	Falsifiable condition	Evidence category
KEY-01	All 122 playing keys expose nine contact IDs	Derived structural test
KEY-02	Partial motion can leave a strict subset closed	Behavior test
KEY-03	Repeated rendering preserves event timestamps	Determinism test
KEY-04	Matched traces agree within declared uncertainty	Specimen validation
KEY-05	Changing GUI frame rate leaves audio unchanged	Architecture test
KEY-06	Unmeasured dimensions retain unknown status	Provenance test

CHECKS AND DECISIONS

- No assertion of universal 9 mm travel is permitted from the single published C-3 specimen.
- Do not declare the subsystem faithfully calibrated until held-out hardware traces are evaluated.

SOURCES S005-01 Hammond Organ Service Manual, factory manual compilation / S005-02 Moro, McPherson and Sandler, Dynamic temporal behaviour of the keyboard action on the

HARDWARE EVIDENCE

Count physical assemblies before coding oscillators

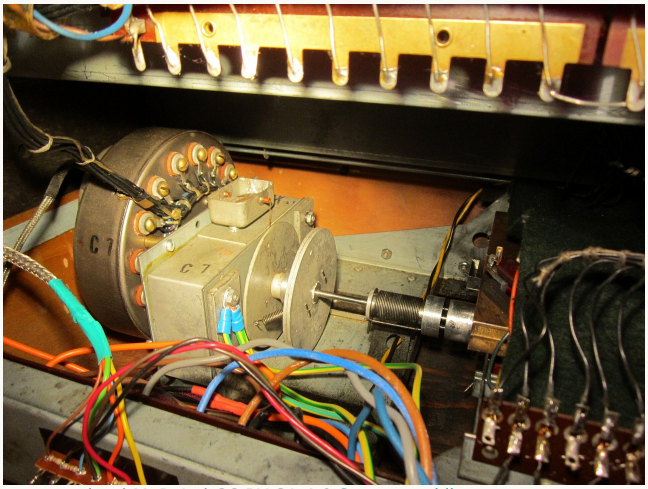
Tone generator mechanical assembly

The factory account specifies 48 rotating subassemblies and 24 driving gears in 12 sizes. There are 91 active tonewheels within 96 disk positions; five positions contain balancing blanks. Steel partitions divide the generator into shielded compartments. The documented wheel pairs are 2/32,4/64,8/128, and 16/192 teeth, with five 16-tooth wheels paired with blanks. [S006-01]

ORIGINAL DESIGN PROPOSAL. Keep three identifiers distinct: the numbered electrical tone, the rotating shaft assembly, and the physical compartment. A tone index is not a keyboard note, and an empty balancing position is not an inaudible oscillator. Construct an explicit relationship table before writing synthesis code. This prevents an apparently plausible 91-voice design from losing the shared mechanical relationships that measurements may later reveal.

The data contract should allow an unverified relationship to remain unknown. An engineer can initially render independent electrical sources while retaining shaft and compartment fields for later calibration; that approximation must remain visible in the profile metadata. An attractive exploded model should reference the same identifiers, so selecting a visible wheel can open its evidence record rather than a guessed frequency.

Record	Required content	Evidence state
Electrical source	Generator number, waveform reference, pickup ID	Documented index plus measured response
Shaft assembly	Two disk positions, mate IDs, gear ratio	Transcribe from chosen revision
Compartment	Associated shafts and shielding relationship	Documented topology
Blank position	Balance-only role; no electrical source	Explicitly excluded from audio graph



P009 | Hammond synchronous motor & scanner.jpg | MePaJa | CC BY-SA 4.0 Source and license

CHECKS AND DECISIONS

• The profile contains 91 active source records and five explicitly represented blank positions when using this documented assembly.

• Every electrical source resolves to exactly one pickup and a physical disk record.

SOURCES S006-01 Hammond Organ Service Manual, early console models

PHYSICAL ATLAS AND ORIGINAL DESIGN

Preserve manufacturing clues without inventing dimensions

Tone generator mechanical assembly

The service description identifies steel tonewheels, Bakelite gears, magnetic pickup assemblies, spring couplings, and protective lacquer. Its approximately 2-inch wheel diameter is descriptive, not a tolerance drawing. The motor and scanner have oil-distribution components; the starting-motor figure labels lubricating wicks. [S006-01]

Use the original motor photographs to establish installation context and visible geometry. Their captions identify particular photographed parts, which does not establish interchangeability across every B-3. The run-motor/scanner photograph and starter photograph are both attributed to MePaJa under CC BY-SA 4.0. [S006-03][S006-04] Keep an untouched source copy and record every crop or annotation separately.

ORIGINAL DESIGN PROPOSAL. Model materials at two different levels. The visual mesh needs surface appearance, dimensions, fasteners, and occlusion. The audio model needs inertia, compliance, damping, magnetic coupling, and bearing-condition effects only where evidence shows audible consequences. A polished brass-colored mesh is not evidence for a particular alloy; a visible wick does not justify inventing an oil-viscosity coefficient. Request measured dimensions or clear markings when either affects the implementation.

P008 | Hammond shaded pole starter motor.jpg | MePaJa | CC BY-SA 4.0 Source and license

CHECKS AND DECISIONS

- Every textured component retains a source or an explicit artist-estimated label.
- No scale is inferred from a perspective photograph without a known reference in the same plane.

SOURCES S006-01 Hammond Organ Service Manual, early console models / S006-03 Hammond synchronous motor and scanner original photograph / S006-04 Hammond shaded pole starter motor original photograph

ORIGINAL MODEL DERIVATION

Translate a compliant drive into a reduced mechanical model

Tone generator mechanical assembly

The documented 60 Hz run motor operates at 1200 rpm; the 50 Hz version is described at 1500 rpm. Couplings between the motor, drive sections, and wheel assemblies reduce transmission of speed pulsations. [S006-01] The earlier Hammond patent provides useful drive-construction context, but its drawings must not be substituted for the chosen B-3 revision. [S006-02]

ORIGINAL DESIGN PROPOSAL. Define theta_j as the mechanical angle of shaft j in radians and omega_j as its angular velocity in radians per second. A disk with N_i teeth produces electrical phase N_i times theta_j, plus a fixed pickup-reference offset. This preserves relationships between disks on one shaft while allowing different tooth counts. The 16/192 pair therefore has a derived frequency ratio 12, not an octave-only ratio.

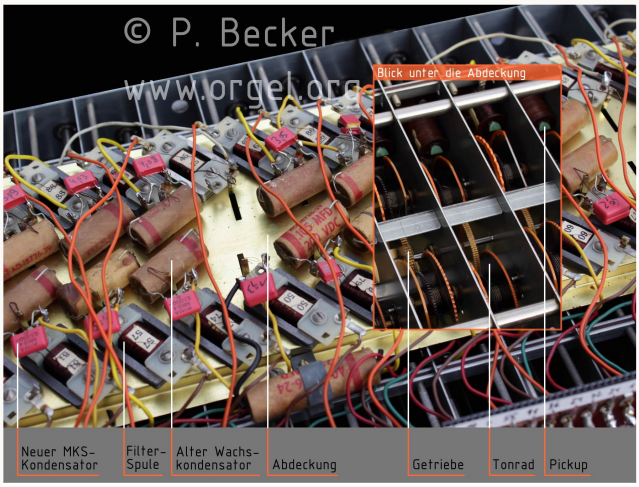
A full rigid-body simulation is optional. Begin with verified mean ratios and continuous phases. If captured sidebands demand a dynamic model, add a small set of torsional modes with positive damping and fitted forcing. Stiffness, inertia, friction, and ripple amplitude remain unknown until identified; do not assign realistic-looking constants and call them measured. Express each rotational coordinate in a consistent reference frame before coupling gears with different ratios.

MODEL EQUATIONS

phi_i = N_i * theta_j + alpha_i; f_i = N_i * omega_j / (2*pi)

J_j * theta_j_ddot + b_j * theta_j_dot + sum_k K_jk*(theta_j - r_jk*theta_k) = tau_j

J: inertia (kg m^2); b: damping (N m s/rad); K: stiffness (N m/rad); r: signed ratio; tau: torque (N m).



P014 | Generator Blick ins Innere.jpg | Peterbecker2000 | CC BY-SA 4.0 Source and license

CHECKS AND DECISIONS

• The final coupled model derives reaction torques consistently; the displayed equation is a modeling scaffold, not a verified generator netlist.

• Freeze mechanical parameters and compare long-run pitch and sideband structure with the reference capture.

SOURCES S006-01 Hammond Organ Service Manual, early console models / S006-02 Laurens Hammond US1956350A Electrical musical instrument

IMPLEMENTATION

Build the generator as persistent state owned by the instrument

Tone generator mechanical assembly

ORIGINAL DESIGN PROPOSAL. Load the mechanical graph outside the audio callback. Validate all relationships, convert units once, and publish an immutable profile containing rational gear ratios, shaft assignments, tooth counts, and evidence status. Allocate persistent state for each represented rotating assembly. A note event changes contacts downstream; it does not construct or reset the generator.

Implement the constant-speed case first. It provides a deterministic baseline against which modulation can be evaluated. Preserve phase across buffer boundaries, tempo changes, and silent periods. A transport seek is a host action rather than a mechanical restart; make any restart behavior an explicit product decision. The optional startup presentation should use its own state machine, independently of host transport.

For a reduced dynamic model, choose an integration method and control rate from stability tests, then interpolate the resulting phase or velocity continuously to audio rate. Check the modulation spectrum before lowering its update rate. If detailed waveform harmonics are evaluated, their band limit depends on instantaneous frequency; acceleration is not permission to let harmonics alias. Expose diagnostics through a lock-free snapshot, while photo lookup, mesh loading, and model fitting remain off the audio thread.

Interface	Ownership
GeneratorProfile	Immutable topology and provenance
GeneratorState	Persistent angles and optional dynamics
SourceFrame	Electrical outputs consumed by contacts
MechanicalSnapshot	Read-only telemetry for the UI

```
prepare(profile, sampleRate):
  verifyGraphAndUnits(profile)
  allocatePersistentRotors(profile)
  compileSourceToRotorMap()
processBlock(events):
  for each audio sample:
    advanceMechanicalState()
    for each active source i:
      phase = toothCount[i]*rotorAngle[rotor[i]]
      source[i] = bandlimitedShape(phase, i)
    contacts.route(source, events)
  publishBoundedDiagnosticSnapshot()
```

P018 | Tone Wheel Assembly.jpg | Ekupp at English Wikipedia | CC BY 3.0 Source and license

MEASUREMENT PLAN

Acquire evidence that separates mechanics from electronics

Tone generator mechanical assembly

ORIGINAL DESIGN PROPOSAL. Document the generator in its existing condition before changing anything: identification marks, repairs, lubrication history if known, mounting condition, region, supply frequency, temperature, and visible couplings. Ask a qualified organ technician to define safe access and signal acquisition. The research objective is calibrated evidence, not unrecorded adjustment of pickup gaps or bearing condition.

Capture rotational information and electrical output against a common timebase whenever practical. A non-contact tachometer can establish a rotational reference; a technician-approved isolated audio acquisition path can capture the resulting signal. A microphone recording alone mixes generator sound with valve, speaker, room, and cabinet behavior. Retain both a mechanically informative recording and a separately characterized electrical path when available.

Measure stable operation, startup, controlled shutdown, and repeated warm sessions. Fit the mean frequency before analyzing modulation residuals. Compare residual phase across sources that the physical graph places on the same shaft, in the same compartment, and on different drive sections. Correlation supports a common mechanism; it does not by itself identify a unique spring stiffness or inertia. Include instrument drift and acquisition-clock error in the uncertainty record.

Acquisition	Question resolved	Required metadata
Scale-referenced photographs	Actual geometry and repair state	View, scale, part ID, untouched original
Synchronous tach and audio	Rotation-to-phase relationship	Clock, calibration, speed reference
Repeated steady captures	Stable versus variable defects	Temperature and elapsed runtime
Start/stop captures	Transient rotational behavior	Switch event timing and signal path

P019 | Tongenerator Unterseite.jpg | Peterbecker2000 | CC BY-SA 4.0 Source and license

CHECKS AND DECISIONS

• Independent repeated sessions reproduce fitted mean frequency within a predeclared measurement uncertainty.

• A proposed dynamic model must outperform the constant-speed baseline on held-out recordings, not only training captures.

ACCEPTANCE AND HANDOFF

Accept a measured mechanical abstraction and disclose its limits

Tone generator mechanical assembly

ORIGINAL DESIGN PROPOSAL. Deliver a generator topology file, an image-to-part index, a deterministic rendering harness, a parameter provenance report, and a comparison report for the selected specimen. Each parameter must identify whether it is documented, measured, derived, proposed, or unknown. Unknown physical coefficients can remain absent when a simpler validated abstraction is sufficient.

First verify topology and arithmetic: tooth counts, blank positions, source ownership, signed gear ratios, and unit conversions. Then verify temporal behavior using identical event streams at multiple block sizes and sample rates. Define numerical tolerances before running comparisons. The important criterion is convergence toward the same continuous behavior; bit identity across all sample rates is neither implied nor required.

Accept added mechanical complexity only when it explains repeatable observations. A model that sounds old because arbitrary independent flutter was added to every note has not demonstrated the intended mechanism. Keep optional wear and instability controls separate from the selected reference profile. This chapter does not provide a complete production gear-tooth table, factory mechanical tolerances, measured inertias, or certified material specifications. Obtain these through the targeted source and acquisition tasks rather than filling gaps with invented precision.

Gate	Falsifiable condition
Topology	No orphan source, duplicate pickup, or audible blank position
Phase continuity	Note and block boundaries do not reset rotating state
Regional variant	50/60 Hz profiles use their own verified motor and gear data
Model stability	No unbounded state under supported controls and rates
Evidence completeness	Every supplied coefficient has an origin and unit
Fidelity claim	Scope names specimen, condition, captures, and remaining unknowns

CHECKS AND DECISIONS

• A source pair assigned to one shaft preserves its tooth-count-derived phase relationship under shared speed modulation.

• New mechanical detail must pass an explicit held-out comparison before entering the reference preset.

HARDWARE EVIDENCE AND ORIGINAL DATA CONTRACT

The complete drawbar surface

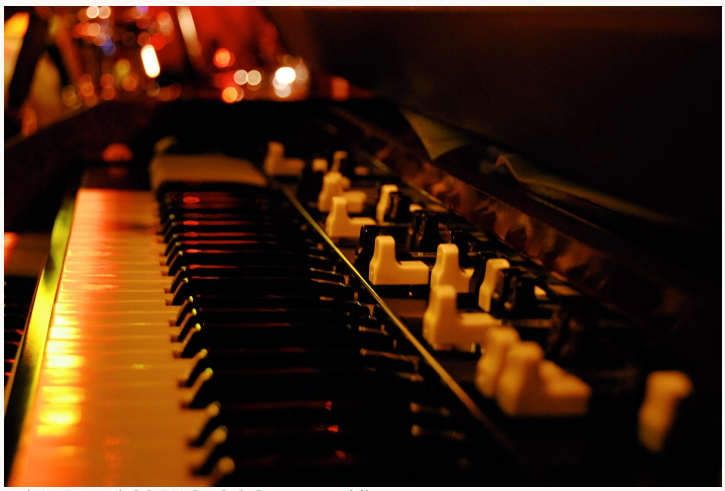
Drawbars presets and console switch hardware

The vintage console has four banks of nine manual drawbars and two central pedal drawbars: 38 controls. The two left manual banks serve Swell; the two right banks serve Great. Positions 0 through 8 describe the registration scale. The standard cap colors and nominal footages below identify function, not calibrated material color. [S007-01] [S007-02] [S007-03]

ORIGINAL DESIGN PROPOSAL: implement every bank even when a compact view displays only the selected one. Store all 38 settings independently; changing the selected bank must not overwrite the inactive bank. The table describes nominal musical relationships. Actual wheel assignment, gear tuning, and foldback belong to the mapping subsystem; multiplying a fundamental by these ratios is not a complete tonewheel engine.

Give each control a stable semantic identity comprising manual, bank, and footage. Keep this identity independent of screen coordinates or color. The nine-character registration string follows left-to-right drawbar order; parse each digit separately and reject missing or out-of-range digits. Display an explicit value for keyboard and assistive-technology users.

Index	Footage	Cap	Nominal ratio to 8 ft
1	16 ft	Brown	1/2
2	5 1/3 ft	Brown	3/2
3	8 ft	White	1
4	4 ft	White	2
5	2 2/3 ft	Black	3
6	2 ft	White	4
7	1 3/5 ft	Black	5
8	1 1/3 ft	Black	6
9	1 ft	White	8



P003 | Hammond B3 & Drawbars.jpg | ArtBrom | CC BY-SA 2.0 Source and license

CHECKS AND DECISIONS

- Exactly 36 independent manual settings and 2 pedal settings survive serialization.
- A footage identifier remains unchanged when the user rearranges or scales the interface.

SOURCES S007-01 Hammond Organ Service Manual early models factory scan / S007-02 Hammond B3/C3 service compilation historical section / S007-03 Hammond B3 mk2 owners manual manufacturer edition

HARDWARE EVIDENCE AND ORIGINAL STATE MODEL

Preset keys select a route through the instrument

Drawbars presets and console switch hardware

For each manual, C cancels; C# through A select nine wired registrations; A# and B select the left and right drawbar banks respectively. Selection normally releases the previously latched preset key. Upper-manual B is the percussion-eligible selection. These are registration switches rather than additional sounding keys. [S007-01] The factory mechanism also permits simultaneous latching of multiple registration keys; C is a momentary mechanical cancel without electrical contacts. Chapter 019 specifies the optional physical latch-mask model.

ORIGINAL DESIGN PROPOSAL: represent selection as one mutually exclusive enum per manual, not twelve independent Boolean buttons. Keep physical bank positions, wired preset values, and selected route separate. A vintage fixed preset must continue sounding its wired values while the player prepares an inactive drawbar bank. Selecting that bank then exposes its current settings. This distinction supports real performance gestures and prevents a common simplified-interface error.

Use enharmonic aliases only for display: the A# adjust selector can be shown as Bb in a lesson, but both labels resolve to the same internal selector. Treat the depressed key as a visual state. A preset-selection command must never allocate a tonewheel voice. Offer modern total-state snapshots in a separately named software layer so they cannot silently rewrite vintage preset semantics.

Selection	Registration source	Upper percussion eligibility
C	No selected registration	No
C# D D# E F F# G G# A	Associated wired nine-value preset	No
A# / Bb	Left bank for that manual	No
B	Right bank for that manual	Yes, if percussion enabled

P007 | Hammond ottava dei preset.jpg | http://it.wikipedia.org/wiki/Utente:Salli | CC BY-SA 3.0 Source and license

SOURCES S007-01 Hammond Organ Service Manual early models factory scan

DOCUMENTED CONTROLS AND ORIGINAL IMPLEMENTATION RULES

Switch meanings and routing truth tables

Drawbars presets and console switch hardware

The four percussion tablets select on/off, normal/soft level, fast/slow decay, and second/third harmonic. Vibrato has separate Swell and Great enable tablets and a shared six-mode selector. Great vibrato also applies to the pedals in the factory topology. Overall Normal/Soft volume is separate from percussion volume. [S007-01] [S007-02]

ORIGINAL DESIGN PROPOSAL: encode switches as enums with named states. Never infer meaning from a generic button index. The renderer must show the active legend and expose the same semantic label to automation. The shared vibrato selector controls one scanner configuration; upper and lower routing flags determine which signals enter it. Depicting two independently adjustable scanner rates would introduce an extension.

Expression is a separate normalized mechanical-position input into the expression subsystem. Preserve its position when changing registration. START and RUN can drive an optional mechanical-start simulation, while a normal ready-to-play software mode initializes the instrument deterministically. Leslie speed controls belong to the selected external cabinet configuration and must not be mistaken for the console scanner selector.

Swell enable	Great enable	Selected scanner-mode routing
Off	Off	Neither manual nor pedals
On	Off	Swell only
Off	On	Great and pedals
On	On	Swell, Great, and pedals

MODEL EQUATIONS

PercussionEligible = (manual == Swell) AND (preset == B) AND percussionOn

P004 | Hammond B-3 START-RUN.jpg | MePaJa | CC BY-SA 4.0 Source and license

CHECKS AND DECISIONS

- Exercise all four enable combinations for each V1, V2, V3, C1, C2, C3 setting.
- The global Soft tablet and percussion Soft tablet change distinct parameters.

SOURCES S007-01 Hammond Organ Service Manual early models factory scan / S007-02 Hammond B3/C3 service compilation historical section

HARDWARE EVIDENCE AND ORIGINAL ALGORITHM

Continuous contact motion and real-time event handling

Drawbars presets and console switch hardware

The factory manual specifies continuous-contact drawbars for B-3/C-3 assemblies: two contacts joined through a 1-ohm resistor avoid gaps between adjacent buses. The printed 0-8 scale therefore must not be interpreted as eight mechanically isolated detents. [S007-01]

ORIGINAL DESIGN PROPOSAL: separate normalized physical travel from registration markings. The first playable implementation may use nine calibrated gains with a short bounded transition; label this as an approximation. A refined model can evaluate contact overlap and transformer loading against travel. Do not claim that a smooth linear amplitude slider reproduces the contact network. Store both nominal registration display and physical travel when the fidelity mode requires them.

Queue timestamped semantic events from the editor or host. The audio thread owns active state and publishes a read-only snapshot for drawing. An inactive bank can receive edits without affecting current audio. Preset selection chooses the appropriate vector at the requested sample offset; it must not reset shared generator phases. Smoothing should suppress unintended digital discontinuities while preserving deliberately modeled physical switching transients.

State field	Suggested representation
physicalTravel	Normalized bounded value with profile-specific travel law
markedValue	0-8 display value; not the entire circuit model
selector	Cancel, fixed preset, adjustLeft, adjustRight
profileRevision	Immutable evidence-linked calibration identifier

```
onControl(event):
  applyAtSampleOffset(event.offset)
  if event.kind == DrawbarTravel:
    bank[event.manual][event.bank][event.index] = event.travel
  if event.kind == PresetSelect:
    selector[event.manual] = event.selection
  registration = resolve(selector, banks, wiredPresets)
  eligible = upperSelector == B and percussionOn
  publishReadOnlySnapshot()
```

P010 | Hammond B3 - Volume (normal,soft), Vibrato Swell (on,off), Vibrato Great (on,off), Vibrato and Chorus (v-1,c-1,v-2,c-2,v-3,c-3) - ONKIO HAUS (2014-06-01 by JacoTen @ Flickr 14356834922).jpg | JacoTen from Tokyo, Japan | CC BY-SA 2.0 Source and license

SOURCES S007-01 Hammond Organ Service Manual early models factory scan

ORIGINAL ACQUISITION AND ASSET PLAN

Build the visual atlas from inspectable control photographs

Drawbars presets and console switch hardware

The existing photographs provide complementary views: P003 shows drawbar-cap shape and keyboard context; P007 shows the preset octave; P004 shows the START/RUN plate. They have verified Creative Commons licenses but different creators and license versions. Preserve those records for every derivative. P003 has colored stage lighting, so its pixels are unsuitable as calibrated material-color measurements. [S007-04] [S007-05] [S007-06]

ORIGINAL DESIGN PROPOSAL: acquire frontal, orthographic, side, and oblique macro photographs of each control family with a scale and neutral color reference. Capture the same drawbar fully in, at every numbered mark, and fully out. Document an underside contact carriage only during an authorized unpowered session conducted by a qualified technician. Photograph complete tablet legends before removing anything; replacements can erase evidence of the original orientation.

Measure cap width, projection, stem section, row spacing, tablet pivot location, and switch-plate radius. Track uncertainty and perspective calibration per measurement. Use separate meshes for stem, cap, plate, and legends so travel and replacement variants remain editable. Require a front-view annotation layer whose labels come from the semantic control table rather than being baked into a generated image.

Asset	Credit and license	Use and limitation
P003	ArtBrom; CC BY-SA 2.0	Shape/context; colored-light photograph
P007	Salli; CC BY-SA 3.0	Preset arrangement; 351x337 limits detail
P004	MePaJa; CC BY-SA 4.0	Switch legends; specimen-specific

CHECKS AND DECISIONS

- Every shipped photograph has creator, source URL, license URL, and modification note.
- Every dimensional assertion has a scale-bearing image or documented measurement.

SOURCES S007-04 Hammond B3 and Drawbars photograph / S007-05 Hammond ottava dei preset photograph / S007-06 Hammond B3 START RUN photograph

ORIGINAL TEST PLAN AND OPEN QUESTIONS

Acceptance gates for a trustworthy control surface

Drawbars presets and console switch hardware

ORIGINAL DESIGN PROPOSAL: finish this subsystem when the visible console, serialized state, and audio routing agree under every transition. Test interactions before refining wood grain or brass highlights. A beautiful drawbar bank with swapped footages is a functional defect. A preset key that appears selected while a different registration sounds is equally unacceptable. The interface must remain operable with animation disabled.

Use scripted gestures that move inactive banks, change selectors during sustained chords, and toggle both volume-related tablets independently. Verify outcomes through engine-state traces and controlled renders. Compare physical drawbar motion against recordings before choosing an overlap law; continuous contact alone does not specify an amplitude interpolation curve. Retain uncertainty when a specimen has replacement caps, modified percussion wiring, or a nonstandard Leslie switch.

The modern B-3 mk2 manual describes editing a recalled preset by moving drawbars. Treat that as modern behavior requiring an explicit compatibility decision; it does not prove vintage wired presets followed moving banks. [S007-03] Hand off the semantic schema to the preset, MIDI, percussion, expression, and UI chapters. Exact control dimensions, overlap distance, and specimen-specific preset wiring remain acquisitions to complete.

Test	Required observable outcome
Move inactive bank	Current wired or other-bank registration stays unchanged
Select A# then B	Correct independent bank becomes active
Press Cancel	Manual disconnects without a sounding preset note
Percussion eligibility	Only enabled upper B admits percussion
Great vibrato	Pedal routing follows selected vintage topology
Restore project	Banks, presets, tablets, and expression restore coherently
Move drawbar slowly	No unintended zero-output gaps in continuous-contact mode

CHECKS AND DECISIONS

- Document pass/fail evidence; these proposed tests have not been executed against hardware.
- Resolve dimensions and wiring against the selected specimen before claiming dimensional fidelity.

SOURCES S007-03 Hammond B3 mk2 owners manual manufacturer edition / S007-01 Hammond Organ Service Manual early models factory scan

HARDWARE EVIDENCE

Choose the cabinet family before choosing its sound

Leslie cabinet families and physical atlas

The reference build uses a specifically identified vintage 122. The factory122-family manual distinguishes the full-height122 from the shorter142; the 145/147 manual independently documents the corresponding two cabinet heights. Treat these as documentary exterior envelopes, not complete CAD drawings. [S008-01] [S008-02]

The manufacturer distinguishes balanced 122 and unbalanced 147 interfaces. The 21H manual describes rotor on/off control arrangements, while the modern 122H/142H uses brushless servo motors. These are separate implementation targets; a similar wooden exterior does not establish identical electronics or motion. [S008-03] [S008-04]

ORIGINAL DESIGN PROPOSAL: create an immutable family record before importing any measurements. Record cabinet badge, amplifier chassis identity, region, rotor assemblies, driver labels, modifications and source revision separately. A cabinet containing a replacement amplifier remains a documented hybrid specimen; do not silently rename it to whichever component was photographed. Unidentified components stay unknown until inspected.

Family	Documented envelope or distinction	Modeling consequence
122 /147	41 in high x 29 in wide x 20.5 in deep in cited vintage editions	Separate amplifier profiles within a tall cabinet class
142 /145	33 in high x 29 in wide x 20.5 in deep in cited editions	Separate short cabinet geometry and acoustic calibration
21H	On/off motor arrangements described	Do not insert122 slow-speed behavior
122H /142H	Modern servo drives and selectable response	Separate modern product configuration

P020 | Leslie 122 (isometric).jpg | Hammond b3 con leslie 122.jpg: Salli at it.wikipedia derivative work: Leslie 122.jpg: Ritchie333 derivative work: Clusternote | CC BY-SA 3.0 | Retouched/transformed; excluded from geometry evidence. Source and license

CHECKS AND DECISIONS

- Store all exterior dimensions with source edition and scope; measure the chosen specimen independently.
- A model number may select a candidate profile but must not certify a particular cabinet as original.

SOURCES S008-01 Electro Music Leslie 122 122V 122RV 142 222 222RV operating and maintenance manual / S008-02 Electro Music Leslie Organ Speaker Models 145 and 147 manual / S008-03 Leslie Speaker 21H owners manual / S008-04 Hammond Leslie Heritage Series 122H and 142H Owners Manual V1.00 220713

MECHANISM AND PHYSICAL ATLAS

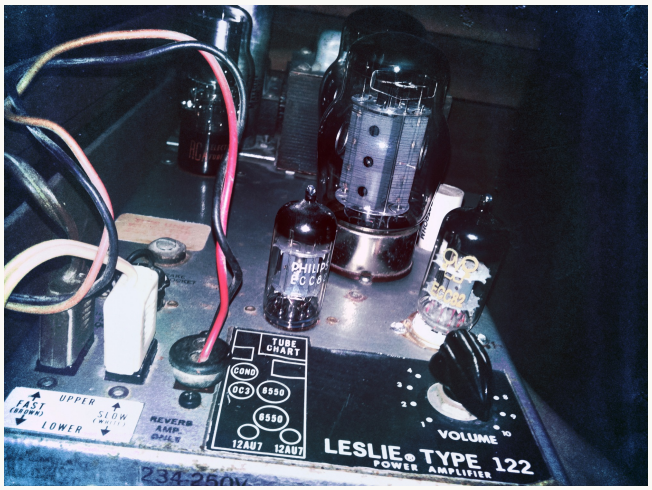
Separate the stationary drivers from rotating acoustic parts

Leslie cabinet families and physical atlas

The vintage 122 manual shows the amplifier feeding a nominal 800 Hz dividing network, then separate treble and bass paths. The compression driver feeds the upper horn assembly; the 15-inch bass driver feeds the lower rotor path. Two drive assemblies control the rotor motion. The reverb variants add a separate branch rather than turning every 122 into a reverberant model. [S008-01]

ORIGINAL DESIGN PROPOSAL: maintain an assembly graph with independent objects for chassis, crossover, drivers, horn, drum, motor assemblies, belts, baffles, back panels and openings. A graphical exploded view should preserve attachment relationships and highlight which nodes actually rotate. A renderer that spins the woofer cone around the cabinet is a mechanically misleading teaching model.

Use this graph to plan the later DSP boundaries. Electrical loading belongs to the amplifier/crossover/driver system; radiation and propagation belong to the rotating geometry and surrounding space. The modular representation does not assert that every block is acoustically independent. Preserve coupled loading where measurements or circuit analysis show that a one-way chain would lose a significant interaction.



P021 | Leslie 122 Power Amplifier.jpg | Mads Bødker | CC BY 2.0 Source and license

CHECKS AND DECISIONS

- Assign driver motion and rotor motion to different model variables.
- Exclude the optional reverb branch from the baseline 122 unless the selected specimen contains it.

SOURCES S008-01 Electro Music Leslie 122 122V 122RV 142 222 222RV operating and maintenance manual

ORIGINAL MODEL

Make variant differences explicit in the digital contract

Leslie cabinet families and physical atlas

ORIGINAL DESIGN PROPOSAL: expose a CabinetProfile that references an amplifier model, crossover record, driver pair, rotor-dynamics record, cabinet geometry and measurement set. Keep the identifiers independent but validate their compatibility. A cabinet-height switch should not secretly change the valve circuit; an amplifier change should not silently stretch the cabinet mesh.

The modern manufacturer block diagram is a useful warning about naming: its 11-pin and line inputs traverse additional input circuitry, while the vintage-style input follows another route. Accordingly, modern input measurements cannot be imported as an unidentified vintage 122 transfer function. The interface, input route and reference level must accompany every captured response. [S008-04]

Store units at the ingestion boundary and convert to SI internally. Record a physical quantity and its uncertainty instead of burying a guessed dimension in a renderer constant. Unknown geometry can drive an explicitly illustrative mesh; it must not produce a measurement-certified CAD export. Publish the profile schema before implementing variant selectors so audio, UI and tests consume the same identities.

Profile field	Required representation
identity	Family, specimen ID, badge photograph, chassis revision and modification notes
geometry	Meters, coordinate frame, panel/opening IDs and measurement provenance
electrical	Input route, circuit revision, driver and crossover identifiers
motion	Separate horn/drum states, radians and rad/s, command semantics
evidence	Documented, measured, derived, proposed or unknown for each field

MODEL EQUATIONS

omega_rad_s = 2*pi*rpm/60; rpm is rotational speed in revolutions per minute.

length_m = 0.0254*length_in; the conversion is exact, while the measured length may be uncertain.

P023 | Lesliehorn.jpg | DukeHammond | Public domain Source and license

CHECKS AND DECISIONS

• Reject profiles containing an unlabeled mixture of modern input circuitry and vintage capture data.

SOURCES S008-04 Hammond Leslie Heritage Series 122H and 142H Owners Manual V1.00 220713

VISUAL SOURCE SELECTION

Build a photographic atlas with evidence labels

Leslie cabinet families and physical atlas

The public-domain photographs of a treble horn and lower deflector by DukeHammond can support structural illustration. Their source pages do not establish a serial-numbered 122 specimen or calibrated dimensions. Label them as component references and retain their original descriptions, creator credit and file hashes. [S008-05] [S008-06]

The widely circulated Leslie 122 isometric image is explicitly retouched: its rear-left edge was extrapolated and its perspective changed. It is suitable for contextual illustration under its stated CC BY-SA3.0 terms, but must not be used to recover cabinet depth, edge geometry or joinery. Credit the derivative chain and retain the transformation disclosure. [S008-07]

ORIGINAL DESIGN PROPOSAL: give each image two independent permissions: may_display and may_support_geometry. A visually attractive image can satisfy the first and fail the second. Store annotated overlays as separate files, keep originals intact, and link every callout to an assembly ID. For screws, belts, bearings and terminals, an unresolved identification should be visible in the caption rather than resolved by a plausible-looking generated detail.

Image class	Evidence use
Original overview with badge	Identify cabinet and arrangement; dimensions still need scale
Scaled orthographic photograph	Candidate geometry evidence after lens and plane checks
Component close-up	Part markings, texture and visible connections
Retouched isometric image	Context only; extrapolated regions excluded
Generated UI rendering	Art direction only; no hardware or circuit authority

P024 | Lowerrotor.jpg | DukeHammond | Public domain Source and license

CHECKS AND DECISIONS

- Every photograph has an author, source URL, rights record and explicit modifications field.
- Mark the isometric122 asset as retouched in the printed caption.

SOURCES S008-05 Lesliehorn original photograph by DukeHammond / S008-06 Lowerrotor original photograph by DukeHammond / S008-07 Leslie 122 isometric derivative photograph

MEASUREMENT PLAN

Acquire the missing geometry and assembly measurements

Leslie cabinet families and physical atlas

ORIGINAL DESIGN PROPOSAL: define the origin at the center of the cabinet floor footprint, with the front direction and positive rotation convention recorded in a reference photograph. Measure the outer envelope first, followed by panel thicknesses, chamber boundaries, baffle openings, rotor centers and microphone reference points. Do not scale a short cabinet uniformly from a tall one; establish which internal distances actually differ.

Photograph six exterior views with a ruler in the measured plane, then obtain separate close-ups of labels, back-panel fasteners, louvres, feet, hinges where fitted, motor mounts, pulley grooves, belts, bearings, driver labels and crossover markings. A qualified technician should obtain internal circuit and live operating measurements; the emulation dataset needs the resulting evidence and calibration, not an improvised connection procedure.

Acquire audio and motion data after the specimen inventory is stable. Record horn and drum angular speed separately, motor commands, microphone coordinates, room condition and input level. Film a visible reference marker only when it can be observed without changing the operating geometry. Treat any disassembly, replacement part or belt adjustment between sessions as a new specimen-state revision.

Acquisition	Result required by the implementation agent
Exterior and interior geometry	Dimension table with units, uncertainty and matching photograph IDs
Part identity	Readable label images and unresolved-part register
Rotor motion	Separate speed traces and command timestamps
Acoustic condition	Microphone poses, room description and cabinet placement
Revision history	What changed, who recorded it and which captures remain applicable

P025 | Half-Moon Switch.jpg | CJ Sorg from Hendersonville, United States | CC BY-SA 2.0 Source and license

CHECKS AND DECISIONS

- At least two independent scale observations agree within their stated measurement uncertainty.
- Acoustic captures reference the exact specimen-state revision used for geometry and electrical records.

IMPLEMENTATION AND ACCEPTANCE

Implement the family selector only after evidence validation

Leslie cabinet families and physical atlas

ORIGINAL DESIGN PROPOSAL: implement the baseline 122 profile first. Load its evidence manifest, build a low-detail assembly mesh, connect its separate rotor-state objects, then attach the electrical and radiation models developed in later chapters. Add 147 as an explicit amplifier/interface variation with its own calibration. Add short cabinets only after their acoustic records exist. Keep 21H and modern H-series as separately gated targets.

Before publishing a selectable model, compare its declared family against the cabinet, amplifier, driver and motor evidence. Automated validation should reject missing units, missing provenance, duplicate part IDs, impossible rotations and incompatible input routes. A profile with provisional values may remain useful for development, but its status must travel into the preset browser and exported diagnostic report.

The vintage scans are valuable reading sources, but their host explicitly restricts redistribution. Keep original manual links, page locators and independently authored diagrams in the deliverable; obtain permission before bundling those scan images. This documentary gate is separate from audio validation: a well-sourced picture does not establish acoustic fidelity, and a convincing recording does not establish complete mechanical knowledge. [S008-08]

```
profile = loadCabinetProfile(id)
validateUnitsAndSourceRecords(profile)
validateVariantCompatibility(profile)
geometry = buildAssemblyGraph(profile.geometry)
motion = createIndependentRotorStates(profile.motion)
audio = buildCabinetModel(profile.electrical)
attachRadiationModel(audio, geometry, motion)
runProfileIdentityTests(profile)
exportEvidenceStatus(profile)
```

CHECKS AND DECISIONS

- Selecting 142 or145 never silently substitutes tall-cabinet impulse responses.
- Changing horn speed leaves drum speed unchanged unless an explicit control command targets both.
- A retouched photograph cannot pass the geometry-evidence gate.
- No unknown component becomes documented merely because it received a plausible default.
- Release claims state which specimens, variants and tests were actually completed.

SOURCES S008-08 Captain Foldback Leslie manual scan index and reproduction terms

HARDWARE EVIDENCE AND MODEL CONTRACT

Separate tone identity from terminal position

Tonewheel frequency and phase system

The factory manual numbers electrical frequencies1 through 91 in increasing order, while the B-3 generator cover places output terminals in a nonsequential physical arrangement. Its generator drawings also distinguish tooth count from electrical frequency number. A terminal location, tonewheel identity and MIDI note therefore require separate fields. [S009-01]

ORIGINAL DESIGN PROPOSAL. Make a canonical source registry before writing a keyboard mapper. Each record owns a stable one-based tone ID, a zero-based array index, a pitch reference, its physical pickup and shaft relationships, and separate documented, computed and measured frequency fields. Never use a physical connector position as an array offset. The upper and lower manuals must reference the same source identity when their wiring requests the same generator output.

For debugging, adopt a written pitch-naming convention and numeric MIDI anchors. In this chapter the equal-tempered reference assigns tone 1 to MIDI24, conventionally C1, and tone 46 to MIDI69 at 440 Hz. These are indexing anchors for a reference calculation; they do not assert that every gear-generated frequency equals equal temperament. Keep manual key number, MIDI note and generator number visible in diagnostic output so an octave or semitone error is immediately detectable.

Identifier	Required interpretation
tone_id1…91	Electrical source ordered by frequency; remains stable across routing.
array_index0…90	Software storage only; conversion is explicit.
terminal_location	Physical location transcribed from the selected generator cover.
shaft_id and tooth_count	Mechanical relationship, not a note name.
reference MIDI note	Derived anchor for test displays; MIDI= tone_id+23 here.

CHECKS AND DECISIONS

• A profile rejects duplicate tone IDs, duplicate physical pickup assignments and unclassified index conventions.

• Tone 1,13,46,73 and 91 are tested independently at the source registry boundary.

SOURCES S009-01 Hammond Organ Service Manual early console models

SOURCE COMPARISON AND DERIVED REFERENCE

Use rational gearing without inventing factory provenance

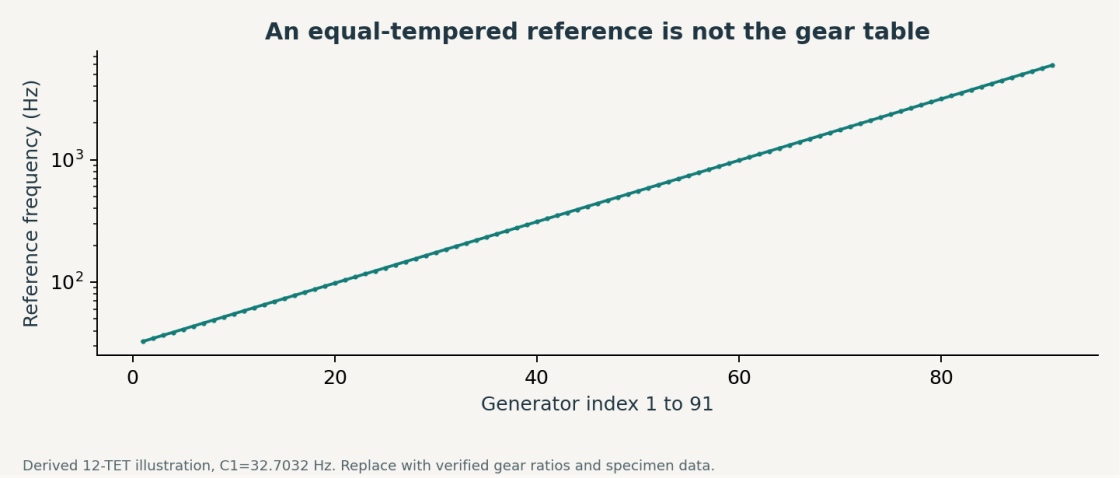
Tonewheel frequency and phase system

Hammond’s 1934 patent includes a detailed gear, tooth and frequency table, but it describes an earlier invention. Its physical gear factors must not be silently transferred to a production B-3. Equal products can use different gear counts and wheel tooth counts. [S009-02]

The table below is a derived initialization reference from the pinned setBfree60 Hz implementation: for each listed ratio, f=20 revolutions/second ×2 teeth ×ratio. It is an auditable software comparison, not a newly verified factory gear inventory. That distinction allows an implementation to proceed while preserving the requirement to confirm its selected physical specimen. [S009-04]

ORIGINAL DESIGN PROPOSAL. Store numerator and denominator as integers and preserve the source’s factorization separately. Reduced rational ratios describe speed, but cannot identify a replacement gear’s actual tooth count. Generate higher-octave provisional frequencies algorithmically only after the source-selection rule is explicit. The seven upper192-tooth outputs require their own rule and must not be treated as an ordinary256-tooth octave. Resolve disagreements between published tables through original drawings or measurement, not majority voting among copied websites.

Pitch class	Candidate ratio	Derived lowest frequency Hz
C	85/104	32.692308
C#	71/82	34.634146
D	67/73	36.712329
D#	35/36	38.888889
E	69/67	41.194030
F	12/11	43.636364
F#	37/32	46.250000
G	49/40	49.000000
G#	48/37	51.891892
A	11/8	55.000000
A#	67/46	58.260870
B	54/35	61.714286



Original explanatory calculation. Illustrative parameters are not measured instrument data.

SOURCES S009-02 Laurens Hammond US1956350A Electrical musical instrument / S009-04 setBfree tonegen.c pinned source implementation

ORIGINAL DIGITAL MODEL

Keep mechanical phase and electrical phase distinct

Tonewheel frequency and phase system

ORIGINAL MODEL. Define a shaft angle theta_j in radians and a tooth count T_i for source i. Electrical phase is T_i times the shaft angle plus an unknown pickup-reference offset alpha_i. This is a kinematic derivation: each passing tooth advances one electrical cycle. A shared shaft therefore creates a specific phase relationship between its wheels, while flexible couplings can introduce additional relative motion. Use the mechanical graph documented in chapter 6 rather than assuming all sources have one rigid phase.

Initialize unknown offsets reproducibly from a stored seed or explicit profile values, and label them modeled. Do not present random offsets as measured hardware phases. Once initialized, keep sources advancing through silence, note release and subsequent note attack. A key event changes routing and contacts; it does not restart the wheel phase. A host transport reset is not inherently a physical organ restart.

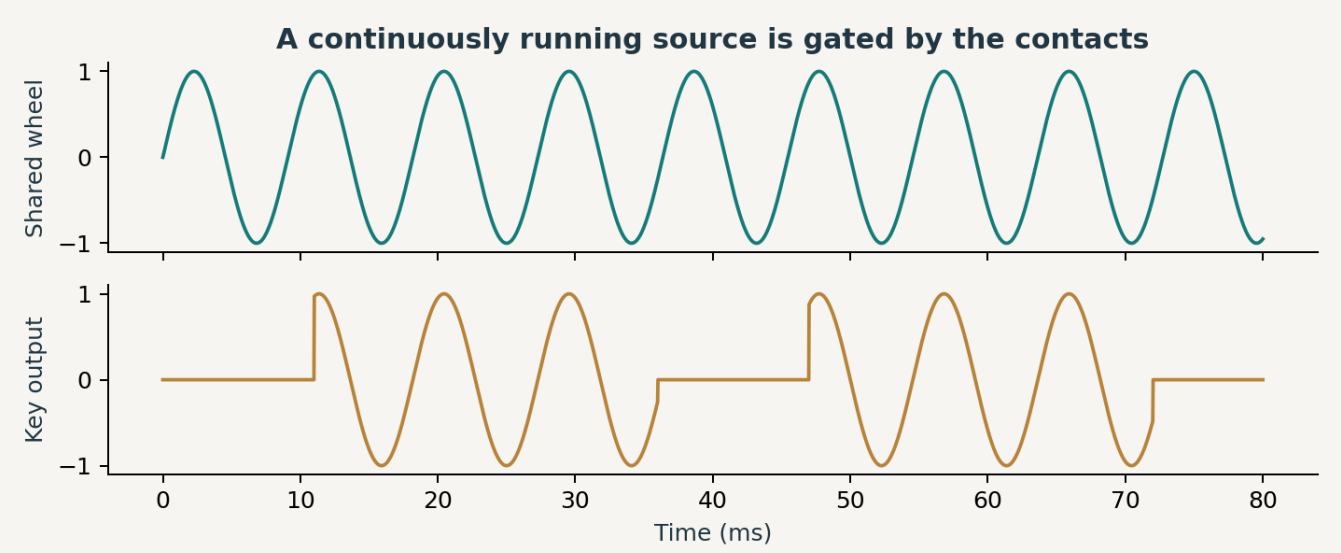
Mains-derived pitch drift can first be represented as a common time-base perturbation, with separately identified shaft residuals added later. These are candidate parameterizations, not measured deviations. Preserve phase continuity when changing tuning or sample rate. Integrate frequency into phase; directly replacing an accumulated phase with a new tuning formula creates discontinuities unrelated to the modeled mechanism.

MODEL EQUATIONS

f_i(t) = T_i * omega_j(t) / (2*pi), where omega is mechanical rad/s.

phi_i(t) = T_i * theta_j(t) + alpha_i; theta_j evolves continuously.

phi_i[n+1] = wrap(phi_i[n] + 2*pi*f_i[n]/Fs); Fs is samples/s.



Original 110 Hz simulation; chosen gates are not measured contact timings.
Original explanatory calculation. Illustrative parameters are not measured instrument data.

IMPLEMENTATION SEQUENCE

Compile an explicit frequency profile

Tonewheel frequency and phase system

ORIGINAL DESIGN PROPOSAL. Build a preparation-time compiler that resolves the chosen profile into91 source definitions. Validate positive rates, integer tooth counts, finite frequencies, unique identities and valid references before publishing the immutable result to the audio thread. Keep rational frequency generation, optional measured overrides and a12-TET test mode as distinct configurations rather than overlapping flags.

Use the region-specific motor and gearing together. The factory describes1200 rpm for the 60 Hz run motor and 1500 rpm for the 50 Hz version; this does not justify combining either speed with the other variant’s unverified gear table. [S009-01] An emulator’s50 Hz table can itself be estimated, as setBfree explicitly records. [S009-04] Require evidence metadata for those entries before promoting a50 Hz profile to a calibrated product model.

Render each source once per sample or equivalent vector block, then route its output. Preserve oscillator state across buffer boundaries. Use double-precision accumulated phase or a tested fixed-point representation and bound long-run numerical error. More detailed waveform harmonics need an anti-aliasing policy based on their highest retained frequency, even though the fundamental source lies within the audio band. Scheduling, profile compilation, file parsing and measurement fitting remain outside the callback.

```
prepare(profile, Fs):
  for tone in profile.sources:
    verifyIdentityUnitsAndEvidence(tone)
    f = resolveDocumentedOrMeasuredFrequency(tone)
    delta[tone] = 2*pi*f/Fs
    harmonicLimit[tone] = chooseBandLimit(f, Fs)
processSample():
  for tone in allSources:
    phase[tone] = wrap(phase[tone] + delta[tone])
    field[tone] = waveform(phase[tone], tone)
  contactsAndBuses.consume(field)
```

CHECKS AND DECISIONS

• Offline preparation fails on an unknown required gear entry rather than substituting12-TET silently.

• A waveform harmonic above the configured Nyquist margin is suppressed or handled by verified oversampling.

SOURCES S009-01 Hammond Organ Service Manual early console models / S009-04 setBfree tonegen.c pinned source implementation

MEASUREMENT AND CALIBRATION PLAN

Measure frequency and phase with a shared clock

Tonewheel frequency and phase system

ORIGINAL ACQUISITION PLAN. Request calibrated electrical recordings from a qualified technician using an identified B-3 generator and a common acquisition clock. Record source identity, operating conditions, instrument warmup, capture duration and sample-clock accuracy. Simultaneous multi-channel capture is essential when the goal is relative phase or correlated drift. Sequential single-channel recordings can estimate mean frequency but cannot recover a lost common time origin.

Estimate each source’s mean frequency with a phase-slope or sinusoidal fitting method, then examine residual phase over time. Use multiple capture lengths to separate finite-record estimation error, sample-clock error and genuine modulation. Retain raw recordings and fitting scripts so another engineer can repeat the calculation. Avoid reading the largest FFT bin as an exact frequency; bin spacing is a property of the analysis window, not of the generator.

For paired sources, compare electrical phase after accounting for their tooth-count ratio and fixed offsets. Examine residual correlations across a shared shaft, compartment and distant assemblies before adding independent random modulation. A reproducible phase seed is useful for testing but does not establish authentic mechanical phase. Report uncertainty and observation duration alongside every fitted frequency; repeat captures after restart to determine which parameters are stable and which are initial-condition dependent.

Capture	Question resolved
Simultaneous source pair	Shared-shaft phase and frequency relationships.
Multiple tones and clock reference	Common drift versus independent residual motion.
Repeated startup sessions	Stable calibration versus restart-dependent offsets.
Long and short windows	Mean frequency versus modulation and estimator limits.
Known digital reference through ADC	Acquisition-clock and analysis-pipeline error.

CHECKS AND DECISIONS

- A saved fit includes the original waveform hash, clock assumptions, residual statistics and uncertainty.
- No estimated phase relationship is certified from recordings without a common time reference.

ACCEPTANCE AND DOCUMENTARY AUDIT

Make indexing and pitch errors impossible to hide

Tonewheel frequency and phase system

A useful warning appears in the 2016 Hammondizer paper: its stated one-based range and printed Equation 2 use a45 offset. That expression does not produce the declared C1 anchor at index1 under the convention adopted here. The paper concerns an inspired effects architecture, so transcribing its formula into a B-3 source table without independent anchor tests is inappropriate. [S009-03]

ORIGINAL DERIVATION. For this chapter’s one-based C1-to-F#8 reference,12-TET frequency is440×2^((w−46)/12). This yields tone 1≈32.703196 Hz, tone 46=440 Hz and tone 91≈5919.910763 Hz. It is only a reference; the rational profile and measured profile are separate. Name octaves through the explicit MIDI convention because historical organ labels and software conventions differ.

Proposed completion requires91 documented identities, a complete applicable frequency profile, passing anchor tests, persistent source phases, block-size invariance and a retained comparison report. Compare manual and pedal paths that reference the same source using their known downstream transfer functions; do not mistake differing filtering for differing oscillators. No percentage-authenticity claim follows from these tests. The unresolved production gear inventory, specimen offsets and dynamic residuals remain acquisition tasks until supported by traceable evidence.

Proposed test	Pass evidence
Index anchors	Tone 1/MIDI24; tone 46/MIDI69; tone 91/MIDI114.
Rational arithmetic	Independent high-precision calculation agrees with compiled frequencies.
Shared sources	Equivalent source taps retain correlated phase through both manuals.
Silent interval	Phase advances as if audio output had remained enabled.
Buffer partitioning	Same event timeline renders within the defined numerical tolerance.

MODEL EQUATIONS

f_ET(w) = 440 * 2^((w-46)/12), w in 1…91, with MIDI note = w+23.

error_cents = 1200 * log2(f_candidate / f_reference).

SOURCES S009-03 Werner and Abel Modal Processor Effects Inspired by Hammond Tonewheel Organs

HARDWARE EVIDENCE

Record the pickup before assigning its sound

Magnetic pickup and waveform formation

The factory description places a magnetized rod and coil beside each active wheel, with a sharpened pole end near its perimeter. It gives approximate rod dimensions of 4 in length and 1/4 in diameter, describes different coil sizes across frequency regions, and identifies copper rings on certain low-frequency assemblies. These are documentary construction facts, not an individual pickup's measured tolerances. [S010-01]

Separate three objects in the model: the moving geometry, the induced source voltage, and the loaded terminal voltage. A recording taken after a generator filter cannot independently identify all three. Preserve its exact capture node and load instead of calling it an unqualified raw tonewheel sample. The filter model belongs to chapter 11; coupling and mechanical modulation belong to chapters 12 and 13.

Original design proposal: give every pickup a provenance record linked to its wheel, shaft and output terminal. Permit unknown entries for winding turns, resistance, inductance, magnetic material, remanence, gap and pole geometry. A developer can implement a measured-voltage source while those fields remain unknown; filling them with plausible constants would create fictitious physical precision.

Record	Evidence requirement
Pole and wheel geometry	Scaled original photograph or dimensional survey; distinguish teeth from surface damage.
Magnetic and winding properties	Measured or manufacturer-documented values, including method and uncertainty.
Electrical capture node	Before/after filter, probe impedance, terminal identity and calibration.
Specimen condition	Replacement history, corrosion, physical alignment and stable operating condition.

CHECKS AND DECISIONS

- Every exported waveform declares whether its level is calibrated volts or normalized amplitude.

SOURCES S010-01 Hammond Organ Service Manual, early console models

ORIGINAL PHYSICAL MODEL

Model changing flux linkage rather than moving audio

Magnetic pickup and waveform formation

Original derivation: describe mechanical angle theta in radians and a geometry vector q in metres. Let lambda(theta,q) be coil flux linkage, measured in weber-turns. Faraday induction produces the source electromotive force from its time derivative. With fixed geometry, frequency follows tooth passage; with changing gap or lateral displacement, the additional derivative terms can change both waveform shape and modulation. This representation is a proposed reduction, not a solved field model of a particular B-3.

For a wheel with T teeth, electrical phase is T times mechanical angle. Do not discard mechanical angle after constructing that phase: unequal teeth or eccentricity may recur once per revolution. A one-tooth waveform alone cannot represent such longer-period structure. Keep shared shaft angles supplied by the mechanical model and allow sparse full-revolution components where captures justify them.

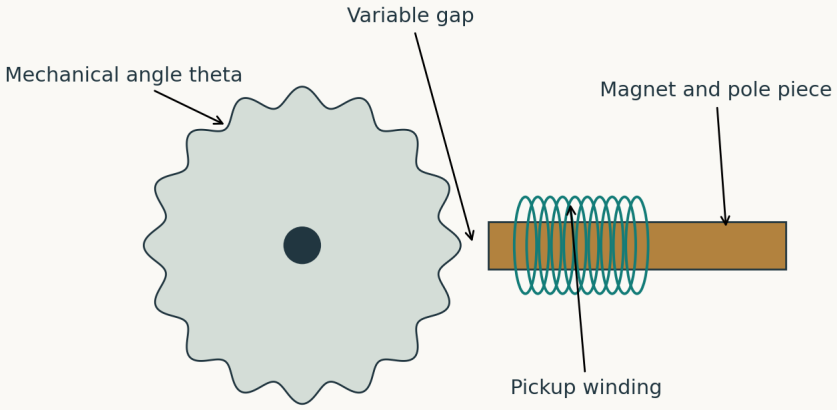
The ISMA study combines pickup recordings with high-speed motion tracking on a 1938 Model A and explores electromagnetic models. Its measurement method is a valuable precedent; its specimen, geometry and simulated material choices are not universal B-3 parameters. [S010-02]

MODEL EQUATIONS

e(t) = -d lambda/dt = -(partial lambda/partial theta)*omega - sum_j[(partial lambda/partial q_j)*qdot_j]

f_tooth = T*omega/(2*pi); omega is mechanical rad/s and f_tooth is Hz.

A tonewheel changes magnetic flux through the pickup



e(t) = -N dPhi(t)/dt

Original conceptual drawing. Tooth count, proportions and field geometry are illustrative, not production dimensions.

Original explanatory calculation. Illustrative parameters are not measured instrument data.

SOURCES S010-02 Muenster and Pfeifle, Non-linearities of the mechano-electrical tonegenerator of the

ORIGINAL DIGITAL MODEL

Choose a waveform representation with a declared boundary

Magnetic pickup and waveform formation

Original design proposal: implement three interchangeable source modes behind one interface. A documented nominal mode provides a provisional sinusoid. A calibrated mode stores complex Fourier coefficients of the observed terminal voltage. A physical mode stores flux-linkage coefficients and applies the angular derivative before the electrical network. These modes have different evidence requirements and must not be presented as equivalent physical solutions.

For fixed geometry, expand flux linkage over a full mechanical turn. Differentiation multiplies coefficient l by minus j*l*omega. This makes speed dependence explicit and avoids noisy finite differences of a sampled flux table. Keep measured full-turn asymmetry when it survives repeated captures; keep tooth-harmonic and revolution-sideband terms distinguishable in metadata.

Apply the bandwidth limit before evaluating the oscillator. Harmonics retained at one sample rate may exceed Nyquist at another; a narrow transition margin is an implementation choice that must be tested. When parameters vary, interpolation itself can add sidebands. Nonlinear pickup models also require alias control, as the research literature notes. [S010-03]

Representation	Prevent this mistake
Normalized voltage table	Do not infer coil turns or absolute volts from normalization.
Loaded terminal coefficients	Do not apply the same measured filter again downstream.
Flux-linkage coefficients	Do not omit the derivative or confuse mechanical and tooth phase.
Whole-turn residual	Do not replace repeatable eccentricity with unrelated white noise.

MODEL EQUATIONS

lambda(theta) = sum_l Lambda_l*exp(j*l*theta); e(theta) = sum_l[-j*l*omega*Lambda_l]*exp(j*l*theta).

Retain only components whose time-varying frequency and modulation bandwidth fit the selected internal sample rate.

SOURCES S010-03 Werner and Abel, Modal Processor Effects Inspired by Hammond Tonewheel Organs, Applie

IMPLEMENTATION

Compile pickup profiles outside the audio callback

Magnetic pickup and waveform formation

Original implementation sequence: first define immutable PickupProfile records with source boundary, units, wheel ID, mechanical tooth count, coefficient basis, observation load, sample-rate policy and evidence IDs. Next write an offline importer that validates conjugate symmetry, removes acquisition DC when justified, preserves phase origin and produces the selected band-limited tables. Save an import report alongside the profile.

The audio engine receives the shared mechanical state and evaluates one source per wheel. Key events never instantiate a fresh pickup oscillator or restart its phase. Process any verified electrical impedance/filter model exactly once, then expose a generator-terminal bus for contacts and leakage routing. A profile swap uses a bounded transition without reallocating tables in the callback.

For a future physical solver, use offline magnetic-field calculations to investigate geometry sensitivity and produce reduced coefficients. Running a finite-element solve inside a plugin is not the first implementation milestone. Keep that research model independent of the production evaluator so measured and physical sources can be compared at identical ports and levels.

```
prepare(profile, rate):
  validate_units_and_boundary(profile)
  coeffs = import_with_phase_reference(profile)
  tables = compile_bandlimited_sets(coeffs, rate)
  publish_immutable(tables)
render(sample, mechanical_state):
  for pickup in active_generator:
    theta, omega = shared_shaft_state(pickup)
    source = evaluate_profile(pickup, theta, omega)
    terminal[pickup.id] = electrical_port(source)
  return terminal
```

CHECKS AND DECISIONS

- Unknown material constants never receive silent defaults in a profile labeled physically calibrated.
- One cached source value is reused by every key path selecting that wheel at the same sample.

MEASUREMENT PLAN

Capture the evidence that separates geometry and filtering

Magnetic pickup and waveform formation

Original acquisition proposal: commission a qualified technician to record selected pickup and generator-terminal signals with documented loading and calibrated gain, while an optical reference records mechanical rotation. Do not alter pickup adjustment merely to obtain a more convenient waveform. Retain operating condition and before/after photographs so a profile remains tied to its actual instrument.

Begin with representative low, middle and high generator regions, then expand to all active sources. Capture enough complete revolutions to distinguish repeatable tooth shape from drift and noise. Estimate both mechanical rotation and electrical tooth frequency; their ratio is an independent consistency check. Use coherent angle resampling rather than aligning every captured cycle independently and erasing the modulation being studied.

Collect repeated sessions and hold some out from fitting. A single spectrum cannot identify magnet material, air gap, coil turns and load simultaneously. Where those parameters remain unidentifiable, prefer a measured source model with a declared port and uncertainty. The purpose is a predictable digital instrument, not an unsupported reconstruction of hidden manufacturing data.

Capture	What it resolves
Synchronized angle and voltage	Phase origin, tooth frequency, once-per-turn structure and correlated modulation.
Documented pre/post-filter observations	Which waveform features belong to the source versus the electrical network.
Repeated calibrated recordings	Stable coefficients, session drift and acquisition noise floor.
Scaled geometry and visible markings	Pole orientation, wheel identity and material evidence; hidden properties remain unknown.

CHECKS AND DECISIONS

- Store original captures and calibration records; derived coefficients must be reproducible from them.

SOURCES S010-02 Muenster and Pfeifle, Non-linearities of the mechano-electrical tonegenerator of the

ACCEPTANCE AND SOURCE CRITIQUE

Reject plausible constants that fail independent checks

Magnetic pickup and waveform formation

Original acceptance policy: test equations and software before evaluating authenticity. A constant flux linkage must generate zero emf. A single sinusoidal linkage must yield the analytically differentiated amplitude and phase. Doubling angular speed with unchanged flux geometry doubles open-circuit emf amplitude in this reduced model; it does not establish how a loaded, lossy real pickup behaves at that speed.

Check every published number against basic identities. The ISMA paper's section 8 contains a claim that a different diameter explains an original wheel's stated 20,000 rpm speed for a C tone. Diameter alone does not change f=T*rpm/60. Treat that statement as unresolved, and do not adopt the paper's rpm, material or amplitude values as B-3 constants. Its useful observational methods remain separable from this problem. [S010-02]

The Hammondizer study intentionally designs an inspired effects processor and moves nonlinear processing for artistic reasons. Reusing its architecture without that qualification would misrepresent a calibrated pickup reconstruction. [S010-03] Finish this subsystem only when source boundary, units, phase behavior, alias performance and held-out waveform error are reported; numerical fidelity thresholds should be chosen against acquisition repeatability.

Gate	Retained evidence
Mathematical correctness	Analytic derivative, sign, units, periodicity and zero-input fixtures.
Runtime correctness	Identical event-free output across block sizes; continuous phase and bounded transitions.
Spectral integrity	Alias comparison against a higher-rate reference with the same coefficients.
Physical claim	Measured specimen agreement with repeatability, uncertainty and unresolved parameters stated.

CHECKS AND DECISIONS

- No test result, measurement, or calibrated tolerance is claimed as completed by this chapter.

SOURCES S010-02 Muenster and Pfeifle, Non-linearities of the mechano-electrical tonegenerator of the / S010-03 Werner and Abel, Modal Processor Effects Inspired by Hammond Tonewheel Organs, Applie

HARDWARE EVIDENCE

Identify the actual filter family before assigning values

Generator filters and voicing components

The factory description uses a grounded tap on the filter winding and nominal capacitor groups summarized below. It also states that transformers differ and were matched with their capacitors. The printed mf or mfd notation here denotes microfarads, not modern SI millifarads. Normalize it explicitly:0.255 uF=255 nF=2.55e-7 F;0.105 uF=105 nF=1.05e-7 F. [S011-01] [S011-03]

ORIGINAL DESIGN PROPOSAL: represent each wheel output as a distinct electrical port, carrying its physical frequency number, topology variant, pickup impedance, filter parameters, downstream load and evidence record. A group designation provides a starting topology; it does not supply a measured inductance, winding ratio, loss resistance or Q. Leave those parameters unresolved until acquired or fitted.

Keep the generator filter separate from later wiring taper, drawbar attenuation, AO-28 equalization and acoustic response. Otherwise the optimizer can compensate an incorrect generator response with unrelated controls, producing a good fit to one registration and poor behavior elsewhere. Save both the observed terminal spectrum and the chosen decomposition into source and filter so future measurements can revise the interpretation without losing the original evidence.

Frequency numbers	Documented baseline network
1–43	Resistance-wire shunt; no separate reactor/capacitor pair.
44–48	Reactor without the series tuning capacitor.
49–54	Tuned reactor plus nominal 0.255 microfarad capacitor.
55–91	Tuned reactor plus nominal 0.105 microfarad capacitor.

CHECKS AND DECISIONS

- This table is the earlier baseline description; apply the subsequent bulletin as a separate revision.
- Store SI units and original notation together; reject an accidental1000-fold capacitance conversion.

SOURCES S011-01 Hammond factory Service Manual, generator filters and component matching / S011-03 Stefan Vorkoetter, Hammond Organ Tonewheel Generator Capacitor Replacement and Calibr

REVISION EVIDENCE

The 1965 revision changes frequencies 37 through 48

Generator filters and voicing components

Technical Bulletin 16, dated January 8 1965 and reprinted April 1 1966, specifies revised filtering for frequencies 37–48. Its network uses a 50 microfarad,3 V capacitor and a 10 ohm,half-watt resistor. The listed B-3 starting serial is93990. The bulletin permits earlier instruments to receive the change and says existing filter transformers need not be removed. These are documented circuit and applicability facts, not a measurement of the organ being emulated. [S011-02]

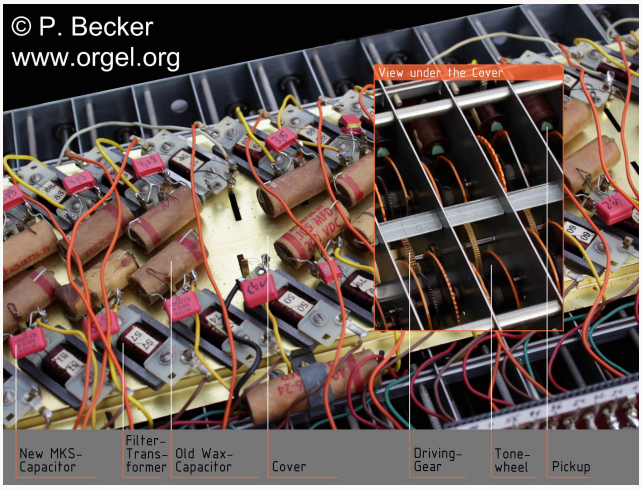
ORIGINAL DESIGN PROPOSAL: make the revision a per-port configuration field rather than a global old/new tone knob. Serial history is a useful discovery clue, but a replaced generator, field retrofit or later repair can invalidate inference from console serial alone. Confirm photographed components and continuity with the selected specimen record. Preserve early and revised networks as separately testable profiles.

The added resistance also changes loading. Model the driving source impedance instead of applying a cosmetic high-pass to an already normalized sine. For an illustrative ideal voltage source,50 microfarads with 10 ohms gives a 318.31 Hz RC corner; that calculated value is not a verified assembled-generator response. Existing impedances and connected circuitry change the response and output level.

MODEL EQUATIONS

H(s)=Zload(s)/(Zsource(s)+1/(s*C)+Zload(s)); all impedances are in ohms.

For Zsource = 0 and Zload=R: fc=1/(2*pi*R*C). This is a limiting calculation, not a measured specification.



P015 | Generator View.jpg | Peterbecker2000 | CC BY-SA 4.0 Source and license

SOURCES S011-02 Hammond Technical Bulletin 16 Improved Generator Filtering, January 8 1965, reprinted

ORIGINAL DIGITAL MODEL

Choose between an electrical model and an identified response

Generator filters and voicing components

ORIGINAL DESIGN PROPOSAL: implement two explicit fidelity levels. The circuit level retains pickup inductance/resistance, capacitor impedance, the grounded winding tap, winding coupling, loss and the external load. Transcribe connection orientation before solving. The identified level fits a stable transfer function to the measured electrical port response. Both can be useful, but an identified fit must not be presented as a complete component model.

For a coupled-inductor representation, retain winding orientation and mutual inductance. The inductance matrix must be physically admissible; arbitrary independently fitted entries can create a network that produces energy. A simple resonance estimate is useful for initial fitting, but the pickup, tapped winding and loading mean that 1/sqrt(LC) alone does not identify every production component.

The proposed low-order band-pass expression below supplies three observable parameters: center frequency, quality factor and peak gain. Add additional poles/zeros only when residuals justify them. Preserve polarity and phase relative to the common generator reference, because magnitude-only matching can produce an incorrect summed waveform. Reject a fit that changes the documented topology merely to absorb a contact fault or an unidentified measurement load.

Field	Implementation meaning
topology_id	Early shunt, reactor, tuned winding, or documented RC revision.
physical_parameters	SI-valued L,C,R,M and load, each with confidence and source.
identified_response	Stable poles/zeros or frequency samples; retain complex phase.
representation_boundary	Raw pickup waveform or already filtered terminal waveform; never both filters.

MODEL EQUATIONS

M=k*sqrt(L1*L2), with signed winding orientation; require abs(k)<=1 and L1,L2>0.

Hfit(s)=g*(s/(Q*w0))/(s*s/(w0*w0)+s/(Q*w0)+1). Here w0 is rad/s, Q is dimensionless, and g is peak gain.

SOURCES S011-01 Hammond factory Service Manual, generator filters and component matching

IMPLEMENTATION

Compute steady filter voicing once where the assumptions hold

Generator filters and voicing components

ORIGINAL DESIGN PROPOSAL: when the source is periodic, the filter is linear and time invariant, and loading is fixed, multiply each complex harmonic coefficient by the filter response at that harmonic frequency. The resulting waveform bank incorporates steady voicing without running a separate recursive filter for every wheel sample. This is an exact steady-state construction under those assumptions, not an exact model of changing load or startup transients.

If startup, time-varying pitch, modulation sidebands or key-dependent loading matters, retain a stateful circuit or a validated hybrid. The filters belong to continuously running generator sources; ordinary note-on must not zero their state. If a captured waveform already represents the filtered terminal, bypass this stage rather than applying the same response twice.

Build coefficients and harmonic banks outside the audio callback. Publish immutable profile data, preserve oscillator phase, and crossfade prepared banks when switching specimens. Bound retained harmonics below Nyquist with a documented transition margin. An offline discretization reference may use SciPy state-space conversion, which supports bilinear and zero-order-hold methods; use it as a numerical comparator, not a runtime dependency or proof of hardware correctness. [S011-04]

MODEL EQUATIONS

Bout[i,h]=Bin[i,h]*H_i(j*2*pi*h*f_i); f_i is the shared wheel frequency in Hz.

```
for port in profile.ports:
    require_verified_units(port)
    if port.waveform_boundary == POST_FILTER:
        bank[port] = bandlimit(port.capture)
    elif port.is_stationary_linear_fixed_load:
        for harmonic in retained_harmonics(port):
            bank[port,harmonic] = source[port,harmonic] * H(port,harmonic)
    else:
        prepare_stateful_network(port)
publish_immutable_profile(bank)
// Note-on changes contact routing; it does not reset this bank.
```

CHECKS AND DECISIONS

• Compare phasor-bank output with a warmed-up stateful reference using the same source and load.

• For any changing source or load, verify the approximation against the stateful result before enabling the shortcut.

SOURCES S011-04 SciPy official documentation, scipy.signal.cont2discrete

ACQUISITION AND CALIBRATION

Measure what the schematic cannot tell you

Generator filters and voicing components

Stefan Vorkoetter provides firsthand before/after observations from his 1962 M-111 showing that capacitor replacement and subsequent level calibration changed generator outputs. His chosen calibration curve belongs to that project; it is not a factory B-3 target. Use this as evidence that condition and prior work matter, not as a universal aging coefficient. [S011-03]

ORIGINAL MEASUREMENT PLAN: commission a qualified technician to document the selected generator and obtain calibrated terminal spectra, repeat captures, relevant source/load impedances and component measurements where appropriate. The aim is evidence acquisition, not an unsupervised restoration recipe. Record instrument identity, component markings, photograph orientation, capture gain, input loading, warmup state and uncertainties with each trace.

Where accessible, compare pickup-side and terminal-side spectra under the same clock reference. A terminal waveform alone does not uniquely identify magnet strength, pickup waveform, filter loss and load. Fit only identifiable quantities; preserve alternative parameter sets when the evidence cannot distinguish them. Use simultaneous reference channels or explicit phase registration rather than unrelated recordings for phase comparisons. Keep unnormalized voltage measurements so automated loudness normalization cannot erase the voicing being investigated.

Observation	What it resolves
Complex spectrum before/after	Filter gain and phase versus an assumed raw waveform.
Impedance and loading record	Whether capture equipment or later circuitry changes the terminal response.
Component mark and measured value	Nominal label versus specimen value; include measurement frequency and uncertainty.
Repeated warm captures	Repeatability floor and effects of time, temperature and setup.
Held-out notes/registrations	Whether the fitted voicing survives a different routing and load.

SOURCES S011-03 Stefan Vorkoetter, Hammond Organ Tonewheel Generator Capacitor Replacement and Calibr

ACCEPTANCE AND OPEN QUESTIONS

Close the filter milestone with falsifiable tests

Generator filters and voicing components

ORIGINAL ACCEPTANCE PROPOSAL: create a filter-port test suite before fitting the complete organ. Fixtures must select an explicit hardware revision and a waveform boundary. They should exercise a representative port from each network family, then every actual port in the acquired dataset. Report both algorithmic agreement and disagreement with physical captures; those are different claims.

Use a continuous-domain or high-rate numerical reference to test magnitude, phase, settling and load response. A proposed initial numerical tolerance is 0.1 dB magnitude and 1 degree phase over the declared useful passband, excluding frequencies below the measurement/noise floor. This is an engineering target for comparing implementations, not a Hammond specification or an achieved result. Tighten or relax it only with a recorded rationale.

The milestone remains provisional until per-port losses, coupling, loaded response, modifications and calibration are sufficiently identified. A stable filter with plausible nominal capacitors does not establish authentic voicing. The deliverable must include an uncertainty table and reproducible residual plots, allowing the next agent to see precisely which behavior is documented, calculated, fitted or still unknown.

Test	Required result
Unit conversion	0.255 uF and 0.105 uF survive serialization in farads without scale changes.
Revision selection	Early and bulletin networks remain distinct; a retrofit overrides a serial-only inference.
Numerical stability	All discrete poles remain inside the unit circle for every supported profile and sample rate.
Boundary consistency	A post-filter capture does not pass through generator filtering again.
Continuous operation	Contact events do not reset source/filter state or produce artificial restart transients.
Real-time and alias checks	No callback allocation; block-size invariance and retained-harmonic limits verified.

SOURCES S011-01 Hammond factory Service Manual, generator filters and component matching / S011-02 Hammond Technical Bulletin 16 Improved Generator Filtering, January 8 1965, reprinted / S011-04 SciPy official documentation, scipy.signal.cont2discrete

HARDWARE EVIDENCE

Treat leakage as structured sound with a location

Generator leakage and crosstalk topology

The factory manual describes four tonewheels within each steel-partitioned magnetic compartment. It also places generator filters at angles intended to reduce interaction. These construction details establish reasons to investigate several coupling paths; they do not provide a numerical leakage matrix. [S012-01]

Hammond describes leakage as a collection of unwanted tone frequencies whose prominence varies with the instrument. Its current digital manuals expose adjustable leakage controls, but those controls are not measurements of an original B-3. A parameter range in a modern keyboard must not become a vintage coupling coefficient. [S012-02]

Original design proposal: classify residual sound by where it enters the signal path before deciding how to synthesize it. Pickup contamination selected by a key contact is different from downstream wiring feedthrough that may remain with no keys held. Mains hum, broadband noise and mechanical sound require separate records. Do not assume that every audible background component is generator leakage.

Retain the source boundary established in chapter 10. A measured terminal waveform may already contain neighboring frequencies and filtering. Adding another generic leakage layer to that recording can count the same mechanism twice.

Candidate mechanism	Evidence needed
Magnetic pickup interaction	Source-correlated residual at the identified pickup or terminal.
Filter or harness coupling	Dependence on electrical node, loading and routing.
Post-contact feedthrough	Residual with relevant contacts open and control states documented.
Noise or mechanical sound	Separate spectral, coherence and acoustic observations.

CHECKS AND DECISIONS

- A leakage profile names its injection point and states which recorded effects it already contains.

SOURCES S012-01 Hammond Organ Service Manual, early console models / S012-02 Hammond SK PRO Reference Guide, Custom Sets

ORIGINAL TOPOLOGY MODEL

Build adjacency from the instrument rather than pitch order

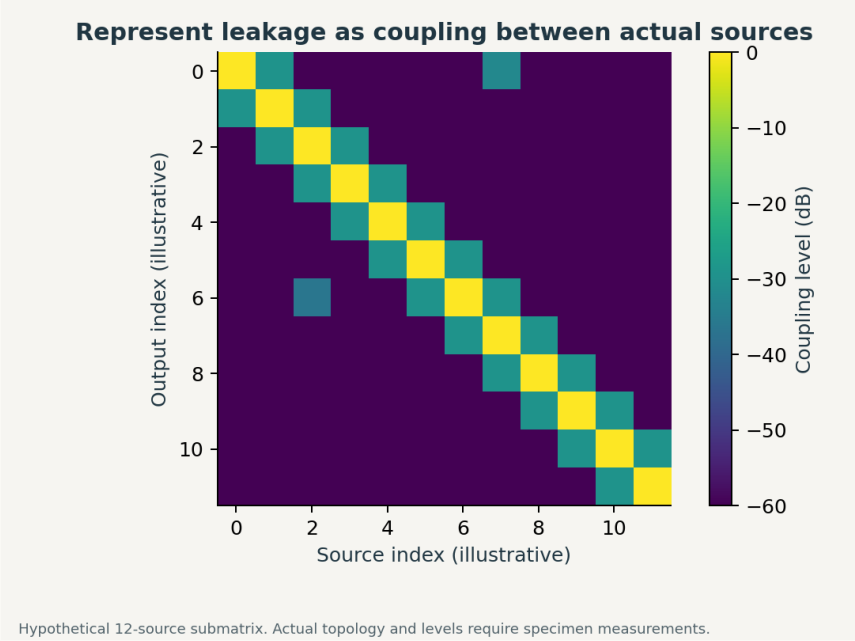
Generator leakage and crosstalk topology

Original design proposal: represent physical location independently of frequency number. Assign each wheel a shaft, compartment, pickup, filter and terminal identity. Add proposed coupling edges only where photographs, the selected wiring diagram, or repeatable measurements provide a reason. Physical neighbors need not be neighboring semitones, and harmonic relationships do not prove spatial proximity.

The DAFx 2011 paper explicitly leaves crosstalk out of its efficient organ model and notes that the relevant frequencies depend on wheel ordering. The later Hammondizer paper uses a semitone-neighbor matrix as an artistic effect. That is a useful software abstraction, not a production generator adjacency table. [S012-03] [S012-04]

Keep magnetic, filter and harness edges in separate namespaces. Their apparent symmetry must be measured rather than imposed: a transfer between loaded terminal voltages is not identical to a reciprocal mutual-inductance parameter. Until physical parameters are available, label the graph an empirical transfer model.

Use a sparse prior as a fitting aid, not a truth claim. A distant source can contribute through a shared electrical route. If the data requires such an edge, record the evidence and uncertainty rather than rejecting it because the generator drawing looks tidy.



Hypothetical 12-source submatrix. Actual topology and levels require specimen measurements.

Original explanatory calculation. Illustrative parameters are not measured instrument data.

CHECKS AND DECISIONS

- No implementation assumes that source index i couples only to indices i-1 and i+1.

SOURCES S012-03 Pekonen, Pihlajamaki and Valimaki, Computationally Efficient Hammond Organ Synthesis, / S012-04 Werner and Abel, Modal Processor Effects Inspired by Hammond Tonewheel Organs, Applie

ORIGINAL DIGITAL MODEL

Define a bounded sparse transfer model

Generator leakage and crosstalk topology

Original model proposal: let s_j be a clean shared source at a declared electrical boundary. A terminal z_i combines its diagonal response with a sum of independently identified cross-path responses. Each edge filter is causal and stable; its gain is dimensionless when source and destination use the same voltage convention. Preserve polarity and phase rather than fitting magnitude alone.

A gain-only matrix is the first reduced model. Add frequency dependence only when measurements justify it. Use stable low-order sections or finite impulse responses, and retain the fitted bandwidth. This is a feedforward approximation of observed transfers, not a solved coupled electromagnetic network.

Apply the leakage amount control only to off-diagonal terms. Its calibrated value is one; zero produces a diagnostic clean profile. Values beyond the fitted range are optional creative extensions, not alternate factory settings. Do not automatically normalize the sum and thereby erase a real level change.

A fixed linear mix cannot create new frequencies. Rapid coefficient movement can create modulation sidebands, while later nonlinear stages can create intermodulation. Smooth user changes with a stated transition policy and verify the complete chain at each internal sample rate.

Stored field	Purpose
Source and destination boundary	Prevents inconsistent units or double filtering.
Gain/filter and phase convention	Makes the transfer reproducible.
Evidence, uncertainty and valid range	Separates calibration from creative scaling.

MODEL EQUATIONS

z_i[n] = D_i{s_i}[n] + alpha * sum_(j in N_i) H_ij{s_j}[n].

y[n] = R(k[n],d[n]) z[n] + b[n]; R is contact/drawbar routing and b is separately evidenced downstream residual.

For gain-only edges: |z_i| <= |D_i s_i|max + |alpha| * sum_j |g_ij|*|s_j|max.

IMPLEMENTATION

Render shared sources once and reuse the coupling graph

Generator leakage and crosstalk topology

Original implementation sequence: first define immutable edge records containing source ID, destination ID, filter representation, gain convention, calibration evidence and state requirements. Compile these into a destination-indexed sparse array outside the audio callback. Preserve a straightforward dense implementation solely as an offline numerical reference.

Render each generator source once for each sample and reuse it throughout the graph. Reading a coupled source must never instantiate another oscillator with a different phase. Keep oscillator state running when keys are released, because contacts change routing rather than the generator state.

A dense91-by91 multiplication contains 8,281 coefficient products per sample; this is a derived operation count, not a CPU benchmark. Sparse evaluation scales with the number of retained edges. Profile actual filter and memory costs before choosing a numerical budget. Fixed spectral transfers may sometimes be compiled into source coefficients, provided their phase and boundary definitions remain equivalent.

Publish profile changes through preallocated snapshots. For filters with memory, transfer or crossfade state using a tested bounded method. A graph update must not allocate, lock, parse files, or expose partially updated indices in the audio callback.

```
prepare(profile):
  validate_boundaries_units_and_ids(profile)
  graph = compile_sparse_edges(profile)
  allocate_filter_state(graph)
  publish_immutable(graph)
render(sample):
  source = evaluate_shared_generator(sample)
  terminal = diagonal_response(source)
  for edge in graph:
    terminal[edge.dst] += alpha * edge.process(source[edge.src])
  return route_contacts_and_drawbars(terminal)
```

CHECKS AND DECISIONS

- Sparse and dense reference implementations agree within the declared numerical tolerance.
- Changing note count does not create extra copies of a generator source or its coupling state.

MEASUREMENT AND IDENTIFIABILITY

Measure residuals without pretending the matrix is uniquely known

Generator leakage and crosstalk topology

Original acquisition proposal: have a qualified technician capture generator terminals and selected downstream nodes with calibrated gain, documented probe loading, a common sample clock and a stable rotation reference. Retain all controls, active contacts, temperature, mains conditions and specimen identifiers. Use repeated sessions instead of a single persuasive spectrum.

Compare isolated contact routes, different drawbar settings, held chords, all-keys-released intervals and unaffected direct outputs. These observations help locate contamination before or after switching. A measured line at another wheel frequency establishes a component in the residual; it does not by itself prove a particular magnetic path.

The inverse problem can be non-unique. A source harmonic may share a frequency with another wheel, and mechanically related sources can have strongly correlated phase. Do not infer separate cross-path coefficients from a rank-deficient regression. Report identifiable combinations, confidence intervals and unresolved alternatives. If necessary, retain an empirical loaded-terminal spectrum rather than inventing a physical decomposition.

Use held-out keys, registrations and sessions to test the fitted profile. Compare complex spectra where phase alignment is supported, and compare line levels or distributions where it is not. Acquisition noise must be measured independently so the fit does not convert interface noise into a permanent instrument feature.

Observation	Question it addresses
Open versus closed contacts	Does the residual follow the selected source path?
Different electrical nodes	Where does the contamination first appear?
Repeated phase-referenced captures	Is it coherent source leakage or unrelated noise?
Held-out chord and registration tests	Does the fitted explanation generalize?

ACCEPTANCE AND OPEN QUESTIONS

Accept a calibrated residual model rather than a noise knob

Generator leakage and crosstalk topology

Original acceptance policy: make structural tests independent of the final listening judgment. With all off-diagonal edges disabled, the diagonal source path must be unchanged. With one test edge enabled, only its declared destination receives the additional signal before routing. A sign inversion must produce the expected phase reversal. Invalid source IDs, non-finite coefficients and unstable filter profiles must fail validation before activation.

Check silence and switching explicitly. If a model injects generator leakage after all contacts, require corresponding captured evidence. If contamination is within a pickup path, opening its route must remove that contribution except for separately modeled downstream memory or feedthrough. Avoid a universal rule that every B-3 either has constant audible leakage or becomes perfectly silent.

Finish with a report separating documented topology, fitted transfer behavior and unknown mechanism. The four cited sources establish useful construction and modeling context; none supplies a complete measured91-source coupling matrix for the chosen specimen. No default decibel level or universal aging range is justified by this research.

Inspect the final output through the organ electronics and Leslie after the dry residual tests pass. Distortion may reveal quiet coherent components that a solo terminal spectrum makes easy to underestimate. Preserve the dry evidence so a downstream tonal adjustment cannot conceal a faulty coupling model.

CHECKS AND DECISIONS

• Leakage changes preserve shared source phase and remain deterministic across block boundaries.

• Profile transitions produce no unexplained discontinuities or alias products above the declared engineering target.

• Validation includes both pre-routing terminals and held-out musical outputs.

• The release profile names missing measurements and makes no unmeasured authenticity percentage claim.

SOURCES S012-01 Hammond Organ Service Manual, early console models / S012-02 Hammond SK PRO Reference Guide, Custom Sets / S012-03 Pekonen, Pihlajamaki and Valimaki, Computationally Efficient Hammond Organ Synthesis, / S012-04 Werner and Abel, Modal Processor Effects Inspired by Hammond Tonewheel Organs, Applie

HARDWARE EVIDENCE

Separate generator irregularity from deliberate modulation

Mechanical modulation and motor imperfections

The factory description assigns speed-pulsation attenuation to spring couplings at the motor and tonewheels, together with flexible drive-shaft couplings. It describes a motor forcing event each mains half-cycle and a separate suspension that limits vibration transfer to the cabinet. The start and synchronous run motors have different roles. [S013-01]

Original modeling inference: half-cycle forcing corresponds to 100 or 120 events per second for 50 or 60 Hz mains. This is a forcing-rate calculation, not a measured pickup sideband amplitude. Coupling compliance, losses and the gear relationships determine what reaches each electrical source. A tonewheel rotational rate, tooth-passage frequency and mains forcing rate are different quantities.

Original design proposal: keep generator irregularities separate from scanner vibrato, rotary-cabinet modulation, electrical hum and room response. Give each mechanism a bypass and observation point. A pleasant global low-frequency oscillator cannot establish which mechanism caused a recorded fluctuation. Define nominal and specimen-calibrated profiles separately, and leave the modulation depth unknown until useful measurements exist.

Mechanism	State and evidence boundary
Common drive-speed variation	Shared mechanical phase; infer from simultaneous frequency residuals.
Shaft torsion or local eccentricity	Local angular or gap coordinates linked to physical assembly IDs.
Electrical hum	Additive electrical interference; do not encode as pitch variation.
Scanner or Leslie movement	Separate intentional mechanisms, validated in their own chapters.

CHECKS AND DECISIONS

- Disabling scanner and Leslie processing must not silently disable the generator state.
- A factory forcing rate must never be labeled a measured modulation depth.

SOURCES S013-01 Hammond Organ Service Manual, early console models

MECHANISM AND ORIGINAL MODEL

Distinguish angular motion from changing pickup gap

Mechanical modulation and motor imperfections

The ISMA 2019 authors tracked wheel motion and observed corresponding pickup-voltage amplitude variation. Their instrument was a 1938 Model A. The observation supports investigating gap-dependent modulation, but its dimensions, motion magnitudes and material parameters cannot be imported as B-3 defaults. [S013-03]

Original design proposal: separate angular irregularity from transverse motion. Angular error changes tooth phase; changing pickup gap can alter amplitude and harmonic shape through the magnetic source model. Chapter 10 already permits a geometry-dependent flux linkage, so a physical implementation should feed the displacement there. Adding a second generic tremolo after that model would count the same mechanism twice.

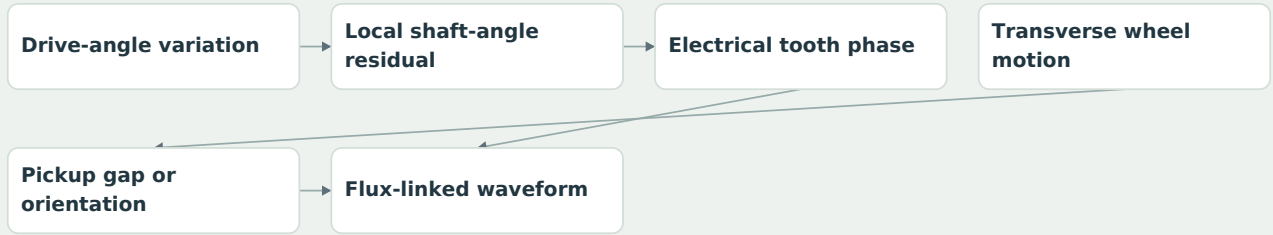
For a voltage-source implementation, a calibrated residual can approximate effects not retained in the waveform table. Preserve components synchronous with a mechanical revolution and those shared by a shaft or drive section. Uncorrelated residual noise is a different term. Spectral sidebands alone do not uniquely distinguish amplitude modulation from phase modulation; retain the complex signal and a mechanical reference when fitting.

MODEL EQUATIONS

Pure AM example: x(t) = A*[1+m*cos(2*pi*f_m*t)]*cos(2*pi*f_c*t).

Derived AM sidebands at f_c +/- f_m each have amplitude A*m/2; m is dimensionless.

Two distinct modulation routes



Original conceptual model. Angular and gap effects may share a mechanical cause, but they enter different mathematical variables.

SOURCES S013-03 Muenster and Pfeifle, Non-linearities of the mechano-electrical tonegenerator of the

ORIGINAL DIGITAL MODEL

Preserve shared phase with a hierarchy of residuals

Mechanical modulation and motor imperfections

Original derivation: let theta_d be drive angle in radians, r_j the signed gear ratio for shaft j, and delta_j its angular residual. A wheel with T_i teeth on that shaft has electrical phase T_i*(r_j*theta_d+delta_j)+alpha_i. Keep alpha_i as a documented or explicitly provisional pickup-reference offset. The shared drive term automatically couples the relative pitch of all downstream sources.

An equivalent reduced common-speed model uses a time perturbation q(t), measured in seconds: phase_i=2*pi*f_i*[t+q(t)]+alpha_i. Its fractional frequency deviation is qdot(t). This model captures common timing error but cannot by itself represent shaft-specific torsion or transverse pickup motion. Choose the smallest hierarchy that explains held-out captures; additional modes must improve identifiable behavior.

When a torsional mode is required, use a stable damped second-order state with positive modal frequency and damping. Fit its transfer response rather than guessing a physical spring constant from one spectrum. A measured residual may be used directly if its capture and looping boundary are documented. Random processes must be seeded and indexed by sample time, so block-size changes do not change the instrument.

Profile field	Contract
Common modulation	Units, spectrum, source evidence and region-specific operating condition.
Local residual	Shaft/compartment ID, angular or geometric basis, correlation relationships.
Model provenance	Measured trace, fitted mode or provisional artistic approximation.
Runtime state	Persistent phase, modal state, sample index and deterministic seed.

MODEL EQUATIONS

phi_i(t) = T_i*[r_j*theta_d(t)+delta_j(t)] + alpha_i.

f_i(t) = (1/(2*pi))*d phi_i/dt; integrate instantaneous frequency, never substitute f_i(t)*t.

delta_ddot + 2*zeta*Omega*delta_dot + Omega^2*delta = b*u(t); Omega in rad/s, zeta dimensionless, b carries the required input units.

IMPLEMENTATION

Implement steady rotation before optional start and coast

Mechanical modulation and motor imperfections

Original implementation sequence: compile nominal gear relationships, source assignments and any modal discretization outside the audio callback. First pass constant-speed and phase-continuity tests. Next add common residuals, then justified local residuals, and finally pickup-gap effects. Each addition needs a separate comparison against the same recordings. Preserve one shared source value for every downstream contact selecting that wheel.

An optional startup mode needs explicit states for stopped, starting, synchronized and coasting. The factory starting sequence establishes the distinction between startup and locked operation, but does not provide a complete torque/friction identification. [S013-01] Fit speed trajectories from captures or label an artistic approximation. Preserve angle when changing state; a start button is not permission to reset every waveform simultaneously.

Keep mechanical startup independent from host transport and separately model electrical availability. A coasting generator alone does not predict a recorded power-off sound, because the amplification and supply states also matter. Prevent control-rate stepping from producing spurious sidebands. Band-limit the source representation for its instantaneous speed, and validate interpolation or oversampling with a higher-rate reference.

```
prepare(profile, rate):
  compile_shared_topology(profile)
  discretize_stable_modes(rate)
  allocate_persistent_states()
renderSample(sampleIndex):
  drive = update_drive_state(sampleIndex)
  for shaft in shafts:
    shaft.advance(drive, local_residual(shaft))
  for wheel in wheels:
    phase = wheel.teeth * shaft.angle + wheel.offset
    source[wheel] = pickup(phase, shaft.gap_state)
  return source
```

CHECKS AND DECISIONS

- The audio callback performs no allocation, file access, mutex locking or coefficient fitting.
- Transport play/stop changes mechanical state only through an explicit product policy.

SOURCES S013-01 Hammond Organ Service Manual, early console models

MEASUREMENT AND CALIBRATION

Measure correlation before fitting attractive imperfections

Mechanical modulation and motor imperfections

Original acquisition proposal: obtain simultaneous generator signals and an independently calibrated rotational reference. Record the capture node, probe loading, timebase accuracy, warmup, supply frequency and instrument condition. A qualified technician should define safe electrical access. Use non-contact motion acquisition where practicable, and preserve the unmodified raw recordings.

Estimate each carrier and then compute residual amplitude and phase. Compare normalized instantaneous-frequency residuals between sources mapped to the same shaft, separate shafts, and different drive sections. Common acquisition-clock error can resemble common mechanical drift; an independent timebase or calibrated reference is needed to bound that ambiguity. Do not infer a physical coupling solely from two similar-looking spectra.

Fit angle-synchronous components using the mechanical reference. Retain real elapsed time as well: resampling every cycle onto an identical period can remove the fluctuation being measured. Repeat steady, startup and shutdown captures in separate sessions. Hold out both sessions and wheel groups during fitting, so a model cannot memorize one noisy recording and claim general mechanical fidelity.

Data product	Question answered
Complex carrier residuals	Are fluctuations predominantly amplitude, phase, or both?
Cross-spectral coherence	Which sources share a reproducible component?
Angle-referenced averages	Which effects repeat with a particular shaft revolution?
Repeated operating transitions	Are startup and coast trajectories stable enough to parameterize?
Clock and gain uncertainty	Could the observed variation originate in the acquisition system?

CHECKS AND DECISIONS

- Retain an identification report showing which parameters remain nonunique.

SOURCES S013-03 Muenster and Pfeifle, Non-linearities of the mechano-electrical tonegenerator of the

ACCEPTANCE AND SOURCE CRITIQUE

Accept measured structure rather than a generic wow control

Mechanical modulation and motor imperfections

The DAFx 2011 work supplies an efficient amplitude-modulation approximation for tonewheel imperfections. It is useful as a low-cost synthesis reference, not a measurement of the selected B-3. [S013-02] Original acceptance policy: a nominal preview may expose an explicitly artistic imperfection control, but a calibrated profile must use supported rates, depths and correlations. Do not turn a convenient paper example into a universal generator constant.

Create analytic fixtures before fitting hardware. A zero-residual profile must reduce to the constant-speed baseline. Pure sinusoidal AM must produce its predicted two sidebands. A sinusoidal phase perturbation must produce the correct instantaneous-frequency deviation; verify complex phase as well as magnitudes. If two sources share exactly the same timing perturbation, their fractional frequency deviations must agree within the declared numerical tolerance.

Validate long sustained chords, silent intervals, rapid note changes and different buffer partitions. Mechanical state must continue while contacts are open, and note-on must not reseed or restart it. Compare modulation spectra and cross-source coherence on held-out captures, with repeatability bounds. No fidelity percentage or measured pass follows from this research chapter; completion requires retained software and acquisition evidence.

Proposed gate	Retained result
State invariance	Same seed and sample timeline produce equivalent results across buffer sizes.
Model isolation	Each residual route can be bypassed without changing unrelated paths.
Spectral integrity	Sidebands agree with analytic fixtures; alias products meet the chosen budget.
Specimen agreement	Held-out residual spectra and correlations meet declared uncertainty-aware tolerances.
Transition integrity	Start/coast preserves angle continuity and uses a separately identified electrical-output model.

SOURCES S013-02 Pekonen, Pihlajamaki and Valimaki, Computationally Efficient Hammond Organ Synthesis,

ORIGINAL DATA CONTRACT

A calibration profile is evidence plus an executable model

Tonewheel calibration data model

The selected vintage B-3 baseline contains 91 numbered electrical sources. The registry must preserve those identities independently of storage order, terminal position and keyboard assignment. The underlying hardware evidence is developed in chapters 6 and 9; this chapter specifies how an implementation agent should store it without converting assumptions into measurements. [S014-01]

ORIGINAL DESIGN PROPOSAL. Separate three documents: immutable observations, an editable model profile, and a compiled runtime bank. Observations retain original recordings and instrument readings. A profile chooses a documented topology and fitted or provisional parameters. Compilation converts that profile into efficient sample-rate-specific tables and coefficients. Never overwrite observations with the fitted answer.

Represent evidence per parameter, not merely once in a file header. One profile can contain a documented capacitor label, an observed output voltage, a derived gear frequency and an unknown winding resistance. A single confidence score conceals these differences. Use categorical provenance and explicit uncertainty definitions instead of an unexplained authenticity percentage.

Layer	Required contents
Observation bundle	Specimen and session IDs; original assets; loading, gain, clock and environmental records.
Model profile	Schema version, hardware identity, topology, source definitions, fitting recipe and evidence references.
Compiled bank	Resolved numeric arrays, band limits, sample rate, compiler version and parent-profile digest.
Playback state	Advancing phases and filter histories; deliberately separate from immutable calibration.

CHECKS AND DECISIONS

- A calibration edit creates a new version; original observations remain recoverable.
- A nominal profile may render a declared reference mode but cannot claim measured specimen status.

SOURCES S014-01 Hammond Organ Service Manual early console models

ORIGINAL SCHEMA DESIGN

Make identifiers units and unknowns unambiguous

Tonewheel calibration data model

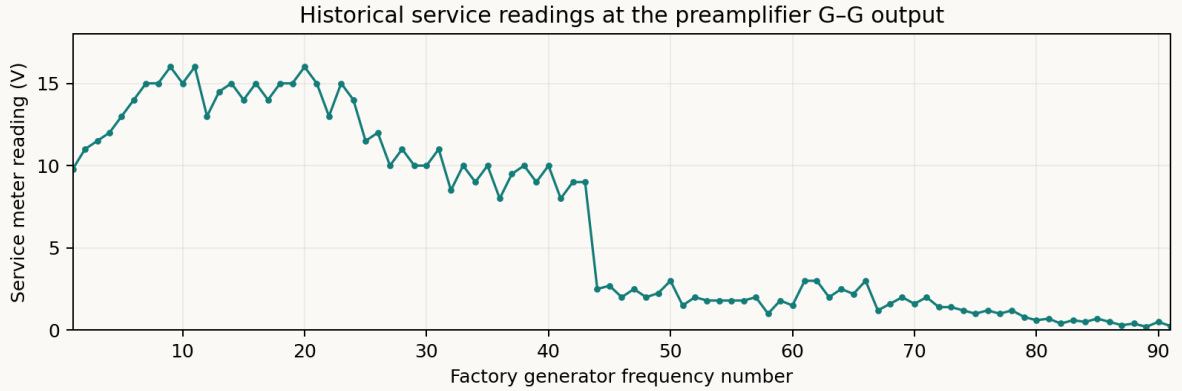
ORIGINAL DESIGN PROPOSAL. Use one-based tone_id values 1 through 91 at the document boundary and perform the zero-based conversion inside the compiler. Store shaft_id, pickup_id and terminal_location as separate references. Preserve unreduced gear-tooth counts when known; a reduced speed ratio alone cannot identify a physical replacement gear.

Each quantity record contains value, unit, evidence_class, source_refs and uncertainty. Use null with an explicit reason when an observation is unavailable. Zero is a numerical result, not a missing-value sentinel. JSON itself excludes NaN and Infinity; reject parsers or writers that silently accept those extensions. [S014-04]

Keep nominal, observed and fitted quantities separate. A fitted frequency can differ from an observed frequency because the fitting procedure compensates for clock uncertainty or parameter coupling. Retain the method and residual rather than replacing the observation. Bounds below are proposed software validity rules, not factory tolerance specifications. Region-specific gearing must remain associated with its motor and applicable evidence.

A concrete boundary trap appears in factory Table V: its voltage readings use a meter across the preamplifier G–G output, under a special service connection. They are not raw pickup amplitudes. The companion preserves all 91 independently checked readings and the stated conditions; meter range and detector type remain unknown. Do not load this curve directly into oscillator gains. [S014-06]

Field	Type and proposed validity rule
tone_id	Integer 1..91; complete and unique across the selected baseline.
frequency_hz	Positive finite number for a running reference; instantaneous stopped-rotor state is separate.
gear_numerator / denominator	Positive integers or explicitly unknown; retain physical tooth counts separately.
amplitude_v_rms	Finite nonnegative value; record electrical port and measurement load.
phase_offset_rad	Finite radians with a declared reference; unknown phase remains null.
R_ohm, L_henry, C_farad	Positive where the chosen topology requires a component; absent component is a topology decision.
uncertainty	Method, units, interval or standard uncertainty definition; null when unassessed.



Source: factory Table V, printed 4-3 / PDF168; meter connection at printed 4-1 / PDF166.
These are documentary test-path values, not raw pickup amplitudes, oscillator gains or measurements made for this suite.

Original visualization of factory Table V. These are preamplifier-output service meter readings under a defined historical test setup, not raw generator amplitudes.

SOURCES S014-04 RFC 8259 The JavaScript Object Notation Data Interchange Format / S014-06 Hammond factory service manual: Table V and its preamplifier-output measurement bound

ORIGINAL REPRESENTATION MODEL

Declare exactly what the waveform already contains

Tonewheel calibration data model

ORIGINAL DESIGN PROPOSAL. Make waveform_boundary a required discriminated field: induced_emf, loaded_pickup_port, or filtered_generator_terminal. A captured terminal waveform already includes upstream filtering under its recorded load. The compiler must not apply that response again. Conversely, a raw magnetic-source model requires the selected electrical network before it reaches the manual harness.

Declare the phase domain. An electrical-cycle table represents one tooth cycle; a shaft-revolution table can retain repeatable differences around the complete wheel. Keep that domain, tooth count, rotation direction and phase origin beside the samples or harmonic coefficients. Do not average away mechanical-period behavior and later describe the resulting table as a complete wheel model.

Store leakage as directed coupling records between named sources and ports. Each record specifies a constant complex gain or a transfer-function representation, its units, and its observation boundary. An absent edge means unmodeled or unobserved according to an explicit policy; it must not automatically mean a measured zero. Keep hum and random noise in separate records so spectral fitting cannot trade them silently against coherent leakage.

Representation	Metadata that prevents ambiguity
Waveform samples	Phase domain, endpoint convention, DC treatment, voltage scale and bandwidth.
Harmonic phasors	Harmonic index, real/imaginary values, peak convention and phase reference.
Filter network	Topology ID, SI parameters, load model and applicable revision.
Leakage edge	Source tone, destination port, transfer representation, observation basis and uncertainty.

MODEL EQUATIONS

x(phi) = a0 + sum_h Re{b_h * exp(j*h*phi)}; b_h is a peak-voltage phasor under this declared convention.

CHECKS AND DECISIONS

• The same waveform cannot be tagged both raw and post-filter.

• Compile a sample-rate-dependent harmonic limit; an archival table is not automatically safe at every rate.

IMPLEMENTATION SEQUENCE

Validate the document before preparing audio data

Tonewheel calibration data model

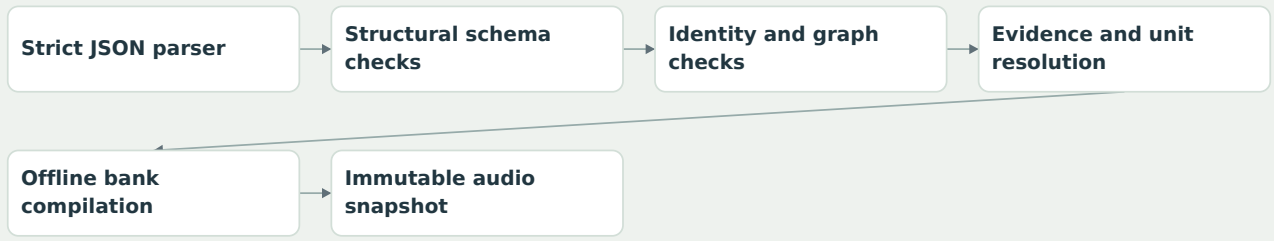
JSON Schema supplies structural assertions for types, required properties, numeric bounds and array lengths. Its uniqueItems keyword checks whether complete array elements differ; two unequal records can still contain the same tone_id. Therefore structural validation must be followed by domain-specific identity and graph checks. Pin the 2020-12 dialect and validator version rather than relying on a tool default. [S014-03]

ORIGINAL IMPLEMENTATION PROPOSAL. Use a strict parser that rejects duplicate object names, then validate the schema, resolve evidence references, and check the exact ID set. Confirm that every shaft, filter and asset reference exists. Enforce parameter conditions for the selected topology and refuse contradictory observation boundaries. Unknown measurement fields may remain in a research profile, but compilation requires an explicit provisional-mode policy.

The record below is an intentionally incomplete schema example, not an acquired observation or a complete compilable profile. Tone 46 and its 440 Hz reference follow the chapter 9 indexing fixture; the null observation honestly records that no specimen measurement has been supplied.

```
{"tone_id":46,
 "reference_frequency_hz":440,
 "reference_class":"derived_index_fixture",
 "observed_frequency":{
   "value":null,"unit":"Hz","evidence_class":"unknown",
   "reason":"specimen capture not acquired"},
 "waveform_boundary":"filtered_generator_terminal",
 "capture_asset":null,
 "profile_state":"research_incomplete"}
```

Original preparation and publication boundary



All file parsing, asset loading and coefficient preparation occur outside the audio callback.

SOURCES S014-03 JSON Schema draft 2020-12 Validation vocabulary / S014-04 RFC 8259 The JavaScript Object Notation Data Interchange Format

ACQUISITION AND REVISION PROVENANCE

Ingest measurements without losing their conditions

Tonewheel calibration data model

Technical Bulletin 16 identifies a generator-filter change for tones 37 through 48 beginning at B-3 serial 93990, while also permitting retrofit to earlier instruments. A console serial therefore cannot conclusively determine the network currently fitted. Store serial-derived expectations separately from photographed or traced topology observations, and retain conflict records for the acquisition team. [S014-02]

ORIGINAL ACQUISITION PLAN. Import untouched recording bytes with channel identity, sample rate, sample-clock characterization, interface gain, electrical load, warmup state, instrument configuration and capture time. A phase-sensitive comparison needs a common reference or a documented alignment method. Record any alignment transform separately so it can be reversed; do not discard relative phase during automatic normalization.

Separate repeatability from common calibration uncertainty. A gain-calibration error can affect many wheel measurements together, so independently perturbing every amplitude is not an equivalent uncertainty model. Permit named uncertainty groups and an optional covariance asset. If the evidence cannot identify pickup, filter and load contributions independently, preserve an effective terminal model rather than inventing component precision.

For reproducible profile identity, choose a documented canonical serialization and hash its output, while hashing original binary assets byte-for-byte. RFC 8785 defines one canonical JSON scheme; an ordinary pretty-print or sorted-key dump is not automatically equivalent. Preserve the original import and the canonicalization-version record. [S014-05]

Retained record	Reason
Original capture hash	Connect every fitted parameter to unchanged source data.
Transformation recipe	Reproduce calibration, trimming, alignment and spectral estimation.
Revision observation	Resolve serial inference versus actual generator configuration.
Fit version and residuals	Explain changes without relabeling estimates as observations.

SOURCES S014-02 Hammond Technical Bulletin 16 Improved Generator Filtering / S014-05 RFC 8785 JSON Canonicalization Scheme

ACCEPTANCE AND HANDOFF

Make data integrity a release gate

Tonewheel calibration data model

ORIGINAL ACCEPTANCE PROPOSAL. Deliver a versioned schema, semantic checker, profile compiler, migration tool and synthetic fixtures before collecting a large dataset. Synthetic fixtures must be labeled as such and exercise the validator; they are not substitute instrument measurements. Include a round-trip test for every supported schema version and preserve the meaning of source IDs across migrations.

Compile only after the chosen policy resolves every runtime requirement. A production profile must not silently replace missing calibration with zeros or generic constants. A separately named development profile may use declared proposals. Publish fully prepared immutable data to the audio engine and reclaim old banks outside the callback. Phase and filter-state migration needs its own compatibility or crossfade rule; changing a profile is not permission to restart every wheel.

Schema validity establishes a coherent representation, not physical authenticity. The final handoff should distinguish validated software invariants, documented hardware facts, fitted observations and open acquisitions. No measurements or runtime implementation have been completed by this chapter; its deliverable is the specification that an agent must implement and prove.

Fixture or test	Required result
Duplicate tone ID with different data	Semantic check rejects it even if uniqueItems passes.
Missing tone, broken shaft or asset ID	Specific validation error; no partially published audio bank.
Null, zero, NaN and Infinity	Unknown is distinct from zero; nonfinite numeric input rejected.
Farads versus microfarads	Explicit conversion fixture catches a million-fold unit error.
Early generator with documented retrofit	Observed topology overrides serial inference with retained evidence.
Raw and already filtered alternatives	Equivalent declared-boundary fixtures do not receive duplicate filtering.
Canonicalization and migration	Defined digests reproduce; semantic changes create new profile versions.

CHECKS AND DECISIONS

- No parsing, disk access, allocation, fitting or destruction occurs on the audio callback.
- Loading an invalid profile leaves the previous valid profile and playback state intact.

SOURCES S014-03 JSON Schema draft 2020-12 Validation vocabulary / S014-04 RFC 8259 The JavaScript Object Notation Data Interchange Format / S014-05 RFC 8785 JSON Canonicalization Scheme

HARDWARE EVIDENCE AND ROUTING CONTRACT

Fix the three index systems before mapping a note

Manual harmonic mapping and foldback

The factory service manual provides a complete61-key, nine-drawbar numerical routing chart. In its B-3 table, the lowest playing C sends both the 16-foot and 8-foot drawbars to tone 13, while its 5 1/3-foot drawbar uses tone 20. At the other end, the top C uses only tone IDs61 through 89, with several repeated assignments. These are wiring choices, not a generic harmonic oscillator formula. [S015-01]

ORIGINAL DESIGN PROPOSAL. Represent playing-key number, electrical tone ID, physical generator terminal and software array index as distinct types. Number the 61 playing keys1–61 from the lowest C; exclude the reverse-colored preset octave. Number drawbars1–9 in physical left-to-right order. Tone IDs remain one-based and convert to zero-based storage exactly once. An event arriving from a MIDI controller first passes through an explicit manual-zone adapter.

The supplied mapping dataset adopts MIDI36 for playing key 1 and MIDI96 for key 61. These are software conventions, not factory MIDI specifications. Retain the numeric indices in diagnostic displays so differing note-name octave conventions cannot silently transpose the model. Load the same factory routing into both manuals, then apply their distinct contacts, presets and downstream paths. A modified specimen can override the factory table only through a separately identified profile.

Quantity	Meaning and boundary
Playing key k	1–61; k=1 is the lowest playable C.
Drawbar d	1–9; footage order is fixed.
Tone ID w	Electrical source1–91; the standard manual table uses13–91.
Source array	Index=w−1; storage convention only.
MIDI adapter	Proposed k=MIDI−35 within the selected61-note manual zone.

SOURCES S015-01 Hammond Organ Service Manual early console models

DOCUMENTED MAP AND ORIGINAL DERIVATION

Map drawbar roles to source offsets

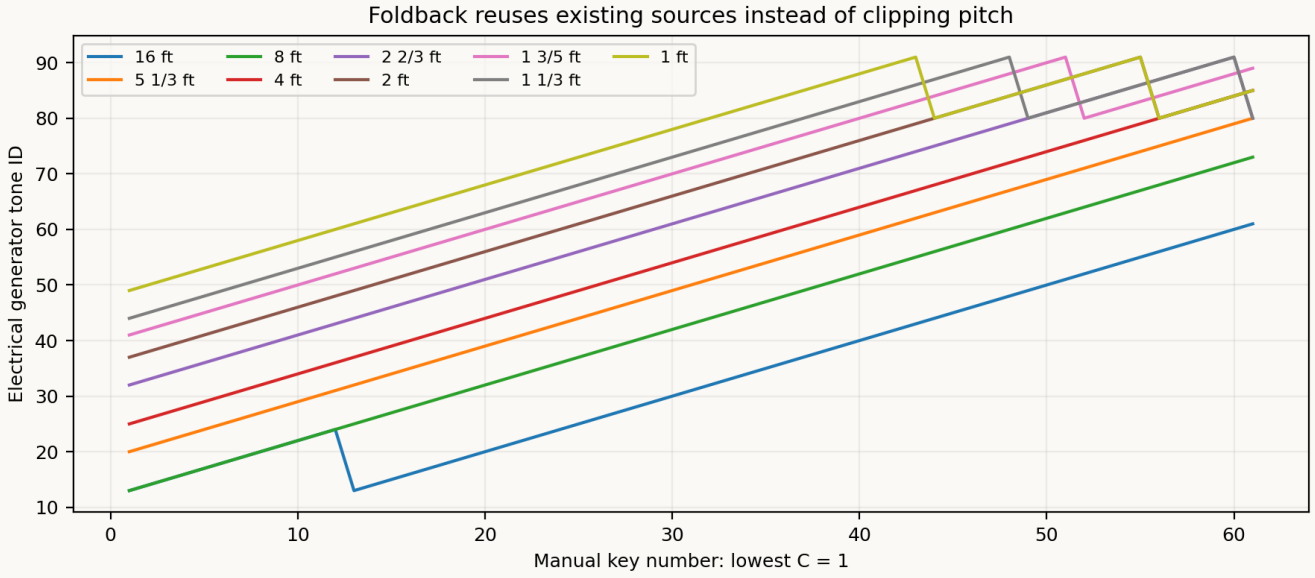
Manual harmonic mapping and foldback

ORIGINAL DERIVATION. Express the source request before foldback as w=k+o[d], with offsets0,19,12,24,31,36,40,43 and 48. This compact sequence reproduces the factory table once the explicit low and high fold rules are applied. The full 61×9 result is provided in data/015-manual-map.json, with independent boundary rows, source locator and the inspected file hash. [S015-01]

The familiar harmonic ratios describe musical roles relative to the 8-foot fundamental. They must not instruct a model to multiply one oscillator frequency by3,5 or6. The hardware selects another member of the shared geared source field. Consequently, a non-octave partial can differ from an exact integer multiple, and its phase belongs to its own physical source. Preserve that distinction even when a simplified12-TET source profile is used for software tests.

Offset generation and measured source frequency are separate responsibilities. The mapping table returns an identity; the generator registry resolves that identity into a waveform. This separation lets a calibrated generator profile replace a provisional one without changing keys, drawbars, or stored registrations.

Drawbar and role	Nominal ratio to8 ft	Offset o[d]
1: 16 ft	1/2	0
2: 5 1/3 ft	3/2	19
3: 8 ft	1	12
4: 4 ft	2	24
5: 2 2/3 ft	3	31
6: 2 ft	4	36
7: 1 3/5 ft	5	40
8: 1 1/3 ft	6	43
9: 1 ft	8	48



Original plot of the factory-reference61x9 mapping. Lower16-foot and high harmonics fold by octave; specimen modifications remain possible.

Original visualization derived from factory Table III. Electrical source numbers are not physical terminal positions; specimen wiring still needs verification.

SOURCES S015-01 Hammond Organ Service Manual early console models

ORIGINAL ALGORITHM VERIFIED AGAINST FACTORY DATA

Fold at the actual boundaries more than once when necessary

Manual harmonic mapping and foldback

ORIGINAL ALGORITHM. For drawbar1 only, add12 to the preliminary tone ID when k is1 through 12. For every drawbar, subtract12 repeatedly while the requested ID exceeds91. Do not clamp out-of-range values to91, silence them, or apply a single subtraction to every case. The top octave of the 1-foot drawbar requires two octaves of correction for some keys; the 1 1/3-foot drawbar also requires a second correction at the top C. [S015-01]

The manufacturer’s modern guide names4G as a foldback setting while describing the preceding F-sharp as the last unshifted key for the 1-foot drawbar. These refer to opposite sides of one boundary. Treating the setting label as the highest available generator pitch introduces a semitone error. Prefer factory key numbers43 and 44, and source IDs91 and 80, in implementation tests. [S015-02]

The low fold is specific to the 16-foot manual route. Applying the same minimum-source rule to a pedalboard would remove its intended low sources. Similarly, optional extended-range modes belong in a separately named modern feature profile, not in the standard vintage compatibility mode.

First upper fold	Factory key number
2 2/3 ft	61
2 ft	56
1 3/5 ft	52
1 1/3 ft	49; another fold at 61
1 ft	44; another fold at 56

```
mapManual(k, d):
  require 1 <= k <= 61 and 1 <= d <= 9
  offsets = [0,19,12,24,31,36,40,43,48]
  w = k + offsets[d-1]
  if d == 1 and k <= 12: w += 12
  while w > 91: w -= 12
  require 13 <= w <= 91
  return ToneId(w)
```

SOURCES S015-01 Hammond Organ Service Manual early console models / S015-02 Hammond New B-3 and New C-3 Owners Playing Guide

REAL TIME IMPLEMENTATION

Preserve source collisions without merging the contacts

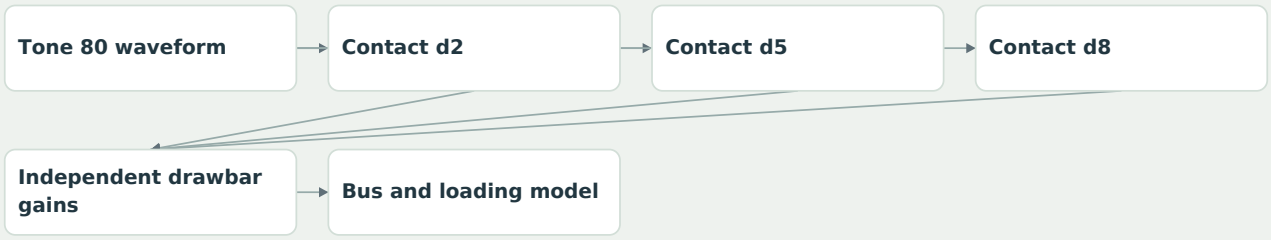
Manual harmonic mapping and foldback

ORIGINAL DESIGN PROPOSAL. Compile the profile’s routing into an immutable61×9 array during preparation. The audio callback reads small integer IDs and never searches a document, parses JSON, allocates a voice, or recomputes a general-purpose pitch conversion. Publish the table together with its source-profile version so an atomic profile change cannot combine old routes with new calibration data.

Foldback deliberately makes distinct drawbars reference the same source. At key 61, drawbars2,5 and 8 all request tone 80; drawbars4,6 and 9 request tone 85. They share the source waveform and phase but retain independent contact timing, bus contribution and drawbar gain. A linear prototype may aggregate their instantaneous coefficients after those states are evaluated. A circuit model must retain any branches whose loading or nonlinear behavior makes that aggregation invalid.

Test routing before adding oscillator accuracy, scanner effects or distortion. Inject a synthetic identification signal into one source ID and check which contacts carry it. Then render the shared source field and compare equivalent routes. This two-stage test prevents leakage, level differences or harmonic content from obscuring a simple lookup error. Mapping itself has no sample-rate dependence; anti-aliasing remains the responsibility of waveform generation and nonlinear downstream stages.

Source identity is shared while paths remain distinct



Original conceptual example for top key 61; shared phase does not imply identical contact or loading behavior.

CHECKS AND DECISIONS

- No key press resets a mapped source phase.
- All549 map entries are immutable during the audio block.

SOURCES S015-01 Hammond Organ Service Manual early console models

MEASUREMENT AND VARIANT ACQUISITION

Verify the factory map against the selected specimen

Manual harmonic mapping and foldback

ORIGINAL ACQUISITION PLAN. Record each manual with one drawbar active at a time, a documented preset selection, percussion disabled and the scanner bypassed. Preserve key number, drawbar number, registration, expression setting, gain calibration and specimen identity. Captures must distinguish a wrong source assignment from a damaged contact or a weak source. Use repeated strokes and both manuals to expose inconsistent connections.

Test all 61 keys for each of the nine drawbars. The table supplies549 expected source identities per manual, but this is not a requirement for 549 unrelated oscillator instances. In each capture, compare the dominant pitch and phase-correlated source signature with the generator registry. Record anomalous extra tones separately: leakage and source waveform harmonics can remain present even when the lookup is correct.

Add focused tests around keys12/13,43/44,48/49,51/52,55/56 and 60/61. These boundaries efficiently reveal octave-wrap and off-by-one errors. If the specimen has a low-frequency manual modification, retain the observed override and document its provenance instead of rewriting the factory baseline. The pedal table is a separate acquisition: the factory continuation contains eight harmonic-contact columns and intentional missing assignments, so a nine-drawbar manual mapper is not a substitute. [S015-01]

Observation	Interpretation to investigate
Same wrong pitch on both manuals	Source registry or shared profile mapping.
One manual only	Contact route, wiring change or manual-specific fault.
Correct pitch with different level	Gain, contact resistance, loading or voicing.
Boundary error only	Wrap rule or index convention.
Pedal mismatch	Independent pedal contact and mixture topology.

SOURCES S015-01 Hammond Organ Service Manual early console models

ACCEPTANCE AND HANDOFF

Make all 549 routes an explicit acceptance gate

Manual harmonic mapping and foldback

ORIGINAL ACCEPTANCE CONTRACT. The factory-table profile passes only when every generated entry matches the retained numerical reference, the specified boundary cases pass, and both manuals resolve identical source IDs for equal key/drawbar requests. Preserve test output with the dataset hash. This establishes routing conformity to the inspected document, not measured acoustic equivalence to an unidentified organ.

The provided dataset retains all 61 manually transcribed factory rows alongside the independently formulated algorithm and focused boundary cases. A numerical comparison found agreement for all 549 entries. This is one-agent documentary verification, not a second independent review or a specimen measurement. Require a separate reviewer before a release declares transcription closed; retain source locators and review status in version control, and never overwrite an exception silently to make a test pass.

A final musical test must include registrations with colliding folded components and chords spanning the boundary. Verify that equal source identities remain coherent through note overlap, silent periods, block partition changes and manual switching. These are proposed software tests, not claims that the engine has already passed them. Continue with drawbar loading, contact timing and percussion logic only after this basic wiring layer is stable.

Test input	Expected tone IDs in drawbar order
k=1	13,20,13,25,32,37,41,44,49
k=13	13,32,25,37,44,49,53,56,61
k=43	43,62,55,67,74,79,83,86,91
k=44	44,63,56,68,75,80,84,87,80
k=56	56,75,68,80,87,80,84,87,80
k=61	61,80,73,85,80,85,89,80,85

CHECKS AND DECISIONS

- The upper-fold algorithm uses repeated octave subtraction; a single subtraction fails required cases.
- Manual sources1–12 remain unreachable in the standard factory profile.
- Pedal routing is independently represented and tested.

SOURCES S015-01 Hammond Organ Service Manual early console models

ARCHITECTURE AND EVIDENCE

Make the generator a shared field rather than a voice pool

Real-time shared oscillator implementation

The DAFx 2011 paper demonstrates a computationally economical additive organ model. Its intentionally simplified attack, click and modulation choices are useful comparisons, but they do not establish a component-exact B-3 implementation. This chapter instead turns the source-identity, mechanical-phase and pickup contracts from chapters 9,10 and 13 into a testable software boundary. [S016-01]

ORIGINAL DESIGN PROPOSAL. Give one GeneratorField instance ownership of every source state. A key, manual or drawbar receives a source index and a sample; none owns a replacement oscillator. Produce each of the 91 source values once for a sample instant, then let contact and bus modules consume the same values. This keeps coincident harmonics coherent across simultaneous keys and both manuals.

Separate immutable compiled data from mutable audio state. Compiled data contains waveform coefficients, table banks, evidence metadata and source-to-shaft relationships. Audio state contains mechanical angles, residual motion and electrical network memory. Preserve the boundary between induced pickup emf and loaded terminal signal so optimization cannot silently add or remove a generator filter. A profile change must declare which boundary its waveform represents.

Object	Responsibility
GeneratorProfile	Immutable identities, units, source boundary and waveform data.
GeneratorState	Persistent drive/shaft angles and dynamic pickup/filter state.
SourceFrame	91 values at one common sample instant.
ContactRouter	References sources; owns closure timing and bus connections.

CHECKS AND DECISIONS

• Two keys referencing the same source read identical source samples before their individual contact processing.

SOURCES S016-01 Pekonen Pihlajamaki and Valimaki Computationally Efficient Hammond Organ Synthesis

ORIGINAL NUMERICAL MODEL

Compile reference and optimized waveform paths from the same data

Real-time shared oscillator implementation

ORIGINAL DESIGN PROPOSAL. Implement a scalar double-precision harmonic-sum renderer first. Treat it as the numerical oracle for later table lookup or SIMD code, not as proof of hardware accuracy. The oracle and optimized renderer must consume the same coefficients, phase reference, amplitude units and bandwidth policy. This separates an optimization error from an inaccurate specimen profile.

A full mechanical revolution and one electrical tooth cycle are different table domains. A tooth-periodic waveform can use electrical phase; once-per-revolution asymmetry needs mechanical angle or an explicitly separate residual model. Store the domain in the compiled asset. An unlabelled single-cycle recording cannot establish either interpretation. In the equations, A and B are amplitudes in the declared source unit; phi and theta are angles in radians; psi and beta are fixed phase offsets; k and l are integer orders; n is the sample index.

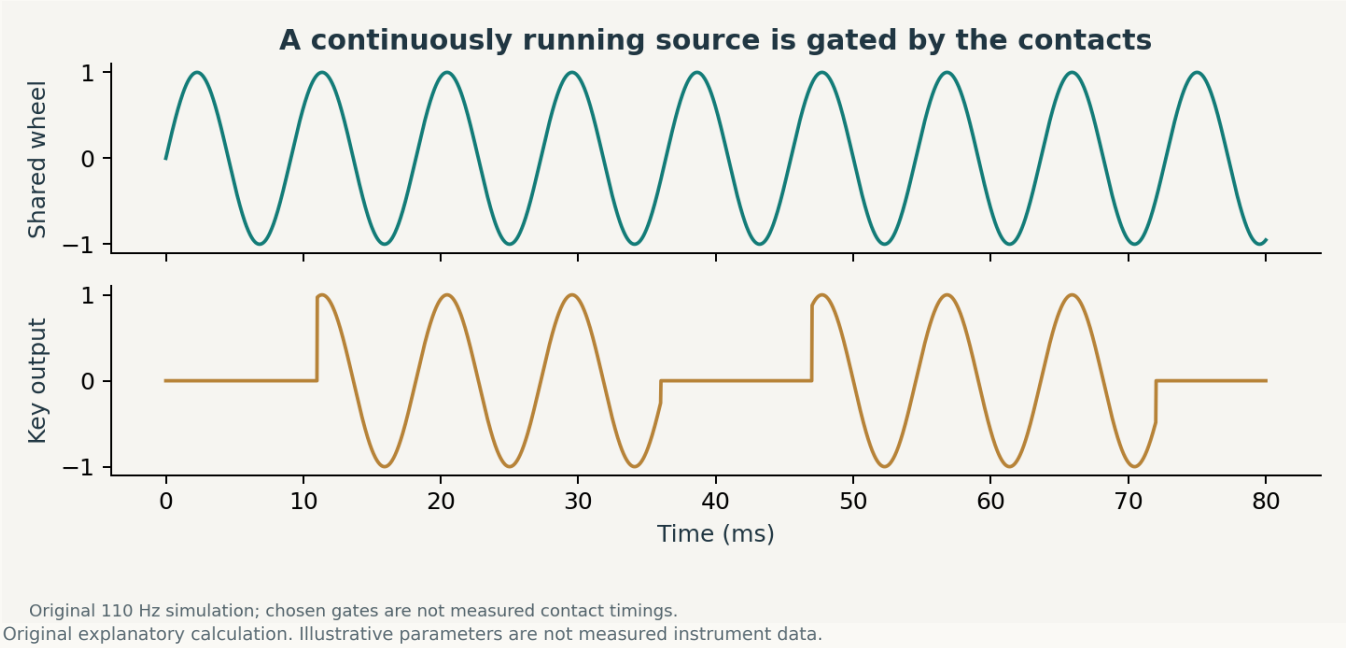
Fractional lookup interpolates between table entries; a larger table or higher-order interpolation changes numerical error, not source provenance. [S016-05] Compile candidate tables outside processing, including a periodic guard sample, and benchmark their errors against the oracle. Choose table length from the required spectral accuracy and highest retained spatial order instead of assigning the same arbitrary size to every wheel.

MODEL EQUATIONS

Electrical-domain model: x_i[n] = sum_k A_ik*cos(k*phi_i[n] + psi_ik).

Mechanical-domain model: x_i[n] = sum_l B_il*cos(l*theta_j[n] + beta_il).

For table length M and phase q in radians: u=M*fract(q/(2*pi)); j=floor(u); a=u-j; y=(1-a)*T[j]+a*T[j+1].



Original 110 Hz simulation; chosen gates are not measured contact timings.
Original explanatory calculation. Illustrative parameters are not measured instrument data.

SOURCES S016-05 Carnegie Mellon Introduction to Computer Music wavetable synthesis

IMPLEMENTATION AND LIFECYCLE

Preserve phase while preparing and changing rate

Real-time shared oscillator implementation

JUCE supplies a general function-based oscillator, but its inspected prepare method resizes internal storage and calls reset; reset clears phase, and ordinary frequency changes use smoothing. Those are library behaviors, not the required physical behavior of a continuously rotating organ. Pin and inspect the chosen version before wrapping it. [S016-02]

ORIGINAL DESIGN PROPOSAL. Own mechanical phase in the engine, independent of any library oscillator lifecycle. Prepare memory and rate-dependent tables while processing is inactive. Convert frequency to a per-sample increment without recomputing phase from frequency multiplied by absolute time. A new sample rate changes integration step size; it does not inherently imply a new rotor angle.

Define initial-condition and resume policies explicitly. An offline render should derive its initial state from the session snapshot or a stored deterministic seed. Host transport movement, a new note and preset selection must not implicitly trigger a physical startup sequence. If the host suspends processing, select a documented sample-time policy; wall-clock catch-up is a separate optional behavior with different reproducibility consequences.

Operation	Proposed phase rule
Note on or off	Change contacts only; preserve generator state.
Drawbar or preset selection	Change routing/gains; preserve generator state.
Sample-rate preparation	Preserve explicit angle snapshot; rebuild rate-dependent data.
Deliberate instrument restart	Use separately defined mechanical and electrical initial conditions.
Offline rerender	Restore the same deterministic state and event stream.

CHECKS AND DECISIONS

- A note-on event never calls an oscillator reset method.

SOURCES S016-02 JUCE official oscillator implementation and API

ORIGINAL REAL-TIME IMPLEMENTATION

Keep the processing loop bounded and independent of block size

Real-time shared oscillator implementation

Steinberg documents variable processing-block lengths, parameter-only calls with zero samples, and audio-thread execution of process and setProcessing. Memory allocation is explicitly discouraged on those paths. [S016-03] Build the source engine so its mathematical state advances by sample count, not by callback count.

ORIGINAL DESIGN PROPOSAL. Use preallocated structure-of-arrays storage for coefficients and angles. SIMD padding is software storage only and must not be confused with the generator’s physical blank disks. Apply sample-offset events before evaluating their affected sample; generate the shared field, process contacts and networks, then advance the mechanical state. Test the chosen convention explicitly.

Publish prepared profiles through a bounded ownership handoff. A pointer swap alone is insufficient if releasing the old object triggers deallocation on the audio thread; reclaim retired data elsewhere. Keep parsing, fitting, logging and GUI updates out of processing. JUCE provides a scoped denormal-control helper, but verify platform behavior and compare low-level numerical tails with the scalar reference. [S016-04]

```
prepare(profile, rate, maximumBlock):
  validateAndCompileOutsideAudio(profile, rate)
  allocateBoundedStateAndScratch(maximumBlock)
process(block, events):
  if block.samples == 0: acceptParameterFlush(); return
  for n in 0 .. block.samples-1:
    applyEventsAtOffset(n)
    field = evaluateAllSources(currentMechanicalState)
    buses = contactsAndNetworks.consume(field)
    output[n] = downstream.process(buses)
    advanceMechanicalStateOneSample()
  publishBoundedDiagnosticsWithoutAllocation()
```

CHECKS AND DECISIONS

- Instrumented processing performs zero allocations, deallocations, locks or file operations.
- A zero-sample call does not advance phase or access an absent audio buffer.

SOURCES S016-03 Steinberg VST3 API documentation processing and threading / S016-04 JUCE ScopedNoDenormals class reference

ORIGINAL ANTI-ALIASING PLAN

Budget waveform bandwidth and conversion error separately

Real-time shared oscillator implementation

ORIGINAL MODEL. A bandlimited coefficient set must fit the actual internal sample rate. Electrical harmonic k occupies approximately k times its carrier frequency, while a mechanical spatial order l occupies l times shaft rotation frequency. Modulation creates additional sidebands, so retaining every coefficient merely because the unmodulated carrier is below Nyquist is insufficient.

Choose a declared stopband margin and test representative modulation extremes against a higher-rate offline reference. Table interpolation is not a substitute for bandlimiting. When switching between table bandwidths, preserve phase and use a validated transition to avoid an unrelated timbral step. Never relabel the resulting crossfade as physical generator behavior.

Contact discontinuities and downstream nonlinear stages can generate new bandwidth after a clean source field. Give those stages their own anti-aliasing strategy; source tables alone cannot protect the complete organ. If generator output is rendered at an internal oversampled rate, preserve that common time base through contact switching and decimate only at the chosen architectural boundary. Report resampler passband error, stopband attenuation and latency separately from source-waveform error.

Error or cost	Proposed isolation test
Harmonic truncation	Compare retained spectrum with declared bandwidth policy.
Lookup interpolation	Compare to the same-coefficient scalar oracle.
Modulation sidebands	Use high-rate reference under maximum supported motion.
Rate conversion	Measure passband, stopband, impulse delay and phase.
Memory example	91 sources ×512 float samples =186368 bytes; derived, not a requirement.

ACCEPTANCE AND PERFORMANCE

Require a numerical report before optimizing further

Real-time shared oscillator implementation

ORIGINAL ACCEPTANCE PLAN. Test the source field independently of musical routing. Render every source, simultaneous source pairs and complete field captures from retained initial states. Include silence intervals between repeated key events so a reset-on-note bug cannot hide inside an isolated-note test. Check correlated mechanical states, source identities and selected waveform boundaries before judging the final audio spectrum.

Partition one event stream into many block sequences and compare against a one-sample reference. Proposed core coverage includes 44.1,48 and 96 kHz plus block sizes 1,32,64,127,512 and 1024 within the configured maximum. These are engineering test cases, not manufacturer requirements. Keep deterministic scalar results separate from floating-point tolerances required by different SIMD reductions.

Profile worst observed callback duration, allocation instrumentation, memory and audio underruns on each supported platform. A 32-sample block at 48 kHz spans 0.667 ms, a derived scheduling interval rather than total system latency. Set a measured engine budget within that interval with room for other modules. Retain compiler flags, CPU model, thermal conditions and test load. No CPU percentage or authenticity result has been measured for this proposed implementation.

Gate	Required retained evidence
Identity and continuity	Stable source IDs; unchanged phase across notes and buffers.
Numerical equivalence	Oracle comparison with justified absolute, spectral and phase tolerances.
Bandwidth	Worst supported tuning/modulation alias report.
Lifecycle	Rate changes, suspension policy and zero-sample flush tests.
Performance	Platform timing report and zero forbidden callback operations.

CHECKS AND DECISIONS

- Do not pass this subsystem because it merely sounds organ-like.
- Escalate missing waveform or mechanical calibration as an evidence gap, not an optimization task.

HARDWARE EVIDENCE / MODEL BOUNDARY

What the drawbar actually controls

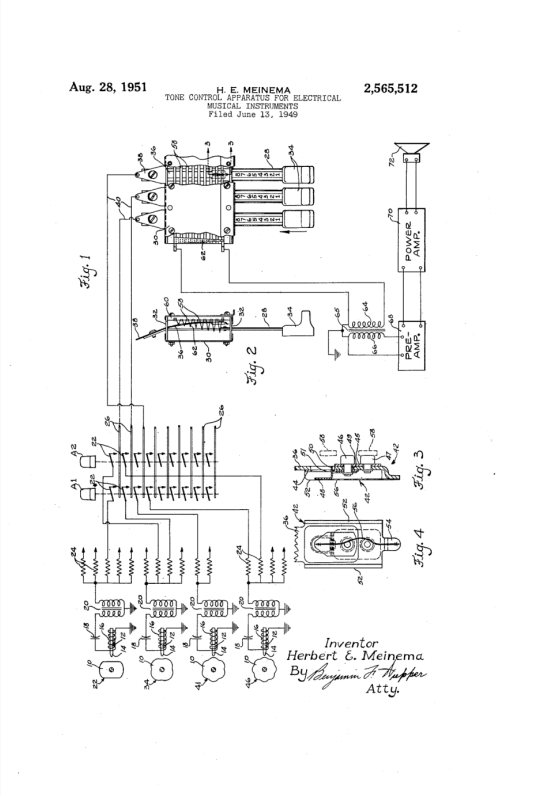
Drawbar attenuation and bus mixing

The factory manual defines the fully inserted drawbar as silent; marked positions 1 through 8 increase intensity. B-3 drawbars contact collector bars connected to matching-transformer taps. The instrument therefore requires an electrical routing model, with 0 = mute and 8 = full, rather than an assumed modern potentiometer law. [S017-01]

The 1934 patent specifies tap turns 6, 8, 11, 16, 22, 32, 45 and 64 and explains an approximately geometric progression. Those historical numbers establish design intent; they do not establish the winding specification of a production B-3 matching transformer. [S017-02]

ORIGINAL DESIGN PROPOSAL: maintain two explicit model levels. NominalMix is a transparent, linear reference with the gain table on page 3. CircuitMix retains source, contact and transformer parameters with provenance. Never label a nominal curve as a measured factory transfer function. Define the input boundary as generator terminal voltages before key resistance wires, or identify a different recorded boundary explicitly; otherwise a sampled loaded waveform and a second loading model can count the same coloration twice.

Profile field	Required interpretation
position	Physical marking 0 through 8; intermediate travel retained separately.
gain_basis	nominal_3 dB, measured_transfer, or solved_network.
reference_load	Actual acquisition load, including following circuitry.
provenance	Specimen, revision, source, units and uncertainty.



1951 drawbar-contact patent US2565512A. The illustrated resistive-divider embodiment is not an exact B-3 tapped-transformer schematic. Source and license

CHECKS AND DECISIONS

- Reject a profile whose control direction makes 0 full and 8 silent.
- Keep unknown winding turns explicitly unknown; the 1934 patent must not silently populate a B-3 production profile.

SOURCES S017-01 Hammond Organ Service Manual, factory scan / S017-02 Laurens Hammond, US1956350A, Electrical musical instrument

ELECTRICAL TOPOLOGY / ORIGINAL ABSTRACTION

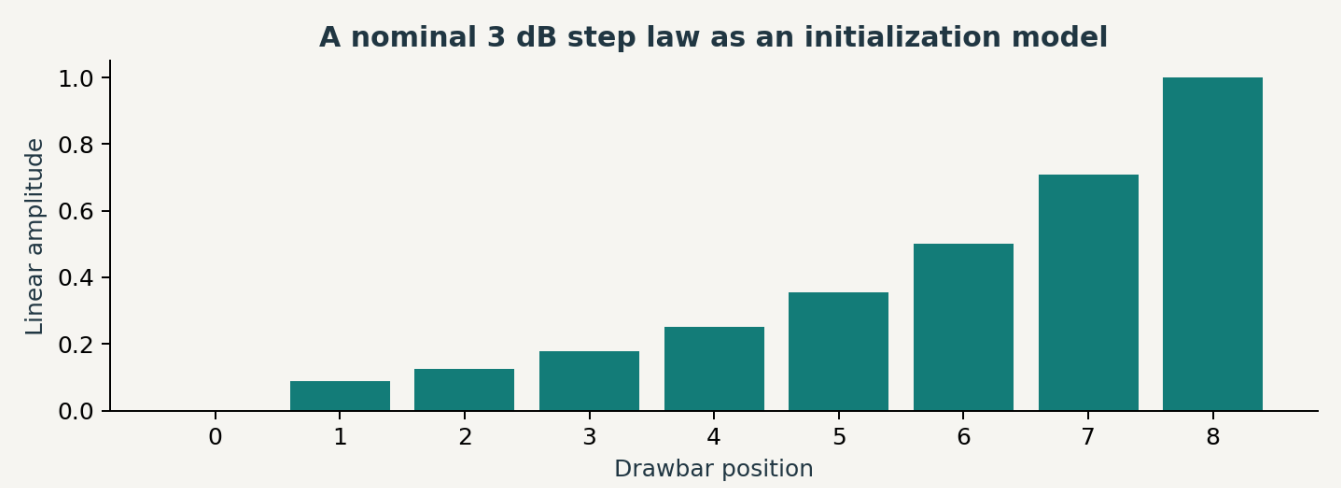
Preserve the branches before collapsing the mix

Drawbar attenuation and bus mixing

Factory resistance wires limit generator loading and help one generator feed multiple key circuits. Preset and drawbar collector bars share matching-transformer taps. There are separate manual transformers; the lower-manual transformer also receives pedals. Their secondaries feed the preamplifier. These documented connections do not quantify residual chord-dependent loading. [S017-01]

The continuous-contact assembly uses two contacts joined through 1 ohm, maintaining a conducting path during movement while avoiding a direct short between adjacent collector bars. The related 1951 patent describes overlap and an approximately 1-ohm element, but its illustrated resistor-divider coupling must not replace the production tapped-transformer topology. [S017-01] [S017-03]

ORIGINAL DESIGN PROPOSAL: keep a branch identifier for every key contact, drawbar/preset route and selected tap until any loading calculation is complete. Model both manual networks independently before downstream combination. Give percussion a separate coupling port; its borrowed harmonic and return route cannot always be represented by adding an independent voice after the mix. Import the detailed percussion topology from its chapter rather than guessing it here.



Derived illustration: 0 is mute; 3 dB steps initialize a model pending specimen calibration.
Original explanatory calculation. Illustrative parameters are not measured instrument data.

CHECKS AND DECISIONS

• With loading enabled, opening branch B may change branch A through the shared network; any such effect must follow network parameters, not a generic polyphony compressor.

SOURCES S017-01 Hammond Organ Service Manual, factory scan / S017-03 Laurens Hammond, US2565512A, Tone control apparatus for electrical musical instrument

ORIGINAL DIGITAL REFERENCE / DERIVED VALUES

A correct nominal gain law and coherent sum

Drawbar attenuation and bus mixing

Werner and Abel use a simplified 3-dB drawbar model. Their Table 2 displays maximum amplitude at index 0 and silence at index 8. Treat that table as a reversed index convention; copying it directly into a physical registration parser reverses the instrument controls. [S017-04]

ORIGINAL DESIGN PROPOSAL: use the ascending amplitude law below as the initial reference. These are derived values, not measured B-3 tap gains. Here p is the physical integer marking; g is a dimensionless voltage/amplitude multiplier. Decibels convert with division by 20. The silent endpoint is explicitly zero, not one further 3-dB decrement.

For each manual, sum contact weights that address the same wheel only after evaluating their independent contact states. M(k,d) is the verified wheel map, c is contact amplitude, x_w is the shared continuously running wheel signal, and n is sample index. Two equal, same-phase paths ideally double amplitude: +6.0206 dB. Summing powers would incorrectly predict +3.0103 dB. Real loading can change the result; the ideal comparison tests routing rather than proving hardware behavior.

Marking p	Proposed relative dB	Derived amplitude g
0	-infinity	0
1	-21	0.0891
2	-18	0.1259
3	-15	0.1778
4	-12	0.2512
5	-9	0.3548
6	-6	0.5012
7	-3	0.7079
8	0	1.0000

MODEL EQUATIONS

g(0) = 0; g(p) = 10^(3*(p-8)/20), p = 1,...,8.

a_w[n] = sum_{k,d: M(k,d)=w} c_kd[n] * g(p_d[n]); y[n] = sum_w a_w[n] * x_w[n].

CHECKS AND DECISIONS

- Do not divide output by the number of held keys. Allocate headroom with a fixed documented scale.
- Preserve shared oscillator phase when several keys or foldback ranks select the same wheel.

SOURCES S017-04 Kurt James Werner and Jonathan S. Abel, Modal Processor Effects Inspired by Hammond T

ORIGINAL DESIGN PROPOSAL / REAL-TIME CONTRACT

Implement the network as a replaceable module

Drawbar attenuation and bus mixing

First implement NominalMix against the factory wheel map. Then introduce a parameterized passive network with source impedances, resistance wires, contact conductances, tap winding sections and the downstream load. Modified nodal analysis is one possible implementation: its unknown vector contains node voltages and necessary branch currents. Ideal transformer constraints require additional equations; a scalar turns ratio applied after summing is not a complete loaded network.

Represent intermediate travel as contact connectivity plus conductance, not a guaranteed equal-power crossfade. Until travel measurements exist, expose a separately named smooth-control surrogate. An overlap can bridge taps through the specified contact resistor; blindly interpolating two already loaded outputs does not solve that circuit. Keep the stationary-marking reference mode available for comparisons.

Prepare fixed-size storage and topology metadata off the audio thread. Process control changes at timestamped sample offsets; changing topology must preserve relevant capacitor/inductor states. Use a bounded solver with a defined failure response. Add magnetic nonlinearity only after identifying it from data. Evaluate oversampling for nonlinear stages and abrupt contact transitions, and include its latency and antialias filter response in the model contract.

MODEL EQUATIONS

G(q) * z + C * dz/dt = b(x): z contains electrical unknowns; q is contact/tap state; x is generator excitation. G and C are the assembled system matrices, not measured B-3 constants.

```
prepare(profile, sampleRate):
  validate_units_topology_and_provenance(profile)
  allocate_fixed_workspaces()
  prepare_solver_and_resampling()
process(events, generators):
  for timestamped_segment in events:
    update_contact_state(timestamped_segment)
    update_bounded_network_coefficients()
    solve_or_render_nominal_reference()
    publish_output_without_allocating()
```

CHECKS AND DECISIONS

• No memory allocation, file reads, locks or unbounded iteration occur in the audio callback.

• Compare passive-network solutions with the same network in a trusted offline solver before adding nonlinearity.

SOURCES S017-01 Hammond Organ Service Manual, factory scan / S017-03 Laurens Hammond, US2565512A, Tone control apparatus for electrical musical instrument

ORIGINAL ACQUISITION AND CALIBRATION PLAN

Measure the law instead of tuning by impression

Drawbar attenuation and bus mixing

Use a documented specimen and repeatable acquisition boundary. Record single-wheel registrations at all nine markings, several frequencies and both manuals, with fixed expression, percussion off and vibrato bypassed. Retain raw captures, noise floor and analysis windows. If access requires internal electrical probing, a qualified technician should choose the safe measurement points and instrumentation; the software dataset must record the resulting load.

For each frequency calculate the fundamental amplitude ratio to marking 8, and separately retain harmonic distortion and noise. A single downstream recording cannot uniquely distinguish pickup impedance, tap loading, transformer response and preamplifier gain. Capture additional boundaries or impedance data before assigning those effects to components. Never infer winding turns directly from one observed output ratio.

Compare a same-wheel collision against the sum of separately captured branches, then repeat with branches selecting different wheels. Record relative phase and identical gain settings. This separates coherent summation errors from loading or later nonlinearity. Repeat slow drawbar travel in both directions with travel position recorded; any hysteresis, contact bounce or dead region must be attributed to that specimen rather than presumed universal.

Capture	Parameter resolved
Isolated tone; p=0 through 8	Gain-versus-frequency table and mute residual.
Two branches; common wheel	Coherent addition and interaction residual.
Multiple different wheels	Chord loading and intermodulation candidates.
Position-synchronized travel	Overlap connectivity and transition duration.
Documented source/load tests	Impedance model and parameter identifiability.

CHECKS AND DECISIONS

• Report measured ratios in dB with repeatability and noise bounds; proposed tolerances must be labeled engineering targets.

• A parameter is not identified if different plausible networks explain the same captures equally well.

SOURCES S017-01 Hammond Organ Service Manual, factory scan / S017-03 Laurens Hammond, US2565512A, Tone control apparatus for electrical musical instrument

ORIGINAL VERIFICATION / RELEASE CONTRACT

Acceptance gates and honest fidelity labels

Drawbar attenuation and bus mixing

Deliver the drawbar module only after deterministic synthetic checks pass. In the nominal reference, all-zero registration removes the intended drawbar contribution; separately modeled leakage and electronic noise may remain. Marking 8 has unit relative gain. A test with two identical input branches must match their algebraic sum. These checks validate software conventions, not the physical instrument.

Preserve inactive bank settings without allowing their movement to change the active bank. Switching banks must not restart wheel phase. A preset uses its actual routing data rather than a fabricated nine-digit registration when the hardware network differs. Confirm that changing sample rate or block size does not alter stationary gains, event timing or network stability.

The acceptance report must distinguish implemented approximations, measured fits and unresolved parameters. Production winding turns, frequency-dependent impedances, interwinding leakage, source loading and the travel-to-contact map remain unmeasured here. Reserve fields for them now; improve the profile when captures exist. Do not invent transformer saturation merely to make loud chords sound warmer. Preserve model versions so a later calibrated network does not silently change existing projects.

Gate	Evidence to retain
Direction and silence	0/8 endpoint test and muted-signal residual.
Coherent collisions	Shared-wheel algebraic comparison before nonlinear stages.
Route ownership	Upper/lower, pedal and percussion-port test traces.
Numerical behavior	Bounded worst-case solve cost and finite output.
Calibration boundary	Raw recordings, load and source-profile identity.
Claim strength	Nominal reference or measured specimen fit; never unsupported 1:1.

CHECKS AND DECISIONS

- Publish unresolved measurements alongside the profile instead of substituting historical patent values.
- Retain the linear reference as a regression oracle even after the circuit model is added.

SOURCES S017-01 Hammond Organ Service Manual, factory scan / S017-02 Laurens Hammond, US1956350A, Electrical musical instrument / S017-03 Laurens Hammond, US2565512A, Tone control apparatus for electrical musical instrument / S017-04 Kurt James Werner and Jonathan S. Abel, Modal Processor Effects Inspired by Hammond T

HARDWARE EVIDENCE AND MODEL BOUNDARY

Make the transient an outcome of switching

Keyboard contact timing and key click

The factory manual routes generator signals through resistance wires and nine playing contacts to harmonic busbars. Registration circuitry then combines the bus signals. This makes the contact network a meaningful electrical boundary for modeling attack and release. It does not justify attaching an unrelated noise sample to each note. [S018-01]

The 2017 study finds that gesture and contact timing interact, with early and sometimes later bouncing. Its preliminary resistance measurement includes the resistance wire: the reported closed value must not be copied as a universal bare-contact resistance. The measured instrument was one 1967 C-3. [S018-02]

ORIGINAL DESIGN PROPOSAL: retain three separate records: mechanical gesture, contact transition sequence and electrical output. A click can change because one of these changed, so fitting only an output envelope creates an ambiguous model. Keep keyboard clatter recorded by room microphones separate from electrically generated transients. Percussion has its own circuit and trigger state; a short audible transient is not automatically percussion.

Implement an ideal-contact reference first. Add contact-specific onset delays, measured bounce and release behavior as separately switchable refinements. Every refinement must explain an observed difference against the reference. A convenient random process is a provisional synthesis choice until the capture data supports its statistics.

Record	Required identity
Gesture	Manual, key, trajectory source and timestamp basis
Contact	One of nine physical contacts, mapped source and make/break events
Electrical trace	Probe boundary, gain, loading and bandwidth
Acoustic trace	Microphone and room; mechanical noise remains distinguishable

CHECKS AND DECISIONS

- A note-on must never reset the shared tonewheel phase.
- Contact click and percussion must have independent diagnostic bypasses.

SOURCES S018-01 Hammond Organ Service Manual, factory manual compilation / S018-02 Moro, McPherson and Sandler, Dynamic temporal behaviour of the keyboard action on the

ORIGINAL DIGITAL MODEL

Preserve source phase at every contact edge

Keyboard contact timing and key click

ORIGINAL DERIVATION: in a fixed-load behavioral approximation, multiply each continuously running source by its contact connection state and calibrated bus contribution. The resulting transient depends on the source value at the actual edge time. Two identical contact sequences triggered at different generator phases therefore need not produce identical electrical attacks. Here u and s are voltages, a is dimensionless effective gain, c is a dimensionless connection state, and t is seconds.

For an instantaneous connection change, the output jump is proportional to the source value at that instant. A source zero crossing removes this value jump, but can leave a derivative discontinuity. Consequently, correcting amplitude steps alone is not a complete antialiasing argument. Include slope behavior or compare the entire switched signal with a sufficiently resolved reference.

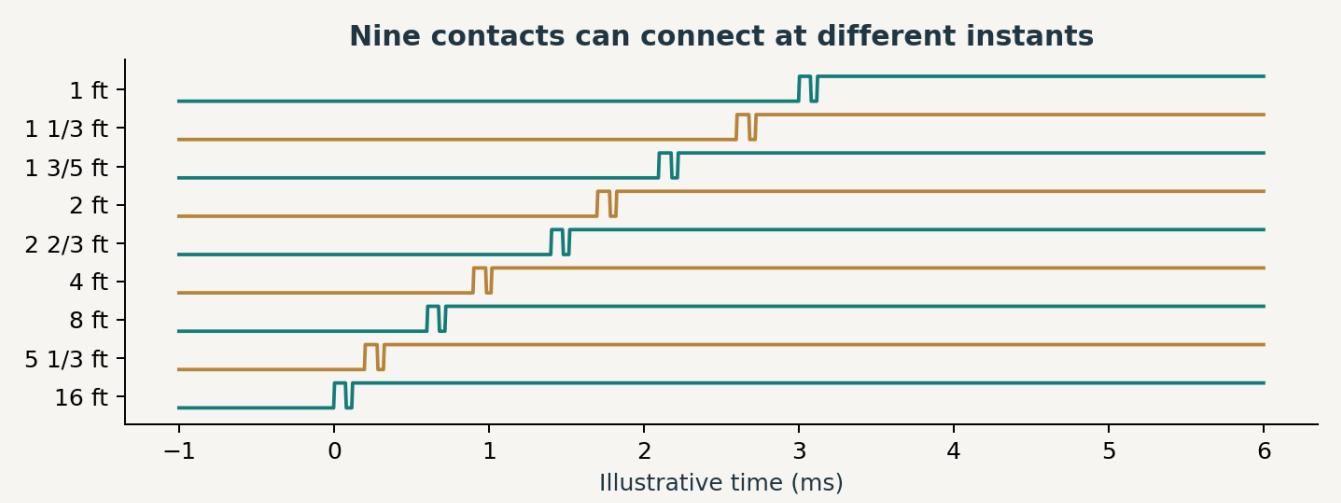
The sum below is an explicit approximation, not a complete switched-circuit solution. If changing contact states materially changes loading, send the conductance states into the verified network solver instead. Put resistance wires and transformer loading in that network exactly once. A sample recorded after the circuit cannot be treated as an unloaded source without accounting for the already embedded response.

MODEL EQUATIONS

u(t) = sum_j a_j * c_j(t) * s_w(j)(t)

Delta u_j = a_j * Delta c_j * s_w(j)(t_edge)

If Delta u_j = 0, Delta du_j/dt can still equal a_j * Delta c_j * ds_w(j)/dt.



Hypothetical timing only: actual ordering, bounce, and velocity dependence require captures.
Original explanatory calculation. Illustrative parameters are not measured instrument data.

CHECKS AND DECISIONS

- Test closure at a source peak and at a zero crossing without restarting the source.

SOURCES S018-01 Hammond Organ Service Manual, factory manual compilation

ORIGINAL STATE AND DATA DESIGN

Keep nine histories and a shared mechanical cause

Keyboard contact timing and key click

ORIGINAL DESIGN PROPOSAL: each key owns nine connection histories, but they should not automatically be independent random streams. Represent a common key-motion state, per-contact thresholds and residual contact variation. That structure can express correlated late disturbances while preserving different closure points. Fit correlations only when simultaneous measurements support them.

Store make and break models separately. Release is not necessarily a reversed attack trajectory or the same bounce distribution. The cited 2017 experiment concentrates on note onsets and explicitly identifies its single-instrument limits; it does not supply a complete release calibration. [S018-02]

A continuous-position controller supplies a measured gesture after calibration. Ordinary MIDI initiates a named synthetic trajectory with declared assumptions. Velocity may select its transient behavior without scaling sustained organ loudness. Preserve the selected trajectory class in presets so a later controller change does not silently redefine the instrument.

Use absolute sample coordinates plus a fractional-sample field for events. Define stable ordering when edges share a timestamp. Seed any stochastic residual from a persistent profile seed and key-event identity, never from block size or GUI time. Partial depression must be able to leave some contacts connected while others remain open.

Field	Units or rule
on/off threshold	Position units; individual contact; unknown until calibrated
edge time	Absolute frame plus fractional frame
bounce sequence	Ordered make/break events with provenance
gesture adapter	Measured position or named MIDI surrogate
random seed	Deterministic profile/event identity
connection state	Open, making, closed or breaking; electrical output derived separately

CHECKS AND DECISIONS

- Replaying the same event trace and seed must reproduce the same connection histories.

SOURCES S018-02 Moro, McPherson and Sandler, Dynamic temporal behaviour of the keyboard action on the

IMPLEMENTATION AND NUMERICAL CONTROL

Render the switching before reducing its bandwidth

Keyboard contact timing and key click

ORIGINAL DESIGN PROPOSAL: choose either a high-rate switching reference or an event-based bandlimited renderer, then document its approximation. Upsampling an already aliased base-rate click does not remove the error. Resolve source phase and contact transitions before the first bandwidth reduction; keep all following circuit stages at a compatible time basis.

An oversampled implementation evaluates the source field and switching at the internal rate, filters and decimates the result, and retains all filter memory across blocks and notes. Fractional event timing remains necessary when an edge lies between internal samples. An edge-correction implementation needs both jump magnitude and the relevant derivative behavior, including interacting edges close together.

JUCE documents FIR/IIR oversampling choices, preallocation and latency reporting. Its constructor factor specifies a power of two. Those API facts do not establish that any particular rate is sufficient for measured contact behavior. Pin the dependency and measure numerical convergence, phase response and total delay. [S018-04]

Preallocate contact state, event storage, correction tails and diagnostic buffers. Never allocate or resize for a chord. Add explicit overflow behavior for malformed or excessive event streams. State serialization must distinguish a musical preset from a diagnostic snapshot containing running phases and pending edges.

```
prepare immutable mapping and contact profiles
prepare bounded event storage and filters
for each audio block:
  ingest timestamped gesture events
  update or replay nine contact histories
  for each required internal time:
    evaluate shared sources without note reset
    render fractional contact edges
    solve or evaluate the bus network
    advance shared mechanical state once
  lowpass and decimate if applicable
  retain filters, edges and contact state
```

CHECKS AND DECISIONS

- Report and compensate the actual processing delay.
- A block boundary must not truncate an edge correction or bounce sequence.
- A generic keyboard debouncer must not discard the contact bouncing being modeled.

SOURCES S018-04 JUCE dsp Oversampling class reference

MEASUREMENT AND CALIBRATION PLAN

Capture enough variables to identify the mechanism

Keyboard contact timing and key click

ORIGINAL ACQUISITION PLAN: have a qualified technician establish a documented measurement configuration that preserves or explicitly models loading. Record continuous key position, all nine contact states, a phase reference and a dry electrical output on a common or accurately synchronized clock. Document sensor latency and bandwidth before interpreting relative onset times.

Gather pressed, struck, partial and repeated gestures across both manuals and representative white and black keys. Include release and rearticulation while other keys remain held. Capture individual drawbars before complex registrations. Reserve different takes and keys for validation rather than fitting and testing on the same recordings.

Fit the contact event model first against observed states. Then compare the resulting electrical prediction using the measured source phase and circuit profile. This separates errors in mechanical timing from errors in source waveform or electrical filtering. Treat a residual noise layer as a named empirical residual only after those errors have been investigated.

The 2017 study does not provide an exhaustive audio-spectrum calibration or a universal production tolerance. [S018-02] Adopt its measurement-oriented questions, then acquire the missing specimen data. Preserve raw captures and uncertainty; do not overwrite recordings with aligned, denoised or normalized derivatives.

Experiment	Discriminates
One contact / one rank	Source phase, edge timing and downstream bandwidth
Partial key travel	Thresholds and contact subset behavior
Make versus break	Distinct attack and release processes
Held chord plus rearticulation	Shared-source and network interactions
Repeated identical command	Deterministic mechanics versus stochastic residual
Different acquisition bandwidth	Sensor limits versus actual contact timing

CHECKS AND DECISIONS

- Retain clock calibration, probe boundary and processing history with every take.

SOURCES S018-02 Moro, McPherson and Sandler, Dynamic temporal behaviour of the keyboard action on the

ACCEPTANCE AND FAILURE MODES

Pass mechanism tests before tuning a pleasing click

Keyboard contact timing and key click

ORIGINAL ACCEPTANCE PROPOSAL: use synthetic fixtures to verify mathematics before comparing hardware. At a fixed source phase and event sequence, change host block boundaries and confirm equivalent output after accounting for declared latency. Compare optimized processing against the scalar reference, including simultaneous contacts and repeated foldback sources.

As a proposed numerical gate, require normalized RMS error no greater than1e-4 against a converged high-rate reference for agreed synthetic fixtures, after gain and latency conventions are fixed. This is an engineering target, not a Hammond specification or a demonstrated result. Convergence and real-time cost must both be recorded.

The 2011 DAFx paper offers a sixth-harmonic envelope as a deliberately simplified click approximation. [S018-03] It is useful as a comparison baseline, but cannot establish the contact mechanism. Likewise, adding the same white-noise burst to every note can conceal phase, registration and partial-travel errors.

Hardware acceptance requires held-out gestures, transient spectra and timing distributions, plus controlled listening. Approve the keyboard subsystem only when its remaining discrepancies are localized and documented. Leave the calibration label provisional until recordings support it; do not translate a successful synthetic test into a one-to-one authenticity claim.

Fixture	Required result
Isolated muted drawbar path	Its contribution is zero with percussion and leakage disabled.
Peak versus zero-crossing edge	Phase-dependent value and slope behavior preserved
Same source on different contacts	Shared phase; separate connection histories
Note release during held chord	No source or shared-filter reset
Partial depression	Only the selected contact subset conducts
Different block sizes	Same timestamped histories and declared numerical agreement

MODEL EQUATIONS

For nonzero-reference fixtures: E = RMS(candidate - reference) / RMS(reference). Proposed E <= 1e-4.

SOURCES S018-03 Pekonen, Pihlajamäki and Välimäki, Computationally Efficient Hammond Organ Synthesis,

HARDWARE EVIDENCE

Model the cradle, not twelve independent programs

Preset matrix and registration system

Each manual has twelve reversed-color keys. C releases the latch and immediately returns; it has no electrical contacts. C#–A select nine wired registrations; A# and B select the left and right drawbar groups. Ordinary selection releases the previous key, but simultaneous depression can latch two keys. [S019-01]

ORIGINAL DESIGN PROPOSAL: expose a simple selector in normal playing mode while retaining a physical latch mask in the fidelity model. The mask contains the eleven contact-bearing keys, never C. Treat Cancel as an action that empties that mask; its animation returns to rest. A screen that permanently depresses C would contradict the mechanism even when the audio is silent. Assign mask bits0–8 to C#–A, bit9 to A#, and bit10 to B.

The simultaneous-latch exception qualifies chapter 007's convenient single-selector abstraction. Do not silently claim that the abstraction covers every mechanical state. An optional physical-interaction mode can represent overlapping contacts, but its sound requires the electrical network below. A naive sum of two preset outputs is an unverified approximation. Keep ordinary MIDI program selection mutually exclusive unless the user explicitly chooses this extension.

Action	Proposed state transition
Press fixed key C#–A	Release ordinary prior latch; engage selected nine-contact route.
Press A# or B	Expose that bank's existing positions without copying values.
Press C	Release every preset latch; preserve playing-key and generator states.
Simultaneous physical press	Retain possible multiple-latch state with explicit fidelity status.
Release finger from latched key	Selection remains until another selection or Cancel.

CHECKS AND DECISIONS

- Cancel changes registration connectivity, not the generator phase or held-note inventory.
- Cancel on either manual leaves the other manual and the independent pedal registration intact.

SOURCES S019-01 Hammond Organ Service Manual, factory early-console compilation

PRIMARY NUMERICAL DATA

A complete documented matrix, with the correct provenance

Preset matrix and registration system

TableIV specifies an approved chapel arrangement; it is not evidence that every surviving B-3 has these registrations. The numerical transcription below was visually checked against printed4-1. [S019-01]

ORIGINAL DATA PROPOSAL: name this profile factory_chapel_table_iv. Store each registration as nine digits in the established physical drawbar order. Preserve leading zeros and keep upper/lower records independent. The matrix contains 2×9×9=162 tap assignments; its digits designate connections, not linear amplitude multipliers. Resolve their gains through the chosen transformer/drawbar calibration. Digit order is16,5⅓,8,4,2⅔,2,1⅗,1⅓,1foot; factory lead colors in that order are brown, red, orange, yellow, green, blue, violet, gray, white. [S019-01]

Treat the table as an immediately usable documentary baseline, with reviewStatus=single_visual_transcription. A specimen profile supersedes individual assignments only when a photographed or measured connection supports the change. Include the source page, document hash, revision date, and reviewer for every imported matrix. Do not replace this evidence tag with the vague label original factory presets.

Preset key	Upper: nine tap numbers	Lower: nine tap numbers
C#	005320000	004545440
D	004432000	004423220
D#	008740000	007373430
E	004544222	004544220
F	005403000	006644322
F#	004675300	005642200
G	005644320	006845433
G#	006876540	008030000
A	327645222	427866244

P007 | Hammond ottava dei preset.jpg | http://it.wikipedia.org/wiki/Utente:Salli | CC BY-SA 3.0 Source and license

CHECKS AND DECISIONS

- Require exactly nine characters per registration, each in 0–8; reject numeric conversion that drops leading zeros.
- A second reviewer must compare all 162 digits with the source before release.

SOURCES S019-01 Hammond Organ Service Manual, factory early-console compilation

ORIGINAL ELECTRICAL MODEL

Follow nine buses through contacts into transformer taps

Preset matrix and registration system

The preset contacts collect the nine manual buses. Fixed leads terminate on numbered preset-panel bars; adjust leads reach drawbars. Both paths feed transformer primary taps. The B-3 has separate manual matching transformers; the lower path also receives pedals. [S019-01]

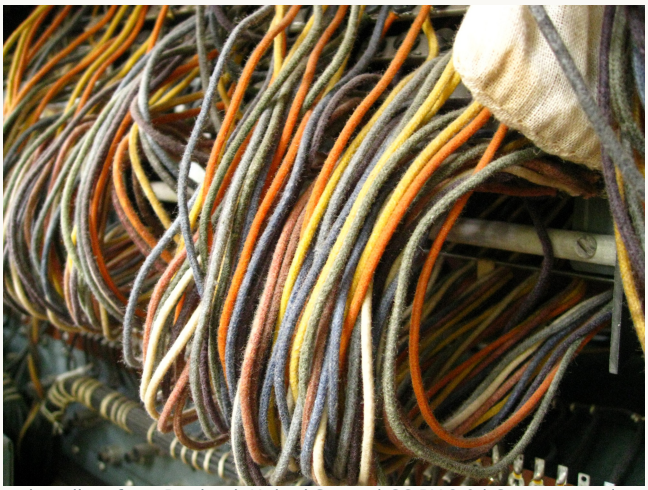
ORIGINAL MODEL: build a nine-pole selector feeding the harmonic-to-tap matrix, then the loaded transformer model. A fixed registration and its equivalent drawbar setting should share the same downstream calibration. Chapter 015 supplies source identities before these buses; this chapter never remaps a wheel simply because a preset changes.

For a provisional linear model, let u[m,d,n] be the signal on manual m, harmonic bus d, at sample n; let R[m,p,d] be its selected tap0–8. G[m,j] is a calibrated dimensionless gain. The expression below approximates one active registration. It does not solve transformer loading or simultaneous latch states. Keep the topology available so a circuit solver can later replace the gain-vector surrogate without changing preset data.

Multiple closed routes can connect the same bus to different taps. Derive their behavior from branch impedances and contact states; do not equate mechanical concurrency with summing independent audio programs. Unknown contact resistance and switching sequence remain profile parameters, not invented constants.

MODEL EQUATIONS

y[m,n] = sum(d=1..9, G[m,R[m,p,d]] * u[m,d,n]) [single-route linear approximation]



P039 | Hammond preset patch cables.jpg | eyeliam from Portland, United States | CC BY 2.0 | Context specimen; exact model/revision requires confirmation. Source and license

SOURCES S019-01 Hammond Organ Service Manual, factory early-console compilation

IMPLEMENTATION

Make percussion eligibility a circuit transition

Preset matrix and registration system

Upper B enables the percussion route. Its auxiliary preset switch bypasses the percussion compensation resistor R1 when another upper preset is selected; percussion switching also repurposes the B-group1-foot route for keying. [S019-01]

ORIGINAL IMPLEMENTATION: registration selection must notify the percussion network, not merely change nine gains. Apply the B interlock, selected harmonic routing, and keying-contact connectivity together at the event sample. The percussion envelope remains owned by the percussion subsystem. Do not reset it automatically on every program change or infer eligibility from the visible tablet alone.

Changing away from B while keys remain held can alter the keying circuit. Re-entering B must therefore be evaluated from electrical charging and contact history. A global rule that percussion rearms only after all MIDI notes are released is insufficient evidence for this transition. Measure the actual specimen before choosing transition constants.

Compile preset matrices outside the audio callback. Queue timestamped selector events with deterministic ordering; mutate audio-owned state only at the assigned sample. The following sequence is an original interface proposal. Any click-suppression ramp must be explicitly optional, because smoothing an actual switch transient changes the reference behavior.

```
onPresetEvent(event):
  advanceAudioTo(event.sampleOffset)
  oldMask = latch[event.manual]
  newMask = applyCradleEvent(oldMask, event)
  route = compiledProfile.resolve(newMask)
  contactNetwork.setTarget(route)
  percussion.updateInterlock(newMask, contactNetwork)
  latch[event.manual] = newMask
  publishControlSnapshot()
  // Never restart generators or synthesize note retriggers.
```

CHECKS AND DECISIONS

- Switch upper A#↔B during a held chord with percussion on; record audio and keying-state traces.
- Lower selections must not independently enable upper percussion.

SOURCES S019-01 Hammond Organ Service Manual, factory early-console compilation

MEASUREMENT PLAN

Acquire the missing connections and transition timing

Preset matrix and registration system

ORIGINAL ACQUISITION PLAN: ask the instrument technician for a complete panel photograph, orthogonal close-ups of every key group, and readable tap-number and manual labels. Record instrument serial identity, known modifications, and which viewpoint defines left/right. Photograph before any cleaning or movement. Internal inspection and continuity work belong to a qualified technician on an appropriately isolated, unpowered assembly.

Create one record per harmonic lead: manual, preset key, harmonic index, observed wire color, terminal/tap number, photo region, confidence, and reviewer. Avoid inferring a termination from a wire merely passing a screw. Where occlusion prevents confirmation, retain null and request another angle. This produces an auditable162-assignment specimen matrix without pretending that a color alone proves connectivity.

For audio comparison, hold playing keys constant while changing registration, with scanner and percussion disabled initially. Match the fixed registration using the corresponding drawbar bank and compare spectrum, level, and switching transient. Then add percussion tests and measure the B-interlock transitions separately. Preserve audio-interface gain, sample rate, clock, microphone or electrical capture point, and every panel setting.

Capture	Question resolved
All18 fixed registrations versus equivalent banks	Matrix transcription and loading equivalence.
C while a chord remains held, then A#	Cancel release timing and return without note retrigger.
A#↔B with changing inactive positions	Correct bank association and independent memory.
B exit/re-entry at several held-key durations	Percussion charge history and interlock transients.
Authorized simultaneous-latch observation	Contact overlap and network behavior; not assumed mixing.

CHECKS AND DECISIONS

- Publish raw captures and metadata alongside conclusions; label unperformed captures as pending.

STATE AND ACCEPTANCE

Serialize physical settings separately from modern memories

Preset matrix and registration system

The mk2 supports Combination Presets that recall parameters beyond registrations. That modern feature is not the vintage wired panel. [S019-02] ORIGINAL DESIGN PROPOSAL: provide explicit namespaces for instrument settings, wired-panel edits, and optional total-state software snapshots. A reverse-key selection in vintage mode must not recall Leslie speed, expression, scanner settings, or a new generator profile.

Persist every inactive drawbar bank as well as the active selection. Preserve continuous travel where the drawbar model supports it; a rounded digit alone loses intermediate positions. Save the wiring profile identifier and content hash plus explicit overrides, so an updated default library cannot silently change an existing project. Validate enum ranges and matrix dimensions before publishing a replacement state atomically.

Ordinary project restore should initialize with no held playing keys, preventing phantom sustained notes. A deterministic offline checkpoint is a different artifact: it additionally includes contacts, envelope state, generator phase, and downstream histories under a versioned engine-state contract. Do not market ordinary parameter recall as sample-exact continuation.

Acceptance requires round-trip equality of persistent settings, correct inactive-bank behavior, and equivalent results across supported audio block partitions. Confirm that fixed presets ignore drawbar movements until their adjust bank is selected. Keep the simultaneous-latch approximation and unmeasured percussion-transition behavior visible in the release limitations.

State group	Persisted contents
panelProfile	ID, revision, hash, two9×9 matrices, evidence tags.
manual[2]	Selected route; optional physical latch mask.
drawbars[2][2][9]	Independent travel and calibrated display values.
globalControls	Separate tablets, expression and cabinet settings.
extensionSnapshots	Explicit modern recall scope and version.
runtimeCheckpoint	Separate optional full-engine state, never confused with a preset.

SOURCES S019-02 Hammond B-3 mk2 Owner's Manual

HARDWARE EVIDENCE

One shared control process, behind the upper B preset

Percussion trigger and retrigger logic

Documented: factory percussion is available on the upper manual with the B adjust preset selected. The eighth-harmonic bus provides keying through terminal K. The control capacitor changes state while a key is held; releasing all upper keys allows recharge through a finite charging path. The factory also describes a preset-operated interlock associated with upper-manual level compensation. These are separate functions, not evidence for an envelope per note. [S020-01]

Original design proposal: make percussion a manual-wide subsystem downstream of physical contact and preset routing. Its state must survive voice allocation, notes joining a chord, and registration edits. Chapter 21 supplies the control-circuit transfer, gain, decay selection and harmonic extraction. This chapter supplies detector history and event ordering. Both modules must share the same timeline; passing an occasional boolean trigger between unrelated envelopes would discard the history that matters.

Baseline scope is an unmodified vintage reference instrument. Modern conveniences belong to named extension modes. A velocity-scaled attack, polyphonic percussion envelope, or sustain pedal that holds percussion eligibility can be useful, but it must not silently redefine the reference behavior.

State or connection	Implementation responsibility
Upper physical contacts	Persist independently of audible registration.
Upper B selection	Resolve an electrical route, not a MIDI channel label.
Shared control state	Store once per organ instance.
Selected harmonic audio	Feed the shared envelope from current contacts.
Lower manual and pedals	Do not enter the upper detector aggregation.

CHECKS AND DECISIONS

- Expose detector, routed eligibility and control state separately in diagnostic traces.

SOURCES S020-01 Hammond factory service-manual compilation: percussion operation and system schematic

ORIGINAL MODEL

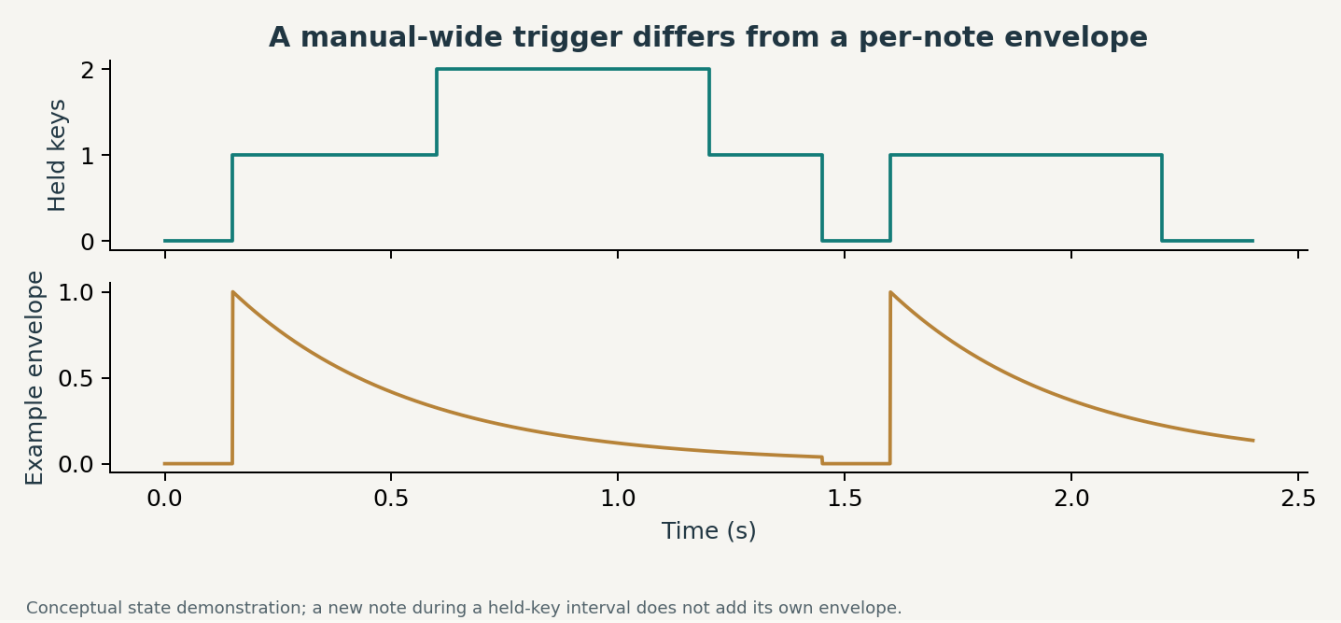
Follow the contact that reaches terminal K

Percussion trigger and retrigger logic

The factory describes the keying bus as the eighth harmonic: this identifies the 1-foot harmonic, not the eighth drawbar counted from the left. The B adjust contact lies in that route. [S020-01] A contact study found gesture-dependent closure and bounce behavior on its 1967 C-3 specimen. Those measurements motivate contact-aware timing; they do not establish a universal B-3 percussion reset threshold. [S020-02]

Original design proposal: maintain a contact vector for each physical upper key and derive the keying connection from the relevant harmonic contact. The simplest prototype reduces closed contacts to a logical OR. A more detailed model passes equivalent detector conductance to the circuit solver, allowing resistance, bounce and loading to be introduced without redesigning the host event layer. Do not infer this contact from whether the audible 1-foot drawbar contributes sound.

Keep switch routing as its own function. While keys remain down, changing preset or the percussion tablet can connect or disconnect the detector network. Preserve control-circuit state across that change. The audible result must follow traced switch topology and subsequent circuit evolution, with reference captures resolving transition details; neither an unconditional retrigger nor an unconditional suppression is established here.



Conceptual state demonstration; a new note during a held-key interval does not add its own envelope.
Original explanatory calculation. Illustrative parameters are not measured instrument data.

CHECKS AND DECISIONS

- A silent drawbar setting must not erase the physical keying contact.

SOURCES S020-01 Hammond factory service-manual compilation: percussion operation and system schematic / S020-02 Moro, McPherson and Sandler: Dynamic temporal behaviour of the keyboard action on the

ORIGINAL MODEL

Rearm is elapsed-time history, not a note counter reset

Percussion trigger and retrigger logic

Original design proposal: use a continuous readiness variable q as a temporary surrogate while the circuit model is built. Define q = 1 as fully recovered and q = 0 as exhausted. These are dimensionless model coordinates, not measured voltages or a claim that audible gain is linear in capacitor charge. Let g indicate a resolved keying connection, and let positive time constants be fitted parameters in seconds.

Advance q toward zero while g is active and toward one while the detector is released. The exact exponential update below is bounded for nonnegative elapsed time and positive time constants. It avoids block-length-dependent Euler drift. The gain function F(q) and switch-dependent targets are replaced by chapter 21’s validated circuit representation when available. A brief release recovers only part of the state; a zero-duration opening recovers none.

Feed current selected-harmonic audio through the shared gain. A newly added chord tone can therefore enter an already-decaying percussion sound without receiving its own fresh attack. Do not freeze a sample of the first note. Keep oscillator phases continuous and keep control-state integration separate from antialiasing of switched audio. A circuit surrogate remains an approximation until recorded recharge curves, sustained-note residuals and switching transients agree.

MODEL EQUATIONS

q_next = q_target + (q - q_target) * exp(-dt / tau)

g = 1: q_target = 0, tau = tau_decay; g = 0: q_target = 1, tau = tau_recover

y_percussion(t) = F(q(t)) * h_selected(t); dt and tau are seconds; h_selected is routed harmonic audio

CHECKS AND DECISIONS

- For dt = 0, q_next equals q exactly.
- After increasing silent release gaps, predicted recovery is monotonic under fixed settings.
- Never reset q merely because another upper note starts.

SOURCES S020-01 Hammond factory service-manual compilation: percussion operation and system schematic

IMPLEMENTATION

Translate host notes into physical-key ownership

Percussion trigger and retrigger logic

VST3 events provide a bus index and sample offset relative to the processing block. Note-on data include channel, pitch and an optional note identifier. [S020-03] [S020-04] Original implementation proposal: normalize these into timestamped ownership changes before updating contacts. Use available note identities to match releases. For unidentified repeated notes, select and document a bounded policy such as FIFO ownership; it is software behavior, not an organ mechanism.

Multiple owners of one physical pitch should keep that virtual key down until the chosen policy releases its final owner. A duplicate NoteOn must not automatically create another percussion attack. Separate controller ownership from optional contact gestures: an explicit re-strike extension can synthesize release and depression only when its timing is defined. Allocate fixed-capacity storage outside the audio callback and expose overflow counters.

Preserve deterministic ordering for events with equal offsets, including preset automation. Advance the shared circuit only for elapsed samples. An off/on pair at one timestamp may change connectivity in event order, but it supplies no recharge duration. Route sustain, panic and transport-reset policies explicitly; never let a generic synthesizer voice manager reset this subsystem by accident.

```
for event in stable_time_order(events):
  render_and_advance(cursor, event.offset)
  update_owners_or_switches(event)
  derive_physical_key_targets()
  apply_due_contact_changes(event.offset)
  detector = resolve_keying_network()
  circuit.set_detector(detector)
  cursor = event.offset
render_and_advance(cursor, block_end)
# No envelope reset on NoteOn.
# Contact subevents share this sample clock.
```

CHECKS AND DECISIONS

- Partitioning one event stream into different block sizes preserves detector and control-state traces.

SOURCES S020-03 Steinberg VST 3 Interfaces: Event structure / S020-04 Steinberg VST 3 Interfaces: NoteOnEvent structure

MEASUREMENT

Capture the release window and preset transitions

Percussion trigger and retrigger logic

Proposed acquisition protocol: identify the instrument, preamp revision, modifications, percussion controls and recording point before collecting comparisons. Use a stable dry recording path so rotary motion does not obscure envelope recovery. Capture key movement or contact timing alongside audio where practicable. A MIDI sequencer driving another keyboard does not by itself prove when the vintage detector contact opened.

First obtain repeatable isolated notes with generous release gaps, then shorten the all-keys-open interval across a logarithmic sweep. The sweep limits and repetitions are experimental choices, not published Hammond timing specifications. Measure onset amplitude relative to a fully recovered reference and retain the entire envelope. Repeat across pitches and gestures to separate control recovery from contact variability. Keep traces with incomplete release rather than discarding them as poor takes.

Next collect held-note routing transitions, including B to another preset and back, percussion off/on, and cancellation. Compare detected connectivity, control-state prediction and audible output. The goal is to identify which transitions change the charge path and which only change audio routing. Any internal electrical acquisition belongs to a technician-qualified measurement plan; this chapter does not prescribe live chassis probing.

Capture family	Question resolved
Legato overlap, then full release	Does one remaining key prevent recovery?
Same-note repeats with varied gaps	How much readiness returns before the next attack?
Second note during decay	Does it join the existing envelope as predicted?
Held key, B / other / B	What do preset transitions do to detector state?
Held key, percussion off / on	What routing and transient behavior actually occurs?
Lower or pedal notes held	Confirm independence from upper release.

SOURCES S020-01 Hammond factory service-manual compilation: percussion operation and system schematic / S020-02 Moro, McPherson and Sandler: Dynamic temporal behaviour of the keyboard action on the

ACCEPTANCE

Acceptance tests must distinguish facts from policies

Percussion trigger and retrigger logic

Original acceptance proposal: use separate suites for established routing behavior, software bookkeeping, and specimen-dependent transitions. The first suite checks structural invariants. The second checks declared host policies. The third remains pending until reference captures exist. A passing MIDI test does not validate contact physics, and fitting one recovery curve does not establish every preset-switch transient.

Deliver a timestamped event fixture format, a fixed-capacity key-ownership table, detector traces, and an adapter to the circuit module. Each fixture records initial control state, switch settings, event ordering, contact realization and expected evidence class. Save captured traces alongside sample rate and recording provenance. No implementation or measured fidelity is claimed by this research chapter.

Fixture	Required result or disposition
Overlapping upper notes	No independent envelope reset for the joining note.
Last upper key released	Recovery progresses with elapsed open-contact time.
Another upper key still down	No false all-released condition.
Repeated NoteOn, same identity	Follow duplicate-event policy without forced reattack.
Two owners, one release	Key remains owned under the declared ownership policy.
Off/on at one sample offset	No invented positive recharge interval.
Preset or tablet change while held	Circuit history retained; compare with pending captures.
Block-size and sample-rate changes	Equivalent event timing; bounded, stable control state.
Panic and state restoration	Explicit deterministic release/reset policy; no stuck ownership.

CHECKS AND DECISIONS

- Mark unmeasured switching cases as unresolved, not passed.
- Keep per-note retrigger and velocity-emphasis extensions disabled in the reference preset.

SOURCES S020-01 Hammond factory service-manual compilation: percussion operation and system schematic / S020-03 Steinberg VST 3 Interfaces: Event structure / S020-04 Steinberg VST 3 Interfaces: NoteOnEvent structure

HARDWARE EVIDENCE

Freeze the percussion circuit identity

Percussion circuit and envelope

The production reference is Figure 24, printed 3-50, PDF page 129: AO-28, with V5/V6 designated 6C4 and V7 12AU7. Its no-code AO-28-1 annotations qualify components and the P-to-D filter. The supplementary 1995 redraw is a readability aid, not the factory revision date. [S021-01] [S021-02]

ORIGINAL IMPLEMENTATION DECISION: create a percussion manifest before writing the envelope. Store the photographed chassis stamp, schematic fingerprint, transformer identifiers, repair history and every adopted component value with its provenance. Keep early and later transfer networks as distinct profiles. An unreadable stamp must produce an unresolved profile, not silently select whichever drawing is easiest to parse.

The table contains schematic nominal values, converted into SI units for computation; they are not measurements of the target instrument. Preserve both the literal drawing notation and the interpreted unit. A program should reject an unqualified capacitor value such as 0.33, because interpreting it as farads or picofarads would destroy the result. Specify voltage references separately from normalized digital envelope coordinates. Do not infer transformer parasitics from their symbols.

Reference	Documented nominal	Software representation
C31	0.33 microfarad	3.3e-7 F
R57 / R58	1.5 / 4.7 megohm	1.5e6 / 4.7e6 ohm
R55 / R56	82 / 22 kilohm	8.2e4 / 2.2e4 ohm
V7 characteristic	Unmeasured specimen	Calibrated gain map
P-to-D network	Revision qualified	Named transfer profile

CHECKS AND DECISIONS

- Manifest distinguishes documented, interpreted, proposed and measured fields.
- No production date is inferred from the 1995 redraw title block.

SOURCES S021-01 Hammond Organ Service Manual: percussion operation and AO-28 schematic / S021-02 B3/C3 service-manual compilation with dated AO-28 redraw

HARDWARE PATH / ORIGINAL ABSTRACTION

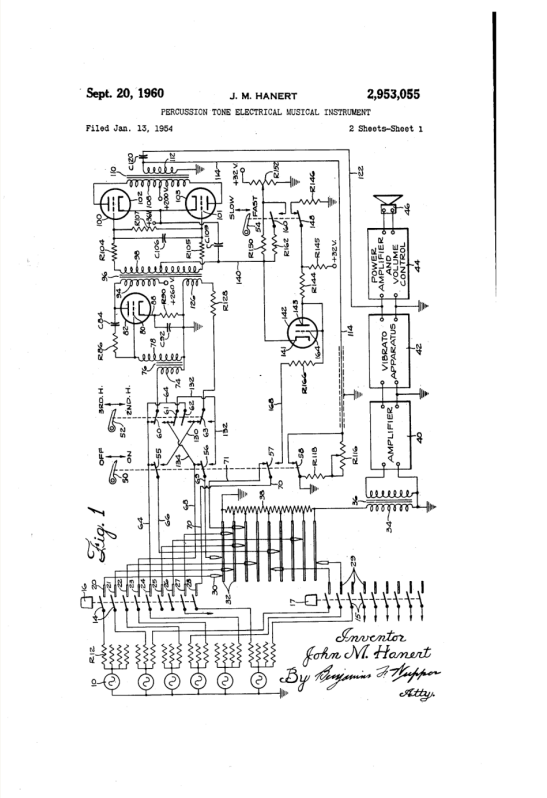
Preserve the borrowed harmonic and coupled switches

Percussion circuit and envelope

Selected second/third-harmonic signal enters H through T4/V5; T5 returns signal through R50/J to its drawbar while driving V7. T6 and filtering feed D. Soft shunts T6 through R46 and bypasses upper attenuation R1. Normal retains R1 only with upper B selected. Fast parallels R57/R58 and changes bias; the manual describes approximately 50% greater signal. Percussion on repurposes upper B's eighth-harmonic, 1-foot connection. [S021-01]

Hanert describes a return winding that preserves sustained registration, and a percussion path without scanner vibrato; the patent also couples faster decay with greater initial intensity. Treat these as design principles, not substitute production component values. [S021-03]

ORIGINAL MODEL PROPOSAL: expose separate sustained-return and percussion outputs from a common source selection. Moving the selected drawbar should change its sustained contribution without becoming a percussion-volume knob. Implement the normal-mode attenuation upstream of the shared organ output so lower-manual and percussion levels are not inadvertently attenuated together. Name percussionVolume and consoleVolume separately. Chapter 020 supplies the eligible routing state and control-contact events; this chapter supplies their continuous electrical consequence. Model the missing 1-foot contribution at the upper-B routing boundary, with preset switching explicitly covered by integration fixtures.



Historical patent plate US2953055A_p01; invention embodiment, not a verified production drawing for the selected organ or cabinet. Source and license

CHECKS AND DECISIONS

- Drawbar movement cannot directly multiply the percussion envelope in the baseline profile.
- Preset changes resolve the upper-B route before applying its attenuation or 1-foot mask.

SOURCES S021-01 Hammond Organ Service Manual: percussion operation and AO-28 schematic / S021-03 John M. Hanert, Percussion tone electrical musical instrument, US 2,953,055

ORIGINAL DIGITAL MODEL

Model capacitor voltage before modeling loudness

Percussion circuit and envelope

ORIGINAL DERIVATION: begin with a piecewise first-order capacitor surrogate. Let v be the envelope-control voltage in volts, v_inf the mode-dependent asymptote, tau its time constant in seconds, and Fs the audio sample rate in hertz. Select the charging or discharging branch from chapter 020’s contact state. For constant controls during one sample, the exponential update is exact for this simplified RC model; it is not an exact solution of the complete valve circuit.

Using the listed nominal parts gives isolated discharge products of 1.551 seconds for R58*C31 and approximately 0.375 seconds for (R57 parallel R58)*C31. These are calculated electrical time constants, not audible decay lengths. The simple charging product (R55+R56)*C31 is 0.03432 seconds; retain it only as a starting hypothesis because diode conduction, loading and bias can change recharge behavior.

Map voltage to nonnegative gain with a fitted monotonic function G(v,mode,cutoff). The cutoff coordinate and the fast-mode initial bias must remain independent parameters. A bare exponential audio multiplier cannot independently represent both. Fit the transfer function on training captures, then test unseen notes and chords. Keep v continuous when controls change unless the circuit model explicitly predicts charge redistribution; do not reset it merely because a new audio block begins.

MODEL EQUATIONS

v[n+1] = v_inf + (v[n] - v_inf) * exp(-1 / (Fs * tau))

R_parallel = (R57 * R58) / (R57 + R58)

p[n] = H_out{ G(v[n], mode, cutoff) * H_in{s[n]} }

CHECKS AND DECISIONS

- H_in and H_out are identified transfer models; s is the coherently summed selected-harmonic source.
- Electrical tau and measured time to a specified audio threshold are separate exported metrics.
- Envelope state remains finite and continuous through sample-rate changes.

SOURCES S021-01 Hammond Organ Service Manual: percussion operation and AO-28 schematic / S021-02 B3/C3 service-manual compilation with dated AO-28 redraw

ORIGINAL IMPLEMENTATION CONTRACT

Build the module in a testable order

Percussion circuit and envelope

ORIGINAL BUILD PLAN: first implement the routing fixtures with a unity envelope, then the capacitor state, then the voltage-to-gain curve, then input/output coloration. Integrate revision-dependent transformer loading only after the simpler model can reproduce independently measured behavior. This order keeps a wrong harmonic map from being concealed by equalization or distortion tuning.

Input data should include the selected harmonic bus, sustained-return source, the chapter 020 contact state, timestamped tablet changes and an immutable specimen profile. Outputs should include percussion audio, the return-path audio or correction, upper-path attenuation, and debug-only voltage/gain traces. Do not create a new oscillator for percussion: reuse the phase-continuous wheel signals already supplied by the generator engine.

Prepare coefficients and profile data off the audio thread. Use bounded state, fixed storage and deterministic event ordering. The initial linear transfer model needs no nonlinear oversampling by itself; a later nonlinear valve model requires an explicit alias assessment and any oversampling filters must report their latency. Preserve contact-generated transients from chapter 018 without injecting an additional arbitrary click into every percussion onset. Expose artistic overrides as a separate mode from the historical profile.

```
process sample:
  apply timestamped route and tablet events
  state = triggerModule.controlContactState()
  mode = profile.mode(tablets, route)
  v = rcAdvance(v, state, mode, sampleRate)
  source = harmonicBus.readSharedSources(mode.harmonic)
  returnAudio = returnPath.process(source)
  percussion = outputPath.process(gainMap(v, mode) * inputPath.process(source))
  emit(percussion, returnAudio, mode.upperAttenuation)
```

CHECKS AND DECISIONS

- No allocation, locking or file access occurs inside the sample/block processing callback.
- Block-size changes preserve the same sample-indexed state trace.
- The returned sustained signal is not added twice to the selected drawbar.

SOURCES S021-01 Hammond Organ Service Manual: percussion operation and AO-28 schematic / S021-03 John M. Hanert, Percussion tone electrical musical instrument, US 2,953,055

ORIGINAL MEASUREMENT PLAN

Capture what the schematic cannot determine

Percussion circuit and envelope

ORIGINAL ACQUISITION PROPOSAL: record a documented reference organ through an appropriate isolated audio interface at a fixed expression position, initially without Leslie motion or room capture. Photograph tablet settings and record warm-up conditions for each session. Any internal measurement belongs to a qualified valve-equipment technician; audio-only capture can establish the behavioral model without exposing the researcher to the chassis.

Use repeated recordings to estimate repeatability before choosing tolerances. Fit a narrowband analytic amplitude or sinusoidal regression at the selected source frequency, inspect the residual, and preserve the raw waveform. A broadband RMS window can mistake contact noise or sustained leakage for envelope energy. Record the exact analysis bandwidth and window length with every result.

A baseline with all drawbars closed isolates percussion more effectively; then add the corresponding sustained drawbar and check phase-sensitive interaction. Separate training and validation notes across the compass. Do not normalize each take independently, because that would erase the normal/soft and fast/slow level relationships. Normal attenuation needs a paired steady-state measurement after percussion has decayed, while restoration through J needs a separate sustained-path comparison. No captures in this chapter have yet been performed.

Capture set	Quantity to identify
Second / third; multiple notes	Source mapping and frequency response
Normal / soft; fixed gain	Relative percussion level and loading
Fast / slow; same playing sequence	Initial gain and full envelope curve
Percussion on / off; upper B	Steady upper-path attenuation
Selected drawbar swept	Return-path level and phase
Short release gaps	Recharge curve; chapter 020 integration
Held notes; cutoff variation	Residual leakage floor
Two-note and larger chords	Gain compression and intermodulation

CHECKS AND DECISIONS

- Every fitting result points to raw audio, settings, revision and analysis configuration.
- A fit is rejected if its gain error improves only by silently changing acquisition normalization.

ORIGINAL ACCEPTANCE GATES

Accept behavior, then claim measured fidelity

Percussion circuit and envelope

ORIGINAL ACCEPTANCE POLICY: publish separate results for routing correctness, numerical stability and measured resemblance. The first two can be tested before owning reference recordings; the third cannot. A consistent mathematical model is useful, but it does not establish the sound of an identified AO-28. Avoid a single similarity percentage that hides one correct isolated note and an incorrect chord response.

Define quantitative acceptance tolerances only after measuring the hardware's repeatability and establishing the project's perceptual target. Report envelope error over time, onset-level error, source-frequency error, steady leakage, and spectral residual independently. Evaluate at several host sample rates and event positions. A failed case should identify the responsible layer: routing, recharge, gain curve, transfer network or shared downstream amplification.

Keep uncertain specimen parameters visible in the build report: transformer response, replacement capacitor values, valve imbalance, cutoff setting and exact early/late filter configuration. A useful initial release may label these as calibrated behavioral approximations. Component-exact and one-to-one claims remain blocked until topology, values and measured responses justify them.

Fixture	Falsifiable criterion
Upper B versus other presets	Routing matches the approved switch truth table
Selected drawbar changes	Sustained return changes without direct envelope scaling
Normal / soft transition	Upper attenuation and percussion gain follow distinct controls
Fast / slow transition	Rate and bias are independently represented
Repeated short gaps	Shared voltage state follows chapter 020 events
Vibrato enabled	Percussion branch bypasses the scanner model
Silence and long holds	Finite output; measured leakage remains a separate parameter
Offline versus real-time render	Equivalent timestamped events produce equivalent output

CHECKS AND DECISIONS

- All numerical pass thresholds are tagged project targets, never manufacturer specifications.
- No photographic asset or patent embodiment is treated as proof of a measured component response.

SOURCES S021-01 Hammond Organ Service Manual: percussion operation and AO-28 schematic / S021-03 John M. Hanert, Percussion tone electrical musical instrument, US 2,953,055

HARDWARE EVIDENCE / SCOPE

The pedal division is a shared-source mixture

Pedal generator mapping and shaping

The B-3 pedal assembly has 25 notes, eight contact positions per note and four collecting buses. The two drawbars receive mixtures through a reactor/resistor network; they are not two independent sine generators. The B-3 generator includes twelve complex low-frequency wheels. These are documented construction facts, distinct from an assumed spectral calibration. [S022-01]

ORIGINAL DESIGN PROPOSAL: make PedalDivision a consumer of the same continuously running GeneratorBank used by both manuals. Give it its own contact topology, four electrical bus outputs and two drawbar positions. Feed its resulting signal into the lower-manual mixing/routing model, preserving the Great vibrato relationship established elsewhere. Never implement an unrelated bass synthesizer and merely match its fundamental frequency.

The reference profile is the factory B-3/C-3 wiring represented in this chapter's table. Earlier console pedal arrangements, 32-note concert pedal systems, and later electronic pedal voices belong to separate profiles. Preserve the chosen instrument serial number, generator identity, loom modifications and filter revision beside the data. The supplied matrix is documentary evidence, not a continuity measurement on a surviving instrument.

Boundary	Required implementation decision
Generator source	One shared phase and waveform per generator number
Pedal actuation	Independent contact edges; all played pedals may contribute
Two drawbars	Coupled electrical mixture, calibrated by position
Other divisions	Share source identity; retain separate contact ownership

CHECKS AND DECISIONS

- No new oscillator is allocated when a pedal is pressed.
- A pedal with both drawbars at zero does not stop or reset generator state.

SOURCES S022-01 Hammond Organ Service Manual, original console models

DOCUMENTED MATRIX / DERIVED RULE

Encode the unusual contact map literally

Pedal generator mapping and shaping

Table III’s pedal columns are fundamental, third-contact, second, fourth, sixth, eighth, tenth and twelfth. The third-contact column does not uniformly carry a third harmonic: rows1–7 repeat the second-harmonic generator; rows8–12 have dashes; rows13–25 use the expected higher source. Preserve these exceptions instead of repairing them into a textbook harmonic series. [S022-01]

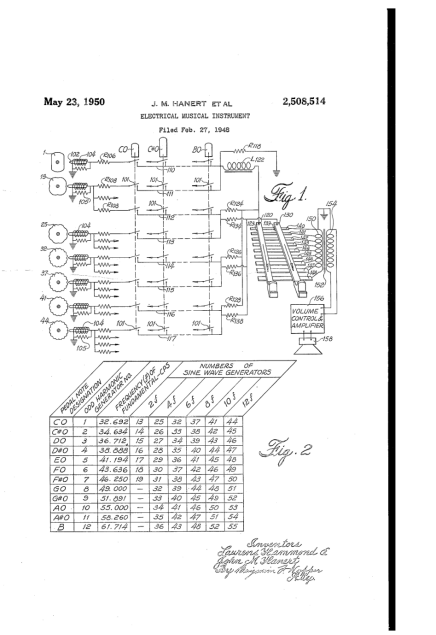
ORIGINAL DERIVATION: for pedal p numbered1–25, the regular source offsets are0,12,24,31,36,40,43. The exceptional column follows the piecewise rule below. All source numbers are one-based. The companion JSON provides200 positions:195 connected entries and five documented disconnections. Null means an absent connection, never generator zero. Two contacts receiving the same generator remain distinct conductance paths.

Keep physical bus membership separate from the ordered table columns. The verified diagram groups fundamental/third on brown, second/fourth on red, sixth/eighth on orange and tenth/twelfth on yellow. Color is a diagram identifier; a repaired loom still requires physical verification. [S022-01]

Pedal	Ordered generator-source vector
1	1,13,13,25,32,37,41,44
7	7,19,19,31,38,43,47,50
8	8,null,20,32,39,44,48,51
13	13,32,25,37,44,49,53,56
25	25,44,37,49,56,61,65,68

MODEL EQUATIONS

third(p) = p+12 if p<=7; disconnected if8<=p<=12; p+19 otherwise



Historical patent plate US2508514A_p01; invention embodiment, not a verified production drawing for the selected organ or cabinet. Source and license

CHECKS AND DECISIONS

- Compare every row with the original scan during independent transcription review.
- Do not deduplicate repeated source numbers before evaluating contact loading.

SOURCES S022-01 Hammond Organ Service Manual, original console models

ORIGINAL SPECTRAL MODEL

Separate intrinsic partials from borrowed tones

Pedal generator mapping and shaping

The bass patent describes complex sources rich in odd partials, combined with other generator outputs for even-harmonic reinforcement. Its stated purpose includes avoiding undesirable phase relationships when pedal and manual tones share frequencies. This is historical design evidence; the patent's illustrative harmonic strengths are not measured B-3 coefficients. [S022-02]

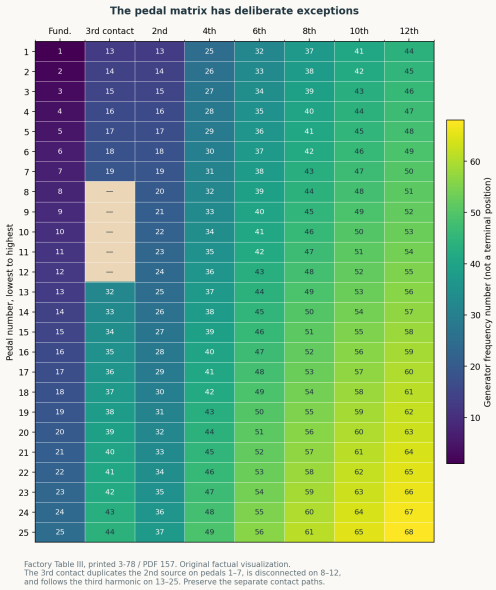
ORIGINAL MODEL: represent each low wheel by a calibrated periodic waveform or Fourier coefficients referenced to that wheel's phase. Intrinsic partial m follows m times its instantaneous frequency. A separate borrowed generator follows its own gearing-derived frequency and existing phase. Consequently an intrinsic fifth partial and a nominal fifth-partial source need not null perfectly. Keeping this distinction permits natural beats without inventing per-note random detuning.

Use a band-limited waveform family or harmonic summation truncated below the internal Nyquist limit. Preserve coefficient phase, not merely magnitude. Distinguish a pickup-terminal capture already containing analog filtering from an induced-voltage model requiring that filtering afterward. Otherwise the same spectral shaping is applied twice. Measured residual even harmonics may be retained; do not force a mathematically odd waveform simply because the design emphasizes odd harmonics.

MODEL EQUATIONS

x_w[n] = sum_m A[w,m] cos(m*phi_w[n] + psi[w,m])

phi_w[n+1] = wrap(phi_w[n] + 2*pi*f_w[n]/Fs)



Original visualization derived from factory Table III. Electrical source numbers are not physical terminal positions; specimen wiring still needs verification.

CHECKS AND DECISIONS

- Changing pedal ownership cannot randomize any shared source phase.

SOURCES S022-02 US2508514A, Electrical musical instrument, Hanert and Hammond

IMPLEMENTATION / ORIGINAL DESIGN PROPOSAL

Solve the pedal network before simplifying it

Pedal generator mapping and shaping

Build the electrical reference as a small linear network: contact series resistances, four bus nodes, resistor mixing panel, filter reactor and its loss, two drawbar connection states, and transformer input loading. Use the production diagram for connectivity; do not substitute patent resistor examples. [S022-01][S022-02] An unreadable or unmeasured reactor inductance stays an explicit unknown, not a guessed factory constant.

For a first implementation, identify a stable four-input/two-output transfer model at each control state. Retain a full network solver offline as an oracle. Precompute matrices when the hardware profile or sample rate changes; handle contact events without heap allocation. If the solver models generator source impedance, share the corresponding source-loading calculation with manual paths. A unilateral bus sum is an approximation whose error should be quantified.

When interpolation between drawbar settings is used, interpolate an appropriate stable representation and check passivity or pole locations. Switching a linear network can still create discontinuities. Preserve stored inductor energy and avoid arbitrary state reset. Mechanical contact timing belongs to the contact layer; artificial release envelopes must not conceal a mapping or switching error.

Model field	Units / required provenance
contactSeriesResistance[p,c]	Ohms; measured or documented per profile
pedalReactor	Henries and series ohms; currently unknown
drawbarLoad[d,position]	Electrical model or fitted transfer; state-specific
busToOutputTransfer	Complex response, reference levels and load recorded

```
sources = generatorBank.renderShared(block)
for event in orderedPedalContactEvents:
    updateContactConductance(event)
for sample in block:
    busExcitation = accumulateMappedSources(sources)
    output = pedalNetwork.advance(busExcitation, drawbars)
    lowerMix.add(output)
```

CHECKS AND DECISIONS

• Offline and real-time network forms agree on impulses and sustained mixtures within the declared numerical tolerance.

SOURCES S022-01 Hammond Organ Service Manual, original console models / S022-02 US2508514A, Electrical musical instrument, Hanert and Hammond

MEASUREMENT / PROPOSED PROTOCOL

Capture pedal color at identifiable boundaries

Pedal generator mapping and shaping

Acquire dry, calibrated recordings before Leslie coloration obscures the bass network. Record instrument identity, control positions, preamplifier tap, interface calibration, source loading, sample clock, temperature and mains frequency. A qualified technician should choose appropriate measurement interfaces for internal circuitry. External recordings alone can support an end-to-end fit, but cannot uniquely identify every internal resistor and inductor.

ORIGINAL PROTOCOL: for each of 25 pedals, capture both drawbar axes at positions0–8 with the other drawbar at zero, counting the silent0/0 state once. That produces17 states per pedal,425 total. Add4/4 and 8/8 for each pedal to test mixture interaction, giving475 proposed states. These counts describe a research plan, not an existing dataset. Record onset, stable sustain and release in every case.

Add repeated strikes, slow releases, adjacent-pedal overlap and selected pedal/manual unisons. Fit complex spectra from long stable windows and preserve phase alignment to a reference generator channel where available. Compare combined recordings against the sum of isolated recordings to expose loading or downstream nonlinearity. Separate noise estimates from harmonic estimates; a narrow spectral bin alone cannot identify which physical path produced energy.

Capture	Question resolved
Individual low generator	Intrinsic partial magnitude and phase
Four buses / known injection	Mixing and reactor transfer
Drawbar axes plus paired states	Control law and coupling
Pedal with manual unison	Shared-source phase and load interaction
Slow pedal motion	Contact order, bounce and release timing

CHECKS AND DECISIONS

- Store raw recordings, calibration files and analysis settings together.
- Reserve held-out pedals and mixtures for validation rather than fitting every recording.

ACCEPTANCE / HANDOFF

Pedal acceptance gates and remaining unknowns

Pedal generator mapping and shaping

ORIGINAL ACCEPTANCE PLAN: validate the wiring matrix independently of audio first. All195 connected entries must match the factory-reference transcription, five absent entries must remain absent, and every active source must lie within 1–91. Inspect the transition across pedals7/8 and 12/13 explicitly. Those boundaries catch errors that an ascending scale of fundamentals cannot reveal.

Next test shared phase by rendering a pedal, a related manual note, and both together from identical generator state. A purely linear fixture should equal the sum of its component renders within floating-point tolerance; a loaded circuit fixture should instead agree with the full network oracle. Declare which fixture applies. Do not require arbitrary waveform nulls against unsynchronized recordings, and do not let a chord limiter mask gain mistakes.

The handoff consists of the factory map, provenance, parameter schema, raw capture manifest, fitted coefficients, offline reference solver and deterministic fixtures. The map has received one visual review here; independent review and specimen continuity checks remain open. Reactor values, actual contact resistances, low-wheel spectra and measured transfer functions are unfilled calibration fields. The result can therefore support faithful architecture immediately while keeping the acoustic identity claim conditional on measurement.

Gate	Pass condition
Mapping	All200 positions checked, including five nulls
State	Held pedals survive unrelated manual events correctly
Phase	Shared generator trajectories remain continuous
Spectrum	Held-out harmonic/phase errors reported, not invented
Real time	No audio-thread allocation; bounded event processing
Provenance	Every fitted value identifies recording and analysis version

CHECKS AND DECISIONS

- A low-note sine-only shortcut must be labeled a reduced model.
- Optional monophonic pedal modes or sustain controls are extensions, outside the vintage reference profile.

SOURCES S022-01 Hammond Organ Service Manual, original console models

HARDWARE EVIDENCE / SCOPE

One pedal, a capacitive transfer network

Expression mechanism and response

The factory description identifies two stator sets, a pedal-linked rotor, a direct loud feed and a compensated soft feed. Their overlap changes coupling between the first and second V4 sections. The AO-28 drawing also places factory-set C22 feedback around the receiving stage. Figure 12 depicts the pedal shaft, arm, connecting rod and spring. These are the production references for this chapter. [S023-01]

Original design proposal: treat expression as part of the common preamplifier, with separate mechanical position, electrical transfer and controller adaptation. Reserve a downstream master-output control for monitoring. Moving expression changes the drive delivered to later nonlinear stages; moving that final master control need not. Keeping the controls separate makes drive calibration interpretable and allows quiet listening without silently changing the emulated operating point.

The 1940 Hammond–Meinema patent describes capacitor-controlled amplifier feedback, but its circuit is a historical embodiment rather than an AO-28 wiring authority. Do not import its capacitor ranges or tremulant mechanism into this baseline. [S023-02] The modern B-3 mk2 documents selectable curves and separate minimum low/high-frequency levels; these belong to a modern comparison profile. [S023-03]

Record	Required distinction
Physical position u	Normalized heel-to-toe travel; proposed coordinate, not decibels
Electrical profile	AO-28 revision, fixed network, measured rotor capacitances
Controller profile	Pedal endpoints, polarity and response; independent of hardware model
Output master	Downstream monitoring gain; never used to fit expression response

CHECKS AND DECISIONS

- Record the selected AO-28 revision before fitting any expression curve.
- Keep modern curve presets labeled as product options.

SOURCES S023-01 Hammond Organ Service Manual: B-3/C-3/RT-3 expression and AO-28 / S023-02 US2221188A: Electrical musical instrument, Hammond and Meinema / S023-03 Hammond B-3 mk2 / C-3 mk2 / B-3P mk2 Owner's Manual

ORIGINAL MECHANICAL AND ELECTRICAL MODEL

Model the overlap and its loading

Expression mechanism and response

Original design proposal: describe the mechanism with pedal angle theta, linkage mapping u=L(theta), and two measured capacitance functions C_L(u) and C_S(u), in farads. Include residual coupling at both endpoints. A linear crossfade assumes complementary capacitances and negligible loading; neither follows merely from the existence of two stators. Preserve that assumption only as an explicitly provisional prototype.

Separate fixed capacitance from variable overlap, and attach each parasitic to its actual pair of circuit nodes. A rotor-to-chassis capacitance affects the receiving node differently from a stator-to-chassis capacitance. Store forward and backward angle sweeps separately until repeatability establishes whether one curve suffices. If mechanical play is measurable, model it in L(theta), rather than disguising it as arbitrary audio smoothing.

The original diagram below is a signal-responsibility view, not a replacement schematic. The loud and soft branches share an upstream source, so their impedances interact. A final circuit implementation must solve those coupled nodes and retain the receiving-stage feedback. First validate stationary pedal positions; only then compare moving-pedal captures against a time-varying model.

P005 | Hammond B3 25-note pedalboard, Museum of Making Music.jpg | Hammond_B3,_Museum_of_Making_Music.jpg: doryfour derivative work: Clusternote | CC BY-SA 2.0 | Retouched/transformed; excluded from geometry evidence. Source and license

CHECKS AND DECISIONS

- Capacitance curves stay nonnegative and retain documented measurement units.
- A fitted linkage curve reproduces both endpoint positions without changing electrical gain normalization.

SOURCES S023-01 Hammond Organ Service Manual: B-3/C-3/RT-3 expression and AO-28

ORIGINAL DERIVATION

A useful transfer function, with explicit limits

Expression mechanism and response

For a stationary-pedal analytical surrogate, let $V_L=H_L(s)V_{in}$ and $V_S=H_S(s)V_{in}$ represent the two branch voltages. Define $Y_L=sC_L$, $Y_S=sC_S$, grid loading Y_g, and feedback admittance Y_f. Approximate the receiving amplifier by $V_o=-A(s)V_r$. Kirchhoff current balance gives the expression below. All admittances use siemens; s has units of inverse seconds and A is dimensionless.

This equation shows why multiplying by a single gain cannot generally reproduce the network: pedal position changes frequency-dependent numerator weights and the loaded denominator. It also exposes a boundary of the surrogate: prescribed branch voltages ignore their loading by the rotor. Replace that assumption with a coupled modified-nodal model when constructing the component-level AO-28. Include feedback coupling components in Y_f rather than assuming it is an isolated ideal capacitor.

For movement, capacitor current is the derivative of charge. A frozen-position coefficient model drops the v dC/dt term. Treat this as a controlled approximation, compare it with a charge-based reference solver, and retain the simpler model only when measured gestures cannot distinguish it within the declared error budget. No perceptual threshold is claimed here.

MODEL EQUATIONS

$$V_r / V_{in} = (Y_L H_L + Y_S H_S) / [Y_L + Y_S + Y_g + (1 + A)Y_f]$$

$$H_{expr}(s,u) = V_o / V_{in} = -A(s) (V_r / V_{in})$$

$$i(t) = d[C(t)v(t)]/dt = C(t)\, dv(t)/dt + v(t)\, dC(t)/dt$$

CHECKS AND DECISIONS

- Compare the surrogate against the coupled circuit at endpoints and several interior positions.
- Do not apply a separate expression EQ after a netlist that already contains the expression network.

ORIGINAL IMPLEMENTATION CONTRACT

Implement the control without changing the instrument

Expression mechanism and response

Expose ExpressionModel.process(input, position, profile) at the documented preamp boundary. The profile carries capacitance/transfer data, provenance, revision and fit residuals. Keep calibration loading, coefficient generation and file access outside the audio callback. The runtime receives immutable prepared data and advances its existing filter state; expression movement must not reset generator phase, percussion state or scanner phase.

Original controller proposal: choose assignable CC11 as a default, map the connected pedal’s calibrated minimum and maximum to u in [0,1], and offer inversion before the instrument curve. Higher-resolution host automation should reach the same u coordinate. Resolve simultaneous GUI, MIDI and host updates with the central timestamp/ownership policy, then reconstruct one continuous trajectory. Do not smooth each input source independently and sum conflicting positions.

For a fitted-filter implementation, interpolate physical parameters or use a specifically verified stable realization. Smooth parameter motion sufficiently to control unintended switching artifacts, but measure the added gesture latency. A moving control can create modulation sidebands even in a linear audio network; nonlinear downstream stages add further aliasing risk. Test the complete oversampled path, keeping controller time constants expressed in seconds across sample-rate changes.

State	Persistence policy
Controller calibration	Device/profile setting, versioned
Expression position	Host parameter; explicit preset-recall policy
Filter history	Continuous during gestures; defined reset on prepare
Specimen profile	Content hash, units and source identifiers

```
onEvent(event):
  target = controllerMap(event.value, calibration)
  trajectory.schedule(event.sampleOffset, target)
for each audio sample:
  u = trajectory.nextPosition()
  p = preparedProfile.evaluate(u)
  y = expressionState.step(input, p)
  output = downstreamPreamp.step(y)
```

ORIGINAL ACQUISITION PROTOCOL

Measure position, spectrum and downstream drive

Expression mechanism and response

Create a capture manifest before collecting audio: instrument serial, AO-28 revision, component changes, supply condition, warm-up interval, output load, interface calibration and pedal-angle reference. Use a qualified technician for internal measurements. A recording of a Leslie alone combines expression, amplifier, crossover, radiation and room response; it cannot identify the expression network uniquely.

Acquire repeatable stationary positions spanning the complete travel, including closely spaced endpoint samples. Use identical tone material and a fixed acquisition gain. Record both absolute output level and spectral ratios relative to a chosen reference position. Establish the noise floor before interpreting a nearly closed position: a ratio dominated by hum is not a measured bass shelf. Phase-sensitive measurements need coherent excitation or a reference channel; unrelated tonewheel takes cannot provide reliable phase ratios.

Then capture matched slow and quick heel-to-toe gestures in both directions, with a synchronized position trace. Repeat at several input levels and registrations. A level-dependent difference may belong to subsequent amplification rather than the passive overlap itself. Fit the stationary transfer first, the mechanical mapping second, and any nonlinear interaction last; retain withheld gestures for evaluation.

Capture	Purpose
Stationary frequency/position grid	Magnitude, phase and endpoint residual coupling
Repeated angle sweeps	Linkage repeatability, play and timing
Several signal levels	Separate small-signal response from downstream drive
Silence at each position	Noise/hum dependency and measurement floor
Withheld registrations/gestures	Detect overfitting to one chord or controller

CHECKS AND DECISIONS

- Save raw voltage/audio calibration alongside normalized plots.
- Publish confidence limits and excluded noise-floor bins with every fitted curve.

ORIGINAL ACCEPTANCE AND HANDOFF

Acceptance gates for a convincing swell response

Expression mechanism and response

The initial deliverable is a bounded provisional profile, not a claim of vintage accuracy. Release the calibrated profile only with source identity, raw captures, fitting script, stationary response plots and withheld-gesture comparisons. Set numerical tolerances from repeatability and project listening goals before evaluating the model; do not select thresholds afterward to hide the residuals.

Test expression at the preamp boundary and at final output separately. A downstream master-gain change should leave internal distortion measurements unchanged, while an expression change may alter them. Verify that upper, lower, pedals and percussion follow the intended common path without double attenuation. Treat any independently added reverb or external-input bypass as an explicitly routed feature, not an accidental consequence of where the gain multiply was placed.

Document unresolved parameters individually: rotor geometry, capacitance against angle, specimen end-stop range, parasitic loading and fixed-feedback setting. A photoreal pedal animation cannot supply those values. The next agent should integrate this stateful block with chapters 31–38, then run the dry-organ validation before adding Leslie acoustics. UI chapter 70 can expose a useful position meter while retaining a separate final-output meter.

Acceptance fixture	Required observation
Stationary positions	Response agrees with the declared reference and uncertainty
Rapid automation and reversal	No state reset, NaN, unstable burst or unexplained discontinuity
Sample-rate/block changes	Same time trajectory within the specified timing budget
Repeated offline render	Identical profile and event stream reproduce output
Endpoint controller calibration	Hardware travel reaches intended limits without clipping early
Profile reload	Versioned data restored; missing measurements remain marked unknown

HARDWARE EVIDENCE / ORIGINAL ARCHITECTURE

One instrument state, several input adapters

Console control state machine and MIDI semantics

The B-3 has independent manual preset routing, two drawbar groups per manual, and continuous-contact drawbars. C cancels and returns without electrical contacts; two simultaneously pressed preset keys can latch. Upper B provides percussion eligibility. [S024-01]

ORIGINAL DESIGN PROPOSAL: separate persistent control positions, physical key/contact states, and evolving DSP state. Inputs request changes; the audio engine owns the applied state. Neither a GUI redraw nor a host program selection may rebuild the generator. A manual is a routing destination, not an independent oscillator synthesizer. Use numeric pitches and explicit physical-key indices so differing octave-name conventions cannot alter wiring.

Provide one canonical command format to VST3, standalone MIDI, keyboard-screen gestures and replay fixtures. Keep the input origin in each command. A held note retains its original manual destination when channel routing changes, so its later release cannot become stranded on a different manual. The interface returns an applied sequence number and read-only display snapshot. A control display can indicate a pending gesture until the engine acknowledges it.

State domain	Proposed sole owner / contents
Console controls	Audio control layer: four nine-drawbar banks, pedal pair, selectors, tablets, expression.
Physical playing keys	Ownership/contact layer: separate upper, lower and pedal key arrays.
Electrical and mechanical history	DSP modules: source phases, scanner state, percussion charge, filters, rotor angles/speeds.
Host and GUI	Adapters and views: mappings, presentation and timestamped requests; no direct DSP mutation.

SOURCES S024-01 Hammond Organ Service Manual: early consoles and B-3

IMPLEMENTATION CONTRACT

Resolve note ownership before making contact gestures

Console control state machine and MIDI semantics

VST3 provides event-bus identity and sample offsets; note data include channel, pitch and an optional noteId, with -1 indicating absence. [S024-02] [S024-03] ORIGINAL POLICY: record source, bus, channel, note identity or fallback serial, destination manual, physical key and press sequence. Match releases against this record before consulting the current input mapping.

For events without identities, use a documented FIFO ownership policy per source/channel/pitch. Multiple owners can hold one physical key; only transitions between zero and nonzero owners request physical depression or release. A second owner does not add another copy of the wheel audio or force a percussion attack. An optional repeated-note re-strike mode must synthesize an explicit release interval; label that as a performance extension.

Preserve note-list ordering at equal timestamps. Reject out-of-range playing keys instead of silently folding their MIDI pitches. Pitch foldback inside a valid key is chapter 015’s harmonic-routing operation. Keep velocity available to the contact-gesture model; do not turn it into an undocumented sustained-volume law. The standalone raw-MIDI adapter and the VST3 event adapter must each implement their own input semantics: never assume every normalized zero-velocity VST3 NoteOn is raw MIDI.

Input situation	Original deterministic response
Two controllers hold upper middle C	Maintain two owners and one depressed virtual key.
One controller releases it	Remove that owner; key remains down.
Mapping changes while held	Future presses use new mapping; existing releases use saved destinations.
Unmatched release	Ignore safely, count diagnostic; do not release another owner.
Owner capacity exhausted	Apply documented bounded recovery; never allocate in the audio callback.

P025 | Half-Moon Switch.jpg | CJ Sorg from Hendersonville, United States | CC BY-SA 2.0 Source and license

SOURCES S024-02 VST3 Event structure / S024-03 VST3 NoteOnEvent structure

AUTOMATION / SWITCHING

Apply routing changes on the same timeline as notes

Console control state machine and MIDI semantics

VST3 parameter queues describe piecewise-linear automation, including implicit prior values and within-block points. Its MIDI-mapping interface asks the host to deliver mapped controllers as ordinary parameters. [S024-04] [S024-05] ORIGINAL IMPLEMENTATION: reconstruct continuous control travel per sample; resolve discrete selectors with a documented step policy. Do not read only the last point of each block.

Keep physical drawbar travel normalized separately from its electrical gain law. Maintain all four banks when hidden. A#/B selection changes connectivity without copying positions. Cancel empties the selected manual’s eleven-bit latch mask while leaving ownership and generator phases intact. The optional simultaneous-latch mode passes branches to the loading model, not to a sum of independent presets. Upper-B changes also update percussion routing without resetting charge.

Define an explicit tie rule because note events and automation arrive in separate host containers: apply discrete routing changes first at an equal sample offset, then preserve note-event order; scheduled contact changes use stable insertion order. This is a product policy, not a claim about sub-sample hardware timing. Never advance a recharge integrator for a zero-duration off/on pair. Treat noise suppression as an optional output treatment, not hidden modification of contact history.

```
for t in merged_event_boundaries:
  render_and_advance(cursor, t)
  apply_continuous_curve_values(t)
  apply_discrete_routes_in_stable_order(t)
  apply_note_ownership_events_in_order(t)
  schedule_physical_contact_gestures(t)
  apply_due_contact_transitions(t)
  resolve_manual_and_percussion_networks()
  publish_applied_sequence_if_due()
  cursor = t
render_and_advance(cursor, block_end)
```

CHECKS AND DECISIONS

- An empty automation queue retains the previous value.
- Changing a drawbar while holding a key changes that routed bus without a note retrigger.

SOURCES S024-04 VST3 IParamValueQueue / S024-05 VST3 IMidiMapping / S024-01 Hammond Organ Service Manual: early consoles and B-3

ORIGINAL STATE MACHINES

Separate virtual power, rotor command and host transport

Console control state machine and MIDI semantics

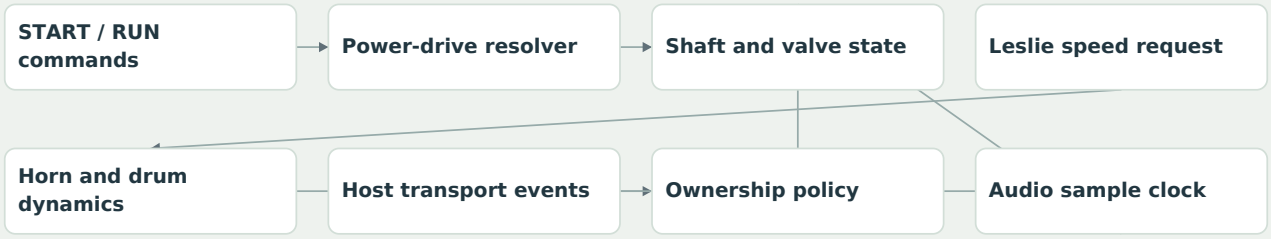
The factory starting sequence uses a momentary START action followed by RUN; releasing START disengages its motor after the run motor takes the load. [S024-01] ORIGINAL MODEL PROPOSAL: keep switch positions separate from mechanical angular speed and amplifier readiness. A click on RUN is not an instantaneous phase reset or a measured warm-up curve.

Offer two named policies: an instant-ready performance initialization and a mechanical-start simulation. The latter needs measured acceleration, synchronization and warm-up behavior; an arbitrary cinematic ramp must remain labeled illustrative. Both use the same steady-state engine. Model shutdown as removal of drive followed by the chosen mechanical/electrical evolution, rather than clearing every state variable immediately.

Leslie speed controls request targets; horn and drum modules retain their own actual speeds and angles. Supported commands come from the selected cabinet/control profile. Do not infer a stop position solely from a modern three-position user interface. Host play/stop and tempo changes are independent of these virtual switches.

Use processed audio time for deterministic rendering. If the host suspends processing, the default proposed policy freezes the model until processing resumes; an optional wall-clock continuation mode cannot promise identical offline replay. Disclose this distinction. A transport seek may clear host note ownership under a documented policy while leaving source phase and rotary motion continuous.

Orthogonal controls and continuously evolving state



Original control architecture. There is no inferred numerical startup or rotor time constant; chapter 044 supplies the rotary dynamics contract.

SOURCES S024-01 Hammond Organ Service Manual: early consoles and B-3

SERIALIZATION / REAL-TIME DESIGN

Snapshots preserve controls; checkpoints preserve history

Console control state machine and MIDI semantics

ORIGINAL DATA CONTRACT: a user patch stores configuration and control positions; a deterministic replay checkpoint additionally stores dynamic history. These are distinct files or distinctly tagged record types. A patch must not accidentally resurrect notes that were held when it was saved. Applying a registration during performance should preserve current ownership under an explicit live-load policy.

Version the schema, parameter identifiers and physical profile separately. Store all drawbar banks, latch masks, preset wiring profile, scanner/percussion controls, expression calibration, cabinet identity and input mapping. Preserve unknown future fields only where the migration policy supports them. Validate bounds and references before publishing an immutable prepared configuration to the audio thread.

A replay checkpoint also needs sample clock, owner records, pending contact events, random-generator state, generator phases, scanner reactive state, percussion circuit state, nonlinear solver memory and rotary delay buffers. Bitwise continuation is a proposed guarantee only for compatible engine version, sample format, execution mode and sample rate. Otherwise report a migrated approximation.

Compile profiles, allocate buffers and resolve external resources away from audio processing. Transfer a ready configuration through a bounded message or pointer-exchange mechanism with safe reclamation. A visual snapshot reports the actual applied state; it is not allowed to become a second control authority.

Record	Required distinction
Patch schemaVersion / profileHash	Identifies format and the exact calibration/reference profile.
Control state	All visible and inactive banks; named selector values rather than label text.
Runtime checkpoint	Complete time-dependent state plus compatibility manifest.
Replay event log	Absolute sample time, origin, stable order, payload and application outcome.
Migration report	Changed defaults, missing profile fields and unsupported behavioral extensions.

CHECKS AND DECISIONS

- Loading the same prepared patch twice leaves held keys and source phases unchanged under live-load policy.

ACCEPTANCE / EVIDENCE GATES

Test panic behavior and event ordering as audible features

Console control state machine and MIDI semantics

MIDI distinguishes All Sound Off120, Reset All Controllers121 and All Notes Off123; the published overview is not the full implementation specification. [S024-06] [S024-07] ORIGINAL POLICY: release addressed ownership for All Notes Off, immediately suppress addressed note contributions for All Sound Off, and reset only the declared controller defaults for Reset All Controllers. Keep shared tonewheel phase advancing. A global emergency Panic is a separate action that can silence the complete output and clear shared tails.

A shared nonlinear amplifier or reverberator cannot necessarily attribute its residual state to one channel. Document the channel-mode tail policy and test that muting one manual does not silence another. A sustain extension is disabled in the vintage profile; if enabled, define its release behavior separately. Do not confuse that extension with physical percussion recharge.

Validate both adapters. VST3 hosts map controllers through parameter delivery, so test repeated channel-mode actions rather than assuming identical zero-valued messages always survive host coalescing. Provide a reliable independent user Panic command. Deliver machine-readable traces and replay fixtures; these verify software contracts, not measured historical fidelity.

Proposed regression fixture	Required observation
Same event trace, varied block sizes	Equal control/contact histories and audio within declared numerical tolerance.
B to fixed preset to B while held	No automatic envelope reset; circuit topology and charge history remain observable.
Same-sample release and press	Stable order; no invented recharge time.
Upper/lower same pitch; upper All Sound Off	Lower ownership and its source remain intact.
Queue overflow, malformed state, lost release	Bounded recovery and diagnostics; no callback allocation or stuck ownership.
Checkpoint and continuation	Compatible replay reproduces state; incompatible restore reports its limitation.

SOURCES S024-06 MIDI Association: Summary of MIDI1.0 Messages / S024-07 MIDI Association: Control Change Messages / S024-05 VST3 IMidiMapping

HARDWARE EVIDENCE AND NOMINAL VALUES

Freeze the line revision before choosing a delay algorithm

Scanner vibrato delay line circuit

The factory manual distinguishes several vibrato circuits and warns that their components are not generally interchangeable. Its production drawing is the revision anchor; a historical patent remains separate mechanism evidence. [S025-01] [S025-03] The 2016 paper explicitly studies a late-B-3 circuit. Its figures and component table contain18 inductors,18 capacitors and 19 tap voltages; its Section 2.1 sentence claiming nineteen LC stages is inconsistent with those figures. [S025-02]

The following values were visually checked in the paper's Table 1. They are documented nominal model values, not measurements of a selected organ. Use a paper-specific symbol namespace: these labels must not silently become factory assembly callouts or AO-28 schematic designators. Convert units once during profile import and retain the printed representation alongside the SI value.

Published model element	Documented nominal	SI interpretation
L1–L18	500 mH each	0.5H
C1–C17	0.004 µF each	4e-9F
C18	0.001 µF	1e-9F
R1+ / R1−	27 /68 kΩ	27000 /68000Ω
R2+ / R2−	56 kΩ /0.15 MΩ	56000 /150000Ω
R3+ / R3−	39 kΩ /0.15 MΩ	39000 /150000Ω
R4+ / R4−	33 kΩ /0.18 MΩ	33000 /180000Ω
R5+ / R5−	18 kΩ /0.18 MΩ	18000 /180000Ω
R6+ / R6−	12 kΩ /0.18 MΩ	12000 /180000Ω
Rc / Rt	22 /15 kΩ	22000 /15000Ω

CHECKS AND DECISIONS

- Reject a profile that turns0.004 µF into0.004F or500 mH into500H.
- Do not promote nominal model values to specimen-measured status.

SOURCES S025-01 Hammond factory service manual: vibrato system / S025-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus / S025-03 US2560568A: Vibrato apparatus

TOPOLOGY AND INTERFACES

Preserve the loaded network and its common return

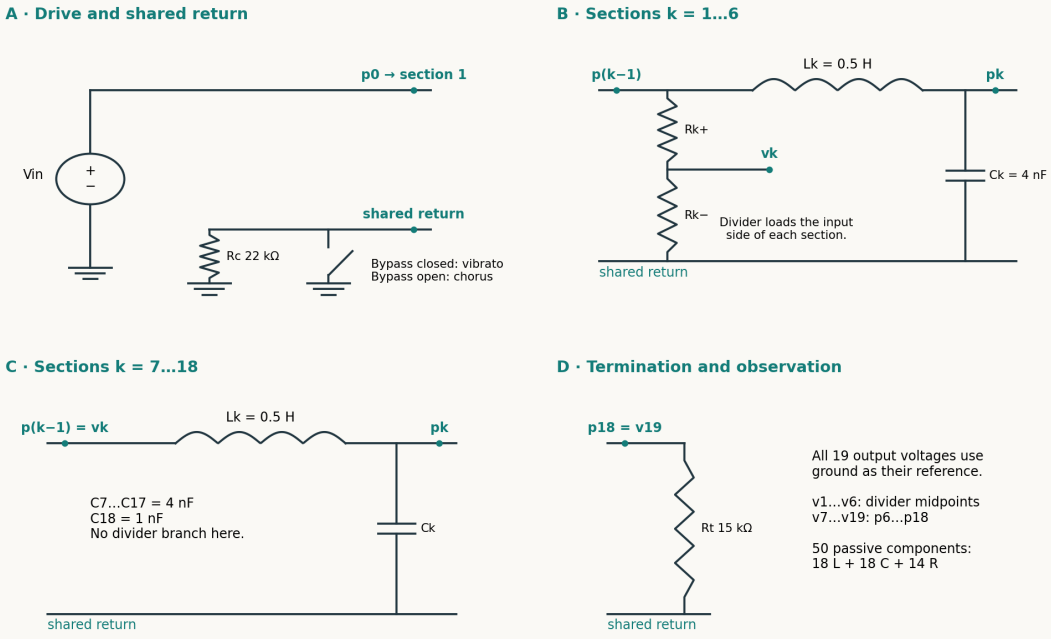
Scanner vibrato delay line circuit

The factory chorus arrangement includes a resistor in the line's return path that is shorted for vibrato. The published model also includes compensated divider outputs for the first six taps and a terminating resistance. [S025-01] [S025-02] These are connected circuit elements: their loading cannot be reproduced merely by assigning different numerical delay lengths to taps.

ORIGINAL DESIGN PROPOSAL: create an explicit circuit graph with named source reference, common line return, section junctions, divider junctions and termination. Distinguish the first six divider outputs from the unattenuated junctions that follow. Store the 19 observable voltages relative to the receiver's reference; do not accidentally export a voltage measured relative to the moving common return.

Treat scanner coupling as a selectable boundary model. A first implementation may assume high impedance and consume the tap vector without feeding current back. A later coupled model can accept angle-dependent loading. These two interfaces represent different approximations and must have different profile identifiers. Keep the AO-28 driving and receiving impedances outside the nominal paper circuit until the complete electrical boundary has been traced. The companion includes both the complete 50-component JSON graph and an original full-network drawing with every passive component labeled.

Scanner ladder: original circuit decomposition



Topology and nominal values: DAFx 2016, Figure 1 / Table 1. This is the published idealized late-B3 model, not a measured specimen.
Every section shares the same return; pk feeds the next section. Coils are ideal here. Connected AO-28 and scanner loading require separate evidence.

Original circuit decomposition of the DAFx2016 published model; see the companion graph for all component edges and nominal values. Not a measured specimen circuit.

CHECKS AND DECISIONS

- Export tap reference polarity with every offline trace.
- Verify changing termination changes upstream responses as the circuit predicts.

SOURCES S025-01 Hammond factory service manual: vibrato system / S025-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

ORIGINAL DERIVATION

Use a lossless section only as a numerical sanity check

Scanner vibrato delay line circuit

The following calculation is an original diagnostic for an ideal uniform LC ladder. It intentionally omits divider loading, end conditions, coil resistance, stray coupling and the final smaller capacitor. With the nominal 0.5H and 4 nF values, its low-frequency section delay is44.72µs, characteristic impedance approaches11.18 kΩ, and uniform-lattice propagation band ends near7.12 kHz. These are derived checks, not measured B-3 bandwidth or group delay.

The test is useful because an incorrect capacitance prefix moves the implied frequency scale drastically. It is not a substitute for solving the loaded finite network. In particular, group delay is the derivative of phase, so a model may match magnitude reasonably while scanning the wrong time trajectory. Compare complex transfer functions at every observable tap.

ORIGINAL DESIGN PROPOSAL: use18 inductor currents and 18 capacitor voltages as an initial physical-state accounting exercise. The 19 observed tap voltages are outputs, not19 independent delay buffers. A chosen numerical formulation may use additional algebraic unknowns, but it must preserve the same energy-bearing components and boundary conditions.

MODEL EQUATIONS

Ideal uniform section: tau0 = sqrt(L*C); Z0_low = sqrt(L/C).

Uniform-lattice dispersion: cos(theta) = 1 - omega^2*L*C/2; f_edge = 1/(pi*sqrt(L*C)).

For each actual tap k: H_k(j*omega)=V_k/V_in; group_delay_k=-d(arg(H_k))/d(omega).

Stored ideal energy: E = sum(0.5*L_i*i_i^2) + sum(0.5*C_i*v_i^2), in joules.

CHECKS AND DECISIONS

• Plot amplitude, unwrapped phase and group delay together.

• Use energy decay with drive removed to expose sign errors; account for supplied energy when boundaries change.

SOURCES S025-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

ORIGINAL IMPLEMENTATION PLAN

Build one circuit description and two independent solvers

Scanner vibrato delay line circuit

Create one canonical graph containing element terminals, units, nominal or measured value, uncertainty, source locator and revision. Generate an offline frequency-domain nodal solve from it. Separately implement the real-time formulation described in chapters 028 or029. Their agreement should be assessed on complex tap responses, not only the final musical output, which can hide compensating errors.

Prepare matrices or wave-port coefficients outside the audio callback. The process interface consumes calibrated input samples, persistent circuit state and the current return-switch condition; it produces a fixed-size tap vector. Switching V1/V2/V3 changes selection downstream, while a vibrato/chorus change also changes the electrical return condition. Preserve capacitor voltages and inductor currents where the formulation permits; do not erase stored energy just because the player changes a mode.

Add measured losses only after nominal parity is demonstrated. A coil's series resistance is not interchangeable with arbitrary output damping. Keep optional stray or mutual-coupling elements named and evidence-backed. Coefficient rebuilding, profile parsing and memory allocation belong to preparation; sample processing should have bounded work and no file access.

Deliverable	Required content
LineProfile	Graph identity, SI values, source records, unresolved boundary parameters
LineState	Persistent reactive states and current electrical mode
TapFrame	19 voltages with fixed ordering, units and reference polarity
Offline oracle	Complex response at every tap and documented source/load conditions
Solver report	Sample rate, discretization, numerical error and proposed tolerances

CHECKS AND DECISIONS

- A block-size change preserves the same state trajectory within stated numerical tolerance.
- A frozen scanner position does not change the line's electrical mode.

SOURCES S025-01 Hammond factory service manual: vibrato system / S025-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

ORIGINAL CAPTURE AND CALIBRATION PLAN

Measure the line before fitting the rotating output

Scanner vibrato delay line circuit

Request a qualified instrument technician's measurement campaign that identifies the exact line assembly and its connected AO-28 revision. Obtain an underside photograph, a terminal-to-schematic crosswalk and a record of component changes. A photograph alone cannot establish inductance, parasitic loss or the loaded frequency response. Internal acquisition requires suitable isolated instrumentation and qualified handling; this chapter specifies evidence outputs rather than live servicing steps.

Capture complex transfer from the agreed drive reference to each accessible tap under both return-switch conditions. Record probe impedance and the driving and receiving boundaries. Compare repeated measurements before fitting tolerances: a probe that changes the response can otherwise be mistaken for authentic component spread. Retain raw channels and calibration data so phase can be reconstructed rather than inferred from magnitude.

Fit the smallest identifiable parameter set first: coherent global LC scaling, then documented load boundaries, then supported section deviations. Reserve some frequencies and switch states for validation. If several parameter combinations explain the observations equally well, store that uncertainty rather than selecting an apparently precise component value. Follow with rotor-running recordings only after the stationary line model is understood.

Acquisition	Question it resolves
Assembly identity and wiring photographs	Which production circuit and repair history are present?
Per-tap complex transfer	Where do dispersion, attenuation and reflections originate?
Probe/loading comparison	Is the measurement altering the network?
Component measurements with conditions	Which losses and tolerances are supported?
Held-out switch-state recordings	Does one parameter set explain both boundary conditions?

CHECKS AND DECISIONS

- No measurement fields are populated from the illustrative calculations on the preceding page.
- Report instrument repeatability before setting a model-error acceptance threshold.

SOURCES S025-01 Hammond factory service manual: vibrato system

ACCEPTANCE AND UNRESOLVED PARAMETERS

Exit gates for a trustworthy scanner-line module

Scanner vibrato delay line circuit

ORIGINAL ACCEPTANCE PLAN: first establish documentary consistency, then numerical equivalence, then physical agreement. A correct implementation of a published nominal circuit is a valuable milestone, but does not establish a measured replica of a particular line box. Keep the solver implementation version separate from the physical calibration profile so corrections to one do not silently rewrite the other.

The passive linear line should not generate new harmonics for fixed boundaries and steady sinusoidal excitation. Time-varying scanner mixing and mode changes introduce different spectral questions; assess those in the complete scanner tests. Discretization can still misplace resonances and phase even without nonlinear aliasing. Compare across sample rates and, if prewarping is used, state which feature is prioritized and which residual errors remain.

Unresolved coil losses, coupling, tap loading and source impedance remain explicit profile fields. The practical exit deliverable is a reproducible tap-response report, a stable real-time module and a list of measurements still required. Musical similarity alone cannot close those electrical questions, and no pass status is asserted by this research chapter.

Gate	Falsifiable condition
Documentary	18L/18C/19 outputs and correct units; revision and source preserved
Network	Independent nodal and real-time solutions agree within a declared numerical budget
State	Mode changes do not cause unexplained energy creation or reset artifacts
Rate	Complex response errors are reported at every supported sample rate
Boundary	Source/load changes match the selected graph, including common-return reference
Physical	Held-out hardware captures meet thresholds chosen from actual repeatability

CHECKS AND DECISIONS

- Resolve the paper's nineteen-stage prose inconsistency by its explicit figures, table and partition, not by creating a nineteenth LC section.
- Leave physical acceptance pending until reference captures exist.

SOURCES S025-01 Hammond factory service manual: vibrato system / S025-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

HARDWARE EVIDENCE

Count the physical plates separately from the circuit nodes

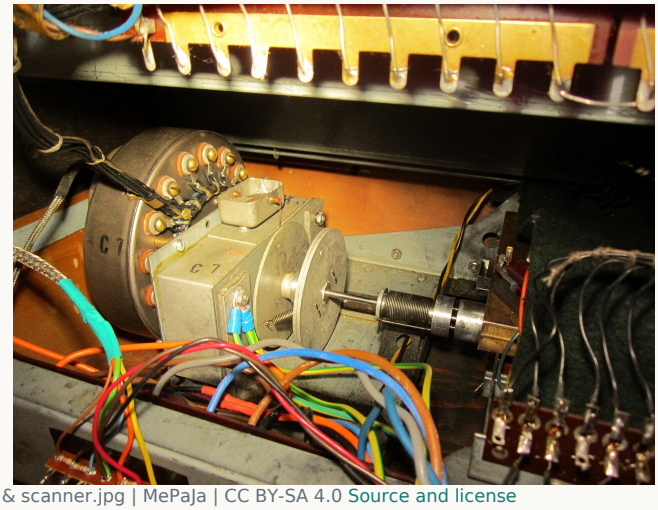
Scanner capacitive rotor mechanism

The factory scanner description specifies16 stationary plate groups and a nominal 412 revolutions per minute, driven from the organ running motor. Its output uses two side brushes and one end brush. Figure 5-6 is the most useful factory construction photograph in this source. These are documented descriptions, not measurements of the reference specimen. [S026-01]

The late-B3 research model connects those16 positions to9 scanner terminals. Its ladder drawing contains 18 LC sections and 19 node voltages; one sentence in Section 2.1 instead says19 stages, conflicting with the drawing, component table and subsequent partition. Use the explicitly enumerated circuit, and record this discrepancy. [S026-02]

ORIGINAL DESIGN PROPOSAL: put physical positions, terminal connections and ladder-node selection in three separate arrays. Their different lengths are intentional. Do not infer wiring from the visual position of a plate or from the fact that a software array happens to have19 entries. Store a specimen identifier, source locator and confidence with each mapping.

Object	Evidence-scoped count or rate	Implementation meaning
Stator positions	16 documented	Circular mechanical index
Scanner terminal voltages	9 in late-B3 paper	Repeated around the circumference
Ladder outputs	19 in selected research topology	Not19 rotating contacts
Factory nominal speed	412 rpm; derived6.8667 Hz	Provisional source nominal, not precision calibration
Patent line embodiment	24 sections plus termination	Separate historical profile [S026-03]



P009 | Hammond synchronous motor & scanner.jpg | MePaJa | CC BY-SA 4.0 Source and license

CHECKS AND DECISIONS

- Reject a profile that conflates physical stator index with ladder-node index.
- Keep nominal speed uncertainty; do not round the documented rate to an unexplained6 Hz.

SOURCES S026-01 Hammond factory service manual: scanner theory and construction / S026-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus / S026-03 US2560568A: Vibrato apparatus, Laurens Hammond and John M. Hanert

MECHANISM AND CONSTRUCTION

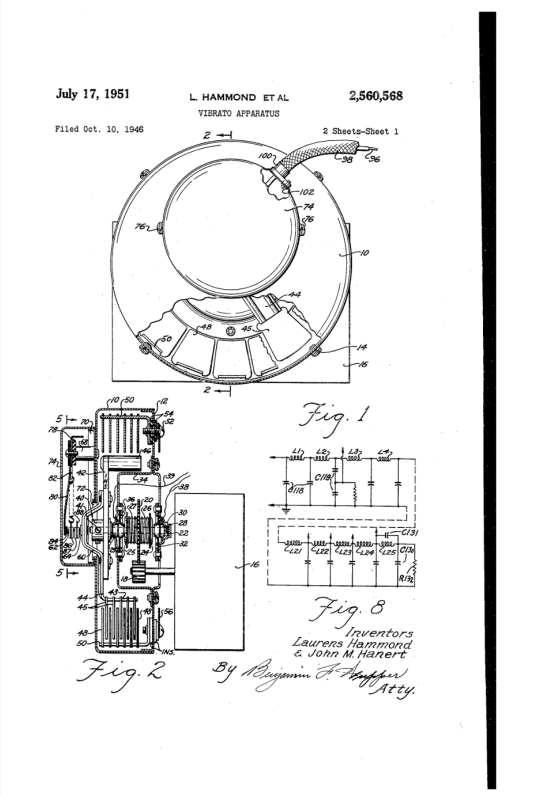
Follow the signal through a capacitive pickup

Scanner capacitive rotor mechanism

Stationary and rotating plates interleave without becoming a sequence of metallic tap switches. The patent describes an insulated, counterbalanced rotor arm, stator mounting hardware and a separate output pin/brush system. Its dimensions and material examples describe that patented embodiment; they do not establish every later production assembly. [S026-03]

The important modeling distinction is between capacitive transfer at the plates and conductive collection at the output brushes. Rotating overlap changes the coupling from neighboring stators. A brush defect would interrupt or contaminate the already combined signal. Treating each stator crossing as a contact click puts the disturbance in the wrong place and manufactures switching artifacts.

ORIGINAL DESIGN PROPOSAL: the physical inspection model should expose rotor angle, plate groups, insulating supports, output pin and brush locations as separate objects. Its drawing needs an explicit schematic scale unless acquired dimensions exist. Fringing fields, finite gaps, parasitic capacitance and receiver impedance remain electrical parameters; photorealistic plate geometry alone does not identify them.



Historical patent plate US2560568A_p01; invention embodiment, not a verified production drawing for the selected organ or cabinet. Source and license

CHECKS AND DECISIONS

• No periodic hard switch appears at ordinary stator transitions in the nominal model.

• Optional brush impairment acts after the capacitive combination, with independently supported parameters.

SOURCES S026-01 Hammond factory service manual: scanner theory and construction / S026-03 US2560568A: Vibrato apparatus, Laurens Hammond and John M. Hanert

ORIGINAL MODEL AND DERIVATION

Choose an overlap model with an explicit loading assumption

Scanner capacitive rotor mechanism

The DAFx scanner model assumes two significant neighboring capacitances, negligible loading of the ladder and an approximately constant sum of overlap capacitances. This yields a linear crossfade, with scanner terminals visited from 1 to9 and back. It is a deliberately simplified physical approximation, not a measured capacitance map. [S026-02]

ORIGINAL DESIGN PROPOSAL: implement that approximation first, then retain a charge-based extension. Let Cj(theta) connect stator voltage vj to rotor voltage vo; Cg represents rotor capacitance to ground and Gload a resistive receiver approximation. Capacitances are farads, voltages volts, charge coulombs and Gload siemens. The charge equation below follows directly from capacitor charge and current conservation. It includes changing capacitance through differentiation; replacing d(Cv)/dt with C dv/dt discards a term.

In the floating, zero-charge, zero-parasitic limit, the equation reduces to normalized capacitance weights. With loading or stray capacitance, arbitrary normalization can erase real angle-dependent gain. Use measured coupling profiles when available; preserve the simpler model as a named approximation so calibration results remain interpretable.

Parameter	Status in this edition
Cj versus angle	Unknown; acquire capacitance or transfer measurements
Plate gaps and fringing	Unknown; no uncalibrated photo-derived numbers
Output loading	Must include the identified recovery-stage input
Normalized triangular weights	Proposed baseline approximation, not hardware measurement

MODEL EQUATIONS

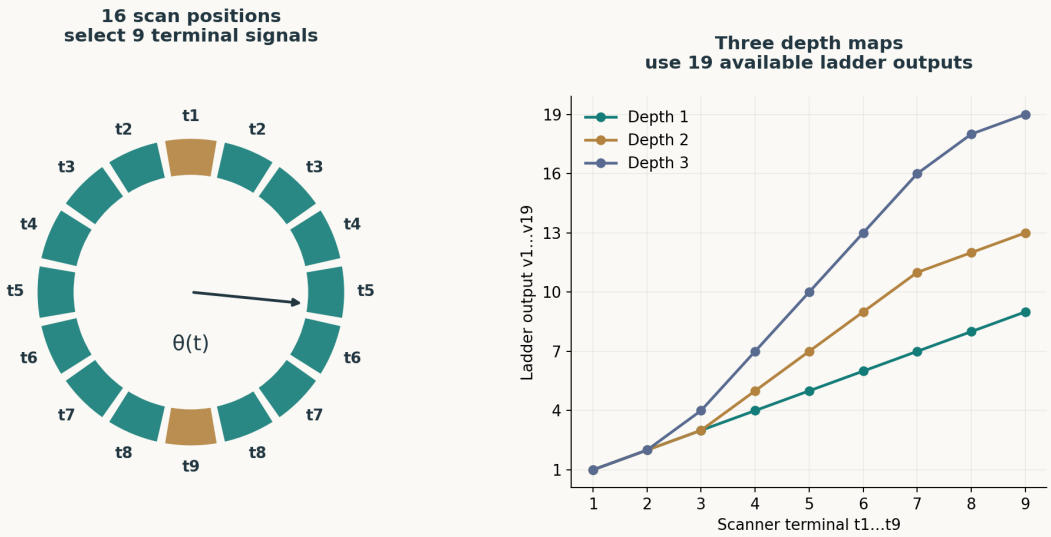
qout = sum_j Cj(theta)*(vo-vj) + Cg*vo

dqout/dt + Gload*vo = 0 [proposed lumped receiver model]

vo = sum_j Cj*vj / sum_j Cj [zero-charge, unloaded limiting case]

Cj(theta) approximately epsilon*Aoverlap,j(theta)/gap + Cfringe,j(theta) [geometric initialization only]

Keep physical positions, terminals and circuit nodes separate



Left: proposed uniform-angle indexing convention; it is not a mechanical drawing or measured capacitance curve.
Right: original plot of the published DAFx2016 Table 2 mapping. V/C at the same depth share the map but change the common-return circuit.

Original explanatory calculation. Illustrative parameters are not measured instrument data.

SOURCES S026-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

IMPLEMENTATION CONTRACT

Implement a continuously rotating, bounded interpolator

Scanner capacitive rotor mechanism

ORIGINAL DESIGN PROPOSAL: define zero angle at the center of terminal1 and a positive direction around the stator ring. Use the 16-entry terminal order below; each depth mode separately supplies its nine selected ladder-node voltages. This coordinate convention is an implementation choice. Reverse direction or rotate the initial angle when comparing to a measured reference, without changing the electrical mapping.

Advance rotor phase at the internal audio rate, including when no notes sound or when a manual bypasses vibrato. Mode changes update routing at their scheduled sample without resetting angle. Allocate arrays during preparation; the audio callback performs bounded arithmetic and reads an immutable profile. The running-motor model should eventually supply angular velocity with the verified regional gear ratio.

The crossfade is continuous but its slope changes at plate centers. Evaluate modulation sidebands against an offline high-rate reference before selecting an internal rate. Do not add arbitrary smoothing or substitute allpass interpolation solely to remove coloration: those changes alter the model. Preserve phase across sample-rate changes, and distinguish angular state from a musical transport position.

```
order = [0,1,2,3,4,5,6,7,8,7,6,5,4,3,2]
for each internal audio sample:
    nodes = ladder.process(input)
    terminals = modeMap.select(nodes)
    u = 16 * phaseTurns
    k = floor(u)
    fraction = u - k
    a = terminals[order[k]]
    b = terminals[order[(k+1) % 16]]
    output = (1-fraction)*a + fraction*b
    phaseTurns = wrap01(phaseTurns + rotorHz/internalFs)
```

CHECKS AND DECISIONS

- Same event stream and initial phase produce equivalent results across block partitions.
- At the wrap boundary, terminal2 blends back to terminal1; no spurious jump to terminal9 occurs.
- Any frozen-scanner diagnostic mode is explicitly separated from physical running behavior.

SOURCES S026-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

MEASUREMENT AND IDENTIFICATION

Measure overlap independently of the ladder response

Scanner capacitive rotor mechanism

ORIGINAL MEASUREMENT PLAN: identify angular transfer separately from the ladder before fitting a combined vibrato recording. Otherwise, an error in ladder phase can be hidden by a distorted overlap curve. Record the exact scanner, running motor, mains region, preamp revision and measurement-chain loading. Measurements at internal circuitry belong to a qualified technician; this plan defines required evidence rather than a live servicing procedure.

Acquire synchronized rotor-angle and output observations over repeated revolutions. Where independent stator excitation is practicable under a professionally prepared test setup, estimate each transfer function as a function of angle and frequency. Otherwise jointly fit a constrained model from known ladder-node signals, explicitly reporting the weaker identifiability. Preserve raw traces, phase reference and calibration uncertainty.

Do not demand perfect cyclic symmetry from a worn specimen. Estimate deterministic angular variation first, then characterize nonrepeatable residue separately. A single audio sine wave cannot uniquely determine absolute plate capacitance, input impedance and gain; impedance or independent loading information is needed to resolve those ambiguities.

Acquisition	Parameter it constrains	Required record
Rotor period and angle reference	Mean speed, jitter, phase	Timebase calibration and region
Angle-resolved transfer	Overlap shape and neighbor leakage	Frequency, amplitude, receiver loading
Independent loading evidence	Absolute coupling versus divider ratio	Probe and circuit impedance
Repeated revolutions	Repeatable asymmetry versus noise	Unaveraged synchronized channels
Photographs with scale	Plate count and geometric constraints	Camera pose, scale, specimen ID

CHECKS AND DECISIONS

- Reserve complete revolutions and unseen test frequencies for validation.
- Fit uncertainty intervals; do not replace an unidentifiable capacitance with a falsely precise number.

SOURCES S026-01 Hammond factory service manual: scanner theory and construction / S026-03 US2560568A: Vibrato apparatus, Laurens Hammond and John M. Hanert

ACCEPTANCE AND OPEN QUESTIONS

Acceptance gates for the scanner mechanism

Scanner capacitive rotor mechanism

ORIGINAL ACCEPTANCE PROPOSAL: begin with synthetic invariants before comparing hardware. For the unloaded normalized model, equal terminal voltages must produce that same output at every angle. Exciting one terminal reveals its expected angular support; endpoints appear once per cycle and intermediate terminals twice. These tests validate routing and interpolation, not whether the assumed overlap matches a particular scanner.

Then compare complete cyclic transfer surfaces and modulation sidebands with held-out recordings. Report both gain and phase; a pleasant pitch wobble can conceal incorrect amplitude modulation. Tolerances should be chosen from measurement repeatability and the product fidelity goal, not presented as historical factory specifications. Keep optional faults and aesthetic exaggeration outside the nominal profile.

The principal unresolved measurements are exact coupling-versus-angle, receiver loading, residual neighbor coupling, rotational asymmetry and specimen speed. Deliver the implementation with these fields explicitly provisional. A source nominal provides a usable starting point, while the validation gate remains open until reference evidence exists.

Test	Falsifiable condition	Failure suggests
Common terminal signal	Angle-independent output in normalized model	Weights do not sum to one
Full revolution	Correct forward-and-return terminal order	Mapping or wrap error
Bypass and silence	Rotor phase continues	Note- or mode-triggered phase reset
Rate and block changes	No phase discontinuity; bounded numerical error	State migration defect
High-rate comparison	Sideband error meets declared product budget	Insufficient rate or altered weighting
Hardware holdout	Predicted cyclic gain and phase match uncertainty target	Unmodeled loading or geometry

SOURCES S026-01 Hammond factory service manual: scanner theory and construction / S026-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

HARDWARE EVIDENCE / CONTROL CONTRACT

Separate the routing tablets from the six-position selector

Vibrato and chorus switching matrix

The B-3 uses Swell and Great tablets to select the affected material; Great also includes the pedal signal. Its six-position selector chooses V1, V2, V3, C1, C2 or C3 and has no OFF setting. Percussion joins the audio after the vibrato system. These are different routing responsibilities, not seven presets of one generic effect. [S027-01]

ORIGINAL DESIGN PROPOSAL: define U as the upper sustained bus, L as the lower sustained bus, P as the shaped pedal bus, and Q as the separately generated percussion contribution. Use two Boolean routing states plus a semantic mode identifier. The following logical matrix describes destinations; it does not assert equal voltage scaling or erase the transformers and preamplifier gains. Resolve those electrical quantities in the AO-28 model.

Keep one shared scanner phase and one physical wet-channel history. Separate per-manual effects can be an extension, but independent phases would change the reference behavior. Even when both tablets are off, the mechanical scanner keeps its phase; an implementation should continue its clock and preserve network state. A performance bypass may mute its audible contribution without resetting it.

Swell / Great	Scanner input material	Bypass material
OFF / OFF	None selected	U + L + P
ON / OFF	U	L + P
OFF / ON	L + P	U
ON / ON	U + L + P	None selected

CHECKS AND DECISIONS

• Q remains outside the scanner-selection matrix in every row.

• The visible selector and serialized state contain exactly six historical modes; bypass belongs to the two tablets.

SOURCES S027-01 Hammond Organ Service Manual: selective vibrato and vibrato-system schematic

FACTUAL TRANSCRIPTION / REVISION SCOPE

Encode depth as a documented connection map

Vibrato and chorus switching matrix

The published late-B-3 circuit assigns nine scanner terminals to nineteen labeled ladder-node voltages. V and C at the same depth use the same terminal map. The table below is a visual transcription of the paper's Table 2, not a newly measured instrument. Its node numbers are mathematical schematic labels, not manufacturer wire colors or physical lug positions. [S027-02]

ORIGINAL DESIGN PROPOSAL: store the map as three fixed arrays of nine one-based source indices, with an explicit conversion at the C++ array boundary. Associate each profile with its cited circuit revision. Do not interpolate the integer connection map when automating between depths: a selector changes electrical connections. Optional continuous depth belongs to a separately named extension with its own behavior and tests.

Keep nominal map validation separate from specimen validation. The factory diagram identifies the switching mechanism, while the paper provides a clear numerical reference for this particular model. A wire-by-wire comparison against the chosen instrument is still required before assigning production-level certainty. Never merge a historical longer-line switch harness into this profile simply because it also offers six named effects.

Mode pair	t1 … t9 select these node numbers	Evidence
V1 / C1	1, 2, 3, 4, 5, 6, 7, 8, 9	Published late-B-3 model
V2 / C2	1, 2, 3, 5, 7, 9, 11, 12, 13	Published late-B-3 model
V3 / C3	1, 2, 4, 7, 10, 13, 16, 18, 19	Published late-B-3 model

P010 | Hammond B3 - Volume (normal,soft), Vibrato Swell (on,off), Vibrato Great (on,off), Vibrato and Chorus (v-1,c-1,v-2,c-2,v-3,c-3) - ONKIO HAUS (2014-06-01 by JacoTen @ Flickr 14356834922).jpg | JacoTen from Tokyo, Japan | CC BY-SA 2.0 Source and license

CHECKS AND DECISIONS

- Every map contains nine indices in the inclusive range 1–19.
- V1, V2 and V3 must not be implemented merely as three scaled sinusoidal delay depths.

SOURCES S027-01 Hammond Organ Service Manual: selective vibrato and vibrato-system schematic / S027-02 Werner, Dunkel and Germain — A Computational Model of the Hammond Organ Vibrato/Choru

CIRCUIT INTERPRETATION / ORIGINAL MODEL

Chorus changes the electrical reference, not just the mix knob

Vibrato and chorus switching matrix

The factory explanation places an unmodulated contribution across a resistor associated with the line; the resistor is short-circuited for vibrato. In the paper's explicit circuit, Rc is 22 kOhm and the line return floats through it when chorus is selected. [S027-01] [S027-02]

ORIGINAL MODEL DERIVATION: call the line-return voltage r and a local tap voltage relative to that return z_k. A ground-referenced scanner terminal then carries v_k = r + z_k. For normalized scanner weights, their weighted sum already contains r once. Adding an extra dry copy after reconstructing these absolute node voltages would count that contribution twice. Conversely, processing only z_k and discarding r would lose the chorus contribution.

The return voltage depends on the line impedance and drive, so a universal fifty-fifty dry/wet slider is not a circuit identity. Solve the return connection together with the ladder, or identify both transfer paths from that solution. Preserve phase, source impedance and recovery-stage loading. A reduced model can neglect scanner loading only as a declared approximation; compare its error against the loaded solution before adopting it. Avoid a reference-node naming scheme that silently treats every local return as chassis ground.

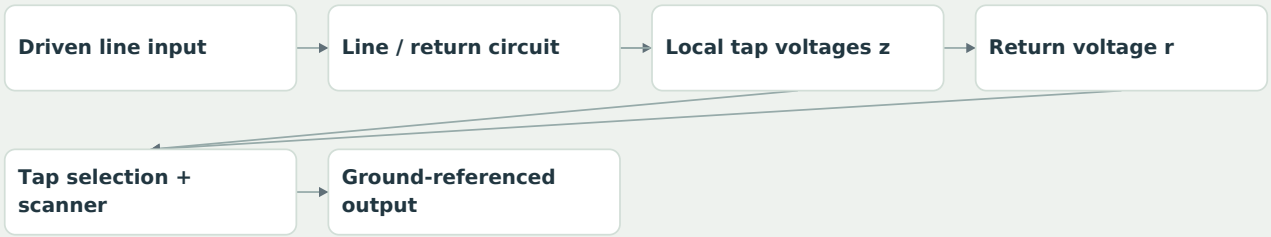
MODEL EQUATIONS

v_k[n] = r[n] + z_k[n]

y_scan[n] = sum_k w_k(theta[n]) * v_map(k)[n]

If sum_k w_k = 1, then y_scan = r + sum_k w_k * z_map(k).

Absolute voltage reconstruction in the chorus model



Original explanatory graph of the proposed implementation; not a replacement for the complete factory schematic.

SOURCES S027-01 Hammond Organ Service Manual: selective vibrato and vibrato-system schematic / S027-02 Werner, Dunkel and Germain — A Computational Model of the Hammond Organ Vibrato/Choru

IMPLEMENTATION / REAL-TIME CONTRACT

Apply switching without destroying physical state

Vibrato and chorus switching matrix

ORIGINAL DESIGN PROPOSAL: represent a mode as {depthMap, chorusReturnEnabled}. Precompute all topology-dependent operators outside the audio callback. At each event sample, update the semantic switches, route the manual buses, advance the selected circuit and read the scanner at its continuing phase. Neither note-on nor mode selection resets the rotating mechanism. A mode change and a note change at the same sample need a documented deterministic event order.

Maintain physical capacitor voltages and inductor currents across an idealized topology change wherever the chosen solver permits. If a WDF stores waves instead, changing port resistances may require a state conversion; copying coefficients while reusing incompatible wave states can inject an artificial impulse. An output crossfade can conceal this bug, so validate physical state continuity first. A separately selectable smooth-switch policy is a product extension, not evidence of the hardware's actual contact transient.

For the shared linear block, combine routed inputs before the solve. Preserve the real order of nonlinear preamplification if that stage is included; separate nonlinear processing followed by summation is generally a different model. Allocate arrays during preparation, reject out-of-range modes, and expose the active evidence-profile identifier in diagnostic snapshots. Host parameter smoothing must never turn a Boolean electrical state into an accidental intermediate resistance.

```
onEvent(e):
  updateSemanticControls(e)
  selectPrecomputedTopology(mode.family)
  selectTapMap(mode.depth)
for each sample n:
  u = routeToScanner(U[n], L[n], P[n], tabs)
  d = routeToBypass(U[n], L[n], P[n], tabs)
  nodes = line.step(u, preservedPhysicalState)
  theta = scanner.advanceClock()
  scanned = scanner.read(nodes, tapMap, theta)
  out[n] = ao28.combine(d, scanned, Q[n])
```

CHECKS AND DECISIONS

• Random block partitions produce identical event timing and the same circuit trajectory within declared floating-point tolerance.

• Switching the selector during sustained input does not create a phase restart or an undocumented equal-power dry mix.

SOURCES S027-01 Hammond Organ Service Manual: selective vibrato and vibrato-system schematic / S027-02 Werner, Dunkel and Germain — A Computational Model of the Hammond Organ Vibrato/Choru

MEASUREMENT PLAN / REQUIRED EVIDENCE

Measure routing and mixing as separate experiments

Vibrato and chorus switching matrix

ORIGINAL CAPTURE PLAN: use a documented direct-output chain before the Leslie so cabinet motion cannot obscure scanner behavior. Begin with isolated U, L and P stimuli and exercise the four tablet combinations. Repeat with percussion only and then with percussion plus sustained drawbars. Record the exact registration, preset selection, expression position, selector setting, warm-up condition and specimen revision with every take. Internal electrical observation belongs to a qualified technician using an approved measurement setup.

For each of the six modes, collect low-level broadband excitation and long steady tones at several frequencies. A single static frequency response is insufficient for a rotating, time-varying block: preserve rotor phase or estimate it from synchronized modulation features, and analyze phase-binned response and sidebands. Record repeated selector changes to characterize contact timing separately from the settled response.

Fit routing errors before tonal differences. If Great incorrectly leaves pedals dry, a later spectral equalizer cannot repair the architecture. Compare chorus at matched excitation and documented gain, retaining an unnormalized copy; automatic loudness normalization can hide the loading difference being investigated. The captures described here are required future evidence. No measured gain ratio, switching duration or pass score is supplied by this chapter.

Experiment	Primary observation	Confound to hold fixed
Isolated upper / lower / pedal	Correct tablet destination	Other source buses muted
Percussion-only sequence	Post-scanner independence	Correct upper-B routing
V versus C at one depth	Absolute voltage and loading change	Identical input and phase reference
Three depths within one family	Different selected tap span	Same circuit profile and scan rate
Selector change while held	Transient and state continuity	Recorded event time and contact behavior

SOURCES S027-01 Hammond Organ Service Manual: selective vibrato and vibrato-system schematic / S027-02 Werner, Dunkel and Germain — A Computational Model of the Hammond Organ Vibrato/Choru

ACCEPTANCE / HANDOFF

Release gates for the switching subsystem

Vibrato and chorus switching matrix

ORIGINAL ACCEPTANCE CONTRACT: the implementation agent should deliver a revision-tagged routing profile, six-mode unit fixtures, an electrical reference solve and a capture comparison report. Documentary correctness and measured sonic agreement are different gates. The supplied JSON map can establish the former for the published circuit, while only acquired specimen data can establish the latter. Report each gate as passed, failed, not run or blocked, with an evidence path.

Keep the chosen reference model small and inspectable. It should expose the two input routes, active return topology, nine selected nodes, scanner phase and reconstructed output reference. These diagnostics make a wrong ground, index conversion or duplicated dry path observable. They also allow later scanner and AO-28 work to change numerical methods without silently changing the control semantics.

Unresolved hardware questions remain the switch harness correspondence for the chosen specimen, contact make/break behavior, actual line loading, component drift and the gain/polarity relationship of the bypass and recovery paths. Assign these to acquisition tasks; do not bury them as guessed constants. Historical line revisions and modern independent-pedal vibrato controls require different profiles. A successful software test of a proposed topology is not proof that an uninspected organ has that topology.

Gate	Falsifiable requirement
ROUTE-01	All four tablet combinations route U/L/P as the documented logical matrix.
MODE-01	Each semantic mode selects its exact nine-node map and intended return state.
REF-01	Chorus return is included once; local and chassis-reference voltages are distinguishable.
STATE-01	No note or selector event resets scanner phase.
PATH-01	Percussion remains outside scanner modulation unless an explicit extension is enabled.
CAPTURE-01	Measured comparison remains not run until traceable reference recordings exist.

SOURCES S027-01 Hammond Organ Service Manual: selective vibrato and vibrato-system schematic / S027-02 Werner, Dunkel and Germain — A Computational Model of the Hammond Organ Vibrato/Choru

EVIDENCE AND IMPLEMENTATION BOUNDARY

Build a circuit with nineteen observable node voltages

Wave digital filter scanner implementation

The selected research schematic contains eighteen LC sections: six with additional divider networks and twelve without them, followed by termination. It exposes nineteen node voltages. A sentence calling this nineteen stages conflicts with the drawing and component enumeration. The source/switch combination needs a coupled root solution. [S028-01]

ORIGINAL DESIGN PROPOSAL: implement the electrical ladder as a reusable multi-output module before connecting rotor interpolation. Import the reviewed component registry from chapter 025, preserving resistor branches and explicit voltage reference nodes. Generate the WDF graph from connectivity, not symbol placement. A chain of independent low-pass filters would discard the reverse electrical influence of downstream loading. The companion scanner-network data can scaffold connectivity; it is not a measured specimen model.

Expose nineteen volts-valued outputs and an input impedance contract. Keep the ideal-source research configuration separate from a loaded AO-28 boundary. The simplified scanner in chapter 026 may read these outputs without changing them; a loaded scanner must participate in the circuit solve. This distinction prevents receiver loading from being counted twice.

Assign circuit topology, scanner depth mapping and rotor angle to separate state owners. This lets a depth gesture change selection without resetting stored electrical energy. The preparation report should include graph hashes and component counts so a later optimization cannot silently replace the reviewed network with a different structure.

Deliverable	Required content
Circuit profile	Revision, SI values, source locators and unresolved parasitics
Port graph	Node pair, current direction and polarity per port
Prepared network	Positive port resistances and coefficients for one internal rate
Runtime state	Reactive memories and electrical switch topology
Probe interface	Nineteen named node voltages, optional currents

CHECKS AND DECISIONS

- Count eighteen inductors and eighteen capacitors; node count is a separate invariant.
- Reject undocumented global polarity inversions.

SOURCES S028-01 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

ORIGINAL MATHEMATICAL CONTRACT

Derive each leaf from one wave convention

Wave digital filter scanner implementation

ORIGINAL DERIVATION: define current i entering the positive-voltage terminal and a positive reference resistance Rp. Incident and reflected voltage waves then reconstruct physical voltage and current unambiguously. The power identity below diagnoses a passive adaptor; it is not a universal stability proof for an arbitrarily switched or incorrectly connected network.

With bilinear substitution s=K(1-z^-1)/(1+z^-1), adapted capacitors and inductors become signed one-sample memories. Component values remain in their port resistances. Do not mistake the compact recurrence for evidence that capacitance or inductance no longer affects the sound: adaptor coefficients depend on the impedances propagated from their descendants.

A polarity inverter changes both exchanged wave signs. Derive adaptor relations from Kirchhoff constraints under the declared orientation, testing each connection through voltage/current reconstruction. An existing WDF library is useful, but inspect its port directions and prepare/reset semantics before adopting it. [S028-02]

Leaf	Adaptation / reflection under this convention
Resistor R	Rp=R; b=0
Capacitor C	Rp=1/(K*C); b[n]=a[n-1]
Inductor L	Rp=K*L; b[n]=-a[n-1]
Ideal voltage source Vs	b=2*Vs-a
Open / short circuit	Open: b=a; short: b=-a

MODEL EQUATIONS

a=v+Rp*i; b=v-Rp*i; v=(a+b)/2; i=(a-b)/(2*Rp)

v*i=(a*a-b*b)/(4*Rp); Rp in ohms, v in volts, i in amperes

K=2*fs for unwarped bilinear discretization; K has units s^-1

CHECKS AND DECISIONS

- A short reconstructs zero volts for arbitrary incident waves.
- An open reconstructs zero current independently of its mode label.

SOURCES S028-02 Chowdhury DSP Wave Digital Filters library and ideal voltage source implementation

ORIGINAL ROOT ELIMINATION

Solve the source and switch without invented delay

Wave digital filter scanner implementation

ORIGINAL DERIVATION: partition the root scattering matrix into internal ports i, connected to source/switch elements, and external ports e, connected to the remaining tree. Define the root element law ai=Phi*bi+Psi*x. Here x contains independent source values; ai and bi face the adaptor. These definitions determine the algebra, including multiplication order.

Substitution gives the system below. Factor M off the audio thread for each fixed switch topology, then prepare Gamma and Theta through linear solves. Per-sample processing becomes a bounded matrix-vector operation. If M is singular or poorly conditioned, inspect connectivity and physical constraints instead of inserting an arbitrary unit delay.

The printed equation 7 does not reproduce this general elimination; source-sign statements in equations 13 and 15 also disagree. These are visible publication inconsistencies, not evidence that the authors software necessarily used the printed errors. [S028-01] Preserve an independent derivation and residual tests as implementation authority.

MODEL EQUATIONS

bi=S11*ai+S12*ae; be=S21*ai+S22*ae

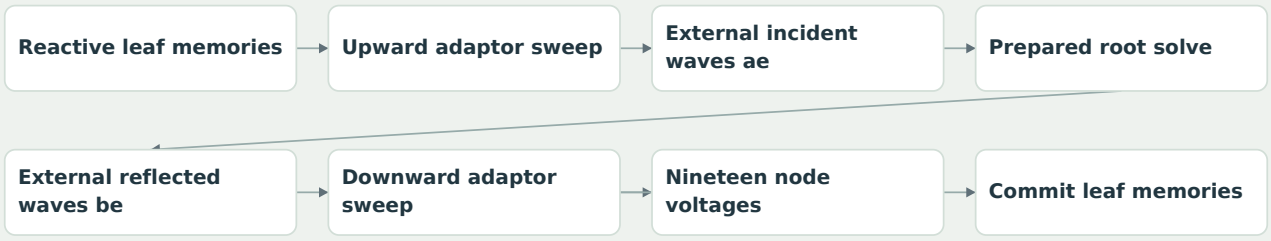
M=I-Phi*S11

ai=solve(M,Phi*S12*ae+Psi*x)

Gamma=S21*solve(M,Phi*S12)+S22

Theta=S21*solve(M,Psi); be=Gamma*ae+Theta*x

A coupled root closes instantaneous electrical constraints



Original algorithm diagram; the root contains the source and electrical switch, not the mechanical scanner.

CHECKS AND DECISIONS

- Evaluate both original root equations after substitution and retain the residual.
- Do not swap Phi and S11; their products generally do not commute.

SOURCES S028-01 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

DISCRETIZATION AND RUNTIME

Prepare at the internal rate and execute two passes

Wave digital filter scanner implementation

ORIGINAL DESIGN PROPOSAL: select the internal rate independently of host block size. Compare unwarped, one-frequency-prewarped and oversampled processing against the same analog reference. Prewarping fixes one frequency, not every resonance. Use omega0=2*pi*f0 in radians per second, with f0 explicitly in hertz.

The publication’s printed prewarp example has a Hz/radians inconsistency. [S028-01] Independently calculated at 44100 samples/s, f0=7075 Hz gives effective period 2/K approximately 24.8133 microseconds. Substituting 7075 as radians/s instead gives 22.7245 microseconds. These are arithmetic examples, not chosen hardware calibration constants.

Prepare fixed-size arrays, traversal order and both root states before processing. Read old reactive memories before committing new ones. Rate changes need an explicit policy: reconstruct compatible state from physical capacitor voltages and inductor currents, or perform a documented controlled restart. Carrying old wave memories unchanged across changed port resistances is not automatically a physical state-preserving operation.

MODEL EQUATIONS

K=omega0/tan(omega0/(2*fs)); use K=2*fs as omega0 approaches zero

```
prepare(profile, internalFs):
  validateSIValuesAndPortOrientations()
  adaptLeavesAndPropagateImpedances()
  factorAndPrepareBothRootTopologies()
processSample(inputVolts, switchState):
  oldReflections = readReactiveMemories()
  ae = upwardSweep(oldReflections)
  be = rootSolve(ae, inputVolts, switchState)
  downwardSweep(be)
  taps = reconstructNodeVoltages()
  commitAllReactiveMemories()
  return taps
```

CHECKS AND DECISIONS

- Changing host block length must not change sample-level circuit results.
- Scanner phase uses the actual internal processing rate; downsampling has an explicit boundary.

SOURCES S028-01 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

OFFLINE VERIFICATION PLAN

Compare every tap with an independent circuit solve

Wave digital filter scanner implementation

Ngspice provides AC and transient analyses; its transient output increment and maximum internal step are different controls. Initial-condition and operating-point choices also affect transient comparison. [S028-03] Pin simulator version and settings alongside results. A visually similar final waveform alone does not verify the ladder.

ORIGINAL DESIGN PROPOSAL: assemble a reference netlist independently from the reviewed electrical connectivity. For each electrical switch state, compute complex transfer functions to every tap. First compare the WDF at digital angular frequency omega with the analog solution at s=j*K*tan(omega/2); this isolates topology and sign errors from intentional bilinear warping. Then compare at identical physical hertz to quantify discretization error.

Use zero-state impulse, step and finite-bandwidth excitation traces as separate fixtures. Reduce maximum integration step until the reference converges, and compare integration settings where appropriate. Include nonzero initial energy to exercise memories. Fit winding loss or receiver loading only after numerical agreement is established; otherwise calibration can conceal a coding error.

For deterministic comparisons, use volts rather than arbitrary plugin amplitude until the AO-28 interface gain is fixed. Save all tap traces with shared sample timestamps and identify any resampler delay explicitly. A reference waveform shifted separately at every tap would conceal the propagation relationships the circuit model is supposed to preserve.

Comparison	Evidence to retain
All taps, both states	Complex responses, absolute error and meaningful phase
Warp-matched grid	Topology error before frequency-mapping error
Same-hertz grid	Response and group-delay deviations versus processing rate
Transient convergence	Step refinement and initial-state metadata
Physical measurements later	Specimen, probe loading, source impedance and confidence

CHECKS AND DECISIONS

- Do not grade phase at deep zeros without a stated magnitude floor.
- No SPICE or physical-scanner pass is claimed here; these are execution gates.

SOURCES S028-03 Ngspice User Manual, version43

ACCEPTANCE AND SOURCE AUDIT

Separate algebra, passivity and sound-matching gates

Wave digital filter scanner implementation

An independent numerical check for this chapter used 100 deterministic, well-conditioned synthetic root matrices. The derived elimination matched simultaneous defining equations within 1.11e-16 maximum absolute error in double precision. This checks algebra on those systems; it does not validate a completed eighteen-section circuit, SPICE model or physical instrument.

ORIGINAL DESIGN PROPOSAL: make sign, units and state tests blocking gates before listening evaluation. Passive fixed linear subgraphs must dissipate or conserve energy according to their actual resistance; switches and ideal sources need a separate energy account. Reject non-finite coefficients, nonpositive component values and unexplained DC growth. Do not add damping merely to hide a bad orientation.

The source remains useful, but the inductor expression in equation 19 has the wrong DC short-circuit limit, and equation 23 coefficients do not yield the stated positive-inductance adaptation. Equation 14 switch labels conflict with open/short constraints under the declared voltage-wave convention. [S028-01] Re-derive these leaves, retain the discrepancy log and verify the resulting circuit against independent simulation.

A second reproducible algebra fixture is supplied as reference_code/wdf_algebra_reference.py. It compares root elimination with a simultaneous system solve, verifies adapted reactive-leaf identities and records explicit prewarp units. Its random matrices are mathematical examples, not a completed scanner circuit or a reconstruction of the authors' software.

Gate	Required result
Leaf constraints	Correct open/short/source voltage and current
Root residual	Agreement with simultaneous equations in both states
Reactive signs	Capacitor +delay; inductor -delay under stated convention
Rate and units	Explicit radians/hertz; impedances match internal fs
Stored state	Repeatable restart; documented rate-change policy
Circuit acceptance	All nineteen outputs match independent reference
Sound acceptance	Separate scanner measurements and listening tests remain pending

CHECKS AND DECISIONS

- Keep completed algebra checks separate from not-run circuit and hardware tests.
- Publish graph, profile, discretization and library revision with future comparisons.

SOURCES S028-01 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus / S028-02 Chowdhury DSP Wave Digital Filters library and ideal voltage source implementation / S028-03 Ngspice User Manual, version43

EVIDENCE BOUNDARY AND MODEL CHOICE

Keep the circuit graph before reducing it to states

State space scanner alternative

The selected published late-B-3 model contains 18 LC sections and 19 observable tap voltages, with compensated early taps, termination and a switched common-return resistance. Its explicit schematic and component enumeration take precedence over its inconsistent nineteen-stage sentence. [S029-01] Chapter 025 holds the nominal component table; chapter 027 holds the depth mapping.

ORIGINAL DESIGN PROPOSAL: maintain one auditable circuit graph and derive two artifacts from it: an offline modified-nodal reference and an optimized runtime realization. MNA introduces branch-current unknowns where ordinary nodal variables alone are inconvenient, including voltage sources and inductors. [S029-02] Preserve component IDs and node incidence through every reduction so a response error can be traced back to a physical branch.

Distinguish circuit unknowns from independent stored-energy states. Nineteen exported voltages do not imply nineteen independent delay memories. An unconstrained 18-capacitor/18-inductor network suggests up to 36 energy coordinates, but the actual rank and constraints determine the usable state dimension. Source impedance, receiver loading and added parasitics can change that realization. Treat an unloaded scanner and a capacitively coupled scanner as different model boundaries.

Representation	Purpose	Boundary rule
Circuit graph	Provenance and connectivity	Retain common return and output polarity
Descriptor unknowns	KCL plus branch equations	Algebraic variables may have no memory
Physical energy coordinates	Safe migration and inspection	Capacitor voltage and inductor current
Reduced runtime state	Efficient processing	Keep reconstruction map and units

CHECKS AND DECISIONS

- Reject a topology assembled from nominal values without endpoint nodes.
- Do not create an independent ladder for each played note.

SOURCES S029-01 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus / S029-02 Ho, Ruehli and Brennan: The Modified Nodal Approach to Network Analysis, IEEE Transac

ORIGINAL CIRCUIT DERIVATION

Stamp a descriptor system with explicit signs and units

State space scanner alternative

ORIGINAL DERIVATION: choose one electrical datum. Each branch-incidence column has +1 at its positive endpoint and −1 at its negative endpoint; positive branch current follows that orientation. Let v contain node voltages, iL inductor currents and iV ideal-voltage-source currents. A_C, A_R, A_L and A_V are the respective incidence matrices. External current injection is positive into a node. These equations follow directly from KCL and the branch laws.

C_N is a nodal capacitance matrix in farads, G a conductance matrix in siemens, L an inductance matrix in henries and R_L the modeled winding resistance in ohms. Include divider, termination, common-return and identified load resistances in G. The supplied 025-scanner-network.json names the early divider attachment points explicitly: divider k branches before Lk; Ck follows Lk. [S029-01] Do not shift those taps by one section.

E can be singular because some nodes and source-current unknowns are algebraic. Do not compute inverse(E). Solve the descriptor system directly or eliminate algebraic variables with verified rank and conditioning. Represent an ideal short by a constraint or node contraction, not infinite floating-point conductance. Any measured mutual-inductance extension must leave L physically admissible. Here B maps the input vector u into the equations; P reconstructs the referenced taps and D represents instantaneous input feedthrough.

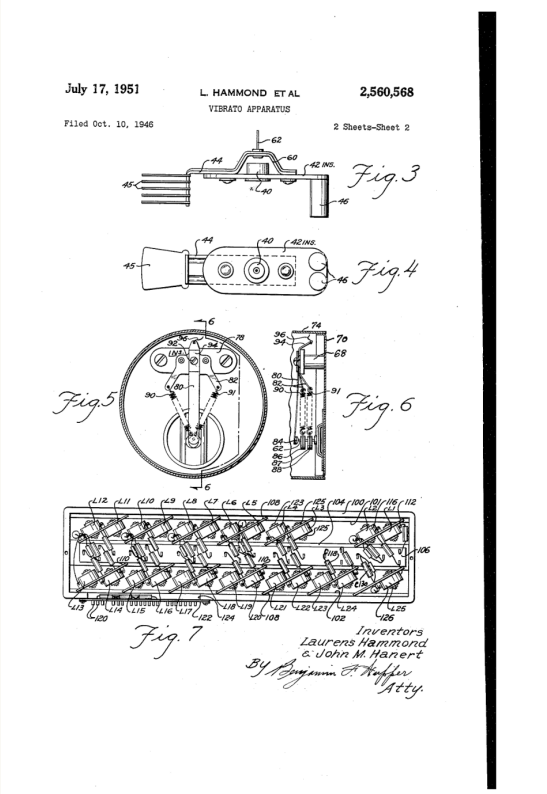
MODEL EQUATIONS

C_N = A_C diag(C) A_C^T; G = A_R diag(1/R) A_R^T

C_N dv/dt + Gv + A_L iL + A_V iV = i_injected

L diL/dt + R_L iL - A_L^T v = 0; A_V^T v = uV

E dz/dt + Kz = Bu; z = [v; iL; iV]; y = Pz + Du



Historical patent plate US2560568A_p02; invention embodiment, not a verified production drawing for the selected organ or cabinet. Source and license

SOURCES S029-01 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus / S029-02 Ho, Ruehli and Brennan: The Modified Nodal Approach to Network Analysis, IEEE Transac

ORIGINAL NUMERICAL MODEL

Discretize the complete loaded network together

State space scanner alternative

ORIGINAL DERIVATION: for fixed E, K and B, trapezoidal integration gives the two-endpoint recurrence below. h is the internal sample interval in seconds. Factor M once; each sample solves Mz_new = rhs using the previous state and both input endpoints. Initial algebraic constraints must be consistent with the initial input. This formulation preserves coupling across the ladder instead of cascading independently designed section filters. Eliminate or explicitly enforce algebraic constraints: a raw trapezoidal descriptor recurrence can retain alternating constraint-error modes that are not physical resonances.

The bilinear substitution maps analog angular frequency to digital frequency nonlinearly. SciPy documents that its bilinear function performs no prewarping. [S029-04] A chosen matching frequency f_star can define kappa as below, replacing 2/h consistently in every reactive branch. This is a global frequency mapping, not permission to retune individual components until one plot looks right. One matching frequency cannot align every resonance.

After valid reduction to ordinary state space, SciPy cont2discrete can provide independent zero-order-hold or bilinear comparisons. [S029-03] These methods imply different input reconstruction; compare like assumptions. A stable discretization does not prevent scanner modulation sidebands from aliasing. Internal rate and output filtering still need a separate error budget.

MODEL EQUATIONS

M = E/h + K/2; N = E/h - K/2

M z_new = N z_old + B*(u_new + u_old)/2

s = kappa*(1-z_delay)/(1+z_delay), with z_delay = z^(-1)

kappa = 2*pi*f_star / tan(pi*f_star/fs_internal); default kappa = 2*fs_internal

CHECKS AND DECISIONS

- Apply one discretization convention to all capacitors and inductors.
- Record f_star as a proposed numerical choice, not a factory specification.

SOURCES S029-03 SciPy official documentation: scipy.signal.cont2discrete / S029-04 SciPy official documentation: scipy.signal.bilinear

IMPLEMENTATION CONTRACT

Prepare matrices off the audio thread and preserve energy state

State space scanner alternative

ORIGINAL DESIGN PROPOSAL: compile and validate each supported electrical topology during preparation. Store scaled matrices, a bounded solve plan, output reconstruction and physical-state conversion maps in an immutable profile. Use double precision initially; benchmark a reduced implementation before accepting lower precision. Do not form a matrix inverse or perform symbolic elimination in the audio callback.

A mode change can alter tap selection, electrical termination or both. A changed tap selector alone must not reset the line. When topology changes, preserve capacitor voltages and inductor currents where new constraints permit, then reconstruct algebraic variables. Detect incompatible ideal constraints; use an explicitly modeled finite switch transition if needed. Arbitrarily interpolating stable matrices does not establish a stable or physical transition.

Sample-rate changes require rebuilding coefficients while retaining physical state and scanner angle. Initialize a cold model with an explicit charge/current condition and optional documented pre-roll; never silently zero a running line at a block boundary. Allocation, matrix compilation and profile destruction belong outside real-time processing. A failed validity check needs a bounded, reported containment policy rather than an unbounded retry.

```
prepare(profile, internalFs):
    graph = validate_and_stamp(profile)
    models = compile_fixed_topologies(graph, internalFs)
    verify_constraints_and_output_polarity(models)
processSample(u, scheduledMode):
    if electrical_topology_changes:
        state = migrate_physical_energy_state(state, scheduledMode)
    rhs = N*state + B*(u + previousInput)/2
    state = bounded_factor_solve(Mfactor, rhs)
    taps = P*state + D*u
    previousInput = u
    return scanner.read(taps, scheduledMode)
```

CHECKS AND DECISIONS

- The solve plan has fixed maximum work and no callback allocation.
- Block partition changes preserve event order and numerical trajectory.

SOURCES S029-02 Ho, Ruehli and Brennan: The Modified Nodal Approach to Network Analysis, IEEE Transac / S029-03 SciPy official documentation: scipy.signal.cont2discrete

MEASUREMENT AND CALIBRATION

Separate numerical agreement from hardware identification

State space scanner alternative

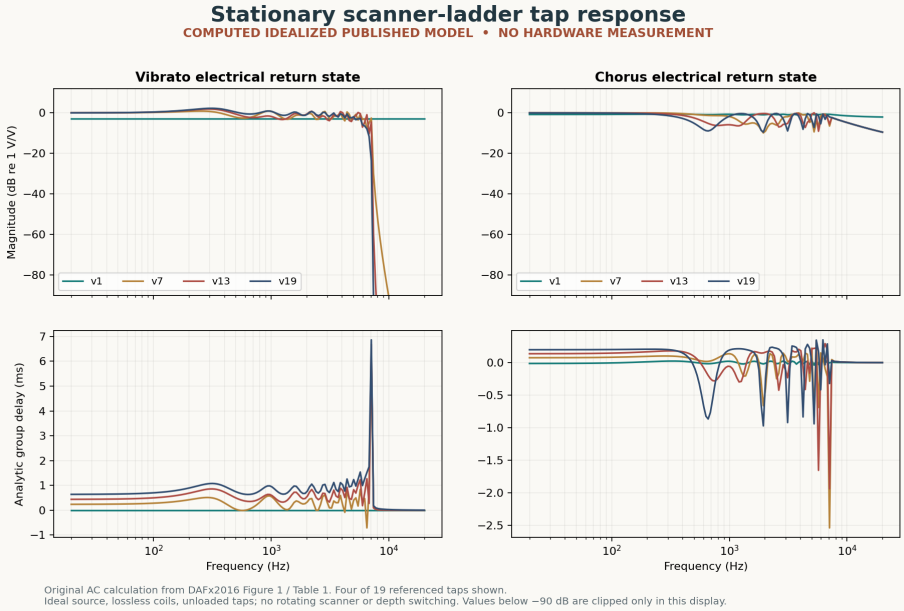
ORIGINAL VERIFICATION PLAN: compare all referenced taps against an independently assembled frequency-domain circuit solve. Evaluate (j*omega*E + K)z = Bu using the declared parameters, then compare complex gain and group delay. Comparing only scanned output can hide a polarity error or trade incorrect ladder phase against incorrect overlap. A WDF implementation provides a useful second realization if it does not share faulty stamping code.

Next compare transients with a higher-rate offline reference under declared input reconstruction. Test impulse, step and multisine excitation, mode changes and silence decay. Under fixed parameters and zero source power, a passive line should not gain stored energy. Apply that test to the fixed ladder: a coupled time-varying capacitor can exchange energy with the mechanical drive and needs a broader energy balance.

Only then fit specimen data. Acquire referenced tap transfer functions, source/load impedances and repeated mode measurements through a professionally prepared setup. Fit winding losses and component variation with evidence-supported ranges; retain unidentifiable combinations as uncertainty. Reserve frequencies and entire modes for holdout tests. Numerical agreement validates implementation consistency, not hardware parameters.

The companion supplies scanner_ac_reference.py and scanner_nominal_response.json: an executed idealized nodal AC calculation for all 19 taps in both electrical return states. Its topology, limiting-circuit and residual checks are software evidence; it has not been compared with SPICE, WDF or hardware.

Comparison	What it can establish
Independent AC solve	Signs, loading, topology and complex response
Higher-rate transient solver	Discretization and transition error
Fixed-network energy decay	Passive numerical behavior under stated conditions
Measured tap responses	Specimen parameter identification
Held-out scanned audio	Combined line, mapping and scanner prediction



Original explanatory calculation. Illustrative parameters are not measured instrument data.

SOURCES S029-01 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus / S029-02 Ho, Ruehli and Brennan: The Modified Nodal Approach to Network Analysis, IEEE Transac

ACCEPTANCE AND DECISIONS

Acceptance gates and the limits of a reduced model

State space scanner alternative

ORIGINAL ACCEPTANCE PROPOSAL: retain a readable descriptor reference after optimizing the runtime model. Every optimization must preserve the input/output contract, physical state migration and declared error budget. A high-order numerator/denominator polynomial is an avoidable intermediate; SciPy warns about coefficient overflow in sufficiently high-order bilinear conversion. [S029-04] Structured network arithmetic keeps internal quantities available for diagnosis.

For a well-conditioned, double-precision synthetic fixture, a normalized linear-solve residual below 1e-9 is a proposed software target. Scale the residual by matrix, state and right-hand-side norms so mixed units do not masquerade as accuracy. Set spectral, phase, transition and CPU limits from the reference and product requirements. No hardware pass is claimed in this edition.

Remaining circuit uncertainties include source and receiver loading, coil loss, parasitic coupling, component revision and finite switch behavior. Until acquired, deliver clearly named nominal and provisional profiles. Keep a faster approximation only when its deviation from the chosen reference is documented; efficiency alone does not establish fidelity.

Gate	Required evidence
Topology and rank	Audited endpoints; no unintended floating reference or singular constraint
Numerical solve	Scaled residual and conditioning report
Tap response	All 19 polarities and complex transfers checked
Transitions	Physical-state continuity or documented finite-switch model
Real-time behavior	Worst-case profiling with bounded work and no allocation
Specimen fidelity	Held-out measurements, uncertainty and declared tolerances

CHECKS AND DECISIONS

• Evaluate stability on the reduced dynamic subspace; do not interpret algebraic constraint modes as stored-energy poles.

SOURCES S029-01 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus / S029-04 SciPy official documentation: scipy.signal.bilinear

REFERENCE CONTRACT / CAPTURE DESIGN

Define the measurement boundary before judging the sound

Scanner measurements and perceptual calibration

The manufacturer routes Great and pedals together and adds percussion after the vibrato system. Its scanner nominal is 412 rpm, or 6.8667 revolutions per second by calculation. [S030-01] The research model in chapter 025 has 18 LC sections and 19 observable voltages; its scanner approximation is separate. [S030-02] Neither source is a new measurement of the selected instrument.

ORIGINAL CAPTURE PROPOSAL: distinguish a scanner-module experiment from an organ-output experiment. The first records the actual drive reference and recovered scanner output under documented loading. The second includes source waveforms, upstream gain and later AO-28 stages. Retain both; do not label a difference in the whole organ as a scanner error without local evidence. Keep the Leslie outside this campaign.

For every take record specimen, circuit revision, mains region, sample clock, input/output gain calibration, registration, preset, expression, tablet routing, selector state and thermal condition. Capture several repeat takes before fitting. Obtain internal observations through a qualified technician and an approved measurement setup; measurement loading is part of the declared circuit boundary. Preserve unnormalized voltage or calibrated digital amplitudes alongside any listening copies.

Capture family	States and purpose
Routing isolation	Upper, lower, pedal and percussion cases; all four tablet combinations.
Settled modulation	V1, V2, V3, C1, C2, C3 with identical excitation and level.
Reference and residue	Matching bypass, no intentional excitation, repeated runs.
Control transitions	Timestamped mode/tablet changes during sustained input.

CHECKS AND DECISIONS

- A capture manifest states whether the measured boundary is module-level or complete AO-28 output.
- No numerical model demonstration is accepted as a recording of hardware.

SOURCES S030-01 Hammond factory service manual: selective vibrato and scanner mechanism / S030-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

ORIGINAL ANALYSIS / STATIC NETWORK

Establish complex line response before rotating-response fitting

Scanner measurements and perceptual calibration

The published paper compares its ladder implementation with a SPICE circuit simulation. That establishes numerical-model comparison, not hardware agreement. [S030-02] ORIGINAL MEASUREMENT PLAN: first compare the real-time solver against an independent circuit solution using the same topology, source impedance, receiver load and reference-node polarity. Then compare that nominal circuit with acquired electrical traces.

Collect synchronized input and accessible tap channels under both return conditions. Use a documented repeatable broadband stimulus; estimate complex response from cross-spectra with an explicit averaging convention. If channels must be acquired in separate runs, retain a simultaneous input reference and verify clock stability. Do not derive phase from magnitude alone. Probe loading and any disconnected scanner boundary require separate identifiers.

For the linear, time-invariant line boundary, report magnitude, unwrapped phase, group delay and residual coherence. Mask frequencies with insufficient excitation or deep response nulls before differentiating phase; otherwise numerical spikes can masquerade as meaningful dispersion. The complete rotating system does not have one ordinary group-delay curve. Its angle-dependent response belongs to the next experiment.

Quantity	Recorded convention
X, Y_k	Synchronized complex spectra of input and tap k.
H_k	Dimensionless voltage ratio; phase in radians.
Group delay	Seconds; only where response and uncertainty support it.
Low coherence	Investigate noise, nonlinearity, drift or unmodeled motion; not an automatic fault diagnosis.

MODEL EQUATIONS

H_k(f) = mean(Y_k(f) * conjugate(X(f))) / mean(abs(X(f))^2).

gamma_k^2(f) = abs(S_yx(f))^2 / (S_xx(f) * S_yy(f)).

tau_g,k(f) = -(1 / (2*pi)) * d unwrap(arg(H_k(f))) / df.

CHECKS AND DECISIONS

- A frequency response uses the same electrical reference at its input and output.
- Model errors are reported before specimen component fitting.

SOURCES S030-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

ORIGINAL DERIVATION / PERIODIC SYSTEM

Identify a full rotation through complex modulation sidebands

Scanner measurements and perceptual calibration

ORIGINAL MODEL-IDENTIFICATION PROPOSAL: for a steady sinusoidal input, represent the settled output by complex coefficients on the rotor harmonic grid. This retains both amplitude and phase modulation. Use measured angle theta(t) when available; otherwise estimate one shared rotational trajectory with an explicit phase-origin ambiguity. A separate fitted phase curve for every frequency would hide model errors.

Fit the coefficients by least squares against the recorded carrier and rotor angle. Start with a bounded harmonic order, extend it until remaining components reach the measured noise limit, and retain the fit conditioning. Long tones spanning many revolutions resolve slow sidebands; a short spectrogram can blur them. Compare signed sidebands, their phase relationships and the reconstructed angular response, not merely total spectral width.

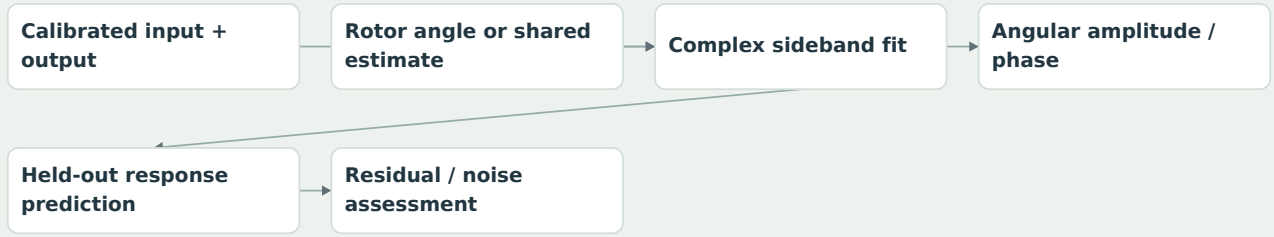
The paper highlights plate-scale spectral structure associated with 16 times its modeled rotation rate. [S030-02] Do not mistake that structure for the fundamental vibrato rate. The full-cycle harmonic grid remains spaced by the rotation frequency; some orders may be weak. Use the actual measured rate rather than substituting the paper demonstration rate or the factory nominal.

MODEL EQUATIONS

y(t) = Re{X * sum[m=-M..M] H_m(f) * exp(j*(2*pi*f*t + m*theta(t)))}.

For constant rotor rate f_r: theta(t)=theta_0+2*pi*f_r*t, so sidebands occur at f+m*f_r.

One phase reference, several diagnostic views



Original identification design. Coefficients describe a measured periodic transfer; they are not supplied as measured values here.

CHECKS AND DECISIONS

• Changing the analysis window does not turn plate-passage structure into an apparent 16-times-faster rotor.

SOURCES S030-01 Hammond factory service manual: selective vibrato and scanner mechanism / S030-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

ORIGINAL DIAGNOSTIC EXPERIMENTS

Separate intended modulation from hum, wear and switching residue

Scanner measurements and perceptual calibration

ORIGINAL EXPERIMENT DESIGN: acquire quiet-state records and repeated driven records with the same gain chain. Keep raw recordings before denoising, mains notching or phase averaging. A line at a familiar frequency does not establish its origin; use input coherence, excitation dependence, angular synchrony and repetition to separate plausible causes. When mains and rotor motion are correlated, phase locking alone cannot uniquely identify the source.

Compare measured input and output modulation. Tonewheel or upstream gain variations already present at the input must not be credited to scanner imperfection. Analyze amplitude scaling across several low drive levels: a linear response should scale with excitation, while an additive contaminant may not. Treat departures as evidence to investigate rather than immediately adding a nonlinear or random-noise parameter.

Preserve settled runs and switch-event runs as separate datasets. For transitions, retain the control timestamp, pre-event phase, previous mode and circuit state. A fixed transient pasted onto every change can conceal contact timing and stored-energy behavior. Repeated irregular clicks, dropouts or asymmetry require a specimen-condition record; they should not silently become the nominal personality of every B-3.

Observation	Discriminating comparison
Carrier-linked sidebands	Compare input/output complex modulation and excitation level.
Fixed-frequency residue	Quiet versus driven records; measured mains-related reference if available.
Angular irregularity	Repeat at the same rotor phase; separate repeatable structure from random residue.
Switch transient	Align by control event and retain pre-event rotor phase.
Broadband excess	Check gain-chain noise and clipping before fitting hardware aging.

CHECKS AND DECISIONS

- The nominal calibration and optional fault profile have separate identities.
- A residual spectrum includes its calibrated floor and analysis settings.

SOURCES S030-01 Hammond factory service manual: selective vibrato and scanner mechanism / S030-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

ORIGINAL CALIBRATION / PERCEPTUAL PLAN

Fit interpretable parameters, then listen to withheld examples

Scanner measurements and perceptual calibration

ORIGINAL FITTING ORDER: freeze routing, polarity and six-mode maps first. Fit the stationary line response next, then angular overlap and rotor trajectory, then documented loading interactions. Chorus must retain its return-voltage contribution exactly once, as chapter 027 derives. Do not repair a doubled dry path with an arbitrary output equalizer or amplitude normalization.

Use one shared physical parameter set across all six modes. Begin with only parameters supported by the available observations; several combinations of component scale, loading and gain may explain similar outputs. Keep correlated parameter uncertainty instead of publishing unjustified individual component precision. Reserve frequencies, registrations and complete repeat takes for validation before optimization begins. A held-out recording must not be repeatedly used to tune the model.

Prepare randomized paired listening examples of sustained single drawbars, richer registrations, chords and mode changes. Use identical musical input and document any allowed overall gain or time alignment. Preserve a calibrated comparison as well as a level-matched listening version, because normalization removes evidence about real gain differences. Ask separately about pitch trajectory, cyclic brightness, chorus body and transition behavior. These are diagnostic judgments; a pleasing preference does not prove electrical equivalence.

Fit stage	Evidence that must already be fixed
Line parameters	Topology, boundary impedance and complex-response calibration.
Scanner overlap	Shared angular reference and independently checked line response.
Combined six modes	Exact switching maps and chorus reference reconstruction.
Perceptual refinement	Withheld inputs, disclosed alignment and unchanged physical scope.

CHECKS AND DECISIONS

- No mode receives an unexplained private equalizer merely to improve its fit.
- Report physical uncertainty separately from optimization residual.

SOURCES S030-01 Hammond factory service manual: selective vibrato and scanner mechanism / S030-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

ACCEPTANCE / AGENT HANDOFF

Deliver calibration evidence with explicit gates and unresolved tolerances

Scanner measurements and perceptual calibration

ORIGINAL HANDOFF CONTRACT: supply raw captures, manifests, calibration files, phase estimates, analysis scripts, fitted profiles and a reproducible comparison report. The report must distinguish documentary correctness, numerical equivalence and specimen agreement. None of the required hardware captures has been acquired for this chapter; all acoustic acceptance states remain not run.

Choose acoustic tolerances from repeated-capture uncertainty and the declared fidelity goal before final evaluation. Report complex sideband error, phase-aligned cyclic gain error and frequency-dependent delay error where identifiable. Include confidence or repeatability intervals and exclude bins only by a documented rule. An aggregate score must not conceal a failed mode, incorrect route or unstable transition.

Exercise every supported sample rate and representative block partition. Compare the complete rotating output with an independently converged higher-rate reference, including resampling filters and their latency. This separates discretization and modulation aliasing from component calibration. The audio callback should only consume the resulting immutable profile; fitting, file loading and uncertainty analysis remain offline. Publish exactly which gates ran, which failed and which are blocked by missing evidence.

Gate	Falsifiable deliverable
SCAL-01	Six modes and all routing cases have traceable manifests and correct source destinations.
SCAL-02	Tap reference polarity and shared chorus return pass independent circuit comparison.
SCAL-03	Rotation estimate, sideband coefficients and angular response reproduce withheld records within predeclared uncertainty-aware limits.
SCAL-04	Quiet-state residue and optional condition defects remain separately reported.
SCAL-05	Rate/block changes preserve physical timing; offline reference bounds numerical error.
SCAL-06	Listening results name the specimen, stimulus set and remaining limitations.

CHECKS AND DECISIONS

- Missing measurements cannot be converted to a passed gate by substituting a published simulation.

SOURCES S030-01 Hammond factory service manual: selective vibrato and scanner mechanism / S030-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

HARDWARE EVIDENCE AND REVISION CONTRACT

Identify eight sockets before assigning digital stages

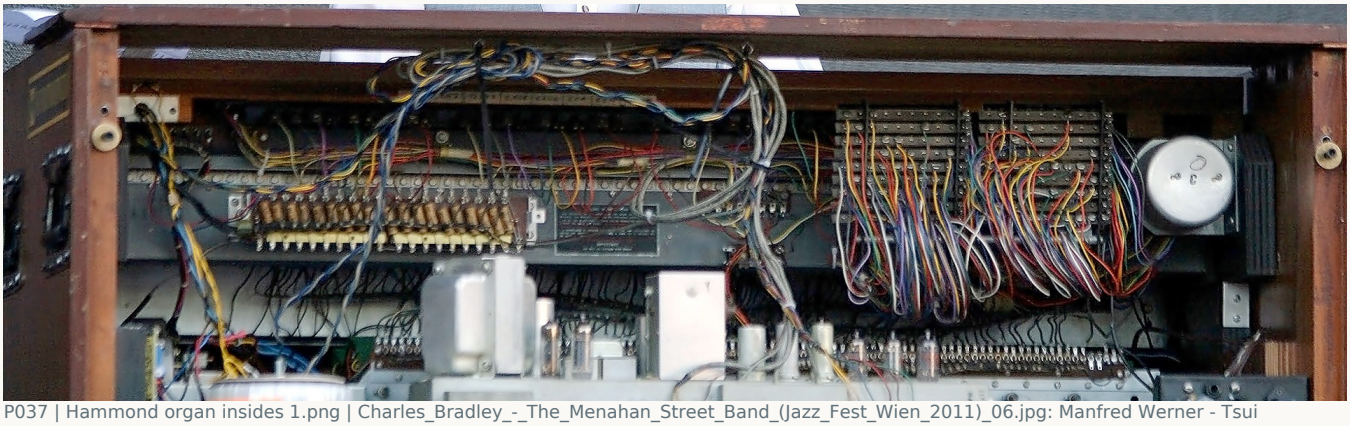
AO28 valve complement and topology

Factory Figure 24 identifies the following eight AO-28 sockets. The clearer 1995 redraw agrees, sometimes adding an A suffix to a tube family. Treat suffixes and fitted brands as specimen fields. This is a preamplifier and transformer-output chassis; its output is not the Leslie loudspeaker power stage. [S031-01] [S031-02]

ORIGINAL DESIGN PROPOSAL: freeze a documentary profile before compiling any valve model. Record chassis designation, code letter, mains variant, schematic sheet, installed valve markings and known modifications. A tube family chooses a candidate device model; it does not establish its actual operating point, production spread or remaining emission.

Use the socket identifier as a persistent physical identity and the section identifier as a computational identity. This prevents a later replacement of V3 from changing only one of its modeled sections. Preserve separate electrical connections even when two sections share one glass envelope. The companion stage registry is a topology scaffold, not a completed circuit netlist or approved replacement-parts list.

Socket	Documented family	Role as drawn
V1	6AU6	Non-vibrato input amplifier
V2	6AU6	Vibrato input amplifier
V3	12BH7	A: line drive; B: transformer output
V4	12AX7	Common gain before and after expression
V5	6C4	Percussion audio amplification
V6	6C4	Percussion keying/charge control
V7	12AU7	Two percussion control sections
V8	6X4	Power-supply rectification



P037 | Hammond organ insides 1.png | Charles_Bradley_-_The_Menahan_Street_Band_(Jazz_Fest_Wien_2011)_06.jpg: Manfred Werner - Tsui derivative work: Clusternote (talk) | CC BY-SA 3.0 | Context specimen; exact model/revision requires confirmation. | Source derivative is cropped and sharpened. Source and license

CHECKS AND DECISIONS

• Inventory validation requires exactly V1 through V8 for this selected drawing; other Hammond amplifier rosters are different profiles.

• A missing specimen code letter remains unresolved rather than defaulting silently to the no-code-letter variant.

SOURCES S031-01 Hammond factory service manual: AO-28 Figure24 and theory / S031-02 AO-28 schematic redraw dated April20,1995, attached to B-3 compilation

SOURCE-TRACED TOPOLOGY / ORIGINAL GRAPH

Keep the three signal entries and scanner loop distinct

AO28 valve complement and topology

The schematic separates non-vibrato A, vibrato B and percussion H paths. V3A drives the external line/scanner route; V3B drives T3. V4 surrounds the expression network. Percussion joins the returning path downstream of the scanner. [S031-01] [S031-02]

ORIGINAL DESIGN PROPOSAL: retain these branches as observable test boundaries. Feeding a single mixed organ bus through one saturation block would erase the distinction between driven and bypassed scanner paths. Likewise, processing percussion through the scanner merely because it sounds with the upper manual would change the documented routing.

The graph below shows functional dependencies, not component-level connectivity. Preserve coupling capacitors, feedback and loading in the detailed model. The scanner connection is an electrical boundary: the driver, line and receiver must agree on reference potential and impedance. The main output model should expose its transformer/load contract; a floating-point sample alone cannot describe the analog load that shaped it.

P031 | 12AX7 GE 1.png | Oliviosu | CC BY-SA 3.0 | Tube type illustration; fitted specimen brand is unverified. Source and license

CHECKS AND DECISIONS

- A scanner-only perturbation must affect the routed branch, with any cross-coupling justified separately.
- The T3 output boundary is tested with the intended cabinet input model.

SOURCES S031-01 Hammond factory service manual: AO-28 Figure24 and theory / S031-02 AO-28 schematic redraw dated April20,1995, attached to B-3 compilation

ORIGINAL MODEL DESIGN

Resolve shared valves without inventing shared distortion

AO28 valve complement and topology

V3 contains two sections serving different signal paths. The factory theory paragraph 5-46 describes a driver V3 followed by a12BH7 output tube, but Figure 24 identifies V3 itself as12BH7. Use the drawing and section connections for this profile; do not create an extra socket from that prose. [S031-01] [S031-02]

ORIGINAL DESIGN PROPOSAL: represent each section with grid, plate and cathode terminals, then associate the sections with one physical valve record. A shared envelope does not by itself prove audible section-to-section crosstalk. Add heater coupling, thermal drift or internal leakage only when supporting data exist. Conversely, explicit shared supply and bias connections belong in the electrical graph even before such refinements.

Separate terminal voltages in volts from normalized audio samples. Introduce one documented conversion at each capture or host boundary. Unknown source gain must not be absorbed into several unrelated drive controls; otherwise matching a spectrum may conceal the wrong operating region. Begin with DC operating points and small-signal transfer, then enable device nonlinearities while retaining the same topology and load.

Identity layer	Required data	Mistake prevented
Physical valve	Socket, family, fitted markings, revision	Treating V3A and V3B as unrelated parts
Electrical section	Terminal nodes and model parameters	Confusing the scanner driver with output drive
Shared network	Supply/bias connections and state	Adding arbitrary global sag
Host boundary	Volts-to-sample scale and polarity	Untraceable gain staging

MODEL EQUATIONS

For section j: i_j = F_j(v_grid-v_cathode, v_plate-v_cathode, state_j).

For shared supply node s: sum(currents_into_s) = sum(currents_out_of_s); dynamic elements retain their state.

P033 | Hitachi 12BH7A.JPG | Olli Niemitalo | CC0 | Tube type illustration; fitted specimen brand is unverified. Source and license

SOURCES S031-01 Hammond factory service manual: AO-28 Figure24 and theory / S031-02 AO-28 schematic redraw dated April20,1995, attached to B-3 compilation

IMPLEMENTATION SEQUENCE

Build a topology-correct preamp before adding aging controls

AO28 valve complement and topology

ORIGINAL DESIGN PROPOSAL: import the socket registry and terminal graph, reject duplicate designators, and resolve every component edge. Confirm the exact percussion-filter revision before stamping matrices. The 1995 redraw labels an AO-28-1 no-code-letter alternate network; that annotation is a revision branch requiring verification, not permission to combine the most convenient values from both drawings. [S031-02]

First deliver a linearized reference with explicit input and output impedances. Add the expression network from chapter 023 and scanner interface from chapters 025–030. Preserve physical state when routing or expression moves. Introduce nonlinear device models in chapter 033 only after linear gain, polarity and phase agree with the selected circuit reference.

Compile immutable graph structures off the audio thread. Keep solver scratch space, filter state and oversampling buffers preallocated. A stage cannot safely be reordered merely because its linear filters commute at one operating point: nonlinear drive, feedback and loading can invalidate that equivalence. Treat convergence failure as a bounded error path with diagnostic counters, never an unbounded retry inside an audio callback.

Deliverable	Required before proceeding
Stage registry	Eight sockets; section identities and sources
Linear reference	Loaded complex transfer and DC consistency
Integrated dry preamp	Expression and scanner boundaries match
Nonlinear model	Bounded solver and alias characterization

```
profile = loadVerifiedTopology(profileId)
assert validateNodesAndRevision(profile)
linear = buildLinearReference(profile)
state = initializeConsistentOperatingPoint(linear)
for sample in block:
    events = orderedEventsAt(sample)
    updateControlTargets(events)
    preserveEnergyStateAcrossSwitches()
    solveCoupledElectricalGroups(state, inputs)
    observeScannerAndOutputPorts(state)
    publishBoundedDiagnostics()
```

SOURCES S031-02 AO-28 schematic redraw dated April20,1995, attached to B-3 compilation

MEASUREMENT PLAN / NOT RUN

Capture the branches and load needed to identify the model

AO28 valve complement and topology

ORIGINAL MEASUREMENT PROPOSAL: document the selected chassis and valve markings before recording. Record control states, warm-up condition, output load and acquisition-chain calibration with every take. Internal-node work belongs to a qualified technician using an approved measurement plan; the software team should request named signals and metadata rather than improvise live chassis connections.

Use separate non-vibrato, vibrato and percussion exercises, then combinations. Capture low-level complex response before level-dependent sweeps. Repeat expression endpoints and intermediate positions under the same source and load. A complete B-3-plus-Leslie microphone recording cannot uniquely identify AO-28 transfer because the downstream amplifier, crossover, rotors and room contribute their own changes.

No reference preamp was measured for this edition. Accordingly, this chapter supplies no verified output wattage, absolute gain, clipping threshold or fitted valve parameters. Resolve those quantities through a defined loaded measurement. A test recording without its load and level reference may remain useful for listening, but it cannot become a calibrated circuit target.

Capture family	What it identifies	Required context
A and B branch separately	Relative gain, polarity and scanner-drive differences	Routing states; linear input range
Expression positions	Loaded transfer and downstream drive	Mechanical position; source level
Percussion plus sustained tone	Joining point and level interaction	Preset, decay, harmonic and volume states
Output into known load	T3 and receiving-stage boundary	Differential polarity; impedance and level
Repeated warm sessions	Drift versus repeatability	Valve identities and environmental notes

CHECKS AND DECISIONS

• Reject absolute-gain fitting when capture-chain gain or output load is unknown.

• Keep a held-out combined-path recording to test whether independently fitted branches still interact correctly.

ACCEPTANCE AND HANDOFF

Require structural tests before claiming preamp fidelity

AO28 valve complement and topology

ORIGINAL ACCEPTANCE PROPOSAL: first prove documentary and computational consistency. Verify each socket assignment against the source drawing and each modeled section against its physical owner. An internal diagnostic should identify the active revision, observation nodes, volt-to-sample scaling and enabled approximations. These checks can pass before any physical recording exists; they do not demonstrate acoustic equivalence.

Then compare the loaded reference model and implementation with the same drive and controls. Exercise all routing combinations, expression motion, percussion overlap and realistic output loads. Preserve deterministic state when the host divides an identical event stream into different block sizes. Characterize any intentional crossfade or solver approximation instead of allowing it to masquerade as a historical circuit property.

The handoff consists of the stage registry, revision manifest, detailed component graph from chapter 038, linear and nonlinear validation evidence, and the unresolved acquisition list. Only the structural registry is supplied here. Component values and fitted curves need their own evidence records; photographic resemblance and correct tube names alone cannot establish the required sound.

Test	Pass condition	Current status
Socket and section audit	V1–V8 roster; correct V3/V4 split roles	Documentary review completed
Revision exclusivity	Exactly one selected alternate network	Specimen selection pending
Loaded signal routing	Correct branch destination and polarity	Engine test not run
State continuity	Equivalent event stream across block sizes	Engine test not run
Acoustic comparison	Held-out captures meet agreed uncertainty limits	Capture and test not run

CHECKS AND DECISIONS

- Do not infer a production speaker-power rating from the 12BH7 designation.
- Do not add an extra output valve to reconcile the inconsistent theory prose.
- Keep device-family substitutions and optional aging profiles separate from the selected reference.

SOURCES S031-01 Hammond factory service manual: AO-28 Figure24 and theory / S031-02 AO-28 schematic redraw dated April20,1995, attached to B-3 compilation

HARDWARE EVIDENCE / MODEL BOUNDARY

Establish the loaded reference before adding distortion

AO28 linear network and transformer model

Factory Figure 24 separates vibrato and non-vibrato inputs, routes the common signal through V4 and its expression network, and shows the V3 output section driving T3. C22 belongs to feedback and R40 to the tone-control network. Printed variants prevent treating every AO-28 as one interchangeable circuit. [S032-01] The 1995 redraw clarifies labels but is a later transcription, not independent factory evidence. [S032-02]

ORIGINAL DESIGN PROPOSAL: construct a small-signal reference that preserves these electrical connections before fitting saturation. A fixed frequency-response curve is valid only for its source impedance, receiving load, control positions and operating point. The same nominal components can produce different transfer functions when a connected stage loads them. Therefore store an electrical boundary contract with every response plot and filter profile.

Keep the scanner as an explicitly connected subsystem. First validate the non-vibrato path and stationary electrical boundaries. A rotating scanner is time-varying; one static transfer curve cannot describe its complete modulation. Reserve Leslie power-stage and loudspeaker loading for their own models, connected through the selected interface.

Boundary or setting	Required profile record
Inputs A/B	Source impedance and voltage reference, not just sample amplitude
Scanner C/D boundary	Driver and receiver loading; chosen coupling approximation
Expression / tone / volume	Physical setting and circuit revision; tone is a service calibration
T3 / cabinet connection	Both output-leg voltages, return, cable and receiving load

CHECKS AND DECISIONS

- Reject a response profile without its source/load conditions.
- Do not assign a Leslie transformer model to AO-28 T3 merely because a designator matches.

SOURCES S032-01 Hammond Organ Service Manual: AO-28 Figure24 and preamplifier theory / S032-02 B-3 service compilation: explicitly dated 1995 AO-28 redraw

ORIGINAL MODEL / MATHEMATICAL CONTRACT

Linearize the connected circuit at a declared operating point

AO28 linear network and transformer model

Modified nodal analysis retains node voltages plus selected branch currents, allowing voltage sources and coupled elements to remain explicit. [S032-03] Official ngspice documentation describes AC analysis as a linearization around the DC operating point. [S032-04] Use this as the offline numerical reference, with documented tube models and bias; a solver does not make an unverified tube model physically correct.

ORIGINAL DERIVATION: collect operating conditions in q, including bias, stationary expression position, service tone setting and load. Assemble descriptor matrices E and G from the complete connected network. Linearized valve derivatives belong in G; reactive energy belongs in E. Each input column represents a separately defined electrical excitation. Do not multiply unloaded stage responses when feedback or finite interstage impedance couples their nodes.

For a triode, the incremental plate current can be represented by transconductance and plate conductance at q. A pentode adds screen dependence where required. Preserve cathode bypass and supply-decoupling impedances when they affect the measured band. An ideal AC-ground supply is an explicit approximation; it removes shared-rail coupling that later tests may reveal.

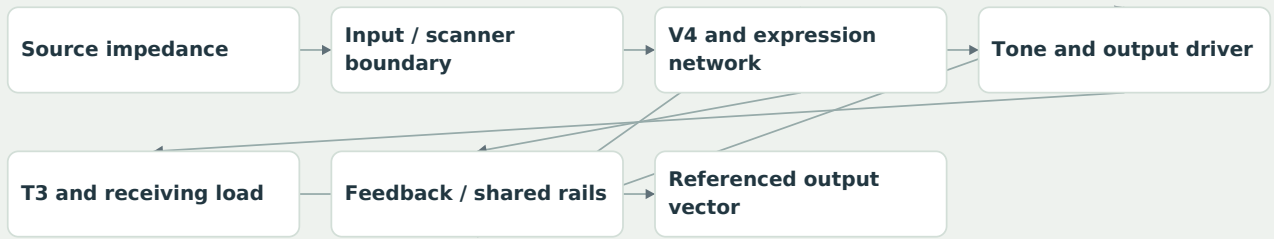
MODEL EQUATIONS

$E(q)\,\dot{x} + G(q)\,x = B(q)\,u;\quad y = C(q)\,x + D(q)\,u.$

$H(s;q) = C(q)\,[s\,E(q) + G(q)]^{-1}\,B(q) + D(q).$

$\Delta i_p = g_m\,\Delta v_{gk} + g_p\,\Delta v_{pk};\quad g_p = 1/r_p.$

One connected small-signal problem



Original dependency view, not a substitute for component connectivity. Each block retains its electrical ports.

CHECKS AND DECISIONS

- Record volts, amperes, farads, henries and siemens explicitly in the netlist contract.
- Keep the two V3 sections electrically distinct while retaining any modeled shared supply interaction.

SOURCES S032-03 Ho, Ruehli and Brennan: The Modified Nodal Approach to Network Analysis / S032-04 Ngspice User Manual, version47+ development edition

ORIGINAL TRANSFORMER MODEL

Give T3 an identifiable linear transformer model

AO28 linear network and transformer model

ORIGINAL DESIGN PROPOSAL: begin with a passive multiport transformer model containing winding resistance, magnetizing behavior, leakage and the minimum capacitance needed to explain measured complex response. Do not invent a T3 turns ratio or inductance from its drawn symbol. A resistance ratio is not a turns ratio: conductor gauge and winding geometry are additional unknowns.

Choose one coherent representation: coupled self-inductances with mutual coupling, or an ideal ratio plus an equivalent magnetizing/leakage network. Combining both without derivation double-counts energy. Ngspice documents winding-dot conventions and coupling-matrix constraints. [S032-04] For a split secondary, preserve both winding sections and their polarity; fit differential and common-mode behavior separately when the available measurements support it.

Fit the smallest identifiable model against more than one known receiving load. A single magnitude curve can conceal compensating errors in source impedance, magnetizing inductance and capacitance. Keep saturation, hysteresis and level-dependent permeability outside this linear baseline; their absence is a scope limit, not a statement that the hardware lacks them.

Parameter group	Evidence needed
Ratio and winding polarity	Phase-coherent voltage ratios with specified loading
Copper resistance	Winding-specific measurements and temperature record
Magnetizing / leakage terms	Complex impedance and response under multiple loads
Capacitance / loss terms	High-frequency magnitude and phase; constrained fit
Secondary balance	Both leg voltages relative to the documented return

MODEL EQUATIONS

M = k sqrt(L_p L_s); the complete inductance matrix must be positive semidefinite.

Ideal transformer only: Z_reflected = a^2 Z_load, with a = N_primary / N_secondary.

CHECKS AND DECISIONS

• Nonnegative stored magnetic energy is a numerical acceptance condition.

• Unknown transformer parameters remain null in the evidence record; provisional fit values carry a separate status.

SOURCES S032-04 Ngspice User Manual, version47+ development edition

ORIGINAL IMPLEMENTATION SEQUENCE

Prepare a real-time realization without discarding its boundaries

AO28 linear network and transformer model

First produce the revision-specific netlist and a reproducible DC/AC analysis script. Compare the full circuit with deliberately isolated sections to locate loading and feedback contributions. Then derive a reduced state-space or rational-filter realization of the connected transfer matrix. Match phase, gain and input/output impedances over the declared useful band; matching only one output magnitude is insufficient for later integration.

Prepare immutable coefficients and factorizations outside the audio callback. At runtime, advance bounded state with explicit electrical scaling. Controls that change the circuit require a validated coefficient/state transition, not clearing history. Arbitrarily interpolating polynomial coefficients can destabilize a filter; use a verified stable parameterization or a state-aware transition between prepared models.

A fixed linear block does not generate new harmonics, but its discretization can alter high-frequency phase and gain. Choose the mapping and rate by comparison with the analog reference. Fast expression or tone movement creates a time-varying system; test those sidebands separately. Account for resampling response and latency when this block sits inside an oversampled nonlinear chain.

```
prepare(profile, rate):
  validateRevisionUnitsAndBoundary(profile)
  q = solveDeclaredOperatingPoint(profile)
  analog = assembleConnectedLinearization(profile, q)
  reference = solveComplexResponses(analog, testGrid)
  runtime = discretizeAndReduce(analog, rate)
  compareResponses(runtime, reference)
  publishImmutable(runtime)
process(inputVector, controlEvents):
  applyValidatedStateTransitions(controlEvents)
  outputVector = advanceBoundedState(inputVector)
  return outputVector
```

CHECKS AND DECISIONS

- No allocation, file access or matrix-size growth occurs in the audio callback.
- Pin the simulation program and model-library versions; a rolling documentation URL is not a reproducible executable.
- Publish reduction residuals and the retained frequency range with each prepared profile.

SOURCES S032-03 Ho, Ruehli and Brennan: The Modified Nodal Approach to Network Analysis / S032-04 Ngspice User Manual, version47+ development edition

ORIGINAL ACQUISITION PROTOCOL

Measure gain, phase and loading as separate observables

AO28 linear network and transformer model

Create an acquisition manifest with AO-28 identity, modifications, tube set, supply condition, warm-up interval, control positions and the receiving connection. Internal electrical work belongs to a qualified technician. Preserve an appropriate simultaneous input reference and both output legs; a microphone recording of the Leslie also contains power amplification, drivers, rotation and room response.

Establish the small-signal range experimentally. Repeat the same excitation at several amplitudes and compare normalized complex responses, residual harmonics and noise. Select a range where response remains level-independent within the measured uncertainty. An output buried in noise cannot establish a corner frequency, and a visibly distorted sweep cannot calibrate the linear baseline.

Use stationary expression and tone settings first, then several documented receiving loads and cable conditions. Separate the direct path from the scanner-return contribution. Measure both differential and common-mode output where relevant; independently normalized legs can hide imbalance. Save raw traces and instrument calibration so later fitting does not depend on an undocumented gain correction.

Capture family	What it identifies
Complex frequency response	Gain, phase, polarity and group delay
Known load variations	Source/output impedance and transformer damping
Several stationary controls	Feedback, expression and service-tone interactions
Amplitude ladder	Small-signal validity and onset of level dependence
Quiet-state capture	Noise floor and excluded frequency bins

MODEL EQUATIONS

V_diff = V_G1 - V_G2; V_common = (V_G1 + V_G2)/2, relative to the same reference.

CHECKS AND DECISIONS

- Retain absolute voltage calibration; do not normalize every take independently.
- Use held-out loads and control positions to detect an overfitted equalizer.

ORIGINAL ACCEPTANCE / HANDOFF

Require two distinct passes: numerical agreement and hardware agreement

AO28 linear network and transformer model

The first gate compares the runtime model with its declared circuit reference. Verify transfer magnitude, phase, port impedance, initial-state behavior and settling after control changes. A matching SPICE result establishes consistency with that netlist and device model. It does not establish that the netlist matches the selected instrument.

The second gate compares with physical captures and their uncertainty. Set project tolerances before evaluating held-out data, and retain the signed residual curves. Investigate a narrow response discrepancy before compensating it with an unrelated EQ: it may indicate wrong loading, an incorrect capacitor unit, reversed winding polarity or a revision mismatch. Preserve failed cases in the regression suite.

Handoff includes source-located netlist, operating-point record, transformer evidence, prepared runtime model, fitting code and both validation reports. All hardware-dependent acceptance items remain not run in this research edition. Subsequent nonlinear work must reproduce this baseline as signal level tends toward the fitted small-signal region; it should not quietly replace the already established frequency response.

Fixture	Acceptance evidence
Superposition at fixed controls	Linear model obeys additivity within declared numerical tolerance
Connected versus reduced circuit	Complex response and port behavior agree within reduction budget
Load and control changes	No unstable state, reset click or unexplained gain renormalization
Reference captures	Held-out residuals satisfy predeclared uncertainty-aware limits
Nonlinear integration	Small-signal limit converges to this calibrated baseline

CHECKS AND DECISIONS

- Report numerical, documentary and physical verification separately.
- Missing T3 parameters or a source/load mismatch block a specimen-level fidelity claim, while a clearly labeled provisional model may proceed.

HARDWARE EVIDENCE AND MODEL SCOPE

Bind each valve model to its socket, source and voltage domain

Valve nonlinear models and solver design

Use chapter 031’s verified AO-28 socket registry to select a device family. The manufacturer sheets below are modeling priors, not evidence that the chosen instrument contains that brand or those exact characteristics. Preserve suffix, publication date, test conditions and provisional-data markings. The inspected Tung-Sol6AU6 and 12BH7 sheets explicitly identify their data as tentative. [S033-02] [S033-03]

ORIGINAL DESIGN PROPOSAL: define a device profile containing terminal names, current sign convention, parameter units, fitted voltage domain, source hashes and uncertainty. Pass terminal voltages relative to the cathode, with volts and amperes at the circuit boundary. A separate calibrated volts-per-full-scale conversion connects the circuit to audio samples; a normalized waveshaper parameter must not quietly stand in for a physical bias voltage.

As a documented unit check, GE lists a12AX7 condition of 250V plate,−2V grid,1.2 mA plate current,1600micromhos transconductance and 62500 ohms plate resistance. Their product gives the listed amplification factor100. These are datasheet conditions, not the AO-28 operating point. [S033-01]

Family prior	Inspected evidence	Implementation decision
12AX7 / GE	Plate, transfer and differential curves	Fit current and local slopes together.
6AU6 / Tung-Sol	Plate and screen-current families	Retain screen voltage/current ports.
12BH7 / Tung-Sol	Independent per-section plate curves	Use separate coefficients from 12AX7.
6C4 / Tung-Sol	Plate and grid-current curves [S033-09]	Fit grid current against plate and grid voltage.
12AU7 / Tung-Sol	Family-specific sheet [S033-10]	Separate section profiles where evidence warrants.

CHECKS AND DECISIONS

- Reject missing units or undocumented family substitutions.
- Treat the 6X4 rectifier separately in chapter 035.

SOURCES S033-01 General Electric 12AX7, ET-T509B, June1953 / S033-02 Tung-Sol 6AU6/6AU6A tentative data, September1956 / S033-03 Tung-Sol 12BH7 tentative data, December1950 / S033-09 Tung-Sol 6C4 manufacturer data / S033-10 Tung-Sol 12AU7 manufacturer data

DOCUMENTED MODEL AND ORIGINAL IMPLEMENTATION CONTRACT

Make the triode law numerically explicit

Valve nonlinear models and solver design

Koren’s triode law is a phenomenological plate-current surface. The equations below restate its positive-plate operating region using an explicitly stable softplus. They do not model every terminal current, every capacitance or arbitrary reverse-voltage operation. Its published example subcircuit adds grid-current and capacitance branches separately. [S033-04] [S033-05]

ORIGINAL DESIGN PROPOSAL: return plate current, grid current, terminal charges and their voltage derivatives from one side-effect-free device evaluator. Evaluate the fractional power only after checking its argument; multiplying an invalid power by a zero sign term can still produce NaN. Store a declared extension outside the fitted domain, and count every excursion. Never silently extrapolate an attractive curve across uncontrolled grid conduction.

Dimensional analysis requires triode KVB to carry volts squared in this formula; KP and mu are dimensionless, and KG1 carries volts-to-the-X per ampere. The identically named pentode knee parameter in Koren’s other equation has voltage units. Version these parameter schemas separately. Grid-to-plate capacitance belongs in the connected network so its feedback effect emerges from the solved voltages.

MODEL EQUATIONS

sp(z) = max(z,0) + log1p(exp(-abs(z)))

z = KP*(1/mu + Vgk/sqrt(KVB_V2 + Vpk*Vpk))

E1 = (Vpk/KP)*sp(z); Ip = 2*pow(E1,X)/KG1 for E1 > 0

At each valid bias: gm = dIp/dVgk; rp = 1/(dIp/dVpk).

P034 | Toshiba 6AU6A.JPG | Olli Niemitalo | CC0 | Tube type illustration; fitted specimen brand is unverified. Source and license

CHECKS AND DECISIONS

- Do not evaluate exp(z) directly for unrestricted positive z.
- Preserve the factor2 convention when importing fitted KG1.

SOURCES S033-04 Norman Koren: Improved vacuum tube models for SPICE, Part1 / S033-05 Norman Koren: Improved vacuum tube models for SPICE, Part2

MECHANISM AND MODEL SELECTION

Preserve the pentode screen port and loaded grid behavior

Valve nonlinear models and solver design

The inspected6AU6 sheet includes separate plate-current families at 100V and 150V screen voltage; its 100V chart also plots screen current. These conditions are coordinates in a device model, not permission to fix the actual AO-28 screen node at either value. [S033-02] A fit using only one plate-current trace can conceal an incorrect screen dependency.

ORIGINAL DESIGN PROPOSAL: evaluate Ip and Ig2 jointly as functions of plate, control-grid and screen voltages relative to cathode. Retain the verified suppressor/shield connection and document any neglected suppressor current. Enforce terminal-current conservation, including displacement currents. A pentode must not be reduced to a triode curve followed by a clipping function merely because both sound distorted.

Treat Koren’s pentode form as one candidate, with separate validation near the knee and cutoff. Its simplified screen-current treatment is a stated compromise, not specimen evidence. [S033-04] Positive-grid current also loads the preceding network and can charge coupling capacitors; model its branch inside the solve. A proposed smooth rectifying law needs independently fitted onset, slope and curvature, rather than a universal diode drop. The 6C4 sheet plots grid current against plate voltage at several positive grid biases; a grid-only branch therefore needs a restricted validity claim. [S033-09]

Candidate	Useful property	Required qualification
Analytic plate/screen surfaces	Compact evaluation and derivatives	Check all operating coordinates and slope errors.
Shape-preserving current tables	Directly follows acquired curves	Validate interpolation derivatives and boundary policy.
Analytic model plus fitted residual	Adds local detail efficiently	Constrain residuals outside supported data; retain provenance.
Single-input static waveshaper	Cheap stylistic approximation	Cannot alone represent changing screen bias, loading or stored charge.

CHECKS AND DECISIONS

- A screen-voltage perturbation must affect the connected screen supply consistently.
- Adding a common voltage to every terminal must leave device currents unchanged.

SOURCES S033-02 Tung-Sol 6AU6/6AU6A tentative data, September1956 / S033-04 Norman Koren: Improved vacuum tube models for SPICE, Part1 / S033-09 Tung-Sol 6C4 manufacturer data

ORIGINAL REAL-TIME ALGORITHM

Solve the connected nonlinear network with a bounded failure path

Valve nonlinear models and solver design

Ngspice documents Newton-Raphson circuit solving and several operating-point convergence aids. [S033-08] Use a circuit simulator as an offline numerical reference, while designing the audio callback around a fixed cost bound. A simulator’s successful DC solve does not establish the timing behavior or accuracy of a real-time implementation.

ORIGINAL DESIGN PROPOSAL: form residuals from the selected network and an implicit treatment of reactive elements. Scale voltage and current equations before comparing residual norms. Compute analytic or independently verified derivatives, warm-start from the previous accepted state and use a bounded damping search. A small Newton step alone is insufficient: the physical equations must also be satisfied.

Keep capacitor and inductor histories unchanged during trial iterations. Commit them only after acceptance. Exceeding the budget invokes a documented bounded fallback and records a diagnostic counter outside the audio output. Holding a prior state or switching to a linear surrogate may avoid non-finite samples, but is a degraded result that fails the nominal fidelity gate. Initialization continuation and parameter fitting belong off the audio thread.

Constraint	Implementation rule
Numerical shunts / limiting	Record as solver aids; test whether they alter the response.
Audio callback	No allocation, blocking, fitting or unbounded retry.
Oversampling	Update every nonlinear and coupled reactive state at the selected internal rate.

```
candidate = previousAcceptedVoltages
for iteration in boundedIterationBudget:
    F, J = assembleResidualAndJacobian(candidate, history)
    if scaledResidualOK(F) and updateOK: accept; break
    delta = solveScaledLinearSystem(J, -F)
    alpha = boundedDampingSearch(candidate, delta)
    candidate += alpha * delta
if accepted:
    commitReactiveHistory(candidate)
else:
    runBoundedFallbackWithoutPartialCommit()
    incrementNonRealtimeDiagnosticCounter()
```

SOURCES S033-08 Ngspice User Manual, version47+ development edition

PARAMETER IDENTIFICATION AND EVIDENCE

Fit documented priors, then measure the selected instrument

Valve nonlinear models and solver design

Koren describes estimating model parameters through optimization, and SciPy supplies bounded least-squares tools with explicit residual, Jacobian and scaling interfaces. [S033-06] [S033-07] These are fitting mechanisms; neither supplies a measured AO-28 profile. No curve digitization, device fitting or instrument capture was completed for this chapter.

ORIGINAL DESIGN PROPOSAL: first digitize the selected manufacturer curves with axis calibration and line-width uncertainty. Keep milliamperes, micromhos and historical capacitance notation alongside converted SI values. Hold out complete bias traces or complete screen-voltage slices; randomly withholding neighboring points from the same smooth trace provides a weak generalization test. Fit positive parameters in transformed coordinates and inspect correlated parameter directions.

Then request qualified reference measurements covering the actual circuit’s bias and signal excursions. Separate device variation from supply, source impedance, load and coupling-capacitor effects. Add grid-current and charge data only when their measurement conditions are known. Validate the connected stage with level-dependent sweeps and transients after the static surfaces pass; harmonic matching at one level cannot uniquely identify the device.

Fit evidence	Store with each record
Manufacturer curve	Sheet revision, page, plotted condition, extraction uncertainty.
Specimen trace	Tube identity, section, thermal condition, calibrated voltages/currents.
Chosen parameters	Model version, units, domain, covariance or identifiability notes.
Held-out result	Untuned residuals by current, slope and voltage region.

MODEL EQUATIONS

Proposed offline objective: J(theta)=sum_j wj*((Ij_model-Ij_data)/sigma_j)^2 + lambda*R(theta).

Ij is a selected terminal current in amperes; sigma_j includes acquisition or digitization uncertainty. R and lambda are declared regularization choices.

CHECKS AND DECISIONS

• Confirm the GE example’s1600micromhos becomes0.0016S. [S033-01]

• Branchwise abs/max kernels need checked derivatives; complex-step differentiation is not automatically valid.

SOURCES S033-01 General Electric 12AX7, ET-T509B, June1953 / S033-06 Norman Koren: Finding SPICE tube model parameters / S033-07 SciPy least_squares official reference

ACCEPTANCE CONTRACT — IMPLEMENTATION NOT YET RUN

Gate device accuracy, circuit accuracy and runtime behavior separately

Valve nonlinear models and solver design

ORIGINAL DESIGN PROPOSAL: maintain three independent results: device surfaces against their source data, the connected digital network against an offline circuit reference, and the complete organ against physical captures. Agreement with the same equation implemented twice is a numerical consistency check, not evidence that the selected valve or amplifier has been reproduced.

Use the matrix below as required tests. Set numerical tolerances from precision and conditioning experiments; set hardware tolerances from measurement uncertainty and the chosen product objective. The GE typical-value calculation on page 1 is only a dimensional check. Actual model coefficients, validated voltage domains and solver budgets remain open deliverables.

Nonlinear products require the oversampling and alias checks in chapter 037. Converged Newton iterations do not guarantee an alias-free output. Supply and bias memory belong to the connected circuit rather than a second arbitrary saturation block. Keep the 8-socket AO-28 identity from chapter 031 while allowing independently evidenced section variation.

Test	Required result	Current status
Terminal conservation	All conduction and charge currents balance under the declared signs.	Not run
Jacobian / unit invariance	Analytic derivatives agree with independent perturbations; common-mode shifts change nothing.	Not run
Domain boundaries	Finite values and declared behavior through cutoff, grid conduction and source/load extremes.	Not run
Connected small signal	Bias, gain, phase and impedance match the offline reference at selected conditions.	Not run
Dynamic recovery	Burst and overload recovery retain capacitor/bias memory and correct polarity.	Not run
Real-time bound	Worst-case processing meets its budget; fallback count is zero in accepted fixtures.	Not run

CHECKS AND DECISIONS

- Reject a profile whose only evidence is a plausible sound or a copied generic coefficient set.
- Report every numerical aid and every unresolved measured parameter in the release evidence.

HARDWARE EVIDENCE / ORIGINAL INTEGRATION CONTRACT

Assign headroom to the routed circuit, not the plugin output

AO28 nonlinear interactions and headroom

The AO-28 drawing separates non-vibrato, vibrato and percussion entries, places expression between the V4 sections, and includes feedback and shared bias/supply connections. V3A drives the scanner line; V3B drives T3. A printed operating voltage is not a clipping threshold. [S034-01] [S034-02]

ORIGINAL DESIGN PROPOSAL: define headroom as a property of a specified excitation, routing, control state and receiving load. Preserve calibrated voltages at each electrical boundary. A normalized audio sample has no intrinsic plate-voltage meaning, and an arbitrary drive knob cannot identify the actual instrument operating region.

Keep the linear reference from chapter 032 as the limiting behavior of the nonlinear model. First enable verified valve laws inside the same connected graph; then add further mechanisms only when measurements justify them. Do not compensate a wrong input scale by simultaneously changing valve parameters, output gain and transformer saturation. Those adjustments can fit one recording while destroying identifiable physical meaning.

Record	Required meaning
Electrical scale	Volts per normalized unit at A, B, H and the selected output boundary
Headroom condition	Excitation spectrum, crest factor, registration, expression, routing and load
Calibration state	Documented nominal, provisional estimate or specimen measurement
Output gain	Host presentation level, kept separate from electrical operating level

CHECKS AND DECISIONS

• No universal AO-28 clipping voltage or dBFS headroom is asserted without a declared measurement condition.

• A change to host output gain must not silently change internal valve drive.

SOURCES S034-01 Hammond factory service manual: AO-28 Figure24 and amplifier theory / S034-02 AO-28 retrospective schematic redraw dated April20,1995

ORIGINAL DERIVATION / SHARED SIGNAL PATHS

Preserve nonlinear interactions after signals meet

AO28 nonlinear interactions and headroom

ORIGINAL DERIVATION: a nonlinear operation applied to a sum generally differs from summing separately processed signals. The polynomial below is a diagnostic example, not a fitted AO-28 characteristic. Its cross-terms explain why per-note saturation cannot reproduce a common amplifier driven by several notes or manuals.

Build each physical input bus before advancing its nonlinear stage. Preserve generator-phase relationships, repeated-source foldback and actual drawbar contributions; peak drive is not determined by the number of held keys alone. When separate branches later meet, retain their common downstream stage and feedback. Research on circuit block composition likewise treats interstage loading as part of the connection. [S034-03]

Use controlled combinations to locate interactions. Compare A-only, B-only and simultaneous excitations under identical state and loading. A nonzero combined-minus-separated residual is informative only after linear timing, noise and scanner modulation have been accounted for. Percussion routing and its Normal/Soft compensation must remain correct before attributing a level change to saturation.

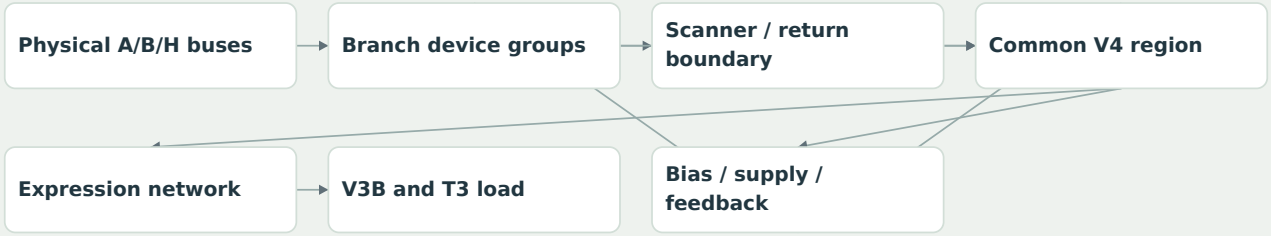
MODEL EQUATIONS

Diagnostic polynomial: $F(x)=c1*x+c2*x^2+c3*x^3$, with $F(0)=0$.

$F(a+b)-F(a)-F(b)=2*c2*a*b+3*c3*a*b*(a+b)$.

Here a and b are simultaneous input signals; c1,c2,c3 are illustrative coefficients, not measured tube parameters.

Shared drive and electrical return paths



Original dependency graph, not a replacement schematic. Shared nonlinear processing follows actual connection points; no additional coupling magnitude is invented.

CHECKS AND DECISIONS

- The implementation must retain cross-products generated by its declared shared nonlinear network.

SOURCES S034-03 Yeh and Smith: Simulating Guitar Distortion Circuits Using Wave Digital and Nonlinear

SOURCE-TRACED NETWORK / ORIGINAL MODEL PROPOSAL

Treat bias, stored charge and feedback as connected state

AO28 nonlinear interactions and headroom

The legible redraw returns V1 and V4 cathodes to the V2-associated bias reference and shows its R17/C10 network. It also makes C22 feedback and the expression connections explicit. Resolve these references as connected nodes; do not replace every printed bias annotation with an independent ideal voltage source. [S034-02]

ORIGINAL DESIGN PROPOSAL: retain the relevant capacitor charges, inductor currents and bias-node dynamics in one instance-level electrical state. Candidate effects include level-dependent bias shifts, asymmetric clipping and history-dependent recovery. Their audible magnitude is unknown here. A device law plus stored charge can produce memory even when the instantaneous device-current law is memoryless.

Do not add generic sag, hysteresis or a slew limiter merely because the instrument contains valves and transformers. Identify each proposed effect with a circuit element, a boundary assumption and an observable test. Shared glass alone does not establish coupling between V3A and V3B. Conversely, a shared traced bias or supply node must not be separated to simplify the code without a quantified approximation.

Mechanism candidate	Model boundary and evidence needed
Grid conduction and coupling charge	Use a supported grid-current law; measure post-burst recovery before claiming blocking behavior
Cathode / shared bias movement	Stamp the connected return network; measure dependence on combined drive
Supply modulation	Use chapter 035 supply impedance or state an ideal-rail approximation
C22 and expression interaction	Keep feedback inside the coupled solve; compare several fixed pedal positions
T3 nonlinear magnetics	Require identified magnetic behavior; retain the linear transformer baseline meanwhile

CHECKS AND DECISIONS

• An observed transient must not be assigned to one mechanism solely because that mechanism can imitate its shape.

SOURCES S034-02 AO-28 retrospective schematic redraw dated April20,1995

ORIGINAL ALGORITHM / REAL-TIME CONTRACT

Solve bounded nonlinear groups on one electrical timeline

AO28 nonlinear interactions and headroom

State-space formulations separate stored-energy state from nonlinear device currents, while preserving their implicit dependence. [S034-03] Ngspice normally establishes a DC solution before transient analysis. [S034-04] Use a consistent operating-point initialization, then compare the proposed runtime solver with a reproducible offline circuit reference.

ORIGINAL DESIGN PROPOSAL: start verification with the charge-based backward-Euler residual below. It is a deliberately explicit reference formulation, not a claim that this integrator is the best production sound model. Let z contain node voltages and necessary branch currents, q contain charge/flux terms, u the electrical inputs, p the controls, and h the internal sample interval. Retaining previous control values in q prevents an abrupt capacitance change from silently rewriting stored charge.

Preallocate graph structures, Jacobian storage and oversampling buffers. Scale residual rows by physical units, limit iterations and monitor both residual and update size. A bounded fallback prevents an audio-thread stall; any fallback activation fails the fidelity gate. Keep every connected feedback group on the same internal timeline and retain resampler delay/state in comparisons.

MODEL EQUATIONS

$$r(z_n)=[q(z_n,p_n)-q(z_{previous},p_{previous})]/h + g(z_n,u_n,p_n)=0.$$

```
prepare: resolve graph; initialize consistent DC state
for each internal sample:
  apply sample-stamped control targets
  candidate = previousAcceptedState
  for iteration in boundedIterationBudget:
    residual, jacobian = assembleCoupledSystem(candidate)
    if weightedResidualAndUpdateConverged: break
    step = solvePreallocated(jacobian, -residual)
    candidate = boundedDampedUpdate(candidate, step)
  if converged: commitStateAndObservePorts(candidate)
  else: countFailureAndUseDeclaredBoundedFallback()
  advanceSharedResamplerAndDiagnosticState()
```

CHECKS AND DECISIONS

• Choose integration rate and iteration budget from measured convergence and alias error; report them with the profile.

SOURCES S034-03 Yeh and Smith: Simulating Guitar Distortion Circuits Using Wave Digital and Nonlinear / S034-04 Ngspice User Manual, version47+ development edition

ORIGINAL ACQUISITION AND IDENTIFICATION PLAN

Capture a level-and-history matrix before fitting character

AO28 nonlinear interactions and headroom

Have the acquisition technician provide calibrated direct-preamp fixtures with instrument identity, source/load conditions and voltage references. Record the intended output legs synchronously. A Leslie microphone recording includes additional nonlinear and acoustic stages and cannot alone identify AO-28 headroom. Preserve absolute scale; independently normalizing each take hides compression.

Begin with fixed expression, registration and routing. Establish a small-signal reference, then increase a fixed-spectrum excitation over a documented amplitude ladder. Repeat at several frequencies and pedal positions. Use bursts followed by a small probe to expose recovery or bias memory, and paired inputs to expose shared-stage interaction. Keep quiet-state and repeat captures to estimate uncertainty.

Define reported thresholds before fitting. For example, a one-decibel fundamental compression point is a proposed project metric, not a Hammond specification. Report harmonic and intermodulation spectra, polarity, waveform asymmetry and recovery alongside that scalar. A low-frequency rolloff or loading error must not be fitted away as a nonlinear character parameter.

Capture family	Identification purpose
Fixed-spectrum level ladder	Gain compression, harmonic growth and asymmetry
Two-tone / paired-route signals	Shared nonlinear interactions and path dependence
Burst, silence, small probe	History-dependent recovery and baseline return
Expression positions and motion	Drive redistribution versus final presentation gain
Held-out registrations / loads	Generalization beyond the fitted waveform

CHECKS AND DECISIONS

- Publish input calibration, peak and RMS levels, crest factor, repeatability and the fitting/holdout split.

ORIGINAL ACCEPTANCE / DELIVERABLES

Accept the connected model before exposing a drive control

AO28 nonlinear interactions and headroom

Deliver a revision-specific graph, device-model provenance, operating-point record, electrical scaling contract, bounded solver and an offline reproduction script. Keep numerical agreement with the chosen circuit separate from agreement with hardware. The latter remains untested until physical captures exist.

Reject a model that matches a loud chord but loses its calibrated small-signal response, fails after a control transition, or depends on host block size. Compare high-rate references with latency accounted for; nonlinear aliasing can resemble upper harmonics while having the wrong pitch dependence. Chapter 037 supplies the detailed anti-aliasing budget.

A reduced or hybrid model is acceptable only within a documented validity region supported by held-out tests. Record which interactions it preserves and which it approximates. A creative overdrive extension can exceed that region as a separate feature; it must not silently change the baseline instrument profile or become evidence of vintage accuracy.

Gate	Required evidence
Small-signal limit	Convergence to chapter 032 response at declared controls and load
Combined-input response	Correct interaction residual against the chosen circuit reference
State and control history	Repeatable burst recovery and transitions without state resets
Numerical behavior	Bounded work; no NaN/Inf; zero fallback events in accepted fixtures
Rate and block size	Residuals within predeclared budgets after timing alignment
Specimen fidelity	Held-out physical captures pass uncertainty-aware limits; currently not run

CHECKS AND DECISIONS

• No completed AO-28 nonlinear implementation, physical headroom measurement or fidelity score is claimed by this chapter.

HARDWARE EVIDENCE / PROFILE BOUNDARY

Treat the supply as a connected circuit, not a hum preset

Power supply hum grounding and noise

The factory AO-28 drawing shows T7 feeding the V8 6X4 rectifier, a reservoir and several resistor-capacitor supply branches, together with heater connections. RCA identifies6X4 as a full-wave rectifier and presents operating curves with specified filter and source impedances. Tube identity alone therefore does not define the loaded rail response. [S035-01] [S035-03]

ORIGINAL DESIGN PROPOSAL: import the supply graph with the audio netlist. Store each rail in volts, each branch current in amperes and each capacitor state separately. Printed DC annotations are documentary operating references, not measurements of the chosen specimen. Record transformer resistance, capacitor condition and connected stage loads as unresolved until evidence establishes them.

The 1995 redraw is a useful visual cross-check but appears to repeat the C56 identifier in a later filter branch. Resolve that label against the factory sheet and chapter 038 inventory before assigning persistent component IDs. Do not manufacture a missing label from a neighboring number. [S035-02]

Documented block	Required model contract
T7 and heater winding	Mains profile; winding source impedances and coupling remain specimen data.
V8 and first reservoir	Rectifier current law or measured rail surrogate; identify which is active.
RC supply branches	Preserve individual states and their connected stage currents.
Reference/chassis connections	Ideal schematic reference is distinct from measured interference transfer.

CHECKS AND DECISIONS

- Reject duplicate circuit identifiers during graph import.
- Do not substitute Leslie power-stage sag for AO-28 supply behavior.

SOURCES S035-01 Hammond factory service manual: AO-28 Figure24 power supply / S035-02 AO-28 retrospective schematic redraw, April20,1995 / S035-03 RCA Receiving Tube Manual RC30: 6X4/12X4

ORIGINAL MECHANISM MODEL

Keep mains-related components coherent across the instrument

Power supply hum grounding and noise

ORIGINAL DERIVATION: an ideal symmetric full-wave rectifier charges on both mains half-cycles, so its basic ripple repetition is twice the mains frequency:100 Hz for 50 Hz, or120 Hz for 60 Hz. This predicts candidate frequencies, not their amplitudes at the organ output. Imbalance, heater coupling and other paths can also contribute at the mains fundamental. The actual supply topology supplies the boundary conditions. [S035-01] [S035-03]

ORIGINAL DESIGN PROPOSAL: share one electrical phase reference across modeled mains-related sources, with separately calibrated path phases and gains. A heater-related term, rail ripple and magnetic pickup are different injection paths. Their correlation should not be recreated with independent oscillators that drift accidentally. Mechanical modulation from chapter 013 remains separately identified even when it shares a mains reference.

Represent grounding effects as documented or fitted transfer paths between physical references and signal nodes. Preserve the distinction between the instrument, its connecting cable, the Leslie input and the recording chain. A hum component introduced by the capture installation must not silently become a permanent characteristic of the organ profile.

P036 | FRec var.jpg | RJB1 | CC BY-SA 4.0 | Tube type illustration; fitted specimen brand is unverified. Source and license

CHECKS AND DECISIONS

• Changing note count cannot create additional independent mains oscillators.

SOURCES S035-01 Hammond factory service manual: AO-28 Figure24 power supply / S035-03 RCA Receiving Tube Manual RC30: 6X4/12X4

ORIGINAL EQUATIONS / NUMERICAL DESIGN

Model loading and rail ripple without counting either twice

Power supply hum grounding and noise

ORIGINAL DESIGN PROPOSAL: choose either a coupled supply solver or an explicitly identified reduced model. In the coupled version, rectifier current, stored charge and instantaneous stage currents determine the rails. In a reduced version, use measured periodic rail traces plus identified small-signal load responses. Adding a second decorative ripple signal to a supply that already produces ripple would double-count the same mechanism.

For reservoir voltage V and capacitance C, conservation of charge gives the first equation below. The simple ripple estimate assumes nearly constant load and short recharge intervals; it is a diagnostic, not a fitted AO-28 specification. Distributed RC branches and time-varying valve currents invalidate a single universal sag time constant.

Start with steady operating points, then a periodic steady state, and finally coupled audio loading. Resolve rectifier conduction and nonlinear interaction at a rate justified by convergence tests. Oversampling only the final waveshaper does not repair aliasing already created by supply-dependent gain or clipping. Keep initialization, warm-up proposals and shutdown behavior explicitly separate from measured instrument behavior.

MODEL EQUATIONS

C*dV/dt = I_rectifier - I_local_load - sum(I_branch); C in farads, V in volts, currents in amperes.

Small-ripple diagnostic: DeltaV_pp approximately I_load/(2*f_mains*C).

Reduced model: deltaV_s(f) = sum_j Z_sj(f)*deltaI_j(f); Z_sj is a signed load-to-rail transimpedance in ohms.

Periodic surrogate: V_s(t)=Vbar_s+sum_m A_sm*cos(m*theta_mains(t)+phi_sm); s indexes rails, m harmonics; A is volts, phases radians.

CHECKS AND DECISIONS

- Compare candidate internal rates against a higher-rate offline reference.
- A rail perturbation acts at its connected stages, not automatically on every audio branch.

SOURCES S035-01 Hammond factory service manual: AO-28 Figure24 power supply / S035-03 RCA Receiving Tube Manual RC30: 6X4/12X4

IMPLEMENTATION / STOCHASTIC MODEL

Generate noise with physical units and reproducible state

Power supply hum grounding and noise

Ideal resistor thermal-noise density depends on resistance and absolute temperature; integrated noise depends on the receiving bandwidth. These general relationships do not establish the AO-28 output noise floor. [S035-04] ORIGINAL DESIGN PROPOSAL: keep each physical source in its appropriate voltage or current units and propagate it through the connected circuit or an identified equivalent transfer.

The discrete variance below assumes a flat one-sided voltage power spectral density S1 through Nyquist, before additional filtering. When internal sample rate changes, rescale the source and preserve the final physical bandwidth. Otherwise an oversampling option changes the hiss level. For colored noise, include finite low-frequency and high-frequency limits; an unbounded1/f model has no usable stationary variance.

Use a deterministic source identifier and seed, retaining generator and filter state across blocks. Independent device sources may become correlated after passing through shared networks. Conversely, copied noise buffers should not accidentally correlate unrelated channels. A reduced covariance model must be positive semidefinite and fitted to appropriate captures; arbitrary stereo widening is not an electrical-noise model.

State or parameter	Implementation rule
Mains phase	Continuous across notes and blocks.
Random-source identity	Stable per physical source; explicit deterministic seed.
Noise filter memory	Persistent; initialized outside the audio callback.
Unknown amplitude	Null in evidence data; any audition default is labeled proposed.

MODEL EQUATIONS

S1 = 4*k_B*T*R [V^2/Hz]; k_B is Boltzmann's constant, T is kelvin, R is ohms.

For the stated flat-band convention: sigma_discrete = sqrt(S1*Fs_internal/2).

After filtering: variance_out = integral_0^(Fs_internal/2) S1(f)*abs(H(f))^2 df.

SOURCES S035-04 Texas Instruments SBOA345: Understanding Op Amp Noise in Audio Circuits

PROPOSED CAPTURE AND IDENTIFICATION

Measure quiet-state behavior before assigning noise to parts

Power supply hum grounding and noise

ORIGINAL CAPTURE PLAN: obtain calibrated direct-output recordings under a documented load, alongside the recording system’s own noise baseline. Internal rail or heater measurements belong to a qualified technician’s isolated measurement plan; this chapter does not specify live probing or alterations to protective grounding. Record the completed measurement chain rather than reconstructing it from an unexplained audio file.

Collect repeated quiet windows with keys released and the generator running. Vary expression position, drawbar state, scanner routing and percussion selection in a controlled matrix. Compare which spectral components follow each control. That dependence helps identify an injection region, but it does not uniquely prove a particular faulty component. Preserve room and microphone noise separately from the electrical direct signal.

Estimate mains frequency from the recording or a suitable isolated reference, rather than forcing exact50/60 Hz FFT bins. Fit narrow deterministic components before estimating the stochastic floor. Retain complex cross-spectra where simultaneous channels exist. Use repeat windows and held-out states to distinguish stationary noise, drift, intermittent defects and changing acquisition conditions.

Capture record	Why it is required
Control and routing snapshot	Identifies the transfer acting on each noise source.
Gain, load and volts-to-sample scale	Makes comparisons meaningful across sessions.
Mains estimate and window duration	Separates narrow harmonics from spectral leakage.
Repair state, temperature, elapsed run time	Keeps condition-dependent behavior tied to its specimen.

CHECKS AND DECISIONS

- No numerical noise-floor claim is accepted without a calibrated bandwidth and measurement record.

SOURCES S035-01 Hammond factory service manual: AO-28 Figure24 power supply / S035-04 Texas Instruments SBOA345: Understanding Op Amp Noise in Audio Circuits

ACCEPTANCE CONTRACT

Separate nominal residue, specimen aging and recording faults

Power supply hum grounding and noise

ORIGINAL DESIGN PROPOSAL: deliver SupplyProfile, NoiseProfile and their provenance together. The supply interface returns named node voltages and consumes connected stage currents; the noise interface returns identified port disturbances. Runtime state includes mains phase, capacitor states, random-source counters and noise-filter memories. Profiles may describe a documented restored instrument or a particular aged specimen, but their labels must match the evidence.

Keep intermittent crackles, degraded filtering and external ground-loop contamination in explicit condition profiles. Their presence in one recording is not evidence that every B-3 should reproduce them. Likewise, a perfectly silent audition mode is a product option, not a measured claim about the original hardware. The unmeasured baseline remains visible until a capture campaign supplies it.

Release these modules only after numerical fixtures and held-out recordings answer different questions. Circuit and stochastic fixtures establish that the code obeys its chosen model; reference recordings establish whether that model describes the selected instrument. Neither a visually plausible spectrum nor a small solver residual substitutes for the other.

Acceptance fixture	Required result; status here
Zero interference coefficients	No added hum or hiss; implementation test pending.
Block-size permutations	Identical deterministic output for equal events and state; pending.
Sample-rate changes	Same integrated physical noise band within estimated statistical uncertainty; pending.
Rail-load synthetic fixture	Charge conservation and convergent response; pending.
Held-out quiet/control captures	Matches measured lines, floor and correlations within declared uncertainty; no captures yet.

SOURCES S035-01 Hammond factory service manual: AO-28 Figure24 power supply / S035-04 Texas Instruments SBOA345: Understanding Op Amp Noise in Audio Circuits

EVIDENCE AND SCOPE

Define condition as a dated state, not a birth-year slider

Aging tolerances and specimen profiles

Hammond Bulletin 16 documents a filtering change for generator sources 37–48, with B-3 serial 93990 as a production threshold and provision for retrofitting earlier instruments. Consequently, production date alone cannot identify the installed electrical network. [S036-01] This is a revision and service-history issue, not evidence that an older component gradually becomes the newer circuit.

ORIGINAL DESIGN PROPOSAL: describe an instrument through four separate records: factory identity, inspected configuration, dated condition observations and service events. An as-found profile represents a captured state. A restored profile represents another documented state after specified work. Neither label establishes how the instrument sounded when newly manufactured. Keep unobserved original values unknown.

Calendar age may be searchable metadata, but it must not silently drive gain loss, pitch instability, extra leakage or distortion. Begin the emulator with a selected documentary topology and explicit provisional parameters. Replace those parameters with specimen observations as evidence becomes available. A cosmetic wear control may exist as an artistic extension, with a name that does not imply calibrated physical age.

Record	Required distinction
Identity	Model, serial, chassis and mains variant
Configuration	Installed networks and modifications
Condition	Dated observations with test conditions
Service event	What changed; before/after evidence
Artistic extension	Proposed effect, outside measured profile

CHECKS AND DECISIONS

- A pre-threshold serial with a verified retrofit selects the inspected topology.
- A profile named restored must identify its service event or remain explicitly hypothetical.

SOURCES S036-01 Hammond Technical Bulletin 16: Improved Generator Filtering

COMPONENT EVIDENCE / MODEL OWNERSHIP

Assign each deviation to a mechanism and observation boundary

Aging tolerances and specimen profiles

Manufacturer guidance describes electrolytic-capacitor characteristics as dependent on temperature, frequency and construction; resistor stability likewise requires stated test conditions. [S036-02] [S036-03] These references explain what to measure. Their modern product families do not supply a universal aging curve for the unidentified components in a particular AO-28 or generator.

Arnold distinguishes magnetic-output variation from losses associated with thermal or demagnetizing stress. It does not identify the selected Hammond magnet material. [S036-04] A weaker recorded tone therefore cannot, by itself, prove magnet degradation: pickup geometry, loading and downstream transfer are competing explanations. Record the observed amplitude first; assign a physical cause only when independent evidence supports it.

ORIGINAL DESIGN PROPOSAL: attach each condition parameter to one model boundary. A captured loaded generator waveform already incorporates the components upstream of that recording point. Applying another capacitor-aging filter to the same response would count one deviation twice. The same rule applies to amplifier noise or distortion already embedded in a measured residual model.

Candidate condition	Useful observation	Avoid assuming
Capacitor change	Impedance versus frequency and temperature	One drift law for every dielectric
Resistor change	Resistance and operating conditions	Tolerance equals aging amount
Pickup variation	Waveform, geometry and loading	Amplitude uniquely measures magnet strength
Contact condition	Resistance and gesture-tagged transitions	Every difference is wear
Mechanical condition	Shaft/rotor timing and repeated runs	Independent random detuning per note

CHECKS AND DECISIONS

• Every modeled deviation declares whether it is component-level or already present in a measured transfer/waveform.

SOURCES S036-02 TDK: Capacitors, Part7 — Electrolytic Capacitors [2] / S036-03 Vishay Beyschlag: Basics of Linear Fixed Resistors / S036-04 Arnold Magnetic Technologies: Stabilization & Calibration

ORIGINAL STATISTICAL MODEL

Preserve correlations without inventing a population

Aging tolerances and specimen profiles

ORIGINAL DESIGN PROPOSAL: represent repeatable specimen differences separately from run-to-run fluctuations and parameter uncertainty. A fixed pickup-amplitude pattern belongs to the specimen. A recorded warm-up trajectory belongs to an operating run. Uncertainty describes what the researcher does not know; it should not automatically become audible modulation.

If a sufficiently varied dataset later supports generated specimen profiles, use shared latent factors for related components or mechanisms, plus smaller local residuals. Estimate their covariance from the appropriate population and preserve its uncertainty. One instrument cannot establish between-instrument distributions. A component tolerance bound also does not determine a Gaussian distribution, independence or a particular aging trajectory.

For positive component parameters, the proposed log-domain form below prevents negative resistance, capacitance or inductance. The scale values and covariance remain unset until justified. A user-requested fictional profile may use explicitly proposed distributions, but it must be saved under an artistic profile identity. Draw that profile once; do not redraw its physical components for every note.

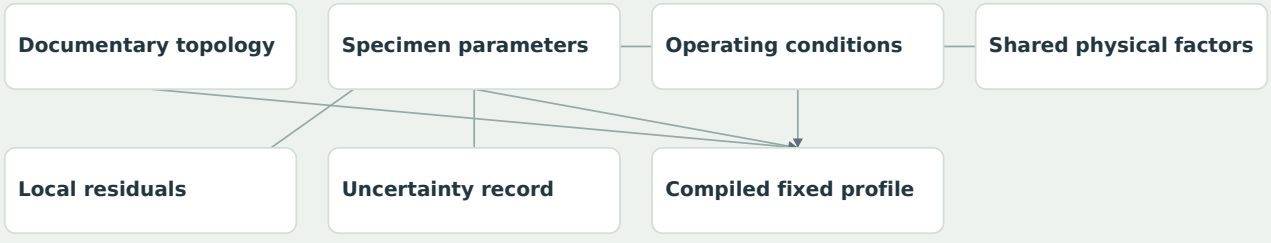
MODEL EQUATIONS

Proposed positive parameterization: p_i = p_ref,i * exp((B*z)_i + e_i).

z: shared latent factors; e_i: local residual; B: fitted coupling matrix. No numerical distribution is established here.

Feature sensitivity J_ki = partial feature_k / partial parameter_i. Nearly dependent columns indicate ambiguous attribution.

Separate the sources of variation



Original proposed data model. Uncertainty is metadata unless an explicitly selected simulation uses it.

CHECKS AND DECISIONS

- Unknown covariance stays unknown; it is not replaced by an undocumented random seed.

IMPLEMENTATION CONTRACT

Make service history reversible in the software

Aging tolerances and specimen profiles

ORIGINAL DESIGN PROPOSAL: store the documentary base and each dated observation as immutable records. Express a service comparison as a new profile with explicit replacements or measured deltas, retaining links to the original capture and part inventory. Do not overwrite the as-found dataset when preparing a restored profile. Registration presets should reference a profile identifier rather than duplicating its component values.

Compile validated profiles away from the audio thread. Resolve units, topology dependencies, waveform boundaries and missing values before publishing a complete immutable snapshot. Reject physically invalid values and unsupported circuit combinations. A topology change is not safely represented by interpolating every old parameter toward a new one; retain separate compiled networks where needed.

For a permitted live comparison, preserve continuous generator and scanner phase and align any parallel rendering paths. Transfer circuit states only through a defined mapping; otherwise use a controlled transition between independently initialized engines. Chapter 029 covers state constraints. This transition is an implementation policy, not a claim about what a technician changing a physical component would sound like.

Persist	Keep separate
Profile/version/hash	Current drawbar registration
Replacement provenance	User interface wear appearance
Measurement conditions	Host buffer and device configuration
Evidence and uncertainty	Artistic randomization settings

```
profile = loadImmutableProfile(profileId)
verifyEvidenceLinksAndUnits(profile)
resolveTopologyAndObservationBoundaries(profile)
rejectMissingRequiredCalibration(profile)
compiled = buildAndValidateOffAudioThread(profile)
plan = chooseCompatibleStateTransition(active, compiled)
publishSnapshotAtDeclaredSampleBoundary(compiled, plan)
retainPreviousProfileForReversibleComparison()
```

CHECKS AND DECISIONS

- Reloading the same profile produces the same fixed instrument parameters.
- Profile publication performs no file access, allocation or unbounded solve in the audio callback.

MEASUREMENT PLAN — NOT YET RUN

Acquire paired evidence before attributing a sound to aging

Aging tolerances and specimen profiles

The published contact study used a recently serviced 1967 C-3 and found gesture-dependent contact transients. [S036-05] It is useful evidence that technique and condition must be separated, but it does not define what an unserviced B-3 of the same age should do. Do not fit a wear coefficient to recordings made with different key gestures.

ORIGINAL ACQUISITION PROPOSAL: arrange dated, paired captures before and after a documented service event, with qualified personnel handling internal instrument work. Keep the external measurement chain, loading, controls and environment matched. Record thermal history and repeat runs sufficiently to estimate within-session variation. Declare the repetition plan and stability criterion before examining the desired result.

Photograph relevant assemblies and record changed part identities without assuming unchanged parts are original. Compare generator outputs, direct organ outputs and rotary acoustics at separate boundaries. Keep raw level calibration; a loudness-matched listening copy is a derived artifact, not a replacement for the electrical recording. Additional instruments are necessary before generalizing a measured difference into a family profile.

Comparison	Control or record
Before / after service	Same topology unless the service changes it explicitly
Generator capture	Pickup identity, load, phase reference and temperature
Keyboard capture	Key, gesture, repetition and contact path
Amplifier capture	Input level, supply conditions, expression and load
Leslie capture	Microphones, room, driver state and rotor trajectory

CHECKS AND DECISIONS

• A tonal change following replacement is associated with that intervention; causation among simultaneously changed components remains unresolved.

SOURCES S036-05 Moro, McPherson and Sandler: Dynamic temporal behaviour of the keyboard action on the

ACCEPTANCE AND OPEN DECISIONS

Accept a specimen match separately from an aging model

Aging tolerances and specimen profiles

ORIGINAL ACCEPTANCE CONTRACT: first reproduce the selected dated specimen under its recorded conditions. Only then evaluate whether a profile family generalizes to unseen instruments, service states or operating conditions. Keep whole instruments and complete sessions out of fitting when they are used for that evaluation; adjacent excerpts from the same capture are weak independence evidence.

Set numerical tolerances from reference repeatability, measurement uncertainty and the project's declared perceptual criteria. No universal drift percentages, allowable spectral errors or aging-rate coefficients have been established here. A failed identification test should reduce the number of independently fitted causes, not encourage compensating parameter changes that only match one recording.

Optional faults and extreme wear effects belong in explicitly selectable extensions. Their presence is not a requirement for vintage authenticity. Preserve the calibrated profile as a stable reference and make every artistic alteration reversible. The result of this chapter is a profile and evidence contract; neither a measured population of B-3 organs nor a verified aging simulation has been produced.

Gate	Required evidence or result
Boundary ownership	No response duplicated in waveform and circuit layers
Attribution	Distinguishable parameter effects or grouped uncertainty
Reproducibility	Same fixed profile across block sizes and reloads
Numerical validity	Finite outputs and valid passive parameters across supported settings
Specimen agreement	Held-out captures under matched conditions
Population claim	Independent instruments and documented service histories

CHECKS AND DECISIONS

- Reject a calendar-age preset presented as measured when its parameters are proposed.
- Retain missing calibration as a visible blocker rather than reporting the acceptance gate as passed.

DIGITAL MECHANISM / ORIGINAL ERROR BUDGET

Locate aliasing before choosing an oversampling factor

Nonlinear oversampling and anti aliasing

Nonlinear processing can generate frequencies above the processing Nyquist limit; oversampling moves that limit but does not generally make an unbounded harmonic series disappear. [S037-01] Numerical convergence of a valve solver therefore does not establish spectral accuracy.

ORIGINAL DESIGN PROPOSAL: assign separate error budgets to source generation, contact/control modulation, nonlinear circuit solving and final decimation. A base-rate contact discontinuity can already contain folded energy before an oversampled amplifier receives it. Upsampling that damaged sequence cannot identify which in-band components were aliases. Render the modeled contact event at the required internal resolution or use a justified bandlimited event representation.

Keep intended key-click bandwidth and actual circuit filtering in the reference. Arbitrarily smoothing a contact edge can reduce an alias metric while changing the instrument. Likewise, a correct output lowpass cannot remove an alias that already occupies a legitimate passband frequency. Inspect internal boundaries as well as the final output, and distinguish physical nonlinear harmonics from sampling artifacts.

Origin	Required control
Waveform generation	Bandlimit each source representation at its actual evaluation rate
Contact and parameter motion	Preserve event time and the modeled physical transition
Valve / magnetic nonlinearity	Choose rate or admissible antialiasing method from measured numerical error
Rate reduction	Filter before every decimation boundary

MODEL EQUATIONS

For a generated frequency f, its folded location is f_alias = abs(((f+Fs/2) mod Fs)-Fs/2). Fs is the sampling rate at which aliasing first occurs.

CHECKS AND DECISIONS

• Do not describe a fixed2×,4× or8× setting as universally alias-free.

SOURCES S037-01 Parker, Zavalishin and Le Bivic: Reducing the Aliasing of Nonlinear Waveshaping Using

ORIGINAL MULTIRATE ARCHITECTURE

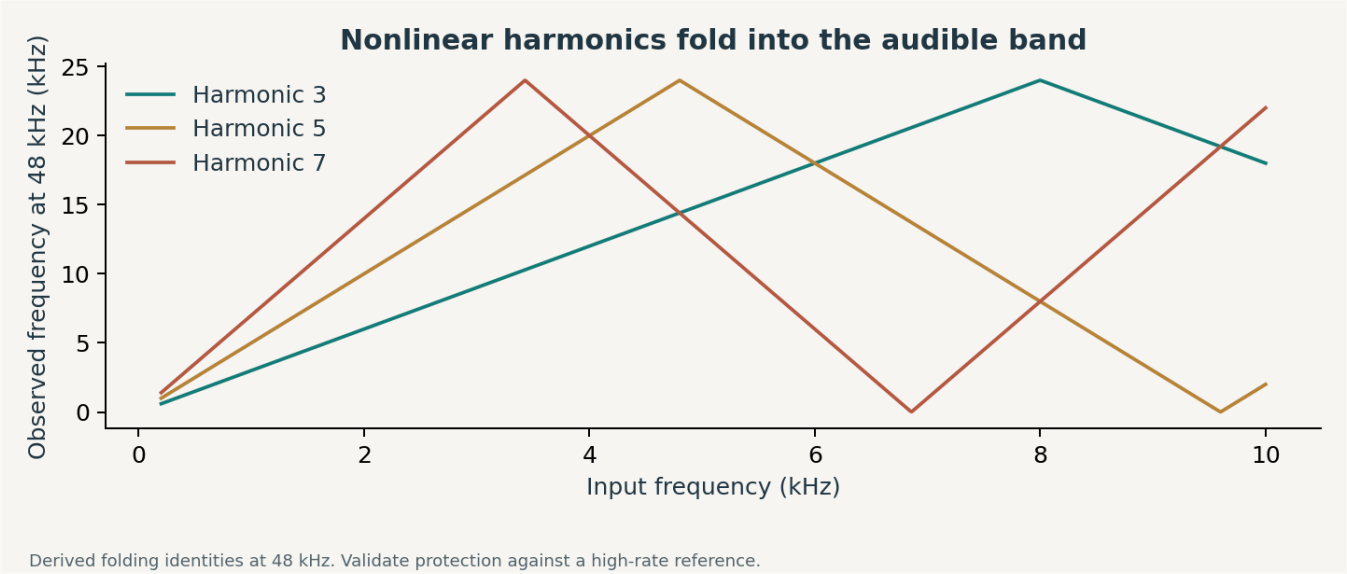
Keep connected feedback groups on one internal clock

Nonlinear oversampling and anti aliasing

ORIGINAL DESIGN PROPOSAL: let the internal rate be Fint=R*Fs and the interval h=1/Fint. Advance a directly generated tonewheel phase by frequency/Fint, not frequency/Fs on every internal tick. Importing existing base-rate audio instead requires an interpolation filter. Keep contact events, controls and all states inside a connected feedback group on the same declared timeline.

Recompute discretization coefficients from physical units whenever the rate changes. Preserve capacitor charge and inductor current through a documented state mapping; clearing histories creates a different transient. Control interpolation, envelope time constants and noise power per unit bandwidth must retain their intended physical interpretation at every R.

JUCE offers FIR and IIR oversampling variants, preparation APIs and latency reporting; its constructor factor is an exponent, so a factor argument3 requests8×. [S037-04] Pin the implementation version. Prepare buffers and replacement configurations outside the audio callback, then measure transport delay and align parallel paths. Physical circuit group delay remains part of the model.



Derived folding identities at 48 kHz. Validate protection against a high-rate reference.
Original explanatory calculation. Illustrative parameters are not measured instrument data.

CHECKS AND DECISIONS

- Benchmark proposed2×,4×,8× and 16× candidates against the same reference and workload.
- Preserve shared oscillator phase when the quality setting changes.

SOURCES S037-04 JUCE dsp::Oversampling official class reference

ORIGINAL DERIVATION / MEMORYLESS CANDIDATE

Treat first-order ADAA as a different discrete-time operator

Nonlinear oversampling and anti aliasing

For a suitable scalar memoryless function f, first-order ADAA averages the function along the line segment between successive inputs, using an antiderivative F. The published construction also discusses near-equal arguments and its linear filtering effect. [S037-01] It is a candidate numerical operator, not a generic substitute for a coupled tube model.

ORIGINAL DERIVATION: applying the formula to f(x)=x gives the two-sample average below. Its half-sample low-level delay and cosine magnitude response must be included when comparing or mixing paths. Calling this operator transparent at small signal would hide a frequency-response error.

Use a units-aware near-equality test and a stable divided difference. The midpoint limit preserves the intended timing; blindly returning f(x_n) changes that limit. For tanh, evaluate log(cosh(x)) through abs(x)+log1p(exp(-2*abs(x)))-log(2), avoiding overflow at large finite arguments. Test branch continuity and approximation error over the actual scaled domain. Multivariable device currents and stored charge require a separate derivation. If x is voltage and f returns current, F carries ampere-volt units; retain that scaling in any implementation.

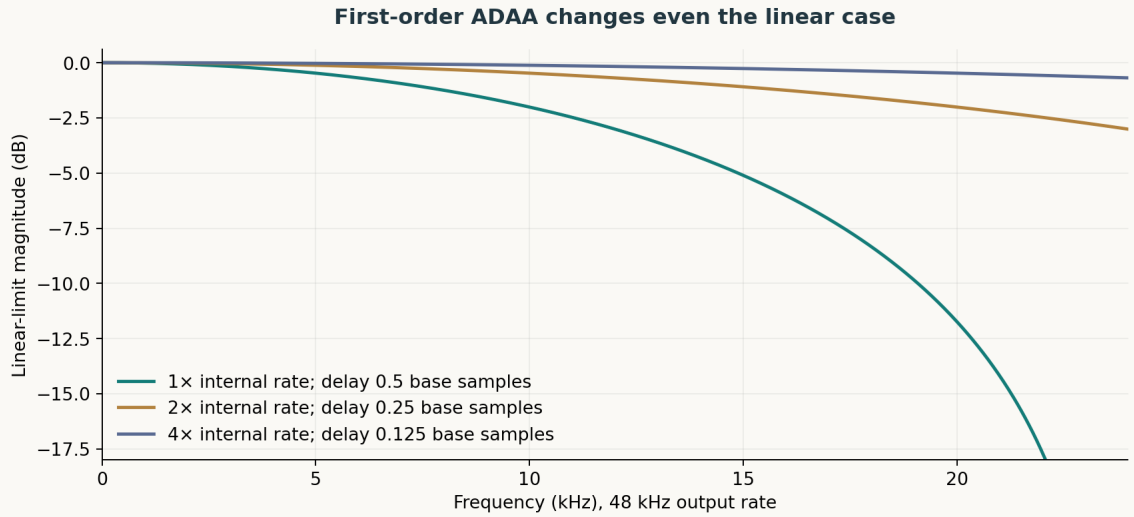
MODEL EQUATIONS

F′(x)=f(x); y_n=[F(x_n)-F(x_(n-1))]/[x_n-x_(n-1)].

For nearly equal arguments: y_n approaches f((x_n+x_(n-1))/2).

For f(x)=x: H(z)=(1+z^-1)/2; |H(frequency)|=|cos(pi*frequency/Fint)|.

Cubic fixture f(x)=x^3: y=(a^3+a^2*b+a*b^2+b^3)/4, where a and b are successive inputs.



Original mathematical calculation: y[n] = (x[n] + x[n−1])/2 for f(x)=x.
The internal-rate moving average adds both filtering and fractional latency. This is not an AO-28 response or an oversampling quality verdict.

Original explanatory calculation. Illustrative parameters are not measured instrument data.

CHECKS AND DECISIONS

• Validate against independent numerical integration, including equal, nearly equal and opposite-sign inputs.

SOURCES S037-01 Parker, Zavalishin and Le Bivic: Reducing the Aliasing of Nonlinear Waveshaping Using

STATEFUL METHODS / IMPLEMENTATION GATES

Do not insert ADAA delay blindly into an implicit circuit

Nonlinear oversampling and anti aliasing

The 2020 WDF construction explicitly treats dynamic circuits with one nonlinear element. [S037-02] The 2022 compensation paper addresses changes to response and stability introduced by standard antiderivative methods in stateful systems. [S037-03] Neither statement validates arbitrary substitutions throughout the selected AO-28 graph.

ORIGINAL DESIGN PROPOSAL: establish the connected oversampled circuit reference first. Introduce an antialiasing substitution only after deriving its incident/reflected variables, extra histories and feedback consequences. Verify the complete small-signal network and transient stability. A low alias floor does not excuse shifted poles, changed bias or missing shared-stage interactions.

Keep the production implementation bounded: preallocate, retain accepted physical histories during solver trials and count failures. Prepare a quality change as a new configuration, migrate compatible state and align paths before a controlled transition. Host latency must match the delivered signal; JUCE can add compensation to make its resampler latency integral. [S037-04] Do not simulate a negative live delay by discarding samples.

Candidate block	Decision
Standalone scalar waveshaper	ADAA may be evaluated with explicit response and delay accounting
Coupled multi-valve group	Require a derived stateful formulation or retain oversampled solving
Linear LC network	Check discretization and modulation separately; ADAA is not needed merely because it is a circuit
Contact switching	Model event timing and bandwidth; a downstream waveshaper solution is insufficient

CHECKS AND DECISIONS

• A successful nonlinear solve and a passive-looking component are not substitutes for the complete spectral and stability tests.

SOURCES S037-02 Albertini, Bernardini and Sarti: Antiderivative Antialiasing in Nonlinear Wave Digita / S037-03 La Pastina and D’Angelo: Antiderivative Antialiasing with Frequency Compensation for / S037-04 JUCE dsp::Oversampling official class reference

ORIGINAL NUMERICAL VALIDATION PLAN

Construct a converged reference and identify what the residual means

Nonlinear oversampling and anti aliasing

Increase an offline reference rate and tighten integration tolerances until successive results converge within a declared budget. A selected32× or64× rate is a proposed starting experiment, not proof of an exact reference. Filter before decimation, retain the same physical excitation and startup state, and record the reconstruction and output filters.

Compare total waveform error after known transport-delay alignment, with fixed calibrated gain. That residual includes aliasing, integration error, changed linear response and state mismatch. For steady coherent tones, inspect expected harmonic/intermodulation bins separately. For contact transients or incommensurate organ tones, use matched windows and time-frequency residuals; spectral leakage must not be mislabeled as aliasing.

A computed diagnostic was executed here: a unit9 kHz sinusoid through f(x)=x+x^3 at 48 kHz produces a21 kHz folded component of amplitude0.25. Evaluating at 2× and applying an ideal offline lowpass before decimation removes that component to numerical precision. This finite cubic example demonstrates the mechanism; its noncausal FFT filter and successful2× rate are not production AO-28 results.

Fixture	Purpose
Linearized path / impulse	Resampler response, technical latency and ADAA coloration
Coherent swept single tones	Folded harmonics versus physical harmonics
Two tones and dense registrations	Intermodulation and shared-stage drive
Contacts, percussion and automation	Nonstationary events and time alignment
Rate doubling and tighter solver	Reference convergence rather than assumed truth

CHECKS AND DECISIONS

• A matched-linear-response diagnostic may isolate mechanisms, but retain the uncompensated total-error result as well.

ORIGINAL ACCEPTANCE / HANDOFF

Release an error budget, not an oversampling badge

Nonlinear oversampling and anti aliasing

Deliver the internal-rate map, reconstruction/decimation designs, state-transfer rules, event-time conversion, host latency policy and a reproducible comparison script. Record passband magnitude/phase error, stopband leakage, solver residuals, transient error and worst-case execution time separately. Choose numeric limits before fitting to the reference; this chapter supplies no measured instrument noise floor or universal audibility threshold.

The narrow cubic ADAA identity was also checked against independent16-point quadrature for 104 input pairs, including equal and near-equal values. Maximum absolute error was8.89e-15 or less. That verifies a mathematical fixture, not a stateful tube implementation. Complete AO-28, resampler, WDF and hardware acceptance remain not run.

Keep physical calibration and numerical validation separate. A highly oversampled implementation of an incorrect circuit remains incorrect; a correct circuit with unresolved folded products remains numerically incomplete. Quality presets should preserve registration, physical parameter values and event ownership, while changing only the documented numerical approximation and its latency policy.

Gate	Required outcome
Numerical safety	Finite output and zero solver fallback events in accepted fixtures
Reference convergence	Successive reference refinements remain inside declared bounds
Spectral fidelity	Total error and alias-specific evidence both satisfy predeclared limits
Parallel paths	Measured technical delay alignment; physical phase response preserved
Rate / block changes	No unintended phase reset, event loss or state discontinuity
Runtime	Bounded worst-case work at target buffer sizes; no audio-thread allocation

CHECKS AND DECISIONS

- Store all test settings and distinguish executed toy checks from proposed engine tests.

DOCUMENTARY INVENTORY AND SCOPE

Start with a traceable inventory, then identify every circuit instance

Circuit netlists component provenance and extraction

The accompanying 038-ao28-bom.json retains 85 records from the factory assembly list, including 84 numbered stock rows and the assembly heading. Seventy part-number fields are resolved through visual readings and explicitly identified manufacturer-catalog matches;15 remain unresolved. The original list and the cleaner B-3/C-3 catalog use different drawing namespaces. These are documentary records, not measurements of installed hardware. [S038-01] [S038-02]

ORIGINAL DESIGN PROPOSAL: treat this inventory as the acquisition layer. A repeated resistor type may serve several circuit positions, while one electrolytic can contains several electrical sections. The number of inventory rows therefore cannot establish a complete production bill of materials, component quantity, state count or netlist size. Keep quantity null whenever the document does not state it.

Extend the supplied value registry into a connectivity-complete schematic-instance graph linked to the stock inventory. Each electrical element needs reviewed terminal connections and revision applicability. Build the corresponding offline circuit only after those links pass review. A legible catalog can resolve a part number without resolving which of several identical resistors a photograph shows.

The separate ao28-component-values.json registry adds124 electrical placements and valve sections across the documented alternatives. It records readable nominal values, units, variant membership and diagram regions. These entries supplement the 85 stock records; neither count is the number of installed physical packages, and neither dataset claims complete connectivity. [S038-01] [S038-02]

Deliverable	What it establishes	What remains separate
Assembly inventory	Documented types, callouts and readable stock numbers	Per-instance quantity and connectivity
Schematic-instance registry	Electrical identifiers and terminal roles	Actual fitted values and modifications
Specimen record	Observed chassis and measured component data	Generalization to other B-3 organs

CHECKS AND DECISIONS

- Preserve all 85 records, including variants and repeated callouts.
- Do not convert null quantity to one or call this a manufacturing-complete BOM.

SOURCES S038-01 Hammond factory service manual: AO-28 parts and Figure24 / S038-02 Hammond B-3/C-3 original manufacturer repair-parts catalog

IDENTIFIER AND PROVENANCE CONTRACT

Keep assembly callouts, stock numbers and electrical labels separate

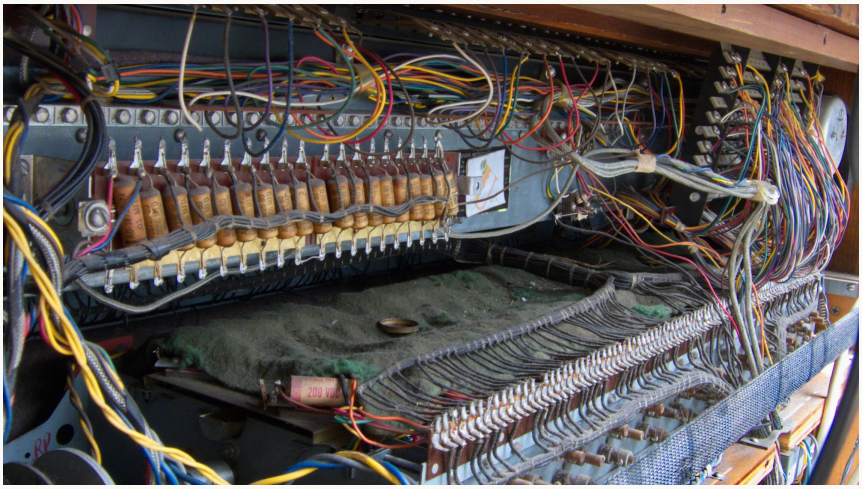
Circuit netlists component provenance and extraction

Factory 8-x callouts and the cleaner catalog’s6-6-x references locate assembly items; they are not R, C, T or V schematic designators. The inventory preserves uncertain old glyphs as null and stores a candidate separately. Transformer labels are retained only where the list itself identifies them. A known manufacturer number still does not identify a particular physical instance. [S038-01] [S038-02]

ORIGINAL DESIGN PROPOSAL: assign each source document a stable namespace and checksum, then key observations by document, page, figure and item. Record a crosswalk as an assertion with evidence and reviewer status. Never overwrite the older observation with the cleaner transcription: a later source can contain a correction, a production change, or a new error.

Use a separate component-instance key for the simulator. Link both halves of a dual valve to their shared physical envelope and heater record while retaining distinct plate, grid and cathode nodes. Preserve transformer winding identity and reference polarity. A lettered terminal on the chassis is a port contract, not permission to assume that every similarly named node is electrically identical.

Namespace conditional components before export. The solid P–D network and the no-code AO-28-1 replacement are alternatives; repeated C32 labels can denote different values in different profiles. Preserve the unresolved primary R62 glyph and secondary-only expression-capacitance annotation rather than silently merging them into one nominal circuit. [S038-01] [S038-02]



P038 | Hammond C3 color coded wires, SJSF 2012.jpg | Henry Zbyszynski | CC BY 2.0 | Context specimen; exact model/revision requires confirmation. Source and license

CHECKS AND DECISIONS

- Reject duplicate instance keys within a selected circuit namespace.
- Allow repeated stock numbers and repeated assembly callouts without merging electrical instances.

SOURCES S038-01 Hammond factory service manual: AO-28 parts and Figure24 / S038-02 Hammond B-3/C-3 original manufacturer repair-parts catalog

VALUE IMPORT AND DIMENSIONAL CHECKS

Normalize units before generating any simulator text

Circuit netlists component provenance and extraction

The older parts sheet uses MF notation; matching entries in the cleaner factory catalog explicitly state microfarads. Preserve both printed strings and a canonical SI value. Capacitor voltage ratings, resistor power ratings and resistance tolerances belong in separate fields; none is an operating-point measurement. [S038-01] [S038-02]

Ngspice interprets M and m as milli and uses meg for a factor of one million. Consequently, copying a vintage label directly into a SPICE deck can silently change a value by orders of magnitude. Emit reviewed SI numbers in exponent notation and test the exporter against known conversions. Do not rely on a simulator accepting a decorative unit suffix as dimensional validation. [S038-04]

ORIGINAL DESIGN PROPOSAL: make unit resolution an explicit import step with three outcomes: accepted, contradictory, or unreadable. Only accepted values enter a numerical build. Multi-section cans produce multiple electrical capacitances tied to one physical container record. Record terminal commonality from the circuit drawing; the inventory’s slash notation alone does not establish shared-node polarity.

Documentary notation	Canonical interpretation	Export guard
0.33 MF, corroborated as µF	3.3e-7 F	Never0.33e-3 F
4.7K pF	4.7e-9 F	Apply both kilo and pico factors
40/450 capacitor section	40e-6 F;450V rating	Do not divide40 by450
1 MEG resistor	1e6 ohm	Never emit ambiguous1M
Unreadable or conflicting glyph	null with alternatives	Block numerical export

CHECKS AND DECISIONS

- Round-trip values through the chosen solver and compare parsed SI magnitudes.
- Fail the build on unresolved values on any active circuit path.

SOURCES S038-01 Hammond factory service manual: AO-28 parts and Figure24 / S038-02 Hammond B-3/C-3 original manufacturer repair-parts catalog / S038-04 Official ngspice User Manual,47+ development edition

ORIGINAL IMPLEMENTATION SEQUENCE

Trace connectivity before reducing the circuit for real time

Circuit netlists component provenance and extraction

ORIGINAL DESIGN PROPOSAL: begin with a literal graph of the chosen schematic. Store component pins, electrical nets, source polarity, winding orientation and switch alternatives before simplifying any stage. A crossing without a junction must remain disconnected; a repeated voltage annotation must not create an independent supply source. This distinction preserves the connected bias behavior discussed in chapter 034.

After a second trace agrees, construct the offline reference with documented source and receiving loads. Derive the linear network and nonlinear device partitions from that graph. Any reduction to state space, a wave-digital structure or a measured surrogate must preserve explicit ports and an auditable mapping back to the unreduced network. Unknown transformer parameters remain exposed parameters rather than fabricated turns ratios.

Generate runtime coefficients outside the audio callback. Validate dimensions, finite values, model domains and factorization success before publishing an immutable profile. A topology-changing control needs a deliberate state-transfer policy; deleting capacitor voltages or inductor currents during a mode change can create an implementation transient that has no physical basis. Compare the reduced model with the offline reference before judging its resemblance to recordings.

```
observations = load_source_records()
assert namespaces_and_units_are_valid(observations)
instances = apply_reviewed_crosswalks(observations)
graph = trace_pins_nets_and_switches(instances)
assert independent_trace_agrees(graph)
assert active_values_are_resolved(graph)
reference = build_offline_loaded_circuit(graph)
compiled = reduce_and_factorize(reference)
assert ports_and_state_units_match(compiled, graph)
assert comparison_suite_passes(compiled, reference)
publish_immutable_profile(compiled)
```

CHECKS AND DECISIONS

• Keep the literal graph, reduction script and generated profile under the same versioned evidence manifest.

• No allocation, file access or matrix rebuilding is required in the audio callback.

SOURCES S038-01 Hammond factory service manual: AO-28 parts and Figure24 / S038-04 Official ngspice User Manual,47+ development edition

INSPECTED CONFLICTS AND REVIEW QUEUE

Resolve a source conflict without erasing its history

Circuit netlists component provenance and extraction

The original Figure 24 supply inset identifies the later 10/350 capacitor as C59. The 1995 redraw prints C56 at that position, duplicating its reservoir label, although its top-view legend still lists C59. This was independently checked visually. For the selected factory circuit, retain C59 and preserve the conflicting redraw observation in the provenance record. [S038-01] [S038-03]

The cleaner repair catalog also prints6-6-43 against both a100 pF capacitor and a1200ohm resistor. The older list places those descriptions at adjacent42 and 43 positions. A separate resistance glyph near item47 is too ambiguous to safely reconcile with the later catalog’s1500ohm entry. These are import-review tasks; they are not evidence that a surviving organ contains an impossible component. [S038-01] [S038-02]

ORIGINAL DESIGN PROPOSAL: attach competing observations to a single issue record, describe the consequence of each interpretation, and acquire the smallest piece of evidence that separates them. A clearer source or a qualified continuity/component identification session can resolve the mapping. Preserve the rejected reading and its reason so that later agents cannot accidentally restore it during an automated refresh.

Issue	Current handling	Resolution evidence
C56/C59 redraw collision	FactoryC59 retained in its namespace	Original inset inspected; specimen revision still needed
6-6-43 used for two types	Crosswalk flagged; no instance merge	Independent drawing/physical correspondence
Item47 resistance glyph	Normalized value and accepted stock number null	Clear source plus verified circuit position
No-code-letter AO-28 branch	Separate alternative graph	Chassis identification and actual connections

CHECKS AND DECISIONS

- A source conflict must survive serialization and remain visible in the service view.
- Do not resolve an ambiguous glyph by the most convenient simulation result.

SOURCES S038-01 Hammond factory service manual: AO-28 parts and Figure24 / S038-02 Hammond B-3/C-3 original manufacturer repair-parts catalog / S038-03 AO-28 retrospective redraw dated April20,1995

ACCEPTANCE AND REMAINING ACQUISITION

Gate the circuit handoff on evidence and reproducibility

Circuit netlists component provenance and extraction

ORIGINAL DESIGN PROPOSAL: split acceptance into documentary, numerical and specimen gates. The supplied inventory has been structurally checked for 85 records, preserved variants, source references and explicit unresolved fields. This validates the delivered data format, not a complete AO-28 model. No whole-AO-28 simulation, hardware comparison or acoustic fidelity result is asserted by this chapter.

Before implementation promotion, a second reviewer must reconcile each active circuit instance with its drawing and resolve any unit or identity conflict. Then verify the offline circuit's connected operating point, frequency response and state behavior using the same source and load assumptions as the proposed runtime model. Compare currents, voltages and port power as appropriate; a plausible output waveform alone can conceal a wrong connection.

The final gate requires an identified instrument: chassis code, mains variant, modifications, fitted components, transformer characteristics and acquisition-chain calibration. Store measured values beside their documentary nominal values with units and uncertainty. A future restoration or model-profile update should create a new evidence version rather than silently changing an earlier reference. Chapters 057–061 define the capture and comparison campaign needed to close this gate.

Gate	Required artifact	Current scope
Documentary structure	Inventory JSON and source/collision records	Executed structural checks
Per-instance circuit	Independent complete trace and reviewed values	Not completed here
Numerical equivalence	Offline/runtime comparisons with fixed loads	Not run for full AO-28
Physical correspondence	Calibrated captures and specimen manifest	Not measured here

CHECKS AND DECISIONS

- All active numerical inputs have an accepted provenance state before release.
- Report unrun tests as unrun; no score substitutes for missing reference captures.

SOURCES S038-01 Hammond factory service manual: AO-28 parts and Figure24 / S038-02 Hammond B-3/C-3 original manufacturer repair-parts catalog / S038-04 Official ngspice User Manual,47+ development edition

EVIDENCE AND ORIGINAL DESIGN CHOICES

Choose a representation at a declared physical boundary

Hybrid circuit and measured model choices

The scanner study combines a circuit-based ladder with a simplified scanner mechanism. It does not establish that one modeling method should represent the entire instrument. [S039-02] MNA and wave-digital/state-space research offer different numerical formulations; choosing a formulation does not supply missing component characteristics or specimen measurements. [S039-03] [S039-04]

ORIGINAL DESIGN PROPOSAL: keep a high-detail reference and a candidate runtime representation for each justified boundary. Choose the candidate from the required controls, memory, loading and observable errors. A reduced model is useful when it preserves the tested behavior within a stated domain. Additional component detail is useful when it resolves a measured discrepancy or enables a required interaction.

The table is the proposed starting architecture. Every substitution needs an evidence record. A measured waveform or transfer function must identify what hardware was already included in its capture chain.

Subsystem	Starting representation	Essential qualification
Generator	Continuous source bank; tables or harmonics	Capture boundary, mechanical angle and loading retained
Contacts and routing	Stateful contact/conductance network	Preset, drawbar and percussion interactions preserved
Scanner	WDF or state-space ladder plus rotor coupling	Referenced nodes; loading assumption explicit
AO-28 and power stage	Connected nonlinear electrical clusters	Bias, feedback, source and load remain connected
Stationary acoustic path	Measured FIR or parametric response	Only the declared linear, stationary condition
Unresolved repeatable error	Optional measured residual	Same ports, valid domain and held-out evidence

CHECKS AND DECISIONS

- Record the chosen approximation and its replacement criterion before optimizing it.

SOURCES S039-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus / S039-03 Ho, Ruehli and Brennan: The Modified Nodal Approach to Network Analysis / S039-04 Yeh and Smith: Simulating Guitar Distortion Circuits Using Wave Digital and Nonlinear

BOUNDARY CONTRACT AND COUPLING

Partition electrical networks without discarding their loads

Hybrid circuit and measured model choices

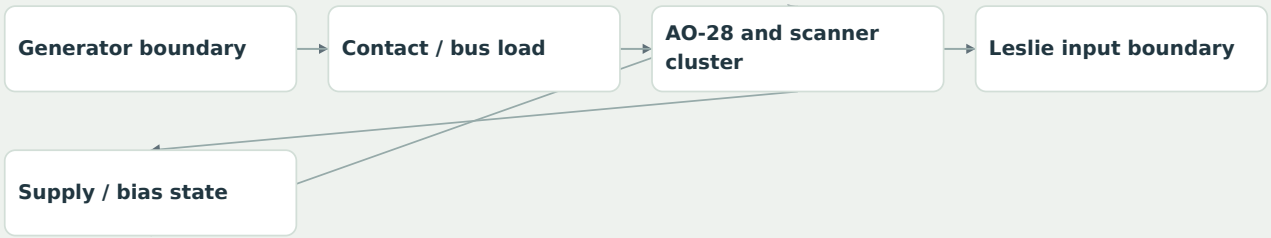
The AO-28 drawing connects its stages through expression, feedback and shared electrical references. [S039-01] Yeh and Smith explicitly discuss two-way interaction between circuit blocks; a one-direction audio cascade is therefore an approximation when the following block changes the preceding block's load. [S039-04]

ORIGINAL DESIGN PROPOSAL: define each electrical boundary using voltage and current signs, reference node, units, source impedance and receiving load. Keep strongly interacting stages inside one solved cluster. Alternatively, exchange wave variables through a derived interconnection. A measured output-voltage map alone does not provide the missing input-current behavior required to reconnect that block under arbitrary loads.

For an intentionally one-direction interface, quantify the loading approximation under the selected configuration. If an amplifier recording already includes its output transformer and speaker load, preserve that combined boundary or obtain a justified separation. Do not append those same responses again. Similarly, a loaded pickup recording cannot silently become an ideal unloaded voltage source.

The diagram describes modeling boundaries, not an additional physical circuit. A block may contain many stages; its internal detail can change while the declared external contract remains fixed.

Electrical information can cross a boundary in both directions



Original interface diagram. Bidirectional arrows require a derived coupling solution or a quantified approximation; they are not independent audio cables.

CHECKS AND DECISIONS

• Changing the declared load must affect the connected reference where its equations predict an interaction.

SOURCES S039-01 Hammond factory service manual: AO-28 Figure24 / S039-04 Yeh and Smith: Simulating Guitar Distortion Circuits Using Wave Digital and Nonlinear

ORIGINAL HYBRID MODEL PROPOSAL

Use measured corrections without hiding the cause of error

Hybrid circuit and measured model choices

Neural amplifier research demonstrates learned feedforward and recurrent model classes on guitar devices. Its training duration, CPU results and listening outcomes are not a Hammond acceptance threshold. [S039-05] A learned model may be considered here only after appropriate B-3 or Leslie captures exist.

ORIGINAL DESIGN PROPOSAL: first align the physical-model output and reference at the same calibrated ports. Resolve time offset, polarity, gain and capture-chain response before computing a residual. Otherwise the learned correction may compensate for an avoidable alignment error, an incorrect load or a microphone response that belongs elsewhere.

Select either a restricted output residual or a correction to identified physical parameters. Do not fit both without an identifiability argument. A small output error cannot uniquely identify internal component values: source gain and downstream gain may compensate for one another. Independent level calibration and controlled interventions reduce that ambiguity.

Separate deterministic discrepancies from stochastic noise. Keep learned hidden state across audio blocks, document its initialization, and expose the supported control and level domain. Outside that domain, use a declared fallback. A residual inserted inside feedback requires a separate stability analysis; output-fit accuracy alone does not establish a safe interconnection.

MODEL EQUATIONS

Aligned residual: r[n] = y_reference[n] - M_theta(u,c,z)[n].

Proposed hybrid: y_hat[n] = M_theta(u,c,z)[n] + alpha*R_phi(u,c,h)[n].

u: calibrated input; c: controls; z: physical state; h: learned state; alpha: declared enable/domain policy.

CHECKS AND DECISIONS

• Ablate the correction and test its improvement on complete held-out sessions, not adjacent training excerpts.

SOURCES S039-05 Wright, Damskägg, Juvela and Välimäki: Real-Time Guitar Amplifier Emulation with Deep

IMPLEMENTATION CONTRACT

Compile a consistent runtime plan before processing audio

Hybrid circuit and measured model choices

ORIGINAL DESIGN PROPOSAL: compile a ModelPlan from the specimen profile, port contracts, host sample rate and quality setting. Resolve units, topology, rate conversion, latency and state ownership off the audio thread. Each physical mechanism has one owner, including any response already embedded in a waveform, impulse response or learned block.

A fixed convolution y[n]=sum_k h[k]*u[n-k] is linear and time invariant by construction. It cannot alone reproduce level-dependent clipping or a rotating scanner. A directional impulse-response family with moving geometry is a different, time-varying model; chapter 053 specifies that implementation.

Keep nonlinear anti-aliasing and learned-model rate assumptions compatible. Increasing the operating sample rate of a discrete recurrent model changes the physical duration represented by its state updates unless the model or interface accounts for that change. Preserve the qualified model rate through a documented conversion, or qualify another rate-specific model. Profile changes need defined state transfer and latency alignment.

```
compile(profile, rate, quality):
    verifyProvenanceAndUnits(profile)
    resolvePhysicalOwnership()
    constructCoupledElectricalClusters()
    selectQualifiedReducedBlocks()
    allocateStatesAndRateConverters()
    alignAllParallelPathLatencies()
    return immutablePlan
process(events, buffers):
    applySampleStampedEvents(events)
    advanceSharedPhysicalStates()
    solveAndRenderDeclaredBoundaries()
```

CHECKS AND DECISIONS

- No training, file access, allocation or unbounded iteration in the audio callback.
- Changing host block size must preserve the chosen model’s event and state semantics.

SOURCES S039-03 Ho, Ruehli and Brennan: The Modified Nodal Approach to Network Analysis / S039-04 Yeh and Smith: Simulating Guitar Distortion Circuits Using Wave Digital and Nonlinear / S039-05 Wright, Damskägg, Juvela and Välimäki: Real-Time Guitar Amplifier Emulation with Deep

MEASUREMENT AND MODEL SELECTION

Compare candidate models with experiments that distinguish them

Hybrid circuit and measured model choices

ORIGINAL EXPERIMENT PROPOSAL: compare each reduced candidate against the selected connected reference, then compare the qualified implementation against physical captures. These answer different questions. Matching an idealized circuit proves consistency with that model; it does not establish that the model describes the selected instrument.

The supplied scanner AC program and nominal response data are useful for the first comparison. They calculate a published idealized network, with assumptions explicitly recorded. They do not validate rotating coupling, actual coil losses or the complete organ. Use reference_code/scanner_ac_reference.py and retain its model identity with every derived result.

Rank candidates using a declared quality and cost budget: complex response, transient error, nonlinear products, aliasing, worst-case callback time and memory. The weights and thresholds remain project decisions informed by repeatability and listening tests. Add complexity where an experiment demonstrates a meaningful improvement; retain failed cases as regression fixtures.

Experiment	What it distinguishes
Source/load changes	Unloaded cascade versus connected boundary behavior
Same spectrum, different preceding history	Memoryless mapping versus required state
Control transitions while notes remain held	Static fits versus valid control-dependent dynamics
Dense registrations at calibrated levels	Shared nonlinear processing versus independent per-note shaping
New sessions and unseen musical gestures	Repeatable mechanism versus capture-specific compensation

CHECKS AND DECISIONS

- Keep identification data, model-selection data and final evaluation captures separately labeled.

SOURCES S039-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus / S039-05 Wright, Damskägg, Juvela and Välimäki: Real-Time Guitar Amplifier Emulation with Deep

ACCEPTANCE AND HANDOFF

Promote a model only with its evidence and limitations attached

Hybrid circuit and measured model choices

ORIGINAL ACCEPTANCE CONTRACT: every runtime block ships with its boundary definition, provenance, validity domain, persistent state, numerical method and test results. The contract follows the block when its implementation changes. A fast model and a detailed model should be comparable at the same ports, under the same controls and with aligned latency.

The implementation agent should first complete the direct organ path from chapter 040, then add the separately qualified Leslie chain. Preserve the detailed reference as an offline comparison even when a reduced representation is selected for realtime use. Quality options may change computational approximations, but must not silently select different specimen tuning, registrations or captured room responses.

No complete hybrid instrument, trained residual or hardware comparison was produced in this chapter. The deliverable is a selection and qualification procedure. Missing measurements remain explicit acquisition tasks; attractive audio examples, solver precision and a visually correct interface cannot fill those gaps.

Required record	Acceptance question
Boundary and ownership	Which physical responses are included exactly once?
Source and control domain	Which loads, levels, rates and settings are supported?
State and transition policy	How are history, startup and profile changes handled?
Numerical and runtime evidence	Are error, aliasing and worst-case cost qualified?
Physical evaluation status	Which specimen captures support the claimed match?

CHECKS AND DECISIONS

- Reject undocumented compensation and unsupported extrapolation from the baseline profile.
- Leave physical-fidelity tests marked not run until the required captures and evaluations exist.

SOURCES S039-01 Hammond factory service manual: AO-28 Figure24 / S039-04 Yeh and Smith: Simulating Guitar Distortion Circuits Using Wave Digital and Nonlinear / S039-05 Wright, Damskägg, Juvela and Välimäki: Real-Time Guitar Amplifier Emulation with Deep

REFERENCE BOUNDARY / ORIGINAL VALIDATION PLAN

Define the electrical finish line before adding the cabinet

Organ subsystem validation before Leslie

Here, dry organ means the signal before Leslie and room processing; scanner vibrato and chorus remain part of the organ under test. Figure 24 identifies the AO-28 output-transformer boundary. Record the actual receiving load, interface and revision when comparing that boundary with software. An unloaded capture and a loaded model are different experiments. [S040-01]

ORIGINAL DESIGN PROPOSAL: expose diagnostic outputs at source, contact/bus, scanner-return and final-organ boundaries. Use immutable profiles and one retained event script. Software may provide every internal trace; physical access requires an appropriate calibrated interface and qualified service personnel. Do not turn a software observation port into an improvised live probing instruction.

Keep documentary, numerical and specimen comparisons distinct. The published scanner response comparison uses SPICE as its circuit reference. [S040-03] This suite’s chapter 029 AC calculation and chapter 037 nonlinear toy fixtures establish narrower numerical properties. They do not pass the complete organ gate. A failure should be localized before later cabinet filtering obscures it.

Reference tier	What can pass at that tier
Documentary	Correct revision, source IDs, wiring map and units.
Circuit/numerical	Agreement with an independently checked model under identical input and loading.
Physical specimen	Agreement with calibrated captures, repeated conditions and stated uncertainty.
Musical	Performance behavior and held-out listening comparisons after the preceding gates.

CHECKS AND DECISIONS

- The run manifest identifies the exact output boundary and receiving load.
- All complete-engine and physical tests in this chapter are proposed, not run.

SOURCES S040-01 Hammond factory service manual: AO-28 and Table III / S040-03 Werner, Dunkel and Germain: Hammond vibrato/chorus WDF model

FACTORY MAP / ORIGINAL TEST FIXTURES

Make routing and coherent source reuse observable

Organ subsystem validation before Leslie

Factory Table III supplies the source assignments used by chapters 015 and 022. For manual key 61, the nine source numbers are 61, 80, 73, 85, 80, 85, 89, 80, 85. These are frequency identifiers, not Hertz. [S040-01] Run the complete 61-by-nine map on both manuals with percussion disabled, then test the conditional percussion routes separately.

ORIGINAL DESIGN PROPOSAL: begin with a deterministic ideal-source fixture whose frequency and phase are known. A source identity collision must reuse the same evolving waveform while retaining separate contact paths and gains. Then enable voicing, leakage and connected loading one mechanism at a time. A spectrum alone cannot identify which of several contacts selected the same source.

Track each source in a continuous record or with a simultaneous phase reference. Independently recorded notes cannot be added coherently merely by aligning their onset peaks. Permit documented acquisition latency and initial phase alignment; do not independently retune or time-warp each recording to hide incorrect source frequencies.

Fixture	Required observation
Low manual collision	Keys1 and 13 at 16-foot both select source 13; keep contact identities distinct. [S040-01]
High foldback	Key 61 matches all nine factory assignments, including repeated sources.
Pedal exceptions	Pedals1–7 duplicate the second source in the third-contact column;8–12 disconnect it. [S040-01]
Drawbars and quiet state	Sweep the continuous control, preserve generator phase through rests, and distinguish configured leakage from erroneous note ownership.

CHECKS AND DECISIONS

- The source-field trace never restarts because a note is pressed or released.
- Mapping equality is exact; audible gain and waveform tolerances require a separate calibration record.

SOURCES S040-01 Hammond factory service manual: AO-28 and Table III

TEMPORAL TEST DESIGN

Exercise contact timing and percussion as histories

Organ subsystem validation before Leslie

The contact study found gesture-dependent behavior on a particular 1967 C-3; a single velocity value does not describe the complete touch. [S040-02] ORIGINAL DESIGN PROPOSAL: test recorded contact trajectories separately from the MIDI-to-gesture approximation. Repeat each trajectory at several known generator phases, and retain attack and release windows without level-normalizing away their differences.

Percussion tests must operate on the routed contact history and continuous recharge state defined in chapters 020–021. Hold one eligible upper key while adding and releasing others; then release all keys for varying gaps before playing again. Store detector state, control voltage surrogate and percussion output together. An envelope that sounds plausible in isolation can still be triggered by the wrong event.

Use the same source phases and registration when comparing settings. Switching preset or percussion controls while keys remain held needs a physical oracle; deterministic software behavior alone is not evidence of the original transition. Mark these cases awaiting capture until the electrical state change has been established.

Sequence	Discriminating result
Legato overlap	Additional held keys do not create an undocumented fresh percussion envelope.
Release-gap sweep	Proposed gaps 0, 1, 2, 5, 10, 20, 50, 100, 250, 500, 1000 ms; extend until recovery reaches a measured plateau.
Repeated MIDI pitch	Owner accounting cannot manufacture a second physical key press while the same key remains held.
B / other / B while held	Compare routed keying and charge history against an acquired reference; no guessed audible pass criterion.
Slow/fast, normal/soft, second/third	Test each setting and changes during an existing decay, not only freshly initialized notes.

CHECKS AND DECISIONS

- Release-gap values above are proposed experimental settings, not factory timing specifications.

SOURCES S040-02 Moro, McPherson and Sandler: Dynamic temporal behaviour of the keyboard action

MODULATION AND CONNECTED NETWORK TESTS

Validate all scanner modes without hiding routing errors

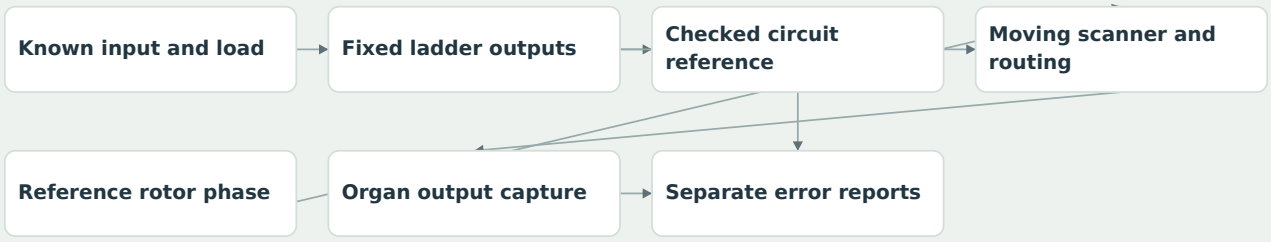
Organ subsystem validation before Leslie

The late-B-3 research model includes six selector modes and a circuit whose common-return connection changes between vibrato and chorus. [S040-03] ORIGINAL DESIGN PROPOSAL: first compare the fixed ladder outputs with the checked circuit reference; then enable scanner motion. Matching stationary magnitude curves does not establish the rotating amplitude, phase or switching behavior.

Cross all six modes with the four Swell/Great tablet combinations. Excite upper sustained, lower sustained, shaped-pedal and percussion paths separately before combining them. Use chapter 027’s routing contract, including the pedal association and downstream percussion destination. A combined chord can mask a wrong branch or an extra dry contribution.

For moving tests, retain rotor phase and inspect the complete modulation spectrum around known inputs. Compare magnitude and phase trajectories, rather than only one apparent modulation-rate peak. Keep circuit state through selector changes. Acquisition-clock drift and rotor-phase uncertainty belong in the report; arbitrary phase resetting or a corrective output chorus must not conceal an incorrect internal model.

Two scanner comparisons with different oracles



Original test architecture. The circuit-reference comparison and the physical moving-scanner comparison have separate completion states.

CHECKS AND DECISIONS

- No additional dry mix is silently added to already ground-referenced chorus output.
- Mode changes do not reset scanner angle or discharge every state variable by policy.

SOURCES S040-03 Werner, Dunkel and Germain: Hammond vibrato/chorus WDF model

CONNECTED AO-28 CALIBRATION

Separate expression, loading and nonlinear error

Organ subsystem validation before Leslie

The AO-28 schematic places expression inside the V4 signal path. [S040-01] ORIGINAL DESIGN PROPOSAL: measure several repeatable pedal positions using the same input levels, load and control settings. Compare frequency response and later-stage drive together. Do not fit a master gain curve to compensate for an incorrectly loaded expression network.

Begin nonlinear discrimination with fixed topology and known injected software signals. Playing another key also changes contact selection and potentially loading; consequently, a chord-minus-isolated-notes residual is not automatically valve distortion. Align source phases and account for the changed network before attributing the residual to saturation. On hardware, any electrical injection or internal measurement belongs to a qualified, documented acquisition setup.

Use small-signal, increasing-level and recovery sequences. Record source spectra, output harmonics, non-harmonic products, mean level and quiet-state residue. Fit only on designated training conditions; reserve different registrations, note combinations and expression trajectories. A nonlinear fit that improves one loud chord while altering all quiet registrations has not passed this gate.

Variable	Keep or report explicitly
Electrical level	Calibration in volts, interface gain, headroom and receiving impedance.
Expression	Position or linkage coordinate, direction, settling and rate of movement.
Nonlinear analysis	Input content, state initialization, window, noise floor and unresolved source collisions.
Aliasing	Compare converged high-rate references and matched linear filtering; an oversampling label is not an error bound.

MODEL EQUATIONS

For fixed configuration c: Delta_c = F_c[a+b] - F_c[a] - F_c[b] + F_c[0].

A linear fixed network gives Delta_c=0; changed topology, phase or state invalidates that inference.

CHECKS AND DECISIONS

- Do not change the input scale independently for every registration to improve the fit.

SOURCES S040-01 Hammond factory service manual: AO-28 and Table III

IMPLEMENTATION / ACCEPTANCE

Issue a dry-organ gate report before rotary integration

Organ subsystem validation before Leslie

VST3 event offsets are relative to the current processing block. [S040-04] ORIGINAL DESIGN PROPOSAL: translate one absolute event timeline into many block partitions, including events at the first and last frame. At a fixed sample rate, contact and control transitions must retain identical sample positions. Audio differences must remain within a declared implementation-level floating-point budget.

Use proposed rates 44.1, 48 and 96 kHz and block sizes 1, 17, 64, 127, 512 and 1024 frames as a diagnostic matrix, then include the actual supported product configurations. Cross-rate comparisons require appropriate reconstruction and latency accounting; they are not byte comparisons. Revisit anti-aliasing and solver stability when the internal rate changes.

Export a gate manifest with test IDs, reference tier, specimen/profile hashes, event script, input/output hashes, calibration, analysis settings, tolerances, result and unresolved causes. A numerical pass permits continued integration with its stated model scope. A specimen-fidelity claim requires the physical evidence. Preserve the direct-organ render after adding Leslie so later regressions remain localizable.

Gate	Release meaning
D40-D documentary	Exact routing and revision checks have retained evidence.
D40-N numerical	Implementation agrees with its verified reference model.
D40-H hardware	Held-out calibrated captures satisfy declared uncertainty-aware criteria.
Current edition	No complete dry-organ engine or hardware campaign was executed; these implementation gates remain not_run.

```
for fixture in dryOrganSuite:
    reference = resolve(fixture.referenceTier, fixture.hash)
    for config in supportedTestConfigurations:
        engine = initialize(fixture.profile, fixture.initialState)
        traces = renderAbsoluteTimeline(engine, fixture.events, config)
        assertEventTimingAndOwnership(traces)
        metrics = compareAtDeclaredBoundary(traces, reference)
        saveReport(metrics, provenance, tolerances, executionStatus)
```

SOURCES S040-04 Steinberg VST3 Event structure

HARDWARE EVIDENCE

Freeze the 1971 rotor-amplifier identity

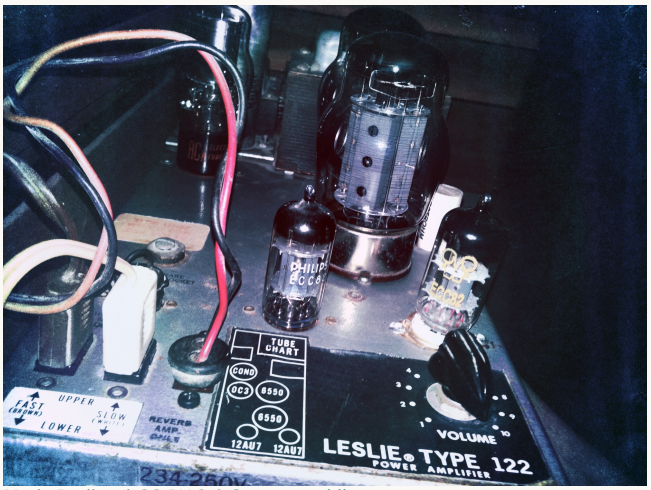
Leslie 122 amplifier input and control separation

Baseline evidence is Electro Music publication044578, June 1971: printed19 is the split117V schematic; printed20 lists036616 at 60 Hz and 053231 at 50 Hz. The selected reference is036616 without optional reverb. This identifies a documented configuration, not every surviving122. [S041-01]

ORIGINAL DESIGN PROPOSAL: encode publication, assembly, mains-frequency family, installed modifications, and physical specimen as separate metadata fields. Reject an imported parameter set when its assembly identity is missing. Photograph the specimen label and underside before assigning a measured profile; replacement components can make a correctly labelled chassis differ from the drawing.

The modern 122H has a different input/preamp and servo-motor architecture. Its historical comparison confirms balanced122 versus unbalanced147 inputs, but its modern specifications must remain a separate profile. [S041-02] Archive scans remain viewing references; the archive restricts redistribution. [S041-03]

Documented item	Role / boundary
V1:12AU7A, half used	Rotor-control valve
V2:12AU7A	Balanced audio driver
V3/V4:6550	Push-pull output pair
V5:0C3	Gas regulator
D1–D4: silicon,800PIV,1A	Main rectifier bridge
Q1/Q2 andV6–V9	Optional reverb drawing; excluded here



P021 | Leslie 122 Power Amplifier.jpg | Mads Bødker | CC BY 2.0 Source and license

CHECKS AND DECISIONS

• Require five main-amplifier valve positions and four main rectifiers in the evidence inventory; do not count optional reverb circuitry as baseline122 hardware.

• Record whether an as-built specimen actually matches036616 before claiming component-level fidelity.

SOURCES S041-01 Electro Music: Leslie 122/122V/122RV/142/222/222RV General Operating and Maintenance / S041-02 Hammond Suzuki Leslie122H/142H Owner's Manual, 00457-40211 V1.00-220713 / S041-03 Captain Foldback: Download scans of Leslie manuals — archive provenance and terms

CIRCUIT INTERPRETATION

Separate audio transmission from rotor control

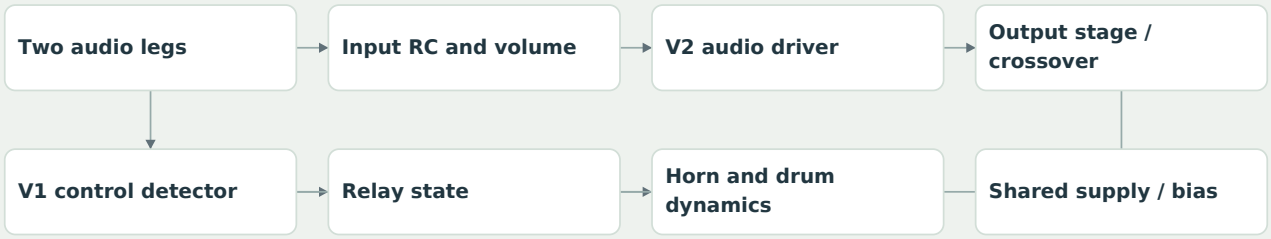
Leslie 122 amplifier input and control separation

The source shows R1/R3=10 kΩ, shunt volume R2=100 kΩ, C6/C8=0.015 µF and inter-grid C7=0.001 µF. V1 receives one input leg through R4=1 MΩ; its plate circuit operates Rel1. V2 handles audio, while C6/C8 exclude input DC. [S041-01]

ORIGINAL CIRCUIT INTERPRETATION: do not implement the control valve as another audio gain stage. Its one-leg sensing is also not a literal mathematical average of both inputs. A common-mode coordinate describes the interface conveniently, but a component model must retain the actual sensing connection and shared bias dependencies.

ORIGINAL DESIGN PROPOSAL: model three interacting systems: electrical audio, electrical command detection, and mechanical rotation. A relay transition selects a drive condition; it does not reset horn angle, drum angle, or oscillator phase. Keep a reduced control model replaceable by a circuit-derived detector once source-interface and specimen measurements establish thresholds, delays and disturbance rejection.

Three responsibilities, explicit coupling



Original conceptual model derived from the circuit; no connector wiring or measured timing is represented.

CHECKS AND DECISIONS

• Trace one differential audio stimulus and one control transition separately in the simulator; label every node that can receive both.

• Do not enable an ideal common-mode-only detector while calling it a literal transcription of V1.

SOURCES S041-01 Electro Music: Leslie 122/122V/122RV/142/222/222RV General Operating and Maintenance

ORIGINAL DIGITAL MODEL

Define volts, loading and gain consistently

Leslie 122 amplifier input and control separation

ORIGINAL DERIVATION: let va and vb be instantaneous input-leg voltages in volts relative to the same reference. Differential voltage vd is the full voltage between legs; common-mode voltage vc is their average. These coordinates prevent a factor-of-two amplitude mistake when the upstream AO28 model supplies opposing legs.

For an ideal low-impedance source, ignore coupling-capacitor reactance and valve input loading temporarily. A variable shunt resistance Rsh between two equal series resistors Rs gives the approximate differential divider below. This is a useful limiting check, not the complete frequency response. Restore both grid loads, the inter-grid capacitor, source impedance and volume-law geometry in the implemented network.

ORIGINAL DESIGN PROPOSAL: carry the input as a two-port loading problem. A linearized admittance matrix Y(f), in siemens, can return input currents to the upstream model. Near nonlinear operation, solve a coupled port or identify a controlled approximation. A one-way voltage interface is acceptable only when its source-impedance assumption is declared and its error is tested. Never infer a measured common-mode rejection ratio from nominal component symmetry.

MODEL EQUATIONS

vd = va - vb; vc = (va + vb)/2; all voltages in V.

va = vc + vd/2; vb = vc - vd/2.

Hd ≈ Rsh / (2*Rs + Rsh); Hd dimensionless; resistances in ohms.

[ia, ib]^T = Y(f) * [va, vb]^T; currents in A, Y in S, f in Hz.

CHECKS AND DECISIONS

• Reconstruction must recover both original legs within numerical round-off for arbitrary audio plus DC vectors.

• The simplified divider must approach zero when Rsh approaches zero; test the full model separately at the physical control endpoints.

• Swapping input legs must reverse differential polarity in the symmetric linear limit while preserving its magnitude.

ORIGINAL IMPLEMENTATION PLAN

Implement the interface before acoustic modulation

Leslie 122 amplifier input and control separation

ORIGINAL DESIGN PROPOSAL: first implement a passive input-network reference using documented values and an explicitly chosen source impedance. Compare its transfer function with an independent offline nodal calculation. Next attach the V2/audio stage to the power-stage work in chapter 043. Preserve the input-load contract when replacing a linear stage with a nonlinear solver.

Represent a speed request as a timestamped event with an explicit command-to-relay mapping. Keep detector integration state, relay state, horn angular velocity and drum angular velocity separate. Preserve angular phase across slow/fast changes. The standard source describes slow and fast operation; a stop extension requires its own accessory/profile identity. [S041-01]

Render audio and control on sample-aligned timelines. A reduced detector may run more slowly only after its bandwidth and event-time errors are bounded. Preallocate state, parameter queues and oversampling buffers. Apply anti-aliasing around nonlinear audio stages and verify solver convergence under abrupt input/volume changes. Shared supply or switching transients belong in a measured or circuit-derived coupling model; adding an arbitrary click whenever speed changes would conceal uncertainty rather than resolve it.

```
for each sample n:
  event = commands.atSample(n)
  ports = source.solveWithInputLoad(loadState)
  audio = inputNetwork.step(ports, volumeState)
  detector = controlModel.step(ports, biasState)
  relay = relayModel.step(event, detector)
  horn.step(relay)
  drum.step(relay)
  output = powerStage.step(audio)
  render(output, horn.phase, drum.phase)
```

CHECKS AND DECISIONS

• Deliver an offline input-network reference and deterministic real-time module before connecting room or microphone simulation.

• Profile the worst supported sample rate and buffer size; no allocation, lock, disk access or blocking call may occur in the audio callback.

SOURCES S041-01 Electro Music: Leslie 122/122V/122RV/142/222/222RV General Operating and Maintenance

ORIGINAL MEASUREMENT PLAN

Acquire evidence that separates control from sound

Leslie 122 amplifier input and control separation

ORIGINAL MEASUREMENT PROPOSAL: ask a qualified technician to provide calibrated, isolated captures from a documented fixture. This chapter specifies observables rather than live connector or mains procedures. Retain raw recordings, gain settings, instrument serials, calibration, uncertainty and an unambiguous diagram of the measurement boundary.

Acquire electrical behavior before fitting microphones. Otherwise cabinet directivity and room reflections can be mistaken for input filtering or amplifier distortion. Capture both legs simultaneously so differential and common-mode signals are derived from the same clock. Repeat the quiet condition with control held steady and during transitions. Keep physical acoustic relay noise distinct from an electrical switching artifact.

No capture described here has been performed. Until these data exist, store unknown parameters as null with provenance and a substitute model identifier. A usable prototype can accept abstract slow/fast commands; it must not report a physically calibrated detector threshold, relay delay, or switching-ripple level.

Required capture	What it identifies
Dual-leg sweep; several volume settings	Input frequency response, imbalance and attenuation law
Source/load comparison	Loading error at the AO28 boundary
Steady command plus differential audio	Detector susceptibility to musical signal
Control transition plus amplifier output	Electrical disturbance and detection delay
Independent horn/drum tachometry	Relay-to-motion delay and separate inertia
Quiet fixed state at several drive levels	Hum/noise versus signal-dependent supply coupling

CHECKS AND DECISIONS

• Reject captures without a shared time reference when estimating detector-to-relay-to-rotor delay.

• Require repeated transitions in both directions and record variability; one event is not a timing distribution.

• Freeze microphone and room metadata only for the later acoustic comparison.

ACCEPTANCE AND HANDOFF

Make fidelity claims conditional on evidence

Leslie 122 amplifier input and control separation

ORIGINAL ACCEPTANCE PLAN: classify results as topology verified, numerically checked, or hardware matched. These are separate statuses. The inspected schematic supports the first for this interface; no physical instrument comparison currently supports the third. Archive the exact parameter file with every render so a later calibration does not silently change an earlier reference.

Hand the power-stage agent differential drive, source loading, control state and shared-bias requirements. Hand the motor agent timestamped relay transitions, without imposing an instantaneous rotor speed. Hand the user-interface agent requested speed and actual estimated rotor speeds as distinct states; visual motion should follow the latter.

If an ideal symmetric model rejects common-mode audio perfectly, treat that as a mathematical diagnostic. Introduce measured mismatch only through identified component or port parameters. Neither a generic noise generator nor a modern 122H specification resolves missing vintage evidence.

Gate	Pass condition / evidence
Identity	036616 reference and 1971 drawing retained; optional reverb separate
Units	Full differential volts and per-leg volts round-trip correctly
Linear port	Independent circuit calculation agrees within declared numerical tolerance
Control separation	Musical input does not spuriously change modes over tested envelope
Transition continuity	No artificial angle reset or discontinuous rotor-speed jump
Measured match	Threshold, loading and transition captures attached before claim
Variant boundary	147 and modern 122H parameter files fail vintage-profile validation

CHECKS AND DECISIONS

• Set numeric error tolerances from measured uncertainty and product requirements; proposed tolerances must never be labelled manufacturer specifications.

• Do not release a component-exact claim while source-interface impedance, detector switching polarity or relay timing remains unverified.

HARDWARE EVIDENCE / VARIANT IDENTITY

Identify the 147 before comparing its sound

Leslie 147 amplifier differences

The vintage 147 requires its own electrical profile. Hammond’s historical comparison distinguishes its unbalanced audio input and AC speed-control interface from the 122 arrangement. Similar six-contact connectors do not establish electrical compatibility. Its qualitative description of a brighter 147 is useful context, not a numerical response curve. [S042-04]

ORIGINAL DESIGN PROPOSAL — define the baseline as the particular vintage 147 drawing on printed page 6 of the 145/147 manual. Preserve the October 1979 publication as a second evidence set until differences have been reconciled. Never assemble an undocumented circuit by selecting convenient parts from both scans. The latter publication includes labeled underside photographs that can support a future visual inventory. [S042-01] [S042-03]

Assign an immutable circuit-profile identifier, source revision and inspection status to every model instance. Keep cabinet identity, amplifier identity and organ-to-cabinet adapter identity separately selectable. An amplifier comparison should hold the remaining identities constant; an historically configured full-system comparison should record all three. Neither comparison has been measured here.

Evidence boundary	147	122 comparator
Audio port	Single-ended signal and return	Differential signal pair
Vintage valve count	12AU7 ×1; 6550 ×2; 0C3 ×1	12AU7 ×2; 6550 ×2; 0C3 ×1
Extra control triode	Absent from inspected 147 circuit	Separate relay-control 12AU7 section
Factory evidence	145/147 p.6; 1979 PDF p.4	1971 p.19; 1979 PDF p.4

CHECKS AND DECISIONS

- Store both publication identifiers; leave the undated manual date unknown.
- The 147 roster contains no 6AU6 and no modern 12BH7 preamp.

SOURCES S042-01 Electro Music: Leslie 145/147 owner and service manual, vintage scan / S042-02 Electro Music: Leslie 122 family owner/service manual, June 1971, publication 044578 / S042-03 Electro Music/CBS: Leslie models 122–147 Service Information / Parts List, October 19 / S042-04 Hammond Suzuki: Leslie Heritage 122H/142H Owner’s Manual, V1.00–220713

HARDWARE EVIDENCE / ORIGINAL FUNCTIONAL DRAWING

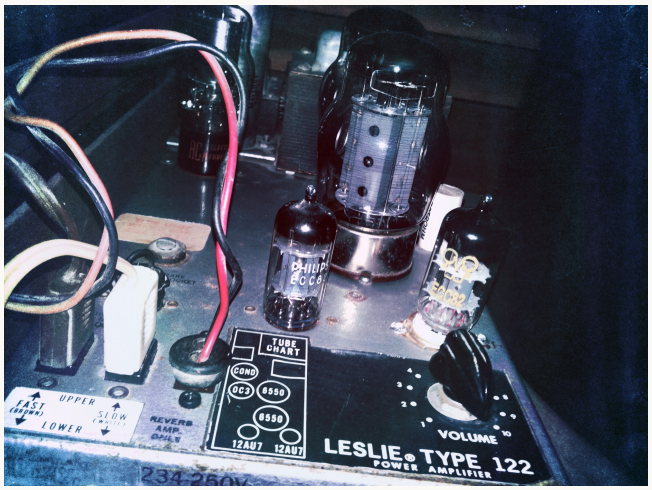
Trace phase inversion, bias and transformer feedback

Leslie 147 amplifier differences

The inspected 147 uses a 10-kilohm input control, a dual-triode phase-driving stage and push-pull 6550s. Transformer winding connections enter the output-valve cathode circuit. The second 12AU7 grid receives a resistor-divided drive signal; this must not be confused with a transformer-secondary feedback return to that grid. The 122 comparison has different cathode and feedback connections. [S042-01] [S042-02] [S042-03]

ORIGINAL DESIGN PROPOSAL — implement the 147 as a connected network, with phase inversion, power stage, transformer winding map and bias participating in one electrical solution. A serial tube-shaper/transformer-filter chain may serve as a surrogate, but cannot be labeled component exact. Verify winding polarity against the selected drawing before fitting any transformer parameter; a convenient block-diagram arrow does not specify physical winding orientation.

Keep rectification, stored supply energy and the 0C3 series-drop path in the model boundary or explicitly document their approximation. The regulator is not an independently fixed screen-voltage source. [S042-01] For an initial model, expose supply and transformer parameters with an unmeasured status. Later measurements must constrain them jointly with the load; otherwise a fitted transformer resonance can conceal an incorrect feedback connection.



P021 | Leslie 122 Power Amplifier.jpg | Mads Bødker | CC BY 2.0 | 122 comparison photograph; use the separately cited 147 schematic for this chapter. Source and license

CHECKS AND DECISIONS

- A winding-map review must distinguish electrical junction dots from line crossings.
- The control event never becomes audio-rate AC mains in the software interface.

SOURCES S042-01 Electro Music: Leslie 145/147 owner and service manual, vintage scan / S042-02 Electro Music: Leslie 122 family owner/service manual, June 1971, publication 044578 / S042-03 Electro Music/CBS: Leslie models 122–147 Service Information / Parts List, October 19

ORIGINAL DESIGN PROPOSAL / INTERFACES

Make the digital port describe a physical boundary

Leslie 147 amplifier differences

The old 147 drawing provides OPEN, 8-ohm and 16-ohm console-load positions. These describe source loading, independently of the cabinet's speaker load. [S042-01] Model the selected branch together with the input control and source impedance; OPEN removes that branch, not every input impedance. A perfectly stiff normalized audio source can conceal loading effects that matter when an adapter or organ output stage is present.

Use the following small-signal relation as a boundary check, not a substitute for a nonlinear coupled solve. Vs is unloaded source voltage, Zs source impedance, Zin total amplifier-side input impedance and Vin the resulting port voltage. Complex impedances are functions of frequency; a time-domain implementation needs causal states. If the upstream model already solves this load, pass the loaded voltage and declare that ownership so the attenuation is not applied twice.

The public control contract carries semantic rotor commands and sample timestamps. Chassis connector details remain evidence metadata. An explicitly documented organ adapter converts the organ output model to this single-ended boundary; simply taking one conductor of a 122 differential model has no established calibration.

Field	Units / meaning	Required provenance
inputVoltage	Instantaneous volts relative to signal return	Loaded versus unloaded boundary
sourceImpedance	Ohms or causal impedance model	Adapter and fixture identity
consoleLoad	Open / 8 / 16 branch selection	Selected drawing and actual setting
volumePosition	Dimensionless shaft position	Taper and endpoint calibration
rotorCommand	Slow / fast, timestamped	Independent motor profile
electricalProfile	Immutable revision identifier	Source and parameter status

MODEL EQUATIONS

$$V_{in}(f) = V_s(f) * Z_{in}(f) / (Z_s(f) + Z_{in}(f))$$

$$I_{in}(f) = V_s(f) / (Z_s(f) + Z_{in}(f))$$

CHECKS AND DECISIONS

- A linear boundary fixture reproduces the complex voltage divider before tube nonlinearities are enabled.
- Every amplitude conversion declares volts per full-scale sample.

SOURCES S042-01 Electro Music: Leslie 145/147 owner and service manual, vintage scan

ORIGINAL DESIGN PROPOSAL / IMPLEMENTATION ORDER

Build a separate profile without cloning the whole engine

Leslie 147 amplifier differences

First transcribe and review the selected topology. Then create an offline linear operating-point model, add nonlinear valve currents, and only then optimize for real-time use. Share verified component-model code with the 122, while keeping connections, input boundary, control detector and parameter data distinct. The 1979 parts sheet lists a common output-transformer part, 055103, but its paired drawings use different winding connections. Shared inventory therefore cannot justify a shared complete amplifier transfer function. [S042-03]

Prepare all topology, solver and oversampling resources off the audio thread. Bound iteration count; log residual and convergence failures through a nonblocking diagnostic channel. Choose oversampling by comparing alias products and timing cost against an offline reference. Smooth ordinary user controls in their intended physical coordinate, but treat a circuit-profile change as replacement of a stateful system, with an explicit transition policy.

The pseudocode proposes parallel preparation and a short output transition. Its duration is a product decision, not a vintage mechanical property. Separate rotor phase from amplifier state when the cabinet remains selected. Save enough identity and parameter metadata to make a preset reload reproducible; do not silently migrate an old 147 profile to a newer transcription.

```
prepare(profile):
  validate_source_and_winding_map(profile)
  solve_dc_operating_point(profile)
  allocate_solver_and_rate_filters(profile)
  return prepared_instance
process(block, events):
  dispatch_timestamped_rotor_events(events)
  solve_declared_input_boundary(block)
  upsample_into_preallocated_buffer()
  solve_amp_with_bounded_iterations()
  downsample_with_matching_filter()
  report_nonfinite_or_residual_failure()
change_profile(next):
  stage_prepared_instance_off_audio_thread(next)
  transition_outputs_by_documented_policy()
```

CHECKS AND DECISIONS

- Repeated preset loads retain the same revision and parameter hash.
- No callback allocation, unbounded solve, or NaN propagation during overload and profile changes.

SOURCES S042-03 Electro Music/CBS: Leslie models 122–147 Service Information / Parts List, October 19

ORIGINAL DESIGN PROPOSAL / CALIBRATION PLAN

Measure electronics before attributing a cabinet difference

Leslie 147 amplifier differences

Commission qualified bench captures at declared electrical boundaries using professionally selected interface and load fixtures. Record source impedance, input voltage, amplifier-control position, output load, supply condition and specimen identity with each capture. This is a measurement specification, not a cable or live-service procedure. The vintage manual’s schematic voltages were taken with a 20,000-ohms-per-volt meter; they are conditional references, not exact targets for every later measurement instrument. [S042-01]

Preserve two comparisons. The raw-voltage comparison diagnoses gain and clipping threshold; a separately level-aligned listening comparison investigates character without loudness dominating. Capture electrical amplifier output before interpreting microphone results. Where a cabinet is included, document driver revision, crossover, rotor state, microphone geometry and room; otherwise those changes can be mistaken for a 147 amplifier response.

Use several amplitudes and frequencies, followed by two-tone and transient stimuli. Fit the simplest candidate model to a declared training subset and reserve additional stimulus combinations for validation. Keep initial parameters marked proposed until captures exist. A single equalized sweep cannot determine input loading, nonlinear distortion, dynamic bias and transformer behavior independently.

Capture family	Question resolved	Saved evidence
Small-signal sweep	Gain, phase and source-loading behavior	Complex transfer and boundary calibration
Stepped sine levels	Compression and harmonic evolution	Fundamental plus harmonic amplitudes
Two-tone combinations	Intermodulation outside training material	Input/output waveform and spectrum
Bursts and recovery	Bias or supply memory	Time trace with identical gain setting
Slow/fast command	Control-to-motion timing	Timestamp plus independent rotor trace

CHECKS AND DECISIONS

- No comparison silently normalizes the input drive before a distortion test.
- Training and held-out measurements carry specimen, fixture and uncertainty records.

SOURCES S042-01 Electro Music: Leslie 145/147 owner and service manual, vintage scan

ORIGINAL DESIGN PROPOSAL / ACCEPTANCE AND OPEN QUESTIONS

Accept the variant only when the distinctions survive

Leslie 147 amplifier differences

Acceptance has two separate meanings: the runtime reproduces its declared numerical reference, and the chosen reference reproduces an identified physical amplifier. Passing the first is necessary evidence for implementation quality; it provides no independent proof of the second. Publish electrical residuals, alias measurements, worst-case callback time and held-out capture errors separately. Establish tolerances from intended use and acquisition uncertainty before fitting.

Variant reconciliation is a deliverable. The older drawing labels two console-load resistors 8 ohms / 4 watts, whereas the October 1979 schematic and parts list show 8 ohms / 10 watts. [S042-01] [S042-03] Keep the documents distinct; resistor power ratings alone do not establish an audible change or a serial-number boundary. Obtain dated factory change records or inspect the actual specimen before assigning chronology to additional differences.

Unresolved transformer ratios, winding parasitics, valve curves, control-pot taper, specimen tolerances and adapter behavior remain named acquisition tasks. Parameterize them with explicit status rather than inventing universal values. The instrument UI may offer 122 and 147 as intentional software choices, but its documentation must identify which blocks change and must not imply physical connector interchangeability.

Gate	Pass evidence	Failure implication
Topology	Reviewed source-to-model connectivity	Correct the model before fitting
Numerics	Reference agreement and finite bounded solve	Revise solver or approximation
Port behavior	Verified gain and loading units	Recalibrate adapter boundary
Variant separation	Auditable 122/147 profile differences	Reject generic brightness switch
Hardware fidelity	Held-out captures within agreed tolerance	Retain unvalidated status
Release provenance	Revision, parameter hash and limitations	Block a fidelity claim

CHECKS AND DECISIONS

- The product never presents an unmeasured 147 preset as one-to-one validated.
- A 147 selection does not inject modern Heritage preamp or servo-motor data.

SOURCES S042-01 Electro Music: Leslie 145/147 owner and service manual, vintage scan / S042-03 Electro Music/CBS: Leslie models 122–147 Service Information / Parts List, October 19

HARDWARE EVIDENCE / ORIGINAL DESIGN PROPOSAL

Freeze the power-amplifier revision before choosing a model

Leslie power stage: 6550 pair, transformer, feedback and supply

Use the 1971 117 V Leslie 122 rotor amplifier as the baseline. Its V3/V4 output valves are 6550s; V2 is the 12AU7A audio driver, whereas V1 serves rotor control. The optional reverb output amplifier is another circuit. The 1979 147 drawing instead identifies one 12AU7A, two 6550s and a 0C3, with transformer feedback entering the 6550 cathodes. There is no 6AU6 in that roster. [S043-01] [S043-02]

ORIGINAL DESIGN PROPOSAL: represent a power stage as a revision-specific electrical network with named driver, supply and load ports. Its input is the pair of loaded grid-drive connections, not an arbitrary normalized distortion amount. Its output is terminal voltage and current delivered to the crossover. Retain an explicit volts-per-digital-unit calibration so organ expression, preamplifier gain and power-stage overdrive remain causally related.

Give every fitted parameter a source state: documented, derived, measured, proposed or unknown. A shared valve type or transformer stock number cannot establish an identical closed-loop response. Archive the selected source drawing identifier with every profile and invalidate cached coefficients when topology changes.

Documented distinction	Implementation consequence
122: common cathode-bias branch	Shared bias state couples the output pair.
122: plate-to-grid RC feedback	Preserve local feedback and its loading.
147: cathode winding feedback	Use explicit winding ports and signed connections.
Optional reverb: separate amplifier	Do not add its valves to the rotor path.

CHECKS AND DECISIONS

- The profile reports its exact schematic revision and a complete port map before any audio comparison.

SOURCES S043-01 Electro Music Leslie 122/122RV/222/222RV service manual, publication 044578, June 197 / S043-02 Electro Music/CBS Leslie 122 and 147 manual, publication 136370, October 1979

ORIGINAL MODEL / TOPOLOGY

The transformer and speaker load belong inside the network

Leslie power stage: 6550 pair, transformer, feedback and supply

ORIGINAL DESIGN PROPOSAL: begin with a linear multiwinding transformer. Store winding resistance, a reciprocal inductance matrix, winding orientation and all external connections. Require nonnegative stored magnetic energy and nonnegative resistance. Add leakage and winding capacitance only at locations supported by a fitted equivalent circuit. Keep the crossover's electrical impedance connected to this model; a sound-radiation impulse response applied after it does not provide electrical loading. Coupled amplifier/load models have published precedents, but their guitar-speaker parameters are not Leslie data. [S043-05]

The ideal reflected-load identity is a useful unit test. Declare whether a turns ratio uses the complete plate-to-plate primary or one half-primary. Doubling the referenced primary turns multiplies reflected impedance by four. This is a derived transformer identity, not evidence of an actual Leslie ratio. Preserve terminal polarity when tracing 147 cathode feedback; a sign error can turn stabilizing feedback into regeneration.

Use a measured load-impedance function or a passive fitted circuit from chapter 044. The amplifier then sees both crossover branches simultaneously. Keep the later acoustic horn/drum processing separate from this electrical interface.

MODEL EQUATIONS

v = Rw*i + d(lambda)/dt; lambda = L*i. v: winding voltages in V; i: oriented currents in A; Rw: resistance matrix in ohms; L: inductance matrix in H; lambda: flux linkages in V*s.

Zprimary = n^2 * Zsecondary; n = Nprimary/Nsecondary. This ideal relation uses explicitly named complete windings; N denotes turns and Z is impedance in ohms.

P035 | Tung Sol 6550.jpg | bbear66 | Public domain | Tube type illustration; fitted specimen brand is unverified. Source and license

CHECKS AND DECISIONS

• Ideal-transformer tests conserve power and reproduce the declared turns-ratio convention; passive load fits cannot generate energy.

SOURCES S043-01 Electro Music Leslie 122/122RV/222/222RV service manual, publication 044578, June 197 / S043-02 Electro Music/CBS Leslie 122 and 147 manual, publication 136370, October 1979 / S043-05 Jaromir Macak and Jiri Schimmel, Simulation of a Vacuum-Tube Push-Pull Guitar Power A

ORIGINAL MODEL / STATE VARIABLES

Fit the 6550 currents, then solve shared bias and screen supply

Leslie power stage: 6550 pair, transformer, feedback and supply

The original Tung-Sol sheets distinguish plate and screen currents, with separate charts for different connection modes. Start from the beam-power curves, not the triode-connected chart or coefficients borrowed from a 12AX7. Fit separate current functions for each output valve using voltages measured relative to its cathode. Datasheet curves initialize a model; they do not identify the fitted valves in a particular cabinet. [S043-03]

ORIGINAL DESIGN PROPOSAL: solve the DC operating point before transients. For the baseline 122, the schematic documents 150 ohms and 200 microfarads in the shared cathode branch, with nominal labels of 25 V cathode, 415 V plate and 310 V screen relative to chassis. Thus the corresponding nominal electrode differences are derived as 390 V plate-to-cathode and 285 V screen-to-cathode. They are not newly measured targets. Include any signed current from the rotor-control branch at the bias node. [S043-01]

Preserve screen-current demand and coupling-capacitor charge during overload. The 0C3 has an approximately 105 V operating drop, but the Leslie places it in a series-drop path: screen voltage follows the surrounding supply circuit rather than remaining an independent ideal rail. [S043-01] [S043-04] Model reservoir and choke dynamics before adding an arbitrary sag control. Whether bias shift, grid-current charging or supply motion is perceptually significant remains a measurement question.

MODEL EQUATIONS

Ip = Fp(Vgk,Vpk,Vsck); Isc = Fsc(Vgk,Vpk,Vsck); Ig = Fg(Vgk,...). Currents are in A; all electrode voltages are relative to the corresponding cathode in V.

Ck*dVk/dt = Ik1 + Ik2 + Iaux - Vk/Rk. This reduced common-bias equation is for the 122; Ik includes electrode currents and Iaux is signed current entering the bias node from other branches. Ck in F, Rk in ohms, t in s.

CHECKS AND DECISIONS

• A 147 profile cannot inherit the reduced 122 cathode equation without adding its winding-feedback connections.

SOURCES S043-01 Electro Music Leslie 122/122RV/222/222RV service manual, publication 044578, June 197 / S043-02 Electro Music/CBS Leslie 122 and 147 manual, publication 136370, October 1979 / S043-03 Tung-Sol 6550 original manufacturer data and characteristic curves, 1963 sheets / S043-04 Tung-Sol 0A3, 0B3, 0C3, 0D3 voltage-regulator data, August 31, 1944

ORIGINAL DESIGN PROPOSAL / IMPLEMENTATION

Build the numerical reference before optimizing the plug-in

Leslie power stage: 6550 pair, transformer, feedback and supply

Implement a double-precision offline circuit reference with explicit capacitor, inductor and device states. First establish a converged DC solution, then reproduce linearized gain and phase, then enable nonlinear tube currents. Fit the linear transformer/load network before experimenting with core saturation. A Jiles-Atherton formulation is a candidate for magnetic memory, not evidence that a particular Leslie needs an audible hysteresis effect; its published implementation also makes geometry assumptions. [S043-06]

For a nonlinear core, flux linkage evolves from winding voltage. Core dimensions, turns and magnetization parameters must be identified or retained as unknowns; do not copy a paper's example values. Distinguish a physical parameter fit from a phenomenological transfer fit when measurements cannot uniquely identify internal quantities.

Oversample the complete coupled nonlinear region, including its feedback and reactive load. Filter before downsampling and measure aliasing against a higher-rate converged reference. Select the factor from a declared error budget. The real-time implementation must preallocate storage, bound solver work and report failed convergence; silently clipping invalid states would conceal a defective model.

```
profile = load_verified_topology()
reference = build_offline_network(profile)
state = solve_dc(reference)
verify_linearized_response(reference, state)
fit_devices_and_load(training_captures)
for each oversampled_step:
    candidate = bounded_implicit_solve(state, input)
    if converged(candidate) and finite(candidate):
        state = candidate
    else:
        record_failure_for_validation()
        use_prevalidated_bounded_fallback()
    output = read_load_terminal(state)
downsample_with_antialias_filter(output)
```

CHECKS AND DECISIONS

• Halving the offline integration step must converge the chosen waveform metrics; the same test is repeated after each model reduction.

SOURCES S043-05 Jaromir Macak and Jiri Schimmel, Simulation of a Vacuum-Tube Push-Pull Guitar Power A / S043-06 Martin Holters and Udo Zolzer, Circuit Simulation with Inductors and Transformers Bas

ORIGINAL DESIGN PROPOSAL / CALIBRATION

Acquire measurements that separate the possible causes

Leslie power stage: 6550 pair, transformer, feedback and supply

No hardware captures were made for this chapter. Commission a qualified technician to acquire calibrated electrical measurements using appropriate isolated instrumentation and loads. Record cabinet and amplifier identifiers, transformer markings, fitted valves, service history, warm-up state and the electrical measurement boundaries. Voltage and current calibration must accompany every capture; a microphone recording alone cannot uniquely identify the power stage.

Fit the smallest defensible model first. Small-signal gain and phase constrain the linear network; level dependence constrains the active devices; recovery and intermodulation constrain stateful behavior. Keep the acoustic recording chain, crossover loading and amplifier input scale separately documented. A published guitar-amplifier study illustrates the importance of load interaction, but supplies no measured Leslie parameters. [S043-05]

Partition captures by excitation and operating condition before fitting. Reserve held-out combinations so a model cannot pass by memorizing one sweep. If internal transformer parameters remain unidentifiable, deliver the measured terminal behavior, fitted range and uncertainty instead of invented construction details.

Proposed capture	Identification purpose	Required context
Low-level transfer and impedance	Gain, phase, electrical resonances	Defined input/output ports and load
Level-dependent sinusoidal sweeps	Compression and harmonic evolution	Calibrated terminal volts and load current
Low-frequency bursts and recovery	Separate bias/supply/core memory hypotheses	Repeated initial state and settling record
Two-tone and chord signals	Intermodulation and dynamic interaction	Independent validation material
Documented load comparison	Check reciprocal amplifier/load behavior	Approved bench load and connected crossover

CHECKS AND DECISIONS

• Every fitted coefficient traces to a capture or is marked proposed; no training capture is counted as independent validation.

SOURCES S043-05 Jaromir Macak and Jiri Schimmel, Simulation of a Vacuum-Tube Push-Pull Guitar Power A

ORIGINAL DESIGN PROPOSAL / ACCEPTANCE

Define acceptance without claiming an unmeasured one-to-one match

Leslie power stage: 6550 pair, transformer, feedback and supply

Maintain separate numerical, electrical and perceptual evidence. Agreement with an offline solver establishes implementation consistency. Agreement with calibrated hardware captures establishes fidelity over those measured conditions. Listening tests assess the resulting musical behavior. None alone establishes universal equivalence across all vintage cabinets, replacement valves and transformer revisions.

Deliver a profile manifest, connected netlist, calibrated port scales, device fits, transformer/load fit, solver configuration and held-out comparison report. Store each result with sample rate, oversampling filter, input level and topology hash. For reduced models, publish the validated input and load range. A normalized soft clipper plus a fixed filter can remain an explicitly approximate option, but should not be labeled a component-exact 6550 power stage.

Unresolved priorities are the actual transformer's winding ratios and parasitics, specimen device curves, supply impedance and crossover/driver load. Obtain those before assigning supposedly authentic numerical defaults. Keep unsupported core hysteresis and sag parameters outside the historical baseline until their contribution is identified. Source manuals establish circuit identity; datasheet examples and research algorithms do not finish the calibration. [S043-01] [S043-03] [S043-06]

Gate	Pass condition
Topology	Independent source trace confirms electrode and winding connections.
Numerics	Declared error bound holds against converged reference; no hidden solver failures.
Audio quality	Aliasing, latency and state transitions meet the project's published targets.
Physical comparison	Held-out captures meet predeclared tolerances with uncertainty reported.
Honesty	Unmeasured parameters and untested operating ranges remain explicitly labeled.

CHECKS AND DECISIONS

• Power ratings, datasheet example loads and attractive distortion plots cannot substitute for specimen measurements.

SOURCES S043-01 Electro Music Leslie 122/122RV/222/222RV service manual, publication 044578, June 197 / S043-03 Tung-Sol 6550 original manufacturer data and characteristic curves, 1963 sheets / S043-06 Martin Holters and Udo Zolzer, Circuit Simulation with Inductors and Transformers Bas

HARDWARE EVIDENCE

Four reactive values, two documented assembly identities

Crossover and driver electrical response

The 1971 manufacturer manual identifies divider 052803 and labels its application 800 Hz, 16 ohms. Its drawing provides actual component values; these do not need to be guessed from the nominal crossover label. The listed treble driver is 020305 and the 15-inch bass driver is 052308, both nominally 16 ohms. C16 is a dual paper capacitor, part 052852, rated 50 V with 5 percent tolerance. These are documented nominal specifications, not measurements of a surviving cabinet. [S044-01]

The October 1979 publication lists crossover assembly 113860 for both 122 and 147, retaining the same reactive values. Preserve separate dated profiles: matching component labels do not establish identical losses, wiring hardware, driver condition or serial applicability. The later parts illustration is particularly useful for locating the two inductors, dual capacitor and terminal strip. [S044-02]

ORIGINAL DESIGN PROPOSAL: initialize a source-backed component registry before designing filters. Store the nominal value, unit, source locator and verification status separately from any future specimen measurement. Keep unavailable parasitics null. Replacing a null with zero silently turns an unknown into a fictional factory specification.

Element	Documented nominal value	Documented component identity
L2, treble branch	3.2 mH	052837 in 1979 parts list
L3, bass branch	5.2 mH	052845 in 1979 parts list
C16A	7.8 microfarads	Section A of 052852
C16B	12.5 microfarads	Section B of 052852

CHECKS AND DECISIONS

- Do not replace the four values with generic 800 Hz textbook values.
- Identify assembly 052803 or 113860 explicitly in every exported cabinet profile.

SOURCES S044-01 Electro Music: Leslie 122 family manual, June 1971, publication 044578 / S044-02 Electro Music/CBS: Leslie models 122–147 Service Information / Parts List, October 19

SOURCE-TRANSCRIBED TOPOLOGY / ORIGINAL INTERPRETATION

Trace the floating treble branch before assigning grounds

Crossover and driver electrical response

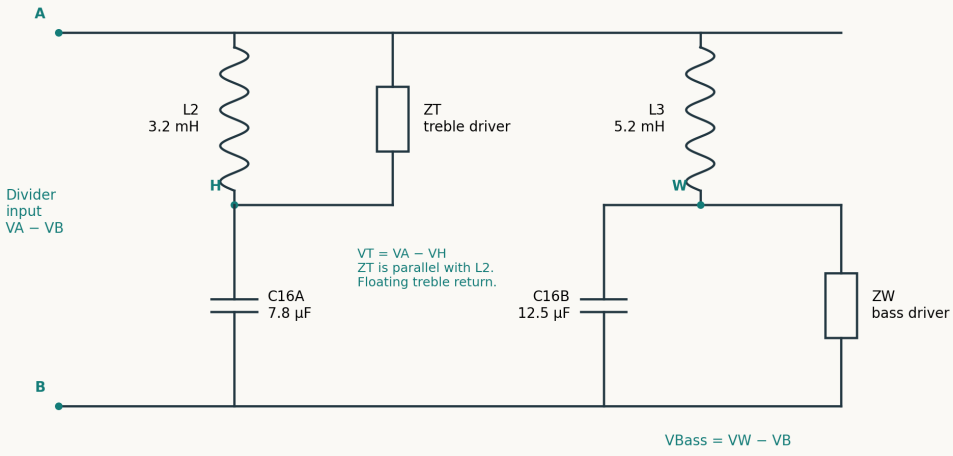
Define A as the divider input black lead, B as its red lead, H as the C16A red section connection, and W as the C16B black section connection. The following compact node list is transcribed from the published dividing networks. The capacitor common, white, connects to B. The treble load lies across L2; its other connection is A. This is easy to misread as an inductor feeding a grounded tweeter. [S044-01] [S044-02]

For the implementation, define Vin=VA−VB, VT=VA−VH and VW=VWnode−VB. These are mathematical voltage orientations. Preserve separate speaker-polarity metadata: a pressure sign cannot be inferred from the shape of a loudspeaker symbol. The 1979 122 drawing marks driver dots, while the selected 1971 drawing does not establish their acoustic sign. Resolve the specimen orientation by documented terminal identification and an acoustic impulse capture.

ORIGINAL MODEL: branch currents add at the amplifier port. Therefore the power amplifier sees a frequency-dependent combined load. Its source impedance changes the voltage reaching both branches. Feed that load back into chapter 043; two independent filters driven from an ideal voltage source are only a declared approximation. The 147 console-load selector belongs to the input interface, not this divider.

Connection	Element or load	Electrical role
A ↔ H	L2 and treble impedance ZT, parallel	Treble voltage measured across this pair
H ↔ B	C16A	Series capacitor in treble current path
A ↔ W	L3	Series inductor in bass path
W ↔ B	C16B and woofer impedance ZW, parallel	Bass voltage measured across this pair

Leslie 122 passive divider: electrical node relationships



Original redraw from the 1971 / 1979 manufacturer documents. B is the electrical reference, not protective earth.
Nominal component values; driver impedances are frequency-dependent. Electrical voltage polarity does not establish acoustic polarity.

Original electrical redraw of the manufacturer divider topology. The treble driver has a floating return. Nominal values; no specimen measurement.

CHECKS AND DECISIONS

- A continuity-derived node graph must agree with the selected drawing before profile promotion.
- Do not ground H or connect the treble return directly to B in the source-transcribed graph.

SOURCES S044-01 Electro Music: Leslie 122 family manual, June 1971, publication 044578 / S044-02 Electro Music/CBS: Leslie models 122–147 Service Information / Parts List, October 19

ORIGINAL CIRCUIT DERIVATION

Derive the loaded response, then expose the approximation

Crossover and driver electrical response

Use s=j2πf and parallel(x,y)=xy/(x+y). ZT and ZW are complex driver terminal impedances. With ideal reactive elements, define PT=parallel(sL2,ZT), PW=parallel(1/(sC16B),ZW), BT=PT+1/(sC16A), and BW=sL3+PW. The equations below follow from the preceding node graph. Add winding resistance, capacitor losses and fitted driver dynamics only as explicitly documented or measured parameters.

A useful reproducible calculation substitutes two constant 16-ohm resistors and an ideal source. This is an illustrative electrical model, not an acoustic prediction or measured Leslie. Its unloaded LC natural frequencies are approximately 1007.39 Hz for treble and 624.26 Hz for bass; neither equals an asserted exact 800 Hz cutoff. At 800 Hz, the calculated input-impedance magnitude is 23.83 ohms. The numbers demonstrate why a nominal driver rating is insufficient to specify the full amplifier load.

Keep phase referenced to the declared port orientations. Do not force the voltage responses to sum to unity. Acoustic summation additionally requires driver transfer functions, horn/drum radiation, microphone position and propagation delay. Those dependencies belong to later acoustic chapters; compensating them here would count the same effect twice.

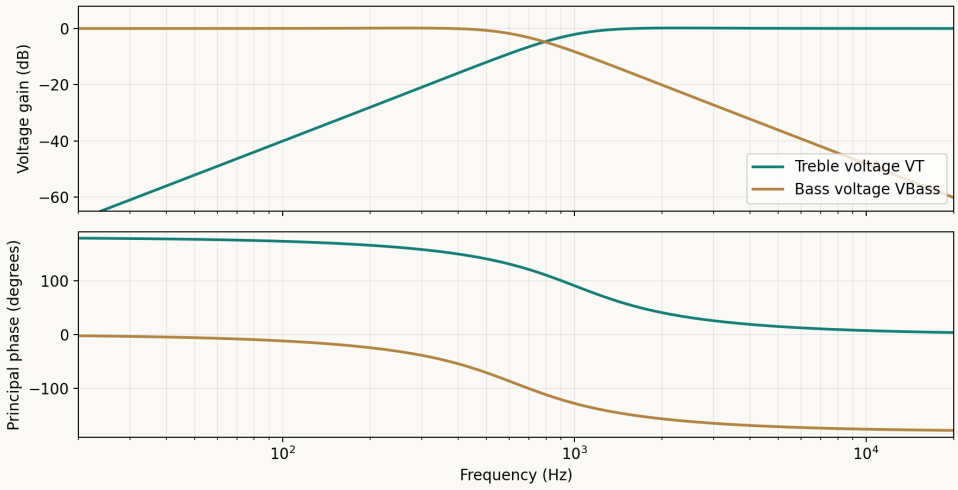
Derived test frequency	Treble magnitude / phase	Bass magnitude / phase
400 Hz	−15.88 dB / +149.17°	−0.06 dB / −54.19°
800 Hz	−4.60 dB / +110.17°	−4.89 dB / −111.46°
1600 Hz	+0.002 dB / +52.86°	−16.20 dB / −149.60°

MODEL EQUATIONS

HT(s)=VT/Vin=PT/BT; HW(s)=VW/Vin=PW/BW.

Zin(s)=parallel(BT,BW); Iin=Vin/Zin.

Calculated divider response with ideal 16 Ω resistive loads



Computed reference fixture, not a measurement. Ideal voltage source, lossless L/C, fixed resistors in place of both loudspeakers. This plot cannot establish the acoustic crossover, real driver loading, cabinet response or amplifier feedback behavior.

Computed ideal divider response with fixed 16-ohm resistors replacing the drivers. Not an acoustic response or a physical measurement.

CHECKS AND DECISIONS

- Numerical fixture: values above assume exactly 16-ohm resistive loads, zero losses and source impedance zero.
- Recalculate after changing either driver impedance; never preserve an unrelated fixed crossover coefficient set.

SOURCES S044-01 Electro Music: Leslie 122 family manual, June 1971, publication 044578 / S044-02 Electro Music/CBS: Leslie models 122–147 Service Information / Parts List, October 19

ORIGINAL DESIGN PROPOSAL / IMPLEMENTATION

Implement one electrical model with explicit acoustic ownership

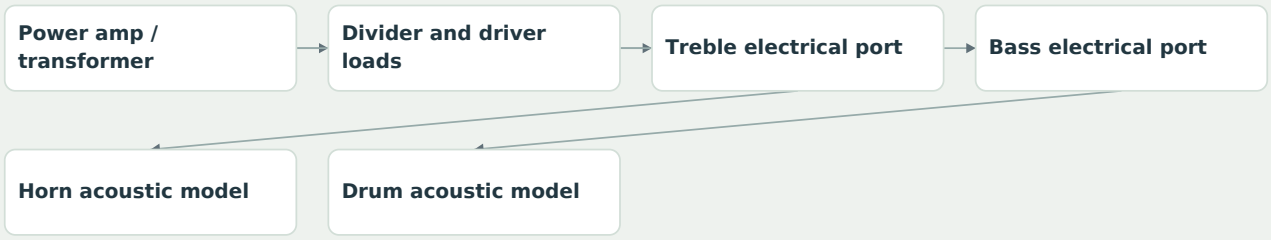
Crossover and driver electrical response

First implement the four-element network with resistive loads and match the analytic fixture. Next replace each load with a stable, passive fit to a measured complex impedance. A single resistance plus inductance misses a driver resonance; a compression driver need not share the woofer model structure. Every fit must include its frequency range, stimulus level, installed configuration and residual error.

Construct a state-space model from the node graph. Candidate states are two capacitor voltages, two inductor currents and the driver-load states. Use consistent SI units and an offline matrix factorization. A bilinear discretization is a proposed starting point; inspect its frequency warping against the analog reference at every supported sample rate. Static linear filtering creates no new harmonics, but a coupled nonlinear amplifier/driver path needs the oversampling and anti-aliasing strategy of chapters 043 and 055.

Expose Vin/Iin to the amplifier solver and VT/IT/VW/IW to downstream driver models. Choose whether the acoustic block consumes terminal voltage, current or cone velocity, then document that choice. Do not simulate the same electromechanical resonance in both the impedance fit and a separately cascaded driver-response filter without a consistent shared formulation. Preallocate states and swap prepared profiles away from the audio callback; preserve state continuity or use a defined transition.

Electrical ports and model ownership



Original implementation architecture. The feedback arrow represents electrical loading, not an added acoustic feedback effect.

CHECKS AND DECISIONS

- The audio callback performs no allocation, file reads or coefficient fitting.
- A sample-rate change cannot silently retain coefficients derived for another rate.

SOURCES S044-01 Electro Music: Leslie 122 family manual, June 1971, publication 044578 / S044-02 Electro Music/CBS: Leslie models 122–147 Service Information / Parts List, October 19

PROPOSED MEASUREMENT / ALL CAPTURES PENDING

Capture the divider as a measured multiport system

Crossover and driver electrical response

Acquire one identified cabinet before claiming specimen fidelity. Record its assembly numbers, driver markings, wiring photographs, repairs and room conditions. Qualified bench work should measure the isolated passive network and drivers without turning this software specification into instructions for exposed energized valve circuitry. Preserve the installed acoustic loading when measuring driver impedance, or clearly label any removed-driver measurement.

Use calibrated voltage/current channels and a known stimulus to estimate complex input impedance and both terminal transfer functions. Preserve absolute timing and polarity. Measure several safe signal levels chosen for the actual driver and fixture; an arbitrary universal test voltage could invalidate the capture. Record channel calibration, cable effects, source impedance and noise floor with the raw data.

Capture the electrical behavior with rotors stopped at documented positions before fitting moving acoustic effects. Separate thermal drift and amplitude-dependent loading from repeatability. A measured multiport surrogate is a valid intermediate model when component parasitics are unavailable, provided its source/load conditions and domain are explicit. Do not label a fitted terminal response a recovered component-exact circuit.

Capture	Required output	Resolves
Component inspection / metrology	L, C, winding R, capacitor loss with conditions	Nominal versus specimen values
Installed driver impedance	Frequency, Re(Z), Im(Z), level, orientation	Frequency-dependent loading
Loaded divider sweep	Vin, Iin, VT, IT, VW, IW, shared clock	Magnitude, phase and input load
Polarity / acoustic impulse	Terminal identity and signed pressure arrival	Electrical-to-acoustic sign
Repeat captures	Raw files plus uncertainty estimate	Fit error versus fixture variation

CHECKS AND DECISIONS

- Every measured field must link to an immutable raw capture and calibration record.
- No capture is presently available; retain measured fields as null.

ORIGINAL ACCEPTANCE PLAN / KNOWN LIMITS

Acceptance gates that prevent a convincing but wrong split

Crossover and driver electrical response

Gate one is source fidelity: the machine-readable registry must contain the verified four values, dated assembly identity and node graph. Gate two is mathematics: independently compare the digital model with the analytic resistor-load calculation over frequency, including input admittance and both complex outputs. This catches a grounded treble branch, swapped capacitor sections or a phase convention error that a listening test might miss.

Gate three is numerical behavior: exercise supported sample rates, long decays, silence, denormal handling and profile changes. Check passivity of fitted loads and stability of the coupled amplifier model. Track an energy balance under the declared port signs; passive internal models must not supply sustained net energy. Explicitly distinguish reactive voltage magnification from power generation.

Gate four is hardware agreement after acquisition. Set tolerances from capture uncertainty and the intended fidelity budget, not an invented factory specification. Compare held-out levels and driver configurations as well as fitting data. Finally validate acoustic integration with the independent horn/drum models. A useful electrical match does not prove microphone-position-independent acoustic identity.

Deliver the registry, analytic-response fixture, source-linked node list, calibrated load profiles and regression report. Outstanding data include winding resistance, capacitor loss/leakage, actual driver impedances, installed polarity and serial applicability of assembly changes. A 50 V capacitor rating is not a digital clipping threshold. The nominal 800 Hz label remains an identity clue, not a universal measured crossing.

Companion executable: reference_code/crossover_ac_reference.py reproduces the 121-point ideal-resistor fixture and checks it against independent node-voltage equations. This is an electrical algebra check; real loudspeaker loading and acoustic fidelity remain outside its scope.

Status	Deliverable	Release consequence
Documented	Four nominal reactive values and source topology	Suitable for a declared idealized network
Derived	Resistor-load transfer and input-impedance fixture	Suitable for implementation verification
Unknown	Specimen losses, impedance and acoustic polarity	Blocks specimen-verified fidelity claim
Proposed	Measurement and acceptance procedures	Execute before claiming hardware agreement

CHECKS AND DECISIONS

- Never substitute an equal-power or Linkwitz–Riley split solely because the cabinet is labeled 800 Hz.
- No claim of a measured 1:1 recreation passes while the specimen-data fields remain null.

SOURCES S044-01 Electro Music: Leslie 122 family manual, June 1971, publication 044578 / S044-02 Electro Music/CBS: Leslie models 122–147 Service Information / Parts List, October 19

HARDWARE EVIDENCE

Identify the horn assembly before fitting its sound

Horn rotor and compression driver acoustics

The baseline is the vintage Leslie 122 family, with an identified specimen still to be acquired. The factory manual separates the treble driver, rotating horn, reflector and spindle; its mechanical view and parts list are printed pages 13–14. Preserve those identities in the asset manifest instead of assigning every visible horn-shaped part the same acoustic role. [S045-01]

Leslie’s historical patent describes one fed horn and an opposite horn isolated by a wall for balancing. Henricksen independently describes the stationary compression driver and dummy side in his technical account. These establish the mechanism; the patent does not establish the dimensions or materials of every production 122. [S045-02] [S045-04]

ORIGINAL DESIGN PROPOSAL: start with a HornAssemblyProfile containing specimen ID, photographs, installed driver markings, horn and reflector condition, measured geometry, and source confidence. The available P023 photograph illustrates appearance, but its unidentified cabinet and motor cannot certify the baseline revision. A photograph without a scale must not silently become a dimensioned CAD model. [S045-05]

Factory entry, printed 14	Documented identity	Agent action
020305	Treble driver, 16 ohms	Verify actual fitted driver markings
050807	Treble horn assembly	Record intact assembly and alterations
050823	Treble-horn reflector assembly	Photograph presence and mounting
050906	Treble-horn spindle assembly	Measure axis and coupling geometry

P023 | Lesliehorn.jpg | DukeHammond | Public domain Source and license

CHECKS AND DECISIONS

- Reject a profile that labels its dummy side as a second electrically driven horn.
- Keep unknown dimensions nullable; never infer millimetres from an unscaled image.

SOURCES S045-01 Electro Music: General Operating and Maintenance Instructions, Leslie 122/122V/122RV/ / S045-02 Donald J. Leslie: US2489653A, Rotatable tremulant sound producer / S045-04 Clifford A. Henricksen: Unearthing the Mysteries of the Leslie Cabinet, Recording Eng / S045-05 DukeHammond: Lesliehorn.jpg

ORIGINAL MODEL

Follow the acoustic path through the rotating assembly

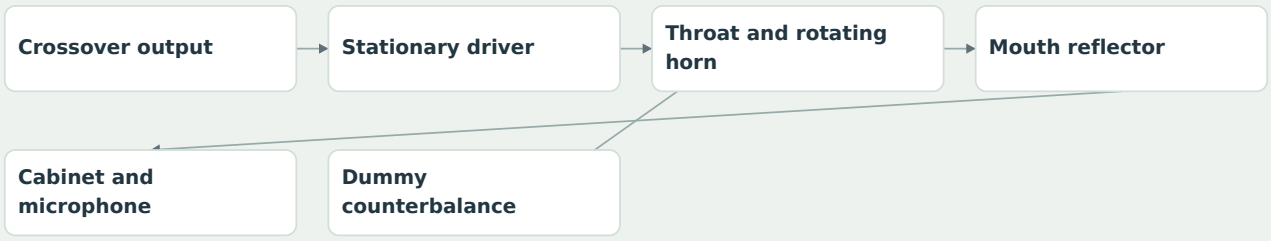
Horn rotor and compression driver acoustics

The useful physical distinction is a stationary sound-producing driver feeding a rotating acoustic projector. The opposite balancing horn is not a second driven radiator. Its possible scattering, aerodynamic and inertial effects remain separate modeling questions; “dummy” does not require pretending that the physical object is absent. [S045-02] [S045-04]

Henricksen reports changed angular radiation with the mouth diffuser removed and a change in apparent acoustic source position. His response example uses a University ID-40 driver, and the described measurements exclude the complete reflecting cabinet. Therefore those curves cannot be renamed a stock 122/Jensen response. Treat the diffuser state as a distinct calibrated configuration. [S045-04]

ORIGINAL DESIGN PROPOSAL: declare whether each measured transfer function includes the driver, horn, free-space propagation, cabinet and microphone. If a complete measured response already includes a subsystem, do not apply an additional model of that subsystem by default. A voltage-to-pressure surrogate is useful for a first acoustic model, but a circuit-coupled crossover also needs the driver’s electrical input impedance and its loading assumptions. Assign that boundary explicitly with chapters 39, 44 and 55.

One driven acoustic route



Original conceptual topology, not a measured drawing. The dummy branch carries mechanical coupling, not a duplicate audio feed.

CHECKS AND DECISIONS

- Every response asset declares included subsystems and reference plane.
- A diffuser preset must select an angular response profile, not merely a global treble control.

SOURCES S045-02 Donald J. Leslie: US2489653A, Rotatable tremulant sound producer / S045-04 Clifford A. Henricksen: Unearthing the Mysteries of the Leslie Cabinet, Recording Eng

ORIGINAL MODEL

Build a moving-mouth approximation with an honest boundary

Horn rotor and compression driver acoustics

The DAFx model explicitly treats a circular moving source as an approximation and adds measured angular responses. Its measurements concern a Leslie 600 horn, so their existence supports a modeling method, not numerical calibration of the target 122. [S045-03]

ORIGINAL DERIVATION: use a common rotor angle theta in radians, axis centre o and mouth radius r in metres. Rz is a rotation about the vertical axis; s is mouth position, u its outward unit axis, and m a microphone position. Distance d is metres; c is sound speed in metres per second. Alpha is the off-axis angle. For a three-dimensional measured response, retain elevation and signed azimuth instead of collapsing everything to alpha.

Define complex Hrad(f, angle, profile) as the source transfer after removing only documented external propagation. Its frequency f is in hertz; its gain and phase carry radiation coloration. For a free-field far-field pressure approximation, multiply by d_ref/d and delay by d/c. This pressure law is not an intensity law. Finite mouths, nearby louvres and near-field microphones can invalidate the scalar approximation; introduce measured position dependence when held-out data demonstrate the need.

MODEL EQUATIONS

s(theta) = o + Rz(theta) * [r, 0, 0]

u(theta) = Rz(theta) * u0; d(theta) = norm(m - s(theta))

alpha(theta) = acos(clamp(dot(u(theta), (m-s(theta))/d(theta)), -1, 1))

t_receive = t_emit + d(theta(t_emit))/c

CHECKS AND DECISIONS

• Use emission-time geometry for the reference moving-source solution; document any low-speed approximation.

• Do not equate photographed mouth radius with a frequency-independent acoustic phase centre.

SOURCES S045-03 Smith, Serafin, Abel and Berners: Doppler Simulation and the Leslie, DAFx 2002

IMPLEMENTATION

Make the horn module reproducible and real-time safe

Horn rotor and compression driver acoustics

ORIGINAL DESIGN PROPOSAL: build a deterministic offline horn renderer before integrating host automation. Input is calibrated driver-terminal voltage, mechanical angle/state, microphone poses and an immutable profile. Output is microphone pressure or a declared pressure-to-digital conversion. The motor model owns theta; every microphone reads that same state. Independent left/right LFOs would represent different rotors.

Precompute angular filter banks or a compact fitted basis away from the audio callback. Keep a shared time reference across impulse responses. Removing arrival delay is allowed only when the removed delay is stored and reconstructed exactly once. Preserve angular gain differences; normalizing every orientation would erase the amplitude modulation being modeled.

For the first implementation, interpolate periodic angular weights over stable fixed filters and use a separately validated moving-source delay stage. Check the interpolation bandwidth and resampling alias energy at the highest supported speed and input frequency. Angle wraparound must be continuous. Reserve memory, bound filter lengths, and swap complete profiles at a controlled transition. These are proposed engineering requirements, not claims of an implemented renderer.

Interface field	Meaning	Constraint
profile_id / provenance	Immutable physical configuration	Unknowns remain explicit
theta / omega	Radians / radians per second	One mechanical owner
Hrad and reference plane	Angular complex response	No duplicated propagation
microphone poses	Metres and orientation	Same scene for all channels
pressure_scale	Pascals per digital unit	Declared calibration or unknown

CHECKS AND DECISIONS

- Changing audio block size must not change the offline result beyond documented numerical tolerance.
- Static theta must reduce to the selected static transfer, with no autonomous modulation.

MEASUREMENT

Acquire the radiation data the photographs cannot supply

Horn rotor and compression driver acoustics

The DAFx experiment measured a Leslie 600 horn at fixed angular increments of 15 degrees using Golay-code pairs, and separated an angle-independent component attributed to base leakage. This is a useful acquisition precedent; its angular spacing is not a guaranteed resolution for every frequency or cabinet configuration. [S045-03]

ORIGINAL MEASUREMENT PLAN: capture a fixed-angle sweep with the exact driver, horn and diffuser identified. Record electrical drive reference, calibrated microphone response, distance, height, temperature, angular zero and time alignment. Measure repeatability before fitting a model. Acquire the isolated assembly and installed cabinet separately, with enough additional angles near sharp lobes to resolve their variation.

Use multiple microphone positions and elevations, not just one on-axis magnitude trace. Preserve complex responses and uncertainty. Hold back angles and positions from fitting. Then record rotation with a synchronized angle or tachometer reference to test the dynamic model. Choose safe stimulus level and bandwidth from the actual driver’s limits and a qualified measurement setup. This project has not performed these captures.

Capture	Separates	Required metadata
Fixed rotor, low level	Angular linear radiation	Voltage, angle, pose, common clock
Diffuser configurations	Geometry-dependent coloration	Photos and separate profile IDs
Several drive levels	Driver nonlinearity	Terminal signal and level history
Free field and cabinet	Radiator versus enclosure	Boundary and panel configuration
Rotating, held-out scene	Motion and interpolation errors	Synchronized rotor angle

CHECKS AND DECISIONS

- Repeat one orientation after the sweep to quantify drift.
- Never independently time-align every measured angle and discard the removed delays.

SOURCES S045-03 Smith, Serafin, Abel and Berners: Doppler Simulation and the Leslie, DAFx 2002

ACCEPTANCE

Accept the horn model by predictions it can fail

Horn rotor and compression driver acoustics

ORIGINAL ACCEPTANCE PLAN: first verify topology, units, phase conventions and static responses; then compare held-out angular spectra, arrival timing and rotating sidebands. Establish numerical tolerances from measurement uncertainty and the product’s fidelity budget before examining the test set. No fidelity score is justified without those reference captures.

Inspect failures by subsystem. Wrong modulation depth can arise from directivity, microphone placement, reflections or geometric delay; it does not uniquely identify the rotor radius. Multiple level maxima per revolution do not prove two driven horns. A plausible sound from one stereo microphone setup is insufficient evidence for a transferable physical model.

The release decision is therefore conditional: a parameterized moving-mouth and angular-filter surrogate can be implemented now, while “matched to specimen” remains blocked by geometry and acoustic measurements. Acquire actual driver identity, reflector configuration, phase-preserving angular responses and synchronized rotation captures before promoting this module to the calibrated tier. Keep the photograph, CAD appearance and acoustic profile linked, but allow each to carry its own evidence status.

Test or symptom	Decision rule
One rotation and angle wrap	No discontinuity or unintended extra audio source
Static versus moving renderer	Zero speed reproduces the static response
Several microphones	One shared rotor produces all channel trajectories
Near-field mismatch	Extend measured position dependence; do not hide with EQ
Diffuser change	Validate held-out angle and phase behavior
High-frequency sidebands	Separate physical modulation from numerical aliases

CHECKS AND DECISIONS

- Publish hardware-calibrated, literature-derived and proposed parameters as distinct categories.
- Archive the exact profile and render settings for every acceptance recording.

HARDWARE EVIDENCE

The driver stays mounted; the deflector turns

Bass drum rotor and stationary woofer acoustics

The factory mechanical drawing places the bass loudspeaker above a separate shaft-supported lower rotor. Its parts list identifies the speaker, wooden rotor, cloth cover and complete rotor assembly individually. The cabinet therefore must not be modeled as a woofer cone orbiting the shaft. Printed pages 13–14 correspond to viewer pages 15–16 in the 26-page PDF. [S046-01]

ORIGINAL DESIGN PROPOSAL: create separate scene and DSP objects for the fixed driver, rotating deflector, fixed cabinet openings and microphone. Speaker excursion belongs to the driver model; angular motion belongs to the deflector. The renderer should animate the drum and shaft while retaining the loudspeaker basket at its mounting plane.

The component photograph makes that separation legible, but its exact specimen is unidentified. Use it for orientation, not scale. [S046-05] Preserve factory nominal labels as evidence fields; do not turn them into measured clearances, acoustic center positions or a complete manufacturing drawing.

Documented item	Factory identifier / label	Modeling consequence
Bass loudspeaker	052308; 15 in, 16 ohm	Fixed frame; electrical/acoustic response from chapter 044.
Wooden rotor	032292; size label 17 in x 9 in	Geometry nominal only; survey aperture and interior contour.
Cloth cover / complete rotor	018713 / 051300; assembly label 17 in	Treat cloth state and deflector as separate evidence fields.
Shaft and pulley	051706	Rotational reference; dynamics belong to chapter 047.

P024 | Lowerrotor.jpg | DukeHammond | Public domain Source and license

CHECKS AND DECISIONS

- Scene review: changing rotor angle cannot move the woofer basket or electrical terminals.
- Photograph captions must identify generic versus reference-specimen imagery.

SOURCES S046-01 Electro Music: Leslie 122 family General Operating and Maintenance Instructions, publ / S046-05 DukeHammond: Lowerrotor.jpg, original lower-rotor photograph

MECHANISM / ORIGINAL MODEL

Follow sound through the changing opening

Bass drum rotor and stationary woofer acoustics

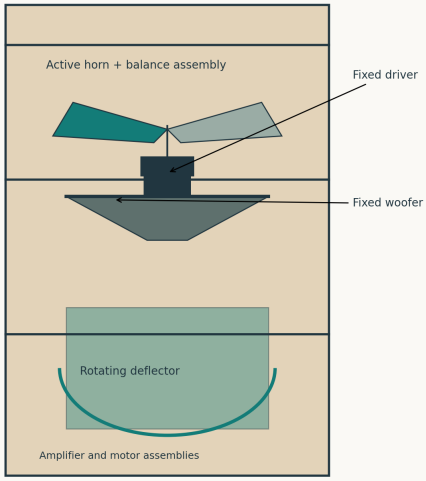
Leslie’s 1949 patent describes a downward-facing stationary low-frequency driver above a rotating channel with a side opening. That historical embodiment explains the principle; it does not establish every dimension or construction detail of the 1971 production 122. [S046-02]

ORIGINAL DESIGN PROPOSAL: represent the lower system as a fixed acoustic excitation coupled through an angle-dependent deflector into fixed cabinet vents. “Shutter” means changing radiation and obstruction, not a perfect on/off valve. Permit a nonzero minimum response and frequency-dependent changes. A scalar tremolo cannot represent changing phase or interference between paths.

Keep three effects distinguishable: orientation-dependent transmission, any changing propagation delay, and cabinet/room interference. Their separation is a modeling choice, not automatically a uniquely identifiable physical decomposition. A measured total transfer may already contain all three; adding another cabinet response would count some effects twice.

The 2002 DAFx paper discusses amplitude modulation and angle-dependent filtering for the woofer port, while explicitly omitting its bass measurement analysis. Its published horn fits therefore provide an algorithmic lead, not a calibrated bass model. [S046-03]

The rotating acoustic elements are not the loudspeakers



Conceptual channel separation, not a Leslie122 construction or dimensional drawing. Consult the actual cabinet assembly plates.

Original explanatory calculation. Illustrative parameters are not measured instrument data.

CHECKS AND DECISIONS

- At frozen angle, the module becomes a stationary acoustic filter and still emits bass.
- Do not force every radiation path to share the same delay modulation.

SOURCES S046-02 Donald J. Leslie: US2489653A, Rotatable tremulant sound producer, 1949 / S046-03 Julius Smith, Stefania Serafin, Jonathan Abel and David Berners: Doppler Simulation a

ORIGINAL MODEL

Geometry, wavelength and the Doppler approximation

Bass drum rotor and stationary woofer acoustics

ORIGINAL DERIVATION: use dimensionless geometry before choosing a bass Doppler depth. Let f be acoustic frequency in Hz, a a selected geometric scale in metres, c sound speed in metres/second, and deltaL a candidate path-length excursion in metres. The quantity ka compares object scale with wavelength; beta compares path excursion with acoustic phase. Neither is a measured 122 parameter here.

For an ideal narrowband signal evaluated at time t minus a changing delay tau(t), instantaneous frequency follows the derivative below. With a fixed geometric excursion, absolute phase modulation decreases as f decreases. However, the fractional frequency shift of that ideal path depends on delay slope, not directly on f. “Bass Doppler is zero because bass is low” is therefore not a valid derivation.

For a real rotating deflector, the effective radiating region can be distributed and frequency dependent. Its fitted delay radius need not equal the drum radius, and certainly is not the cone’s orbital radius. Treat a rotating-point outlet as a reduced approximation; compare its residuals against a complex angle-frequency response before accepting it.

Diagnostic	Interpretation	Forbidden shortcut
ka much smaller than 1	Compact scale relative to wavelength	Do not infer zero amplitude modulation or zero loading change.
ka near or above 1	Spatial phase variation needs attention	Do not assume a single point now describes the whole outlet.
Small beta	Small phase excursion for this proposed path	Do not confuse weak phase excursion with absent directivity.

MODEL EQUATIONS

k = 2*pi*f/c; ka = 2*pi*f*a/c (dimensionless).

beta = 2*pi*f*deltaL/c (phase-modulation index, radians).

y(t) = cos(2*pi*f*(t-tau(t))); f_inst(t)/f = 1 - d(tau)/dt.

CHECKS AND DECISIONS

• For a synthetic sinusoid and known smooth delay law, recover the predicted instantaneous-frequency ratio away from amplitude nulls.

IMPLEMENTATION

Build the bass module around one rotor state

Bass drum rotor and stationary woofer acoustics

ORIGINAL DESIGN PROPOSAL: first implement a frozen-angle response bank H_m(f,theta), where m identifies a microphone and theta the bass angle. Build the bank offline from calibrated impulse responses or explicitly labeled provisional filters. Store sample rate, pressure/voltage reference, microphone position, panel configuration, angular convention and whether propagation delay is included.

Then interpolate periodic angular responses with persistent filter states. A real-valued FIR bank or fitted low-rank FIR basis provides an inspectable starting point. Complex frequency-domain interpolation must preserve conjugate symmetry and use phase/arrival alignment deliberately. Crossfade outputs, not unrelated recursive-filter coefficients. Preserve physical gain differences; per-angle loudness normalization would erase the effect being modeled. Frozen-angle interpolation is a quasi-static approximation; validate it during rotation before treating it as a complete retarded-time propagation model.

Only introduce an explicit fractional-delay stage after identifying and removing the corresponding delay from the response bank. Otherwise the same motion is reproduced twice. The motor module supplies one continuous bass angle to every microphone; stereo must follow a common physical event, with microphone-dependent paths. Chapters 049–053 refine interpolation, directivity, microphones and time-varying convolution.

```
prepare(profile, sampleRate, microphones)
load preallocated periodic response basis
for each audio sample:
    theta = bassMotor.angleRadians()
    excitation = bassDriver.process(crossoverBass)
    basisSignals = persistentFIRs.process(excitation)
    for each microphone m:
        weights = periodicWeights(m, theta)
        y[m] = dot(weights, basisSignals[m])
    bassMotor.advanceOneSample()
return y
```

CHECKS AND DECISIONS

• Angle wrap at 2pi must be continuous in output and interpolation weights.

• Audio processing allocates no memory and reads no files.

• Check modulation sidebands against an offline higher-rate reference before deciding oversampling is unnecessary.

MEASUREMENT

Acquire bass evidence without confusing interference with motion

Bass drum rotor and stationary woofer acoustics

A 2008 study measured a 122A cabinet and found complicated interference in an 800 Hz experiment; a horn-only check still showed the effect. That is useful enclosure evidence, but neither a bass-only transfer dataset nor proof of a vintage 122 bass Doppler law. [S046-04]

ORIGINAL MEASUREMENT PROPOSAL: acquire lower-branch responses with a documented safe isolation arrangement, unchanged cabinet panels and a recorded reference electrical signal. Capture angle-locked stationary sweeps, then continuous rotation with a synchronized optical angle reference. Keep microphone coordinates, polarity, gain, air conditions and orientation convention in every record. Use closed-cabinet captures for the target sound; open-cabinet inspection is a distinct configuration.

Fit on some angles and validate on withheld angles and speeds. At very low frequencies, a short time gate cannot cleanly separate room reflections while retaining frequency resolution; use an appropriate room, longer controlled capture or an explicitly modeled room boundary. Examine complex transfer and angle-locked sidebands alongside level envelopes. Exclude amplitude-null neighborhoods when interpreting instantaneous frequency: unstable phase there can mimic spectacular Doppler excursions.

Capture	Resolves	Required companion
Stopped-angle bass sweeps	Magnitude, phase and angular periodicity	Electrical reference; same acoustic geometry.
Steady rotation tones / broadband	Modulation spectrum and phase evolution	Tachometer and no automatic level control.
Speed transitions	Acoustic evolution during changing angle rate	Independent motor trace from chapter 047.
Repeated microphone placements	Geometry sensitivity and repeatability	Coordinates and repeated baseline capture.

P041 | 122A cabinet open and closed | R. Kronland-Martinet and T. Voinier | CC BY 2.0 | Photograph of the 122A specimen used by Kronland-Martinet and Voinier (2008); not a vintage 122 dimensional reference. Source and license

CHECKS AND DECISIONS

- Deliver raw recordings, calibration metadata, analysis scripts and fitted coefficients together.
- A fit to training angles alone is not a fidelity result.

SOURCES S046-04 R. Kronland-Martinet and T. Voinier: Real-Time Perceptual Simulation of Moving Source

ACCEPTANCE

Release gates and the evidence still missing

Bass drum rotor and stationary woofer acoustics

ORIGINAL ACCEPTANCE PROPOSAL: approve this subsystem in stages. The first gate verifies fixed-driver geometry, stable angle interpolation and deterministic processing. The second verifies agreement with held-out bass captures. The third checks integration with crossover, horn, microphone and room modules. A musically convincing tremolo is useful during development, but it cannot establish acoustic equivalence.

Store provisional and calibrated profiles separately. A reference profile requires a specimen identifier, bass-driver identification, deflector photographs, panel state, verified rotation direction, mechanical survey and licensed measurement set. Record uncertainty in acoustic center, leakage fraction, cloth transmission and driver loading instead of hiding it behind a single “authenticity” control.

The completion artifact for the implementing agent is BassRotorProfile plus a real-time module, a frozen-angle response inspector, reproducible renders and the measurement manifest. Unknowns should remain explicit nullable fields until acquired. This chapter supplies a defensible model boundary and tests, not a claim that the bass rotor has been numerically reconstructed one-to-one.

Gate	Falsifiable check	Interpretation
Static behavior	Zero angular speed preserves the chosen response without muting	Tests acoustic/state separation.
Periodic geometry	Angle 0 and 2pi agree; reversing angle retraces frozen responses	Tests convention and interpolation.
Delay separation	Setting effective path excursion to zero removes only explicit geometric FM	Directional filtering may remain.
Specimen validation	Held-out complex responses and sidebands meet declared uncertainty-based tolerances	Tolerances set from repeatability, not invented factory limits.
Integration	Combined bass/horn response retains measured crossover phase and microphone coherence	Avoid independent stereo tremolos or duplicated room/cabinet filtering.

CHECKS AND DECISIONS

- Publish residual plots and measurement uncertainty; do not substitute a single similarity percentage.
- Freeze acceptance tolerances before evaluating held-out recordings.

HARDWARE EVIDENCE

Identify two independent mechanical drive systems

Leslie motor and belt dynamics

The 1971 manual documents separate upper and lower two-speed motor assemblies. Its mechanical description is the baseline evidence; a cabinet name alone does not identify installed motors, pulleys or modifications. [S047-01] The 1979 exploded views provide another dated parts configuration. [S047-02]

ORIGINAL DESIGN PROPOSAL: maintain a mechanical profile beside the amplifier profile. Record cabinet serial, source publication, motor labels, pulley groove, belt type/condition, regional supply and any stopping accessory. Keep motor shaft speed distinct from acoustic rotor speed. A plug-in preset should identify both horn and drum profiles, even when one switch commands them together.

The modern 122H uses brushless DC servo motors and adjustable motion parameters. Its published defaults must not fill missing vintage measurements. [S047-05] An original Leslie Brake Accessory also exists, but it is an optional configuration rather than proof that every vintage 122 has a stop control. [S047-04]

Documented identity	Evidence or implementation consequence
1979 upper 117V 60 Hz /117V 50 Hz	140019 /140020 motor assemblies; retain exact revision [S047-02]
1979 upper 60 Hz /50 Hz pulley	050500 /050559; three-step part identities [S047-02]
1979 lower 60 Hz /50 Hz pulley	018309 /014928; separate ratio metadata [S047-02]
Vintage baseline control	Slow and Fast; optional modes need separate provenance
Unknown on the selected specimen	RPM, direction, inertia, friction and transition curves remain null

CHECKS AND DECISIONS

- A 50 Hz profile cannot inherit a 60 Hz pulley identity silently.
- No modern 122H speed or timing is labelled a vintage 122 measurement.

SOURCES S047-01 Electro Music Leslie 122/122V/122RV/142/222 service manual, publication 044578, June / S047-02 Electro Music/CBS Leslie 122/147 service publication 136370, October 1979 / S047-04 Leslie Brake Accessory, instructions038398, for 122/142/222 series / S047-05 Hammond Leslie Heritage 122H/142H owner manual, V1.00-220713

MECHANISM AND ORIGINAL GEOMETRY MODEL

Follow torque through contact, pulley and belt

Leslie motor and belt dynamics

In Chorale, the energized small motor engages the large motor through its rim wheel; its spring withdraws the shaft when inactive. The horn belt uses a spring-loaded idler and three-step pulley: the middle groove is normal, the lower faster, the upper slower. [S047-01] The 1966 patent illustrates axial engagement and a friction ring, but its example speeds and alternative controls are not production 122 specifications. [S047-03]

ORIGINAL DERIVATION: define positive speed magnitudes in the local torque direction. Let rm and rr be effective motor-pulley and rotor-pulley radii; rs and rc are small-shaft and rim-contact radii, all in metres. Equality of tangential velocities gives the ideal ratios below. These equations require nonslipping contacts; elastic deformation and sliding make nominal diameter ratios insufficient for exact timing.

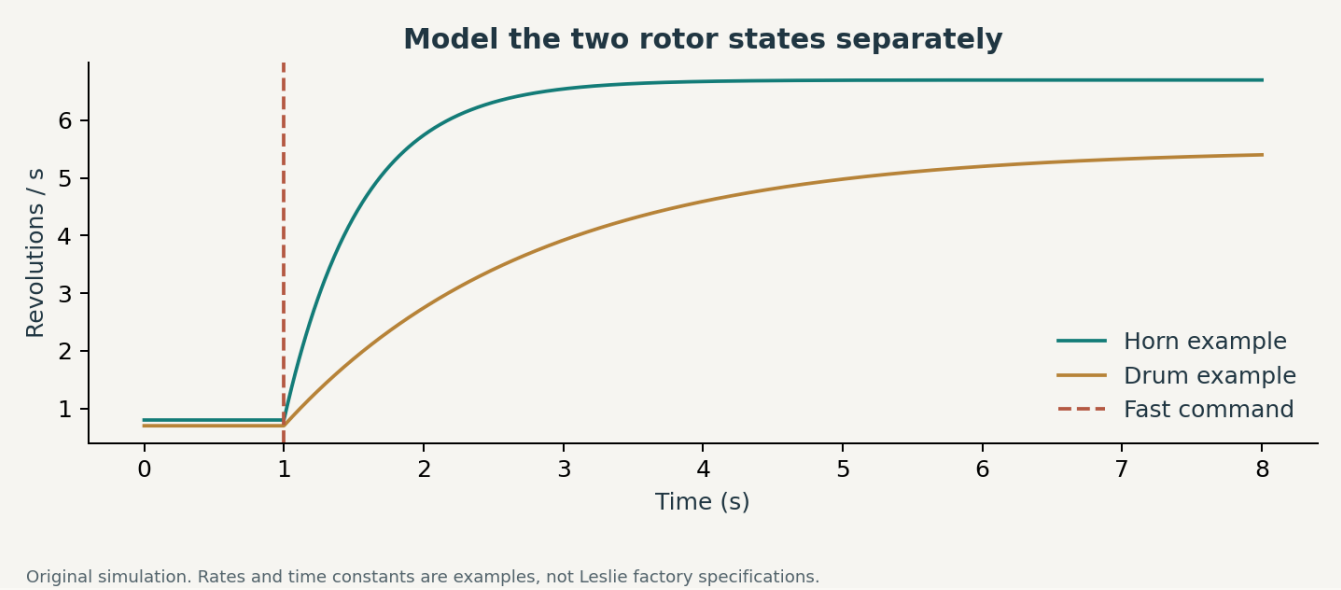
ORIGINAL DESIGN PROPOSAL: photograph each belt route and groove with scale, then establish effective ratios from simultaneous shaft/rotor observation. Store the acoustic direction separately as sigma=+1 or−1, with a declared top-view convention. Do not deduce that sign from an isolated drawing viewed from below. Contact force and belt tension belong to the identified assembly, not a generic age knob.

MODEL EQUATIONS

b = rm/rr; Omega_rotor_fast = b * Omega_main (ideal nonslip).

Omega_rotor_slow = b * (rs/rc) * Omega_small (ideal nonslip).

f_rot = RPM/60; omega = 2*pi*f_rot; theta_dot = sigma*omega.



Original simulation. Rates and time constants are examples, not Leslie factory specifications.
Original explanatory calculation. Illustrative parameters are not measured instrument data.

CHECKS AND DECISIONS

- A shaft-speed reading is never stored in a rotor-RPM field.
- Photographic direction is annotated with camera viewpoint.

SOURCES S047-01 Electro Music Leslie 122/122V/122RV/142/222 service manual, publication 044578, June / S047-03 Donald J. Leslie, US3245284A, Rotor drive for pulsato apparatus, April12 1966

ORIGINAL PHYSICAL MODEL PROPOSAL

Use a torque model when evidence can identify it

Leslie motor and belt dynamics

Model each motor stack and driven rotor with separate inertias. The proposed equations use local positive angular velocities: Omegas for the small motor, Omegam for the main motor, and Omegar for the acoustic rotor. J has units kg m²; torque T and losses L are N m. Radii retain the previous definitions. F is tangential force in newtons; slip velocity v is m/s. The contact engagement variable eta lies between zero and one.

The smooth force law is a numerical surrogate, not a measured rubber law. Fmax limits transmitted force; v0 controls the transition around rolling contact. The signs make slipping dissipate energy: Fc*vc and Fb*vb cannot be negative. Fast and slow motor torques are functions of their electrical commands and speed. They need identified curves; no universal synchronous shaft RPM, inertia or friction coefficient is supplied.

Fit only parameters supported by observations. Rotor angle alone generally cannot separate every inertia, torque and friction parameter. Begin with the reduced model on the next page; promote to this coupled model only when shaft measurements or residual structure justify the additional states. A stiff friction approximation can demand substeps or an implicit solver; validate convergence before placing it in the audio callback.

Unknown parameter group	Evidence needed before identification
Motor torque and engagement	Shaft motion and drive-state timing
Belt transmission and slip	Synchronized main-shaft and rotor angles
Rotor inertia versus losses	Additional independent mechanical evidence; speed curves alone may be insufficient

MODEL EQUATIONS

vc = rs*Omegas - rc*Omegam; Fc = eta*Fcmax*tanh(vc/vc0).

vb = rm*Omegam - rr*Omegar; Fb = Fbmax*tanh(vb/vb0).

Js*dOmegas/dt = Tslow - rs*Fc - Ls.

Jm*dOmegam/dt = Tfast + rc*Fc - rm*Fb - Lm.

Jr*dOmegar/dt = rr*Fb - Lr; dtheta/dt = sigma*Omegar.

CHECKS AND DECISIONS

- With applied drive removed, the numerical model cannot create mechanical energy.
- Repeated integration at smaller steps bounds numerical rather than hardware error.

ORIGINAL REAL-TIME IMPLEMENTATION

Implement continuous phase before detailed mechanics

Leslie motor and belt dynamics

Implement a measured-speed surrogate first. Each rotor owns omega, theta, slow/fast target speeds and separate upward/downward time constants tau. All speeds are rad/s and times seconds. The exponential assumption is proposed; retain fitted residuals and replace it with a monotone speed-dependent acceleration curve if it fails. A time constant is not a rise-time specification: the derived 10–90% interval is ln(9)*tau for this particular model.

The closed-form update integrates both speed and angle over a constant-command interval. Split blocks at timestamped control events, including any separately identified relay delay. A command reversal begins from current speed and phase. Select tau from the sign of target minus current speed, and never restart the horn or drum angle when a switch changes. Use expm1 for accurate small-step evaluation.

Preallocate state and event storage. Feed the same angle trajectory into Doppler, directivity, cabinet response and UI animation; independent free-running modulation oscillators would disagree. The UI reads snapshots and cannot drive the audio clock. Downstream fractional-delay interpolation and anti-aliasing remain required by the acoustic renderer; smooth mechanical motion alone does not establish alias-free audio.

```
for each event-bounded interval dt:
  for rotor in [horn, drum]:
    w0 = rotor.omega
    wt = profile.target(command)
    tau = profile.tau(sign(wt - w0))
    assert tau > 0 and isfinite(tau)
    q = -expm1(-dt / tau)
    dtheta = wt*dt + (w0-wt)*tau*q
    rotor.theta += rotor.sigma*dtheta
    rotor.omega = w0 + (wt-w0)*q
  publish phase trajectories
```

CHECKS AND DECISIONS

- Whole-block and subdivided renders agree within declared floating-point tolerance.
- A speed change produces no jump in angle or speed.
- A zero-duration interval leaves state unchanged; phase wrapping preserves continuity.

ORIGINAL ACQUISITION AND CALIBRATION PLAN

Measure acceleration and deceleration separately

Leslie motor and belt dynamics

The 1971 bass-belt adjustment procedure gives about 5–8 seconds for the bass rotor to reach Chorale after changing from Tremolo. This is a documented service criterion, without a numerical settling threshold; it is neither a universal exponential tau nor a horn specification. [S047-01] No specimen measurement has been performed for this chapter.

Request synchronized noncontact rotor observations and audio from an appropriate professional capture setup. Capture the command timestamp, observed drive switching, horn angle and drum angle on a common timebase. A single unambiguous rotor marker avoids interpreting multiple visual or acoustic lobes as multiple revolutions. Calibrate video-clock drift and unwrap angle before differentiating; retain the raw data and the smoothing method.

Repeat both speed transitions from settled states, then interrupt ramps at several speeds. Record cold/warm conditions, supply family and actual measured frequency, belt groove, specimen history and accessories. Use held-out repetitions to distinguish reproducible asymmetry from noise. Define settling and 10–90% thresholds explicitly. Do not time-warp audio to conceal an incorrect mechanical model.

Capture	Estimated quantity and limitation
Steady Slow / Fast for both rotors	Rotor speed distribution and phase ripple; no universal RPM
Slow→Fast and Fast→Slow	Distinct trajectories, delay and acceleration/deceleration residuals
Command / observed drive switch	Control latency separated from subsequent mechanical response
Simultaneous shaft and rotor tracking	Effective pulley ratio and transient slip
Accessory Stop, only if installed	Actual timed-drive/coasting sequence; separate profile

CHECKS AND DECISIONS

- Deliver timestamped angle CSV, raw recordings, identity photos, calibration and uncertainty.
- Fit targets are linked to capture IDs and never overwrite the documented-parts record.

SOURCES S047-01 Electro Music Leslie 122/122V/122RV/142/222 service manual, publication 044578, June

ORIGINAL ACCEPTANCE PLAN

Gate stopping modes and verify motion independently

Leslie motor and belt dynamics

A real optional brake kit lists a 6C5T thermal relay. [S047-04] The 1966 patent describes timed slow-motor operation followed by release; the 1980 patent describes a later mechanical braking arrangement. These are distinct evidence trails, not interchangeable proof of the selected cabinet's stop behavior. [S047-03][S047-06]

ORIGINAL DESIGN PROPOSAL: expose Slow/Fast in the baseline. Offer coast or accessory brake only under named extensions with their own control sequences, fitted timing and provenance. Coast removes applied drive and lets losses govern speed; braking adds an identified retarding path. Neither means setting angular velocity immediately to zero, and neither implies automatic parking at a chosen angle.

Deliver a versioned motion-profile schema, deterministic reference renderer, calibration notebook and residual report before acoustic fitting. Unknown physical values stay null; an explicitly named synthetic prototype profile may support development. Publish separate statuses for source topology, numerical verification and measured agreement. The tests below are proposed gates, not results already achieved. Final rotor-speed and phase tolerances must follow capture uncertainty and the product's audible-error budget.

Proposed gate	Failure it detects
Independent horn/drum profiles	One shared ramp or accidental phase locking
Rapid mid-ramp reversals	Angle resets, discontinuous speed or discarded momentum
Sample-rate/block partition comparison	Control-rate staircase or host-buffer dependency
Held-out speed and phase residuals	Correct endpoint RPM but wrong transition timing
Finite positive parameters; bounded state	Unstable integration and invalid imported profiles
Disabled accessory produces no Stop path	Modern or modified controls leaking into baseline

CHECKS AND DECISIONS

- No hardware-matched claim is emitted without identified captures.
- UI speed follows the actual model state, not merely the selected switch position.

SOURCES S047-03 Donald J. Leslie, US3245284A, Rotor drive for pulsato apparatus, April12 1966 / S047-04 Leslie Brake Accessory, instructions038398, for 122/142/222 series / S047-06 Donald J. Leslie, US4198880A, Rotor drive for pulsato apparatus, April22 1980

HARDWARE EVIDENCE

Lock the cabinet identity before drawing it

Cabinet construction and acoustic resonances

The baseline is a vintage 122 with a documented back-panel configuration and mains variant. The 1971 factory manual identifies hardwood veneers and lacquer, but does not establish core species, panel thickness, joinery or damping. Its nominal outer dimensions establish a bounding envelope; they cannot determine internal air volume. The independent 145/147 factory manual supplies the comparable exterior sizes. [S048-01][S048-02]

Original design proposal: create a specimen passport before CAD. Record cabinet serial, amplifier assembly, installed drivers, rear panels, repairs, added handles and replacement rotors. Store visible appearance separately from acoustic geometry. A convincing rendering can be completed while hidden dimensions remain null; those nulls must never become undocumented manufacturing facts.

The companion inventory contains 122 mechanical list rows, 52 amplifier/divider list rows and two amplifier headings. Crucially, 020305 is the treble driver; 020313 is its spacer ring, not another driver revision. Printed 13/14 belong to the 122 family; printed 15/16 belong to 222. [S048-01]

Model	Documented nominal H x W x D	Scope
122,122V,122RV	41 x 29 x 20.5 in	1971 manual [S048-01]
142	33 x 29 x 20.5 in	Same source; different cabinet
147	41 x 29 x 20.5 in	145/147 manual [S048-02]
145	33 x 29 x 20.5 in	Different height, not a scaled 122

P022 | Leslie 122.jpg | Salli | CC BY-SA 3.0 Source and license

CHECKS AND DECISIONS

- Reject a CAD profile without specimen identity, measurement units and back-panel state.
- Do not treat equal external dimensions as evidence of identical internal transfer functions.

SOURCES S048-01 Electro Music/CBS Leslie operating and maintenance manual, publication 044578, June 1 / S048-02 Electro Music Leslie Organ Speaker Models145 and 147 owner manual, cover 2MU14

ACOUSTIC MECHANISMS

Separate air paths, panels and the listening room

Cabinet construction and acoustic resonances

Henricksen describes upper horn, middle driver/crossover and lower rotor/amplifier compartments, with upper and lower louvred outlets. [S048-06] Treat this as a structural guide and verify actual partitions and openings on the chosen cabinet. In the 122A experiment, anechoic measurements still showed complicated interference; the authors modelled direct and reflected moving sources. Their implementation removed the upper rear wall and used five first-order image sources. That condition is not a measured transfer function for an intact vintage 122. [S048-03]

Original design proposal: assign every audible path an owner. The air model handles internal reflections, cavity storage, exit diffraction and any measured leakage. A structural model handles vibrating panels only if those contributions have not already entered measured kernels. The room receives position- and direction-dependent cabinet outputs. Room reverberation should not be embedded again in a cabinet preset.

Bastyr and Capone measured another loudspeaker design, not a Leslie, and found that cabinet vibration amplitude alone did not determine radiated pressure. [S048-05] This justifies measuring acoustic output alongside panel motion; it does not justify adding an arbitrary wooden resonance to every 122 preset.

P020 | Leslie 122 (isometric).jpg | Hammond b3 con leslie 122.jpg: Salli at it.wikipedia derivative work: Leslie 122.jpg: Ritchie333 derivative work: Clusternote | CC BY-SA 3.0 | Retouched/transformed; excluded from geometry evidence. Source and license

CHECKS AND DECISIONS

• Each capture/kernel lists whether it includes driver, horn/drum, cabinet, propagation, microphone and room.

• Back-panel removal changes profile identity and invalidates the previous boundary model.

SOURCES S048-03 Kronland-Martinet and Voinier (2008), Real-Time Perceptual Simulation of Moving Sourc / S048-05 Bastyr and Capone (2003), On the acoustic radiation from a loudspeaker cabinet / S048-06 Clifford A. Henricksen (1981), Unearthing the Mysteries of the Leslie Cabinet, author

ORIGINAL MODEL

Build a model that can absorb better measurements

Cabinet construction and acoustic resonances

Proposed implementation: start with a low-level linear acoustic model downstream of the nonlinear amplifier and driver models. Define separate horn-to-microphone and drum-to-microphone complex kernels, indexed by rotor angle and microphone pose. Summing these branches assumes that scattering by the other moving rotor is negligible. Test that assumption; a two-angle residual kernel remains an explicit extension if the separated model fails.

For an interpretable prototype, decompose each response into delayed, filtered paths. Multiple path filtering has a direct precedent in the 2002 paper, whose measured horn was from a Leslie 600 outside the target cabinet. [S048-04] Use the source method without importing its measurements as122 constants. Reflected paths follow the same physical rotor trajectory, evaluated at each path's emission time. Solve emission time against reception time; an instantaneous-angle shortcut must be labeled an approximation.

For a residual panel mode, use a damped resonator with fitted frequency, decay and source/microphone coupling. The equation below is a proposed stable digital form. Here fs is samples/second, f0 is hertz, tau is amplitude-decay seconds, and u is the mode drive. Unknown physical resonances stay disabled until measured or explicitly selected as an artistic extension.

Representation	Useful property	Limit
Angle-indexed complex FIR	Preserves measured interference	Requires phase/time alignment
Filtered geometric paths	Editable dimensions and motion	Approximate scattering
Damped panel modes	Compact residual resonances	Requires coupled measurements

MODEL EQUATIONS

y_m(t) = sum over b,k of {h_bkm(theta_b) * x_b}(t - tau_bkm(theta_b))

t = t_emission,bkm + tau_bkm(theta_b(t_emission,bkm))

r = exp(-1/(fs*tau)); omega0 = 2*pi*f0/fs

q[n] = u[n] + 2*r*cos(omega0)*q[n-1] - r*r*q[n-2]

CHECKS AND DECISIONS

- Require 0 < r < 1, finite states and sufficient headroom for resonant gain.
- Do not add explicit path delay when a measured kernel already contains it.

SOURCES S048-04 Smith, Serafin, Abel and Berners (2002), Doppler Simulation and the Leslie

IMPLEMENTATION

Give the implementation agent a concrete sequence

Cabinet construction and acoustic resonances

Original design proposal: first implement the data contract and render one fixed-angle impulse response. Next load two calibrated source branches and verify their signed sum at one microphone. Then interpolate angle-dependent kernels, retain continuous rotor phase, and add microphone movement. Only after this works should the agent introduce internal geometric reflections, structural residuals or a room renderer. Each addition must explain a remaining measured error.

Keep geometric propagation and residual filtering distinct. Align captured kernels to a declared physical reference plane; store removed absolute delays as metadata. Interpolate complex responses with a periodic angle convention rather than independently normalizing their peaks or converting everything to minimum phase. Smooth parameter changes while preserving legitimate modulation. A slow control update needs audio-rate interpolation before it modulates delay.

Compile geometry, fit filters and load assets off the audio thread. Preallocate delay buffers using the largest permitted propagation distance. Audit fractional-delay interpolation at the largest allowed angular speed and source bandwidth; modulation can create out-of-band energy. Select oversampling or bandwidth limits from a comparison with a high-rate reference, not from a presumed cabinet constant.

Required field	Meaning
geometry_revision / panel_state	Acoustic boundary identity
angle_zero / rotation_sign	Mechanical coordinate convention
response_includes / reference_plane	Prevents duplicated components
sample_rate / delay_seconds / polarity	Reproducible complex response
source_id / uncertainty / measured_date	Traceability and qualification

```
profile = load_verified_profile(id)
assert profile.units == "SI"
assert profile.response_ownership_is_unique()
prepare_buffers(profile.max_delay_seconds)
for each audio sample:
    angles = rotor_state.advance()
    horn = render_branch(HORN, angles.horn)
    drum = render_branch(DRUM, angles.drum)
    exits = combine_owned_paths(horn, drum)
    output = microphone_renderer(exits, pose)
    publish_meter_without_allocation()
```

CHECKS AND DECISIONS

- Audio callback allocates no memory and performs no file or geometry operations.
- A full angle wrap produces no kernel discontinuity or phase reset.

MEASUREMENT PLAN

Acquire the missing cabinet evidence

Cabinet construction and acoustic resonances

No instrument was measured for this atlas. Proposed acquisition begins with calibrated photographs and direct dimensions of the selected cabinet. Use scale bars in the same plane, an explicit camera calibration and common coordinate targets. Capture both overall views and details of baffles, rear-panel gaps, louvre slots, horn clearance and rotor clearance. Photographic texture supplies appearance; measured boundaries supply acoustics.

Capture fixed-angle acoustic responses with the intended rear panels fitted, then repeat controlled panel-state comparisons. Separate horn and bass excitation where a qualified technician can provide the required isolated test configuration. Retain level, polarity, microphone calibration, source reference and environmental conditions. Use anechoic measurement or documented gating; a short gate cannot resolve arbitrarily low-frequency decay.

For proposed structural identification, record panel velocity or acceleration together with near-field pressure. Repeat at several drive levels to distinguish stable linear features from loose fittings or nonlinear rattles. Separately acquire rotating, angle-synchronized recordings for held-out validation. The 2008 study provides a useful interference experiment, but its 122A specimen and rear-wall condition must remain explicit. [S048-03]

Capture group	Deliverable	Uncertainty resolved
Exterior orthographic views	Scale-qualified mesh and texture	Silhouette and finish
Internal partitions and openings	Dimensioned cavity/vent map	Volumes and boundary paths
Panel and louvre close-ups	Thickness/material observations	Geometry and construction
Fixed-angle impulse responses	Complex response bank	Cabinet filtering and interference
Synchronized panel/pressure data	Candidate mode parameters	Structural versus air radiation
Rotating hold-out recordings	Time, angle and microphone tracks	Dynamic generalization

CHECKS AND DECISIONS

- Maintain an untouched raw dataset and a reproducible processing script.
- Do not infer veneer substrate or hidden panel thickness from finish photographs.

SOURCES S048-03 Kronland-Martinet and Voinier (2008), Real-Time Perceptual Simulation of Moving Sourc

ACCEPTANCE AND OPEN QUESTIONS

Accept fidelity only at the stated boundary

Cabinet construction and acoustic resonances

Original acceptance proposal: reserve microphone poses, rotor angles and speed transitions that were not used for fitting. Compare complex transfer functions, impulse arrival times, energy decay, sideband structure and signed horn/drum summation at matched input level. A magnitude-only match can hide incorrect delay, polarity and interference. Set error tolerances after quantifying measurement repeatability; no numerical fidelity score is claimed here.

Run controlled ablations. Removing the cabinet branch should remove only the response contribution it owns. Setting both rotors stationary should give a time-invariant acoustic result. Changing a back panel must alter the appropriate profile, not merely the UI mesh. A room bypass must leave cabinet reflections intact. If panel modes already exist in the measured FIR bank, disabling the explicit structural branch should avoid double counting.

The parts companion is a source catalog, not a procurement or manufacturing BOM. Its 176 records have readable stock identifiers, individual page locators and null physical quantities. Alternative motor assemblies, reverb hardware and cabinet backs remain distinguished. Exact serial transitions, wood-core construction, louvre sections, leakage and measured acoustic constants still require target acquisition. [S048-01] Hand off the approved profile, raw measurements, processing code, rendered model and a limitation ledger together.

Gate	Required evidence
Identity	Serial, mains assembly, panels, installed replacements
Static acoustics	Held-out complex kernels with repeatability bounds
Dynamic acoustics	Angle-synchronized modulation and transition captures
Numerics	Stable decay, no NaNs, interpolation/aliasing comparison
Presentation	UI mesh clearly separated from measured CAD
Claim	Name the exact specimen and tested configurations

CHECKS AND DECISIONS

- Reject a universal 1:1 claim based on one microphone or one drawbar registration.
- A changed source driver, panel configuration or microphone model triggers profile revalidation.

SOURCES S048-01 Electro Music/CBS Leslie operating and maintenance manual, publication 044578, June 1

MODEL BOUNDARY / HARDWARE EVIDENCE

Specify a moving radiation path, not an independent pitch LFO

Doppler, retarded time and fractional-delay rendering

The 2002 Doppler paper develops source/listener delay models and a rotating-mouth approximation, then reports horn measurements from a Leslie 600. Its geometry supplies an algorithmic starting point, not measured radius, speed or radiation data for the selected vintage 122. The actual compression driver remains stationary; a moving effective mouth is a reduced acoustic model. [S049-01]

ORIGINAL DESIGN PROPOSAL: define each propagation path by one rotor trajectory, a microphone position, an effective radiation point, sound speed and an explicitly identified signal boundary. The input waveform here is referenced at the effective mouth; internal horn transit and filtering must already be assigned elsewhere. A bass outlet requires its own evidence and cannot inherit the horn radius.

Preserve a single physical trajectory per rotor. Different microphones generally observe different emission times from that same trajectory. Thus “shared rotor phase” does not mean evaluating every microphone at an identical, undelayed angle. A stereo panner alone supplies neither this propagation relation nor its changing spectral consequences.

Quantity	Contract
Position / distance	Metres in the cabinet coordinate frame; microphone orientation belongs to chapter 51.
Time / angle	Seconds and unwrapped radians on the audio clock; angular velocity in rad/s.
Acoustic radius	Measured or fitted with uncertainty; never inferred from an unscaled photograph.
Path ownership	Declare whether driver, horn transit, cabinet reflection and microphone response are included.

CHECKS AND DECISIONS

• Reject profiles with unspecified coordinate conventions or a microphone intersecting the modeled radiation path.

SOURCES S049-01 Julius Smith, Stefania Serafin, Jonathan Abel and David Berners: Doppler Simulation a

ORIGINAL DERIVATION / RETARDED TIME

Solve the emission time before evaluating source motion

Doppler, retarded time and fractional-delay rendering

For a fixed microphone m and a subsonic point trajectory q(e), let t be observation time and e emission time. Sound must traverse the distance r(e) after emission. The equations below follow by differentiating that travel-time constraint; u points from the emission position toward the microphone. The velocity v is evaluated at e. This yields the moving-source frequency ratio, rather than substituting current geometry into a moving-listener formula.

For a circular path of radius a, horizontal microphone distance R and height difference h, the squared distance contains a negative cosine term. The printed DAFx Equation 18 instead contains a positive term despite its stated coordinates; independently expanding the coordinate norm resolves the discrepancy. [S049-01] This is a source audit, not an author-issued erratum.

Using q(t) instead of q(e) is a current-geometry approximation. Keep that option explicitly named and compare it against the retarded reference. An illustrative modulation plot is not evidence that either approximation matches a physical Leslie.

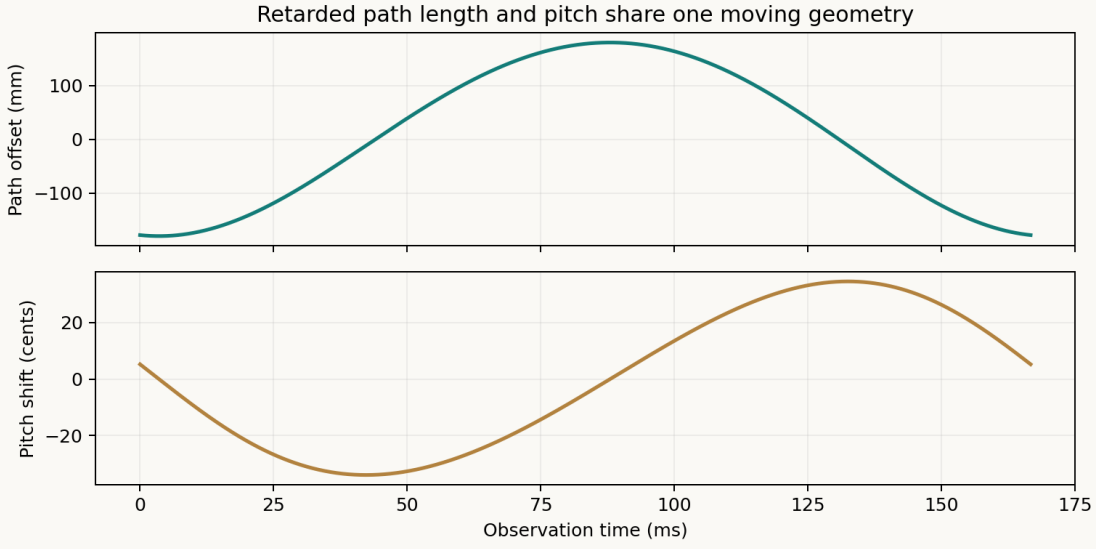
MODEL EQUATIONS

r(e) = ||m - q(e)||; t = e + r(e)/c; u = (m-q(e))/r(e).

F(e) = e + r(e)/c - t; F'(e) = 1 - u dot v(e)/c.

f_received/f_source = de/dt = 1 / (1 - u dot v(e)/c).

r(e)^2 = R^2 + a^2 - 2*R*a*cos(theta(e)) + h^2.



Computed point-path example: radius 0.18 m, microphone radius 1.4 m, 6 rev/s, sound speed 343 m/s. Solve t=e+distance(e)
All geometry is evaluated at emission time e. These are illustrative parameters, not measured Leslie data.

Computed retarded-time point-path model with explicitly illustrative geometry. The same trajectory determines path delay and pitch; not a measured Leslie response.

CHECKS AND DECISIONS

- Require positive distance and F'(e)>0 throughout the admitted operating envelope.
- At zero rotation, require constant delay and frequency ratio exactly one in the mathematical reference.

SOURCES S049-01 Julius Smith, Stefania Serafin, Jonathan Abel and David Berners: Doppler Simulation a

DIGITAL MODEL / FRACTIONAL SAMPLES

Interpolate the waveform without hiding spectral error

Doppler, retarded time and fractional-delay rendering

Write q_s=Fs*e=k+mu, where k=floor(q_s), 0<=mu<1 and Fs is the internal sample rate. Linear interpolation uses weights (1-mu,mu) on samples k and k+1. The DAFx PDF 3 listing reverses those weights: its mu=0 result selects k+1. Verify integer endpoints instead of copying the listing. [S049-01]

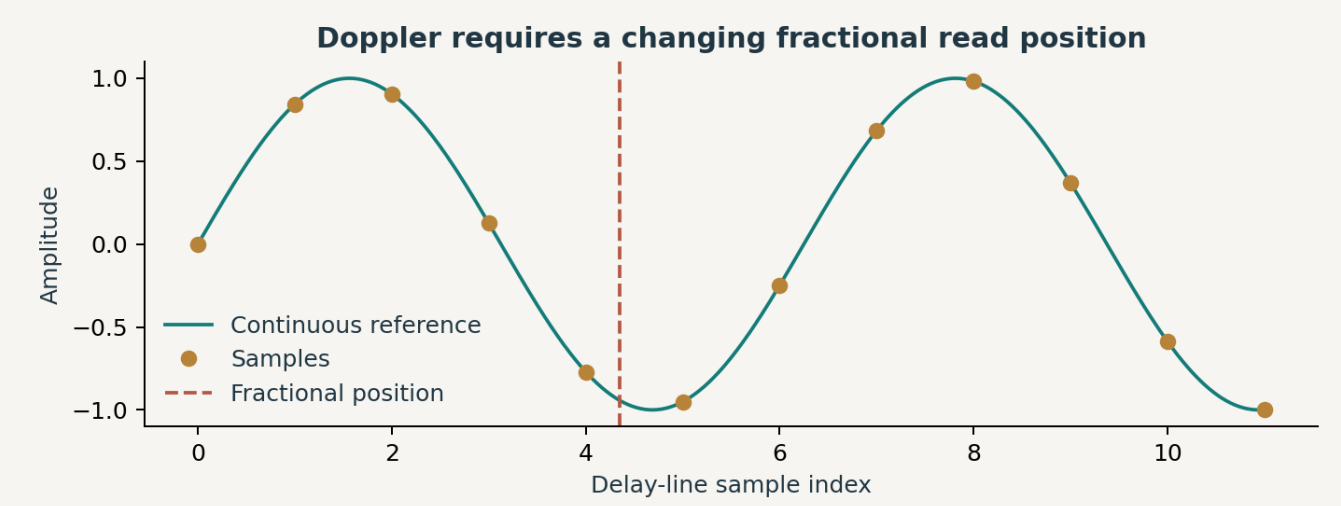
Linear interpolation introduces a moving high-frequency attenuation pattern as mu changes. [S049-03] A four-sample Lagrange interpolator is a useful prototype, with nodes k-1 through k+2; it is not an ideal broadband delay. The polynomial construction and central evaluation convention follow the author's interpolation treatment. [S049-02] Choose the production kernel from measured numerical error across the entire intended passband, fractional position and speed range.

Doppler resampling can move energy upward. A local ratio bound nu_max suggests a prefilter passband below Fs/(2*nu_max), with transition margin; rapidly varying delay also creates modulation sidebands, so this bound alone is not an alias-free guarantee. Compare dynamic spectra with a converged oversampled reference.

Kernel	Engineering tradeoff
Linear	Cheap diagnostic; fractional-position-dependent coloration must be quantified.
Cubic Lagrange	Short support and no recursive state; high-band error still requires testing.
Windowed sinc / polyphase	More work and history; specify bandwidth, stopband and interpolation-table error.

MODEL EQUATIONS

For j in {-1,0,1,2}: w_j(mu)=product over l!=j of (mu-l)/(j-l); y=sum_j w_j*x[k+j].



Original explanatory signal; test interpolation quality across the full audio bandwidth.
Original explanatory calculation. Illustrative parameters are not measured instrument data.

SOURCES S049-01 Julius Smith, Stefania Serafin, Jonathan Abel and David Berners: Doppler Simulation a / S049-02 Julius O. Smith III: Physical Audio Signal Processing, Lagrange Interpolation / S049-03 Perry R. Cook and Gary P. Scavone: STK DelayL implementation documentation

ORIGINAL IMPLEMENTATION PROPOSAL

Use bounded work, historical angles and explicit latency

Doppler, retarded time and fractional-delay rendering

Build an offline bracketed solver first. For known path bounds r_min and r_max, emission time lies between t-r_max/c and t-r_min/c. A runtime solver can predict from the previous root and use a fixed budget of safeguarded Newton steps. Certify its worst-case residual over admitted trajectories; an unconstrained while-loop does not belong in the callback.

Keep enough waveform and rotor-history samples for the longest path and interpolation support. Evaluate the angle at the solved emission time, including any earlier speed-command segment. For a centered cubic kernel, add a fixed processing delay if the physical minimum delay does not provide two future support samples. Report that added algorithmic delay separately from modeled propagation. Allocation and maximum-delay changes occur during preparation; JUCE explicitly warns that its maximum-delay setter can allocate and clear history. [S049-04]

```
prepare: validate geometry, speed bound, history capacity
prepare: select solver budget, interpolation, latency L
for each internal sample n:
  append input and timestamped rotor state to history
  t = (n-L)/Fs
  for each microphone/path:
    lo = t-r_max/c; hi = t-r_min/c
    e = clamp(predicted_emission_time, lo, hi)
    repeat fixed solver budget:
      evaluate F and derivative using rotor history at e
      tighten bracket; take bounded Newton or midpoint
    record residual and evaluate waveform at Fs*e
    use consistent emission angle for directional response
    sum paths with their declared gain/filter ownership
```

CHECKS AND DECISIONS

- No allocation, locks or file access during rendering; retain bounded diagnostic counters.
- Changing host block size must preserve event timing, rotor history and output within the declared numerical tolerance.

SOURCES S049-04 JUCE: juce::dsp::DelayLine class reference

IDENTIFIABILITY / CALIBRATION

Separate propagation delay from measured response banks

Doppler, retarded time and fractional-delay rendering

An angular impulse-response bank may already contain changing arrival time, fixed leakage, internal transit, cabinet reflections and microphone response. The DAFx study explicitly separates a fixed component and delayed angle-dependent horn responses. [S049-01] Applying another geometric delay to an unaligned bank can duplicate the propagation phase. Simply interpolating raw frozen-angle responses also does not prove an exact moving-source solution.

ORIGINAL DESIGN PROPOSAL: choose one ownership convention. Either retain the complete measured response and validate its time-varying rendering, or remove a documented arrival delay, retain that delay as data, and apply it once in the propagation renderer. Do not convert all responses to minimum phase merely to make them convenient: that can discard the phase relationships being emulated.

Acquire synchronized rotor angle, electrical reference and microphone recordings at known coordinates. Estimate a delay trajectory with stated uncertainty and compare its derivative against narrowband demodulation away from amplitude nulls. Validate on withheld angles, microphone positions and transition speeds. Fit effective radius only if the resulting residual supports a single-point model; distributed radiation may require multiple paths or frequency-dependent structure.

Capture field	Why it matters
Time origin / clock calibration	Separates physical transit from recorder latency and drift.
Delay removed from each IR	Prevents a second application of the same propagation.
Phase confidence / null mask	Prevents destructive interference from masquerading as extreme source velocity.

CHECKS AND DECISIONS

• Never label a fitted effective radius a measured horn dimension without an independent geometric survey.

SOURCES S049-01 Julius Smith, Stefania Serafin, Jonathan Abel and David Berners: Doppler Simulation a

EXECUTED SYNTHETIC CHECK / PROPOSED RELEASE GATES

Verify the mathematics before claiming acoustic agreement

Doppler, retarded time and fractional-delay rendering

An original double-precision fixture evaluated 1,001 observation times for a circular point model: a=0.18m, R=1.4m, 6rev/s and c=343m/s. These are synthetic test choices. A 70-step bisection reference gave a maximum travel-time residual of 2.78e-17s. Four Newton steps agreed within 5.56e-17s for this fixture; this does not certify other trajectories or a production runtime.

A centered finite difference of the reference emission time, using a 1microsecond step, agreed with the derived frequency ratio within 2.73e-11. These are mathematical checks, not acoustic measurements. The retained data/049-checks.json records assumptions and results. No finished real-time delay module, anti-alias filter or physical Leslie comparison was tested here.

ORIGINAL ACCEPTANCE PROPOSAL: deliver a deterministic propagation renderer, an offline oracle, the admitted geometry/speed envelope and a residual report. Keep interpolation error, solver error, acoustic-model error and capture uncertainty separately visible. Only held-out physical recordings can support the final fidelity claim.

Companion executable: reference_code/retarded_path_reference.py reproduces the 1,001-time synthetic fixture, comparing bisection, four Newton steps and an independent finite-difference derivative. The stored numerical result is separate from future interpolation, real-time and hardware gates.

Gate	Required evidence
Endpoints and wrap	Integer-delay identity, fractional continuity and ring-buffer boundary fixtures.
Motion	Zero-speed identity; approach raises pitch; retreat lowers pitch; continuous transition phase.
Spectra	Static and modulated kernel errors plus alias residuals against a converged reference.
Integration	One delay owner per path; shared trajectory with microphone-specific emission times.
Hardware	Held-out angle/audio captures, residual budget and configuration manifest.

CHECKS AND DECISIONS

- Treat every gate except the explicitly reported synthetic mathematics as not yet run.

EVIDENCE AND MODEL BOUNDARIES

Treat directivity as a measured transfer function

Rotating directivity and amplitude modulation

The 2002 DAFx study measured a Leslie Model 600 horn in free field, at fixed 15-degree steps, and separated a largely angle-independent component from the moving horn response. Its data support angular filtering, but they are not measurements of the selected vintage 122, its lower rotor or its installed cabinet. [S050-01]

Henricksen reports diffuser-dependent radiation and apparent source position; his response example uses a University ID-40 driver. Keep that configuration separate from stock-driver evidence. [S050-03] The 2008 study identifies a 122A and demonstrates enclosure-related interference. These sources establish mechanisms and useful experiments, not interchangeable calibration files. [S050-02]

ORIGINAL DESIGN PROPOSAL: give every radiation profile an explicit electrical input boundary, acoustic output units, specimen identity and included mechanisms. A driver-terminal-volts-to-pascals response may contain the driver and horn, but it must not silently contain a second copy of the amplifier or crossover. Retain absolute calibration alongside normalized comparison plots. Frequency-dependent radiation cannot be reduced to one broadband volume curve without declaring and testing that approximation.

Profile field	Required meaning
Identity and condition	Cabinet, driver, horn/deflector, diffuser, panels and cloth; unknown values remain null
Input/output boundary	For example Pa/V at identified driver terminals and microphone position
Included response	Driver, propagation, cabinet, microphone and room flags
Valid domain	Frequency, angle, elevation, distance, drive level and uncertainty

CHECKS AND DECISIONS

- Never rename a published 600 or122A dataset as a measured vintage 122 profile.

SOURCES S050-01 Smith, Serafin, Abel and Berners: Doppler Simulation and the Leslie, DAFx 2002 / S050-02 R. Kronland-Martinet and T. Voinier: Real-Time Perceptual Simulation of Moving Source / S050-03 Clifford A. Henricksen: Unearthing the Mysteries of the Leslie Cabinet, Recording Eng

ORIGINAL QUASI-STATIC PATH MODEL

Derive amplitude and delay from the same geometry

Rotating directivity and amplitude modulation

Use one rotor trajectory for propagation and radiation. For a path from acoustic source position s to microphone m, r is distance in metres, u is the unit departure direction, and R(theta) maps rotor coordinates into cabinet coordinates. The local direction q supplies signed azimuth beta and elevation. Evaluate these at the emission time supplied by the propagation solver; reception-time angle is a different quantity.

The equations below are frozen-time bookkeeping, not an exact frequency-domain description of the complete time-varying system. D is a complex pressure response at reference distance r_ref; tau is the separately owned propagation delay. The r_ref/r amplitude law assumes a suitable far-field point-source approximation. Pressure amplitude scales this way; intensity scales with its square. Close microphones require range-dependent measurements instead of an arbitrary gain clamp disguised as acoustics.

Rotating a bare-horn response relative to a microphone can be useful. Rotating a complete cabinet response by merely subtracting microphone azimuth is generally unjustified: the louvers and walls remain fixed. Use a multi-position cabinet profile or separately modeled paths. Sum signed pressure contributions coherently; taking each magnitude first destroys interference. Microphone directionality must use the same arriving-path geometry.

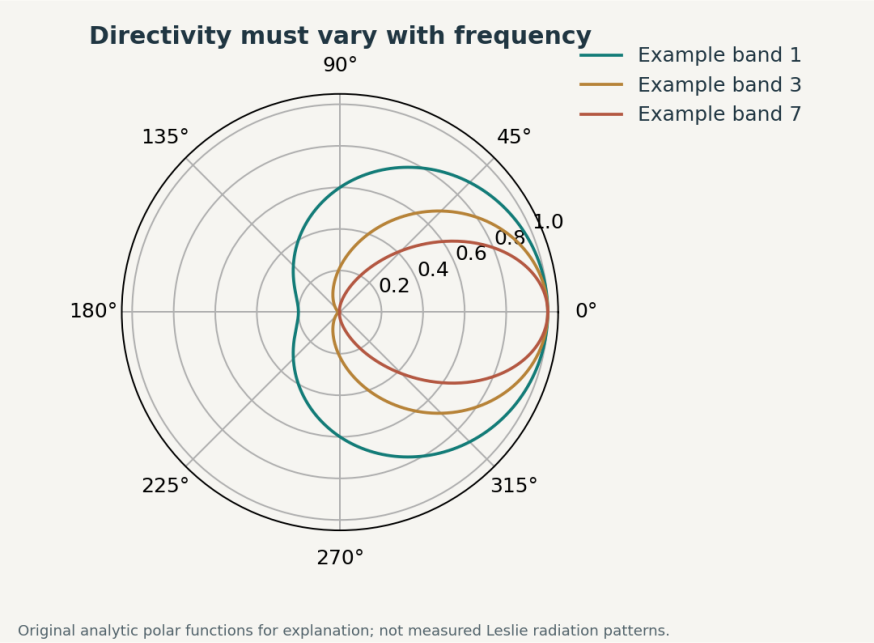
Physical dependency	Implementation consequence
One angle trajectory	Directivity, distance and delay remain synchronized
Fixed cabinet surfaces	Do not rotate a whole-cabinet IR as if it were an isolated horn
Complex path summation	Retain polarity, residual phase and stationary leakage separately

MODEL EQUATIONS

r = norm(m-s); u = (m-s)/r; q = transpose(R(theta))*u.

beta = atan2(q_y,q_x); elevation = atan2(q_z,sqrt(q_x^2+q_y^2)).

G_path(f,t) = D(f,beta,elevation) * (r_ref/r) * exp(-j*2*pi*f*tau).



Original analytic polar functions for explanation; not measured Leslie radiation patterns.

Original explanatory calculation. Illustrative parameters are not measured instrument data.

ORIGINAL DATA AND FILTER DESIGN

Represent periodic angular filters without erasing phase

Rotating directivity and amplitude modulation

Retain the measured complex response or its real impulse response. Separate bulk propagation only when its estimation is documented, then restore any common causal implementation delay explicitly. Independent peak alignment of every capture can erase the spatial timing the model must reproduce. A magnitude-only polar plot cannot determine residual phase, leakage separation or microphone interference.

An implementable proposal is a real FIR whose taps vary periodically with angle. Fit each tap using the Fourier basis below, with angular order M chosen by held-out angles and frequency-weighted residuals. N equally spaced unique angles require N greater than 2M for this unambiguous basis. The endpoints minus 180 and plus 180 degrees represent the same direction; repeated captures there test repeatability rather than adding a new direction.

Alternatively factor the aligned response matrix into a small set of fixed FIR filters and periodic weights. The 2002 study provides a precedent for a reduced basis; its chosen rank must not become a universal constant. [S050-01] Increase rank until residual improvement justifies its CPU cost. Preserve finite real taps, bounded weights, capture uncertainty and the unprocessed originals. Do not force every angle to equal RMS: that would delete the modulation being measured.

Saved artifact	Purpose
Raw IRs and calibration	Recover absolute gain, timing and provenance
Delay decomposition	Prevent propagation from being applied twice
Basis filters and angular coefficients	Deterministic real-time rendering
Held-out residuals and valid ranges	Expose model limits instead of concealing them

MODEL EQUATIONS

h[l,beta] = a[l,0] + sum_m=1..M (a[l,m]*cos(m*beta) + b[l,m]*sin(m*beta)).

Equivalent basis form: h[l,beta] = sum_k w_k(beta)*b_k[l].

For finite FIR bases and bounded weights: |y[n]| <= max|x| * sum_k(max|w_k| * sum_l|b_k[l]|).

SOURCES S050-01 Smith, Serafin, Abel and Berners: Doppler Simulation and the Leslie, DAFx 2002

ORIGINAL REAL-TIME IMPLEMENTATION

Run fixed filter states and read them at emission time

Rotating directivity and amplitude modulation

Advance the fixed FIR basis states continuously for each rotor input. Store their output histories in preallocated buffers. For each microphone path, obtain emission time, geometry and angular weights, then read the basis histories with the same propagation convention. This is a proposed quasi-static rendering approximation; chapter 49 owns the moving-path solver and chapter 53 the longer convolution implementation.

Avoid resetting filters at angular cell boundaries. Periodic Fourier weights provide a continuous wrap; interpolation of arbitrary recursive-filter coefficients needs its own stability argument. Crossfade imported profiles with retained histories and account for implementation latency. Audio processing performs no fitting, allocation or file access. Separate horn/drum inputs retain their crossover phase and actual independent rotor angles.

Angular detail creates temporal modulation. At constant rotation, order m produces coefficient variation at m times rotation frequency; multiplying audio creates sidebands. Design oversampling and interpolation against the highest retained audio frequency and angular order, then compare with a higher-rate reference. Smooth-looking UI motion does not establish an alias-free acoustic renderer. A user-controlled extra tremolo depth is an artistic extension and must not alter the calibrated baseline silently.

```
for bank in profile.basis_banks:
  bank.history.push(bank.firs.process(input[bank.rotor]))
for path in active_paths:
  te = propagation.emission_time(path, receive_time)
  theta = rotors[path.rotor].angle_at(te)
  beta, elevation, range = geometry(path, theta)
  weights = path.profile.angular_weights(beta, elevation)
  basis = path.bank.history.read_fractional(te, declared_latency)
  pressure = dot(weights, basis)
  output[path.mic] += path.profile.range_gain(range)*pressure
```

CHECKS AND DECISIONS

- Changing host buffer partition does not change the physical trajectory or angular lookup.
- Reject zero distance or out-of-domain geometry explicitly; do not produce infinite pressure.

PUBLISHED EXAMPLE AND ORIGINAL ACQUISITION PLAN

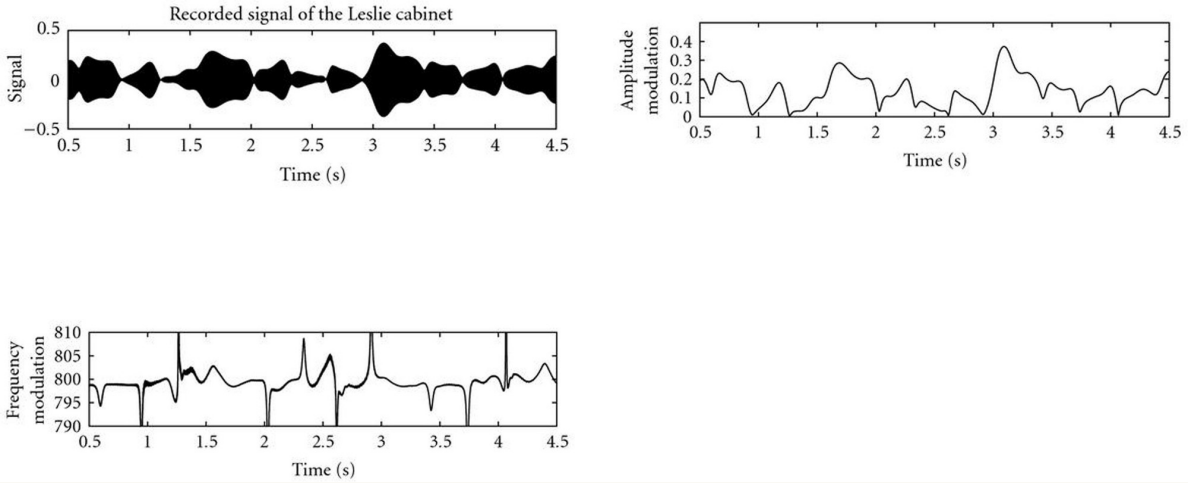
Measure nulls as interference, not motor accelerations

Rotating directivity and amplitude modulation

The reproduced 2008 Fig 5 shows a published 122A measurement with 800 Hz excitation and a microphone 1.2 m away at horn height. Sharp instantaneous-frequency excursions coincide with low amplitude; the paper reports interference persisting with the horn alone. This is external evidence, not a capture made for this suite. [S050-02]

ORIGINAL DESIGN PROPOSAL: acquire calibrated stationary-angle responses for horn and bass separately, at declared levels and multiple microphone positions. Preserve installed-cabinet and isolated-component experiments as distinct datasets. Repeat angular endpoints, measure background noise, and reserve intermediate angles for validation. Record rotor phase zero photographically and retain input voltage, microphone calibration, panel configuration and environmental conditions.

Then capture continuous rotation with independent angle tracking. Compare predicted envelopes and complex narrowband responses across frequency; do not infer mechanical speed from a Hilbert-phase derivative near a cancellation null. Mask estimates according to a declared noise/reliability criterion while retaining the raw signal. Fit motion from the independent angle record, then assess acoustic interference without changing the rotor speed to explain every phase spike.



P042 | Published 122A measured acoustic response at 800 Hz | R. Kronland-Martinet and T. Voinier | CC BY 2.0 | Published author measurement, not a capture performed for this suite. 122A specimen, 800 Hz input, microphone 1.2 m from cabinet at horn height; interference complicates instantaneous-frequency interpretation. Source and license

CHECKS AND DECISIONS

- Deliver held-out angles and rotating captures with synchronized rotor phase.
- No target 122 radiation profile is considered measured until those files and calibration records exist.

SOURCES S050-02 R. Kronland-Martinet and T. Voinier: Real-Time Perceptual Simulation of Moving Source

ORIGINAL VERIFICATION AND RELEASE GATES

Accept the radiation model through falsifiable predictions

Rotating directivity and amplitude modulation

Separate numerical verification from physical agreement. An executed synthetic fixture fitted order 3 angular Fourier coefficients to 24 unique angles and 12 FIR taps. Across 1001 evaluation angles, maximum reconstruction error was 1.78e-15 and wrap discrepancy 4.99e-15 in arbitrary units. This checks the algebraic representation only; it tests no Leslie capture, moving-path renderer or complete plug-in.

For physical validation, report errors on absolute complex response, band-limited envelope, modulation sidebands and microphone cross-spectrum. Weight or exclude phase estimates below a declared reliability floor; retain those bins in the uncertainty report. Do not score each angle after an independent gain normalization. Such normalization can make incorrect modulation depth disappear.

Release a versioned radiation profile, fitting report, raw-data links and domain restrictions. Final acoustic tolerances depend on measurement uncertainty and product listening goals and remain unset here. A sparse convenience table may be a development fixture; it must not inherit a hardware-validated label. If speed-dependent residuals remain after correct geometry and timing, investigate the quasi-static assumption before adding arbitrary compensating oscillators.

Proposed acceptance test	Failure exposed
Angle wrap and derivative continuity	Seam clicks or duplicated-endpoint weighting
Frozen-angle versus source IR	Wrong gain, polarity, timing or filter basis
Held-out angles and microphone positions	Overfit interpolation or invalid cabinet rotation shortcut
Continuous rotation and speed ramps	Incorrect emission-time indexing or quasi-static limits
Higher-rate numerical reference	Angular modulation and delay-interpolation aliasing
Bounded profile and finite-state tests	Invalid imports, unbounded weights or numerical faults

CHECKS AND DECISIONS

- Hardware-fidelity gates remain not_run; only the stated synthetic algebra fixture was executed.

HARDWARE EVIDENCE / ORIGINAL DESIGN PROPOSAL

A microphone scene is part of the reference definition

Leslie microphones, stereo geometry and mono compatibility

A faithful result requires a specified listening or recording scene. The 2008 study measured a 122A in an anechoic room with a microphone 1.2 m from the cabinet at horn-plane height. That establishes one experiment, not the microphone geometry of every vintage 122 recording. The 2002 free-field measurements instead used a Leslie 600 horn with receivers in the rotation plane and on the axis. Keep these specimens and boundaries separate. [S051-02][S051-01]

ORIGINAL DESIGN PROPOSAL: define MicSceneProfile independently of the organ, amplifier and cabinet profiles. Store microphone models, capsule positions, axes, calibration, rear-panel state, room identity and output matrix. Changing a scene changes the reference sound; it must not silently alter generator or amplifier parameters.

Implement a single measured receiver first, then add a correlated pair. A proposed three-channel option may use two upper receivers and one lower receiver, but each still receives any physically present horn/drum leakage. Do not hard-route frequency bands to microphones merely because their names say horn or bass.

Scene option	Explicit implementation boundary
One receiver	One measured pose; mono acoustic reference.
Spaced pair	Two positions and responses; preserve relative delay and phase.
Coincident directional pair	Shared nominal location, different axes; real capsule offsets remain measurable.
Upper pair plus lower receiver	Three pressure signals followed by a declared 2×3 mix matrix.

CHECKS AND DECISIONS

- Every exported preset names its microphone scene and identifies provisional geometry.

SOURCES S051-01 Smith, Serafin, Abel and Berners: Doppler Simulation and the Leslie, DAFx-02 / S051-02 Kronland-Martinet and Voinier: Real-Time Perceptual Simulation of Moving Sources, 200

ORIGINAL GEOMETRY CONTRACT

Separate cabinet azimuth, rotor phase and microphone aim

Leslie microphones, stereo geometry and mono compatibility

PROPOSED RIGHT-HANDED FRAME: origin at the surveyed floor-footprint centre; +x is right when facing the front, +y points toward the rear, +z upward. Metres are mandatory. Azimuth atan2(y,x) starts at +x and increases toward +y. The front is therefore −pi/2. This is a project convention, not a manufacturer coordinate system. Record each rotor axis and its physical zero marker separately.

For a direct path, s is the radiating-point position at emission, m the capsule position, u the source-facing axis, and a the microphone’s forward axis. q points from source to microphone. Consequently the microphone sees the opposite bearing: using a·q instead of a·(−q) reverses front and rear. For a reflected path use its final incoming segment at the microphone and its first outgoing segment at the radiator.

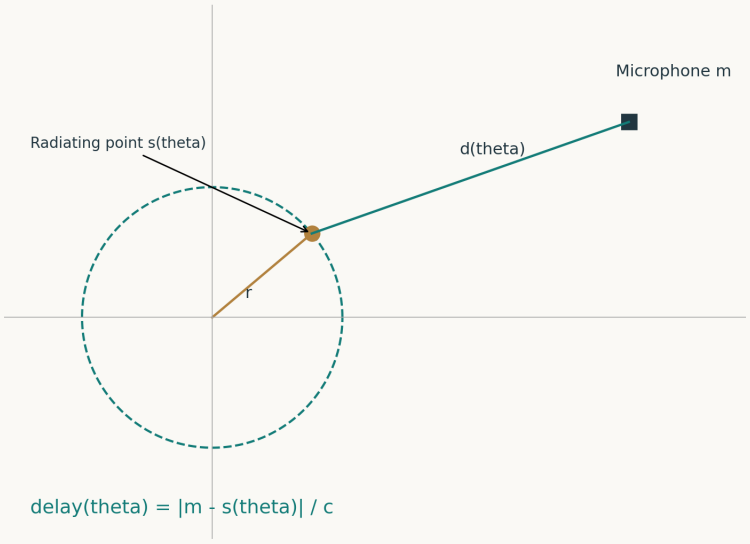
All receivers query the same horn/drum trajectories. Evaluate emission time consistently with chapter 49. A microphone azimuth is neither rotor phase nor microphone aim; pointing toward the cabinet does not move the capsule. Store orientation as a normalized vector or rotation, not an ambiguous single angle.

MODEL EQUATIONS

d = ||m - s||; q = (m - s)/d; d in metres; u,a,q are unit vectors.

cos(source off-axis angle) = u dot q; cos(microphone incidence angle) = a dot (-q).

Microphone geometry defines a time-varying propagation path



Original coordinate illustration. A real horn is an extended directional radiator; this is a baseline propagation model.

Original explanatory calculation. Illustrative parameters are not measured instrument data.

CHECKS AND DECISIONS

• Reject undefined zero-length paths and nonfinite poses; front-facing axial incidence must yield cosine +1.

SOURCES S051-01 Smith, Serafin, Abel and Berners: Doppler Simulation and the Leslie, DAFx-02

ORIGINAL DERIVATION / SIGNAL INTEGRITY

Stereo width changes interference, not only pan position

Leslie microphones, stereo geometry and mono compatibility

DPA distinguishes spaced-pair time differences from coincident directional level differences. [S051-03] For this renderer, derive both channels from the same electrical sources and mechanical states. Give each receiver its own path distance, radiation filtering and microphone response. Independent left/right oscillators, random phase or separate rotor clocks would invent another instrument.

ORIGINAL DERIVATION: average two equal-amplitude copies separated by Delta seconds. Their normalized mono magnitude is |cos(pi*f*Delta)|. The illustrative 0.7 ms difference in the accompanying plot gives a first null near 714.286 Hz. This is calculated interference, not a measured Leslie response. Actual unequal gains, frequency-dependent phase, multiple cabinet paths and two acoustic branches change the cancellation.

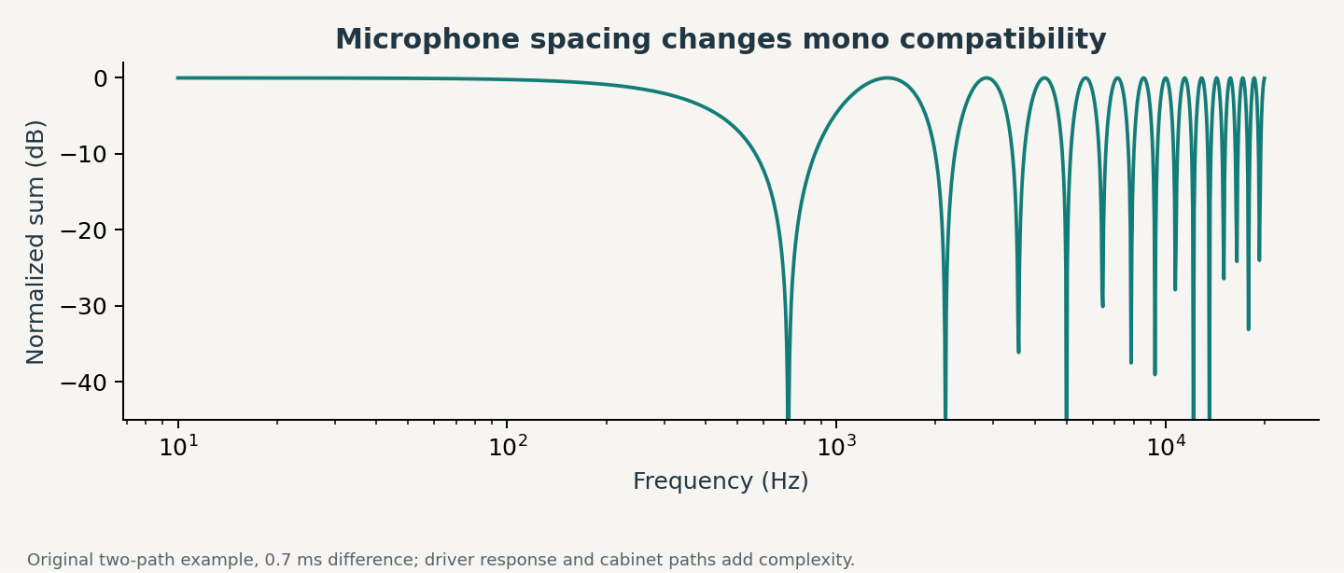
Provide a documented mono fold-down and a separate single-microphone scene. They need not sound identical. A polarity reversal is not a broadband time-alignment correction. A physically accurate spaced scene can have negative correlation or spectral notches; evaluate agreement with its captured reference instead of forcing correlation positive. For an optional ideal M/S scene, equal-gain sum cancels Side, but the Mid microphone still retains its own coloration. [S051-03]

MODEL EQUATIONS

Hmono(f) = (1 + exp(-j*2*pi*f*Delta))/2; f is hertz, Delta is seconds.

First null = 1/(2*|Delta|), for nonzero Delta and equal gains.

Optional normalized decode: L=(M+w*S)/sqrt(2); R=(M-w*S)/sqrt(2).



Original two-path example, 0.7 ms difference; driver response and cabinet paths add complexity.
Original explanatory calculation. Illustrative parameters are not measured instrument data.

CHECKS AND DECISIONS

• With identical coincident channels and equal gains, fold-down equals either channel under the chosen 0.5 normalization.

SOURCES S051-03 Eddy Bøgh Brixen / DPA Microphones: Stereo recording techniques and setups

ORIGINAL IMPLEMENTATION CONTRACT

Implement reception before the output mix matrix

Leslie microphones, stereo geometry and mono compatibility

A microphone is not merely a gain coefficient. The SM57 manual, for example, separately documents frequency response, polar response, proximity behavior, sensitivity and polarity. [S051-04] Do not call an ideal cardioid equation a calibrated SM57 emulation. A measured complete cabinet-to-microphone kernel may already contain microphone coloration and propagation; its ownership flags must disable duplicate processing.

PROPOSED PIPELINE: sum directional paths at each receiver, then apply its preamplifier/conversion and the output mix matrix. A directional microphone weights each arriving path according to that path’s direction; applying one angle-dependent multiplier after summing all directions is generally insufficient. Preserve signed pressure, complex phase and absolute channel timing.

Preallocate delay/filter state. Pose automation needs a declared policy: physical receiver motion produces Doppler; studio repositioning may crossfade two static scenes as an intentional nonphysical convenience. Never silently treat these as equivalent. Compile replacement profiles off the audio thread and retain the old scene through a bounded transition.

```
for each reception sample n:
  trajectory = sharedRotorHistory
  for each microphone j:
    signal[j] = 0
    for each owned acoustic path p:
      state = solvePathAtReception(p, j, n, trajectory)
      signal[j] += renderPathWithMicResponse(state)
    signal[j] = calibratedConversion(j, signal[j])
  output = mixMatrix * signal
  publishReadOnlyPoseSnapshot()
```

CHECKS AND DECISIONS

- A fully measured microphone-inclusive kernel must not receive a second microphone filter.
- Keep geometric acoustic delay separate from fixed algorithmic latency; preserve all relative arrival times.

SOURCES S051-04 Shure SM57 Instrument Microphone User Guide, version 3.6 (2025-A)

MEASUREMENT PLAN / NO NEW HARDWARE CAPTURES

Capture the microphones and the cabinet on one timebase

Leslie microphones, stereo geometry and mono compatibility

DPA’s matching guidance distinguishes sensitivity/frequency-response matching from merely choosing similar microphones; collocated subtraction helps reveal differences. [S051-05] Apply that principle before fitting cabinet asymmetry. Record actual capsule positions rather than stand-clamp positions, channel polarity, preamp gain, filtering, converter alignment and calibration files. A manufacturer nominal specification is not the calibration of the borrowed microphone.

PROPOSED CAPTURE SEQUENCE: record all microphones simultaneously with the electrical excitation reference and independently observed horn/drum angles. Begin with fixed-angle, low-level responses; then record steady rotation, both speed transitions and musical material. Reserve some complete poses and repetitions for validation. Do not independently peak-align channels or normalize every rotor angle: both operations erase information the model must reproduce.

Capture intended near and distant poses and document panel/room states. Measure repeatability before choosing tolerances. If an impulse-response bank embeds the microphone or room, record that boundary explicitly; position interpolation cannot create reliable responses outside the measured region without a separately validated model.

Manifest field	Required meaning
pose_m / forward_axis	Surveyed capsule coordinates and orientation; uncertainty included.
calibration / polarity	Sensitivity, response file, gain, channel sign and acquisition date.
time_reference	Excitation, converter offset, rotor markers and clock-drift treatment.
kernel_includes	Driver, cabinet, propagation, microphone and room ownership flags.
split / provenance	Training or held-out capture; raw hashes and reuse permission.

SOURCES S051-02 Kronland-Martinet and Voinier: Real-Time Perceptual Simulation of Moving Sources, 200 / S051-05 Eddy Bøgh Brixen / DPA Microphones: The difference between matched and selected micro

ACCEPTANCE / EXPLICIT LIMITS

Accept geometry, numerical behavior and measured agreement separately

Leslie microphones, stereo geometry and mono compatibility

Deliver MicSceneProfile, pose validator, receiver renderer, channel-mixing contract, capture manifest and a reproducible stereo/mono comparison report. Establish sample-accurate deterministic renders before fitting microphones. A pair of attractive stereo meters does not establish correct path geometry.

LIMITED CHECKS EXECUTED FOR THIS CHAPTER: front/rear incidence signs and nine ideal two-path comb identities passed numerical assertions. The 0.7 ms fixture’s first-null residual was approximately 1.61e−16 in double precision. These verify small mathematical fixtures only; no real-time receiver module, target cabinet or microphone response was tested.

PROPOSED FIDELITY GATE: compare held-out complex responses, interchannel delay, level difference, cross-spectrum and mono spectra at matched excitation and rotor phase. Downweight regions below the capture noise floor rather than interpreting undefined phase at nulls. Leave final error budgets unset until measurement repeatability and listening sensitivity are established.

Gate	Falsifiable requirement
Coordinate transform	Rigidly transform the complete ideal scene; relative geometry and output remain unchanged.
Receiver identity	Swap complete left/right receiver records; outputs swap exactly.
Shared motion	Block size and UI refresh rate cannot change either rotor trajectory.
Moving delay	Worst supported motion passes bandwidth/alias checks against a higher-rate reference.
Measured fidelity	Held-out poses and transitions meet declared uncertainty-aware limits; currently not run.

SOURCES S051-01 Smith, Serafin, Abel and Berners: Doppler Simulation and the Leslie, DAFx-02 / S051-03 Eddy Bøgh Brixen / DPA Microphones: Stereo recording techniques and setups / S051-05 Eddy Bøgh Brixen / DPA Microphones: The difference between matched and selected micro

REFERENCE SCOPE / ORIGINAL STATE CONTRACT

Give each rotor a persistent physical state

Rotor state, acceleration and deterministic execution

The vintage service manual treats the upper and lower mechanisms separately. Its 5–8-second bass fast-to-Chorale adjustment is a maintenance criterion, not a universal settling definition or exponential time constant. [S052-01] The historical Brake Accessory documents optional stopping; it must not be silently folded into the ordinary two-speed profile. [S052-02]

ORIGINAL DESIGN PROPOSAL: create one horn state and one drum state, each with its own parameters and initial angle. Keep local speed nonnegative and represent direction by a fixed sign relative to the declared cabinet axis. Never infer direction from the apparent rotation in an uncalibrated photograph. A command changes the future trajectory; it does not reset current speed or phase.

Store mechanical angle independently of note activity, musical tempo and GUI frame rate. Choose an explicit initial-phase policy for reproducible rendering. A random initial phase, if offered, needs a retained seed; it is not evidence that the physical instrument changes phase randomly on each note.

State field	Unit and ownership
Angle and turn count	Radians plus integer turns; preserve continuity through wrap.
Angular speed / direction	rad/s and sign; separate values for horn and drum.
Command and actuator mode	Chorale/Tremolo baseline; identified accessory state only when enabled.
Segment anchor	Engine sample timestamp, initial angle/speed, target and model parameters.
Profile provenance	Documented, measured or provisional; no universal RPM or tau assumed.

CHECKS AND DECISIONS

- A note-on, note-off, editor close or repeated unchanged speed command cannot restart a rotor.

SOURCES S052-01 Electro Music / CBS: Leslie 122-family operating and maintenance instructions, public / S052-02 Leslie Brake Accessory installation instructions 038398

ORIGINAL CLOSED-FORM MODEL

Integrate speed and angle from the same segment

Rotor state, acceleration and deterministic execution

Use a first-order speed surrogate only as a declared approximation. For an interval h after a command, omega0 is its initial speed, omegaT the held target and tau a positive time constant. Select upward or downward tau from omegaT minus current speed, not merely from the switch label. A rapid Fast-to-Slow command can still demand acceleration if the rotor has not yet reached Slow.

The equations integrate angle and speed together. At a new event, evaluate the previous segment at that exact time, retain its state, then start the next segment. Anchoring evaluations to command times avoids making the mathematical trajectory depend on host block boundaries.

Evaluate small exponential differences with expm1. Near h/tau=0, the angle increment can suffer cancellation when starting from rest; use the displayed remainder series. For long decay intervals, evaluate the exponential speed residual directly rather than subtracting nearly equal numbers. These numerical choices do not supply missing mechanical calibration.

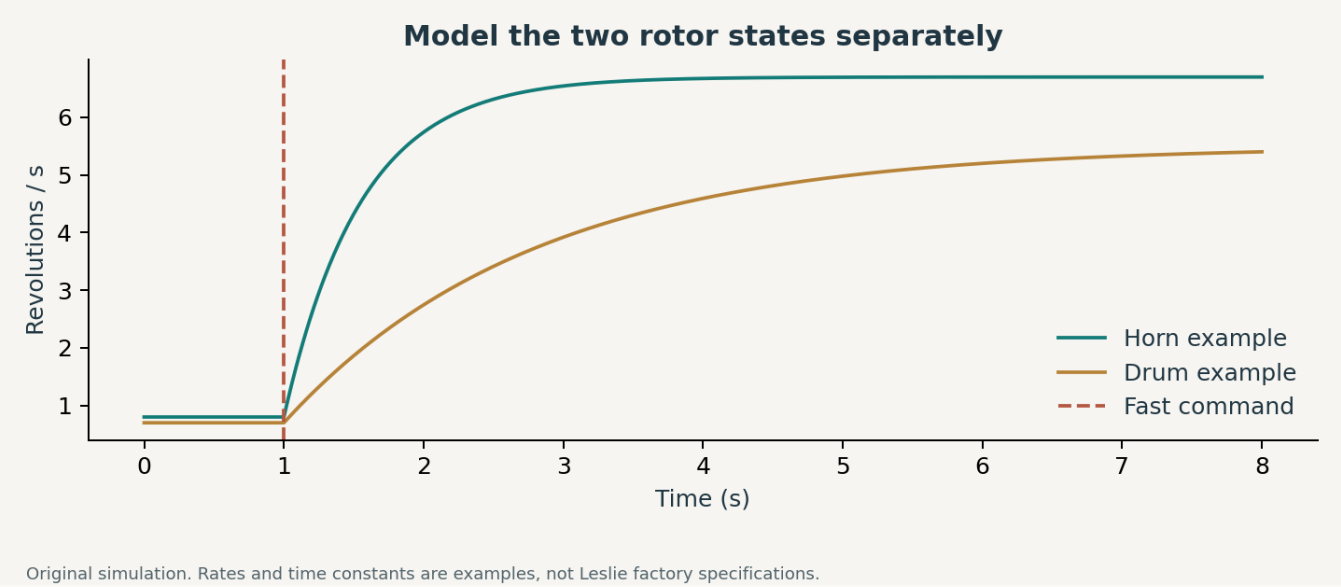
MODEL EQUATIONS

domega/dt=(omegaT-omega)/tau; dtheta/dt=sigma*omega, with sigma=+1 or -1.

omega(h)=omegaT+(omega0-omegaT)*exp(-h/tau).

theta(h)=theta0+sigma*[omegaT*h+(omega0-omegaT)*tau*(1-exp(-h/tau))].

For z=h/tau near 0: g(z)=z²/2-z³/6+z⁴/24-...;
DeltaTheta/sigma=omega0*h+(omegaT-omega0)*tau*g(z).



Original simulation. Rates and time constants are examples, not Leslie factory specifications.
Original explanatory calculation. Illustrative parameters are not measured instrument data.

CHECKS AND DECISIONS

• A command reversal preserves angle and speed at its timestamp; only subsequent acceleration changes.

HOST CONTRACT / ORIGINAL SCHEDULER

Translate host automation into timed commands

Rotor state, acceleration and deterministic execution

VST3 parameter queues contain offset/value points and can omit unchanged values. Continuous curves also have implicit points and linear sections. Reading only the final value of each block loses their timing. [S052-03] Declare the baseline speed selector as a two-state parameter, using step count 1, and handle its discrete automation semantics in the host adapter. [S052-04]

The motor core consumes canonical, sample-stamped command events. Continuous service parameters require their own curve contract; do not accidentally turn a stepped speed switch into a continuously varying target. Retain the previous command when no change arrives. Group simultaneous changes deterministically, applying a validated configuration before its command, and preserve the adapter’s documented tie policy.

Proposed sample convention: an event at frame k starts a new trajectory segment at that boundary; angle and speed at k remain continuous. Preallocate event storage and define an overflow policy with a diagnostic counter. Test the real adapter separately: mathematical block invariance cannot certify a host’s event translation.

```
for each canonical event time k:
  evaluate old horn and drum segments at k
  validate and apply the ordered configuration changes
  resolve new commands without resetting phase
  select each rotor target and signed-error time constant
  append immutable trajectory anchors to bounded history
for each requested sample time t:
  evaluate each owning rotor segment at t
  give acoustic paths read-only trajectory access
  publish a decimated UI snapshot; never integrate in UI
```

CHECKS AND DECISIONS

- Preserve stable parameter IDs and state mappings when adding an optional stop mode.
- Test commands immediately before, on and after block boundaries, including an empty queue.

SOURCES S052-03 Steinberg VST 3 Interfaces: IParamValueQueue / S052-04 Steinberg VST 3 Developer Portal: Parameters and Automation

ORIGINAL OWNERSHIP AND HISTORY DESIGN

One trajectory can serve several observation times

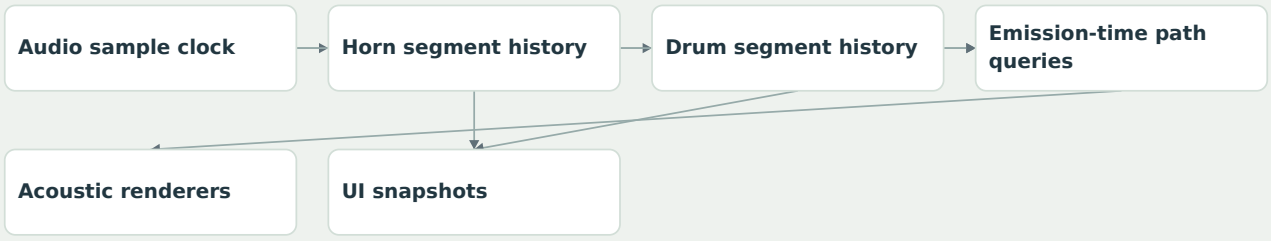
Rotor state, acceleration and deterministic execution

The audio engine owns both rotor clocks. The acoustic renderer queries their histories at emission times determined by chapter 049; two microphones may query different past angles from the same physical trajectory. Keep enough previous command segments for maximum propagation delay and interpolation support. Evaluating every delayed path from only the newest command loses earlier motion.

The UI receives snapshots and may interpolate display motion, but its animation never advances audio state. Silence does not mean the rotor stops. Musical transport changes do not reset mechanical phase. If processing is suspended, adopt and document a deterministic freeze policy; a separate wall-clock catch-up option changes reproducibility and must be explicit.

On a sample-rate change, preserve angle, physical speed and parameters in seconds while creating a new clock epoch. Rebuild or deliberately transition acoustic history; do not reuse old sample indices at the new rate. Keep ordinary musical presets separate from complete offline checkpoints, which also retain trajectory history, timing and random state.

One owner, read-only trajectory consumers



Original software ownership proposal. Different microphones query the same trajectory at their respective emission times; UI animation cannot advance it.

CHECKS AND DECISIONS

- Changing UI refresh rate or closing the editor leaves the rendered trajectory unchanged.

CALIBRATION / STOPPING AND MODEL ORDER

Promote the model only when captures justify it

Rotor state, acceleration and deterministic execution

Acquire synchronized angle markers for both rotors and an electrical command reference. Record steady states, both transition directions and rapid reversals before settling. Fit target speeds from steady intervals first; then estimate delay and ramp behavior with uncertainty. Audio amplitude peaks alone are not reliable angle markers because radiation and interference change with frequency and microphone position.

Compare the exponential residual against a monotone, speed-dependent acceleration model. Escalate to chapter 047’s coupled torque, engagement and slip model only when observations identify the additional states. A fitted trajectory does not uniquely reveal inertia, torque and friction separately. Preserve model identity and held-out residuals in each profile.

An exponential decay toward zero approaches rest asymptotically. It does not implement a finite-time physical brake. An accessory model needs its actual actuator sequence and stopping mechanism. A numerical stop threshold is a declared approximation: retain the final angle and quantify its error rather than calling a rounded speed display proof of stopped hardware.

Candidate model	Evidence needed before acceptance
Exponential speed surrogate	Steady speeds, direction-specific ramps and rapid-switch residuals.
Tabulated acceleration	Coverage of speed and actuator state; monotonicity and integration convergence.
Coupled torque / slip	Additional shaft or engagement evidence; passive-loss and energy checks.
Historical brake accessory	Installed kit identity, relay behavior and measured stopping sequence.

CHECKS AND DECISIONS

• Choose tolerances after establishing capture repeatability; illustrative rotor-ramp graphics are not calibration data.

EXECUTED MATHEMATICAL CHECKS / PROPOSED ACCEPTANCE

Retain a numerical oracle and a separate hardware gate

Rotor state, acceleration and deterministic execution

An original Python reference exercised two synthetic rotor profiles with five timed commands. Four block partitions produced bit-identical sample states at each of 44.1, 48, 96 and 192 kHz. The 841 common-time observations agreed across those rates. These results apply to the event-anchored reference, not an implemented plugin or a physical Leslie.

Closed-form updates were also checked against 80-digit decimal calculations over h/tau from 1e-12 to 100, including start from rest and decay toward zero. Maximum relative error was 2.12e-15. One thousand random split-interval cases checked the speed/angle semigroup relation. The script and detailed unit-labeled results are retained as 052-rotor-checks.py and 052-checks.json.

ORIGINAL ACCEPTANCE PROPOSAL: retain this oracle while implementing the production state service. Then test actual host translation, bounded memory, historical queries, sample-rate transitions and suspension policies. Finally compare withheld physical angle captures. None of those production or hardware gates was executed by this mathematical fixture.

Gate	Required retained evidence
Continuity	No phase/speed reset at commands; correct modulo-angle display.
Time and scheduling	Identical event stream across blocks, rates and editor states.
Bounded work	Fixed storage, admitted event density and explicit overflow behavior.
History	Correct trajectory on both sides of a command during delayed acoustic queries.
Physical fidelity	Specimen-specific captures, fitted uncertainty and held-out residuals.

CHECKS AND DECISIONS

- Synthetic parameter values and test directions must never become factory defaults.

MODEL BOUNDARY / ORIGINAL DESIGN PROPOSAL

Declare what the impulse-response bank already contains

Time-varying convolution and directional filter implementation

A frozen-angle impulse response describes one stationary configuration. It does not, by itself, specify the time-varying system between configurations. General time-varying FIR theory distinguishes input and observation time. [S053-02] The implementation must therefore declare both the kernel boundary and the clock used to select its angle.

ORIGINAL DESIGN PROPOSAL: store kernel units, sample rate, absolute gain, time origin, microphone pose, cabinet panels, driver identity and included mechanisms. A complete microphone response and a propagation-stripped radiator response are different data products. Preserve the original response before any timing removal or factorization; retain the transform that can reconstruct it.

The cabinet walls and louvers remain fixed while the rotors move. Do not rotate an entire cabinet kernel merely by subtracting microphone azimuth. Keep the horn and bass banks separate, and require measured evidence before sharing a basis across receiver poses. Chapters 49–51 define the common trajectories and distinct emission times.

Treat each bank as a level-limited linear approximation. Record its excitation level and stationary electrical settings. Amplifier clipping, driver compression and other level-dependent behavior need their own validated boundary; one linear IR cannot establish those mechanisms merely because its static spectrum resembles a loud recording.

Kernel representation	Required ownership contract
Complete frozen-angle response	Declare included driver, cabinet, propagation, microphone and room; motion still needs a time-indexing model.
Residual radiator/path FIR	Document removed delay and response factors; the separate propagator owns the corresponding travel time.
Fixed directional room response	Feed a declared outgoing source basis; exclude any direct/cabinet response already rendered.

CHECKS AND DECISIONS

• Reject a profile that assigns the same propagation delay or microphone coloration to two processing stages.

SOURCES S053-02 Øyvind Brandtsegg, Sigurd Saue and Victor Lazzarini: Live Convolution with Time-Varyi

ORIGINAL DERIVATION / TWO-TIME KERNEL

Moving the angular weights changes the modeled physics

Time-varying convolution and directional filter implementation

At a chosen discrete interaction boundary, let x[n] be the input, g_r[k] fixed causal basis filters, and b_r(theta) dimensionless angular weights. Define h(theta,k)=sum_r b_r(theta)g_r[k]. The two equations below agree for a fixed angle, but generally differ during rotation. A frozen-angle comparison cannot select the correct convention.

Model A applies the current boundary orientation to stored filter outputs. Model B tags each incoming sample with its orientation before the fixed response spreads that sample through time. For example, a directional excitation entering a fixed room can use B; a proposed quasi-static mouth-response stage before explicit propagation may use A. These are declared approximations, not interchangeable optimizations.

For the residual mouth model, query the resulting emission stream at each path’s retarded time from chapter 49. Evaluate the corresponding angular state on that same emission timeline. If a full moving kernel already accounts for changing transit, adding another variable delay repeats the Doppler mechanism. A long cabinet tail may require a richer two-time model than either frozen-angle approximation.

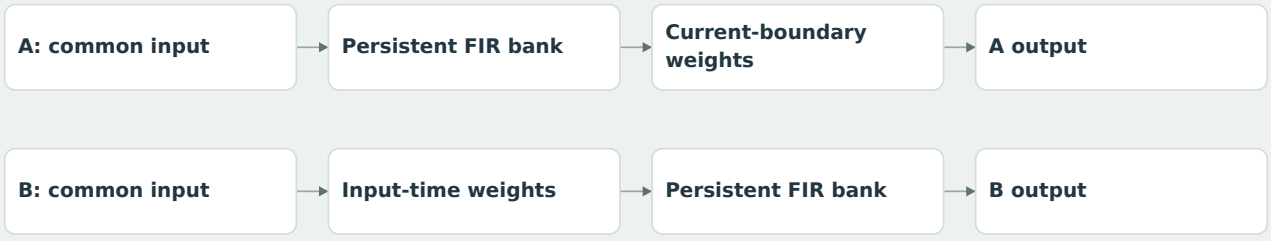
A diagnostic impulse makes the distinction tangible: change angle while its delayed tail is still emerging. Model A applies the new weights to that old excitation; model B retains the weights assigned when the impulse entered. Validate which interaction the modeled stage represents before optimizing its arithmetic.

MODEL EQUATIONS

A: yA[n] = sum_k h(theta[n],k)*x[n-k] = sum_r b_r(theta[n])*(g_r*x)[n].

B: yB[n] = sum_k h(theta[n-k],k)*x[n-k] = sum_r (g_r*(b_r(theta)*x))[n].

Two weight placements, two time-varying systems



Original mathematical alternatives. Identical frozen-angle responses do not prove identical moving behavior.

SOURCES S053-02 Øyvind Brandtsegg, Sigurd Saue and Victor Lazzarini: Live Convolution with Time-Varyi

ORIGINAL IMPLEMENTATION / CONTINUITY

Interpolate phase-preserving residuals and keep their history

Time-varying convolution and directional filter implementation

Keep every active basis filter running continuously. Changing angle then changes periodic weights, without clearing input history or overlap buffers. A cyclic linear stencil should wrap its final interval to the first angle and use weights summing to one. A Fourier angular basis can instead expose the retained angular order and approximation residual. Choose its size from held-out complex-response error.

Interpolate signed IR coefficients or an equivalent complex representation with valid real-signal symmetry. Magnitude-only interpolation discards phase. Blending two differently delayed impulses creates two arrivals; it does not move one arrival smoothly between them. Remove only the delay explicitly owned by the propagator, and preserve the residual phase rather than automatically converting it to minimum phase.

Profile replacement is separate from rotation. Warm the new bank from retained input history or an explicit pre-roll, then crossfade two correctly initialized outputs. Old overlap state depends on the old filter and cannot simply be reinterpreted as new-filter state. Nonuniform partition update timing requires an explicit audible transition schedule. [S053-03]

The angular interpolation grid and audio sample grid are separate. Smooth coefficient variation must be evaluated or interpolated on the audio timeline, not driven by UI refreshes. Bound angular truncation error and modulation bandwidth independently.

Transition choice	Consequence to test
Complementary amplitude fade	w0+w1=1 preserves identical correlated outputs.
Equal-power fade of identical outputs	Calculated midpoint gain is sqrt(2), or +3.0103 dB; not a neutral replacement.
Cold replacement engine	Missing prior input produces a different tail even when the new coefficients are correct.
Independent partition swaps	May produce a hybrid kernel with no intended physical interpretation.

SOURCES S053-03 Christoph Müller-Tomfelde: Time-Varying Filter In Non-Uniform Block Convolution, DAFx

REAL-TIME DESIGN / CALCULATED BUDGET

Partition the computation without changing the time contract

Time-varying convolution and directional filter implementation

Partitioned convolution trades transform work, latency and scheduling complexity; shared input transforms can reduce repeated work. [S053-01] For basis model A, branches share an input. In model B, the weighted input streams differ, so their forward transforms generally differ too. Do not move weights across convolution merely to reuse an FFT.

ILLUSTRATIVE BUDGET: four basis filters, two receivers, 4096 taps, 128-sample partitions and 256-point real FFTs require 32 partitions. Their spectral accumulation totals approximately 12.384 million complex multiply-accumulates per second at 48 kHz, excluding transforms and mixing. This is arithmetic, not a CPU benchmark. Measure worst callback time during two-bank transitions; a 128-sample partition period is 2.667 ms, not necessarily the host block size.

Prepare/decode/allocate outside the callback. A framework handoff must follow its actual synchronization contract: JUCE documents wait-free load entrypoints but forbids interleaving class methods. Explicitly disable unintended IR normalization and trimming; preserve calibrated gain and onset. Query actual algorithmic latency separately from modeled acoustic travel time. [S053-04]

Budget transition banks and input-history retention before publication of a new profile. Attach an absolute activation sample and version identifier to the prepared object. A background worker finishing early or late must not silently redefine the timestamped audio event.

```
off_thread: validate profile, units, timing and ownership
off_thread: build FIR partitions and immutable metadata
prepare: allocate histories, FFT workspaces and transition banks
process: split events on the absolute sample timeline
process: generate the declared weighted or unweighted inputs
process: advance persistent partition histories
process: apply weights only at their declared boundary
process: render each owned propagation path once
process: complete any scheduled bank transition
process: publish bounded read-only diagnostics
```

CHECKS AND DECISIONS

• Compare variable host blocks against one continuous offline reference; include final partial partitions and full tails.

SOURCES S053-01 Frank Wefers: Partitioned convolution algorithms for real-time auralization, 2015 / S053-04 JUCE: juce::dsp::Convolution class reference

SOURCE IMAGE / MEASUREMENT PLAN

Use the published simulation as context, then acquire the target

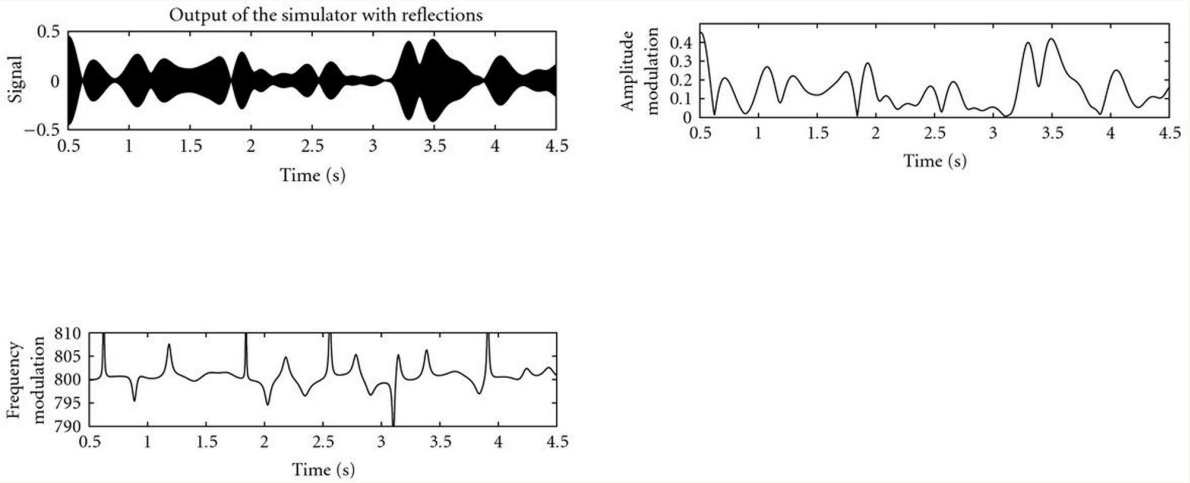
Time-varying convolution and directional filter implementation

The accompanying P043 image reproduces Figure 8 of the 2008 paper: output from the authors' enclosure-image simulation. Figure 5, shown in chapter 50, is their 122A measurement. These must remain visibly distinct; neither establishes a measured match for the chosen vintage 122 or for software implemented from this suite. [S053-05]

PROPOSED ACQUISITION: capture synchronized fixed-angle responses, electrical excitation, rotor markers and microphone channels. Retain absolute gain and interchannel timing. Fit the residual basis on a declared subset of angles, poses and repetitions; reserve complete unseen poses and continuous speed transitions for validation. Record rear-panel state and response ownership with every file.

Test interpolation where lobes or cancellation vary rapidly, not only at favorable angles. Quantify noise and repeatability before interpreting phase near nulls. Increase angular order, IR support or path complexity only when a held-out residual identifies the missing behavior. A longer kernel cannot repair incorrect emission timing or duplicated propagation.

Acquire both rotational directions only where the actual mechanism supports them; otherwise mark unsupported directions explicitly. Retain tachometer zero, angle convention, sweep deconvolution settings and resampling history. A photograph of the cabinet establishes none of those timing fields.



P043 | Published Leslie simulation with enclosure image sources | R. Kronland-Martinet and T. Voinier | CC BY 2.0 | Published authors' simulation result, not a physical measurement and not output from a completed emulator in this suite. Compare with the measured 122A figure only under the article's stated assumptions. Source and license

CHECKS AND DECISIONS

- No target angular IR dataset, measured model fit or production convolver is supplied by this chapter.

SOURCES S053-05 R. Kronland-Martinet and T. Voinier: Real-Time Perceptual Simulation of Moving Source

EXECUTED MATHEMATICAL FIXTURES / PROPOSED GATES

Require numerical equivalence before claiming acoustic agreement

Time-varying convolution and directional filter implementation

EXECUTED CHECK: a synthetic 137-sample input, three 23-tap basis filters and bounded angular weights were evaluated by direct summation and offline partitioned overlap-add. Partition sizes 1,3,8 and 31 agreed over all 159 output samples, including the full tail, with their corresponding A/B reference within 3.34e−16. The two time conventions differed by0.334 in arbitrary units: replacing one with the other was observably wrong. This is an algebra/scheduling fixture, not a real-time engine or physical comparison.

For either stated finite-basis convention, bounded weights and finite FIR norms provide the amplitude bound below. This does not guarantee low aliasing. Angular order M rotating at nu revolutions/second introduces modulation up to M*nu hertz; assess sidebands together with source bandwidth and changing-delay resampling at the internal sample rate.

Deliver a versioned kernel profile, its reversible preprocessing record, an offline reference and retained reports. Final acoustic tolerances remain unset until captures establish repeatability. Hardware-fidelity, deadline and complete-plugin gates are not run here.

Proposed gate	Required comparison
Frozen angle / impulse	Exact reference FIR, gain, sign, onset and complete tail.
Moving angle / wrap	Declared two-time operator; no seam reset or weight discontinuity.
Numerical bandwidth	Oversampled reference under highest supported speed and angular order.
Bank replacement / timing	Warm history, synchronized transition, bounded memory and callback cost.
Physical validation	Held-out complex kernels and rotating recordings at matched boundaries.

MODEL EQUATIONS

||y||inf <= ||x||inf * sum_r (sup_theta |b_r(theta)| * sum_k |g_r[k]|).

Ideal constant-speed modulation only: f_audio_max + M*nu < Fs_internal/2. Ramps and variable-delay resampling require additional bandwidth analysis.

EVIDENCE BOUNDARY / ORIGINAL ARCHITECTURE

Choose the acoustic boundary before adding a room

Room response, spatial rendering and optional122RV spring reverb

A room belongs to the reproduced recording chain, not to every direct organ output. The 2008 Leslie study discusses enclosure reflections and changing coupling to the surrounding space, but its 122A experiment does not define a universal vintage 122 room. Keep the cabinet, microphone scene and external environment as separately identified evidence whenever their measurements permit that separation. [S054-03]

ORIGINAL DESIGN PROPOSAL: every acoustic asset declares its input boundary, output units, included propagation, cabinet panels, microphone response, room conditions and direct-path content. A measured in-room angular response may already contain all those effects. Cascading another cabinet response, microphone filter or room IR then reproduces the same mechanism twice.

Provide distinct product modes: direct organ electrical output; cabinet radiation captured with a declared microphone scene; and an optional studio reverb applied to an existing stereo recording. The last mode can be musically useful, but it is not automatically the physical effect of moving that cabinet into another room. Unknown decomposition is a reason to retain the complete captured response, not to invent a clean separation.

Asset boundary	Permitted interpretation
Electrical input to microphone	Combined cabinet/microphone/room response; inspect what is already included.
Radiation basis to microphone	Spatial room matrix with an explicitly calibrated source representation.
Close-mic stereo to stereo effect	A production effect unless physical equivalence is established.
Room bypass	Requires a genuinely room-free branch or a declared approximation.

CHECKS AND DECISIONS

- Reject an IR with an unidentified direct-path, calibration or microphone boundary.
- A room-off control must not silently remove cabinet reflections.

SOURCES S054-03 R. Kronland-Martinet and T. Voinier: Real-Time Perceptual Simulation of Moving Source

ORIGINAL GEOMETRIC MODEL

Give each reflected path its own emission geometry

Room response, spatial rendering and optional122RV spring reverb

Allen and Berkley construct room responses from image sources and explicitly qualify the nonrigid-wall approximation. Use that construction as an early-reflection model with stated assumptions; arbitrary wall materials, diffraction and low-frequency modal behavior are not made exact by adding more image points. [S054-02]

For a plane n·x=d with unit normal n, reflect a position by q′=q−2n(n·q−d). Each valid reflection sequence produces its own image trajectory and path length. Solve t=e+r_p(e)/c for that path as in chapter 049. Evaluate rotor orientation, outgoing directivity and wall incidence at the appropriate geometry. A wall path must use the ray leaving toward its first reflection, rather than reusing the direct microphone ray.

The published 122A model used five first-order enclosure images with an upper back wall removed. That is contextual evidence for interference, not an instruction to put five reflections into every122 profile. Cabinet reflection paths and room reflection paths need distinct ownership, source identifiers and panel configurations. [S054-03]

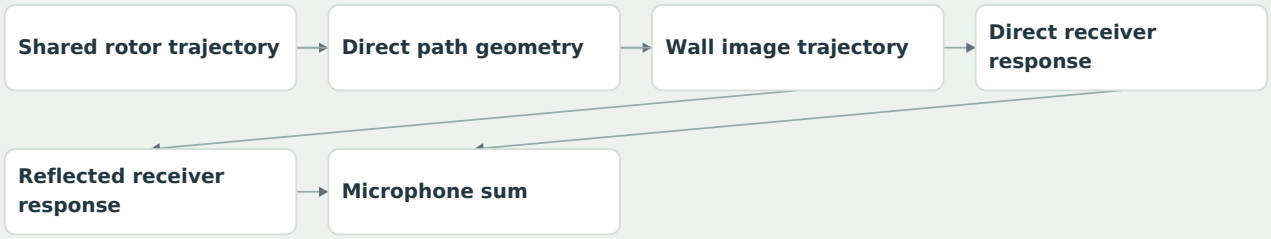
MODEL EQUATIONS

q_image(e) = q(e) - 2*n*(dot(n,q(e)) - d), with ||n||=1.

t = e_p + r_p(e_p)/c; path p has its own emission time e_p.

y_m(t) = sum_p path_filter_p{source}(e_p), under the declared point-path approximation.

Different paths observe different emission times



Original path responsibility diagram. Wall reflection, radiation and receiving responses need calibrated boundaries; this is not a measured room.

CHECKS AND DECISIONS

- At zero rotation, fixed geometry must become a time-invariant response.
- Trace reflection validity and visibility; do not admit a ray through an opaque obstacle.

SOURCES S054-02 Jont B. Allen and David A. Berkley: Image method for efficiently simulating small-roo / S054-03 R. Kronland-Martinet and T. Voinier: Real-Time Perceptual Simulation of Moving Source

ORIGINAL MIMO CONVOLUTION CONTRACT

Preserve cross-channel kernels and absolute timing

Room response, spatial rendering and optional122RV spring reverb

For fixed spatial source bases b_q and microphones m, define h_mq as the impulse response from input q to output m. A two-input/two-output room representation requires four kernels unless symmetry or zero coupling is explicitly established. Two independent convolvers on the diagonal omit crossfeed. A mono source with two receiver IRs is a different, two-kernel contract.

ORIGINAL DESIGN PROPOSAL: encode rotating radiation into calibrated fixed spatial bases before the stationary room matrix, or use the explicit moving-path renderer on the previous page. Do not run both complete models in series. A finite source basis is an approximation whose error must be checked against continuously rotating captures; chapter 053 explains why moving angular weights across a convolution changes the system.

Preserve common capture time zero across every kernel. Independent peak alignment destroys interchannel propagation cues. Declare whether direct arrivals remain inside the matrix. Partitioned convolution adds computational buffering distinct from physical travel time; report the former as algorithmic latency and retain the latter as modeled acoustics. Resampling, truncation and normalization must be reproducible, reversible metadata operations.

Implementation object	Required state
Kernel bank	Input/output indices, common time origin, units and evidence IDs.
Convolver	Persistent history for every active input basis and complete tail.
Asset change	Prepared immutable bank and declared state/crossfade policy.
Host latency	Fixed computational delay, separate from acoustic arrival times.

MODEL EQUATIONS

y_m[n] = sum_q sum_l h_mq[l] * b_q[n-l].

For two inputs: y_L=h_LL*b_L+h_LR*b_R; y_R=h_RL*b_L+h_RR*b_R.

CHECKS AND DECISIONS

- Excite each input separately and verify every output kernel, including off-diagonal paths.
- Do not label equalized, independently normalized kernels as calibrated absolute pressure.

OPTIONAL HARDWARE VARIANT / ORIGINAL IMPLEMENTATION

Keep the 122RV spring channel separate from room reverberation

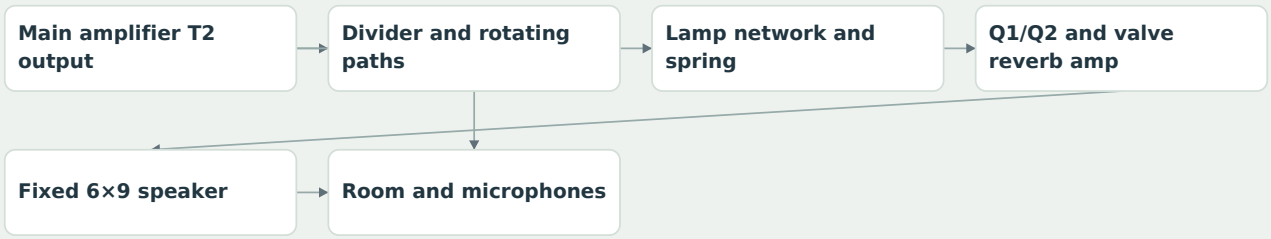
Room response, spatial rendering and optional122RV spring reverb

Factory Figure 2 and the split117V schematic show the main amplifier output feeding the spring branch while continuing to the rotating-speaker divider through the reverb interceptor connection. P12 is the spring input; P13 returns its output to the Q1/Q2 transistor preamplifier. Those transistors receive the spring signal rather than serving as its input driver. The input network includes M1/M2 number19 lamps and R19=15 ohms. [S054-01]

The reverb amplifier uses Q1/Q2=2N3859A, V6=12AU7A, V7/V8=7189 and V9=0B2; T4 feeds a separate fixed6×9-inch speaker in 122RV. The 222RV adds a second6×9 speaker. This radiation is separate from the rotating horn and drum. [S054-01]

ORIGINAL DESIGN PROPOSAL: branch from the modeled main-amplifier output, retain spring-input loading, model the spring and return electronics, then radiate the wet signal from its stationary speaker position. Sum its acoustic arrival with the rotary paths before applying their common environment contract. A generic stereo room IR cannot replace that signal ordering, spring dispersion, amplifier behavior or fixed-speaker radiation. Keep the baseline122 profile free of this optional branch; lamp characteristics, tank response and speaker parameters require acquisition.

Optional spring reverb has its own radiator



Original functional reading of the 1971 schematic and Figure 2. The 122RV extension is separate from baseline122 and from external room reverberation.

CHECKS AND DECISIONS

- Do not apply rotor Doppler to the fixed reverb speaker merely because it is inside a Leslie.
- Retain main-output loading and distinct nonlinear boundaries when modeling the optional circuit.

SOURCES S054-01 Electro Music / CBS: Leslie 122-family General Operating and Maintenance Instructions

ORIGINAL MODEL AND ACQUISITION PLAN

Fit early structure and late decay without duplicating energy

Room response, spatial rendering and optional122RV spring reverb

ORIGINAL DESIGN PROPOSAL: retain measured early reflections explicitly and use either the captured late tail or a labeled statistical late model. If one IR is divided into early and late components, use complementary amplitude windows whose sum is one. An arbitrary overlap of complete early and late responses changes energy and interference. Transition time is a measured/design field, not a universal Leslie constant.

For an optional feedback-delay-network tail, start from a proven lossless structure and introduce controlled attenuation. Schlecht and Habets establish a sufficient unitary-matrix construction and show why the eigenvalues of the feedback matrix alone do not establish stability with arbitrary delays. A proposed implementation should use positive delay lengths, stable bounded loss filters and a retained energy/decay test, rather than a random feedback matrix. [S054-04]

Acquire the room using identified source and receiver poses, calibrated gain and synchronized time. Separate frozen-rotor captures from continuous rotation; a sweep through an unknown changing system cannot simply be treated as an ordinary stationary IR. Record panels, furniture, temperature, occupancy and background noise. Reserve microphone poses, rotor trajectories and input levels that are absent from fitting.

Capture	Question it resolves
Frozen-angle room responses	Arrival times, reflection directions and stationary kernel accuracy.
Continuous rotation	Time ordering and direction-dependent excitation of the room.
Tail and noise captures	Frequency-dependent decay and valid measurement range.
RV-only branch, if present	Stationary speaker response and spring/electronics contribution.

MODEL EQUATIONS

h_early[n]=w[n]*h[n]; h_late[n]=(1-w[n])*h[n]; their sum exactly reconstructs h[n].

For fixed orthogonal A, ||A*v||2=||v||2; additional strictly contractive loss is a design requirement for a decaying tail.

CHECKS AND DECISIONS

- A late model must not regenerate an early reflection already present in a measured kernel.
- Do not normalize each frequency band or microphone independently before measuring spatial decay.

SOURCES S054-04 Sebastian J. Schlecht and Emanuel A. P. Habets: On Lossless Feedback Delay Networks,

EXECUTED SYNTHETIC CHECKS / FUTURE ACCEPTANCE

Accept numerical structure before making a room-fidelity claim

Room response, spatial rendering and optional122RV spring reverb

An original small numerical fixture exercised four paths in a two-input/two-output FIR matrix. Unit impulses recovered the expected off-diagonal and diagonal kernels with zero arithmetic error for the chosen coefficients. Complementary early/late windows recombined within 2.78e-17. The retained data/054-fixture-results.json identifies the synthetic kernels and limited scope; this is neither a production convolver test nor a measured room result.

An exactly duplicated coherent direct path doubles pressure amplitude, a derived6.0206 dB error. Use this as a diagnostic fixture, not as a prediction for arbitrary reflections: differently delayed paths produce frequency-dependent interference. Require silence, impulse, channel permutation, sample-rate, complete-tail and block-partition tests before fitting a physical scene.

The physical gate then compares held-out captures using complex response, arrival timing, banded decay, interchannel correlation and mono summation, with tolerances chosen from repeatability and the intended listening conditions. A scalar decay-time match cannot establish directional or transient accuracy. Save raw captures, processed kernels, scene coordinates, fitting history and the exact renderer build together.

Gate	Evidence required
Boundary correctness	Every direct, cabinet, microphone and room contribution has one owner.
Numerical correctness	MIMO indexing, retained histories, latency and tail checks.
Spatial prediction	Held-out microphone poses and rotor trajectories.
Optional RV fidelity	Separate spring/electronics/fixed-speaker captures.
Release claim	Named specimen, room, microphone setup and tested conditions.

CHECKS AND DECISIONS

- Unmeasured room, spring, lamp and reverb-speaker parameters remain explicit unknowns.
- No physical capture, FDN implementation, real-time deadline or acoustic-fidelity test was performed for this chapter.

HARDWARE EVIDENCE / MODEL BOUNDARY

Identify the fitted drivers before choosing nonlinear parameters

Speaker nonlinearities and system identification

The 1971 parts list identifies020305 as the treble driver,020313 as its spacer ring, and 052308 as the bass speaker. Both driver descriptions carry nominal 16-ohm ratings. Those labels establish stock identity and a nominal electrical class; they do not supply a measured impedance curve, diaphragm mass, compliance, force factor or thermal model. Both drivers remain stationary while the horn and lower deflector rotate. [S055-01]

Published loudspeaker research establishes useful mechanisms, but its fitted numbers belong to its test objects. Klippel's compression-driver example was measured with the horn and phase-plug interface removed. Its resonance, construction and distortion curves must not become Leslie 122 defaults. [S055-02]

ORIGINAL DESIGN PROPOSAL: give each installed driver a versioned electrical-to-acoustic port model. Preserve replacement markings, diaphragm changes, repair history and installation conditions. Start with identified small-signal behavior, then add only nonlinear mechanisms supported by captures. A smooth saturation curve after a cabinet IR cannot establish those mechanisms or reproduce the load seen by the amplifier.

Boundary	Required ownership
Electrical	Terminal voltage/current, polarity and complex load
Mechanical	Diaphragm displacement and velocity, not rotor angle
Acoustic	Declared pressure/volume-velocity source boundary
Thermal	Cold reference, heating history and cooling state
Specimen data	Unknown until the fitted driver is characterized

CHECKS AND DECISIONS

- Reject a generic speaker parameter set labeled as measured 122 data.
- Keep crossover, amplifier and acoustic-transfer versions attached to the driver profile.

SOURCES S055-01 Electro Music/CBS Leslie family manual, publication 044578, June 1971 / S055-02 Wolfgang Klippel, Large Signal Performance of Tweeters, Micro Speakers and Horn Drive

ORIGINAL DERIVATION / COUPLED DOMAINS

Preserve electromechanical reciprocity and stored energy

Speaker nonlinearities and system identification

For an explicitly limited lumped model, define displacement x in metres, velocity v, current i and terminal voltage u. Let L(x) be inductance, Psi(x) permanent-magnet flux linkage, and Bl(x)=dPsi/dx. With positive motion and electrical polarity fixed together, magnetic coenergy Wc=0.5L(x)i^2+iPsi(x) gives both flux linkage and force. This derivation prevents independently chosen electrical and mechanical signs. General driver literature motivates these mechanisms. [S055-02] [S055-04]

The derivative of L contributes to both terminal voltage and electromagnetic force. Dropping one term can create or destroy numerical energy. Define restoring force Fs=dUs/dx directly: a table labeled stiffness must say whether it means Fs/x or dFs/dx. These conventions cannot be interchanged for a nonlinear spring. Here m is moving mass, D(v) is mechanical loss force, and Us is spring potential energy.

Fac is the acoustic reaction force. With effective area Sd, use Q=Sd*v and Fac=Sd*p_load under the chosen orientation. Assign added air mass once, either to effective moving mass or the acoustic load. Higher diaphragm modes and phase-plug flow require an extended model or measured residual; this lumped scaffold is not a complete high-frequency horn model.

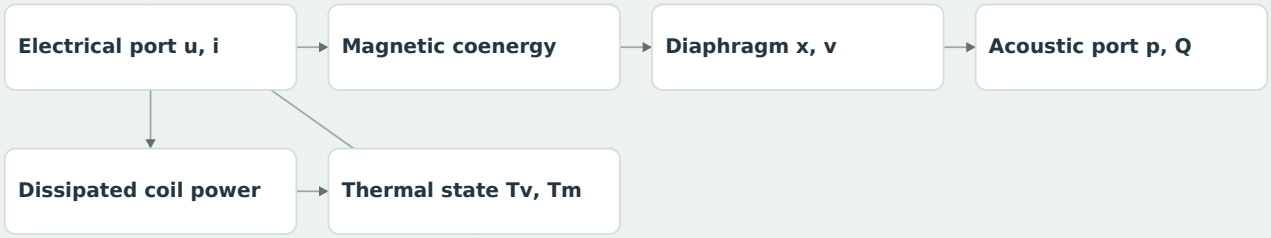
MODEL EQUATIONS

lambda = L(x)*i + Psi(x); u = Re(T)*i + d(lambda)/dt.

u = Re(T)*i + L(x)*di/dt + [Lprime(x)*i + Bl(x)]*v.

m*dv/dt = Bl(x)*i + 0.5*Lprime(x)*i^2 - Fs(x) - D(v) - Fac.

One coupled transducer state



Original responsibility diagram for the proposed model; no measured driver parameters are implied.

CHECKS AND DECISIONS

• Verify instantaneous power balance before discretizing the nonlinear equations.

SOURCES S055-02 Wolfgang Klippel, Large Signal Performance of Tweeters, Micro Speakers and Horn Drive / S055-04 Wolfgang Klippel, Nonlinear Adaptive Controller for Loudspeakers with Current Sensor

THERMAL MODEL / SOURCE AUDIT

Give thermal compression physical memory and calibrated units

Speaker nonlinearities and system identification

Heating changes coil resistance and therefore current, motor force and the load presented through the crossover. Published thermal models distinguish heat-storage and heat-transfer paths, including convection and eddy-current dissipation. Their fitted constants are driver-specific. [S055-03] Resistance-based temperature inference also requires a cold reference and material properties. [S055-05]

ORIGINAL DESIGN PROPOSAL: begin with coil and magnet temperature rises theta_v and theta_m above a constant ambient reference. Positive capacities Cv,Cm use joules/kelvin; positive conductances G use watts/kelvin. Average actual coil loss over a fixed elapsed-time grid, preserving state through note releases and presets. Assign eddy-current heating to its modeled destination once. Add convection dependence only after measurements distinguish it.

Source check: the heat manuscript prints 0.039 for copper on PDF 4. NBS instead gives 0.00393 per degree C at 20 C for standard annealed copper. Do not copy the discrepant coefficient or assume the actual driver uses that material. Store the selected coefficient, reference temperature and calibration evidence explicitly. [S055-03] [S055-06]

Implementation field	Required distinction
Pcoil	Heat dissipated in the coil, not total amplifier output power
Initial state	Declared cold start or retained temperatures
Cooling	Continues during silence; no preset-triggered reset

MODEL EQUATIONS

Cv*d(theta_v)/dt = Pcoil - Gvm*(theta_v-theta_m) - Gva*theta_v.

Cm*d(theta_m)/dt = Gvm*(theta_v-theta_m) - Gma*theta_m + Peddy.

Re(Tv) = Rref*[1 + alpha*(Tv-Tref)]; alpha has units 1/K.

CHECKS AND DECISIONS

• Verify zero-input cooling and nonnegative dissipation within the calibrated range.

SOURCES S055-03 Wolfgang Klippel, Nonlinear Modeling of the Heat Transfer in Loudspeakers, author man / S055-05 KLIPPEL: Voice Coil Temperature measurement guidance / S055-06 US National Bureau of Standards, Copper Wire Tables, Handbook100, February21,1966

IMPLEMENTATION / REAL-TIME CONTRACT

Solve the load with the amplifier and divider

Speaker nonlinearities and system identification

ORIGINAL IMPLEMENTATION PROPOSAL: expose each driver as a voltage/current port plus acoustic source variables. Chapter 044’s floating treble connection must remain intact. The two driver currents contribute to the common amplifier load; nonlinear impedance cannot be represented faithfully by two independent fixed filters following an ideal source.

Build an offline reference first. Validate the small-signal network, then introduce bounded measured curves for restoring force, force factor and inductance. Preserve observed asymmetry instead of forcing every curve even. Use preallocated state and a bounded solver; define a diagnostic fallback for failed convergence. Choosing an oversampling factor requires comparison against a converged high-rate reference, not a universal rule.

Update thermal accumulators on sample-count boundaries independent of host callback size. Keep acoustic-transfer ownership explicit: if an IR already maps terminal voltage to microphone pressure, inserting a second full driver transfer duplicates its response. A residual model must name the linear reference subtracted during fitting.

```
prepare(profile, maximum_internal_rate)
validate_units_curve_domains_and_port_polarities()
for each fixed internal sample:
    apply_sample_stamped_events()
    solve_amplifier_divider_and_driver_ports_together()
    update_electrical_and_mechanical_states()
    accumulate_coil_heat_with_actual_dt()
    if thermal_update_boundary:
        integrate_positive_thermal_network()
    publish_declared_acoustic_source_variables()
    check_finite_states_and_solver_residual()
filter_before_decimation()
```

CHECKS AND DECISIONS

- The same events and elapsed time must produce equivalent states across callback partitions.
- Report solver failures and calibrated-domain excursions; do not silently extrapolate polynomial curves.

MEASUREMENT / SYSTEM IDENTIFICATION

Separate driver distortion from rotation and upstream distortion

Speaker nonlinearities and system identification

ORIGINAL ACQUISITION PLAN: document the fitted parts and begin with stationary rotor angles. Capture calibrated terminal voltage, current and acoustic output on a synchronized clock. Add a displacement reference where needed to resolve mechanical scaling. Preserve installed and isolated-driver experiments as separate datasets; removing a horn or changing a rear volume changes the load being identified.

Fit low-level complex impedance and acoustic response before fitting level dependence. Then use level-controlled tones, two-tone combinations and musical excerpts with declared warmup and rest periods. Reserve frequencies, amplitudes and thermal duty cycles from fitting. A microphone-only residual can include amplifier, crossover, room or propagation errors; it cannot by itself identify a speaker mechanism.

Temperature estimation must separate coil resistance from motional and inductive impedance. [S055-05] Treat candidate parameter sets as uncertain when several combinations fit the same observations. Report sensitivity and residual structure, not merely one optimizer minimum. Regular distortion, intermittent rubbing and loose-part noise should have separate profile identities.

Capture family	Question it resolves
Low-level voltage/current	Complex electrical load and linear reference
Level and two-tone series	Harmonic/intermodulation dependence
Heating and cooling sequence	Thermal memory and resistance change
Mechanical reference	Scale and identifiability of motion parameters
Held-out rotating cabinet	Interaction with subsequent acoustic rendering

CHECKS AND DECISIONS

- Retain excitation files, channel calibration, temperature history and configuration hashes.
- Set capture levels from the inspected hardware and measurement plan, not generic published driver limits.

SOURCES S055-05 KLIPPEL: Voice Coil Temperature measurement guidance

ACCEPTANCE / EXECUTED MATHEMATICAL FIXTURE

Accept mechanisms through independent electrical and acoustic gates

Speaker nonlinearities and system identification

For the conservative magnetic and spring assumptions on page 2, total stored energy is E=0.5L(x)i^2+0.5m*v^2+Us(x). The power identity below supplies an independent implementation check. It does not establish the correct parameter curves for a Leslie driver. In the linear limit, s is Laplace frequency and Zmechanical is total force/velocity impedance; Le and Bl0 are the corresponding linearized values.

Executed original fixture 055-energy-checks.json evaluates 1024 synthetic instantaneous states. Maximum power-balance residual was 1.42e-14 W; deliberately omitting the reluctance term produced errors up to 0.280 W in that synthetic set. A separate 129-frequency linear-limit check preserved positive input resistance. These results verify algebra, not time integration, thermal behavior, aliasing, real-time operation or physical fidelity.

Before promotion, compare retained electrical and microphone captures with measurement repeatability. Keep model-order increases only when held-out improvement justifies their cost. Optional rub/buzz or damaged-diaphragm models require their own specimen evidence. The remaining deliverable is an identified driver profile with validity bounds, not an attractive distortion preset.

Gate	Evidence required
Linear limit	Complex impedance and declared acoustic transfer
Numerics	Convergence, finite states, energy and rate tests
Nonlinear spectrum	Held-out levels, tones and musical material
Thermal history	Held-out heating/cooling and duty cycles
Integrated cabinet	Matched conditions with upstream errors isolated

MODEL EQUATIONS

dE/dt = u*i - Re*i^2 - D(v)*v - Fac*v.

Linear limit: Zdriver(s) = Re + s*Le + Bl0^2/Zmechanical(s).

CHECKS AND DECISIONS

- No numerical fixture is counted as a recorded hardware match.
- Unknown stock-driver parameters remain null until acquisition closes them.

EVIDENCE SCOPE AND ORIGINAL VALIDATION CONTRACT

Define the object that a fidelity result actually describes

Leslie measurement validation and reference suite

The reference is a specified cabinet, electrical interface, operating condition and microphone scene. Record its serial identification, amplifier revision, fitted drivers, crossover, motor assemblies, panel state and modifications before collecting a fitting dataset. A source label alone cannot identify the acoustic object whose response the emulator should predict.

Published experiments provide useful methods, with different scopes. The 2002 study measured a Model 600 horn outside its cabinet; its stationary angles and 2048-sample Golay pairs are an experimental choice, not a 122 sampling prescription. The 2008 paper identifies a 122A measured with a microphone 1.2 m away. Neither is a newly acquired target recording for this suite. [S056-01] [S056-02]

ORIGINAL DESIGN PROPOSAL: write a claim manifest before fitting. Specify admitted input levels, microphone poses, speed commands, temperature/warmup conditions, frequency range and output channels. Separate interpolation inside measured coverage from extrapolation. Add a version whenever a driver, panel, microphone or electrical adapter changes.

Evidence object	Permitted conclusion
Model 600 free-field horn	A published component experiment; no complete122 cabinet acceptance.
Published122A microphone signal	A documented experiment on that specimen and scene.
Target cabinet capture	A possible validation reference after calibration and provenance checks.
Synthetic numerical fixture	A check of stated mathematics; no physical fidelity conclusion.

CHECKS AND DECISIONS

- Reject any reference file whose hardware identity or signal boundary is unknown.
- Do not promote a public synthesis demonstration to measured ground truth.

SOURCES S056-01 Smith, Serafin, Abel and Berners: Doppler Simulation and the Leslie, DAFx 2002 / S056-02 Kronland-Martinet and Voinier: Real-Time Perceptual Simulation of Moving Sources: App

STAGED ACQUISITION

Measure successive boundaries without silently changing the system

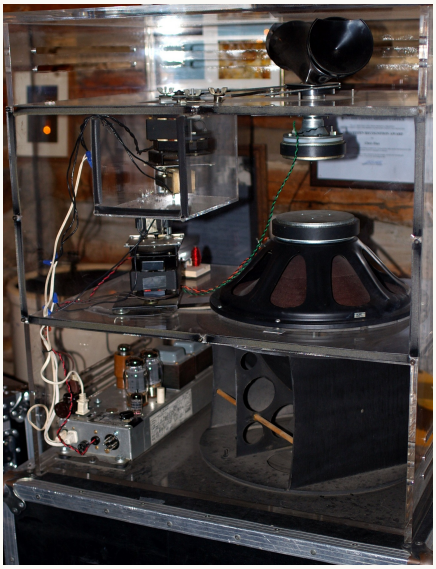
Leslie measurement validation and reference suite

ORIGINAL ACQUISITION PLAN: establish the input interface and amplifier response before fitting the cabinet. Retain differential input voltage, relevant common-mode/control state, source impedance, amplifier output and actual load. Assign electrical fixtures and loading arrangements to a qualified audio technician; an isolated-stage result must carry its changed boundary into the model.

Separate horn and bass contributions with a documented fixture that preserves or explicitly characterizes electrical loading. Disconnecting a branch is not an invisible operation: the common amplifier and divider can respond to the changed load. Verify the complete, reassembled system after component identification. Use near-driver measurements only at their declared receiver positions.

The accompanying acrylic-cabinet photograph reveals mechanisms, but it is not the wooden 122 reference enclosure. Its model and calibrated dimensions are unverified. Do not use an exposed or altered enclosure measurement as the final closed-cabinet acceptance capture. [S056-06]

Boundary	Retain with the result
Electrical input to divider	Voltage/current reference, load, gain and control state.
Isolated radiator branch	Fixture changes, angular origin and receiver geometry.
Complete cabinet to microphones	Both rotor records, panel configuration and complex channel response.
Cabinet in room	Room state and the response already included in each earlier model.



P028 | Leslie Speaker.jpg | Hustvedt | CC BY-SA 3.0 | Cabinet model/revision unverified; contextual photograph only. Source and license

CHECKS AND DECISIONS

- The assembled cabinet must pass after all fixture changes are removed.
- Do not compensate an electrical-load error by tuning microphone directivity.

SOURCES S056-06 Hustvedt: Leslie Speaker.jpg, original photograph of an acrylic cabinet

STATIC TRANSFER AND LEVEL DEPENDENCE

Acquire stationary angular responses as a controlled model boundary

Leslie measurement validation and reference suite

Farina’s exponential-sweep method is formulated for weakly nonlinear, approximately time-invariant systems. That scope does not justify interpreting one continuously rotating sweep as a stationary impulse response. For static identification, use an approved service fixture to hold the required geometry, document residual motion, and separate the capture from normal performance operation. [S056-03]

ORIGINAL DESIGN PROPOSAL: capture low-level complex responses across horn/drum angles and microphone poses, then repeat selected configurations over input level. Refine angular spacing where interpolation fails on withheld intermediate angles; do not assume a published angular increment is sufficient for this cabinet. Include overlapping and crossover bands, where relative phase matters.

Preserve full raw records, excitation, converter loopback, pre-roll and decay. Any gating, delay removal, microphone correction or minimum-phase conversion creates a derived asset with a reversible recipe. Keep absolute gain and arrival time. If independent horn and drum kernels are assumed additive, test that prediction against the complete low-level response.

Acquisition axis	Reason for independent coverage
Angle and microphone pose	Resolve directivity, delay, fixed leakage and enclosure paths.
Input level	Detect departure from the chosen linear approximation.
Repeated placement and run	Estimate repeatability rather than counting repeated FFT bins.
Combined branches	Check relative phase and the adopted factorization.

CHECKS AND DECISIONS

- Stationary-response accuracy is one gate; moving-filter order remains unvalidated.
- Retain unwindowed data so later models can revise propagation ownership.

SOURCES S056-01 Smith, Serafin, Abel and Berners: Doppler Simulation and the Leslie, DAFx 2002 / S056-03 Angelo Farina: Simultaneous Measurement of Impulse Response and Distortion with a Swe

CONTINUOUS ROTATION AND TRANSITIONS

Record independent motion on the same timebase as the audio

Leslie measurement validation and reference suite

ORIGINAL ACQUISITION PLAN: record the applied input waveform, microphone channels, horn angle and drum angle with a shared or calibrated timing relationship. A once-per-revolution pulse gives cycle timing; resolving motion inside that revolution requires additional angular information or an explicitly uncertain interpolation. Store pulse/encoder geometry, direction convention and time-zero photographs.

Capture settled slow and fast operation, both transition directions, short commands that reverse before settling, and multiple initial angle pairs. Replay the measured trajectories into the acoustic renderer first; this isolates radiation/propagation errors from the motor model. Then replace those trajectories with the predicted dynamics and repeat the comparison.

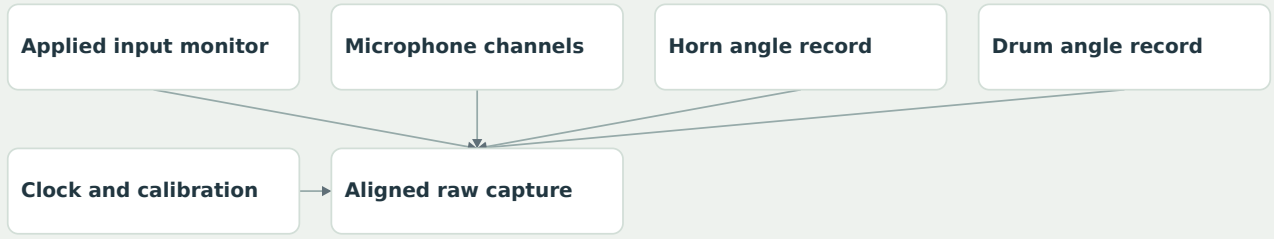
The 122A experiment reports large instantaneous-frequency excursions near amplitude nulls, including a horn-only interference check. Therefore an acoustic phase spike cannot itself establish a mechanical acceleration. Use independent angle evidence and mask unreliable acoustic phase estimates while retaining their excluded coverage. [S056-02]

MODEL EQUATIONS

angle_error(t) = arg(exp(j * (theta_model(t) - theta_reference(t)))) in radians.

speed_error(t) = omega_model(t) - omega_reference(t), in radians/second; estimate reference speed with a declared bandwidth.

One calibrated capture timeline



Original acquisition proposal. Shared physical trajectories are retained; different acoustic paths observe different emission times.

CHECKS AND DECISIONS

- Test continuous motion and transitions; frozen-angle agreement cannot identify a two-time kernel.
- Compare modulation sidebands and timing without fitting a separate pitch LFO to each microphone.

SOURCES S056-02 Kronland-Martinet and Voinier: Real-Time Perceptual Simulation of Moving Sources: App

METRICS, UNCERTAINTY AND FAILURE TRIAGE

Set error budgets only after the measurement can resolve them

Leslie measurement validation and reference suite

NIST distinguishes measurement error from uncertainty and includes covariance when combining uncertainty contributions. Apply that distinction to this project: repeated recordings estimate only part of the uncertainty. Common gain calibration, microphone position and clock errors can correlate many frequencies and channels; they are not independent noise samples. [S056-04]

ORIGINAL ACCEPTANCE PROPOSAL: use a pilot dataset to estimate the measurement floor and select meaningful error budgets, then freeze those budgets before evaluating held-out captures. Where uncertainty is too large to resolve the budget, report inconclusive and improve the acquisition. Increasing uncertainty must not make a model easier to certify.

Compare complex static transfer, calibrated level, arrival time, ramp trajectory, modulation sidebands, stereo coherence and mono sum. Apply only documented acquisition corrections. Arbitrary per-window alignment or dynamic time warping can erase the delay error under test. Report excluded low-SNR/null regions and the tested frequency/angle coverage alongside every score.

Failure pattern	First boundary to investigate
Static response wrong	Identity, loading, gain, driver/kernel or microphone calibration.
Static right; moving wrong	Emission time, filter ordering, interpolation and trajectory history.
Transition only wrong	Command timing, motor dynamics and angle initialization.
Stereo right; mono wrong	Relative delay, polarity and correlated path summation.

MODEL EQUATIONS

E_H = sqrt(sum(w * abs(H_model - H_reference)^2) / sum(w * abs(H_reference)^2)); w is a predeclared nonnegative reliability weight.

CHECKS AND DECISIONS

- A magnitude-only fit cannot certify phase, transient timing or mono compatibility.
- Retain per-condition failures; an average score must not hide an unsupported operating region.

SOURCES S056-04 NIST Technical Note 1297: Section 5, Combined Standard Uncertainty

RELEASE EVIDENCE AND HANDOFF

Freeze the model before the final musical and listening tests

Leslie measurement validation and reference suite

ORIGINAL DATA POLICY: split by independent capture session and physical configuration, not adjacent windows from one recording. Reserve unseen musical phrases, input levels, microphone poses and speed transitions. Store dataset membership before optimization. If held-out failures guide a revision, that revision needs a fresh final evaluation set.

For a sufficiently close candidate, ITU-R BS.1116-3 provides a small-impairment listening method with a hidden reference and double-blind assignment. Adaptation to an emulator requires a stated protocol, suitable listeners, controlled reproduction and a retained statistical analysis; naming the recommendation does not establish compliance. Keep absolute-level validation separate from any documented listening-level adjustment. [S056-05]

Failure to detect a difference is not proof of identity. Report the conditions, sample size, uncertainty and predeclared equivalence question. The handoff requires raw captures, calibration, photos, angle records, processing recipes, profile/build hashes, metric reports and listening results. None of these target-cabinet measurements or listening trials has been performed for this chapter.

```
verify_capture_identity_and_calibration()
freeze_profile_build_and_dataset_split()
run_numerical_regressions()
compare_static_and_measured_trajectory_cases()
compare_predicted_trajectory_and_held_out_music()
if uncertainty_cannot_resolve_budget: status = INCONCLUSIVE
else: apply_predeclared_gates_and_blinded_study()
archive_failures_coverage_and_claim_scope()
```

CHECKS AND DECISIONS

- No measured-fidelity gate is passed by the document or synthetic fixtures alone.
- Release claims name the specimen, signal chain, tested conditions and remaining unknowns.

SOURCES S056-05 ITU-R BS.1116-3: Methods for the subjective assessment of small impairments in audio

ORIGINAL ACQUISITION PROPOSAL

Authorize a defined campaign before acquiring a reference

Physical acquisition campaign and reference capture plan

The first deliverable is a signed-off acquisition charter naming the actual organ, AO-28 revision, mains configuration, rotary cabinet and fitted drivers. Photograph serial plates, component markings, wiring changes, cabinet panels and control states before maintenance. Record ownership and permission to photograph, measure, record and redistribute the resulting assets. A public performance recording is not an interchangeable calibration dataset.

Assign internal electrical access to a qualified service technician using a reviewed measurement interface. The acquisition engineer owns calibration and synchronized recording; the player supplies repeatable gestures; the data steward protects source files and rights records. The AI agent prepares scripts and validates data, but cannot infer that an inaccessible internal node was measured. No improvised live-mains probing procedure is specified here.

Start with an external-output pilot and a documented condition assessment. Retain the instrument’s existing voicing and wear unless a separate restoration decision changes the target. Record warmup trajectories until chosen observables stabilize within a declared repeatability band; an arbitrary elapsed time is insufficient. Any service intervention creates a new configuration record and prompts affected recaptures.

Campaign gate	Required deliverable
Identity and access	Specimen manifest, photographs, permissions and technician-approved scope
Chain qualification	Calibration, loading, bandwidth, timing and clipping report
Subsystem acquisition	Generator, console, circuits and cabinet evidence at declared boundaries
Integration and release	Held-out musical takes, repeatability report and immutable dataset release

CHECKS AND DECISIONS

- Leave inaccessible parameters explicitly unmeasured; preserve the acquisition task.
- A configuration change must not silently overwrite an earlier reference.

DOCUMENTED TRAP / MEASUREMENT CONTRACT

Name the electrical boundary on every recording

Physical acquisition campaign and reference capture plan

Factory Table V’s historical meter connection is across the preamplifier G–G terminals, not directly across a pickup coil. The stated1000 ohms/volt sensitivity needs the selected meter range to determine loading. Its service adjustment criterion is not an emulation-error tolerance. The companion014-factory-service-readings.json preserves this distinction. [S057-01]

Probes introduce resistance and capacitance that can alter the circuit being observed. [S057-02] ORIGINAL PROPOSAL: give every acquisition port a node-pair identity, signal direction, reference polarity, receiving impedance and interface transfer function. Characterize the added load against the expected source impedance; calling an input high impedance is not a quantitative loading budget.

A loaded terminal waveform may already contain pickup harmonics, filtering and coupling. Keep that waveform’s ownership separate from a raw-source model so the implementation does not apply those mechanisms twice. Where separation is not identifiable, publish the effective loaded model and its supported receiving load rather than invented internal parameters.

Boundary class	What the record must identify
Pickup or generator terminal	Exact node, fitted filters, connected network and acquisition loading
Manual bus / scanner return	Routing state, receiving stage and scanner phase where applicable
AO-28 G–G	Output-transformer loading, expression and all relevant controls
Leslie electrical stage	Input/output node pair, amplifier load and driver connection
Microphone pressure path	Pose, orientation, sensitivity, cabinet panels and room

CHECKS AND DECISIONS

- Reject a capture labeled only Hammond output.
- Never substitute the historical G–G table for per-wheel oscillator amplitudes.

SOURCES S057-01 Hammond Organ Service Manual: alignment measurement boundary and Table V / S057-02 Tektronix: How Oscilloscope Probes Affect Your Measurement

ORIGINAL METROLOGY PLAN

Calibrate gain, bandwidth and time as one acquisition chain

Physical acquisition campaign and reference capture plan

Characterize each complete channel: sensor or interface, cables, gain stage, converter, filters and recorder. Retain volts-per-digital-unit or pascals-per-digital-unit calibration, complex response over the analysis band, clipping margin, noise and channel skew. Disable undocumented automatic gain or processing. A floating-point file cannot repair saturation that occurred before conversion.

NTi recommends acoustic sensitivity checks before and after a session. [S057-06] Extend that practice here with retained calibrator readings and environmental conditions; a single-frequency check does not establish the microphone’s entire polar or phase response. Choose sampling rate from measurement bandwidth and anti-alias rejection. A 96 kHz/24-bit audible-path pilot is a proposed starting configuration, not a hardware specification or guaranteed precision.

Record excitation loopback, electrical outputs and microphones on a shared clock where possible; align separately sampled position or tachometer channels with retained synchronization markers and an uncertainty estimate. Loopback can reveal relative timing without proving absolute clock accuracy. For an illustrative 10 ppm mismatch over 60 s, timing disagreement is 600 microseconds, equivalent to 95.04 degrees at 440 Hz. These are calculated examples, not instrument observations.

MODEL EQUATIONS

delta_t approximately epsilon * T; epsilon is relative clock mismatch, T is duration in seconds.

abs(delta_phi) approximately 2*pi*f*abs(epsilon)*T; phase is radians and f is hertz.

CHECKS AND DECISIONS

- Preserve the raw clock record and every alignment/resampling transform.
- A failed end-of-session calibration quarantines the affected interval for review.

SOURCES S057-06 NTi Audio: Room Acoustics Measurements

ORIGINAL ORGAN CAPTURE MATRIX

Acquire organ mechanisms before the musical mixtures

Physical acquisition campaign and reference capture plan

Screen every key and drawbar route on both manuals, then acquire repeated detailed cases across the range, foldback boundaries and anomalous keys. Track a continuous source-phase reference where feasible; isolated notes recorded at unrelated phases cannot simply be summed. Record pedal routes separately and preserve the source assignments in chapters 015 and 022.

The 2017 study relates contact behavior to gesture on a particular 1967 C-3 and deliberately excludes release transients from its listening stimuli. [S057-04] This campaign must acquire both attack and release. Capture key-position trajectories and, where authorized access permits, contact observations alongside audio. Repeat pressed and struck gestures over a useful range; define classes from actual trajectories instead of assuming one MIDI velocity reproduces them.

Use short pilots to determine the repetition count and recording duration needed for each observable. Retain quiet intervals and complete recovery tails. Record control positions photographically or electronically; a handwritten preset name alone does not establish drawbar or switch state.

Mechanism	Required sequence family
Shared generator	All 91 source identities; sustained spectrum, frequency, phase and coupling evidence
Contacts and routing	Single routes, attacks/releases, repeated gestures and coherent collisions
Percussion	Legato, all-up gaps, recharge histories and held-key control changes
Scanner	Six modes, Swell/Great combinations, moving and stationary electrical references
Expression / AO-28	Repeatable pedal positions, level sweeps, chords, load changes and recovery

SOURCES S057-04 Giulio Moro, Andrew P. McPherson and Mark B. Sandler: Dynamic temporal behaviour of t

ORIGINAL LESLIE AND INTEGRATION PLAN

Separate stationary identification from moving-system evidence

Physical acquisition campaign and reference capture plan

Farina’s exponential-sweep method separates linear and harmonic responses under stated approximately time-invariant conditions. [S057-05] Apply an appropriate stationary experiment to a qualified electrical boundary or safely indexed acoustic configuration. A normally deconvolved sweep taken while the rotor changes angle is not automatically a stationary impulse response. Preserve the excitation, inverse-filter recipe, windows and level.

For dynamic cabinet tests, record independent horn and bass angle observations, speed commands and synchronized acoustic/electrical channels. Include both speed directions, interrupted ramps and repeated cycles. Record enough slow revolutions and transition history to separate periodic structure from noise and drift; choose duration from pilot evidence. Do not estimate mechanical acceleration from instantaneous-frequency spikes near acoustic cancellation nulls.

Acquire isolated horn and bass evidence where the approved setup permits, then the installed cabinet and combined organ/cabinet path. Each changed panel, microphone pose, driver connection or room is a new configuration. Optional 122RV wet-speaker radiation remains a separate branch. Finally record player performances that cross registrations, percussion histories, expression trajectories and speed commands, retaining the simultaneous direct-organ reference when available.

Acquisition family	Use in the model
Stationary, several levels	Electrical response, angular kernels and nonlinear discrimination
Continuous rotation with angle	Emission-time and time-varying-filter validation
Independent ramps and commands	Mechanical-state fitting without acoustic inference shortcuts
Unseen integrated performances	Generalization and later blinded listening

SOURCES S057-05 Angelo Farina: Simultaneous Measurement of Impulse Response and Distortion with a Swe

ORIGINAL ACCEPTANCE AND HANDOFF

Release a dataset whose limitations are inspectable

Physical acquisition campaign and reference capture plan

NIST’s uncertainty framework includes correlated inputs and the complete measurement process. [S057-03] Keep repeat-to-repeat spread separate from shared gain, clock, geometry and calibration errors. Repeating a biased chain does not remove its common error. Use the resulting uncertainty budget to choose fidelity tolerances; the factory service threshold does not supply those tolerances.

Freeze training, validation and final-test assignments by physical take or session before fitting. Adjacent windows from the same recording are correlated evidence, not independent held-out performances. Reserve unseen control histories and microphone poses where the intended model claims those capabilities. Keep a clean final listening set inaccessible to iterative parameter tuning.

Deliver untouched recordings, hashes, channel maps, specimen and chain versions, source permissions, event/angle traces, calibration records, analysis recipes and a capture ledger. Failed or ambiguous runs remain quarantined with reasons. Chapters 058–061 own schema, fitting, regression and listening details. This chapter authorizes no claim that the campaign has occurred: all physical acquisitions remain pending.

```
for proposed_run in frozen_capture_plan:
    require identity, permissions, approved_boundary, valid_calibration
    capture raw_channels, controls, synchronization, conditions
    hash raw_files before analysis
    verify clipping, dropout, timing, calibration and metadata
    quarantine failed runs with explicit reasons
    release accepted runs with immutable split and provenance
```

CHECKS AND DECISIONS

- No accepted recording lacks its boundary, load, calibration and raw-file hash.
- No model-fidelity gate passes on synthetic examples alone.
- The final report names unmeasured parameters and unsupported configurations.

SOURCES S057-03 NIST TN1297, Appendix A: Law of Propagation of Uncertainty

ORIGINAL DATA CONTRACT / HARDWARE BOUNDARIES

Preserve what each piece of evidence actually describes

Dataset schema and provenance infrastructure

This chapter supplies two original proposed contracts: capture_manifest.schema.json describes an acquisition and its evidence; calibration_profile.schema.json describes a versioned instrument model assembled from evidence. Neither is a Hammond standard. Keep the original bytes, documentary interpretation, fitted parameters and compiled runtime representation separately identifiable. A valid document can still contain an incorrect physical claim; structural validity is only the first gate. JSON Schema supplies the assertion vocabulary used here. [S058-01]

The shipped capture_manifest.example.json is explicitly fictional planning data. No instrument, session, recording, calibration or permission has been acquired. Its nonexistent timestamps, sample counts, measured values, filenames and hashes are null. Unknown never means numerical zero. Conditional rules prevent that example from claiming recorded status or an acquisition timestamp. [S058-02]

Boundary identity is essential to useful calibration. The factory Table V procedure measures across the G–G terminals through the specified organ setup. Preserve 014-factory-service-readings.json as documentary service evidence; do not copy its readings into raw generator amplitudes. [S058-07]

Boundary token	Quantity	Evidence needed before use
pickup_unloaded	Pickup voltage, V	Pickup identity, probe loading, reference convention
generator_loaded_terminal	Loaded source voltage, V	Terminal mapping, connected network and load
preamp_GG	Differential output voltage, V	Both G terminals, controls, load, complete signal path
acoustic_microphone	Acoustic pressure, Pa	Microphone chain, pose, clock, room and calibration

CHECKS AND DECISIONS

- Refuse a G–G value in the generator amplitude_boundary field. Preserve the actual reference specimen separately from its generic model name.

SOURCES S058-01 JSON Schema Draft 2020-12 validation vocabulary / S058-02 JSON Schema: Conditional schema validation / S058-07 Hammond factory service manual: alignment procedure and Table V

ORIGINAL ACQUISITION CONTRACT

Capture calibration, synchronization and conditions together

Dataset schema and provenance infrastructure

Assign every file channel a persistent channel_id and a distinct zero-based file index. Store the header sample rate separately from the qualified sample_clock_rate_hz: one describes file interpretation, while the other can describe an independently calibrated clock. For separate clocks, preserve synchronization evidence and any resampling transform. A shared label alone is not proof of phase coherence. Unresolved offset or drift must remain visible to the fitting agent.

Every modeled scalar uses value, unit, evidence_class, provenance_refs, uncertainty and unknown_reason. Known values require evidence references; unknown values require null and an explanation. An uncertainty interval and a standard uncertainty are different representations. Their magnitude uses the same unit as the quantity. The semantic checker rejects reversed intervals and nominal values outside the stated interval.

The ADC scale states physical units per full-scale sample. Electrical channels use V/FS; calibrated microphone channels use Pa/FS. A scalar calibration does not describe frequency response, impedance or latency: reference the chain calibration asset for those dependencies. The contract permits uncalibrated recordings to be catalogued, but this does not establish their suitability for physical parameter fitting.

Record group	Fields that must travel with the audio
Actual hardware	Organ, AO-28 and cabinet identities; revisions; identity evidence
Clock and timing	Master clock; per-channel clock and offset; synchronization evidence
Signal chain	Boundary; port reference; load; ADC scale; calibration asset
Conditions	Temperature; mains; warmup; control-state and rotor-command references
Microphone geometry	Coordinate system; position; unit front axis; model and evidence

CHECKS AND DECISIONS

• Check channel count against the actual header and reject duplicate channel IDs. Treat absent calibration values as unresolved, not unity gain.

ORIGINAL PROFILE CONTRACT / HANDOFF TO CHAPTER 063

A model profile with 91 identities and explicit ownership

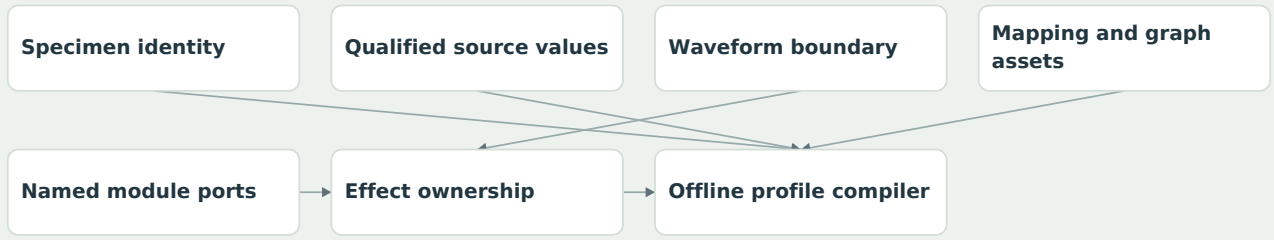
Dataset schema and provenance infrastructure

The generator.sources object contains exactly the string keys 1 through 91. Its identity convention is the factory frequency numbering used by the mapping chapters, not MIDI pitch or physical terminal position. Each source carries qualified frequency, electrical periods per mechanical turn, phase offset and amplitude. All may remain unknown in a proposed draft. A source also declares whether its waveform is unavailable, an asset, or an analytic module.

Declare waveform_boundary before selecting an algorithm: raw flux, unloaded pickup, loaded terminal and normalized waveform have different meanings. effects_already_in_waveform records embedded processing; effect_ownership identifies the module responsible between named ports. The provided checker rejects repeated ownership for the same effect and port pair. A later compiler must also compare waveform effects against the actual module graph; a string registry cannot prove that a filter or propagation delay was applied only once.

document_refs separates mapping tables, scanner graph, scanner routing, AO-28 component registry, AO-28 connectivity netlist and Leslie crossover graph. ao28-component-values.json is a component registry, not a completed AO-28 netlist; leave the netlist reference null until one exists. Proposed, specimen-calibrated and superseded are explicit profile states. A calibrated claim additionally requires actual specimen IDs, evidence, a fit report and held-out manifests; reviewers must still verify those claims.

Profile evidence and effect ownership



Original contract diagram. The compiler is a proposed integration step, not a supplied DSP engine or manufacturer circuit.

CHECKS AND DECISIONS

• Require the full key set 1…91; reject 01 and 92. Preserve null for unresolved circuit connectivity and prohibit silent conversion of unknown parameters to zero.

ORIGINAL IMPLEMENTATION SEQUENCE

Import deterministically, then compile outside the audio callback

Dataset schema and provenance infrastructure

First parse strictly, then validate structure, resolve references, inspect real assets and evaluate model semantics. RFC 8259 excludes NaN and Infinity and explains duplicate-name interoperability problems. This importer also rejects duplicate keys and numbers that overflow its finite runtime representation. Those are deliberate project gates, not a claim that every valid JSON number fits every language. [S058-03]

Hash audio, photographs and circuit-graph files from their actual bytes; compare both digest and size with the manifest. A 64-character hexadecimal string only has the right shape. The supplied checker additionally compares WAV channel count, sample rate and frame count. Preserve originals when deriving crops, normalized tables or resampled audio, and record each parent, operation, tool version and parameter set.

If semantic JSON fingerprints are needed, use a conforming JCS implementation and record that algorithm. RFC 8785 defines more than sorted keys; ordinary json.dumps output is not asserted to be JCS. This package has not implemented canonical serialization. [S058-04] For chapter 063, compile approved data into immutable arrays and coefficients off the audio thread. The future engine should publish a validated snapshot atomically and retain enough provenance to reproduce it.

```
# Proposed importer sequence; compilation is not implemented here.
doc = strict_parse(bytes)
require_supported_schema_version(doc)
validate_structure_and_formats(doc)
resolve_evidence_and_asset_references(doc)
verify_actual_asset_bytes_and_headers(doc)
check_units_boundaries_and_effect_ownership(doc)
check_dataset_partitions_and_declared_permissions(doc)
report_unresolved_model_inputs(doc)
compiled = compile_approved_profile_off_thread(doc)
publish_immutable_snapshot(compiled)
```

CHECKS AND DECISIONS

• Version schema migrations explicitly and retain the source document. Never perform filesystem access, JSON parsing or graph compilation inside the audio callback.

SOURCES S058-03 RFC 8259: The JavaScript Object Notation Data Interchange Format / S058-04 RFC 8785: JSON Canonicalization Scheme

ORIGINAL DATASET GOVERNANCE PROPOSAL

Preserve lineage, permissions and genuinely independent holdouts

Dataset schema and provenance infrastructure

Freeze dataset membership before parameter fitting. Group takes by the physical acquisition relationships that can leak information: the same performance, synchronized channels, rotor run, repeated crop or common parent recording. Every derivative inherits that acquisition group. Choose whether the evaluation tests another take of one specimen or generalization to another specimen, and encode that purpose in grouping_rationale. A final holdout cannot be quietly recycled for tuning.

The supplied checks detect manifest overlap between fitting partitions and a group appearing under different split roles. A complete corpus loader must additionally traverse parent manifests, detect cycles and verify inheritance; that global lineage traversal is not supplied. Keep calibration-reference measurements distinguishable from the audio used to optimize a sound model. Record algorithm version, code revision, random seed and the objective in the fitting record.

Permission evidence travels with the asset independently of its URL. Store, analyze, derive a model and redistribute are separate declared permissions. A publicly viewable schematic or photograph does not fill those fields. Large recordings can use Git LFS, whose repository records point to separately stored objects. [S058-05] A reproducible archive therefore needs the actual permitted objects, a verified inventory and a restore check, not merely pointer files.

Asset kind	Required lineage and interpretation
Raw audio or photographs	Original bytes; actual specimen; acquisition conditions; permission evidence
Crop, resample or normalized table	Parent asset; exact transform; software version; retained original
Circuit component registry	Source sheet and revision; value confidence; no implied connectivity
Fit report and profile	Frozen partitions; objective; qualified parameters; actual reference identity

CHECKS AND DECISIONS

• Reject a shared acquisition group spanning training and final holdout. Verify derive_model and redistribute permissions separately before the relevant release.

SOURCES S058-05 GitHub Docs: About Git Large File Storage

VALIDATION RESULTS / LIMITED CLAIM

Executed checks and the evidence still required

Dataset schema and provenance infrastructure

The delivered validator executed 39 checks successfully: six accepted fixtures or schema checks and 33 intentional rejections. Results are recorded in 058-schema-checks.json. Both schemas passed Draft202012Validator.check_schema. The fictional example and an all-unknown in-memory profile passed their respective contracts; no calibrated profile was fabricated or shipped.

An initial negative timestamp test exposed missing optional format dependencies. The official python-jsonschema documentation distinguishes enabling FormatChecker from installing the dependencies used by particular checks. [S058-06] After installing rfc3339-validator 0.1.4 and rfc3986-validator 0.1.1 alongside jsonschema 4.26.0, malformed date-time and URI fixtures were rejected. The validator now checks that both required format handlers exist and fails immediately if they do not.

A temporary generated silence WAV tested real byte hashing and header agreement; it was deleted afterward and is explicitly synthetic. These checks establish metadata behavior only. No B-3 or Leslie capture, acoustic fit, C++ profile compiler, plugin integration or perceptual comparison was performed. Chapter 063 must consume this proposal through a reviewed importer, and chapter 059 must supply the eventual physical acceptance evidence.

Executed category	Observed result
Structure and identities	Both meta-schemas; exact source keys; required state evidence
Units and ownership	Rejected G–G source amplitudes, wrong phase units and duplicate ownership
Provenance and partitions	Rejected unresolved references and overlapping manifest/group assignments
Actual synthetic file	Accepted correct digest/header; rejected wrong digest and frame count
Not established	Physical truth, full lineage graph, permission validity, sound fidelity

```
# Run from a disposable copy of the full suite.
# These checks regenerate reports; preserve original evidence.
python -m pip install -r reference_code/requirements_data_contracts.txt
python reference_code/validate_data_contracts.py
```

CHECKS AND DECISIONS

• Keep unknown calibration and actual-reference fields unresolved until acquired. Re-run these checks when changing either schema; add physical tests only with genuine captured evidence.

SOURCES S058-06 python-jsonschema 4.26.0: Schema Validation

EVIDENCE / ORIGINAL DERIVATION

Identify what the observations can determine

Calibration objective and parameter fitting

Parameter identification requires a declared specimen, circuit revision, measurement port and operating domain. Structural ambiguity persists even with ideal noiseless observations under the declared experiment; practical ambiguity remains when available data cannot distinguish parameters adequately. Raue and colleagues distinguish these problems in their abstract. Their biological examples provide methodology, not Hammond parameter values. [S059-02]

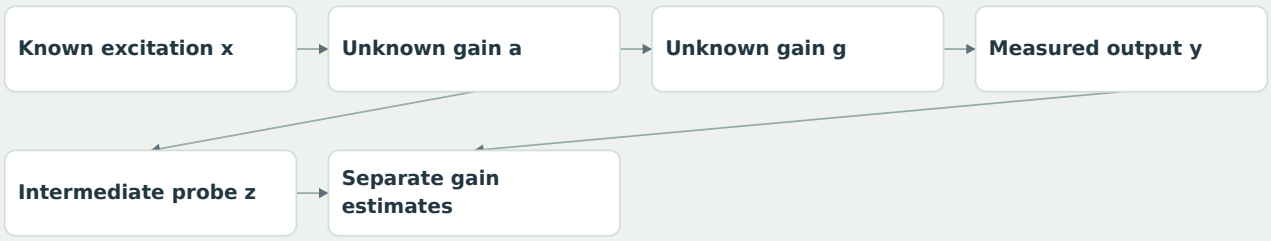
Original derivation: for a known excitation x and two unknown positive gains a and g, output y = agx identifies their product. Replacing a by ka and g by g/k changes neither output nor any output-only fit. A calibrated intermediate observation z = ax breaks this particular ambiguity. A synthetic linear-algebra check gave Jacobian ranks one and two respectively; this establishes only the toy example.

Factory Table V belongs to a service procedure with the meter across preamplifier G–G. Its historical readings are not raw pickup amplitudes. Meter range and detector remain unresolved; its 30% adjustment criterion is not an emulation acceptance tolerance. Do not populate oscillator gains from that table. [S059-01]

MODEL EQUATIONS

y = a*g*x; z = a*x; (a,g) -> (k*a,g/k), k > 0

An additional port breaks a gain ambiguity



Original toy model. The intermediate probe is assumed calibrated and nonperturbing; actual probe loading must enter the circuit model.

CHECKS AND DECISIONS

• Every parameter declares units, boundary, evidence, bounds and fit status. Unobservable component values remain unknown or become explicitly effective parameters.

SOURCES S059-01 Hammond Organ Service Manual: Section IV, Alignment Procedures / S059-02 Raue et al. (2009), Structural and practical identifiability analysis of partially ob

ORIGINAL IMPLEMENTATION PLAN

Fit from identified boundaries outward

Calibration objective and parameter fitting

Use chapter 058 capture IDs and chapter 057 acquisition conditions. First establish wiring, polarity, source assignment and instrument calibration. Treat alternative revisions as discrete hypotheses; an optimizer must not interpolate between incompatible circuits. The following order is proposed, with upstream acceptance required before downstream adjustment.

Characterize connected loading at each electrical boundary. A removed branch or substituted load requires an explicit model. Fit valve stages, divider and drivers jointly where port impedances couple them. Freeze accepted upstream estimates initially, retain their uncertainty, and reopen them only when simultaneous boundary observations support the change. Cabinet equalization must never repair a wrong foldback map or scanner polarity.

Order / parameter family	Required observations	Stage stopping gate
1. Frequency, gain, phase, leakage	Generator-port references; isolated sources; repeated notes	Assignment and phase convention verified
2. Drawbars and linear networks	Low-level complex responses; varied registrations and expression	Common port calibration explains level ratios
3. Contact, percussion, scanner states	Attack/release gestures, repeated triggering, modulation references	Held-out state histories predicted
4. AO-28 and Leslie valve stages	Calibrated input/output pairs at several levels and loads	Gain, distortion and recovery agree together
5. Divider and driver parameters	Voltage/current, impedance, acoustic level and temperature	Electrical and acoustic observations agree
6. Rotor, cabinet and room	Tachometry; static angles; moving captures; surveyed microphones	Motion, directivity and room contributions separated

CHECKS AND DECISIONS

• A stage lacking its required observations stays provisional. End-to-end sound alone cannot certify its internal parameters.

ORIGINAL MODEL / NUMERICAL CONTRACT

Use an objective with visible physical meaning

Calibration objective and parameter fitting

Proposed objective: group calibrated feature errors by capture and boundary. Let p be physical parameters, n bounded acquisition nuisance variables, f measured features, m predictions and L a covariance factor. NIST includes covariance and the measurement process in uncertainty propagation. Preserve shared gain and clock uncertainty instead of treating every sample as independent evidence. [S059-04]

Here e is dimensionless; D scales parameter deviations from prior p0, and lambda weights regularization. Report data error and prior penalty separately. Covariance whitening is used only when supported; otherwise declare heuristic feature weights without likelihood or confidence claims. Correlated time and frequency summaries must not masquerade as independent measurements. Declare volts, amperes, pascals, seconds, hertz and radians before whitening; tie spectral floors to the measured noise floor.

SciPy least_squares supports bounded local optimization and requires real residuals. Stack real and imaginary transfer errors. Scale parameters; use logarithms for strictly positive quantities and separate hypotheses for absent components. Robust losses need an explicit outlier model: intentional click, clipping and recovery are target behavior. [S059-03]

Feature group	Proposed role
Complex transfer; volts and phase	Constrain linear response, polarity and delay
Short/long window spectra	Resolve transients and sustained harmonic structure
Envelopes and event times	Constrain contact, recharge and acceleration behavior
Level, temperature, modulation statistics	Expose saturation, compression and state dependence

MODEL EQUATIONS

e = solve(L, f - m(p,n)); L*transpose(L) = Sigma

J = 0.5*sum(rho(e_i^2)) + lambda*norm(D*(p-p0))^2

CHECKS AND DECISIONS

• Balance registrations, levels and event classes explicitly; a long sustained note must not dominate all short transitions.

SOURCES S059-03 SciPy least_squares official API documentation / S059-04 NIST Technical Note 1297, Appendix A: Law of Propagation of Uncertainty

ORIGINAL MEASUREMENT POLICY

Align acquisition without erasing disagreement

Calibration objective and parameter fitting

Estimate acquisition offset and clock-rate correction from independent synchronization evidence. Archive raw recordings, transformations and uncertainty. A shared capture correction preserves interchannel timing; synchronization alone does not establish absolute clock accuracy. Include acquisition quantities in the measurement model, following the NIST process scope. [S059-04]

Constrain unknown initial generator or rotor phase as explicit shared state, using reference channels where available. Keep physical group delay and propagation in the renderer. Independent phase adjustment for every partial, note or rotor revolution can conceal oscillator, coupling or motion errors. Report fits both before and after permitted nuisance correction.

When contact trajectories or initial phase are unobserved, compare appropriate conditional distributions and envelopes alongside spectra. A pointwise null then has a narrower interpretation. Repeat gestures and sessions to estimate within-condition variability; keep between-session calibration uncertainty separate. Resample complete captures or sessions for uncertainty studies, not overlapping windows assumed independent.

Permitted correction, with evidence	Disallowed compensation
One calibrated channel-group offset	Independent note-by-note time warping
Clock correction with recorded uncertainty	Detuning the source to hide recorder drift
Shared calibrated gain nuisance	Normalizing each registration separately
Constrained initial rotor state	Arbitrary phase reset every revolution
Measured microphone response correction	Unreported equalization of model residuals

CHECKS AND DECISIONS

• Uncertainty reports identify common calibration errors that repetition cannot remove. Unresolved phase histories are disclosed in the comparison scope.

SOURCES S059-04 NIST Technical Note 1297, Appendix A: Law of Propagation of Uncertainty

ORIGINAL OPTIMIZATION AND DIAGNOSIS

Separate convergence from parameter evidence

Calibration objective and parameter fitting

Run fitting offline with fixed capture manifests, controlled seeds and recorded warmup histories. Compare multiple physically feasible initializations. Check derivative calculations and scaled Jacobian columns; nearly collinear columns flag local tradeoffs, not a proof about global identifiability. Repeat selected evaluations at a stricter integration tolerance or higher internal sample rate so numerical error and aliasing cannot become fitted physics.

Profile a suspicious parameter by fixing it across a range and reoptimizing the others. Raue et al. motivate profile likelihood for identifiability analysis. Here profile the data term separately from priors: a bound or strong prior can create apparent certainty unsupported by measurements. A flat profile calls for a new observation, reduced parameterization or a declared unresolved range. [S059-02]

SciPy termination criteria monitor objective change, parameter steps or gradients. Solver success therefore establishes numerical termination, not instrument fidelity. Keep solver status, active bounds, profile ranges and alternate solutions in the calibration report. Proposed stage gates require stable predictions under numerical refinement and no material ambiguity in parameters needed for the claimed operating domain. [S059-03]

MODEL EQUATIONS

P(q) = min over p,n subject to p_j=q of J_data(p,n)

```
freeze(split_ids, feature_spec, nuisance_policy)
for stage in boundary_order:
    assert required_observations(stage)
    fits = solve_from_feasible_starts(stage)
    audit_numerics_and_profiles(fits)
    inspect_first_failing_boundary(fits)
    if stage_gate_fails: stop_and_record_reason()
    freeze_stage_with_uncertainty(fits)
select_using_validation_only()
lock(model, parameters, runtime_profile)
run_final_test_once()
```

CHECKS AND DECISIONS

• Gate thresholds are preregistered engineering proposals based on repeatability and intended use. None is a measured fidelity result in this atlas.

SOURCES S059-02 Raue et al. (2009), Structural and practical identifiability analysis of partially ob / S059-03 SciPy least_squares official API documentation

ORIGINAL ACCEPTANCE PROTOCOL

Test frozen predictions and locate failures

Calibration objective and parameter fitting

Ljung emphasizes fresh-data comparison and residual analysis. Proposed split: training estimates parameters; validation selects structure, weights and regularization; an untouched final test assesses the locked choice. Split by complete take, session and gesture family. Neighboring windows from one recording must not cross partitions. [S059-05]

Include held-out registrations, chords, levels, percussion histories, temperatures and rotor transitions within the claimed domain. Train across levels and registration families rather than allowing independent gains for each. Novel operating conditions remain extrapolations. If final-test results drive revisions, retire that test set and obtain a fresh final assessment.

Report residuals against frequency, source identity, level, angle and temperature; unconditional whiteness is not a universal organ-noise requirement. Archive capture hashes, calibrated transformations, feature definitions, priors, parameter correlations, bounds, code version and runtime profile. Acceptance requires each declared boundary gate and musical validation; missing evidence produces an inconclusive verdict. No physical fit or fidelity pass was performed for this chapter.

Residual pattern	Investigate before adding parameters
Wrong harmonics or note-dependent level	Source map, drawbar routing, upstream loading
Correct magnitude, wrong phase	Polarity, clock, circuit delay, propagation
Errors increasing with level	Valve bias, load, driver nonlinearity, aliasing
Rotor-angle-dependent errors	Trajectory, directivity, moving-filter order
Temperature or history dependence	Thermal state, supply recovery, recharge
Training improves; validation worsens	Leakage between splits, overfit, wrong structure

CHECKS AND DECISIONS

• Release the fitted profile only with its supported domain, uncertainty and unresolved parameters. Test-set failure cannot be repaired by an unreported final EQ.

SOURCES S059-05 Lennart Ljung, Identification of Linear and Nonlinear Dynamical Systems, Theme 4: Pra

EVIDENCE / ORIGINAL TEST ARCHITECTURE

Separate documentary truth from audible fidelity

Regression and adversarial musical tests

The executable delivered here checks data, not a synthesizer. It reads the real package schema, compares all 549 manual assignments with the repeat-fold rule, checks 144 independently read factory cells, and compares every one of 91 service readings with a second visual transcription. All passed. No instrument was measured, and no production audio engine was exercised. The source is the factory chart and historical test procedure, not the similarity of two software implementations. [S060-01]

ORIGINAL DESIGN PROPOSAL: give each future test an evidence level, observable boundary and independent oracle. A successful parser cannot establish contact timing; an exact circuit identity cannot establish a particular cabinet sound. Keep mathematical reference results separate from host integration and measured comparisons. Existing scanner AC, crossover, retarded-path, rotor-state and WDF-algebra utilities are limited reference models; inspect each stated boundary before using its output as a target.

Store one canonical stimulus as absolute event times, instrument variant, initial state and parameter trajectories. Produce intermediate traces at named electrical or acoustic boundaries. When the final stereo signal changes, these traces identify the first divergent stage; a visually convincing interface must never serve as the acceptance oracle.

Evidence level	Oracle	Present status
Documentary	Factory numbers plus independent visual fixtures	Executed; pass
Mathematical	Declared topology, analytic identity or independent solver	Existing limited utilities
Audio engine	Deterministic render and state traces	Proposed; not implemented here
Physical fidelity	Calibrated, held-out specimen captures	No project capture dataset

CHECKS AND DECISIONS

• A report claiming hardware agreement must name the capture dataset, specimen and measurement boundary.

SOURCES S060-01 Hammond factory service manual, Table III and Section IV alignment procedure

EXECUTED DOCUMENTARY CHECKS

Pin the map and the service-meter boundary

Regression and adversarial musical tests

verify_factory_map.py is a standard-library utility with paths relative to its own package. Its literal oracle covers complete manual rows 1,12,13,25,36,37,43,44,48,49,51,52,55,56,60,61: 16 rows and 144 cells independently viewed on the original scans. All 549 cells additionally receive algorithm, range and pitch-class checks; 540 adjacent-column steps are checked. Agreement with the embedded full transcription is labeled internal consistency, not independent corroboration.

The low 16-foot repeat and the upper repeated fold are essential boundaries. At key 61, the nine factory source IDs are 61,80,73,85,80,85,89,80,85. The utility rejects missing rows, wrong ordering, Boolean IDs, altered offsets and plausible in-range miswiring; its 17 deliberately damaged datasets were all rejected. It also passed from another working directory under Python optimization. [S060-01]

All 91 TableV values were independently read before comparison; there were zero numerical corrections. The meter belongs across preamplifier G–G, with 1000 ohms/volt sensitivity. Range and detector type remain unknown. These are historical service indications, not raw pickup amplitudes, modern RMS calibration, or oscillator gains. The stated 30-percent adjustment criterion is not an emulator tolerance. [S060-01]

Check	Executed result	Claim limit
Manual map	549/549 algorithm matches;144/144 visual fixture cells	Only16 rows independently reread
Upper fold	48 folded cells;6 requiring second subtraction	Electrical source IDs, not Hertz
Service readings	91/91 independent transcription matches	No new voltage measurements
Verifier controls	17/17 corruptions rejected	Data validation, not audio tests

```
python reference_code/verify_factory_map.py --json verification.json
```

CHECKS AND DECISIONS

- Retain meter-range and detector fields as unknown until documentary evidence resolves them.

SOURCES S060-01 Hammond factory service manual, Table III and Section IV alignment procedure

ORIGINAL DESIGN PROPOSAL / NOT YET EXECUTED

Stress shared state with musical gestures

Regression and adversarial musical tests

Build a traceable gesture corpus before judging timbre. Address every manual key separately, then folded source-sharing pairs, octave clusters, dense two-manual chords and pedals. Disable leakage and saturation for linear identity tests; enable them separately for interaction tests. The same electrical source should be evaluated once and routed through its contact paths. Check its phase across silent periods and voice allocation changes; do not compare unrelated random initial phases.

Expose contact conductance, shared percussion charge, source phase, routing mask and scanner state through an offline diagnostic interface. These observables make a failed attack explainable. Run slow and fast gestures, partial release, repeated-note overlap and all permutations of events sharing one timestamp. Specify a stable ordering policy for ambiguous digital events; do not assume note-off then note-on commutes with the opposite order. Keep actual physical contact timing a measured input, not an arbitrary MIDI-velocity formula.

Stimulus	State or invariant to examine
Same source through two keys	Unchanged source phase; routing sum in declared linear fixture
Idle, note, release, idle, note	Continuous generator clock; no note-triggered phase reset
Contact bounce / partial release	Bounded conductance; event order; released path settles correctly
Held upper chord plus new key	Single shared percussion history; no per-voice restart
All release, variable short gap	Finite recharge trajectory; no invented instantaneous reset
Preset cancel while held	Routing changes without manufacturing physical key releases
Drawbars0–8, rapid reversals	Continuous requested trajectory; intentional switching distinguished from artifacts
Scanner mode and route changes	Preserved state, correct common return, no duplicate dry addition

CHECKS AND DECISIONS

• Future tests must fail an intentionally note-reset oscillator and an intentionally per-note percussion implementation.

ORIGINAL DESIGN PROPOSAL / HOST CONTRACT

Make time independent of callback boundaries

Regression and adversarial musical tests

JUCE permits variable and zero-length processing buffers and supplies MIDI offsets within each block. VST3 also defines zero-sample parameter flushing. Its parameter queues describe sample-positioned linear segments, including an implicit point from the preceding block; reading only the final value would discard the trajectory. Test the reconstructed control curve independently of the DSP receiving it. [S060-02][S060-03][S060-04]

Render identical absolute events at 44.1,48,88.2,96 and 192 kHz, using blocks1,17,64,257 and a deterministic irregular partition. These are proposed coverage values. At a fixed rate, compare both samples and state checkpoints; require bitwise equality only where arithmetic order is deliberately fixed. Across rates, align physical times and declared filter latency, then apply explicit numerical or perceptual tolerances. Do not require arrays with different sampling grids to match bitwise.

Use counter-addressed random draws for event variation: seed, module, physical source, gesture ID and draw ordinal. Such counter-based generators are available in Random123. [S060-05] Callback number must never enter this address. For continuous noise, either define a shared band-limited physical-time process or compare calibrated spectra across rates; reusing integer sample counters at different rates does not create the same physical realization.

Boundary case	Required behavior
Event at block boundary	Exactly one application at its absolute sample
Zero-frame parameter flush	Update controls; advance no simulated time
Adjacent-sample automation jump	Preserve explicitly specified timing
Unrelated voice added	Existing source/event random draws unchanged
Offline versus realtime	Same stimulus, initial state and quality mode documented

CHECKS AND DECISIONS

- An irregular partition must not change event counts, elapsed sample time or random addresses.

SOURCES S060-02 JUCE AudioProcessor class reference / S060-03 Steinberg VST3 API Documentation / S060-04 Steinberg VST3 IParamValueQueue reference / S060-05 John Salmon: Parallel Random Numbers, As Easy as1,2,3 / Random123

ORIGINAL DESIGN PROPOSAL / CONTROLLED COMPARISONS

Use independent signal and acoustic controls

Regression and adversarial musical tests

Construct tests that can expose a shared implementation error. Compare stationary scanner transfer against the separately assembled AC reference, including absolute polarity and return loading. Compare crossover outputs into declared fixture loads before fitting actual drivers. Test fractional-delay endpoint weights and a static geometry analytically before testing rotation. Match rotor motion at common physical times; different microphones may require different retarded emission times for the same observation time.

For alias testing, render high notes, dense registrations, abrupt contacts and driven amplifier cases against a converged higher-rate reference with documented anti-alias filtering and delay compensation. Increase reference rate until the comparison itself stabilizes. Keep permitted modulation sidebands separate from folded artifacts. A low residual after unrestricted time warping or gain normalization is not an acceptable result; such fitting can conceal pitch, phase and level errors.

Acquire controlled acoustic cases only after electrical boundaries are stable. Preserve room, microphone pose, gains, rotor trajectory and instrument state. Reserve whole registrations, trajectories and microphone positions from fitting. Silence and noise floor are observations; a residual below uncertain capture noise should be inconclusive rather than reported as perfect agreement.

Controlled case	Purpose
Stationary horn / drum positions	Separate angular response from time-varying propagation
Known constant rotation	Check sideband spacing and stereo motion
Slow/fast/reversal trajectory	Check acceleration history and continuous phase
Dry and stationary wet paths	Detect accidental rotary processing of reverb
Repeated identical capture	Estimate drift and measurement repeatability before tolerance

CHECKS AND DECISIONS

- A frozen-angle match alone cannot validate the ordering of angular interpolation and convolution.

ORIGINAL DELIVERY PLAN / ACCEPTANCE

Turn each failure into a reproducible artifact

Regression and adversarial musical tests

Implement the future suite in this order: preserve documentary fixtures; build an event and automation scheduler; instrument generator/contact/percussion state; attach circuit reference comparisons; add rotary geometry and convolution cases; finally run the same corpus through VST3 and standalone wrappers. Keep engine tests runnable without a GUI. Steinberg explicitly allows processing without an instantiated controller. [S060-03]

Every result must record source revision, compiler and floating-point settings, sample rate, block partition, seed, variant, initial state, control-event file and calibration hashes. Store the first divergent sample or event, stage traces, residual audio and the metric definition. Classify pass, fail, inconclusive and not-run separately. A missing capture, unsupported bus configuration or skipped test cannot silently become success. Baseline updates require a reviewed explanation and preserved previous evidence.

Proposed acceptance metrics must be justified per boundary: exact integer mappings, analytic identities within declared numerical precision, spectral/envelope errors for approximate solvers, and capture-aware limits for hardware comparisons. Add allocation and deadline instrumentation to the processing path, and exercise long silence, denormal-prone decay, repeated state restoration and concurrent interface activity. These are future gates; this assignment executed only the documentary utility and its corruption controls.

Deliverable	Required evidence
Current documentary utility	549 assignments,144 independent cells,91 service values,17 rejected mutations
Scheduler regression	Absolute event trace and partition equivalence
DSP integration regression	Stage-local oracle, tolerances and intentional faulty control
Host/standalone regression	Same canonical stimulus through both wrappers
Physical acceptance	Held-out calibrated captures and uncertainty budget

CHECKS AND DECISIONS

- Do not publish a single “1:1” score combining documentary, synthetic and physical evidence levels.

SOURCES S060-03 Steinberg VST3 API Documentation

EVIDENCE AND ORIGINAL STUDY PROPOSAL

State four different questions before recruiting listeners

Listening study and fidelity claim protocol

Every study below is a plan. First identify the physical organ, AO-28 revision, cabinet, receiving load, microphone scene and operating domain through chapters 040, 056 and 057. A listening result describes that comparison and sampled conditions; it cannot establish identity across all B-3 specimens, performances or listeners.

ITU-R BS.1116-3 addresses small impairments, while BS.1534-3 addresses intermediate quality using MUSHRA. [S061-01] [S061-02] Proposed choice: pilot a suitable method, then freeze it. Use an explicitly adapted reference-comparison protocol when instrument behavior prevents strict compliance. A similarity scale, omitted anchors or altered switching rules must be disclosed; a familiar slider interface alone does not make a study standard-compliant. Proposed pilot check: verify anchor audibility for each registration; a low-pass anchor may barely affect bass-dominated material. An ineffective anchor cannot justify rejecting a listener.

For proposed fixed-file ABX, A and B are labeled examples, X duplicates one at random, and the listener identifies its source. Balanced assignment gives chance accuracy 0.5. This detects differences between recordings; it does not determine their cause or desirability. Random contact noise or unrelated takes may supply the cue. Include independent hardware-repeat comparisons and inspect reported cues before attributing discrimination to model error.

Question	Proposed endpoint	Permitted interpretation
Similarity	Rated closeness to the named reference	A defined perceptual judgment; no percentage of physical authenticity.
Discrimination	Correct identification above task chance	A detectable difference under these conditions.
Equivalence	Effect within prespecified practical bounds	Evidence limited to that endpoint, margin and population.
Preference	Choice of the sound or response preferred	Musical appeal; preference can favor an inaccurate coloration.

CHECKS AND DECISIONS

- No score is relabeled “97% faithful”; no finite study promises perceptual identity.

SOURCES S061-01 ITU-R BS.1116-3: Methods for the subjective assessment of small impairments in audio / S061-02 ITU-R BS.1534-3: Method for the subjective assessment of intermediate quality level o

ORIGINAL ACQUISITION AND COMPARISON PLAN

Build a stimulus matrix that exposes the mechanisms

Listening study and fidelity claim protocol

Use physical captures with retained control and gesture histories, then render the corresponding candidate. Pair matching electrical boundaries first; compare microphone outputs only within the same declared scene. Published synthesis examples are not physical ground truth. Preserve pre-roll, quiet intervals and full recovery tails, and reserve entire takes or sessions before tuning.

Moro and colleagues studied gesture-dependent contact behavior on a 1967 C-3; their listening stimuli excluded release transients. [S061-04] The proposed matrix therefore includes attacks and releases. A separate MIDI-controller study evaluates its gesture approximation, without claiming that a scalar velocity supplied the original key trajectory.

Repeat selected hardware cases across sessions and warmup states to estimate specimen variability. Cross candidate and hardware conditions over comparable phase, rotor-start and noise distributions; do not let one source always receive a distinctive take or initial state. If exact trajectories are unavailable, analyze matched gesture classes and disclose residual performance confounding.

Family	Proposed held-out material	Error made audible
Registrations	Single drawbars, mixed chords, both manuals, pedals; low/high foldback regions	Source mapping, balance, leakage and coherent summation.
Contact gestures	Pressed/struck attacks, partial travel, repeated notes and releases	Click spectrum, timing and gesture dependence.
Percussion	Overlapping legato, all-key release gaps, each switch combination	Rearming, shared decay and routing.
Scanner	All six modes; independent manual selections and sustained chords	Modulation, wet/dry relation and mode transitions.
Drive and expression	Quiet/loud chords, pedal sweeps and recovery	Level-dependent color, compression and envelope preservation.
Rotary motion	Settled speeds, both ramps, interrupted commands; stereo and mono	Acceleration, Doppler, spatial motion and cancellation.

SOURCES S061-04 Giulio Moro, Andrew P. McPherson and Mark B. Sandler: Dynamic temporal behaviour of t

ORIGINAL STIMULUS PREPARATION PROTOCOL

Match playback level without removing the instrument behavior

Listening study and fidelity claim protocol

The ITU procedures explicitly address excerpt loudness preparation and controlled reproduction. [S061-01] [S061-02] Proposed implementation: retain a calibrated absolute-level evaluation and a separately labeled listening comparison. Derive each path’s fixed scalar gain from independent calibration material; freeze it across the registrations, expression positions and input levels in its declared block. Document the gain and its uncertainty.

Do not independently normalize each note, attack or sustained segment, compress the candidate, or equalize away tonal disagreement. If a timbre-only experiment needs per-excerpt loudness adjustment, label its restricted question and preserve the untouched dynamics evaluation. A constant gain preserves within-recording amplitude ratios; time-varying normalization need not. Keep the same stereo-channel gain relationship and verify peaks remain unclipped.

For playback comparisons only, remove a documented acquisition or renderer offset using synchronization evidence. Preserve physical onset differences, propagation, scanner motion and rotary timing. Do not warp individual notes or rotor cycles into alignment. Test interactive latency separately. Freeze raw-file hashes and every transformation, including resampling, channel routing and boundary fades.

Calibrate the reproduction chain, record room/headphone configuration and listening gain, and log any participant changes. Use silent margins so switching or loop fades do not truncate the click or recovery being judged. Headphone, loudspeaker and mono conditions receive separate labels and analyses; a favorable headphone result does not certify a room reproduction.

MODEL EQUATIONS

$y(t) = g\,x(t)$, with constant dimensionless gain g.

$R = \mathrm{RMS}(\text{attack}) / \mathrm{RMS}(\text{sustain})$; applying the same nonzero g leaves R unchanged. Windows and RMS definitions are preregistered.

CHECKS AND DECISIONS

- Reject any derived stimulus with undocumented gain, timing changes or clipping.

SOURCES S061-01 ITU-R BS.1116-3: Methods for the subjective assessment of small impairments in audio / S061-02 ITU-R BS.1534-3: Method for the subjective assessment of intermediate quality level o

ORIGINAL PANEL AND BLINDING PLAN

Separate listening expertise from performance evaluation

Listening study and fidelity claim protocol

BS.1116 emphasizes expertise in hearing subtle impairments, controlled presentation and explicit screening criteria. [S061-01] Proposed recruitment therefore records relevant auditory training and Hammond experience separately. No panel has been recruited. Train participants on the task and disclosed artifact classes using development material; do not reveal final condition identities or reward agreement with the developer.

An independent custodian assigns neutral codes and retains the key. Randomize A/B positions and candidate order, counterbalance material and session order, and log the randomization seed. Keep the administrator blind to assignments. Include hidden references, justified diagnostic controls and repeated trials; define screening before collection. Never discard a listener merely for disliking the candidate or disagreeing with the average.

Pilot fatigue and task difficulty, then set session lengths, breaks and the sample plan. Obtain consent and deidentify records. For near-reference comparisons, choose participant and stimulus counts from the planned interval-width and power calculations, accounting for repeated observations. More trials from one player cannot replace independent players. Exclusions, withdrawals and incomplete sessions remain in the accounting.

Role or study	What it evaluates	Boundary of its claim
Trained listener	Blinded recordings; similarity and discrimination	Audio under the specified reproduction conditions.
Experienced performer	Legato, smears, expression, control timing and musical response	Interactive instrument behavior; document controller and latency.
Usability participant	Drawbars, presets and the proposed three-dimensional interface	Workflow and appearance; separate from blinded audio fidelity.

CHECKS AND DECISIONS

- Tactile or visual identification in a live comparison is recorded as a blinding limitation, not concealed.

SOURCES S061-01 ITU-R BS.1116-3: Methods for the subjective assessment of small impairments in audio

ORIGINAL STATISTICAL ANALYSIS PLAN

Predeclare effect sizes, uncertainty and decision rules

Listening study and fidelity claim protocol

Lakens explains why nonsignificance cannot establish equivalence and why bounds must define a meaningful effect. [S061-03] Proposed primary endpoint: a paired rating difference D with justified margins delta_L and delta_U. Select margins before the final data, using the intended musical claim and pilot resolution. Do not enlarge margins after disappointing results. For a compatible two-one-sided-test analysis, the confidence interval below must lie wholly inside both bounds.

ABX uses a different endpoint: probability of correct identification. Estimate it with uncertainty against the defined chance mechanism; low power and a nonsignificant difference are inconclusive. A narrowly bounded detectability claim requires its own prespecified threshold and valid interval procedure. A tiny statistically detectable difference may still meet a separate practical-equivalence criterion; report both outcomes.

Barr and colleagues show why participant and stimulus dependencies matter for generalization. [S061-05] Proposed analysis models listener, item and repeated-session structure, including design-supported condition variation. Choose an appropriate binary model for ABX and an appropriate bounded-rating model or justified alternative for ratings; preregister convergence and assumption-failure handling. Do not treat every repeated trial as an independent participant.

Freeze primary contrasts, critical mechanism families, multiplicity handling, missing-data rules, exclusions and the stopping rule. Report effect estimates and intervals by family and listener group alongside aggregates. A wide interval that crosses a decision boundary is inconclusive. An excellent mean cannot erase a failed percussion or rotor-transition condition.

MODEL EQUATIONS

Equivalence gate: CI_(1 - 2 alpha)(D) is wholly inside (-delta_L, +delta_U).

D is the population paired rating difference; alpha is each prespecified one-sided test level. Interval construction and any multiplicity adjustment must match the analysis.

CHECKS AND DECISIONS

- No observed panel size, p-value, confidence interval or fidelity pass is claimed here.

SOURCES S061-03 Daniël Lakens: Equivalence Tests: A Practical Primer for t Tests, Correlations, and M / S061-05 Dale J. Barr, Roger Levy, Christoph Scheepers and Harry J. Tily: Random effects struc

ORIGINAL AGENT HANDOFF AND ACCEPTANCE PLAN

Release a reproducible report with a bounded claim

Listening study and fidelity claim protocol

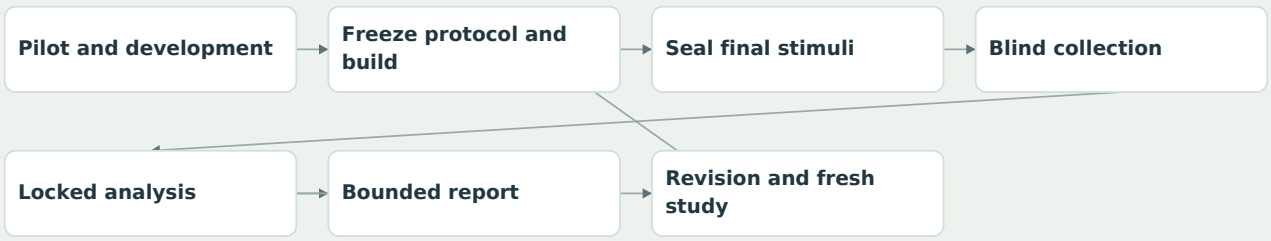
Preregistration distinguishes planned confirmation from exploration; changes and previous knowledge must remain visible. [S061-06] Before the final study, timestamp the protocol, candidate build, profile, stimulus set and analysis script. Lock the model before unsealing held-out takes. Findings used for tuning become development evidence; a changed model needs fresh final evaluation.

Deliver a study manifest containing specimen and chain IDs, control histories, raw/derived hashes, gain and alignment recipes, playback configuration, participant criteria, consent handling, coded trial assignments, permitted repeats and software versions. Store the decoding key separately until data collection and planned analysis are locked. Archive deidentified trial responses, omissions, exclusions, analysis output and every deviation.

The agent produces a condition-by-condition verdict: passed within the declared margin, failed, or inconclusive. State the measured endpoint, reference specimen, sample coverage and uncertainty. Report preference separately. Link failures to reproducible stimuli and their subsystem owners, preserving negative cases. Broader claims require broader evidence; calibration and numerical tests remain separate gates.

Required next inputs are actual physical captures, an eligible panel, pilot variance, justified margins and a reviewed statistical plan. None exists as a completed listening experiment in this chapter. The deliverable is an executable research protocol, not evidence that a digital instrument already matches a B-3.

Preserve the boundary between tuning and confirmation



Original protocol diagram. These are proposed stages; no participants or results are represented.

CHECKS AND DECISIONS

- Release no listening-fidelity claim without actual trial data and an inspectable analysis.

SOURCES S061-06 Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven and David T. Mellor: The pr

OFFICIAL API EVIDENCE / ORIGINAL RUNTIME CONTRACT

Assign the audio clock and budget the actual callback

Performance memory and latency budgets

PortAudio warns against allocation, deallocation, I/O and mutex operations inside its callback. JUCE permits variable and zero-length blocks, and describes its preparation maximum as a hint. These contracts require bounded processing, not a buffer-size assumption. [S062-01][S062-02]

ORIGINAL DESIGN PROPOSAL: one serialized audio execution context owns mutable engine state. The thread identity may change between host calls; ownership still cannot overlap. Consume canonical events at their sample offsets. Evaluate chapter 016’s shared generator field once on its declared internal timeline; chapter 052’s horn and drum trajectory services own physical motion. Renderers query those trajectories, and the UI reads snapshots. Neither a second microphone nor a display frame advances another copy of time.

Let N be supplied frames, Fs samples/second and T measured callback duration in seconds. D=N/Fs is a nominal processing interval, not round-trip latency or a promise that a host grants this whole interval to one plugin. At 48 kHz, 32 frames give 0.6667 ms and 1024 give 21.3333 ms; at 96 kHz, 32 give 0.3333 ms. Track actual host scheduling and underflow indicators separately. Zero-frame calls have no normalized occupancy and advance no simulated time. [S062-09]

State or work	Proposed owner and bound
Generator, contacts, circuits	Audio context; persistent state, fixed compiled topology.
Horn/drum history	Audio trajectory service; bounded retained event horizon.
GUI and analysis	Non-audio threads; consume immutable snapshots.
Unexpected large block	Preallocated internal slices; rebase event offsets without duplicating events.

MODEL EQUATIONS

D = N/Fs seconds; occupancy u = T/D for N > 0.

CHECKS AND DECISIONS

• A GUI close, second receiver, extra key or internal slice boundary cannot change elapsed physical time.

SOURCES S062-01 PortAudio: Writing a Callback Function / S062-02 JUCE AudioProcessor class reference / S062-09 PortAudio public API: portaudio.h

ORIGINAL BOUNDED COMMUNICATION DESIGN

Specify queue capacity and failure behavior before coding

Performance memory and latency budgets

Boost documents wait-free push/pop operations under single-producer/single-consumer restrictions; a full queue rejects a push. Those guarantees do not make an allocating payload or user callback suitable for audio. Use fixed-size trivial records and verified target atomics. The C++ draft explicitly permits potentially blocking non-lock-free atomic operations. A lock-free query also does not prove that an arbitrary retry loop has bounded execution. [S062-03][S062-04]

ORIGINAL DESIGN PROPOSAL: declare maximum admitted event density Emax, queue capacities and maximum records examined per callback. Consume host events through bounded views where possible. Give independent producers separate queues or merge them outside audio; never turn an SPSC queue into a multi-producer queue by convention. Preserve timestamp and tie ordering across queues.

Only untimed, idempotent UI requests may use latest-value coalescing. Sample-positioned automation and musical events retain their trajectories. No finite deadline can accommodate unbounded input. Outside the admitted limit, latch a fault through a verified bounded atomic operation, discard the faulted event epoch and release owned contacts at the next defined boundary. Preserve generator/rotor time, report the discontinuity, and invalidate the render. This is explicit recovery, not successful playback.

Channel	Full-capacity policy
Timed music / automation	Fault and bounded recovery; never silently discard note-offs.
Untimed UI control	Coalesce only where the parameter contract permits it.
Prepared profile handles	Keep current profile; producer retains unpublished ownership.
Retired profile handles	Defer replacement while retirement capacity is unavailable.
Telemetry	Drop diagnostic records, increment loss count; never stall audio.

CHECKS AND DECISIONS

- Force every queue full and verify its declared policy, bounded work and absence of stuck contacts.

SOURCES S062-03 Boost Lockfree: spsc_queue class template / S062-04 C++ working draft: Lock-free property [atomics.lockfree]

ORIGINAL OWNERSHIP AND LIFETIME PROTOCOL

Prepare profiles offline and reclaim them after the last reader

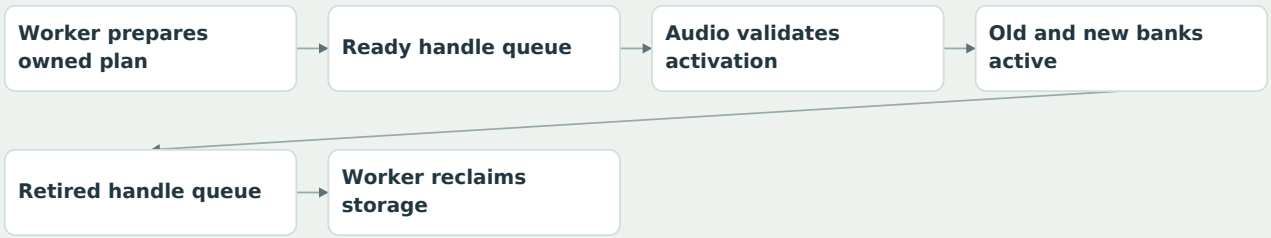
Performance memory and latency budgets

Compile ModelPlan units, topology, waveform tables, solver coefficients, FFT plans, IR spectra and buffer capacities outside processing. Pre-touch required storage and warm replacement histories under a declared schedule. Immutable assets are distinct from mutable runtime state. A prepared handle is not permission for a worker to edit the active graph.

ORIGINAL DESIGN PROPOSAL: transfer a ready handle through a bounded queue. The audio owner accepts only when compatible state migration, transition storage and retirement capacity are available. Reserve a retirement-queue credit before activation; subsequent publications cannot consume that credit. Keep at most the admitted number of concurrent banks. Retire the old handle only after its crossfade, tails and all asynchronous users have finished; a non-audio owner then destroys it. Final shared-pointer release on audio is not an acceptable reclamation strategy. If a timestamped replacement is unready, retain the old profile and report a missed activation; never wait or silently change the requested timestamp.

JUCE’s convolver requires serialized method calls despite its wait-free load entrypoints; background decoding does not authorize concurrent load/process calls. Its queue and externally borrowed data have lifetime requirements. Follow the pinned implementation’s ownership contract, preserve IR gain/onset and account for reported algorithmic latency separately. [S062-05] Unsupported state migration belongs to stopped processing, rather than an unexplained phase reset.

A handle moves; destruction waits



Original lifetime design, not a physical signal diagram. If retirement cannot be guaranteed, defer acceptance; never free an active reader’s storage.

CHECKS AND DECISIONS

- Poison retired storage in a correctness build; no reader may access it after reclamation.
- Destroy/reload profiles while processing; allocation, deallocation and blocking counts inside callbacks must remain zero.

SOURCES S062-05 JUCE dsp::Convolution class reference

DERIVED COUNTS / ORIGINAL OPTIMIZATION WORKSHEET

Calculate work without inventing a CPU benchmark

Performance memory and latency budgets

The table calculates work for explicitly chosen representations. These counts are not timings, CPU percentages or platform capacity claims. Fs is 48000 samples/second; R is oversampling ratio. A nonlinear cluster with iteration cap Imax costs up to R*Fs*Imax solver iterations/second, plus resampling and other work. Iteration counts, fallback frequency and spectral error require measurement; a low cap cannot conceal failed solves.

JUCE exposes oversampler preparation and latency, and SIMDRegister exposes native lane count with alignment requirements. [S062-06][S062-07] Use aligned structure-of-arrays storage and compare the optimized renderer with the scalar oracle. For 91 sources, width 4 needs 23 vector groups and width 8 needs 12; these are packing counts, not 4× or 8× speedup promises. Preserve source identity, mask padding and quantify changes from reassociation or approximate functions.

Keep stateful feedback groups on their declared internal clock. Never reset inaudible sources, double-advance rotors, discard coupled loading, or move angular weights across convolution to reduce cost. Chapter 053’s input-time and output-time operators differ during motion. Share transforms only when the actual input streams coincide. Profile FFT scheduling bursts and two-bank transitions separately from average arithmetic.

Illustrative calculation	Derived result and exclusions
91*Fs source evaluations	4,368,000/s at 1×; 17,472,000/s at 4×; waveform cost unspecified.
4 bases*2 receivers*32 partitions*129 bins*Fs/128	12,384,000 complex MAC/s; excludes FFTs, weighting, histories and transitions.
Same bank: 33024 complex coefficients	264192 bytes with packed pairs of 32-bit floats; excludes workspace, histories and allocation padding.
128-sample partition at 48 kHz	2.6667 ms update period; not automatically plugin latency or host block size.

CHECKS AND DECISIONS

- Record numerical residual, latency and measured worst callback cost for every claimed optimization.

SOURCES S062-06 JUCE dsp::Oversampling class reference / S062-07 JUCE dsp::SIMDRegister class reference

PROPOSED PROFILING CAMPAIGN

Measure tails, bursts and contention with bounded telemetry

Performance memory and latency budgets

Instrument callback entry/exit using a platform-qualified timing primitive, then append fixed-size records to a preallocated ring. Qualify timer overhead with an empty callback and compare instrumented/uninstrumented builds. Retain N, event count, quality, model version, transition state, maximum solver iterations and fault counters. Compute distributions and write files outside audio; telemetry overflow invalidates complete-distribution claims even if a separate maximum counter survives.

Run release builds for performance; use separate allocator, lock, lifetime and race instrumentation for correctness. Guard the actual callback scope rather than assuming one permanent thread ID. Inspect third-party paths and cold first-use behavior. JUCE provides ScopedNoDenormals for scoped denormal disabling. [S062-08] Record supported floating-point modes, restore the caller's environment and compare long decays with the numerical reference; do not add uncalibrated noise merely to hide a subnormal-cost problem.

Retain p50, p99, p99.9, maximum duration, nominal-interval exceedances and available host/device underrun flags. PortAudio exposes underflow/overflow status, while its CPU-load value includes more than the client callback. [S062-09] A low average or absent reported xrun cannot establish worst-case schedulability. Report unavailable counters as unavailable, and separate startup, steady operation and transitions.

Stress fixture	Risk it exposes
Dense two-manual/pedal events, maximum drive	Routing bursts and solver iteration spikes.
Silence following long feedback/convolution tails	Subnormal behavior and incorrect early state suspension.
Simultaneous bank swaps and rapid rotor commands	Transition cost, retained histories and queue pressure.
GUI animation plus background profile preparation	Contention and accidental UI/audio coupling.

CHECKS AND DECISIONS

• Archive CPU, OS, host, driver, compiler flags, library revisions, power/thermal conditions and complete stimulus hashes.

SOURCES S062-08 JUCE ScopedNoDenormals class reference / S062-09 PortAudio public API: portaudio.h

PROPOSED ACCEPTANCE / ENGINE BENCHMARKS NOT RUN

Publish conditional budgets and reject undocumented shortcuts

Performance memory and latency budgets

No complete engine or CPU benchmark exists in this suite. Select minimum supported Windows x86-64 and macOS arm64 machines before approving budgets. An initial planning target is one baseline-quality instance at 48 kHz/64 frames with maximum callback time below 0.40D = 0.5333 ms. This is a proposed target, not measured capability, a manufacturer specification or a guarantee for another quality mode.

Proposed coverage: 48/96 kHz, blocks 32, 64, 128, 256, 512, 1024, plus zero and irregular blocks for correctness; add 44.1/192 kHz if supported. Sweep 1, 4, 8 instances and continue to the measured capacity limit. Include simultaneous profile transitions, distinct profiles, warm steady load and cold first callbacks. Do not infer aggregate capacity by multiplying one instance’s average: host scheduling, shared caches, workers and memory traffic must be exercised together. Share immutable coefficients only with explicit lifetime control; mutable histories remain per instance. Budget peak bytes for shared assets, per-instance state/scratch, queues, telemetry and unreclaimed transition banks; measure resident/private memory and repeated-swap growth.

Run a proposed 30-minute sustained worst-case session after screening each configuration, retaining every miss and the maximum. A finite successful run bounds observed behavior only. Deliver performance.json with hardware/configuration identity, declared budget, metrics, evidence level and pass/fail/not-run status, plus reproducible event fixtures and raw telemetry. Physical-fidelity acceptance remains a separate capture-dependent gate.

Release gate	Reject when
Realtime ownership	Any callback allocation/free, lock, file access or unbounded retry is observed.
Admitted workload	Any lost musical event, missed activation or solver failure is hidden.
Timing budget	Observed maximum exceeds its declared target or attributable audio dropout occurs.
Memory bound	Peak storage or retirement backlog exceeds its declared limit; repeated swaps leak.
State equivalence	SIMD, slicing or scheduling changes source phase, rotor time or moving-filter order.
Evidence completeness	Missing captures or unexecuted benchmarks are reported as passes.

CHECKS AND DECISIONS

• Keep all engine performance results marked not-run until a compiled implementation is profiled on named hardware.

SOURCES S062-01 PortAudio: Writing a Callback Function / S062-02 JUCE AudioProcessor class reference / S062-09 PortAudio public API: portaudio.h

ORIGINAL ARCHITECTURE PROPOSAL

Define the engine independently of its wrapper

C++ engine architecture and module API

The delivered reference_code/engine_contract.hpp defines a C++20 vocabulary for a future implementation. It contains interfaces, units, borrowed views and timing predicates, not tone generation or an audio plug-in. Keep the host adapter responsible for SDK objects, bus negotiation and event conversion; the engine consumes a bounded, framework-independent request. The same graph can then serve VST3, standalone and an offline renderer without reproducing the instrument logic in three wrappers.

Preparation and processing are different lifecycles. JUCE documents variable blocks and occasional zero-sample calls; native VST3 separately defines a maximum that remains fixed while active. The adapter translates its actual host contract rather than assuming one framework description applies everywhere. An oversized callback is subdivided into preallocated segments when supported; otherwise it produces an explicit fault, never an allocation inside rendering. [S063-01][S063-02]

Choose module boundaries according to electrical coupling. The organ circuit owns interacting contacts, percussion, scanner, expression and AO-28 state. The Leslie electrical module owns amplifier, crossover and driver loading. If a required input-current or feedback dependency crosses their proposed interface, fuse that solver region or document and validate the decoupling approximation. A convenient function boundary is not evidence that two physical networks are independent.

Contract	Responsibility
IGraphFactory	Parse, validate and prepare outside callback
IPreparedGraph	Bounded rendering and event dispatch
ISourceBank	Optional shared continuous source implementation
IOrganCircuit	Coupled organ electrical behavior
ILeslieElectrical	Amplifier and loaded crossover
IRadiation	Rotor geometry to microphone pressure

CHECKS AND DECISIONS

- A headless offline client must use the same graph implementation as both audio wrappers.

SOURCES S063-01 JUCE AudioProcessor reference / S063-02 Steinberg VST3 Processing FAQ

DATA CONTRACT AND PHYSICAL PORTS

Compile provenance into explicit signal boundaries

C++ engine architecture and module API

Load chapter 058 authoring JSON on a control or worker thread, preserve unknown values, resolve referenced files and verify their hashes. A schema-valid all-unknown profile is not executable evidence. The compiler must identify every input required by the chosen algorithm and reject unresolved dependencies, or select a clearly named proposed surrogate. Preserve source frequency numbers1–91 while converting to array indices0–90; neither index is a terminal location or a MIDI note.

The header retains source units, waveform boundary, amplitude target, evidence and effect ownership. A voltage-scaled waveform cannot be multiplied by another voltage as though it were dimensionless gain. Decide whether pickup differentiation, filtering and loading are already represented before composing modules. Historical G–G service readings remain documentary output observations; they cannot silently initialize pickup amplitudes.

Import the supplied025 and 027 data using their source namespace. Their nominal scanner has18 LC sections and 19 ground-referenced output voltages; chorus retains a separate return node. Its absolute outputs already contain the return contribution. Import044 with treble voltage VA−VH and bass voltage VW−VB; grounding the treble return changes the network. These are explicit electrical conventions, not acoustic-polarity measurements. [S063-06][S063-07]

Input asset	Compiler invariant
015 /022 mappings	Exact dimensions; disconnected pedal positions remain explicit
025 /027 scanner	Separate node IDs, output numbers and terminal maps
AO-28 registry	Never relabel component values as full connectivity
044 crossover	Keep source and driver loading in declared boundary
058 profile	Preserve evidence, units and missing-value reasons

CHECKS AND DECISIONS

- Reject a profile that applies a generator filter already embedded in its source waveform.

SOURCES S063-06 Werner, Dunkel and Germain: Hammond vibrato/chorus WDF model / S063-07 Electro Music Leslie122 family manual, publication044578, June1971

TIMING CONTRACT

Own each sample and automation segment once

C++ engine architecture and module API

ORIGINAL PROPOSAL: use a monotonic unsigned render-frame counter independent of DAW transport. A positive block owns event frames in[start,start+N); an event exactly at its end stays in the adapter for the next block. Sort accepted events by frame and a stable sequence number that documents cross-stream tie resolution. Splitting blocks must preserve that ordering, source phase and random-counter identities. Reject range overflow before any state mutation.

A zero-frame flush accepts events at the current frame and consumes them once. It commits control targets without stepping oscillators, rotors, envelopes, smoothing, noise or delay lines. It can omit audio buffers. A positive-frame call with disconnected outputs is different: the engine still advances the declared time. Bufferless parameter flush is a documented VST3 case; the precise internal ownership rules here are project proposals. [S063-02]

Steinberg parameter queues describe linear curve sections, including an implicit previous-block point at offset−1. Do not implement that as a jump when the final point arrives. Resolve CurveSpan anchors before entering the engine; preserve the curve over sub-blocks and reject conflicting owners of the same parameter interval. Discrete switches remain discrete; extra smoothing has a separate declared purpose. [S063-03]

MODEL EQUATIONS

p[n] = p_previous + (p_k - p_previous)*(n+1)/(k+1), 0 <= n <= k

```
validate ranges, capacities, ordering and curve ownership
if frames == 0:
    apply each current-frame target exactly once
    return without advancing dynamic state
for each event or curve boundary:
    render preceding half-open segment
    apply ordered boundary events
render remaining samples; retain end-boundary events
```

CHECKS AND DECISIONS

• A curve containing only its final-sample point must match the equivalent fully sampled ramp.

SOURCES S063-02 Steinberg VST3 Processing FAQ / S063-03 Steinberg IParamValueQueue

OWNERSHIP AND LIFETIME PROPOSAL

Publish prepared graphs without freeing them in audio

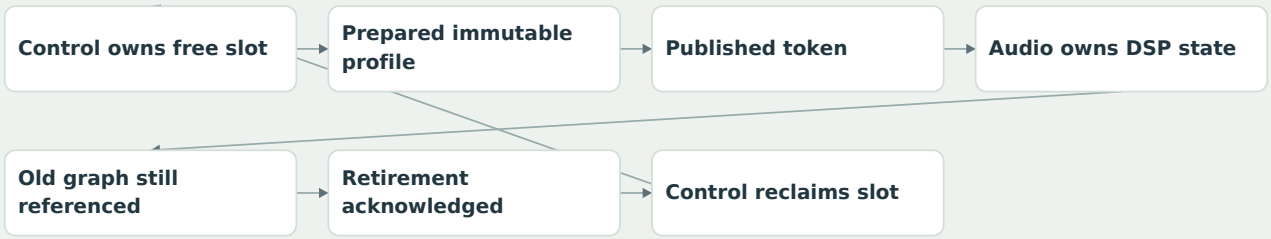
C++ engine architecture and module API

PortAudio excludes allocation, deallocation, I/O and mutex operations from dependable callback practice. This includes less visible work such as a final shared ownership decrement destroying an impulse response, vector storage or graph. A noexcept function declaration does not prove bounded execution. Every implementation behind this header still needs an allocation audit, deadline measurements and target-specific concurrency validation. [S063-04]

The proposed exchange transfers a slot-and-generation token, not ownership through an audio-thread smart pointer. Control prepares a free slot and publishes it. Audio takes sole ownership of its mutable DSP state while all profile data remain immutable. An old graph stays alive through its last sample and any preallocated transition; only a retirement acknowledgement permits control to reclaim its objects. String views, spans, waveforms and all other referenced storage obey the same lifetime.

Bound both publication and retirement queues. If retirement capacity is unavailable, postpone activation; never overwrite an in-use generation or spin until space appears. Shutdown must establish quiescence before reclamation. Immutable sharing and small value messages simplify concurrency reasoning, but publication ordering and reclamation still require a real implementation and review; volatile is not synchronization. [S063-05]

Prepared-slot ownership



Original ownership proposal; no queue or lock-free implementation is supplied.

CHECKS AND DECISIONS

- A forced full retirement queue must delay activation without destroying either active graph.

SOURCES S063-04 PortAudio: Writing a Callback Function / S063-05 C++ Core Guidelines

CONTROL AND STATE DESIGN

Separate musical state from serialization and resets

C++ engine architecture and module API

The proposed ControlState retains four independent nine-drawbar banks, two pedal drawbars and separate upper/lower preset latch masks. Editing the visible bank never changes a parameter’s identity. A physical C-cancel action clears preset routing; it must not masquerade as key release or software panic. Percussion charge, source phase and other continuous state survive ordinary control transitions unless the selected physical model predicts a change.

Use a parameter registry to specify units, normalized mapping and transition behavior. Continuous drawbar movement can use a physical control trajectory; scanner switching changes topology; expression belongs inside its loaded circuit. Do not add an identical low-pass smoother to every value. Host curve reconstruction and model-specific smoothing are separate owners, and both must be deterministic across block boundaries.

Persist a safely copied control snapshot, registry version, profile digest, variant, seed and user geometry outside the callback. A preset is not a full DSP checkpoint. Exact restart of phase, capacitor charge, rotor history and room tails requires a separate versioned stopped-engine format. Host state calls may occur while processing, so serialization must not read live mutable DSP state without a verified transfer protocol. [S063-02]

Situation	Proposed policy
DAW seek	Keep physical state unless explicit reset requested
No process callbacks	Freeze render clock; no assumed wall-clock catch-up
Rate/layout change	Stopped rebuild or separately verified migration
Profile replacement	Prepared compatible transition with retained lifetimes
Last note released	Continue tails and enabled autonomous noise

CHECKS AND DECISIONS

- Saving state during a sustained chord must neither reset the percussion circuit nor race its state.

SOURCES S063-02 Steinberg VST3 Processing FAQ

DELIVERED CHECKS AND FUTURE GATES

Implement and verify in a dependency order

C++ engine architecture and module API

The supplied header and engine_contract_check.cpp were compiled with GCC14.2.0 in C++20 mode, warnings enabled and treated as errors. The translation unit passed19 static assertions covering source identity bounds, half-open timing, zero-frame ownership, integer overflow, trivial-copy control/event records and abstract interfaces. This establishes syntactic and selected type-contract consistency only. No audio graph, profile compiler, exchange queue, host or hardware was executed.

Implement the authoring-profile compiler first, using058 validation and explicit dependency reports. Next implement the event/curve adapter and a deterministic offline scheduler. Only then connect source/contact behavior, the reviewed electrical models, radiation and room processing. Preserve offline mathematical references as independent fixtures; their successful algebra is not proof that a production discretization, anti-aliasing path or solver is correct.

Add concurrency stress with delayed readers, exhausted queues and repeated slot generations before live profile switching. Verify allocations and worst-case processing separately from sanitizer runs. Compare different callback partitions, sample rates, curve subdivisions and flush sequences against one reference timeline. Physical acceptance remains a later capture-and-calibration gate; unknown AO-28 connectivity and unmeasured specimen behavior are not repaired by compiling an interface.

Deliverable	Current state
engine_contract.hpp	Original proposed interfaces; syntax checked
engine_contract_check.cpp	19 compile-time checks passed
063-contract-checks.json	Actual compiler command, result and hashes
Profile compiler / exchange	Not implemented
DSP / plugin host validation	Not run
Instrument matching	No measured reference specimen

CHECKS AND DECISIONS

- Reject any release claim that treats header compilation as real-time or acoustic validation.

CURRENT OFFICIAL EVIDENCE

Freeze the wrapper and its licences

VST3 and standalone host integration

As inspected on 12 November 2026, Steinberg lists SDK 3.8.1, dated 11 August 2026. VST3 changed to MIT licensing with 3.8; advice describing the current SDK only as GPL/proprietary is outdated. Preserve its licence notices, and assess optional VST branding separately. The VST2 API is outside this grant. [S064-01] [S064-02]

JUCE is a separate dependency. Its current master LICENSE.md offers AGPLv3 or a commercial framework licence and points to the JUCE9 agreement, published 17 June 2026. A commercial JUCE licence may be required for the chosen distribution. A JUCE8 agreement applies to that older major version; neither the VST3 MIT grant nor the examples' ISC licence relicenses the JUCE framework. [S064-03] [S064-04]

ORIGINAL DESIGN PROPOSAL. Use the framework-independent engine from chapter 063 behind thin format adapters. JUCE is a practical wrapper candidate, not an acoustic model or mandatory engine dependency. Freeze a reviewed framework revision, SDK revision, compiler, dependency inventory and licence choice before generating binaries. This research chapter does not select or purchase a commercial licence. Keep the DSP tests buildable without a graphical editor.

Proposed build layer	Responsibility
Engine library	Samples, events, physical states and immutable calibration profiles.
VST3 adapter	Host buses, automation, state, lifecycle and latency/tail declarations.
Standalone adapter	Audio/MIDI devices, timestamp conversion and device recovery.
Editor	Control gestures and visualization; never the owner of audio time.

CHECKS AND DECISIONS

• Record exact dependency revisions and their notices; do not label a mutable master URL as a pinned build.

SOURCES S064-01 Steinberg: VST 3 License / S064-02 Steinberg: SDK change history / S064-03 JUCE official repository LICENSE.md / S064-04 JUCE 9 End User Licence Agreement

ORIGINAL ADAPTER DESIGN

Declare buses and translate musical intent

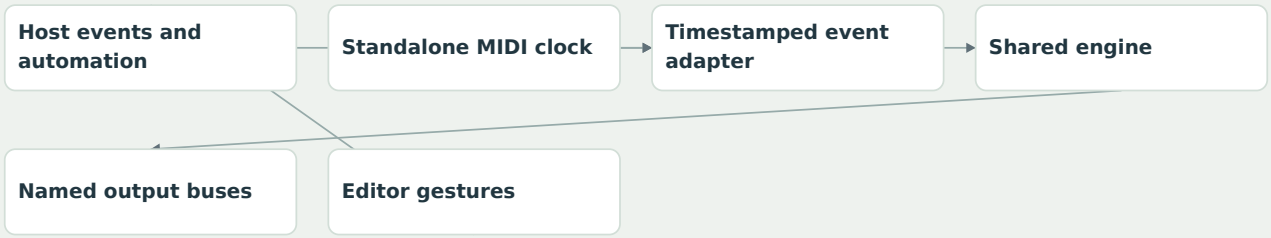
VST3 and standalone host integration

VST3 separates processing from its controller; a host may process without constructing that controller. Buses are negotiated and activated through the component interface. ORIGINAL DESIGN PROPOSAL: the first instrument build exposes one event input and one stereo main output, with no audio input. Optional direct-organ or separate microphone outputs require explicit named auxiliary buses and an agreed routing contract. Never infer a bus identity from the currently visible editor. [S064-05]

Route event input, channel and note identity into upper manual, lower manual or pedals using a persisted map. Keep physical preset-key gestures separate from a host program selection. Preserve ownership for repeated pitches and independent input devices, so one release does not cancel another held key. Velocity may drive the declared contact-gesture approximation; do not silently turn it into a piano-like steady-state loudness law.

Current SDK notes introduce IMidiMapping2 and IMidiLearn2. Implement only the extensions supported by the chosen wrapper and tested host set; retain a documented fallback for ordinary MIDI controllers. Keep controller-to-parameter translation distinct from note events. Note-expression support is an optional modern capability and must not silently change the vintage baseline. [S064-09]

Two adapters, one musical event contract



Original software proposal. Standalone gestures use the same event contract through their own adapter; the GUI does not drive oscillator time.

CHECKS AND DECISIONS

• Process an identical event fixture with the editor absent, open and repeatedly resized; numerical audio must follow the same declared engine contract.

SOURCES S064-05 Steinberg: VST 3 API documentation / S064-09 VST 3.8.0 release notes

SAMPLE-TIMED IMPLEMENTATION

Preserve automation timing and control identity

VST3 and standalone host integration

A VST3 parameter ID must remain associated with the same parameter; stepCount zero declares continuity. Assign independent IDs to all four retained nine-drawbar banks, plus the two pedal drawbars. Switching the editor between A-sharp and B changes its binding, never the identity of recorded automation. The continuous drawbars retain 0–8 markings without forcing hard detents. End a gesture against its original ID before rebinding. [S064-07]

IParamValueQueue describes piecewise-linear curves, including omitted initial points and a previous value at position minus one. An adapter that reads only the final point loses timing. Preserve the curve, with separate discrete-control interpretation, before any explicitly specified physical or anti-zipper smoothing. Do not confuse smoothing with faithful automation delivery. [S064-08]

Hosts may flush parameters through zero-frame calls without audio buffers. Commit control updates without dereferencing audio or advancing oscillator phase, rotor angle, envelopes or delay memory. ORIGINAL DESIGN PROPOSAL: render half-open intervals; a boundary event at offset N is retained for the next interval, not applied one sample early. Reject malformed offsets predictably. Define stable tie-breaking across parameter queues and notes because separate queues provide no universal cross-queue order. [S064-05]

```
decode bounded control changes
if frame_count == 0:
    commit control snapshot; return
merge valid events and automation segments
render intervals [cursor, next_event)
apply boundary changes in declared order
retain offset-N events for next boundary
render remaining samples; advance by N
```

CHECKS AND DECISIONS

• Compare automation ramps across variable block partitions; insert zero-frame flushes and verify that DSP time does not advance.

SOURCES S064-05 Steinberg: VST 3 API documentation / S064-07 VST 3 ParameterInfo reference / S064-08 VST 3 IParamValueQueue reference

PERSISTENCE AND LIFECYCLE

Restore state without restoring pointers

VST3 and standalone host integration

VST3 stores processor state separately from editor-only state; restoring component state synchronizes the controller without emitting ordinary edit callbacks. State access can occur while processing. ORIGINAL DESIGN PROPOSAL: serialize a versioned control/profile document and publish a prepared replacement at a defined boundary. Do not parse files, allocate profiles or destroy their final owner in the audio callback. [S064-05] [S064-06]

Store all retained drawbar values, preset routing, percussion/scanner settings, expression mapping, cabinet/microphone configuration, evidence-qualified profile identity and deterministic noise policy. Store editor camera and expanded panels separately. Absolute sample time, pointer addresses and transient host handles are not portable preset data. Distinguish a musical preset from an optional exact-render checkpoint containing internal phases and filter memories.

The actual VST3 block length can vary within its configured maximum; reconfiguration of that maximum or sample rate follows deactivation/setup. JUCE also warns that callback sizes can differ from the preparation estimate. Provision bounded chunking and validate the wrapper’s actual contract. Prepare oversampling and convolution resources outside processing, and report changed latency through the host mechanism. Old snapshots must remain alive until callback readers retire. [S064-06] [S064-13]

Proposed transition	Required behavior
Preset load	Validate schema and resources; publish atomically at a declared boundary.
Editor recreation	Read current state; do not reset the sound.
Rate / quality change	Prepare new coefficients and latency contract before activation.
Missing profile	Report a visible unresolved dependency; never pretend an unrelated profile matches.

CHECKS AND DECISIONS

• Round-trip old-version states with both banks edited; verify migration, missing-resource handling and unchanged parameter IDs.

SOURCES S064-05 Steinberg: VST 3 API documentation / S064-06 Steinberg: Processing FAQ / S064-13 JUCE AudioProcessor reference

ENGINE–HOST BOUNDARY

Silence, tails, bypass and suspended time

VST3 and standalone host integration

VST3 silence flags describe channels whose samples are all zero; a note-off is not such evidence. Percussion, filter memory, room decay, modeled hum or noise may remain. Tail reporting distinguishes absent, finite and infinite tails. ORIGINAL DESIGN PROPOSAL: compute the declaration from the selected model and its explicit truncation policy. A continuously audible hum option cannot honestly declare a finite decay merely because no keys are held. [S064-10] [S064-11]

Keep free-running source and rotor state advancing while active blocks are processed, including silent intervals. When a host stops callbacks, no callback-based engine can continue by assumption. Adopt an explicit deterministic suspension policy: freeze simulated time, then resume from a declared state, or fast-forward only where a trustworthy elapsed timeline and validated state evolution exist. Wall-clock-dependent reconstruction must be a separate mode, not an unnoticed offline-render difference.

Bypass handling belongs to the plug-in. For the proposed no-audio-input instrument, ramp the output to mute while retaining the declared event and clock policy; there is no input signal to pass through. A separate rotary-effect product would instead require latency-matched input bypass. Do not reset phases or percussion merely because bypass changes. Respect nullable inactive-bus buffers and generate every active output sample deliberately. [S064-06] [S064-10]

Condition	Proposed reporting/action
All keys released	Continue pending state; inspect actual output.
Persistent modeled noise	No false silence flag; declare unbounded generation as appropriate.
Output muted	Zero samples after ramp; internal time follows declared policy.
Callbacks suspended	No invented elapsed sample count; use documented resume policy.

CHECKS AND DECISIONS

• Render note release, long decay, hum enabled, mute and suspend/resume fixtures; check output flags and retained state independently.

SOURCES S064-06 Steinberg: Processing FAQ / S064-10 VST 3 AudioBusBuffers reference / S064-11 VST 3 IAudioProcessor reference

ORDERED ACCEPTANCE PLAN

Build the standalone and prove the adapter

VST3 and standalone host integration

JUCE’s CMake API can create VST3 and Standalone targets from shared code. Fix product identifiers deliberately instead of accepting regeneration-dependent defaults. AudioDeviceManager manages standalone audio/MIDI devices and their saved setup; keep that machine-specific setup outside musical presets. The instrument needs output audio, not microphone input by default. Reopening a device must reprepare sample-rate-dependent engine resources. [S064-12] [S064-14]

ORIGINAL DESIGN PROPOSAL. First run the engine headlessly; next test a minimal VST3 adapter; then add device handling and the editor. Timestamp incoming standalone MIDI against the audio clock with bounded queues and an explicit late-event policy. JUCE’s collector documents timestamp-to-block conversion, but thread safety alone proves neither lock freedom nor a worst-case deadline. Inspect the pinned implementation before choosing it for the required realtime contract. [S064-15] Consume or clear JUCE MIDI appropriately when no MIDI output is intended; retained messages otherwise become output. [S064-13]

Produce separate reports for SDK validation, actual DAW sessions, offline renders and standalone devices. Name every tested host/version, operating system, architecture, rate, buffer, precision, bus layout and plugin revision. A successful binary build does not validate percussion, Leslie acoustics or musical fidelity. No production VST3, standalone audio device or host compatibility test was executed for this research chapter.

Proposed gate	Evidence required; currently not run
Host lifecycle	Scan, instantiate, headless render, repeated activation and teardown.
Data contract	Zero/variable blocks, automation, boundary events, inactive or aliased buffers, unintended MIDI echo.
State integrity	Save/reload, migrated presets, both retained banks, missing resources.
Standalone recovery	Device disconnect/reconnect, rate change, MIDI loss and stuck-note recovery.
Output agreement	Wrapper versus headless fixture under declared precision and latency.

CHECKS AND DECISIONS

• Publish failures and unsupported configurations explicitly; do not convert a proposed compatibility matrix into passing results.

SOURCES S064-12 JUCE AudioDeviceManager reference / S064-14 JUCE official CMake API / S064-15 JUCE MidiMessageCollector reference / S064-13 JUCE AudioProcessor reference

HARDWARE EVIDENCE / ORIGINAL CONTRACT

Give every retained control a permanent identity

Parameter automation and preset compatibility

The factory console provides four manual drawbar groups and two pedal controls. Its later continuous-contact mechanism bridges adjacent buses; the 0–8 markings do not imply isolated detents. The Cancel preset key releases the latch without electrical contacts. These hardware distinctions determine the software state model. [S065-01]

ORIGINAL DESIGN PROPOSAL. Assign each retained bank its own semantic path and numeric ID. Editing upper A-sharp must never overwrite upper B or change the meaning of an existing automation lane. The supplied parameter_registry.json contains 62 proposed parameters, including all 36 manual and two pedal drawbars. Its numbers, ranges and startup values are software proposals, not manufacturer calibration data.

VST3 requires stable IDs and reserves the upper numeric range for hosts. Reserve retired IDs permanently; changes of meaning or normalized law require compatibility treatment. A JUCE string identifier and AU version hint do not prove that its wrapper exports these proposed numeric IDs. Inspect and freeze the adapter mapping. [S065-02] [S065-03] [S065-08]

Proposed ID range	Retained identity
0x1000–0x1008 / 0x1100–0x1108	Upper A-sharp / upper B; nine footages each.
0x1200–0x1208 / 0x1300–0x1308	Lower A-sharp / lower B; nine footages each.
0x1400–0x1401	Pedal 16-foot and 8-foot controls.
0x 2000–0x2400	Percussion, scanner, expression, volume and rotary command.
0x3000–0x3203 / 0x4000–0x4001	Proposed microphone geometry/gain and software output controls.

CHECKS AND DECISIONS

• Enumerate the built plugin and compare exported IDs and semantics with the approved adapter mapping before recording production automation.

SOURCES S065-01 Hammond factory service manual: drawbars, presets and selective vibrato / S065-02 VST3 ParameterInfo reference / S065-03 VST3 Parameters and Automation / S065-08 JUCE ParameterID reference

PARAMETER SEMANTICS

Freeze the normalized law, not just the label

Parameter automation and preset compatibility

ORIGINAL DESIGN PROPOSAL. Store continuous marked drawbar travel x from 0 to 8, normalized as u=x/8. The profile converts travel into contact overlap and loaded circuit response; x/8 is not an amplitude law. Expression describes physical travel, not final output gain. Preserve the trajectory before applying the physical model.

VST3 stepCount is zero for a continuous control, one for a toggle and five for a six-state list. Declare the scanner selector as one list and its Great/Swell routes as separate toggles. The Great route includes pedals in the selected vintage topology. The four percussion settings remain independent of the console Normal/Soft volume control. [S065-01] [S065-02]

Use the documented VST3 discrete conversion and canonical values k/stepCount. Freeze threshold behavior; later substitution of rounding can move old automation transitions. Microphone bounds and output trim are proposed controls, not measurements or reference-loudness calibration. [S065-03]

Control class	Proposed mapping / downstream ownership
Drawbar	Continuous 0–8; stepCount=0; profile owns contact/gain law.
Percussion and routes	Named two-state controls; stepCount=1; circuit state remains continuous.
Scanner mode	V1, V2, V3, C1, C2, C3; stepCount=5; stable stored order.
Leslie command	Chorale/Tremolo; stepCount=1; motor model owns separate rotor ramps.
Mic position / gain	Meters / decibels; geometry admission and acoustic rendering remain separate.

MODEL EQUATIONS

continuous: u=(x-min)/(max-min); x=min+u*(max-min)

discrete: u=k/N; k=min(N, floor(u*(N+1))), for finite 0<=u<=1

CHECKS AND DECISIONS

• Round-trip every enum value and both sides of each threshold; reject nonfinite stored values and unsupported units.

SOURCES S065-01 Hammond factory service manual: drawbars, presets and selective vibrato / S065-02 VST3 ParameterInfo reference / S065-03 VST3 Parameters and Automation

EDITOR AND HOST INTERACTION

Bind a gesture once, then close it deliberately

Parameter automation and preset compatibility

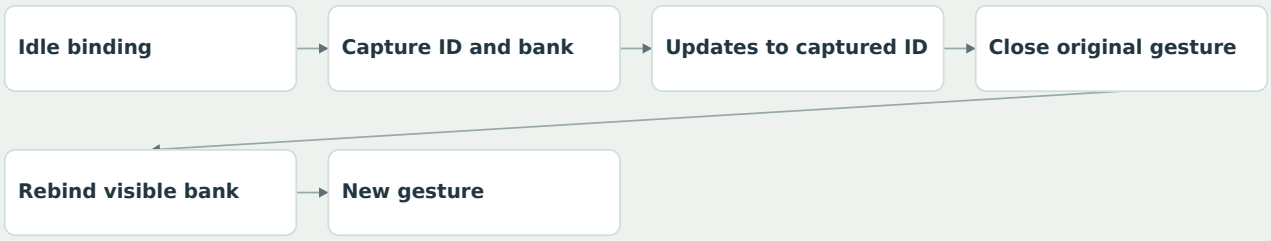
VST3 gestures use beginEdit, performEdit and endEdit; the host delivers changes to processing. UI-thread grouping associates related edits with one timestamp. Playback notifications update the display without generating another gesture. [S065-05] [S065-06]

ORIGINAL DESIGN PROPOSAL. At pointer-down, capture the semantic ID, bank, starting value and transaction token. Every drag update targets that captured ID. If the visible bank changes, finish the original gesture before rebinding; require a fresh pointer-down to edit the new bank. Apply the same closure policy to lost capture, window destruction and keyboard focus loss. Never transfer an unfinished gesture to the parameter newly occupying the same screen position.

Hiding a bank in the custom editor is a presentation choice. Do not set VST3 kIsHidden on its parameters merely because the bank is not visible: that flag also prevents external changes and automation. [S065-02] Maintain all bank parameters in the generic host view. Group a microphone-position drag across its coordinate IDs, while retaining a declared deterministic policy if the host returns separate queues.

Use one undo transaction per gesture; snapshot/profile changes are separate transactions. Undo must not manufacture key releases or reset generators. Respect the host’s automation mode instead of inventing a competing override.

A gesture retains its original bank identity



Original interaction contract. Bank changes never retarget an active host gesture.

CHECKS AND DECISIONS

• Record a drag, switch banks mid-drag, and inspect host lanes: only the original parameter receives that gesture.

SOURCES S065-02 VST3 ParameterInfo reference / S065-05 VST3 IComponentHandler reference / S065-06 VST3 API Documentation

SAMPLE-TIMED IMPLEMENTATION

Preserve curves through every processing boundary

Parameter automation and preset compatibility

The native VST3 queue represents piecewise-linear automation. A block may omit its first point; the previous value is treated at position minus one. Reading only the last point converts ramps into block-dependent steps. Preserve the native curve before applying any declared smoothing or contact mechanics. A framework’s atomic current-value accessor does not retain the queue’s sample positions. [S065-04] [S065-07]

ORIGINAL DESIGN PROPOSAL. Decode continuous segments into the engine contract; quantize discrete controls with the frozen conversion. Hold the final value when no segment follows. Render half-open intervals and defer an offset-N event to the next interval. Reject malformed offsets and overflow predictably. Preserve queue-local order and explicit cross-queue tie-breaking.

A zero-frame host call can flush parameters without audio buffers. Commit control values without dereferencing audio or advancing oscillator phase, percussion charge, rotor angle or delay memory. A changed target may affect the next rendered sample; no elapsed audio time has occurred. [S065-06] Parameter smoothing must not erase deliberately modeled switch transients or quietly add another motor ramp.

```
decode bounded parameter queues
if frames == 0:
    commit controls; publish snapshot; return
carry previous values at sample -1
merge segments and musical events
for each half-open interval:
    evaluate continuous controls at each sample
    apply declared discrete transitions
    render and advance engine time
retain offset-frames events for next boundary
```

CHECKS AND DECISIONS

• Render one ramp with several block partitions and inserted zero-frame flushes; compare control trajectories before comparing audio.

• Name the pinned wrapper path that preserves native automation timing; do not infer sample accuracy from an atomic scalar.

SOURCES S065-04 VST3 IParamValueQueue reference / S065-06 VST3 API Documentation / S065-07 JUCE AudioProcessorValueTreeState reference

VERSIONED STATE AND MIGRATION

Separate wired presets, software snapshots and checkpoints

Parameter automation and preset compatibility

ORIGINAL DESIGN PROPOSAL. Persist all banks independently of selection. Preserve the physical preset latch mask, wired matrix identity and playing-key ownership as different fields. C clears the latch mask without creating note-off contacts. Ordinary preset presses replace the mask with one selected route; documented simultaneous latching needs the fuller network model. The registry therefore describes preset-key commands separately from scalar host parameters instead of squeezing every physical state into a misleading selector. [S065-01]

A modern total-state snapshot changes multiple controls. VST3 warns against automating a program selector while its affected parameters are also automatable. Keep snapshot loading as a non-automated state transaction in the first product, with host notification handled by the wrapper. Vintage route changes do not copy a bank into another bank. [S065-03]

Validate, migrate and prepare resources off the audio thread, then publish a coherent replacement at the agreed boundary. Retire resources after readers release them. JUCE copyState and replaceState use locks and are not realtime safe. [S065-07] Missing profiles or unknown mandatory fields require visible failure, not silent substitution.

Keep immutable migration fixtures and historical normalization; never reuse retired IDs. Reject unsupported future versions while retaining the original. Musical snapshots store controls and profile identity. Exact-render checkpoints additionally require versioned phases, envelopes, filter memories and randomness state; ordinary presets cannot promise sample-identical resume.

State domain	Contract
Control snapshot	62 proposed values, all banks, route masks and profile references.
Instrument evidence	Immutable provenance-qualified calibration and wiring; not overwritten by a preset.
Editor state	Camera, visible bank and panels; does not determine audio routing.
Render checkpoint	Optional complete DSP history, with explicit compatibility and clock rules.

CHECKS AND DECISIONS

• Load old fixtures, save them again, and compare semantics rather than object ordering; preserve a reversible original.

SOURCES S065-01 Hammond factory service manual: drawbars, presets and selective vibrato / S065-03 VST3 Parameters and Automation / S065-07 JUCE AudioProcessorValueTreeState reference

VERIFICATION AND HANDOFF

Prove compatibility before shipping automation

Parameter automation and preset compatibility

Executed static checks verify unique IDs, the four-by-nine bank product, two pedal controls, continuous declarations, enum round-trips and reserved-range exclusions. They validate the supplied metadata only. No production plugin, host automation, audio-state migration or rendered sound was tested.

ORIGINAL ACCEPTANCE PLAN. Compare exported parameters with the approved registry. Record active/inactive-bank gestures, replay across block sizes, and reopen projects with the editor closed. Exercise zero-frame flushes and rate changes; inspect state traces and numerical output separately.

Test stereo microphone gestures, scanner-mode transitions and Leslie speed commands under dense automation. The geometry renderer must own propagation changes, the scanner owns its network transitions, and the motors own acceleration. None of those behaviors is established merely because a scalar parameter reaches its target. Reject invalid geometry before publishing prepared resources; publish the supported automation envelope.

Supply parameter_registry.json, chapter 063 interfaces, chapter 064 wrapper rules and capture/profile schemas. IDs remain provisional until release freeze; afterward they are compatibility promises. Actual wrapper mapping, event capacity, physical calibration and host behavior remain implementation gates.

Required future gate	Observable result; presently not run
Retained banks	Inactive edits survive recall; visible-bank switching never changes lane identity.
Gesture boundaries	Every opened gesture closes once, including lost capture and editor teardown.
Timing	Native ramps and discrete thresholds survive block partition and zero-frame flush.
Preset semantics	Cancel changes routing while physical held-key ownership remains intact.
Compatibility	Old projects restore every control/profile or return an explicit diagnosable failure.

CHECKS AND DECISIONS

• Keep documentary metadata results separate from host, realtime, audio and physical-fidelity pass/fail reports.

SOURCES S065-02 VST3 ParameterInfo reference / S065-04 VST3 IParamValueQueue reference / S065-06 VST3 API Documentation / S065-07 JUCE AudioProcessorValueTreeState reference

CURRENT SOURCE EVIDENCE / ORIGINAL RELEASE PROPOSAL

Freeze the release envelope before building

Cross platform build packaging and release

The deliverables are a VST3 instrument and a standalone application sharing one engine. ORIGINAL DESIGN PROPOSAL: start with Windows x64, macOS arm64 plus x86_64, and Linux x86_64 qualification lanes. These are proposed targets, not tested support claims. Choose minimum operating-system versions, host versions, CPU instruction floors and audio backends only after executing the compatibility matrix.

As inspected on 12November 2026, Steinberg lists SDK3.8.1 and licenses VST3 under MIT since3.8. Current JUCE master refers to the JUCE9 EULA, published 17June 2026; its framework modules remain AGPLv3/commercial. SDK permission does not license the framework, photographs or captured sounds. Record the terms of the exact dependency revisions selected. [S066-03] [S066-04] [S066-05] [S066-13]

Maintain a release manifest beside the source lock: compiler, SDK, CMake, dependency commits, enabled modules, runtime libraries, signing identity reference and model-profile hashes. A moving master branch is documentary context, not a reproducible dependency specification. Inspect the framework’s bundled SDK before replacing it with a newer standalone SDK.

Decision	Required release record
Identity	Stable processor/controller IDs, owned bundle identifier, product version and state-schema version.
Architectures	Explicit native slices, instruction-set baseline and dependency coverage for each lane.
Licences	Chosen framework route, dependency notices, image rights and measured-dataset permissions.
Evidence	Built commit and hashes; separate documentary, numerical, host and acoustic results.

CHECKS AND DECISIONS

• Reject a release manifest containing unspecified dependency revisions or a support claim derived only from successful compilation.

• Do not embed credentials or private signing material in the repository or report.

SOURCES S066-03 Steinberg VST3 licence / S066-04 JUCE framework licence / S066-05 JUCE9 End User Licence Agreement / S066-13 Steinberg SDK change history

ORIGINAL BUILD DESIGN / CONDITIONAL COMMANDS

One engine, reproducible native build lanes

Cross platform build packaging and release

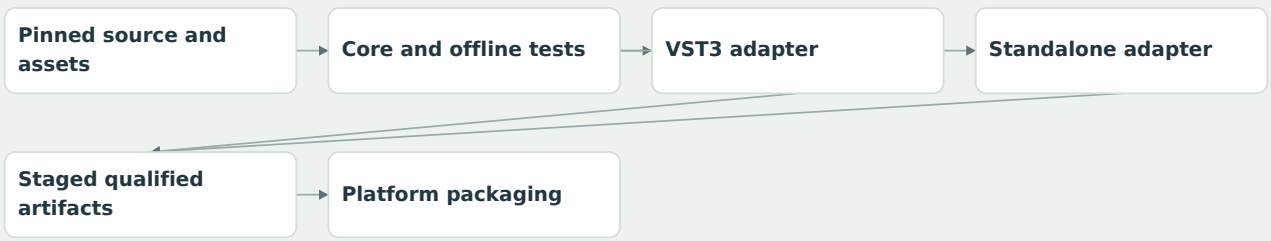
Keep the framework-independent core, offline renderer and engine tests separate from the plugin adapters. JUCE exposes VST3 and Standalone formats through its CMake API; the inspected documentation requires CMake3.22 or later. Its manifest-generation step loads the built plugin, so cross-compilation and signing order need explicit handling. [S066-01]

ORIGINAL DESIGN PROPOSAL. Create versioned configure/build/test presets for each native lane, with matching names and a release configuration. CMake distinguishes shared project presets from developer-local presets; select a preset schema supported by the pinned CMake version. Do not adopt the mutable documentation’s newest schema or prerelease toolchain automatically. [S066-02]

The commands below become runnable only after the implementation agent creates those presets, targets, registered engine_ tests and install rules. They are an ordered example, not commands executed on a finished instrument here. Disable automatic copying into live plugin folders during continuous integration. Stage complete bundles, inspect them, then package the qualified result.

```
# Proposed future Linux preset and JUCE target names
cmake --preset linux-x64-release
cmake --build --preset linux-x64-release --target engine_tests
ctest --preset linux-x64-release -R "^engine_" --output-on-failure
cmake --build --preset linux-x64-release --target TonewheelAtlas_VST3
cmake --build --preset linux-x64-release --target TonewheelAtlas_Standalone
cmake --install build/linux-x64-release --prefix stage/linux-x64
```

Build dependencies and release convergence



Original proposed build graph. Native runners and target names must be implemented; no plugin build is claimed.

CHECKS AND DECISIONS

- Require both adapters to identify the same engine/profile revision.
- Retain failed configuration and test logs; a missing test executable is not a skipped pass.

SOURCES S066-01 JUCE CMake API / S066-02 CMake presets manual

PLATFORM EVIDENCE / PROPOSED DISTRIBUTION PIPELINE

macOS: finish resources, then seal the product

Cross platform build packaging and release

A universal Mac product requires compatible arm64 and x86_64 code throughout its dependencies. Test each advertised native slice; running the Intel slice under Rosetta is a separate compatibility observation. A Mac VST3 is a bundle with executable and resources, not just a renamed library. [S066-10] [S066-07]

Apple specifies signing nested code from the inside out and applying entitlements to main executables, not library code. The standalone application's permissions therefore require separate review from the VST3 loaded inside a host process. Use the applicable DeveloperID workflow, secure timestamps and hardened-runtime configuration; no identity or entitlement set is assumed here. [S066-14] [S066-08]

ORIGINAL DESIGN PROPOSAL. If temporary signing is needed for JUCE's manifest helper to load the module, perform it before generating moduleinfo. Complete all resources before final distribution signing. Retain the notarization result and inspect its log. A ZIP cannot be stapled directly: staple supported contained items and recreate the distribution archive. [S066-01] [S066-09]

Ordered stage	Evidence retained
Assemble	Architectures, libraries, profiles, fonts, notices and final moduleinfo.
Sign and verify	Each code item's identity, timestamp, entitlement decision and signature result.
Notarize	notarytool submission identifier, final status and inspected log.
Staple and package	Ticket validation, final archive hash and exact distributed payload.
Clean-machine trial	Quarantined download, offline first launch, DAW scan and standalone device opening.

CHECKS AND DECISIONS

- A changed resource after final signing requires the signature to be re-established and rechecked.
- Do not report notarization or signature success without actual platform-generated evidence.

SOURCES S066-01 JUCE CMake API / S066-07 VST3 plug-in format structure / S066-08 Apple: notarizing macOS software / S066-09 Apple: customizing notarization / S066-10 Apple: universal macOS binaries / S066-14 Apple: creating distribution-signed code

INSTALLATION EVIDENCE / ORIGINAL PACKAGING POLICY

Windows and Linux: complete bundles and honest compatibility

Cross platform build packaging and release

Steinberg defines architecture-specific binaries inside VST3 bundles and separate user/global search roots. Ship the whole bundle, including resources and moduleinfo. Do not promote a successful scan in a developer’s custom directory into evidence that the installer selected the correct production location. Duplicate copies with the same processor identity must be detected during upgrade testing. [S066-06] [S066-07]

Microsoft SignTool supports signing, time stamping and signature verification. Use explicit SHA256 digest choices in the selected signing workflow; /pa selects normal Authenticode verification rather than driver policy. Sign the actual executable components and installer, then verify the distributed payload. A signature does not establish audio correctness or guaranteed reputation-screen behavior. [S066-11]

ORIGINAL DESIGN PROPOSAL. On Linux, freeze the distribution/toolchain baseline, audit dynamic-library and C++ runtime requirements, and test the chosen display and audio backends on clean systems. Package writable user presets separately from immutable factory assets. Offer a documented per-user installation path where supported; do not silently require administrator access for routine preset use.

Platform / scope	Documented VST3 root
Windows global	%ProgramFiles%/Common Files/VST3/
Windows per user	%LOCALAPPDATA%/Programs/Common/VST3/; documented primarily for development.
macOS per user / global	~/Library/Audio/Plug-ins/VST3/ or /Library/Audio/Plug-ins/VST3/
Linux per user	~/.vst3/; global roots differ across distributions.

CHECKS AND DECISIONS

• Test spaces, non-ASCII user paths, non-admin accounts, absent optional libraries and duplicate old copies.

• Do not bundle development SDKs, signing secrets or unlicensed archival scans in installers.

SOURCES S066-06 VST3 plug-in locations / S066-07 VST3 plug-in format structure / S066-11 Microsoft SignTool

ORIGINAL ACCEPTANCE MATRIX / ALL PRODUCT TESTS PENDING

Qualify the installed artifact, including upgrades

Cross platform build packaging and release

Steinberg’s validator checks VST3 conformity and can run custom tests. Treat it as one gate, then execute the project’s numerical, musical and device tests. A validator pass cannot establish a faithful B-3, a reliable standalone device selector or a stable GPU editor. Record the validator revision and the precise bundle tested. [S066-12]

ORIGINAL DESIGN PROPOSAL. Install the packaged candidate into clean qualification environments and run an advertised host/OS/architecture matrix. Save exact versions, native versus translated execution, sample rate, block size, bus configuration, audio driver and test case. Unsupported or unavailable combinations stay untested; they do not inherit another machine’s result.

Exercise upgrades with retained real project fixtures. Stable IDs and the chapter 065 state contract must preserve all four drawbar banks, routing, model identity and user presets. Reject incompatible future state cleanly. Keep user data separate so repair, rollback or uninstall can preserve the player’s work according to an explicit choice.

Gate family	Required installed-product cases
Discovery	Fresh scan, rescan, missing resources, duplicate versions and editor-free instantiation.
Audio contract	Zero-frame flush, varying buffers, offline rendering, tails, silence and bypass.
State and automation	Save/reopen, older schema migration, retained-bank automation and mid-gesture changes.
Standalone devices	Device loss/reconnect, rate changes, MIDI reconnect, unavailable saved device and safe recovery.
Interface and stress	Scaling, reduced motion, editor reopen, multiple instances and audio during UI load.
Upgrade and removal	Interrupted install, repair, rollback, preserved user data and absent stale bundles.

CHECKS AND DECISIONS

- Compare offline renders against the accepted engine revision after compiler or dependency changes.
- Keep acoustic acceptance separate from installation, signature and host-conformity results.

SOURCES S066-12 Steinberg VST3 validator

ORIGINAL RELEASE CONTRACT / LIMITS OF THIS EDITION

Release evidence is an artifact, not a checklist tick

Cross platform build packaging and release

ORIGINAL DESIGN PROPOSAL. Produce one release record that joins the source lock, build outputs, factory-profile provenance, test reports, licence notices and distributed hashes. Preserve unsigned build identity separately from signed and stapled artifact hashes: platform sealing changes the delivered bytes. Describe reproducibility at the level actually demonstrated; do not promise identical signed archives without a successful comparison.

An initial release may expose a nominal model profile while physical calibration remains incomplete, but that limitation must be visible in its evidence record and claims. No platform package, certificate, notarization submission, DAW session or standalone device test was produced for this chapter. The existing chapter 063 header compilation verifies a separate interface artifact, not a synthesizer build.

Assign a release owner to inspect the complete gate report and remaining limitations. If an advertised platform fails, repair it or narrow the support statement before distributing the candidate. A missing measurement, unexplained numerical change or unavailable test environment remains an explicit open item.

Release attachment	Minimum useful content
Build manifest	Source/dependency commits, toolchains, presets, architectures and options.
Product evidence	Exact installed artifact hashes, host/device matrix and failing or untested cases.
Model evidence	Reference profile, provisional values, calibration lineage and accepted numerical comparisons.
Distribution evidence	Platform signature verification, notarization where applicable and installer checks.
Notices and recovery	Dependency/image notices, migration policy, rollback instructions and retained symbols.
Claim statement	What was measured, computed, tested, proposed and still unknown.

CHECKS AND DECISIONS

- Recompute hashes after final packaging and verify them against the files actually delivered.
- A production release requires executed gates; the proposed procedures in this chapter are not passing results.

SOURCES S066-03 Steinberg VST3 licence / S066-04 JUCE framework licence / S066-09 Apple: customizing notarization / S066-11 Microsoft SignTool / S066-12 Steinberg VST3 validator

ORIGINAL ART DIRECTION / HARDWARE BOUNDARY

A sculptural instrument with readable controls

3D visual design and artistic direction

ORIGINAL DESIGN PROPOSAL. Use a warm walnut frame around a charcoal instrument deck, restrained brass legends, ivory controls and dark green inspection surfaces. The design should feel constructed: shallow panel recesses, narrow highlights, thick key fronts and projected drawbar caps. Its proportions are product-design choices, not dimensions recovered from a photograph. The generated art-direction image establishes atmosphere only; its key counts and legends are not engineering authority.

The vintage evidence requires independent manual drawbar banks, separate pedal controls, percussion tablets and shared scanner selection. Outward drawbar travel increases the marked setting; Great scanner routing includes pedals. The proposed screen follows these semantics while rearranging the deck for a compact editor. [S067-01]

The self-contained interface_prototype.html is an executable control study. It contains no audio renderer, microphone input or MIDI-device connection. A persistent NO AUDIO indicator prevents visual activity from being interpreted as measurement. This chapter establishes the design system; chapters 068–072 extend player workflow, inspection, assets, accessibility and independent implementation review.

Layer	Original design decision
Frame	Warm wood gradients; silhouette carries depth.
Deck	Dark panels with readable ivory and brass legends.
Controls	Continuous drawbar geometry and individually selectable shaped keys.
Inspection	Textual evidence states distinguish documented, proposed and unknown.
Scope	Control prototype; no completed VST3 or acoustic emulation.

AI-generated art direction. Use chapter control specifications, not image text or key counts.

SOURCES S067-01 Hammond factory service manual: controls and presets

STATE AND CONTROL DESIGN

Editing a bank must not silently select it

3D visual design and artistic direction

ORIGINAL INTERACTION CONTRACT. Display one nine-drawbar editor bank for each manual, with explicit A#/B editor buttons. Retain all four banks in state. The reversed preset octave chooses routing independently; a status line states both identities. A musician can prepare an inactive bank without overwriting the active registration. Stable semantic IDs from chapter 065 retain their manual, bank and footage when the view changes.

Each manual displays61 playing keys and 12 reversed preset keys; an expandable drawer displays25 pedals. Preset C is momentary Cancel, not a permanently depressed registration. This prototype uses ordinary exclusive selection. The physical multiple-latch case remains a distinct fidelity requirement from chapter 019. Fixed keys use the documented chapel matrix with its documentary status, not an assertion about the selected specimen.

The two pedal drawbars remain independent of manual Cancel. Four percussion controls expose enable, level, decay and harmonic separately; upper-B eligibility is visible. Scanner mode and Swell/Great routing remain separate. Global Normal/Soft volume differs from percussion Normal/Soft. Expression is labeled mechanical position so a percentage cannot masquerade as a calibrated amplitude law.

UI state	Invariant
Editor bank	Changes displayed values without selecting their route.
A#/B preset	Selects existing values without copying positions.
Fixed preset	Keeps its wired-profile registration while banks are edited.
Cancel C	Clears routing, preserving held keys and bank values.
Percussion	Displays eligibility without simulating envelope recharge.

CHECKS AND DECISIONS

- Inspect retained state after inactive-bank editing and route changes.
- Changing banks during a drag closes the old gesture before rebinding.

SOURCES S067-01 Hammond factory service manual: controls and presets

ORIGINAL MATERIAL AND RENDERER SPECIFICATION

Create depth without turning labels into textures

3D visual design and artistic direction

ORIGINAL RENDERING PROPOSAL. Separate geometry, finish, semantic text and interaction state. A drawbar cap, shaft and recessed track are independent layers. At marking0 the cap sits near the upper edge; at 8 it extends toward the player. Exposed shaft and numerical value follow the same state. Fractional values remain possible; the scale does not assert mechanical detents.

The prototype builds the deck with CSS and the cabinet and microphone plan with inline SVG. It requests no external fonts, textures or libraries. The cabinet is deliberately stylized: its cutaway is a teaching view, not a factory housing drawing. One horn is acoustically active and the opposite structure counterbalances it; the driver stays fixed. A measured asset workflow must replace proposed proportions with qualified geometry. [S067-05]

Use serif lettering for the title and selected headings, then clear interface text for settings, identities and evidence. Do not bake legends into raster artwork: scale, language and state must remain editable. A quiet-surfaces option reduces decorative shadows without altering controls. Proposed colors below define this product treatment; they are not calibrated paint or wood measurements.

Proposed token	Design role
#D6B67C	Warm structure/control highlight, not signal magnitude.
#242725	Low-glare panel background.
#F2E7CF	Primary text and pale control family.
#A5E4DF	Visible focus ring,3CSSpx in the prototype.
Geometry provenance	Original illustration until measured assets replace it.

SOURCES S067-05 Electro Music/CBS Leslie122 family manual044578

ORIGINAL INTERACTION IMPLEMENTATION

One gesture retains one semantic identity

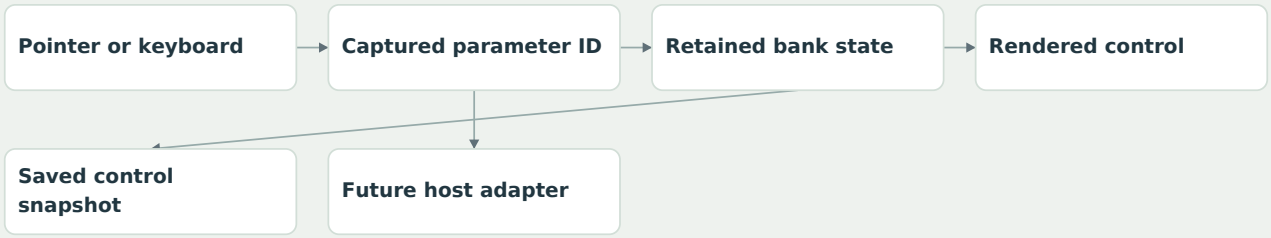
3D visual design and artistic direction

ORIGINAL IMPLEMENTATION. Pointer-down captures manual, bank, index and initial value. Every later movement addresses that captured identity. Lost capture, tab changes or bank changes close the gesture. Vertical movement maps continuously to marked travel, with finer Shift sensitivity; it does not map directly to gain. Keyboard arrows change values and Home/End choose limits. Named slider semantics expose current value and vertical orientation. [S067-02]

Play, Rotary and Service use explicit tab/tabpanel relationships, selected states and arrow navigation. These patterns guide implementation but do not certify screen-reader behavior. [S067-03] Playing keys retain distinct pointer/keyboard owners. Releasing one owner cannot release another owner’s held key. Browser deactivation releases UI-owned notes; ordinary internal focus movement does not manufacture note-offs.

Escape offers visible recovery from held preview keys. Save/restore records controls while excluding transient playing-key ownership. Stored values are validated before application rather than silently coerced. The future native editor must replace direct JavaScript changes with timestamped engine commands and acknowledgments. A browser control interaction cannot establish host automation or audio-thread correctness.

One gesture keeps one identity



Original control contract. Local state exists in the prototype; the host/audio bridge remains future implementation.

CHECKS AND DECISIONS

- Fractional drawbar movement must agree with its cap position and readout.
- Repeated reset or restore must not create extra physical-key ownership.

SOURCES S067-02 W3C ARIA Authoring Practices: Slider Pattern / S067-03 W3C ARIA Authoring Practices: Tabs Pattern

ORIGINAL INSPECTION AND MOTION DESIGN

Reveal the acoustic scene and its evidence

3D visual design and artistic direction

ORIGINAL WORKSPACE DESIGN. Rotary exposes two microphone positions in one cabinet coordinate system. Horizontal position, front distance and height update both coordinates and the plan view. The overview distance control addresses the same microphone-A value, avoiding an inconsistent second rig. The drawing states that it is not to scale. Coordinate bounds alone cannot prove physical clearance or acoustic validity.

Chorale and Tremolo remain the baseline cabinet commands. The visible horn and drum patterns are illustrative, with no RPM or real-acceleration claim. Historical optional brake accessories need their own profiles, so this baseline adds no brake button. Visual motion can be disabled and honors reduced-motion preference; controls stay usable. WCAG guidance supports suppressing nonessential animation, but actual conformance needs broader review. [S067-04]

Service has six substantive views: generator, contacts/routing, scanner, AO-28, Leslie and retained control state. Each separates available data from unmeasured parameters and links into the companion. AO-28 values are not advertised as complete connectivity; the scanner graph is a published nominal model. There is no animated waveform, fake level meter, simulated latency or authenticity percentage. Evidence labels retain textual meaning independently of color.

Workspace	Observable purpose
Play	Operate controls and inspect ordinary vintage routing.
Rotary	Edit proposed microphone geometry and cabinet command.
Service	Inspect source boundaries, evidence status and actual retained state.
Motion disabled	Keep command and numeric state visible while decoration stays still.

SOURCES S067-04 W3C Understanding WCAG2.2: Animation from Interactions / S067-05 Electro Music/CBS Leslie122 family manual044578

DELIVERABLE / INDEPENDENT REVIEW GATES

Validate the controls before connecting DSP

3D visual design and artistic direction

The HTML contains the designed interface, local state, snapshots, original SVG artwork, service panels and keyboard/pointer handlers. Its JavaScript parsed successfully in Node during authoring. That establishes syntax only. Chromium rendering was attempted but the runtime blocked socket creation; the environment’s automatic approval policy rejected escalation. This assignment therefore claims no observed browser layout or screenshot pass. Chapter 072 owns the independent review and records its actual evidence separately.

ORIGINAL ACCEPTANCE PLAN. Inspect1440×1000 and 900×900 layouts, then small-window and zoom behavior. Count generated keys and controls; verify every footage and percussion label. Test pointer release and focus through tabs, Cancel, bank editing, snapshot restore and window deactivation. Observe errors and unexpected requests. Separate control acceptance from engine, host and physical-instrument tests.

In the production renderer, keep decoration and animation off the audio thread. A frame stall must not change source phase, rotor trajectory, parameter identity or a pending note release. Add measured geometry only after acquisition gates pass. The visual layer can report control positions and evidence now; calibrated signal behavior arrives only with the implemented and validated engine.

Required gate	Evidence needed
Inventory	61+61 playing,12+12 preset and 25 pedal keys;38 retained drawbar values.
Bank identity	Inactive edits and mid-drag bank changes preserve ownership.
Accessibility	Keyboard, focus, motion, zoom and assistive-technology checks.
Visual QA	Observed layouts at declared sizes, without clipping or unreadable overlays.
Host and audio	Later independent reports; never inferred from UI success.

SOURCES S067-02 W3C ARIA Authoring Practices: Slider Pattern / S067-03 W3C ARIA Authoring Practices: Tabs Pattern / S067-04 W3C Understanding WCAG2.2: Animation from Interactions

HARDWARE EVIDENCE / ORIGINAL DESIGN

Build the workspace around the player's next action

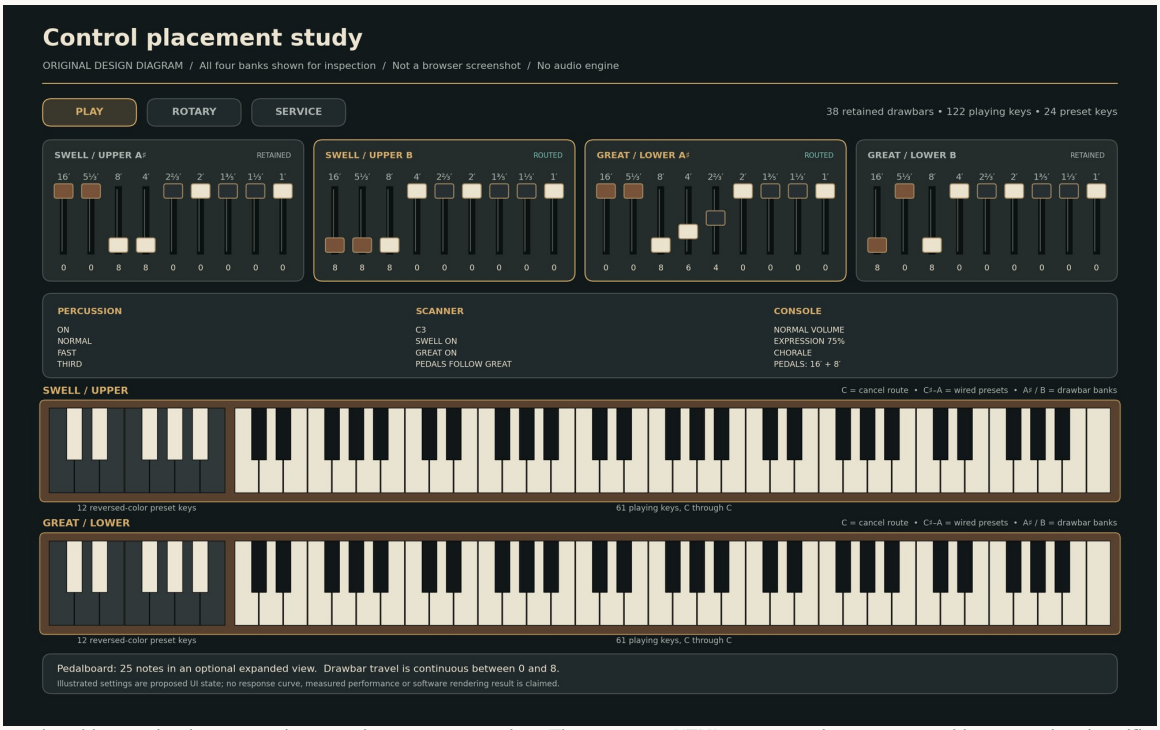
Player workspace interaction design

The reference console has two 61-key playing manuals, twelve reversed preset keys per manual, four nine-drawbar banks, two pedal drawbars and a 25-note pedalboard. Continuous drawbar travel must survive the numbered 0–8 presentation. These documented counts define the control inventory, not mandatory screen dimensions. [S068-01]

ORIGINAL DESIGN PROPOSAL. Keep PLAY focused on choosing a registration, shaping it and changing rotary speed while performing. Show upper and lower manuals together, each with its sounding registration and the identity of the bank being edited. Put the pedal drawer and expression control within reach without forcing a permanent miniature reproduction of the entire bench and floor assembly.

Use PLAY, ROTARY and SERVICE as persistent views over one instrument state. Changing view must never recall a sound. Apply the W3C tab relationships and keyboard pattern to the browser prototype; native implementations need equivalent platform accessibility semantics. Preserve each view's last meaningful focus. Keep NO AUDIO visible in the present interaction prototype; an attractive cabinet animation does not establish an implemented synthesizer. [S068-04]

Workspace zone	Proposed player-facing responsibility
Registration strip	Name sounding route, edited bank and unsaved changes independently.
Manual controls	Nine full-size editable drawbars; inactive-bank summary remains visible.
Performance controls	Percussion, scanner routing, expression and rotary command remain discoverable.
Secondary drawers	Pedalboard, input routing and saved sounds expand without changing sound.



Original control and inspection layout study, not a browser screenshot. The separate HTML prototype has not passed browser visual verification; no audio engine is implemented.

CHECKS AND DECISIONS

- Count playing and preset keys separately; do not count the reversed octave as part of 61.
- Switch among all views while holding notes; persistent control and ownership state must remain unchanged.

SOURCES S068-01 Hammond Organ Service Manual: console controls and manual chassis / S068-04 W3C WAI-ARIA Authoring Practices: Tabs Pattern

ORIGINAL PERFORMANCE WORKFLOW

Prepare an inactive registration without changing the sounding one

Player workspace interaction design

ORIGINAL DESIGN PROPOSAL. Separate editorBank from soundingRoute. A player accompanying on lower B can prepare A# and return to editing B without hearing the preparation. Selecting the physical A# preset then requests the routing change. Display EDIT A# and SOUNDING B together during preparation. A wide layout may show both banks; compact views must preserve their identities.

Use chapter 065's registry paths, such as organ.upper.a_sharp.drawbar.16. The VST3 ID must remain stable; a bank hidden by the custom editor must not acquire kIsHidden, because that host flag prevents external changes and automation. Close an existing edit gesture against its original ID before rebinding a bank, then require a fresh gesture. Host-originated display updates must not create a second edit. [S068-02] [S068-07]

C is a momentary routing cancel, not a note release. Fixed wired presets and A#/B adjust selections have different meanings; simultaneous latches require the explicit physical-state model. [S068-01] Keep modern Save Registration and Save Performance commands outside the reversed octave. Show the scope before recall: registration, complete performance settings, or instrument calibration profile. Copying drawbar positions must never silently copy mic placement or replace the specimen profile.

Bank editing and sounding selection are independent

Original workflow proposal. Opening an editor bank changes no audio routing; preset selection changes routing. Stored positions in both banks survive every transition.

CHECKS AND DECISIONS

- Automate an inactive bank, reveal it, and verify its retained values without a routing change.
- Cancel clears the selected manual's routing mask while held-note ownership remains intact.

SOURCES S068-01 Hammond Organ Service Manual: console controls and manual chassis / S068-02 Steinberg VST3 ParameterInfo reference / S068-07 Steinberg VST3 IComponentHandler reference

ORIGINAL INPUT AND FOCUS CONTRACT

Make keyboard gestures reliable at every window size

Player workspace interaction design

ORIGINAL DESIGN PROPOSAL. Treat the screen manuals as playable controls and routing monitors. Offer a full-range overview plus a magnified playing region when needed. Preserve physical-key indices through zoom and scrolling. Expose octave names as a display preference; they must not alter pitch, wiring or event ownership. The pedal drawer contains 25 indexed keys and both pedal drawbars.

A browser pointer can be captured and later canceled or lose capture. Those events require explicit handling; pointer support does not automatically provide keyboard accessibility. [S068-05] Maintain one ownership record per UI pointer or computer-key press. During a glissando, release that owner's old key before requesting its next key at the defined event time. Closing the editor releases UI-owned notes only; it must not release a note still owned by external MIDI.

Use roving keyboard focus within each manual, with a visible focused key and separate held-state indicator. Capture the original note at key-down so its key-up cannot release a newly focused note. Disable computer-key performance shortcuts inside text fields. Provide a clearly named Panic action distinct from preset Cancel; preserve the ownership and tail policies of chapter 024.

Interaction	Proposed invariant
Pointer exits manual	Capture tracks the gesture; cancellation releases its owned key.
Two pointers, same key	One owner's release cannot release the other's hold.
Focus moves during hold	Release resolves the press record, not current focus.
Resize or zoom	Coordinate mapping changes; physical key identity does not.
Editor closes	UI ownership ends; external ownership and engine clocks follow their own contracts.

CHECKS AND DECISIONS

- Exercise off-window release, multi-touch overlap, text entry and view changes without stranded UI notes.

SOURCES S068-05 W3C Pointer Events Level3 Recommendation

CONTROL FEEDBACK / ORIGINAL DESIGN

Explain control interactions without rewriting their positions

Player workspace interaction design

The four percussion tablets select enable, Normal/Soft, Fast/Slow and Second/Third; eligibility is upper B. Great scanner routing includes the lower manual and pedals. These interlocks should be visible in the interface. [S068-01]

ORIGINAL DESIGN PROPOSAL. Keep a percussion switch's stored position visible even when the current route makes it inactive. Show a concise reason, such as Requires upper B. Where the selected circuit suppresses the upper B bank’s 1-foot audio contribution, annotate the consequence without moving that drawbar to zero or deleting its contact state. An unimplemented prototype must not invent an envelope-charge meter or assert a measured decay time.

Present drawbars as continuous travel with legible markings, editable numerical values and a fine-adjust mode. Pulling toward the player increases extension. Provide named sliders with current value, bounds and orientation; document the chosen keyboard direction and test touch assistive technology. [S068-03] Keep physical expression travel separate from software output trim, because these controls have different model responsibilities. A preset recall must not silently change either unless its displayed recall scope includes them.

Visible situation	Proposed explanation and action
Percussion enabled, upper fixed preset	Show stored enable state and route eligibility; offer explicit upper-B selection.
Wired preset sounding	Label drawbar edits as preparation; do not suggest the fixed registration follows them.
Scanner mode selected, routes off	Retain mode; show that neither manual path currently uses it.
Expression versus trim	Name physical travel and software gain separately; show their different units.

CHECKS AND DECISIONS

• Verify displayed positions against retained state, and audible eligibility against the applied routing snapshot.

SOURCES S068-01 Hammond Organ Service Manual: console controls and manual chassis / S068-03 W3C WAI-ARIA Authoring Practices: Slider Pattern

ROTARY WORKFLOW / ORIGINAL DESIGN

Keep rotary performance immediate and microphone editing deliberate

Player workspace interaction design

The vintage 122 manual describes separate bass and treble two-speed drives, with Chorale and Tremolo commands. [S068-06] ORIGINAL DESIGN PROPOSAL. Keep that two-state command within reach in PLAY and ROTARY. Show requested mode separately from each rotor's evolving state; do not animate both as one rigid object.

ROTARY should expose microphone selection, position, orientation and gain through a cabinet view plus equivalent numerical controls. Use the declared coordinate system and metres. Chapter 065 supplies proposed mic1–mic3 position/gain identities; a pictured microphone's placement does not establish a measured rig. Both rotor paths may reach each microphone. Identify illustrative animation as preview data until it is driven by actual engine telemetry.

Stage controls should support a deliberate sequence: select mic, inspect coordinates, move it, audition, then compare. A/B comparison must state which settings it swaps and retain each rig's provenance. Do not conceal automatic level matching inside a distance control. Add Stop only under an explicitly selected accessory or extension profile; historical brake accessories exist, so the interface must not describe every stop facility as a modern invention. [S068-08]

Action	Proposed state boundary
Choose Chorale/Tremolo	Change target command; retain separate rotor histories.
Select a microphone	Change editing focus only.
Move a microphone	Validate geometry before publishing the new rig; never reset rotor phase.
Open cabinet inspection	Change visualization only; preserve performance settings.

CHECKS AND DECISIONS

- Provide a non-drag alternative for every microphone edit and preserve operation with motion disabled.

SOURCES S068-06 Electro Music Leslie 122-family operating and maintenance manual, 044578, June 1971 / S068-08 Leslie Brake Accessory instructions 038398

PROPOSED ACCEPTANCE / NOT RUN

Accept the workspace through musical tasks and state traces

Player workspace interaction design

ORIGINAL VALIDATION PLAN. Give organists and keyboard users concrete tasks: maintain lower-manual accompaniment while preparing the upper solo bank, switch rotary speed during a held chord, recover from a canceled registration, and restore a saved performance. Record wrong-bank edits, missed targets, accidental recalls, stranded notes, completion time and participant explanations. Do not invent success percentages or infer audio fidelity from usability.

Compare the sculpted interface with its clear two-dimensional presentation at the actual supported window sizes and operating-system scaling. Collect keyboard-only and assistive-technology observations separately from mouse results. Prototype interaction tests and production host automation tests remain different evidence. Chapters 071–072 define the later implementation review; this chapter supplies proposed tasks, not completed participant findings.

Deliver a task script, state-transition trace, annotated failure recordings and a prioritized fix list. Freeze the semantic bindings only after checking the actual wrapper. Preserve visual polish where it improves recognition; remove perspective, motion or decoration where it obscures the current registration or blocks an action.

Proposed acceptance task	Required evidence
Prepare inactive bank	Sounding route unchanged; intended bank's stable IDs alone edited.
Switch banks mid-drag	Original gesture closed once; no transferred gesture or stuck automation.
Cancel while holding chord	Routing cleared; physical-note ownership preserved.
Release outside window	No orphaned UI ownership; external MIDI unaffected.
Restore after editor close	All retained banks and selected routes agree with processor state.
Operate without animation	Every musical control remains identifiable and usable.

CHECKS AND DECISIONS

• Mark every task not_run until its traces or observations exist; no host, musician-panel or acoustic test occurred in this assignment.

SOURCES S068-02 Steinberg VST3 ParameterInfo reference / S068-03 W3C WAI-ARIA Authoring Practices: Slider Pattern / S068-04 W3C WAI-ARIA Authoring Practices: Tabs Pattern / S068-07 Steinberg VST3 IComponentHandler reference

ORIGINAL INTERFACE PROPOSAL

Make SERVICE an evidence workspace with a visible scope

Instrument inspection and service view UX

SERVICE should answer four questions: which physical assembly is represented, which evidence supports it, which model is running, and which observations remain unavailable. Keep the current instrument/profile identity, revision, unsaved-draft state and engine connection visible above the workspace. A convincing valve rendering must never substitute for this information.

Use a selectable assembly tree beside a restrained exploded illustration: generator, manuals, presets, scanner, AO-28, cabinet amplifier, divider, horn and bass assembly. Selection synchronizes a parts list and signal map. Preserve separate identifiers for the physical item and its model module; one valve can contain sections serving different responsibilities. The factory AO-28 drawing supplies circuit roles, not a complete three-dimensional manufacturing model. [S069-01]

Provide Parts, Signal, Profiles and Validation workspaces within SERVICE. Opening an inspection view does not change the sound or enable a diagnostic excitation. Keep performance controls available through a compact strip. In this research prototype, engine-dependent panels must display unavailable states; an optional synthetic illustration must retain its label when enlarged or exported.

Visible state	Meaning and action
No engine connected	Browse evidence and proposed controls; no live signal or health claim.
Nominal model active	Inspect calculated behavior and the exact provisional profile.
Capture loaded	Show the specimen, calibration and acquisition boundary before comparing.
Reference unresolved	Open the missing identity or acquisition record; do not silently select a substitute.

CHECKS AND DECISIONS

- Selecting an assembly leaves all musical controls unchanged.
- Every view identifies whether its contents are documentary, computed, synthetic or measured.

SOURCES S069-01 Hammond factory service manual: AO-28 drawing and service-output boundary

PARTS INSPECTOR AND REVISION HANDLING

Let one component reveal its complete evidence trail

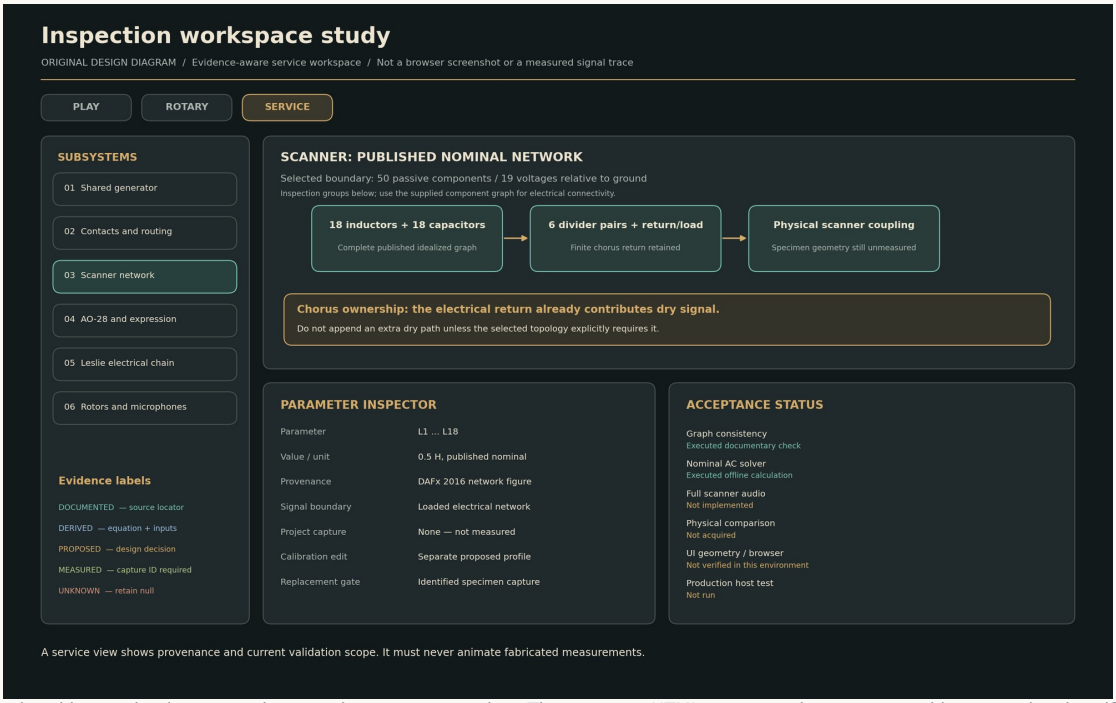
Instrument inspection and service view UX

ORIGINAL DESIGN PROPOSAL: each selectable item opens an inspector containing assembly callout, circuit designator, manufacturer stock number, applicable revision, nominal source value, current model value and unresolved fields. These are separate fields, not interchangeable search aliases. Use data/ao28-component-values.json alongside the stock and stage inventories; their presence does not establish a connectivity-complete AO-28 netlist.

An unresolved value such as the recorded R62 ambiguity opens both candidate readings and their locators. It remains unknown until reviewed evidence resolves it. A user can create a proposed model value in a draft, but cannot overwrite the source transcription. Mutually exclusive revision records appear as alternatives; an inactive alternative is not a physically missing component.

The evidence drawer links source_catalog.json and drawing_locator.html with document identity, printed page, PDF ordinal, inspected-access status and rights note. W3C PROV distinguishes entities, activities and derivation. Apply that distinction to the original image, a transcription, a calculation and a fitted parameter. Preserve their relationships instead of displaying a single authenticity score. [S069-02]

Evidence badge	Inspector behavior
DOCUMENTED	Read-only source fact with variant and exact locator.
MEASURED	Capture, calibration, conditions and uncertainty links.
DERIVED / PROPOSED	Calculation recipe or editable design choice remains explicit.
UNKNOWN	Null value, reason and next acquisition; never a zero-valued control.



Original control and inspection layout study, not a browser screenshot. The separate HTML prototype has not passed browser visual verification; no audio engine is implemented.

CHECKS AND DECISIONS

- A mesh hotspot and its keyboard-accessible list row resolve to the same item ID.
- Retouched/context photographs cannot enable a calibrated dimension readout.

SOURCES S069-02 W3C PROV-DM: The PROV Data Model

SIGNAL INSPECTION AND MEASUREMENT EVIDENCE

Compare traces only after their boundaries agree

Instrument inspection and service view UX

ORIGINAL DESIGN PROPOSAL: a probe selector names both terminals or the microphone position, polarity/reference convention, units, load, model stage and timebase. In the factory service procedure the voltmeter spans the preamplifier G–G output. Its table must not become a raw-pickup overlay merely because both curves are indexed by generator source number. [S069-01]

Overlay a calculated response and a capture only after identity, boundary, gain calibration, timing correction, control state and analysis settings agree. Otherwise display the mismatch beside a disabled comparison action. Preserve raw recordings; windowing, resampling, delay removal and normalization produce derived assets with parent hashes and recipes. A display-only alignment must never alter the stored evidence or silently redefine the tested delay.

NIST distinguishes uncertainty from error and includes covariance where appropriate. Show residuals separately from uncertainty information; unknown uncertainty is not zero. Curve labels and line patterns distinguish origins independently of color. Provide a readable numerical table and export the same data and analysis settings shown by the plot. [S069-03] [S069-08]

Readout	Required qualification
Waveform / spectrum	Origin, units, rate, window, bandwidth and absolute or relative time.
Voltage / pressure	Actual calibration and port reference; otherwise retain normalized units.
Transfer / residual	Input-output pair, load, denominator validity and excluded coverage.
Rotor trajectory	Commanded, simulated or independently recorded angle; direction and timebase.

CHECKS AND DECISIONS

- A model-generated trace never gains a MEASURED badge through an overlay selection.
- Missing or incompatible capture metadata prevents a physical-fidelity result.

SOURCES S069-01 Hammond factory service manual: AO-28 drawing and service-output boundary / S069-03 NIST TN1297: Combined Standard Uncertainty / S069-08 W3C Understanding WCAG2.2 SC1.4.1: Use of Color

CALIBRATION WORKFLOW AND UNDO

Edit a draft, validate it, then activate its exact version

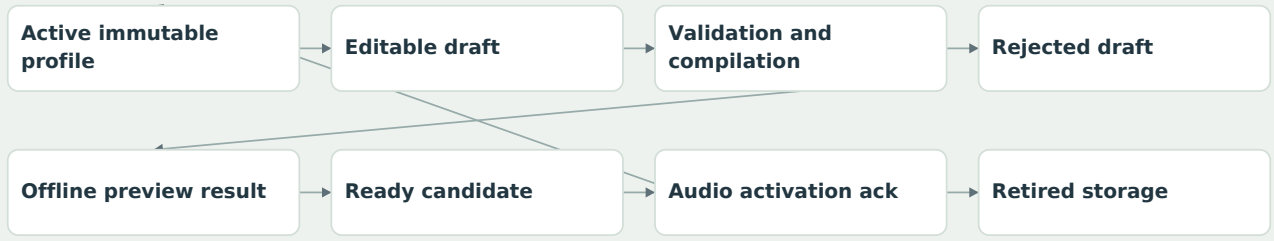
Instrument inspection and service view UX

ORIGINAL DESIGN PROPOSAL: separate the active immutable profile from an editable draft. Editing a component or acoustic parameter records its old value, proposed value, unit, reason and provenance. A documented value remains visible alongside the proposal. Calibration is a reproducible fitting operation with designated captures; moving a slider does not make a value measured.

Validate schema, references, units, revision constraints, effect ownership and algorithm-required fields before compilation. Render an offline comparison using declared initial conditions and an identical event timeline. Show pending, rejected and ready states distinctly. A passing preview validates only the selected tests; it cannot certify physical fidelity. If a newer edit exists when a worker finishes, discard the obsolete result without replacing the active profile.

JUCE UndoManager groups actions into transactions. Use that mechanism, or an equivalent original command log, on the control side for a complete edit gesture and its metadata. Undoing a draft is separate from requesting activation of an earlier profile. Reuse chapter 063’s publication and retirement contract; incompatible state migration requires an explicit stopped rebuild. [S069-05]

One versioned edit transaction



Original proposal. Worker results carry the draft version; audio-thread activation and off-thread storage retirement are separate events.

CHECKS AND DECISIONS

- Undo never mutates raw captures or source documents.
- The visible active profile changes only after the matching activation acknowledgement.

SOURCES S069-05 JUCE UndoManager class reference

REAL-TIME TELEMETRY CONTRACT

Observe the engine without making the display its dependency

Instrument inspection and service view UX

PortAudio documents callback restrictions covering allocation, I/O and potentially blocking operations. ORIGINAL DESIGN PROPOSAL: collect only bounded diagnostic data in the audio path. The interface consumes snapshots; it never traverses live solver objects, formats log messages inside processing, loads images or waits for a GPU fence there. [S069-06]

A snapshot carries profile token, signal-port identity, sample-frame range, units, sequence number, solver-failure counts and telemetry-loss count. Preallocated waveform slots retain only an admitted number of probes. Perform FFTs, plotting, file export and detailed comparison on a worker. Properly filter before signal decimation; min/max display envelopes are not analysis-quality downsampled audio.

JUCE AbstractFifo supplies single-reader/single-writer index management and does not own data storage. Use a separately allocated bounded buffer and one consumer that distributes copies to views. A full telemetry queue may drop a diagnostic packet and report the gap; it must not block audio or discard musical events. Reconfigure buffers only through the qualified lifecycle. [S069-04]

Condition	Required display response
Stale snapshot	Retain last timestamp and visibly mark stale; do not animate continuity.
Sequence gap	Break plotted continuity and show lost diagnostic coverage.
Solver failure	Identify scope and count; do not invent a physical component diagnosis.
Editor closed	Engine timing and musical output remain independent of the view.

CHECKS AND DECISIONS

- Freeze the UI and saturate telemetry transport without delaying the audio callback.
- A CPU average or smooth meter cannot stand in for a deadline or xrun test.

SOURCES S069-04 JUCE AbstractFifo class reference / S069-06 PortAudio: Writing a Callback Function

IMPLEMENTATION ORDER AND PROPOSED ACCEPTANCE

Ship inspection with falsifiable states and reproducible exports

Instrument inspection and service view UX

Implement the evidence-only inspector first, using the supplied inventories, source catalog and exact drawing locators. Next add draft-profile editing against the chapter 058 schema. Integrate offline analysis and the chapter 063 engine bridge only when they exist. Finally add live telemetry and physical-reference comparisons. The static prototype can demonstrate navigation and unavailable states without pretending these later systems have run.

Keep inspection metadata separate from stable performance automation. A component-selection click does not create a host gesture. Advanced profile changes that alter topology, latency or calibration belong to the validated profile workflow, with state migration defined before release. Preserve selection and focus across refreshes; a completed compilation should not move focus into a plot.

W3C status-message guidance supports announcements without taking focus. Announce meaningful outcomes such as rejected profile or missing capture, and provide a persistent text explanation. Do not announce every telemetry update. Export a review bundle containing profile/capture identities, source references, analysis settings, actual results and unresolved gates. A screenshot alone is insufficient evidence. [S069-07]

Proposed gate	Observable result
SC-01 Unknown datum	Null, reason and acquisition task survive display, editing and export.
SC-02 Revision switch	Incompatible records do not merge; active profile remains identified.
SC-03 Late worker result	Obsolete draft result cannot activate or relabel the current graph.
SC-04 Probe mismatch	Different port/unit/clock conventions cannot produce a valid comparison.
SC-05 Accessible status	Keyboard/list selection and meaningful result announcements agree.

CHECKS AND DECISIONS

- These are proposed UI/engine gates, not executed acceptance results in this chapter.
- No inspected part, diagnostic meter or plot implies an implemented audio engine.

SOURCES S069-07 W3C Understanding WCAG2.2 SC4.1.3: Status Messages

HARDWARE EVIDENCE / ORIGINAL DESIGN PROPOSAL

Bind every surface to evidence

3D assets, materials and camera specification

The factory envelope includes bench and pedal keyboard; it is not the bare console bounding box. The walnut-cherry finish designation does not establish substrate species, veneer thickness or coating chemistry. P002 supplies useful furniture relationships; P003's colored lighting cannot establish calibrated albedo. [S070-01] [S070-02] [S070-03]

ORIGINAL DESIGN PROPOSAL: maintain separate specimen geometry and product styling collections. Tag every mesh as documented, measured, inferred or artistic, with specimen, capture state and evidence locator. Exclude retouched P005 and P020 from dimensional reconstruction. Their disclosed extrapolation and perspective/width changes defeat that use. The generated UI art-direction image supplies palette ideas only; chapter 067 and the parameter registry govern controls. [S070-10] [S070-11]

Capture front, rear, side and top views plus overlapping obliques of the same assembly state. Retain raw images, lens calibration, exposure, color target and scale bars in the relevant part plane. Add raking-light macros for relief and direct measurements for glossy, dark or occluded surfaces. Reconstruction confidence must not become an invented dimension. Validate on photographs withheld from fitting.

Capture set	Required output
Console / bench / pedals	Separate envelopes, structural landmarks and assembly states
Keys / drawbars / switches	Travel endpoints, pivots, cap profiles and readable legends
Cabinet / louvers / rotors	Stationary-versus-moving IDs, axis locations and clearances
Wood / cloth / coated surfaces	Color-controlled views, grain direction and scale references

CHECKS AND DECISIONS

• Every accepted dimension has endpoints, units, uncertainty and specimen identity.

• Historical patent illustrations are tagged as invention embodiments, never substituted for production 122 drawings.

SOURCES S070-01 Hammond factory service manual: model identification and B-3 parts / S070-02 Hammond-museum-B-3.png, P002 / S070-03 Hammond B3 & Drawbars.jpg, P003 / S070-10 Retouched B-3 pedalboard derivative, P005 / S070-11 Retouched Leslie 122 isometric derivative, P020

ORIGINAL IMPLEMENTATION CONTRACT

Export a coherent hierarchy and moving pivots

3D assets, materials and camera specification

glTF uses a right-handed frame, metres and radians, with Y upward and the asset front toward positive Z; names need not be unique. [S070-05] ORIGINAL DESIGN PROPOSAL: chapter 004's authoring frame uses x across the console, y rearward, z upward. Apply the conversion below exactly once. It preserves handedness; merely exchanging y and z would reflect the model.

Store persistent asset IDs in the manifest and node extras, with exported node-index bindings. Keep these separate from manufacturer part numbers and automation IDs. For example, asset control.upper.a_sharp.16 binds to parameter organ.upper.a_sharp.drawbar.16. Retain every bank even when its controls are offscreen; never bind parameters by mesh order.

Scene roots b3 and leslie122 own separate stationary, moving and service collections. Shared fastener meshes retain individual transforms. P024 illustrates a stationary speaker above a rotating deflector, but does not identify exact 122 geometry. Renderer animation consumes timestamped audio-state snapshots; it never advances the acoustic rotor trajectory. [S070-04]

Proposed node family	Motion contract
b3/manual/{upper,lower}/key/*	Rotate about measured hinge; independent key identity
b3/control/{manual}/{bank}/*	Translate on local pull axis; retain cap dimensions
b3/pedals/* and expression	Measured local pivots and bounded travel
leslie122/rotor/{horn,bass}	Measured axes; stationary drivers remain siblings

MODEL EQUATIONS

p_gltf = (x_author, z_author, -y_author); conversion determinant = +1

P002 | Hammond-museum-B-3.png | Фікcер | CC BY-SA 4.0 Source and license

CHECKS AND DECISIONS

- A labelled unit cube, front marker, winding and tangent frame survive export.
- Pulling a drawbar toward the player follows its recorded endpoint transform without scaling its cap.

SOURCES S070-04 Lowerrotor.jpg, P024 / S070-05 Khronos glTF 2.0 specification, official AsciiDoc source

ORIGINAL MATERIAL RECIPES

Specify appearance without inventing material chemistry

3D assets, materials and camera specification

ORIGINAL DESIGN PROPOSAL: material names describe appearance, while separate evidence fields identify verified construction. A warm walnut-style finish is a proposed skin; it does not prove solid walnut or any particular veneer. Keep panel grain direction, seams and observed repairs distinct. Do not project P003's stage illumination into every surface. [S070-03]

The following metalness choices are proposed renderer defaults, not measured physical properties. Tune roughness using neutral-light comparison captures, retaining uncertainty. Use a dielectric paint layer over metal; expose a metallic layer only where a documented or deliberately artistic wear mask reveals it. All wear must carry the same evidence status as the base surface.

Proposed material ID	Recipe and evidence boundary
mat.case.finish	Wood-color map, directional grain relief, roughness; metalness 0. Veneer edges only if observed.
mat.keycap / mat.drawbar.cap	Separate ivory-color, black, brown and white appearance assets; metalness 0. Plastic chemistry unknown.
mat.trim.brass / mat.chassis.paint	Bare brass-colored trim: metalness 1; painted face: 0. Neither asserts an alloy or original finish.
mat.cloth / mat.rotor.surface	Cloth weave and rotor coating kept distinct; metalness 0, measured scale pending.
mat.valve.glass	Simplified reflective player-view shell; optional inspection transparency, explicitly a visual approximation.

CHECKS AND DECISIONS

- Example valve photographs never determine fitted branding or a specimen's valve revision.
- Any valve glow is decorative or linked to a declared model variable, never presented as a measured voltage indicator.
- Neutral and product lighting reveal the same material assignments and surface boundaries.

SOURCES S070-03 Hammond B3 & Drawbars.jpg, P003

OFFICIAL FORMAT / ORIGINAL ASSET PIPELINE

Package UVs, textures and levels of detail

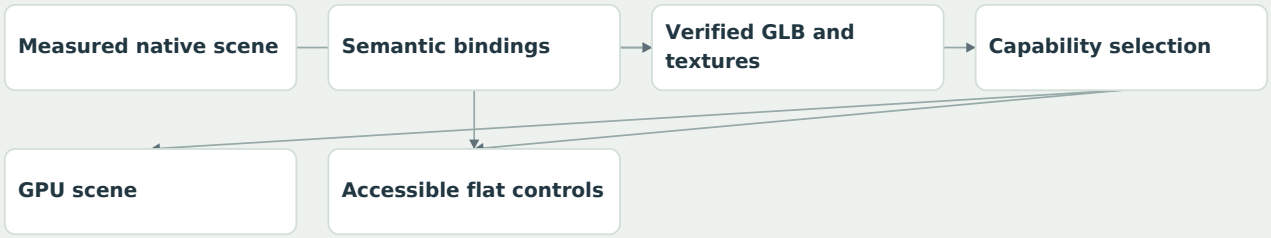
3D assets, materials and camera specification

glTF base-color RGB and emissive textures use sRGB; base-color alpha is linear and RGB is not premultiplied. Factors and vertex colors are linear. Normal maps use linear tangent-space data; roughness occupies green, metalness blue, and occlusion red when packed together. [S070-05]

ORIGINAL DESIGN PROPOSAL: name maps case_finish_basecolor, case_finish_orm and case_finish_normal; store transfer functions in asset_manifest.json. Preserve grain scale per panel, put UV seams at real joins, and keep reflections out of base color. Shade in linear space, apply one recorded display transform, then composite readable UI labels independently of scene exposure. Generate mip levels and inspect cloth/wood shimmer. Use screen-space interaction proxies so simplification cannot shrink usable targets.

Export LOD0 for inspection, LOD1 without closed-case internals, and LOD2 with baked furniture relief; retain identical semantic bindings, active controls and pivots. KHR_texture_basisu supports KTX2/Basis textures transcoded into supported platform formats; conventional fallback textures are possible. [S070-06] Select through a capability manifest, not filename assumptions. Measure decoded resident memory, including mip levels and render targets; compressed download size is not that budget. Pin exporters and compression settings.

One evidence model, multiple render paths



Original proposed delivery architecture; these asset bundles have not been produced or benchmarked.

CHECKS AND DECISIONS

- A channel-test material detects swapped ORM channels and incorrect color decoding.
- Each LOD preserves selection, parameter identity, pivot orientation and declared interaction bounds.

SOURCES S070-05 Khronos glTF 2.0 specification, official AsciiDoc source / S070-06 Khronos KHR_texture_basisu extension

ORIGINAL RUNTIME PROPOSALS

Keep cameras useful and rendering expendable

3D assets, materials and camera specification

ORIGINAL DESIGN PROPOSAL: player mode uses a fixed orthographic camera; inspection uses orbit with a proposed initial 35-degree vertical field of view. Fit clipping planes to visible bounds. Store camera transforms separately from acoustic microphone coordinates. Use a broad soft key, restrained fill and a quiet rim; avoid reflections across legends. A charcoal surround and brass-colored accents can follow the concept image without copying its incorrect controls.

The candidate budgets below are unmeasured starting targets per editor, not hardware specifications or performance claims. Profile actual host, OS, GPU, display scale and instance count. Select quality without changing audio parameters. Supply fully operable flat controls when rendering is unavailable, with matching gestures, values and focus behavior.

JUCE can recreate contexts on hide/show and requires detachment before renderer/component destruction. [S070-07] Rebuild GPU resources from an immutable CPU bundle; keep loading, cleanup and synchronization off the audio callback. Read-only snapshots must tolerate dropped frames. The web preference prefers-reduced-motion expresses reduced nonessential animation. [S070-08] Provide a native equivalent plus an explicit setting; pause decorative rotation and camera motion independently of sound.

Proposed initial budget	Measurement or behavior
100,000 visible triangles; 120 draw calls	Count actual draws, including extra passes
64 MiB resident texture budget	Include mip levels; log uncompressed fallback residency
60 fps normal; 30 fps reduced quality	Report frame-time distribution on named systems
Idle / hidden / motion-reduced	Event-driven refresh; stop decorative repaint requests

CHECKS AND DECISIONS

- Closing and reopening the editor reconstructs resources while audio state persists.
- Changing camera, LOD or lighting cannot change serialized instrument parameters.

SOURCES S070-07 JUCE OpenGLContext class reference / S070-08 W3C Media Queries Level 5, Working Draft 19 February 2026

PLANNED DELIVERABLES / UNPERFORMED GATES

Deliver a reproducible asset pack with acceptance evidence

3D assets, materials and camera specification

ORIGINAL DESIGN PROPOSAL: deliver the named files below only after replacing placeholders with traceable assets. Include hashes, source attribution, alteration history, exporter revision, extension requirements and unresolved geometry. Source photographs are not automatically suitable for shipping as commercial texture maps; preserve each asset's actual license and obtain appropriate rights for new captures.

Use the official glTF Validator for format checking. [S070-09] That does not establish cabinet accuracy, material identity, accessible interaction or runtime performance. Compare held-out silhouettes and calibrated landmarks, separating pose/lens error from mesh error. In the equation, P_i is a measured model-space landmark, q_i its image location, and project includes camera calibration; E_px is pixel RMS residual. Set acceptance bounds from acquisition uncertainty and intended display resolution before fitting.

No physical dimensions, mesh exports or GPU benchmarks were newly acquired here. These are implementation instructions. Integrate with chapter 067's interface and chapter 065's registry, then verify all retained controls and flat-view equivalence. Persist unmeasured pivots as unknowns rather than approving plausible animation.

Future deliverable	Required acceptance evidence
b3_122_master.blend + geometry_ledger.json	Proposed native scene, specimen sources, units, pivots and uncertainty
b3_122_lod*.glb + asset_manifest.json	Stable IDs; hashes; successful format validation
materials/ + cameras.json	Color recipes, UVs, texture channels and neutral/product captures
fallback/ + validation_report.json	Shared bindings, context-recovery and actual timing/memory reports

MODEL EQUATIONS

E_px = sqrt(sum_i(||project(P_i) - q_i||^2) / N), N = matched landmark count

CHECKS AND DECISIONS

- A render stall and unsupported-GPU path leave audio processing independent.
- All blocked geometry and rights questions remain explicit release gates, with no claimed 1:1 verification.

SOURCES S070-09 Khronos glTF Validator

SCOPE / ORIGINAL DESIGN PROPOSAL

Give the instrument a semantic interface

Accessible resizable plugin interface

WCAG2.2 concerns web content; use it to evaluate the browser prototype and as a declared design reference for the native product. Native JUCE controls instead need accessibility handlers that expose hierarchy, actions, state and value interfaces to platform clients. Web ARIA attributes alone cannot establish native-plugin accessibility. [S071-01] [S071-02]

ORIGINAL DESIGN PROPOSAL. Build a semantic control model before either the photorealistic or flat skin. Bind it to data/parameter_registry.json: all 36 manual drawbars retain their bank identities, and editor-bank selection remains separate from sounding preset routing. Keep the chapter 024 Cancel contract: routing changes do not manufacture playing-key releases.

Hide decorative meshes from assistive navigation while exposing their controls through stable names. Concealing an inactive bank in the custom editor must not set VST3 kIsHidden, whose meaning prevents external changes and automation. A searchable control view provides access to every retained bank. [S071-11]

Element	Proposed accessible meaning
Drawbar	Manual, bank, footage, marked travel, editable range; no invented dB conversion.
Percussion tablet	Control name and retained state; eligibility is a separate descriptive field.
Preset key	Momentary Cancel or identified latch action; support declared multiple-latch mode.
Cabinet preview	Decorative geometry plus separate labeled mode, mic and view controls.
Service value	Value, unit, evidence class and read-only/editable status; unknown remains unknown.

CHECKS AND DECISIONS

- Deliver one semantic binding map consumed by both skins and the native accessibility layer.
- Changing visible banks must not rename or reassign a host parameter.

SOURCES S071-01 WCAG 2.2 — W3C Recommendation, 12 December 2024 / S071-02 JUCE AccessibilityHandler / S071-11 Steinberg ParameterInfo

KEYBOARD / ASSISTIVE INPUT

Operate controls without precision dragging

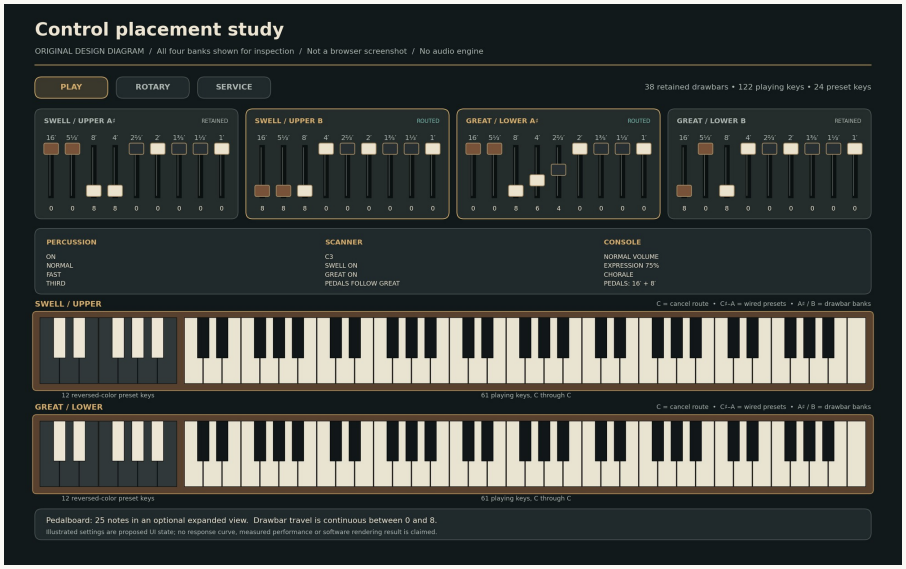
Accessible resizable plugin interface

APG sliders expose a name, current value, range and orientation, with keyboard adjustment. JUCE separately provides numeric/string conversion and ranged-value interfaces. Implement both bindings against the same parameter contract; a drawn handle without those behaviors is incomplete. [S071-03] [S071-04]

ORIGINAL DESIGN PROPOSAL. Use Up/Right to increase marked drawbar travel and Down/Left to decrease it; Home/End select0/8. Proposed keyboard increments are0.1, with Shift+Arrow using0.01 for fine adjustment. These are input resolutions, not physical detents. Provide direct numeric entry and labeled plus/minus buttons; pointer travel still follows the specified pull direction.

Use one tab entry per manual, with internal focus movement that never sounds a note. APG distinguishes focus from selection and navigation between versus within composites. Provide a labeled note inspector with explicit audition Press/Release commands when assistive activation cannot express a held key. That audition facility is a software aid. [S071-05]

Interaction	Required proposed ownership rule
Computer-key press	Capture note/manual owner once; ignore auto-repeat; release the original press record.
Window deactivation or editor close	Release UI-owned audition/playing notes only; internal focus changes preserve original owners. External MIDI owners remain.
Bank switch during edit	Close the original gesture, rebind, then require a new gesture.



Original control and inspection layout study, not a browser screenshot. The separate HTML prototype has not passed browser visual verification; no audio engine is implemented.

CHECKS AND DECISIONS

- Numeric, pointer and accessibility edits must address the same retained parameter.
- Changing focus or pressing Cancel must not create unrelated note-offs.

SOURCES S071-03 JUCE AccessibilityValueInterface / S071-04 WAI-ARIA APG — Slider Pattern / S071-05 WAI-ARIA APG — Developing a Keyboard Interface

VISUAL REQUIREMENTS / PROPOSED TARGETS

Keep labels and states readable on the wood

Accessible resizable plugin interface

WCAG specifies4.5:1 for ordinary text,3:1 for qualifying large text, and 3:1 for necessary non-text control/state information against adjacent colors, subject to its exceptions. Keyboard focus must be visible; the AA obscuration criterion forbids entirely hiding the focused component behind authored content. These are published web criteria, not measured results for this design. [S071-01]

ORIGINAL DESIGN PROPOSAL. Put operational labels on opaque plates, outside reflections and wood-grain highlights. Preserve historic cap colors as appearance, but identify every footage and state through text or shape as well. Focus, latched preset, edited bank and sounding route need distinct cues. Make the focused control fully visible as a stronger product target.

The web minimum pointer-target criterion is24×24 CSS pixels with qualified exceptions, including an equivalent control. A dense musical keyboard does not automatically exempt the entire interface. Prefer a larger44×44 logical-unit product target for ordinary buttons; native units need platform conversion. Provide zoom and an equivalent note inspector where key geometry remains narrow. [S071-06]

Visual risk	Proposed treatment
Gold text on walnut	Solid contrast-qualified label plate; test normal, hover and focused states.
Color-only eligibility	Text such as enabled/ineligible plus a distinguishable symbol.
Hidden explanations	Keyboard-accessible help; essential status remains visible without hover.
Service traces	Labels and line styles distinguish evidence classes without color alone.

CHECKS AND DECISIONS

- Measure final composited colors and hit regions, including scaled and forced-color views.
- Document any exception and its equivalent operation; do not claim conformance from a screenshot.

SOURCES S071-01 WCAG 2.2 — W3C Recommendation, 12 December 2024 / S071-06 W3C Understanding SC2.5.8 — Target Size Minimum

RESIZING / HIGH DPI

Reflow panels and respect host coordinate systems

Accessible resizable plugin interface

Web reflow guidance uses320 CSS pixels for vertical-content width and permits a limited exception where two-dimensional layout is necessary. Its400-percent zoom example is not a native-window sizing rule. Reflow labels, settings, help and inspectors even when the musical key geometry needs its own scrollable or zoomable region. [S071-07]

VST3 ViewRect units depend on the parent type: NSView uses logical units; HWND and X11 embedding use physical pixels. The documented resize sequence negotiates a rectangle and updates the platform view through onSize. Do not generalize those units to every backend or double-apply a host scale factor. JUCE exposes editor scaling, resizability and size-limit hooks. [S071-09] [S071-10]

ORIGINAL DESIGN PROPOSAL. Offer performance, compact and flat layouts while preserving parameter bindings and focus. Collapse the cabinet preview before shrinking control text. Resizing must not select a preset, cancel a held note or reset engine state. Keep accessibility bounds, visible control rectangles and pointer hit tests derived from the same layout transform.

Proposed qualification case	Evidence to retain
Browser1440×1000 and 900×900	No clipped controls; expanded help and pedal drawer remain operable.
Browser zoom/reflow	Reading panels reflow; any keyboard-layout exception is scoped and documented.
Native100/125/150/200/300% scales	Host, OS, backend and monitor scale recorded; hit tests and accessible bounds agree.
Cross-monitor transition	No duplicated scaling, inaccessible dialogs or lost keyboard focus.

CHECKS AND DECISIONS

- Test actual native host embedding separately from the browser prototype.
- Deliver a layout/scale matrix; all proposed combinations remain unqualified until tested.

SOURCES S071-07 W3C Understanding SC1.4.10 — Reflow / S071-09 Steinberg IPlugView interface / S071-10 JUCE AudioProcessorEditor

RENDERING / STATE OWNERSHIP

Reduce motion without changing the instrument

Accessible resizable plugin interface

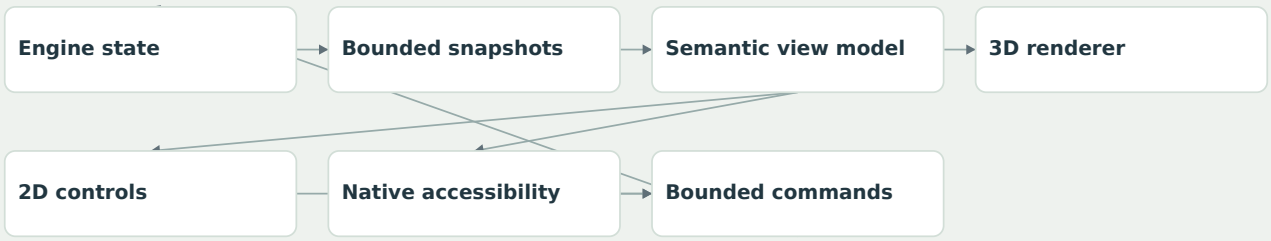
The Media Queries Level5 draft defines prefers-reduced-motion for web preference detection; its draft status is explicit. WCAG2.3.3 addresses disabling interaction-triggered motion at AAA, so do not mislabel that criterion as an AA requirement. This product should still offer a reduced-motion mode independently of a conformance claim. [S071-08] [S071-01]

ORIGINAL DESIGN PROPOSAL. Disable camera sweeps, parallax and decorative rotor spinning when reduced motion is selected. Retain static mode labels and readable model-state summaries. A frozen visual rotor is not an audio brake command. Prototype values must remain labeled illustrative; future engine telemetry must distinguish requested speed, modeled speed and stale display data.

Make the flat renderer a complete control skin, not an error screen. Both renderers consume bounded state snapshots and emit the same commands. Renderer loss, low frame rate or an editor closing must not alter source phase, rotary trajectories or audio processing. Accessibility queries read UI snapshots; they must not synchronously enter the audio callback.

Coalesce display-driven announcements and offer an on-demand state readout. Report user-requested changes and failures without flooding speech with continuously changing RPM, meters or every automation update.

Two skins, one control contract



Original software responsibility proposal. Drawing, accessibility and engine clocks are separate; no production renderer has been validated here.

CHECKS AND DECISIONS

- Switch skins and motion preference during playback without audio-state changes.
- Record GPU-loss recovery and stale-telemetry behavior; do not invent passing results.

SOURCES S071-08 Media Queries Level5 — Working Draft, 19 February 2026 / S071-01 WCAG 2.2 — W3C Recommendation, 12 December 2024

ACCEPTANCE / NOT RUN

Qualify real assistive workflows, not labels alone

Accessible resizable plugin interface

ORIGINAL DESIGN PROPOSAL. Qualify the browser prototype, each native standalone target and each supported host combination separately. Record application/framework versions, backend, OS, screen reader, scale, locale, input device and exact task. An accessibility tree is useful evidence but does not prove that a musician can complete the workflow.

Steinberg requires host-routed keyboard handling and correct handled/unhandled results from the view. Verify that the pinned wrapper preserves host shortcuts instead of adding a second global keyboard interception path. APG also warns that custom sliders need actual testing with touch assistive technology. Neither a desktop mouse test nor native ARIA-like naming covers these cases. [S071-09] [S071-04]

Retain issue reproductions, accessibility-tree snapshots, screen-reader transcripts and event logs. User testing should include people who rely on the relevant access modes. This assignment specifies acceptance work only: no screen-reader session, native host, touch device, audio engine or accessibility-conformance evaluation was executed.

Proposed gate	Observable exit evidence
A11Y-01: navigation	Reach every control and leave every composite/dialog without a keyboard trap.
A11Y-02: semantics	Names, ranges, units, editability and bank state match the actual control model.
A11Y-03: ownership	Internal focus/resize preserves owners; deactivation/close releases only UI owners. External MIDI remains intact.
A11Y-04: presentation	Contrast, zoom, target alternatives and reduced motion pass the declared matrix.
A11Y-05: host integration	Host shortcuts, native accessibility, automation and fallback retain their separate contracts.

CHECKS AND DECISIONS

- Ship an accessibility report with failed, passed, blocked and not-run cases distinguished.
- Do not claim universal accessibility from the browser control prototype or this chapter.

SOURCES S071-02 JUCE AccessibilityHandler / S071-04 WAI-ARIA APG — Slider Pattern / S071-09 Steinberg IPlugView interface

EXECUTED STATE CHECKS / BROWSER VERIFICATION BLOCKED

Independent review: what actually ran

UI implementation review and executable acceptance prototype

The independent review executed the prototype’s actual embedded JavaScript, rather than a second implementation of its state rules. A dependency-free Node harness supplies a deliberately limited DOM, event, pointer-capture and storage test double. Node VM executes trusted project code; it is not a security mechanism. [S072-05] The source hash and runtime version accompany every result.

The initial run passed 25 of 28 scoped checks and exposed three concrete interaction defects. After correction, all 28 passed. This establishes the tested JavaScript transitions under the declared harness assumptions. It does not establish native browser dispatch, CSS layout, hit testing, accessible-tree behavior, audio processing or acoustic similarity.

Browser verification could not run: local Chromium was denied AF_UNIX socket creation, and automatic approval review rejected escalation. The supported cloud browser separately rejected the file URL under its protocol policy. No bypass or substitute screenshot was attempted. The portable browser script remains not_run; it has only passed a JavaScript syntax check.

Evidence	Actual status
Embedded script with test doubles	28 scoped checks passed
Previously failing regression cases	3 corrected and rechecked
Browser test script	Syntax passed; execution not_run
1440×1000 /900×900 layout	Unverified; browser blocked
Audio, VST3 and physical fidelity	Not implemented or tested

CHECKS AND DECISIONS

- Keep data/072-ui-checks.json attached to the exact HTML hash.
- Do not promote a test-double result to a browser or host pass.

SOURCES S072-05 Node.js VM API

IMPLEMENTED CONTROL STUDY / ORIGINAL LAYOUT DIAGRAM

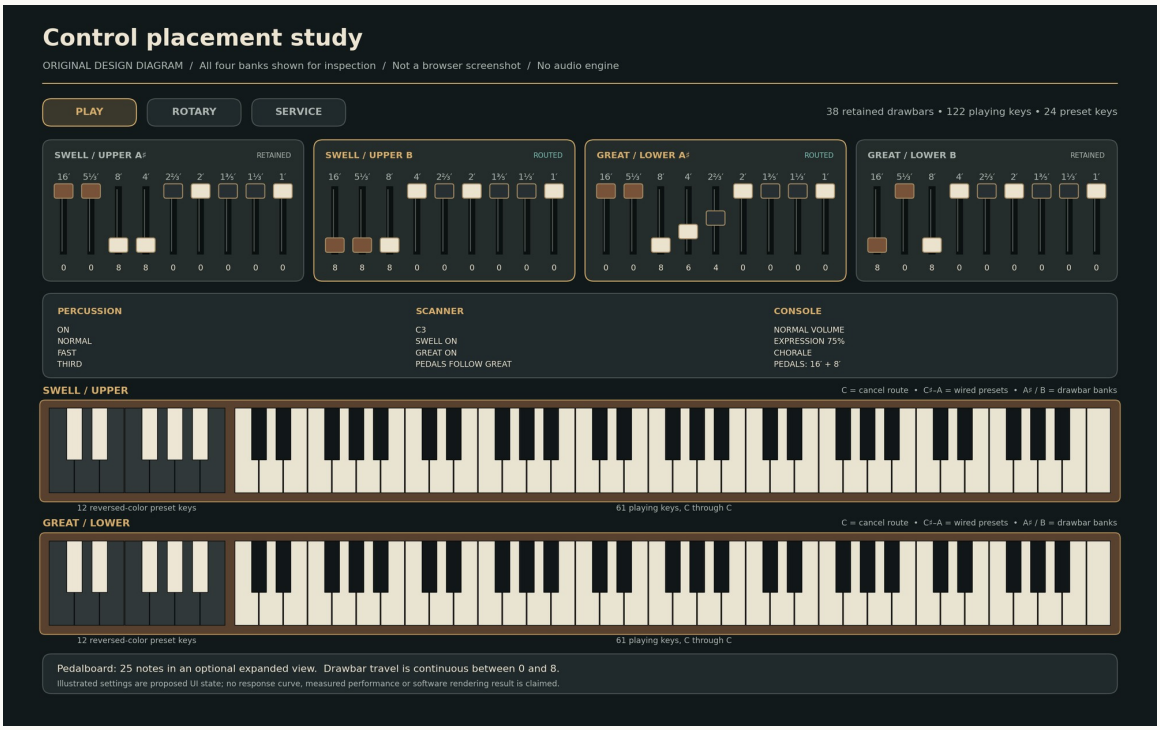
The control surface has explicit boundaries

UI implementation review and executable acceptance prototype

The prototype retains 36 manual drawbar values and two pedal values while displaying one nine-drawbar editor per manual. Each manual builds 61 playing keys and 12 reversed preset keys; the pedal drawer builds 25 keys. The independent execution counted those generated elements, checked pitch endpoints, footages and cap-color order, and checked named vertical-slider attributes. Slider semantics follow W3C guidance, with actual assistive-technology behavior still untested. [S072-01]

Editing A# or B does not select the routed registration. Cancel changes the selected manual's route while preserving held playing-key ownership, the other manual and pedal settings. The factory describes Cancel as a release action and distinguishes manual banks, pedal controls and upper-B percussion eligibility. [S072-07] This prototype intentionally implements ordinary exclusive preset selection; simultaneous physical latch states remain a future engine requirement.

The accompanying original layout diagram is a specification illustration. It is not a screenshot of the HTML and cannot establish responsive rendering. The separate HTML contains the interactive design. Its proposed microphone editor exposes two positions; the wider registry's third microphone, gains and output controls are future integration work.



Original control and inspection layout study, not a browser screenshot. The separate HTML prototype has not passed browser visual verification; no audio engine is implemented.

CHECKS AND DECISIONS

- Count retained identities separately from currently visible editor controls.
- Do not infer native-plugin behavior from a browser control study.

SOURCES S072-01 W3C ARIA Authoring Practices: Slider Pattern / S072-07 Hammond factory service manual

THREE REPRODUCED DEFECTS / ORIGINAL CORRECTIONS

Fix gesture discontinuities and focus ownership

UI implementation review and executable acceptance prototype

The first failure occurred when Shift fine adjustment changed during a drag. The old handler recalculated the whole displacement with the new sensitivity: a value of 2 became 0.3 without further movement. The correction accumulates each new pointer delta against the current value while retaining the captured manual, bank and drawbar identity. The regression now preserves 2 at the modifier transition and reaches 2.15 after the next ten pixels of fine movement.

The second failure allowed the motion checkbox to report enabled even though system reduced-motion CSS suppressed animation. Central preference rendering now clears and disables that control while the system preference applies, then permits deliberate re-enabling after the preference clears. This is a product decision supporting reduced-motion guidance; it is not a WCAG conformance result. [S072-04]

The third failure left the roving Tab entry on an old playing key after pointer focus moved elsewhere. A focus handler now gives exactly one key in that container tabIndex 0. Separately, the shaft-length CSS expression was aligned with cap travel; that algebraic edit remains visually unverified.

```
begin: capture semantic ID, bank, index, lastY
move: deltaY = pointerY - lastY
      lastY = pointerY
      gain = Shift ? fineSensitivity : normalSensitivity
      value = clamp(value + deltaY * gain, 0, 8)
end:  close the captured gesture before rebinding
focus(key): set only that key’s tabIndex to zero
```

CHECKS AND DECISIONS

- A modifier transition without movement must not change the setting.
- Changing the editor bank must end an existing drag before the view changes.

SOURCES S072-04 W3C Understanding WCAG 2.2 SC 2.3.3

ORIGINAL IMPLEMENTATION SEQUENCE / HOST BRIDGE

Preserve semantics when the editor gains an engine

UI implementation review and executable acceptance prototype

First replace direct JavaScript state mutation with an explicit editor-to-engine adapter. Resolve the full semantic path through data/parameter_registry.json; never use the visible slider's index as its permanent identity. Convert drawbar travel, expression position and enumerated controls with the selected registry and wrapper rules. The prototype's proposed defaults and two-microphone ranges are not a frozen production parameter export.

Next map performance actions into the supplied C++ contract: KeyEvent for playing keys, PresetKeyEvent for reversed preset keys, ParameterPoint for an instantaneous parameter target, and CurveSpan for an automation segment. Event carries an absolute frame and deterministic sequence. PanicEvent remains separate from historical Cancel. UI focus, view selection and decorative motion belong to editor state; they must not alter continuous generator or rotor clocks.

Then establish bounded command publication and an acknowledgment path. The audio thread owns musical state; the editor presents acknowledged values and can discard stale telemetry without discarding musical commands. Gesture begin/end must address the original parameter even if a bank or tab changes. Native pointer capture has its own browser-defined behavior, so the test double's capture bookkeeping must be followed by real platform dispatch tests. [S072-03]

Action	Required boundary
Edit an inactive bank	Retained bank identity
Select a preset	Routing event, not playing-key release
Move microphone	Geometry parameter with units
Read diagnostics	Immutable profile and timestamped telemetry
Resize or animate UI	Renderer state; no audio-thread ownership

CHECKS AND DECISIONS

- Verify actual host gestures and normalization before freezing IDs.
- No rendering stall may delay a required note release.

SOURCES S072-03 W3C Pointer Events Level 3

PORTABLE ACCEPTANCE PROCEDURE / NOT RUN HERE

Run the browser checks in a permitted environment

UI implementation review and executable acceptance prototype

The supplied verify_interface_browser.mjs uses official Playwright on an authorized local machine. It opens the extracted HTML, checks 1440×1000 and 900×900 viewports, drives controls, records runtime errors and unexpected network requests, and writes a structured report plus Play and Service screenshots. The script records failure or blockage as failure or blockage; creating a proposed test does not count as passing it.

The procedure exercises real drawbar dragging with a Shift transition, bank retention, keyboard-held notes through Cancel, combined pointer/keyboard ownership, microphone updates, snapshots and reduced-motion behavior. Selected page-width and slider-bound checks are automated. Microphone range changes are explicitly dispatched input events, so they do not establish every touch or pointer interaction.

Inspect the resulting screenshots manually for clipped legends, overlaps, contrast, focus visibility, zoom behavior and panel balance. Re-run on supported platforms with a fixed browser and font environment; Playwright notes that rendering can vary with operating system, browser version and hardware. [S072-06] Screenshots become regression baselines only after review. Screen readers, touch assistive technology and native-plugin DPI conversion require separate sessions.

Acceptance layer	Evidence required
Browser interactions	Executed report; zero unexplained errors
Visual layout	Reviewed screenshots at declared sizes
Keyboard and accessibility	Observed navigation and announcements
Native renderer	Platform-specific coordinate and DPI tests
Audio/host coupling	Production engine and host fixtures

CHECKS AND DECISIONS

- Keep the browser script status not_run until it actually executes.
- Do not label the original control-layout diagrams as observed screenshots.

SOURCES S072-06 Playwright visual comparisons

DELIVERABLES / REMAINING GATES

Release the control study with an honest test record

UI implementation review and executable acceptance prototype

The companion includes interface_prototype.html, verify_interface_state.mjs, verify_interface_browser.mjs, INTERFACE_VERIFICATION.md, the 28-check state report and the before/after change record. The state harness loads the actual HTML and reports its SHA-256. Seven malformed snapshot cases are rejected within one aggregate validation check; they are not seven additional top-level checks. No test count implies acoustic coverage.

The test double deliberately lacks an HTML5 parser, CSS layout, browser default actions, event bubbling, real hit testing, assistive technology and storage-origin behavior. Its pointer and focus callbacks are useful regression fixtures, not implementations of the web platform. W3C’s slider and tab patterns define useful semantics, while actual users and supported assistive technologies remain part of acceptance. [S072-01] [S072-02]

Advance the implementation in this order: reproduce the scoped state checks; execute and review browser acceptance; connect the native renderer to the existing engine contract; validate host gestures and focus ownership; then run production performance and accessibility checks. UI appearance never substitutes for the independent DSP, circuit, recording and listening gates. The current prototype is a functional no-audio control study, with browser visual verification explicitly outstanding.

Retained boundary	Outstanding gate
Ordinary preset mode	Multiple physical latches and electrical loading
Two-microphone control study	Full registry export and rig admission
Scoped JavaScript checks	Actual browser and native interactions
Illustrative cabinet motion	Measured rotor state driven by engine
No-audio interface	Implemented and validated audio path

CHECKS AND DECISIONS

- Keep current artifact hashes and execution scope in the acceptance matrix.
- Treat unrun browser, host and acoustic checks as outstanding.

SOURCES S072-01 W3C ARIA Authoring Practices: Slider Pattern / S072-02 W3C ARIA Authoring Practices: Tabs Pattern

ORIGINAL DESIGN PROPOSAL · REFERENCE LOCK

M0 — freeze the reference, then keep two kinds of progress separate

Ordered implementation milestone plan

Choose an identified vintage B-3, its AO-28 revision and a Leslie 122 as the first target. Record serials, mains/gear variant, modifications, microphone geometry and source-drawing locators; unresolved identity stays unknown. The factory manuals cover multiple circuits, so “B-3” alone is not a component specification. [S073-01][S073-03]

Maintain separate engineering and fidelity statuses for every milestone. Engineering can proceed with explicitly proposed parameters and named surrogate models. A measured profile requires traceable captures at declared boundaries. Neither schema acceptance nor a plausible sound upgrades proposed values to measured ones. Compile a synthetic fixture under its own identity; never silently fill unknown physical values with zero or unity.

Existing research comprises transcriptions, schemas, mathematical references and interface declarations; chapter 067 supplies the visual design direction. There is no delivered audio engine. All new source filenames below are proposed repository targets. Freeze evidence revisions now; acquire measurements in parallel with M1–M7. Missing measurements postpone the relevant fidelity decision, not every useful implementation task.

M0 input	Create / interface	Exit gate
data/calibration_profile.schema.json; data/capture_manifest.schema.json; reference_code/DATA_CONTRACTS.md	Reference lock recording hashes, variants, units, boundaries and unresolved fields	Every selected value has an evidence class; required unknowns identified.
Factory drawings; chapters 058/060/063; data/parameter_registry.json	Separate proposed engineering fixture and future specimen profile	No cross-profile substitution; each artifact declares its validation scope.
Capture requirements from chapters 056–062	Acquisition queue ordered by upstream dependency	Each requested capture names the parameter or uncertainty it can resolve.

CHECKS AND DECISIONS

• M0 freezes evidence and interfaces, not unverified numeric fidelity. A later evidence revision invalidates only affected results.

SOURCES S073-01 Hammond factory service manual: B-3/C-3 circuits, routing and service procedure / S073-03 Leslie Models122/122RV and related-family service manual, publication044578, June1971

ORIGINAL DESIGN PROPOSAL · TESTABLE SKELETON AND GENERATOR

M1–M2 — build an offline clock before building the tonewheel bank

Ordered implementation milestone plan

M1 first implements src/ProfileCompiler.cpp from reference_code/engine_contract.hpp and DATA_CONTRACTS.md. Produce immutable RuntimeProfileView storage plus a dependency/conversion report. Implement tools/render_fixture.cpp with timestamped events, a seed, profile digest and named diagnostic ports. It becomes the common oracle for every subsequent module; silence-producing placeholders do not count as instruments.

Own half-open sample intervals, consume zero-frame controls once without advancing state, and preserve a stable same-frame event order. Reconstruct VST3 curves including the previous-block anchor before unsigned timeline conversion. [S073-04] Test differing callback partitions and preparation limits. JUCE documents variable callback sizes; inspect the selected adapter instead of assuming its nominal block size is exact. [S073-05]

M2 implements one continuously running shared source bank and imports the factory routing maps. [S073-01] Keep source number, physical terminal and MIDI note distinct; preserve mechanical shaft angle separately from tooth phase. data/009-candidate-frequencies.json is a computed candidate, not measured tuning. A proposed sine fixture needs explicit amplitude units and scale; it must not inherit historical G–G readings as pickup gains.

Stage / inputs	Output interface	Exit gate
M1: engine_contract.hpp; schemas; registry	IPreparedGraph / ProcessCall; offline audio and event trace	Partition, flush, overflow and lifetime cases pass; no callback allocations.
M2: data/015-manual-map.json; data/022-pedal-map.json; candidate frequencies	ISourceBank / VoltageSourceBlock; per-source diagnostic render	Every map entry matches the reviewed table, including disconnected contacts; note events never restart source phase.

CHECKS AND DECISIONS

• Parallel: once M1 types freeze, implement generator, profile compiler and event fixtures separately. Require anti-alias tests at supported rates before merging nonlinear source detail.

SOURCES S073-01 Hammond factory service manual: B-3/C-3 circuits, routing and service procedure / S073-04 Steinberg VST3 IParamValueQueue interface / S073-05 JUCE AudioProcessor class reference

ORIGINAL DESIGN PROPOSAL · CONSOLE BEHAVIOR AND SCANNER

M3–M4 — implement physical control state, then the scanner

Ordered implementation milestone plan

M3 turns key gestures into ordered contact trajectories before summing source contributions. Keep four retained manual drawbar banks, pedal controls and preset latch masks. Cancel changes preset routing without manufacturing key releases. Percussion uses shared continuous recharge state; test repeated notes and switching while keys remain held. Expression belongs within the loaded organ circuit. These behaviors follow the reviewed factory evidence; their unmeasured timings remain proposed. [S073-01]

Implement console/contact/percussion behavior behind IOrganCircuit while preserving its diagnostic ports and replaceable model identity. This allows useful M3 work before a complete AO-28 connectivity graph exists. A provisional envelope or gain law is an explicitly named surrogate, never a component-exact circuit. Later M5 integration must retain the same observable event semantics.

M4 imports the reviewed scanner graph and switch map. The published nominal network has 18 LC sections and 19 ground-referenced outputs; the return contribution already present in chorus outputs must not be added twice. [S073-02] Audit the chapter 028 algebra corrections before selecting a WDF formulation. Independent AC agreement is an engineering check, not specimen validation.

Stage / inputs	Create / interface	Exit gate
M3: maps; parameter_registry.json; chapters 016–023	Contact trajectories, preset router, percussion state; IOrganCircuit	Held-key switches, partial releases, bank changes and Cancel preserve declared state.
M4: data/025-scanner-network.json; data/027-vibrato-routing.json; reference_code/scanner_ac_reference.py	Time-domain scanner with named ground/return ports	Every mode agrees with the independent AC oracle under the same loads; transition and stability tests pass.

CHECKS AND DECISIONS

• Parallel: M3 behavioral fixtures and M4 numerical solver can proceed together after M2 routing and port conventions freeze.

SOURCES S073-01 Hammond factory service manual: B-3/C-3 circuits, routing and service procedure / S073-02 A Computational Model of the Hammond Organ Vibrato/Chorus using Wave Digital Filters

ORIGINAL DESIGN PROPOSAL · AO-28, DRY VALIDATION AND LESLIE ELECTRICAL

M5–M6 — close the dry organ before evaluating the Leslie

Ordered implementation milestone plan

M5 first transcribes complete revision-specific AO-28 connectivity from Figure 24’s upper drawing, preserving conditional alternatives. [S073-01] data/ao28-component-values.json contains 124 placement/section records across variants; it is neither a complete netlist nor a count of installed packages. Build and check the loaded linear network before nonlinear tube stages. If current feedback crosses a proposed module boundary, fuse the solver region or quantify the decoupling approximation.

Expose dry G–G voltage as a diagnostic port. Fit and compare source routing, percussion, expression, scanner and amplifier behavior here before downstream cabinet fitting. TableV values describe a historical service-output procedure, not isolated oscillator gains. [S073-01] Oversampling, state migration, operating-point convergence and alias rejection need independent tests; compilation proves none of them.

M6 uses the identified122 drawing and keeps147 as a separate variant. [S073-03] Import data/044-crossover-network.json without grounding its floating treble return: retain A–H and W–B voltage ports and driver loading. The graph is an idealized electrical model. A nominal resistive-load reference cannot establish the real driver’s impedance or acoustic response.

Stage / inputs	Output interface	Exit gate
M5: data/031-ao28-stages.json; data/038-ao28-bom.json; component registry; factory drawing	Connected revision graph; IOrganCircuit; dry render suite	Connectivity and numerical gates pass; dry capture comparison separately reports pending/pass/fail/inconclusive.
M6:122 drawing;044 graph; reference_code/crossover_ac_reference.py	ILeslieElectrical with amplifier/driver loading; branch-voltage renders	Polarity, loaded AC and nonlinear stability pass engineering gates; measured electrical match remains separate.

CHECKS AND DECISIONS

• Parallel: M6 reference transcription and electrical fixtures can start during M5. Do not compensate a dry-organ error using Leslie EQ.

SOURCES S073-01 Hammond factory service manual: B-3/C-3 circuits, routing and service procedure / S073-03 Leslie Models122/122RV and related-family service manual, publication044578, June1971

ORIGINAL DESIGN PROPOSAL · ROTARY ACOUSTICS AND CALIBRATION

M7–M8 — render moving radiation, then fit identifiable parameters

Ordered implementation milestone plan

M7 connects independent horn/drum trajectories to moving-path radiation, cabinet filtering and microphones. Use reference_code/rotor_state_reference.py and reference_code/retarded_path_reference.py as mathematical fixtures. One physical trajectory serves all microphones, while each path has its own emission time. Declare which layer owns propagation delay, directional response and room effects; full cabinet responses cannot be treated as freely rotating source patterns. Optional stationary reverb remains a separate path.

M8 fits upstream quantities first and reserves held-out gestures, levels, rotor transitions and positions for validation. Preserve original captures and alignment metadata; changing topology requires refitting affected downstream stages. NIST combines measurement uncertainties with relevant covariance. [S073-06] Adopt an uncertainty-aware decision rule before examining results: insufficient resolution yields inconclusive, not automatic success. Listening comparisons supplement boundary measurements; they do not identify wrong wiring by themselves.

Acquisition order, begun at M0	Build / output	Exit gate
1: specimen identity; clock, channel and voltage calibration	Capture manifests with traceable units and uncertainty	Boundary and timing are known before fitting.
2: isolated sources/routing; contact/percussion; scanner and dry AO-28	M8 dry profile plus held-out residual report	No downstream correction hides an upstream error.
3: Leslie input/output and driver loading; rotor trajectories; stationary-angle responses	M7 IRadiation / PressureOutput; fitted electrical and motion profiles	Causal/stable moving-path tests; plausible fit checked against independent captures.
4: microphone/room paths and complete performances	End-to-end profile and listening protocol	Declared tolerances tested on held-out data; uncertainty and failures retained.

CHECKS AND DECISIONS

• Parallel: M7 provisional acoustics can run while acquisition continues. M8 may calibrate completed modules incrementally; it cannot claim a measured full instrument from a partial dataset.

SOURCES S073-06 NIST Technical Note1297, section5: Combined Standard Uncertainty

ORIGINAL DESIGN PROPOSAL · HOST, UI AND RELEASE

M9 — add the product around the verified engine contracts

Ordered implementation milestone plan

A minimal VST3 adapter may be smoke-tested after M1/M2; product completion follows the integrated DSP gates. Share one engine with the offline renderer and standalone device adapter. Inspect actual exported parameter IDs against data/parameter_registry.json before freezing them. Publish prepared profiles off the callback, reserve retirement capacity before activation, and verify transitive storage lifetimes. Preset serialization is not an exact DSP checkpoint.

Bind the chapter 067 visual concept to stable parameters only after those contracts settle. Then polish the three-dimensional cabinet, drawbar travel, rotor view and camera without letting editor visibility control audio routing. Prove keyboard operation, reduced motion, native accessibility and rendering with the editor absent. A browser prototype demonstrates design intent; it does not establish plug-in audio or host compatibility.

Steinberg’s validator checks VST3 conformity. [S073-07] Run it alongside actual DAW and standalone sessions, deadline/allocation audits, state migration, latency, tails and bypass tests. Record platform, host, rate, buffer, profile and build identities. Release engineering status and acoustic scope separately; any fidelity claim requires its own evidence. No such product or physical validation has been completed here.

M9 workstream / inputs	Deliverable	Exit gate
Host: chapters 063–066; validated engine	VST3/standalone builds, enumerated IDs, host reports	Same event fixture produces the declared equivalent output across adapters.
UI:067 concept;065 registry	Accessible editor with retained-bank gestures	Automation remains bound to its original ID during bank/camera changes.
Release: licenses, assets, acceptance reports	Reproducible build, notices, installer, evidence-qualified release notes	Every advertised capability has a corresponding passing report.

CHECKS AND DECISIONS

• Immediate next commits: M0 reference lock → M1 profile compiler → offline scheduler → M2 generator. Each agent hands over input hashes, changed files, executable checks, results and remaining uncertainty before downstream integration.

SOURCES S073-07 Steinberg VST3 Validator command line

ORIGINAL DESIGN PROPOSAL / EXECUTION CHARTER

The agent contract: build, prove, disclose

Agent mission statement and operating contract

Give the implementation agent AGENT_MISSION.txt together with this extracted package. Its assignment is to produce working VST3 and standalone software, an inspectable core, reproducible tests, and calibration tools. This chapter summarizes that operating contract; the companion mission contains the detailed sequence and stop conditions. Neither the interface header nor the visual prototype is an existing audio engine.

Lock an identified vintage B-3, AO-28 revision, regional generator configuration, and Leslie 122 reference. The factory console describes physical generator, contact, preset, and amplifier mechanisms rather than an integrated digital chipset. Preset cancellation is a routing operation, so a software panic command must remain separate. [S074-01] Variant boundaries are engineering inputs, not aesthetic labels.

Permit a useful provisional build while the physical corpus is unacquired. Each approximation must name its assumptions, omitted interactions, evidence class, and replacement path. Unknown required inputs block the selected physical algorithm; a documented surrogate may satisfy a different algorithm. Advancement never changes an unknown into a measurement.

Deliverable tier	Required evidence
Documentary atlas	Source locators and explicit uncertainty
Provisional instrument	Working implementation; declared assumptions
Fitted specimen profile	Identified captures; reproducible fitting
Fidelity claim	Held-out acceptance with stated limits
Release candidate	Host, platform, rights and signing gates

CHECKS AND DECISIONS

- Label every reported result by its actual validation scope.
- Do not claim a universal 1:1 match from a useful nominal model.

SOURCES S074-01 Hammond Organ Service Manual, broad console-model compilation

IMPLEMENTATION / DATA CONTRACT

Turn evidence into typed, bounded inputs

Agent mission statement and operating contract

Start with DATA_CONTRACTS.md and the capture/profile schemas. Structural validation can reject malformed records; it cannot authenticate an instrument, a permission, or a successful measurement. [S074-08] The compiler must additionally resolve references, units, asset hashes, effect ownership, graph connectivity, and every algorithm-required quantity. Keep unknown values null with an actionable reason.

The package contains heterogeneous evidence. Candidate frequencies are software-derived, service readings use the G–G boundary, and component placements include alternatives. None may be silently recast as measured pickup calibration or a complete circuit netlist. The scanner paper and Leslie schematic also require their declared terminal references and model assumptions. [S074-02] [S074-03]

Create immutable raw acquisitions and traceable derivatives. Each derivative retains its physical acquisition group, parent identifiers, transformation revision, and rights. Split groups before fitting. Assign one owner to each gain, filter, leakage, saturation, propagation, cabinet, and room contribution so a recorded effect is never applied twice.

Input	Allowed interpretation
009 frequency candidates	Derived source model; not measured table
014 historical readings	Service G–G observations; not pickup gains
025 /027 scanner data	Nominal graph and documented routing
AO-28 value registry	Revision-qualified placements; not netlist
044 crossover graph	Differential ports; nominal load fixture
62-entry parameter registry	Proposed identities; adapter mapping pending

CHECKS AND DECISIONS

- Reject dimensionally invalid amplitude conversion.
- Report required missing inputs before preparing an active graph.

SOURCES S074-02 DAFx 2016 scanner-vibrato physical model paper / S074-03 Leslie 122 family service manual, publication044578, June1971 / S074-08 JSON Schema 2020-12 validation specification

ORIGINAL ARCHITECTURE / REAL-TIME CONTRACT

Own runtime state before adding complexity

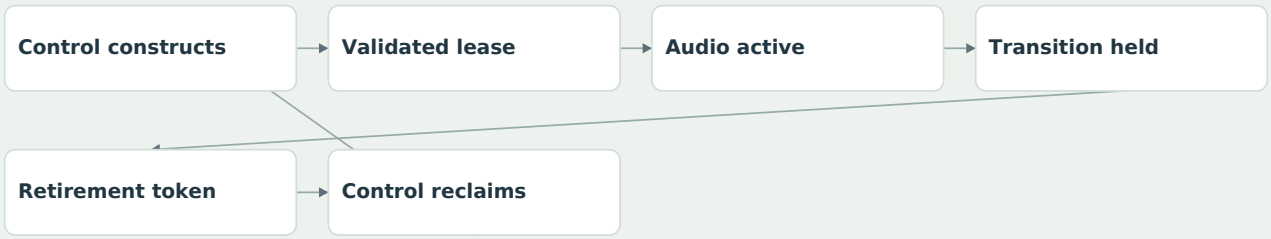
Agent mission statement and operating contract

Implement the actual C++20 interfaces in engine_contract.hpp. ISourceBank owns shared-source advancement; IOrganCircuit owns console and organ interactions; ILeslieElectrical owns amplifier/crossover/load behavior; IRadiation owns motion and pressure outputs. These boundaries do not prove a one-way electrical decomposition. Fuse interacting stages or qualify a loading approximation against an independent model.

Prepare and validate off the callback. PortAudio documents restrictions on allocation, I/O, blocking, and operations with unbounded execution. [S074-05] The original package contract turns that constraint into explicit ownership: control publishes a prepared lease, audio exclusively mutates active DSP, and control reclaims only after retirement acknowledgment. Implement bounded queues; the supplied declarations implement none.

Use monotonic render time, half-open positive blocks, and non-advancing zero-frame flushes. Preserve normalized automation slopes and the native prior-block anchor before conversion to unsigned time. [S074-04] Define same-frame ordering. Copy snapshots to the UI; a mutable DSP reference must never escape merely because a control is visible.

Prepared graph ownership



Original ownership proposal. Transition and retirement storage are bounded; no callback reclamation is implied.

CHECKS AND DECISIONS

- Prove lifetime of all borrowed profile data through retirement.
- Distinguish prevalidation rejection from mid-render solver failure.

SOURCES S074-04 Steinberg VST3 IParamValueQueue interface / S074-05 PortAudio: Writing a Callback Function

IMPLEMENTATION / SEQUENTIAL EXIT GATES

One dependency order, ten milestones

Agent mission statement and operating contract

Use the M0–M9 numbering in chapter 073 and AGENT_MISSION.txt. Every milestone produces executable work, commands actually run, evidence-qualified results, and an unresolved-input list. Gate physical-fidelity claims separately from implementation progress. Acquisition can run alongside development, but upstream errors must not be concealed by downstream equalization.

Build the dry instrument before acoustic fitting. Existing offline fixtures help verify documentary mappings and mathematics, not specimen fidelity. The original graph and code contracts require exact routing and differential voltages; importing a nominal response does not validate a real scanner or cabinet. [S074-02] [S074-03] Preserve this distinction in reports and visible profile metadata.

Milestone	Concrete exit artifact
M0 Reference lock	Identity, variants, assumptions, acquisition plan
M1 Skeleton	Headless renderer, adapters, event/fault tests
M2 Generator	91 shared sources; phase and boundary tests
M3 Console	Verified maps, latch state, contacts, percussion
M4 Scanner	Six modes; routing and numerical comparisons
M5 AO-28 /dry	Revision model; expression; G–G validation
M6 Leslie electrical	Coupled amp/load model; crossover port tests
M7 Rotary acoustics	Independent rotors; paths; pressure outputs
M8 Calibration	Fit scripts, held-out results, limits
M9 Host /UI /release	Actual host mapping, interface, qualified build

CHECKS AND DECISIONS

- Each exit links its inputs, implementation revision and real test output.
- A missing acquisition blocks its claim, not unrelated reversible work.

SOURCES S074-02 DAFx 2016 scanner-vibrato physical model paper / S074-03 Leslie 122 family service manual, publication044578, June1971

MEASUREMENT / CALIBRATION CONTRACT

Acquire what the model cannot infer

Agent mission statement and operating contract

The physical campaign remains future work. Request identified specimen photographs, control trajectories, electrical boundary recordings, driver impedances, rotor speed traces, and acoustic captures with microphone geometry. Use a qualified operator for electrical measurements. Preserve actual clocks, loading, calibration, warm-up, settings, restoration history, and acquisition permissions; a plausible filename is not proof of an observation.

Fit in causal order and retain uncertainty. Source waveform, contact behavior, scanner loading, valve response, driver radiation, fixed cabinet scattering, and room response can compensate for one another if their boundaries are not controlled. Record which parameters are identifiable and which combinations remain ambiguous. An optimization residual is not automatically a unique physical explanation.

Use held-out acquisition groups, matched conditions, and level-controlled comparisons. Include transient and interaction tests, not only steady spectra. Listening contributes perceptual evidence alongside signal metrics. The schema and ancestry checks are prerequisites for this work, not substitutes for it. [S074-08] Publish the exact scope of every passing gate and leave unobserved conditions unclaimed.

Unknown	Acquisition that resolves it
Pickup or loaded-source amplitude	Calibrated boundary recording with loading
Contact and percussion behavior	Repeatable key/control event traces
Scanner losses and loading	Electrical response under stated termination
Rotor transition parameters	Angle/speed versus time in each direction
Cabinet/room separation	Geometry-controlled response measurements

CHECKS AND DECISIONS

- Derivatives cannot cross their parent acquisition partition.
- Every fitted profile links raw evidence, scripts and held-out results.

SOURCES S074-08 JSON Schema 2020-12 validation specification

DELIVERY / HOSTS / USER INTERFACE

Finish a reviewable product, not a promise

Agent mission statement and operating contract

Freeze the real adapter mapping before public state compatibility is promised. The proposed registry keeps all manual banks separately; no actual native/JUCE export is verified yet. Editor visibility differs from VST3 kIsHidden, which affects outside control and automation. [S074-09] Test ramps, gestures, restored state, zero-frame calls, tails, device changes, and multiple instances in actual hosts.

Implement the chapter 067 dimensional interface after the core is stable. Its prototype provides visual/control direction without audio. Use original or licensed wood, metal, keys, and cabinet assets; include readable flat operation, keyboard access, reduced motion, resizing, and telemetry-driven animation. Photo references are evidence, not automatically usable product textures.

At the research date official JUCE material names JUCE9 with AGPLv3/commercial module options; VST3 uses MIT terms separately. Recheck pinned versions before release. [S074-06] [S074-07] Complete native bundle resources and metadata, actual platform qualification, signing/notarization where applicable, installers, and rollback. Never report a signature, host pass, physical match, or corpus that was not produced.

Final handoff	Required contents
Implementation	Source, pinned builds, renderer and actual binaries
Evidence	Profiles, provenance, rights and unresolved quantities
Verification	Commands, results, platforms and not-run gates
Product	UI assets, host IDs, migration, installers and rollback
Completion report	What works; what is provisional; next acquisition

CHECKS AND DECISIONS

- Release claims match the actual artifact and qualification record.
- Missing rights or credentials block publication, not a reviewable candidate.

SOURCES S074-06 JUCE official license source / S074-07 Steinberg VST3 licensing / S074-09 Steinberg VST3 ParameterInfo reference

ORIGINAL DESIGN PROPOSAL · TRIAGE

Prioritize dependencies without stopping useful implementation

Whole-system risk and unknowns register

The most urgent work is to lock a physical reference, establish a calibrated acquisition route, build the deterministic offline skeleton and transcribe the complete AO-28 circuit. These can proceed in parallel. The package supplies documentary and mathematical groundwork; it supplies no new hardware captures or audio engine. An implementation milestone and a fidelity milestone must have separate statuses.

risk_register.json contains 40 open items, each with an owner role, acquisition action, affected M0–M9 milestones, provisional strategy and falsifiable exit criterion. Priorities express dependencies and consequences. No numerical probability, invented budget or completion date is assigned. A blocker names the capability or claim it blocks; missing physical evidence does not prevent a clearly labeled synthetic engine fixture.

Choose one B-3/AO-28/122 configuration before fitting. The factory sources contain revision-dependent circuitry, so their existence does not identify an actual specimen. [S075-01][S075-02] An unselected chassis, unknown microphone or unavailable waveform stays unknown. Never turn a schema example into a capture by filling its null fields with plausible numbers.

Priority	Immediate action	Work allowed meanwhile
P0 reference/evidence	Identify specimen; qualify acquisition and claim criteria	Implement explicit nominal fixtures.
P0 implementation	Build M1 runtime; review complete AO-28 connectivity	Develop replaceable console/scanner interfaces.
P1 integration	Acquire parameters; test native behavior	Keep assumptions and scope visible.
P2 expanded scope	Resolve materials, substitutions and optional families	Release only supported scope.

CHECKS AND DECISIONS

- A blocked physical claim is not a failed software build; a successful build is not physical equivalence.

SOURCES S075-01 Hammond factory service manual, 220-page compilation / S075-02 Electro Music/CBS Leslie manual, publication 044578, June 1971

EVIDENCE GAP · M0 / M5

The AO-28 drawing exists; the solver-ready circuit does not

Whole-system risk and unknowns register

The drawing locator is resolved: upper Figure 24, printed 3-50/PDF 129. [S075-01] The current 124 placement/section records include alternative branches and multiple sections of individual valves. They are neither 124 simultaneously installed packages nor a complete connectivity graph. The registry explicitly contains no nodes or edges. Transcribing connections is a prerequisite for component-resolved M5 work.

Assign the circuit lead a revision-qualified terminal graph, then assign another reviewer continuity, grounding, valve-pin, transformer-polarity and switch-state checks. Preserve conditional alternatives rather than joining repeated designators into one circuit. A readable 1995 redraw is comparison material, but its date and differences remain visible. [S075-09] Resolve the solid-network R62 ambiguity through evidence, not by choosing whichever candidate sounds preferable.

Establish loaded linear behavior and operating points before nonlinear fitting. Transformer identities do not supply magnetic parameters; valve-family labels do not identify installed operating curves. Expression travel and capacitance remain part of the loaded circuit. A replaceable behavioral organ module can advance meanwhile, with its model identity explicit.

Existing artifact	Remaining acquisition	Exit evidence
031 stages; 124 value placements	Selected connectivity and independent review	Terminals, variants and ports reconciled.
038 stock inventory	Instance crosswalk and selected chassis	Source-qualified circuit instances.
Transformer/valve identities	Loaded transfer, bias and multilevel observations	Held-out dry G–G prediction.
Expression sections	Capacitance/travel and surrounding load	Independent transfer-sweep match.

CHECKS AND DECISIONS

• Do not label unloaded cascaded filters a verified AO-28 circuit. Missing required physical fields fail physical-profile compilation.

SOURCES S075-01 Hammond factory service manual, 220-page compilation / S075-09 Hammond B3/C3 service compilation with 1995 redraw appendix

ORIGINAL MEASUREMENT PLAN · M2–M5

Acquire the dry organ in an order that exposes causes

Whole-system risk and unknowns register

Begin with calibrated boundaries and source identity. The 91-frequency file is a computed software-model candidate; it does not supply physical gear teeth, measured tuning, waveforms, shaft phase or leakage. Capture synchronized material that distinguishes electrical tooth phase from mechanical angle and tests simultaneous sources. Preserve whether each recording includes filtering and leakage so later modules do not duplicate them.

Next observe controlled registrations, contact trajectories, percussion recharge and expression. The factory map and its numerical checks can drive routing immediately. They do not establish the chosen instrument’s contact timing, drawbar law or modifications. Retain continuous source state and explicit contact/preset ownership so calibration replaces parameters rather than rewriting semantics.

For the scanner, 025 supplies the published 18-section ladder and 19 outputs, while the paper explicitly uses a simplified scanner model. [S075-03] Its nominal response is engineering evidence; specimen losses, loading and rotating capacitance remain acquisitions. Validate dry G–G output before cabinet EQ or microphone adjustments can compensate for organ errors.

Capture order	Resolve	Provisional meanwhile
Source identity and clocks	Tuning, phase and units	009 candidate frequencies.
Loaded source records	Waveform, filters and leakage ownership	Synthetic waves/coupling.
Key/control sequences	Contacts, drawbars and percussion	Proposed timing/gain laws.
Scanner taps and angle	Losses, return and capacitance	025 nominal idealization.
Dry output with held-out gestures	Coupled organ behavior	Uncalibrated AO-28 profile.

CHECKS AND DECISIONS

• Raw pickup, loaded terminal and G–G readings cannot share a calibration label without an explicit boundary transformation.

SOURCES S075-03 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

ORIGINAL MEASUREMENT PLAN · M6–M8

Resolve Leslie loading before fitting rotating acoustics

Whole-system risk and unknowns register

Use the selected 122 drawing and preserve 147 as a different profile. The 1971 source separates amplifier/divider circuitry from mechanical assemblies. [S075-02] The crossover graph establishes nominal topology and ports; it does not establish actual driver impedance, capacitor loss, installed polarity or cabinet response. Capture the electrical chain before fitting pressure measurements.

Record horn and drum trajectories independently, including repeated transitions. Then acquire complex stationary-angle responses and synchronized rotating observations using identified driver, diffuser, cabinet and microphone geometry. A shared physical trajectory still produces different emission times for different microphones. Give propagation, directivity, cabinet scattering and room response explicit model owners.

Inventories do not supply manufacturing CAD. Panel construction, louvers, rotor axes and horn details require the same specimen’s measurements. Other-model papers and photographs can guide a proposed model, but cannot close a target-specific gap. Optional reverb and historical brake configurations remain separate acquisitions. Fit microphone/room behavior last so those parameters do not absorb earlier errors.

Unknown	Acquisition owner	Closure evidence
Impedance and crossover loss	Electrical lead	Complex loaded branch transfers.
Separate rotor trajectories	Motion lead	Held-out speed/phase transitions.
Cabinet/projector geometry	Geometry lead	Scale-qualified dimensions and pivots.
Angular complex radiation	Acoustic lead	Held-out angles and moving captures.
Microphone and room paths	Recording lead	Reproducible rig and complete paths.

CHECKS AND DECISIONS

• Keep treble A–H floating in the electrical model. Apply each measured or modeled delay/filter exactly once.

SOURCES S075-02 Electro Music/CBS Leslie manual, publication 044578, June 1971

IMPLEMENTATION RISK · M1 / M9

A plausible interface does not resolve product engineering

Whole-system risk and unknowns register

Build the actual compiler, renderer and bounded runtime. The contract’s 19 static assertions and 62-parameter registry checks do not test callbacks, native exports, graph lifetime or saved sessions. PortAudio identifies allocation, I/O and mutex operations among unsuitable callback work. [S075-06] Qualify the implemented path, including overload and profile retirement.

The no-audio HTML prototype exists. Browser execution is environment-blocked at this review; independent 072 code/state checks and local test instructions are pending. This is a UI verification gap, not an acoustic or documentary blocker. Chapters 068–071 propose interaction, inspection, 3D presentation and accessibility; they do not prove telemetry or native usability. Bind production controls to actual stable parameters and qualify visuals in a permitted environment.

Run native standalone/DAW tests. Steinberg’s validator checks VST3 conformity, not sound equivalence. [S075-05] The SDK is MIT-licensed since 3.8; JUCE modules have separate AGPLv3/commercial terms. [S075-07][S075-08] Pin dependencies and audit assets rather than treating one accessible document or license as permission for everything.

Workstream	Still required	Closure evidence
Real time	Scheduling, queues, lifetime and resource checks	Instrumented runtime reports.
Host/state	Native IDs, curves, migration and DAW sessions	Versioned host matrix.
UI/accessibility	Bindings, flat skin and assistive workflows	Interaction/user evidence.
Distribution	Build, packaging, dependency and asset decisions	Reproducible release bundle.

CHECKS AND DECISIONS

• Editor absence, reduced motion and rendering failure must preserve audio state and external MIDI ownership.

SOURCES S075-05 Steinberg VST3 Developer Portal: Validator command line / S075-06 PortAudio: Writing a Callback Function / S075-07 JUCE master LICENSE.md, accessed 12 November 2026 / S075-08 Steinberg VST3 License, accessed 12 November 2026

ACCEPTANCE · REGISTER MAINTENANCE

Close gaps with evidence; keep resolved corrections closed

Whole-system risk and unknowns register

Close a risk with a reviewable source reconciliation, regression test, calibrated capture or scoped acceptance report. Record profile, input hashes, method, uncertainty and reviewer. Reopen affected downstream results when reference identity, topology or boundary ownership changes. Another mirror is not independent corroboration; project-derived paper corrections are not author-issued errata.

Do not repeat completed research as an unresolved blocker. All 91 Table V values passed independent second transcription; the treble-driver/spacer confusion and AO-28 locator are corrected. Remaining gaps concern measurement interpretation, installed parts and connectivity. These are different responsibilities, as the table makes explicit.

Establish acceptance rules before final fitting. NIST distinguishes residual error from uncertainty and includes relevant covariance in combined uncertainty. [S075-04] Use repeated and held-out captures to decide pass, fail or inconclusive within a stated scope. Until that evidence exists, report engineering progress and proposed profiles honestly. A comprehensive source atlas prepares the work; it cannot establish a universal 1:1 instrument.

Resolved issue	Evidence retained	Related open gap
91 Table V values independently matched	060 independent transcription/verification	Meter interpretation and new calibration.
020305 driver; 020313 spacer ring	Corrected 045 and 048 inventory	Installed driver/impedance/condition.
Upper Figure 24 at PDF 129	Drawing locator and component registry	Revision-specific connectivity.

CHECKS AND DECISIONS

• Keep unknowns null. A synthetic fixture, model-generated example, animation or compile result is not a physical capture.

• Begin reference/capture setup, M1 compiler/render harness and AO-28 graph review in parallel; calibrate sources/controls/scanner/dry output before Leslie and room fitting.

SOURCES S075-04 NIST Technical Note 1297, section 5: Combined Standard Uncertainty

INDEPENDENT TECHNICAL AUDIT

Separate resolved corrections from completed emulation

Technical misconception and claim audit

The vintage B-3 has no integrated-circuit chipset. Its generator, switching, scanner and valve circuitry require an identified historical profile. Factory system and AO-28 drawings are the authority; modern digital organs and servo-controlled Leslie models belong to separate profiles. [S076-01] [S076-21] This audit resolves documentary and mathematical questions without claiming a measured instrument or finished plugin.

The AO-28 original is upper Figure 24, printed 3-50/PDF 129 in the 220-page manual. PDF 127 and the lower drawing are wrong substitutes. PDF 71 of the 73-page compilation is a 1995 redraw. Prefer the applicable original when designators conflict, and retain uncertainty where a glyph remains ambiguous. Mirrors of one scan are not independent corroboration. [S076-01] [S076-02]

Use TECHNICAL_AUDIT.md for 29 detailed findings, primary locators and package-relative regression/acquisition gates. Apparent paper errors are independent analyses, not author-issued errata or proof that the authors software was wrong. The 28 patent plates document historical embodiments; the complete scanner SVG depicts a nominal published graph. Neither establishes unmeasured production geometry. [S076-20]

Evidence class	Permitted conclusion
Documentary correction	The named drawing/data interpretation is resolved.
Mathematical fixture	The declared equation holds for the tested assumptions.
Measured comparison	Requires traceable specimen captures; absent here.
Product verification	Requires actual engine, wrapper, runtime and host tests.

CHECKS AND DECISIONS

- M0 records organ, chassis code, generator region, cabinet, drivers and modifications.
- Every downstream artifact keeps revision, evidence class and uncertainty visible.

SOURCES S076-01 Hammond original console service manual, 220-page compilation / S076-02 B-3/C-3 compilation with retrospective 1995 AO-28 redraw / S076-20 Hammond and Hanert: US2560568A, Vibrato apparatus / S076-21 Hammond Leslie Heritage122H/142H owner manual, V1.00-220713

DOCUMENTED MAPS / OPEN CALIBRATION

Audit the routing and states before fitting tone

Technical misconception and claim audit

Table III is the routing authority. [S076-01] The manual map uses repeated octave subtraction above source 91, not one subtraction. Key 61 must produce 61,80,73,85,80,85,89,80,85. The pedal third-harmonic branch duplicates the second source on pedals 1–7, disconnects on 8–12, and resumes ordinary third-harmonic routing on 13–25. Preserve its five intentional nulls.

Keep one running phase per physical source and distinct contact trajectories. A note must not reset the source; a folded assignment must not create another oscillator. Mechanical shaft angle differs from tooth phase. The contact study concerns a 1967 C-3 and does not establish universal timing, velocity gain or release behavior. [S076-17]

Percussion needs continuous shared recharge state; last-key release is not an instant reset. Preserve the 1-foot keying contact when suppressing its audible output. Cancel changes preset routing without manufacturing note-offs. Exact held-note tablet/preset behavior remains a capture/circuit gate. [S076-01] An implementation must expose these separate states so calibration cannot conceal architectural errors.

Trap	Retained boundary
Pitch number versus terminal	data/015-manual-map.json defines the index convention.
Algorithm versus reread	549 assignments checked;144 independently reread manual cells.
Pedal table	200 positions;195 connections;5 explicit disconnections.
Historical Table V	91 reread values at AO-28 G–G; meter range/detector unknown.
Controls	Continuous drawbar travel; expression is a loaded network.

CHECKS AND DECISIONS

• Reject the single-subtraction mutation with reference_code/verify_factory_map.py.

• Run held-key Cancel/retrigger fixtures without resetting phases or charge.

• The total double-fold count is seven; the corrected verifier now enumerates all seven locations and the rerun passed.

SOURCES S076-01 Hammond original console service manual, 220-page compilation / S076-17 Moro, McPherson and Sandler: Dynamic temporal behaviour of the keyboard action on the

INDEPENDENT DERIVATION

Repair scanner count, references and WDF algebra

Technical misconception and claim audit

DAFx 2016 Figure 1/Table 1 defines 18 LC sections and 19 ground-referenced outputs; its prose count conflicts with the drawing. Outputs 1–6 are divider midpoints and 7–19 are p6–p18. The nine-terminal depth map differs from the 16-position scan sequence. [S076-03] Check all 50 passive records against images/scanner_complete_network.svg and data/025-scanner-network.json.

Let r be return voltage and z_k a tap relative to that return. Because v_k=r+z_k, normalized scanner weights already include r once. Adding another dry copy double-counts chorus. A moving capacitor obeys d(Cv)/dt; unloaded crossfading remains a declared approximation. Preserve output references, polarity and state when changing topology.

The equations below independently eliminate the published root variables. The printed Gamma expression substitutes the wrong factor. Source/open/short signs and reactive leaves also require convention-consistent derivation. The prewarp example mixes hertz with radians: at 44100 samples/s and 7075 Hz,2/K is 24.8133 microseconds, not 22.7245. [S076-03] The supplied algebra/AC checks are not a completed WDF, SPICE comparison or hardware validation.

Adapted leaf	Required memory relation
Inductor, R=K*L	b[n]=-a[n-1]
Capacitor, R=1/(K*C)	b[n]=a[n-1]

MODEL EQUATIONS

bi=S11*ai+S12*ae; be=S21*ai+S22*ae; ai=Phi*bi+Psi*x

M=I-Phi*S11; Gamma=S21*solve(M,Phi*S12)+S22; Theta=S21*solve(M,Psi)

a=v+R*i; b=v-R*i; source b=2V-a; open b=a; short b=-a

K=omega0/tan(omega0/(2*fs)); omega0=2*pi*f0; f0 in Hz

CHECKS AND DECISIONS

- Compare all nineteen signed outputs before normalizing any amplitude.
- Use reference_code/wdf_algebra_reference.py; retain its synthetic scope.

SOURCES S076-03 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

DOCUMENTARY CORRECTIONS / OPEN CIRCUIT MODEL

Preserve AO-28 and Leslie revision boundaries

Technical misconception and claim audit

The AO-28 value registry has 124 placement/section records across alternatives; it explicitly contains no connectivity edges. It is not a full netlist or an installed-package count. Retain C59 instead of the redraw duplicate C56 and T7 instead of adding layout T1. Keep solid P–D and no-code alternatives mutually exclusive. Solid R62 and primary expression capacitance remain unresolved. [S076-01] [S076-02]

The 1971 Leslie page 19 left/right sheets are schematic halves. Preserve separate 122/147 input, control and feedback profiles. The confirmed parts correction is 020305 treble driver versus 020313 spacer ring, not a variant conjecture. [S076-05] [S076-06] Stock identity still does not identify the installed driver transfer or repairs.

The treble crossover load floats between A and H; it is not a grounded H–B load. Electrical sign does not establish acoustic polarity. Five-to-eight seconds is a particular bass fast-to-Chorale service adjustment, not universal tau or horn timing. Historical Brake Accessory instructions document an optional thermal-relay stop system. [S076-05] [S076-07] Keep that accessory separate from the baseline and modern servo profile.

Handoff	Blocking gate
data/ao28-component-values.json	Independent values/variant review plus complete pins/nodes.
data/031-ao28-stages.json	Loaded DC/AC/nonlinear model and transformer evidence.
data/044-crossover-network.json	VT=VA-VH; VBass=VW-VB; measured complex loads.
data/048-leslie122-parts.json	Installed cabinet/driver/motor identification.

CHECKS AND DECISIONS

- Do not combine mutually exclusive schematic alternatives.
- Choose rotor ramp direction from target minus current speed, preserving angle.

SOURCES S076-01 Hammond original console service manual, 220-page compilation / S076-02 B-3/C-3 compilation with retrospective 1995 AO-28 redraw / S076-05 Electro Music Leslie 122-family service manual, 044578, June 1971 / S076-06 Leslie 122–147 service information, 136370, October 1979 / S076-07 Leslie Brake Accessory, instructions038398

DERIVED CONSTRAINTS / MEASUREMENT GATES

Protect propagation timing and electromechanical energy

Technical misconception and claim audit

One physical rotor trajectory feeds different emission times at different microphones. With receive time t, emission e, source q, microphone m and sound speed c, solve the retarded path below. The DAFx 2002 experiment concerns a Leslie 600 horn; it does not calibrate an unidentified 122 or its bass. [S076-04] A point-path derivation is also not a complete acoustic field model.

Independent checks find reversed fractional-interpolation weights on PDF 3 and a cosine-sign inconsistency in Equation 18/PDF 4. Use the endpoint-correct interpolation and coordinate expansion below. [S076-04] Preserve source/microphone incidence signs and fixed cabinet geometry. Whole-cabinet angular responses cannot generally rotate by a microphone azimuth offset. Input-time weights w[n-l] and output-time weights w[n] define different changing filters; frozen-angle agreement cannot validate their order. [S076-22]

For variable driver inductance, include both the electrical Lprime*i*v term and mechanical 0.5*Lprime*i^2 force. Otherwise the declared conservative model violates energy balance. General driver papers are mechanism references, not fitted Leslie data. [S076-10] [S076-19] The thermal manuscript prints 0.039, while NBS gives 0.00393 per degreeC at 20C for standard copper; actual coil material/calibration remains open. [S076-08] [S076-09]

MODEL EQUATIONS

t=e+|m-q(e)|/c; u=(m-q)/|m-q|; de/dt=1/(1-u dot v/c)

interpolated=(1-mu)*x[i]+mu*x[i+1]; 0<=mu<=1

distance^2=R^2+a^2-2*R*a*cos(theta) for the declared circular coordinates

CHECKS AND DECISIONS

• Use reference_code/retarded_path_reference.py and driver_energy_reference.py with their stated synthetic scope.

• Acquire moving-cabinet and synchronized driver voltage/current/acoustic data before fidelity promotion.

SOURCES S076-04 Smith, Serafin, Abel and Berners: Doppler Simulation and the Leslie / S076-08 Klippel: Nonlinear Modeling of the Heat Transfer in Loudspeakers, author manuscript / S076-09 National Bureau of Standards: Copper Wire Tables, Handbook 100,1966 / S076-10 Klippel: Nonlinear Adaptive Controller for Loudspeakers with Current Sensor / S076-19 Klippel: Large Signal Performance of Tweeters, Micro Speakers and Horn Drivers / S076-22 Live Convolution with Time-Varying Filters, Applied Sciences2018

RELEASE AUDIT / REMAINING WORK

Keep host, graphics and fidelity gates independent

Technical misconception and claim audit

All 36 manual drawbars need permanent identities across four retained banks. Hiding a bank in the custom editor must not set VST3 kIsHidden, which disables external changes and automation. Inspect actual wrapper export and enum conversion before freezing compatibility. [S076-11] [S076-13] The 62-parameter registry verifies proposals, not a production host interface.

VST3 curves can imply a previous anchor at offset minus one; collapsing a queue to its last value creates block-dependent steps. Zero-frame flushes update controls without advancing modeled time. [S076-12] [S076-23] Serialize JUCE convolution methods with processing despite wait-free loading. [S076-15] The C++ contract and metadata reports do not establish runtime scheduling, resource lifetime or DAW performance.

For the Z-up authoring convention, glTF export (x,y,z) to (x,z,-y) preserves handedness; simply swapping Y/Z reflects geometry. glTF uses metres, radians and Y up. [S076-14] Apply the transform coherently to meshes, pivots, cameras and acoustic coordinates. The UI is a design/control prototype; browser execution was environment-blocked, so no screenshot or visual-pass claim is made.

Proceed through chapter 073 milestones M0–M9 with the mission in 074. A source correction can close a documentary question while the corresponding circuit, acquisition or product gate remains open. Retain explicit evidence classes when reporting progress; no numerical fixture establishes one-to-one sonic fidelity.

Current evidence	Still required
Verified routing metadata	Specimen wiring and calibrated contact/recharge captures.
Scanner algebra and nominal graph	Connected solver and loaded tap measurements.
AO-28 value registry	Complete reviewed connectivity and transformer models.
Acoustic/driver equations	Held-out electrical and moving acoustic recordings.
UI and interface proposals	Native host, runtime, accessibility and rendering tests.

CHECKS AND DECISIONS

• Keep failed, blocked and not-run gates distinct from passed.

• Use TECHNICAL_AUDIT.md to turn each corrected interpretation into a specific regression or acquisition task.

SOURCES S076-11 Steinberg VST3 ParameterInfo reference / S076-12 Steinberg VST3 IParamValueQueue reference / S076-13 Steinberg VST3 Parameters and Automation / S076-14 Khronos glTF 2.0 specification, official AsciiDoc source / S076-15 JUCE dsp::Convolution class reference / S076-23 Steinberg VST3 API Documentation

EVIDENCE SCOPE / ORIGINAL ACQUISITION PLAN

Define what a picture can prove

Photo, drawing and schematic acquisition checklist

The companion data/photo_capture_checklist.json specifies88 acquisition task families for an identified console and cabinet. A task can require several views, states or indexed tiles;88 is not a photograph count. No selected-specimen image has been captured for this edition. Existing gallery assets help locate mechanisms and communicate appearance, while every task retains new_target_specimen_capture_needed=true.

ORIGINAL EVIDENCE CONTRACT. Give every photograph one or more declared uses: identity, topology, geometry, appearance, condition or timing. Acceptance for one use does not establish the others. A sharp capacitor label may establish a printed value but cannot establish capacitance, leakage or installed connectivity by itself. An attractive cabinet photograph cannot identify wood chemistry or acoustic loss.

Two concrete exclusions matter: P005 contains reconstructed pedalboard details and a documented width adjustment; P020 includes extrapolation and changed perspective. Their source records disclose these changes. Keep both as contextual illustrations and request independent dimensional evidence. [S077-06] [S077-07] Preserve an unknown rather than allowing a plausible image to supply it.

Claim sought	Evidence required beyond appearance
Part identity	Readable marking plus assembly context
Electrical connection	Unambiguous terminal-to-terminal trace
Dimension	Defined plane, scale and uncertainty
Motion	Characterized timebase and angular reference
Material or response	Separate qualified measurement

CHECKS AND DECISIONS

- Classify each accepted image by the engineering claims it actually supports.
- Reject generated reconstruction as evidence of hidden physical details.

SOURCES S077-06 P005 original file description and retouching history / S077-07 P020 original Leslie isometric file description

SOURCE MAP / SPECIMEN COVERAGE

Cover assemblies, states and hidden boundaries

Photo, drawing and schematic acquisition checklist

The factory assembly sheets provide the starting hierarchy: cabinet, generator, drawbars, manual action, presets, vibrato line, controls and expression. The Leslie exploded view separates cabinet, rotor and motor hardware. These source categories guide coverage; their callouts do not establish which replacements are installed in a particular instrument. [S077-01] [S077-02]

ORIGINAL CAPTURE ORDER. Record identity and as-found condition first, then external geometry and control legends. A qualified technician documents accessible interiors before any optional reversible access. Capture complementary views only where a named occlusion or alternate state demands them. Record restored condition before subsequent audio work.

The checklist uses task dependencies rather than a universal teardown sequence. All-key contact strips, all-terminal wiring mosaics and all-bank control states require systematic coverage. A representative mechanism closeup is useful for explanation but cannot close those tasks. Optional RV and brake hardware remain conditional on the selected reference; historical brake accessories are documented separately from the baseline cabinet. [S077-08]

Task family	Primary implementation use
GEN / KEY / PED	Source, contact and wiring maps
SCN / AO / LES	Actual network connectivity
ORG / CAB / ROT	Geometry and mechanical hierarchy
RIG / DOC / QA	Capture identity and reproducibility

P027 | Maintaining the Leslie cabinet.jpg | CJ Sorg from Hendersonville, United States | CC BY-SA 2.0 Source and license

CHECKS AND DECISIONS

- A blocked view remains explicitly blocked, with its next acquisition named.

SOURCES S077-01 Hammond factory service manual, broad 220-page collection / S077-02 Electro Music / CBS Leslie manual044578, June1971 / S077-08 Leslie Brake Accessory original instruction sheet

IMAGE QUALITY / ORIGINAL METROLOGY PROPOSAL

Calibrate the image, not its visual plausibility

Photo, drawing and schematic acquisition checklist

FADGI distinguishes reproduction scale, geometric distortion, reference targets and metadata. Its guidance places targets in the relevant image plane and repeats target capture when conditions change. These principles support this workflow, but applying them does not certify a three-dimensional organ measurement or confer a FADGI rating. [S077-04]

ORIGINAL GEOMETRY PROPOSAL. Take separate near-normal views of each required surface and measure physical endpoints independently. Put the ruler in the feature plane; a ruler on the cabinet front cannot scale a recessed linkage without a depth model. Record lens, focus distance, exposure, camera pose, distortion treatment and target identity. For a dimension, retain the point selections, transform, calibration uncertainty and independent cross-check.

Use diffuse light for labels and a separately recorded raking-light view for relief, joints and surface wear. Preserve the original camera file. Lens correction, rectification, focus stacking and color transformations create traceable derivatives. Never conceal missing details with generative fill. Match texture appearance under declared illumination; do not convert perceived gloss or color into an unmeasured material constant.

Acquisition purpose	Required control
Planar dimensions	Same-plane scale and independent endpoints
Internal geometry	Registered views and depth constraints
Readable component value	No glare, full label and location frame
Material appearance	Neutral target, light geometry and processing
Motion record	Timebase, shutter model and frame-loss record

CHECKS AND DECISIONS

- Review original pixels, not only downscaled contact sheets.
- Keep geometric uncertainty separate from pixel resolution.

SOURCES S077-04 FADGI Technical Guidelines for Digitizing Cultural Heritage Materials, third edition,

TOPOLOGY CROSSWALK / ORIGINAL DATA WORKFLOW

Turn photographs into reviewable circuit evidence

Photo, drawing and schematic acquisition checklist

ORIGINAL TRANSCRIPTION WORKFLOW. Start an AO-28 or Leslie network with stable physical-terminal IDs, then attach image regions to each observed connection. Preserve front-versus-rear connector orientation and socket indexing. Join two nodes only when a continuous conductor or a separately documented continuity check establishes the connection. A crossing, matching wire color or familiar circuit pattern is insufficient.

For each designator, retain the as-found label, interpreted value, unit, revision namespace and reviewer. Treat stacked terminals and obscured lead ends as explicit follow-up tasks. The existing AO-28 value registry is not a complete connectivity graph; photographs must close the missing edges without merging alternative revisions into one imaginary chassis.

The generator-filter bulletin explicitly permits field changes on earlier instruments. Capture installed filter topology as well as serial and date evidence. [S077-03] The same discipline applies to replacement valves, drivers, motors and optional accessories. Control-state and motion images should link to chapter 057 capture records; video timing alone does not establish an electrical or acoustic calibration.

```
for observed_terminal in image_region:
    create_or_match_physical_id(observed_terminal)
    attach(master_hash, region, orientation, reviewer)
for proposed_connection in traced_connections:
    if endpoints_and_path_are_supported:
        accept_edge_with_evidence()
    else:
        retain_unknown_and_request_specific_view()
```

CHECKS AND DECISIONS

- A second reviewer can recover every accepted node and value from its cited region.
- No stock-number sequence, OCR suggestion or nominal model fills an unobserved connection.

SOURCES S077-03 Hammond Technical Bulletin16: improved generator filtering

METADATA / ORIGINAL HANDOFF CONTRACT

Package masters, derivatives and rights separately

Photo, drawing and schematic acquisition checklist

Each checklist task carries exact view, engineering use, scale, lighting, identity, dependencies, priority, existing context assets and acceptance criteria. The photo_capture_v1 profile requires specimen, assembly, session, view and state identifiers; camera and lighting settings; coordinate and scale references; processing history; rights records; and reviewer decisions. Unknown required facts remain null with a reason. The checklist contains no invented serials, capture dates, files or hashes.

Keep untouched masters, archival derivatives, annotated evidence and UI textures as separately identified objects. An annotation stores its parent-master hash and image-region coordinates. A coordinate transform names both frames and units; a mirrored photograph cannot silently reverse an amplifier connector. Rights to access an instrument, rights to a photograph and permission to redistribute an old manual are different records.

RFC8493 describes manifests that associate payload paths with checksums. This is useful for transfer-integrity design, but a valid hash proves unchanged bytes rather than truthful measurement or permission. [S077-05] The current checklist is a project JSON plan, not a claim that a complete BagIt package or a capture corpus already exists.

Output class	Required lineage
Camera master	Operator, specimen, state, original bytes
Corrected image	Parent master and declared processing
Annotated topology	Parent region, interpretation and reviewer
3D or UI asset	Source images, dimensions and transforms
Shipped derivative	Creator, permission/license and change notice

CHECKS AND DECISIONS

- Verify hashes after transfer and retain every derivative’s actual parent.
- Carry retouching and specimen limitations into exported assets.

SOURCES S077-05 RFC8493: The BagIt File Packaging Format, version1.0

ACCEPTANCE / IMPLEMENTATION DEPENDENCIES

Close acquisition tasks with evidence, not image volume

Photo, drawing and schematic acquisition checklist

ORIGINAL COMPLETION RULE. A task closes only when its requested feature or state is visible, identity is established, measurements carry their uncertainty, permissions are recorded and an independent reviewer accepts the intended claim. Otherwise use blocked, rejected, reacquire or justified not_applicable. More photographs of the same accessible surface cannot compensate for an unresolved terminal or hidden rotor interface.

Execute identity and quality targets first. Prioritize generator source/terminal mapping, AO-28 connectivity, scanner wiring, Leslie divider connections and the common microphone coordinate frame. Then complete appearance and motion datasets in coordination with the physical acquisition campaign. Capture banks and routed presets as separate state, and preserve pre-access/restored condition so an intervention cannot silently change the reference between recordings.

The delivered JSON was checked for unique task IDs, valid dependencies without cycles, known existing asset IDs and present companion-data paths. These are documentary checks only. All88 task families remain not_acquired, and the total future frame count remains unknown. Physical availability, permissible internal access, calibration uncertainty and final image reuse permissions still require the selected specimen and its custodian.

Gate	Pass evidence
Identity	Console/cabinet/assembly match
Coverage	Every required task accepted or explicitly unresolved
Interpretation	Independent node/value/geometry review
Integrity and rights	Verified lineage plus actual permission record
Model handoff	Mapped dataset fields with declared unknowns

CHECKS AND DECISIONS

• Do not mark a specimen task complete because a gallery picture exists.

• Do not infer acoustic equivalence, a complete netlist or production-ready CAD from photographic coverage.

SOURCES S077-01 Hammond factory service manual, broad 220-page collection / S077-02 Electro Music / CBS Leslie manual044578, June1971 / S077-04 FADGI Technical Guidelines for Digitizing Cultural Heritage Materials, third edition, / S077-05 RFC8493: The BagIt File Packaging Format, version1.0

SOURCE AUDIT / ORIGINAL RELEASE CONTRACT

Keep permission and engineering authority separate

Documentary source and rights audit

The audit covers43 source visual assets and 28 selected drawing plates from seven patents. Fresh primary Commons metadata queries returned all 40 photograph records, and all 40 license labels matched the manifest. The remaining three figures were checked against the publisher's article license and captions. These are recorded inspection results, not a claim that every creator's ownership history was independently investigated.

ORIGINAL WORKFLOW PROPOSAL. Maintain two independent decisions for every asset: whether its declared terms support the intended reproduction, and what engineering claim its content can support. A licensed perspective photograph can be useful context while remaining unsuitable for dimensional inference. A detailed circuit drawing can be technically authoritative while its complete scan remains excluded from redistribution.

rights_audit.json retains source URL, creator, exact license, previous alterations, suite changes, file hash, applicability and inspection scope. Keep that record alongside illustration_catalog.json. A replacement image is a new asset revision: matching its subject or filename does not transfer the previous permission record. No source declaration becomes a measured target-specimen identity.

Decision	Required evidence
Reproduction	Identified license or applicable source-specific basis
Attribution	Creator, supplied title, source, license and alterations
Engineering use	Model/revision, scale, perspective and actual content
Product integration	Intended adaptation, dependency and branding review
Completion	Hash-bound asset and notices in the final artifact

CHECKS AND DECISIONS

- Reject an asset that has a useful subject but no recorded reuse basis.
- Do not assign a dimensional measurement merely because an image is high resolution.

ATTRIBUTION / IMAGE APPLICABILITY

Preserve the history of an edited picture

Documentary source and rights audit

The suite uses several Creative Commons versions, plus CC0 and individual author public-domain releases. Keep each identified version rather than upgrading all credits to one convenient label. Attribution and ShareAlike conditions belong to the applicable asset; preserve required modification notices and avoid suggesting endorsement. [S078-02] [S078-03] [S078-04] [S078-10] [S078-11] [S078-12] [S078-13]

Independent source-content inspection recovered additional alteration details. The table distinguishes ordinary cropping from reconstruction and changed geometry. These differences affect engineering use even when the reuse license is unchanged. The source pages identify derivative authors as well as original photographers; retain both when supplied.

P035’s source machine-readable creator field was empty, but its source prose names bbear66. The final manifest now includes that verified supplement without misidentifying uploader BEG as photographer. [S078-20] ORIGINAL PRESENTATION PROPOSAL: use short captions in the book and complete records in the picture atlas; neither view should erase the source's alteration history.

Asset	Verified prior alteration
P005	Front pedalboard details extrapolated; no geometry evidence [S078-14]
P006	Manuals cropped from source photograph [S078-15]
P012	Warning board removed; region interpolated [S078-22]
P016 / P017	CV image perspective and aspect changed; cropped [S078-16] [S078-17]
P020	Rear-left edge extrapolated; perspective changed [S078-18]
P037	Interior cropped and sharpened [S078-19]

CHECKS AND DECISIONS

• Keep P028/P030 cabinet identity unresolved and example valve photos distinct from fitted parts.

SOURCES S078-03 Creative Commons Attribution-ShareAlike 4.0 International / S078-14 P005 pedalboard source alteration record / S078-18 P020 isometric Leslie source alteration record / S078-20 P035 Tung Sol6550 creator and release

PRIMARY RIGHTS BASIS / SOURCE MEANING

Review patent plates and research figures individually

Documentary source and rights audit

USPTO guidance describes patent text and drawings as typically unrestricted, while expressly retaining exceptions and possible copyrighted or trademarked material. [S078-01] This review searched the seven documents' extracted text and inspected upper/lower margins of all 28 selected plates, alongside the recorded earlier full-sheet review. No notice was identified. Extraction and margin inspection have limits; the result is a documented basis, not a universal legal guarantee.

Patent expiration is not the stated copyright basis. Keep each plate's patent number, original page, source and historical applicability. A scanner invention with alternative networks cannot establish a later production circuit; an early rotor/frequency chart cannot become a measured B-3 tuning table merely because it looks detailed.

The 2008 publisher record links CC BY2.0. The selected122A photograph, authors' measurement and authors' simulation remain different evidence classes. [S078-05] Preserve that distinction in captions and datasets. A permitted figure does not supply the underlying recorded waveform or grant rights to unrelated software downloads.

Asset	What the reused figure establishes
P041 / Figure 4	Authors' open/closed122A cabinet context
P042 / Figure 5	Authors'800 Hz recorded acoustic response
P043 / Figure 8	Authors' modeled output including image sources

CHECKS AND DECISIONS

- No project capture or plugin result may be inferred from P042/P043.
- Preserve notices on any newly acquired plate rather than applying this review to it automatically.

SOURCES S078-01 USPTO website terms: patent information / S078-05 Kronland-Martinet and Voinier: moving sources and Leslie cabinet

CURRENT SOFTWARE SOURCES / ORIGINAL INTEGRATION RULES

License the selected implementation dependencies

Documentary source and rights audit

The official VST3 portal specifies MIT licensing from SDK3.8 and separates optional branding rules. [S078-08] Current JUCE master identifies AGPLv3/commercial licensing for framework modules and links the JUCE9 agreement, published 17June 2026. Older JUCE versions retain their corresponding agreements. [S078-06] [S078-07] These observations are current-source evidence, not a pinned dependency set or completed commercial compliance review.

The inspected pinned setBfree source header declares GPL version 2 or later. [S078-09] Its implementation is not bundled. The companion supplies an original calculation of candidate frequency data from identified numeric ratios; it does not relabel the source program as permissively licensed. Importing upstream implementation later requires a separate dependency decision.

ORIGINAL BUILD CONTRACT. Record revision, license file, notices, transitive dependencies and intended distribution for every imported component. Keep that decision independent of sound-model provenance. Font software also has its own notices; THIRD_PARTY_FONT_NOTICES.txt accompanies the DejaVu-based atlas, while no standalone font files are supplied. [S078-21]

Item	Present scope / next decision
VST3 SDK	Documentation reference; pin actual SDK and notices
JUCE	No framework source bundled; choose applicable route
setBfree	Numeric comparison only; no copied source tree
Fonts / UI assets	Preserve font and individual image conditions
Future recordings	Obtain rights for the new captured corpus

CHECKS AND DECISIONS

- Never infer that SDK MIT permission licenses the framework, photos or recordings.

SOURCES S078-06 JUCE current framework license / S078-07 JUCE9 End User Licence Agreement / S078-08 Steinberg VST3 license / S078-09 Pinned setBfree tonegen.c primary source / S078-21 DejaVu upstream font license

ORIGINAL DATA AND RELEASE PROCEDURE

Make the audit an executable packaging input

Documentary source and rights audit

Use asset IDs as stable joins between chapter placements, visual catalog and rights records. A source URL identifies the origin; a hash identifies the actual included bytes. Preserve both original-source and distributed-file identities when resizing or recompressing. File equivalence checks cannot prove permission, but they can reveal a substitution that invalidates an earlier review.

ORIGINAL PIPELINE PROPOSAL. Build the final manifest from staged deliverables, not the research working directory. Include reviewed images, original illustrations and the finished atlas. Keep entire third-party manual scans, unrelated recordings and studied source trees outside that staging root. Retain viewing links and exact page locators so the implementation agent can still consult the original circuits.

Generated interface art is an appearance proposal. The two static control/inspection studies are original diagrams, not screenshots of the HTML prototype. Their explanatory labels remain necessary even if their style resembles an application. Final browser behavior and geometry verification belong to the separate UI report, whose blocked status cannot be overwritten by an attractive static image.

Packaging input	Relationship retained
illustration_catalog.json	Image, caption, creator and source
rights_audit.json	Permission basis, history and limits
source_catalog.json	Document identity and claim locators
diagram_catalog.json	Original diagram identity and chapter
Final file manifest	Distributed bytes and integrity

```
for file in staged_files:
    identify asset_or_original_artifact
    verify recorded_hash_and_path
    attach applicable_source_notices
    preserve modification_history
    preserve engineering_evidence_class
    reject unexpected_source_scan
write final_manifest_and_exclusion_report
```

CHECKS AND DECISIONS

- A changed hash requires a new revision review; a matching hash does not create legal permission.

EXECUTED CHECKS / REMAINING GATES

Close the audit at the actual release boundary

Documentary source and rights audit

The present checks cover71 source-image package paths and byte comparisons,40 freshly retrieved Commons license records, three publisher-licensed figures, seven patent text searches and 28 selected-plate margin inspections. No unexpected source-scan PDF or read-only research file was present in the inspected staging snapshot. Final archive assembly must repeat the exclusion and integrity checks; an earlier snapshot is not evidence about a later ZIP.

ORIGINAL ACCEPTANCE RULE. Separate completed source inspection from future product release. The project still needs rights for newly acquired specimen photography, recordings and impulse responses; decisions for actual shipped dependency revisions; and review of intended names, marks and adaptations. No licensing purchase, rights-holder contact or commercial-product clearance was performed here.

The permission audit also leaves acoustic truth untouched. A legally reusable photo is not calibrated CAD; a research simulation is not a reference recording; original generated art is not historical documentation. Carry these boundaries into chapter 077 acquisition and chapter 079 acceptance so a future agent cannot transform missing evidence into a pass by changing a label.

Review	Status in this edition
43 source visuals /28 plates	Individually recorded
71 staged source-image files	Present and hash-matched
Specific source alterations	Recorded; final manifest and captions enriched
Final archive scan	Required after assembly
Future commercial dependencies	Unselected / release gate
Physical and browser fidelity	Not established by this audit

CHECKS AND DECISIONS

• Report unresolved permissions by affected asset/use, not a blanket claim about every historical document.

• Keep the final audit and image credits in every distributed companion copy.

ORIGINAL ACCEPTANCE CONTRACT / EVIDENCE SCOPE

A pass belongs to one claim and one boundary

Acceptance matrix and definition of done

The companion acceptance_matrix.json separates recorded documentary, mathematical, syntax and UI-state results from future engine, physical and release gates. Each row states a requirement, setup, metric, threshold origin, milestone, evidence and limitation. An implementation agent must preserve these distinctions when replacing a reference calculation with optimized production code.

ORIGINAL DECISION PROPOSAL: track engineering readiness and specimen fidelity separately. A provisional instrument may become useful before a physical profile is acquired; it must retain that qualification. A component model with missing required connectivity cannot borrow a pass from a surrogate with a different boundary. Inventory completeness and a matching array shape are not acoustic acceptance.

Treat a result as versioned evidence. A changed circuit, source boundary, microphone rig or state contract can invalidate downstream calibration even if an old test still runs. Attach actual hashes and setup records to the accepted report; changing a status field without new evidence does not close a gate. Final book/archive checks remain a separate documentary task performed after integration.

Evidence tier	What a result may establish
Documentary	The inspected source/data claim and stated review coverage.
Mathematical / syntax	The declared equations, fixtures or compile-time contract.
Production implementation	Executed DSP, timing, memory and host behavior.
Physical / perceptual	A named specimen and tested domain within declared limits.

CHECKS AND DECISIONS

- No completed reference fixture automatically promotes an engine or physical gate.
- Record not_run, blocked, failed and inconclusive without converting them to pass.

RECORDED RESULTS / EXACT REVIEW COVERAGE

Retain the checks that were actually executed

Acceptance matrix and definition of done

The matrix reads the supplied reports rather than inferring success from chapter prose. Factory mapping verification covers549 assignments algorithmically and 144 cells through independent second visual reading. All91 historical service readings received an independent second transcription. The final audit corrected a summary count to seven double-fold cases; the routing data itself was already correct.

The schema report records39 expected outcomes with required format handlers enabled. JSON Schema distinguishes structural constraints from configured format behavior; even a successful syntactic check cannot establish that a capture or identified instrument actually exists. [S079-07] The provisional example therefore remains explicitly non-measured.

Compiler and UI results have similarly narrow boundaries. Nineteen C++ static assertions passed, and the 62-entry proposed registry passed its documented checks. The actual HTML JavaScript passed28 state checks using a limited DOM/event/storage double. That execution did not supply a browser layout, native accessibility tree, plugin binary or audio signal.

Recorded artifact	Accepted scope
060 verification	549 algorithmic cells;144 visual cells;91 service values;17 corruptions.
058 schema checks	39 metadata outcomes; no physical-capture claim.
063 /065 reports	C++ syntax and 19 assertions;62 parameter records and normalization fixtures.
072 UI checks	28 scoped state checks; browser rendering remains blocked.

CHECKS AND DECISIONS

- Read exact limits and input identities before reusing a report.
- The seven double-fold locations must remain enumerated in the verification result.

SOURCES S079-07 JSON Schema2020-12 validation vocabulary

ORIGINAL SOFTWARE QUALIFICATION / NO HARDWARE PROMOTION

Give numerical models independent reference gates

Acceptance matrix and definition of done

The supplied scanner AC, wave-digital algebra, crossover, retarded-path, rotor and energy fixtures test different mathematical questions. Small residuals demonstrate agreement within those questions. They do not establish a complete time-domain circuit, stable nonlinear solver, measured rotor or distributed acoustic field. The matrix retains each report with its own assumptions and observed metrics.

ORIGINAL IMPLEMENTATION SEQUENCE: first reproduce the declared nominal graph in an independent formulation; then test the production time-domain algorithm against that reference at comparable frequencies and boundaries. Follow with state migration, complete tails, difficult valid parameters and rate/block changes. Do not compare a bilinear-warped frequency with an analog frequency without declaring the conversion.

Convolution needs moving cases as well as frozen-angle impulses. Input-time and observation-time angular weights describe different operators; profile replacement also needs valid past history. The retained053 report gives numerical observations without a separate machine-readable pass flag, so the matrix labels it a recorded result rather than manufacturing a new acceptance threshold after observing the residual.

Reference evidence	Still required in production
Stationary scanner response	Independent solver, moving coupling, stable time integration.
Ideal16-ohm divider	Measured impedance/loading and physical acoustic comparison.
Retarded point path	Interpolation, bandwidth, multiple paths and geometry limits.
Synthetic rotor / driver	Actual mechanics, thermal behavior and validated integrators.

CHECKS AND DECISIONS

- Retain source, algorithm and measurement boundaries separately.
- Replay assert-based reference scripts with Python assertions enabled.

MEASUREMENT / ORIGINAL DECISION RULE

Choose physical tolerances before the final comparison

Acceptance matrix and definition of done

No universal numerical fidelity threshold is supplied for an unmeasured organ. First quantify repeatability, level/time calibration, loading, background noise and reference drift. NIST guidance calls for the uncertainty components and evaluation method, together with the stated coverage-factor basis where expanded uncertainty is reported. [S079-01] Preserve those records beside each comparison.

ORIGINAL DECISION PROPOSAL: preregister metrics and acceptable bounds for the intended use after a pilot, then freeze them before held-out testing. A mean error inside a tolerance is insufficient when its uncertainty reaches unacceptable values. Report inconclusive when the evidence cannot distinguish the declared acceptable and unacceptable regions; do not replace uncertainty with more displayed decimal places.

Listening addresses a separate question. BS.1116 provides a controlled framework for small impairments, while the actual task must distinguish discrimination, similarity and preference. [S079-03] A nonsignificant difference does not establish equivalence; prespecified meaningful bounds and adequate precision are necessary. [S079-02] Reserve whole performances, sessions and relevant gestures, not merely adjacent samples from fitting takes.

Decision input	Required record
Target	Specimen, circuit revision, microphones, room and operating domain.
Metric	Units, bandwidth, alignment and nuisance-parameter handling.
Bound	Justification fixed before final held-out comparison.
Outcome	Pass, fail or inconclusive with uncertainty and exclusions.

CHECKS AND DECISIONS

- Do not treat factory service tolerances as software-fidelity limits.
- Keep objective, listening and playable-response claims separately qualified.

SOURCES S079-01 NIST TN 1297, section 7: Reporting Uncertainty / S079-02 Daniël Lakens: Equivalence Tests: A Practical Primer, 2017 / S079-03 ITU-R BS.1116-3: Methods for the subjective assessment of small impairments in audio

HOST / INTERFACE / RELEASE GATES

A correct renderer still needs product qualification

Acceptance matrix and definition of done

The official VST3 validator checks interface conformity and supports custom tests. It is one release gate, not proof of every DAW workflow or organ sound. [S079-04] Run actual binaries through the chosen host/device matrix, session restoration, automation, offline bounce, multiple instances and supported lifecycle changes. Preserve the binary hash and complete result.

Processing contracts include varying call lengths, bufferless parameter flushes and restrictions on real-time operations. [S079-05] Test zero-frame control updates without advancing physical clocks, bounded queues, profile retirement, complete tails, bypass and silence policy. A passing C++ header cannot establish scheduling or transitive memory lifetime.

The HTML browser procedure remains blocked here by socket and URL-policy restrictions. Its syntax check and state-double results are retained honestly. Native accessibility, actual pointer dispatch, responsive layout and host coordinates require their own execution. W3C interaction-animation guidance supports disabling nonessential motion, but this single feature cannot establish complete web or native conformance. [S079-06]

Product surface	Separate evidence gate
VST3 / standalone	Actual validator, hosts, devices, state and lifecycle.
Audio callback	Measured deadlines, memory behavior and bounded failure policy.
Interface	Actual browser/native tasks, focus, scaling and reduced motion.
Distribution	Pinned dependencies, rights, packaging and platform verification.

CHECKS AND DECISIONS

- A frame stall must not alter source phase, rotor time or accepted note releases.
- Never substitute a design illustration for a browser screenshot or execution result.

SOURCES S079-04 Steinberg VST3 validator command line / S079-05 Steinberg VST3 Processing FAQ / S079-06 W3C Understanding SC2.3.3: Animation from Interactions

ORIGINAL DEFINITION OF DONE / HANDOFF

Close milestones with evidence, then bound the release claim

Acceptance matrix and definition of done

Execute the matrix in the M0–M9 dependency order while acquisition proceeds in parallel. M0 identifies the reference; M1 provides the compiler and offline renderer; M2–M5 establish the direct organ; M6–M7 establish the cabinet; M8 closes calibration and claim gates; M9 qualifies the delivered product. Read each row’s required scope instead of treating every physical unknown as a reason to stop unrelated engineering.

ORIGINAL CHANGE-CONTROL RULE: a completed report names its code, profile, data, setup, criterion, result and reviewer. When one changes, retain the old report and determine which dependent gates need repetition. A threshold chosen after observing validation data is a revised study, not a retroactive success. A not-applicable decision requires an explicit product exclusion.

For the research package, final PDF/archive integrity remains a separate integration result. Later root checks can close that documentary task without changing browser, host, audio or physical statuses. For the instrument release, supply reproducible artifacts, retained results, resolved rights, supported configurations and known limitations. The agent should finish a useful implementation while making no broader fidelity claim than its evidence supports.

```
for gate in dependency_ready_gates:
    load_exact_inputs_and_scope(gate)
    if required_evidence_missing: record("not_run"); continue
    result = execute_declared_procedure(gate)
    retain(result, identities, setup, decision_rule)
    decide_pass_fail_or_inconclusive(result)
    invalidate_affected_dependents_on_change()
```

CHECKS AND DECISIONS

- A procedure, planned threshold or attractive UI cannot set a gate to passed.
- The release manifest must distinguish provisional engineering from calibrated specimen fidelity.

INTEGRATION REVIEW / ACTUAL ARTIFACT SCOPE

Accept a research handoff with explicit boundaries

Final integration review and research handoff

PACKAGE AUDIT. This suite supplies an implementation mission, chapter data, visual references, source locators, documentary inventories, numerical references and a no-audio control prototype. It does not supply a measured reference organ, finished DSP engine, native plugin binary or demonstrated one-to-one match. The proposed baseline still requires an identified B-3, AO-28 revision and vintage Leslie 122 before specimen-specific calibration.

Completeness has several meanings. A stock list identifies replacement items; a component registry records values and alternatives; a netlist identifies pins, nodes and connections. The factory AO-28 drawing is the connectivity source, not permission to infer wires from adjacency in a parts table. [S080-01] The supplied scanner graph is complete within its named published idealization; that scope does not include every connected physical load. [S080-02]

Use AGENT_MISSION.txt to begin execution, TECHNICAL_AUDIT.md to avoid known mistakes, and risk_register.json to assign missing evidence. INTEGRATION_REVIEW.md records this final research review. Book rendering, export refresh and archive integrity belong to the subsequent assembly gate; the chapter itself cannot certify those later operations.

Artifact inspected	Accepted scope
AO-28 registry:124 records	Placements/sections across alternatives; no connectivity edges.
AO-28 stock list:85 rows	Replacement inventory; quantities and instance crosswalk incomplete.
Hammond/Leslie physical lists	319/176 source rows; not manufacturing-ready BOMs.
Scanner/divider graphs	50/4 passive elements within stated electrical boundaries.
Interface and C++ contract	Executed scoped state checks and syntax checks; no audio engine.

CHECKS AND DECISIONS

- Never convert record count into installed-component quantity.
- An unknown remains unknown after export, fitting or a successful build.

SOURCES S080-01 Hammond factory service manual: AO-28 Figure 24 / S080-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus

ORIGINAL COMPLETION CHECKLIST / M0 TO M5

Finish the missing AO-28 connectivity first

Final integration review and research handoff

ORIGINAL WORK ORDER. Select one documentary AO-28 branch and bind it to chassis evidence before transcribing. Use upper Figure 24, printed3-50/PDF 129, retaining the solid versus conditional P–D alternatives and the starred percussion substitutions. [S080-01] Keep repeated designators namespace-qualified. The 124-record value registry can seed instances, but it cannot generate a faithful circuit merely by sorting designators.

Create an auditable pin-to-node graph with junctions, crossings, connector views, shields, reference returns, external switch states and transformer winding orientations. Review it independently against both the drawing and selected specimen evidence. A second clear scan can resolve a glyph; it cannot establish that a restored chassis still follows that drawing. Preserve disagreements and their applicable profile.

After connectivity, add the model parameters required by the chosen solver. Separate documentary nominal values from measured winding resistance, ratio, losses, valve behavior and control laws. Resolve the null solid-network R62 reading and expression-capacitance law through source or specimen acquisition. A temporary behavioral model remains permissible when explicitly declared; it must not mark this component-level gate complete.

Order	Required output and closure evidence
1 Identity	Selected revision, branch applicability and linked chassis photographs.
2 Instances	One active variant; socket/section identities and component-pin definitions.
3 Connections	Every pin assigned; wire crossings and off-board paths reviewed.
4 Coupling	Bias/supply sharing, feedback and transformer polarity retained.
5 Parameters	Required model values, units, evidence and unresolved inputs.
6 Reference solve	Operating point, loaded AC and transient checks in an independent solver.
7 Physical comparison	Measured intermediate ports and held-out control states.

CHECKS AND DECISIONS

- A clean connectivity graph does not by itself establish measured parameter accuracy.
- No lower-page amplifier or alternate revision enters the selected AO-28 graph silently.

SOURCES S080-01 Hammond factory service manual: AO-28 Figure 24

ORIGINAL SOFTWARE INTEGRATION CONTRACT

Compile evidence into one owned signal graph

Final integration review and research handoff

ORIGINAL DESIGN PROPOSAL. Make the profile compiler the boundary between editable evidence and realtime state. Validate schema, resolve files and hashes, select variants, convert units, build coefficients and allocate storage before publication. JSON Schema describes structural constraints; the project must additionally check physical meaning and cross-file consistency. [S080-04] Preserve the authoring document and a conversion report alongside its immutable runtime view.

Every modeled transfer needs one owner. A loaded-terminal waveform cannot receive the same pickup/filter transfer again. Ground-referenced scanner voltages already include their return contribution. Electrical treble and bass voltages retain the A–H and W–B definitions in the supplied divider graph; those orientations do not determine acoustic polarity. [S080-02] [S080-03]

Use reference_code/ENGINE_CONTRACT.md and reference_code/DATA_CONTRACTS.md for the exact supplied contracts. Permit fused electrical solvers where current loading or shared bias prevents an independent cascade. Keep allocation, graph replacement and resource destruction away from the processing callback. A model with missing required physical inputs must fail compilation or choose an explicitly separate surrogate representation.

Boundary	Integration check
Generator	Source ID, physical port, units and normalization agree.
Scanner	19 output references remain ground-relative.
AO-28/Leslie	Connected load and feedback ownership are explicit.
Microphones	Geometry, propagation, gain and room ownership occur once.

```
read_and_validate_authoring_profile()
resolve_assets_and_verify_digests()
select_exactly_one_variant_per_group()
resolve_units_and_named_port_references()
reject_missing_required_model_inputs()
check_effect_ownership_and_coupled_loads()
prepare_graph_and_all_storage_offline()
publish_immutable_generation_token()
render_with_one_mutable_state_owner()
retire_old_storage_outside_audio_callback()
```

CHECKS AND DECISIONS

- Report the selected approximation and replacement gate with every compiled profile.

SOURCES S080-02 Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus / S080-03 Electro Music Leslie122/222 owner and service manual, publication044578 / S080-04 JSON Schema Core, Draft2020-12

ORIGINAL ACQUISITION PRIORITIES / M0 TO M8

Acquire measurements that distinguish competing models

Final integration review and research handoff

ORIGINAL ACQUISITION PLAN. Begin with identity, geometry, calibrated electrical/acoustic channels and a shared timing reference. Then measure intermediate boundaries before fitting the complete microphone output. Otherwise several gains, filters and delays may compensate for one another, leaving a plausible recording match with incorrect internal behavior. Preserve repeat takes and reserve whole conditions for validation before fitting begins.

data/photo_capture_checklist.json defines88 acquisition task families; all remain not_acquired. Its images identify parts, revisions and topology. They do not measure a transfer function, transformer polarity, key trajectory or acoustic response simply by depicting the assembly. Pair the appropriate shot IDs below with the recording protocols in chapters 057–061 and the capture/profile schemas.

For each result, report calibration, uncertainty components, estimation method and any stated coverage interpretation. NIST’s historical uncertainty guidance makes those reporting responsibilities explicit. [S080-05] An experiment whose uncertainty is too large to distinguish candidate models is inconclusive; it must not become a pass. The factory G–G service table remains a historical indication, not a newly acquired source calibration.

Priority / task family	Measurement that closes the gap
Identity / ID-01, AO-01	Bind chassis, generator and cabinet revisions to the capture.
Generator / GEN-02…07	Shared phase/frequency, waveform boundary, filtering and leakage.
Console / AO-08, EXP-02…03	Percussion recharge/switching and loaded expression behavior.
Scanner / SCN-02…06	Stationary ports, loading, angular coupling and six modes.
Cabinet / LES-02…07	Amplifier/load interaction, driver impedance and divider response.
Motion / ROT-07, RIG-01…05	Independent trajectories, microphone geometry and held-out recordings.

CHECKS AND DECISIONS

- Every acquisition links raw files, conditions, calibrations and unchanged masters.
- A specimen photograph never closes an unperformed electrical or listening test.

SOURCES S080-05 NIST TN1297,1994edition, Section7: Reporting Uncertainty

EXECUTED REPORT REVIEW / RESOLVED CORRECTIONS

Reconcile completed checks without inflating their scope

Final integration review and research handoff

PACKAGE AUDIT. The reports below were read as evidence of their stated checks, not rerun here as audio tests. The independent final audit found seven double-high-fold assignments: six1-foot cases and the top-C1⅓-foot case. Root corrected the summary metadata, derived the explicit seven-location list and reran the verifier. The source routing values were already correct; this issue is closed.

All91 historical service readings have a completed independent second transcription. Remaining meter range/detector uncertainty is a different issue. The treble-driver/spacer correction and AO-28 PDF 129 locator are also resolved. Keep these corrections in history without repeatedly assigning them as missing research.

The interface review executed actual JavaScript with project-specific DOM/event/storage test doubles and fixed three reproduced defects. Browser rendering remained blocked by process and URL restrictions; portable browser tests are supplied but not executed. The original control/service illustrations are design diagrams. Neither those pictures nor28 state checks certify native accessibility, host automation or an audio renderer.

Evidence file / family	Actual completed scope
058-schema-checks	39 metadata acceptance/rejection cases.
060-verification-results	549 algorithm assignments;144 independent visual cells;91 readings;17 corruptions.
063-contract-checks	C++ syntax and 19 compile-time assertions.
065-registry-checks	62 proposed identities;24 enum round-trips;8 threshold fixtures;8 corruptions.
072-ui-checks	28 state checks; browser visual verification blocked_not_run.
077-checklist-checks	88 unique acquisition task families; structure and references only.

CHECKS AND DECISIONS

- Consult each report’s input hashes and limitations before citing it.
- Synthetic circuit, trajectory and energy fixtures remain mathematical evidence.

ORIGINAL FINAL HANDOFF / M0 TO M9

Start implementation with gates that preserve the evidence

Final integration review and research handoff

ORIGINAL EXECUTION ORDER. Run reference acquisition planning and M1 infrastructure in parallel. Implement the profile compiler, offline renderer and event contract first; then build M2 shared sources and M3 console behavior against the supplied maps. Establish M4 scanner and M5 direct-organ contracts before tuning the integrated M6/M7 cabinet chain; independent cabinet reference work can proceed earlier. M8 fits identified captures; M9 qualifies the product and its release.

Native integration must exercise the actual processor lifecycle, bus arrangements, state transitions and audio/UI-thread responsibilities in the selected SDK. [S080-06] Check the exact library contracts used by the implementation: JUCE convolution loading still requires serialization with processing. [S080-07] A compiled interface or correctly shaped parameter JSON cannot stand in for these runtime tests.

Finish each milestone with its source/profile versions, executable fixture, observed result, uncertainty and remaining issue. Use the acceptance matrix for gate definitions and the risk register for ownership. A provisional engineering build may advance with declared approximations; a specimen-fidelity claim advances only with the corresponding physical evidence. Final document/archive checks are a separate delivery record, not an acoustical result.

Workstream	Advance condition
M0–M1 reference and harness	Identity decisions explicit; valid data and deterministic test paths.
M2–M5 direct organ	Routing, state, coupled circuits and numerical behavior verified at named ports.
M6–M7 cabinet	Electrical and moving-path models retain separate measurable boundaries.
M8 calibration	Identified training/held-out corpus; uncertainty-aware comparisons.
M9 release	Actual hosts, standalone devices, UI, persistence, rights and installers qualified.

CHECKS AND DECISIONS

- No not_run, blocked or inconclusive result is silently promoted to passed.
- The final claim names the tested specimen, conditions, outputs and limitations.

SOURCES S080-06 Steinberg VST3 Developer Portal: Audio Processor Call Sequence / S080-07 JUCE dsp::Convolution class reference

Source directory 1 of 8

Clickable source titles. Full access status, verified page locators, claim support and rights notes are retained in the companion source catalog and individual chapters.

Hammond Organ Company service manual scan with supplementary B3/C3 redraws
S001-01, S002-02, S004-99, S007-02, S021-02, S031-02, S034-02, S035-02 ; +4 IDs in catalog | manufacturer manual

Hammond B-3 mk2 C-3 mk2 and B-3P mk2 Owner's Manual
S001-02, S007-03, S019-02 | manufacturer manual

Hammond SK PRO Custom Sets reference chapter
S001-03, S012-02 | manufacturer manual

Leslie Models 122 and 147 Service Information
S001-04, S042-03, S043-02, S044-02, S047-02, S076-06 | manufacturer manual

Hammond Organ Service Manual Early Models, 220-page scan
S002-01, S004-01, S005-01, S006-01, S007-01, S009-01, S010-01, S011-01 ; +40 IDs in catalog | manufacturer manual

Hammond B3 Service Manual alternate-host compilation
S002-03, S032-02 | manufacturer manual

AO-28 preamplifier factory schematic scan
S002-04 | manufacturer manual

AO-28 underside component layout scan
S002-05 | manufacturer manual

Hammond schematic archive source index
S002-06 | archive index

US1956350A Electrical musical instrument — Laurens Hammond
S003-01, S006-02, S009-02 | patent

US2382413A Electrical musical apparatus — John M. Hanert
S003-02 | patent

US2560568A Vibrato apparatus — Laurens Hammond and John M. Hanert
S003-03, S025-03, S026-03, S076-20 | patent

US2565512A Tone control apparatus for electrical musical instruments — Herbert E. Meinema
S003-04, S017-03 | patent

US2508514A Electrical musical instrument — John M. Hanert and Laurens Hammond
S003-05, S022-02 | patent

US2953055A Percussion tone electrical musical instrument — John M. Hanert
S003-06, S021-03 | patent

US2489653A Rotatable tremulant sound producer — Donald J. Leslie
S003-07, S045-02, S046-02 | patent

USRE23323E Rotatable tremulant sound producer — reissue
S003-08 | patent

Terms of Use for USPTO websites — Patent information
S003-09, S078-01 | official docs

US2828659A Electrical musical instruments — unresolved source candidate
S003-10 | patent

Hammond-museum-B-3.png by Φικcep
S004-02, S070-02 | original photograph

Hammond-museum-B-3-back.png by Φικcep
S004-03 | original photograph

Hammond B3 and Drawbars by ArtBrom
S004-04, S007-04, S070-03 | original photograph

Retouched Hammond B3 pedalboard image by doryfour and Clusternote
S004-05, S070-10, S077-06, S078-14 | original photograph

Original Hammond B3 Museum of Making Music photograph by doryfour
S004-06 | original photograph

Vintage Organs first-hand comparison photographs of original and reconstructed B-3 cabinets
S004-07 | firsthand measurement

Moro, McPherson and Sandler, Dynamic temporal behaviour of the keyboard action on the Hammond organ and its perceptual significanc
S005-02, S018-02, S020-02, S036-05, S040-02, S057-04, S061-04, S076-17 | research paper

Hammond B-3 mk2 Owner Manual, Tone-Wheels and Multi-Contact Keys
S005-03 | manufacturer manual

Hammond B3 manuals photograph, Cliff, cropped by Clusternote
S005-04, S078-15 | original photograph

Source directory 2 of 8

Clickable source titles. Full access status, verified page locators, claim support and rights notes are retained in the companion source catalog and individual chapters.

Augmented Instruments Laboratory, Touch on Keyboard Instruments
S005-05 | firsthand measurement

Hammond synchronous motor and scanner original photograph
S006-03 | original photograph

Hammond shaded pole starter motor original photograph
S006-04 | original photograph

Hammond ottava dei preset photograph
S007-05 | original photograph

Hammond B3 START RUN photograph
S007-06 | original photograph

Electro Music Leslie 122 122V 122RV 142 222 222RV operating and maintenance manual
S008-01, S042-02 | manufacturer manual

Electro Music Leslie Organ Speaker Models 145 and 147 manual
S008-02, S042-01, S048-02 | manufacturer manual

Leslie Speaker 21H owners manual
S008-03 | manufacturer manual

Hammond Leslie Heritage Series 122H and 142H Owners Manual V1.00 220713
S008-04, S041-02, S042-04, S047-05, S076-21 | manufacturer manual

Lesliehorn original photograph by DukeHammond
S008-05, S045-05 | original photograph

Lowerrotor original photograph by DukeHammond
S008-06, S046-05, S070-04 | original photograph

Leslie 122 isometric derivative photograph
S008-07, S070-11, S077-07, S078-18 | original photograph

Captain Foldback Leslie manual scan index and reproduction terms
S008-08, S041-03 | official docs

Werner and Abel Modal Processor Effects Inspired by Hammond Tonewheel Organs
S009-03, S010-03, S012-04, S017-04, S076-16 | research paper

setBfree tonegen.c pinned source implementation
S009-04, S078-09 | official docs

Muenster and Pfeifle, Non-linearities of the mechano-electrical tonegenerator of the Hammond organ, ISMA 2019
S010-02, S013-03 | research paper

Hammond Technical Bulletin 16 Improved Generator Filtering, January 8 1965, reprinted April 1 1966
S011-02, S014-02, S036-01, S077-03 | manufacturer manual

Stefan Vorkoetter, Hammond Organ Tonewheel Generator Capacitor Replacement and Calibration
S011-03 | firsthand measurement

SciPy official documentation, scipy.signal.cont2discrete
S011-04, S029-03 | official docs

Pekonen, Pihlajamaki and Valimaki, Computationally Efficient Hammond Organ Synthesis, DAFx2011
S012-03 | research paper

Pekonen, Pihlajamaki and Valimaki, Computationally Efficient Hammond Organ Synthesis, DAFx 2011
S013-02, S016-01, S018-03 | research paper

JSON Schema draft 2020-12 Validation vocabulary
S014-03, S058-01, S074-08, S079-07 | official docs

RFC 8259 The JavaScript Object Notation Data Interchange Format
S014-04, S058-03 | official docs

RFC 8785 JSON Canonicalization Scheme
S014-05, S058-04 | official docs

Hammond New B-3 and New C-3 Owners Playing Guide
S015-02 | manufacturer manual

JUCE official oscillator implementation and API
S016-02 | official docs

Steinberg VST3 API documentation processing and threading
S016-03 | official docs

JUCE ScopedNoDenormals class reference
S016-04, S062-08 | official docs

Source directory 3 of 8

Clickable source titles. Full access status, verified page locators, claim support and rights notes are retained in the companion source catalog and individual chapters.

Carnegie Mellon Introduction to Computer Music wavetable synthesis
S016-05 | official docs

Laurens Hammond, US1956350A, Electrical musical instrument
S017-02 | patent

JUCE dsp Oversampling class reference
S018-04, S037-04, S062-06 | official docs

Steinberg VST 3 Interfaces: Event structure
S020-03, S024-02, S040-04 | official docs

Steinberg VST 3 Interfaces: NoteOnEvent structure
S020-04, S024-03 | official docs

US2221188A: Electrical musical instrument, Hammond and Meinema
S023-02 | patent

Hammond B-3 mk2 / C-3 mk2 / B-3P mk2 Owner's Manual
S023-03 | manufacturer manual

VST3 IParamValueQueue
S024-04, S052-03, S060-04, S063-03, S064-08, S065-04, S073-04, S074-04 ; +1 IDs in catalog | official docs

VST3 IMidiMapping
S024-05 | official docs

MIDI Association: Summary of MIDI1.0 Messages
S024-06 | official docs

MIDI Association: Control Change Messages
S024-07 | official docs

Werner, Dunkel and Germain: A Computational Model of the Hammond Organ Vibrato/Chorus using Wave Digital Filters, DAFx2016
S025-02, S026-02, S027-02, S028-01, S029-01, S030-02, S039-02, S040-03 ; +6 IDs in catalog | research paper

Chowdhury DSP Wave Digital Filters library and ideal voltage source implementation
S028-02 | official docs

Ngspice User Manual, version43
S028-03 | official docs

Ho, Ruehli and Brennan: The Modified Nodal Approach to Network Analysis, IEEE Transactions on Circuits and Systems 22(6), 504–509,
S029-02, S032-03, S039-03 | research paper

SciPy official documentation: scipy.signal.bilinear
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Ngspice User Manual, version47+ development edition
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General Electric 12AX7, ET-T509B, June1953
S033-01 | manufacturer manual

Tung-Sol 6AU6/6AU6A tentative data, September1956
S033-02 | manufacturer manual

Tung-Sol 12BH7 tentative data, December1950
S033-03 | manufacturer manual

Norman Koren: Improved vacuum tube models for SPICE, Part1
S033-04 | research paper

Norman Koren: Improved vacuum tube models for SPICE, Part2
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SciPy least_squares official reference
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Tung-Sol 12AU7 manufacturer data
S033-10 | manufacturer manual

Yeh and Smith: Simulating Guitar Distortion Circuits Using Wave Digital and Nonlinear State-Space Formulations
S034-03, S039-04 | research paper

RCA Receiving Tube Manual RC30: 6X4/12X4
S035-03 | manufacturer datasheet

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Texas Instruments SBOA345: Understanding Op Amp Noise in Audio Circuits
S035-04 | manufacturer application note

TDK: Capacitors, Part7 — Electrolytic Capacitors [2]
S036-02 | manufacturer technical reference

Vishay Beyschlag: Basics of Linear Fixed Resistors
S036-03 | manufacturer application note

Arnold Magnetic Technologies: Stabilization & Calibration
S036-04 | manufacturer technical reference

Parker, Zavalishin and Le Bivic: Reducing the Aliasing of Nonlinear Waveshaping Using Continuous-Time Convolution
S037-01 | research paper

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Wright, Damskägg, Juvela and Välimäki: Real-Time Guitar Amplifier Emulation with Deep Learning
S039-05 | research paper

Electro Music: Leslie 122/122V/122RV/142/222/222RV General Operating and Maintenance Instructions, 044578, June 1971
S041-01, S043-01, S044-01, S045-01, S046-01, S047-01, S048-01, S052-01 ; +11 IDs in catalog | manufacturer manual

Tung-Sol 6550 original manufacturer data and characteristic curves, 1963 sheets
S043-03 | manufacturer datasheet

Tung-Sol 0A3, 0B3, 0C3, 0D3 voltage-regulator data, August 31, 1944
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Jaromir Macak and Jiri Schimmel, Simulation of a Vacuum-Tube Push-Pull Guitar Power Amplifier, DAFx 2011
S043-05 | research paper

Martin Holters and Udo Zolzer, Circuit Simulation with Inductors and Transformers Based on the Jiles-Atherton Model of Magnetizati
S043-06 | research paper

Smith, Serafin, Abel and Berners: Doppler Simulation and the Leslie, DAFx 2002
S045-03, S048-04, S050-01 | research paper

Clifford A. Henricksen: Unearthing the Mysteries of the Leslie Cabinet, Recording Engineer/Producer, April 1981; author-attributed
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Julius Smith, Stefania Serafin, Jonathan Abel and David Berners: Doppler Simulation and the Leslie, DAFx-02
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R. Kronland-Martinet and T. Voinier: Real-Time Perceptual Simulation of Moving Sources: Application to the Leslie Cabinet and 3D S
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Donald J. Leslie, US3245284A, Rotor drive for pulsato apparatus, April12 1966
S047-03 | patent

Leslie Brake Accessory, instructions038398, for 122/142/222 series
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Donald J. Leslie, US4198880A, Rotor drive for pulsato apparatus, April22 1980
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Bastyr and Capone (2003), On the acoustic radiation from a loudspeaker cabinet
S048-05 | research paper

Julius O. Smith III: Physical Audio Signal Processing, Lagrange Interpolation
S049-02 | research monograph by its author

Perry R. Cook and Gary P. Scavone: STK DelayL implementation documentation
S049-03 | official source code

JUCE: juce::dsp::DelayLine class reference
S049-04 | official docs

Eddy Bøgh Brixen / DPA Microphones: Stereo recording techniques and setups
S051-03 | official docs

Shure SM57 Instrument Microphone User Guide, version 3.6 (2025-A)
S051-04 | manufacturer manual

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S051-05 | official docs

Steinberg VST 3 Developer Portal: Parameters and Automation
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Frank Wefers: Partitioned convolution algorithms for real-time auralization, 2015
S053-01 | research monograph by its author

Øyvind Brandtsegg, Sigurd Saue and Victor Lazzarini: Live Convolution with Time-Varying Filters, 2018
S053-02, S076-22 | research paper

Christoph Müller-Tomfelde: Time-Varying Filter In Non-Uniform Block Convolution, DAFx2001
S053-03 | research paper

JUCE: juce::dsp::Convolution class reference
S053-04, S062-05, S076-15, S080-07 | official docs

Jont B. Allen and David A. Berkley: Image method for efficiently simulating small-room acoustics, JASA65(4),943–950,1979
S054-02 | research paper

Sebastian J. Schlecht and Emanuel A. P. Habets: On Lossless Feedback Delay Networks, author preprint2016, journal publication2017
S054-04 | research paper

Wolfgang Klippel, Large Signal Performance of Tweeters, Micro Speakers and Horn Drivers
S055-02, S076-19 | research paper

Wolfgang Klippel, Nonlinear Modeling of the Heat Transfer in Loudspeakers, author manuscript for JAES52(1/2),2004
S055-03, S076-08 | research paper

Wolfgang Klippel, Nonlinear Adaptive Controller for Loudspeakers with Current Sensor
S055-04, S076-10 | research paper

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S055-05 | manufacturer documentation

US National Bureau of Standards, Copper Wire Tables, Handbook100, February21,1966
S055-06, S076-09 | government technical reference

Angelo Farina: Simultaneous Measurement of Impulse Response and Distortion with a Swept-Sine Technique, AES108, paper 5093, 2000
S056-03, S057-05 | research paper

NIST Technical Note 1297: Section 5, Combined Standard Uncertainty
S056-04, S069-03, S073-06, S075-04 | official docs

ITU-R BS.1116-3: Methods for the subjective assessment of small impairments in audio systems
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Hustvedt: Leslie Speaker.jpg, original photograph of an acrylic cabinet
S056-06 | original photograph

Tektronix: How Oscilloscope Probes Affect Your Measurement
S057-02 | manufacturer manual

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S057-03, S059-04 | official docs

NTi Audio: Room Acoustics Measurements
S057-06 | official docs

JSON Schema: Conditional schema validation
S058-02 | official docs

GitHub Docs: About Git Large File Storage
S058-05 | official docs

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S058-06 | official docs

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Lennart Ljung, Identification of Linear and Nonlinear Dynamical Systems, Theme 4: Practical Aspects (Berkeley, 2005)
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JUCE AudioProcessor class reference
S060-02, S062-02, S063-01, S064-13, S073-05 | official docs

Steinberg VST3 API Documentation
S060-03, S064-05, S065-06, S076-23 | official docs

John Salmon: Parallel Random Numbers, As Easy as1,2,3 / Random123
S060-05 | official docs

ITU-R BS.1534-3: Method for the subjective assessment of intermediate quality level of audio systems
S061-02 | official docs

Daniël Lakens: Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses
S061-03, S079-02 | research paper

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Dale J. Barr, Roger Levy, Christoph Scheepers and Harry J. Tily: Random effects structure for confirmatory hypothesis testing: Kee
S061-05 | research paper

Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven and David T. Mellor: The preregistration revolution
S061-06 | research paper

PortAudio: Writing a Callback Function
S062-01, S069-06, S074-05, S075-06 | official docs

Boost Lockfree: spsc_queue class template
S062-03 | official docs

C++ working draft: Lock-free property [atomics.lockfree]
S062-04 | official docs

JUCE dsp::SIMDRegister class reference
S062-07 | official docs

PortAudio public API: portaudio.h
S062-09 | official docs

Steinberg VST3 Processing FAQ
S063-02, S064-06, S079-05 | official docs

PortAudio: Writing a Callback Function
S063-04 | official docs

C++ Core Guidelines
S063-05 | official docs

Steinberg: VST 3 License
S064-01, S066-03, S074-07, S075-08, S078-08 | official docs

Steinberg: SDK change history
S064-02, S066-13 | official docs

JUCE official repository LICENSE.md
S064-03, S066-04, S074-06, S075-07, S078-06 | official docs

JUCE 9 End User Licence Agreement
S064-04, S066-05, S078-07 | official docs

VST 3 ParameterInfo reference
S064-07, S065-02, S068-02, S071-11, S074-09, S076-11 | official docs

VST 3.8.0 release notes
S064-09 | official docs

VST 3 AudioBusBuffers reference
S064-10 | official docs

VST 3 IAudioProcessor reference
S064-11 | official docs

JUCE AudioDeviceManager reference
S064-12 | official docs

JUCE official CMake API
S064-14 | official docs

JUCE MidiMessageCollector reference
S064-15 | official docs

VST3 IComponentHandler reference
S065-05, S068-07 | official docs

JUCE AudioProcessorValueTreeState reference
S065-07 | official docs

JUCE ParameterID reference
S065-08 | official docs

JUCE CMake API
S066-01 | official docs

CMake presets manual
S066-02 | official docs

VST3 plug-in locations
S066-06 | official docs

VST3 plug-in format structure
S066-07 | official docs

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Apple: notarizing macOS software
S066-08 | official docs

Apple: customizing notarization
S066-09 | official docs

Apple: universal macOS binaries
S066-10 | official docs

Microsoft SignTool
S066-11 | official docs

Steinberg VST3 validator
S066-12, S073-07, S075-05, S079-04 | official docs

Apple: creating distribution-signed code
S066-14 | official docs

W3C ARIA Authoring Practices: Slider Pattern
S067-02, S068-03, S071-04, S072-01 | official docs

W3C ARIA Authoring Practices: Tabs Pattern
S067-03, S068-04, S072-02 | official docs

W3C Understanding WCAG2.2: Animation from Interactions
S067-04, S072-04, S079-06 | official docs

W3C Pointer Events Level3 Recommendation
S068-05 | official docs

W3C PROV-DM: The PROV Data Model
S069-02 | official docs

JUCE AbstractFifo class reference
S069-04 | official docs

JUCE UndoManager class reference
S069-05 | official docs

W3C Understanding WCAG2.2 SC4.1.3: Status Messages
S069-07 | official docs

W3C Understanding WCAG2.2 SC1.4.1: Use of Color
S069-08 | official docs

Khronos glTF 2.0 specification, official AsciiDoc source
S070-05, S076-14 | official docs

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S070-06 | official docs

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S070-07 | official docs

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S070-09 | official docs

WCAG 2.2 — W3C Recommendation, 12 December 2024
S071-01 | official docs

JUCE AccessibilityHandler
S071-02 | official docs

JUCE AccessibilityValueInterface
S071-03 | official docs

WAI-ARIA APG — Developing a Keyboard Interface
S071-05 | official docs

W3C Understanding SC2.5.8 — Target Size Minimum
S071-06 | official docs

W3C Understanding SC1.4.10 — Reflow
S071-07 | official docs

Steinberg IPlugView interface
S071-09 | official docs

JUCE AudioProcessorEditor
S071-10 | official docs

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W3C Pointer Events Level 3
S072-03 | official docs

Node.js VM API
S072-05 | official docs

Playwright visual comparisons
S072-06 | official docs

FADGI Technical Guidelines for Digitizing Cultural Heritage Materials, third edition, May2023
S077-04 | official docs

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S077-05 | official docs

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S078-03 | official docs

Creative Commons CC0 1.0 Universal
S078-04 | official docs

Creative Commons Attribution-ShareAlike2.0
S078-10 | official docs

Creative Commons Attribution-ShareAlike3.0
S078-11 | official docs

Creative Commons Attribution2.5
S078-12 | official docs

Creative Commons Attribution3.0
S078-13 | official docs

P016 transformed CV generator detail
S078-16 | original photograph

P017 transformed CV generator underside
S078-17 | original photograph

P037 organ interior source alteration record
S078-19 | original photograph

P035 Tung Sol6550 creator and release
S078-20 | original photograph

DejaVu upstream font license
S078-21 | official docs

P012 removed warning board source record
S078-22 | original photograph

NIST TN 1297, section 7: Reporting Uncertainty
S079-01, S080-05 | official metrology guidance

JSON Schema Core, Draft2020-12
S080-04 | official docs

Steinberg VST3 Developer Portal: Audio Processor Call Sequence
S080-06 | official docs